

BobCAD-CAM

V21

2D & 3D CAD-CAM



OPERATORS MANUAL

BobCAD-CAM V21

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION	9
Section 1. Welcome	10
Introduction to BobCAD-CAM Software	10
Section 2. Technical Support & Software Maintenance	10
Technical Support ON-Demand	10
Section 3. More on Technical Support	11
CAM & Post Processor Questions	11
Toolpath Related Questions	11
CAD File Import and Export Questions	11
General Software Related Questions	11
Support Forum on the web	11
Section 4. Advanced Training	12
Section 5. System Requirements & Software Installation	12
Installation Procedure	12
CHAPTER 2. OPENING THE SOFTWARE & WORKING WITH FILES	15
Section 6. Opening the software	16
New File	17
Open File	18
Section 7. Importing External CAD Files	20
DXF	20
IGES	21
STEP	22
DWG	22
SAT/ACIS	23
Rhino CAD 3DM	23
More file types	24
CHAPTER 3. SOFTWARE INTERFACE	25
Section 8. The basic software interface	26
Main Menus	26
Main Screen Axis	27
Section 9. Customizing the main toolbar	30
Customizing the toolbar lesson	31
Section 10. Selecting entities and solids or surfaces	32
Section 11. Modify Mode	37
 Section 12. The UCS – User Coordinate System	40

UCS training lesson	47
More on the UCS	54
Section 13. The software icons	57

CHAPTER 4. The File menu through the Other menu	61
Section 14. The File menu	62
Environment	64
Tips on chain gap	69
Section 15. The Edit menu	76
Select Entities	76
Select Elements in a dialog	78
Section 16. The Change menu	81
Mirror	81
Tips on snapping entities	85
Rotate.....	86
Translate	88
Reorganize	90
Trim & Extend	92
Break	94
Attributes	95
Dimension	100
Section 17. The Point menu	106
On Entity	108
Section 18. The Line menu	109
Coordinates	110
Join	110
Tips on joining lines	110
Angle	113
Angle with Z axis	116
Chamfer – wireframe	117
Continuous	119
Section 19. The ARC menu	120
Sketch	121
Coordinates	122
Snap Center	123
Tips on snapping arcs	123
Fillet – wireframe	124
3 - entities	128
Ellipse.....	130
Section 20. The Spline menu	130
Cardinal	132
Interpolated	133
Approximate	133
Fit	133
Section 21. The Other menu	135
Rectangle	136
Mode & Corner Type	136
Offset	137
Using Offset Geometry as Toolpath	138
Offsetting multiple shapes	138
Offset circle lesson	140

Hatch	141
Hole Pattern	143
Hole Bolt Center	143
Hole Grid Pattern	145
Machining Hole Patterns	146
Pocket	149
Quick Horizontal Pocket Lesson	150
Spiral Pocket	152
Quick Spiral Pocket Lesson	154
Tapered Wall Pocket	162
Tapered Wall Pocket Lesson	164
Spiral pipe thread	175
Gear	176
Gear Training Lesson	177
Simulate – basic 2D	178
Spline	179
Approach & Depart	180
Face	183

CHAPTER 5. SOLIDS	185
Section 22. Primitives Defined	186
Section 23. The Solids menu	187
Section 24. Primitives	188
Sphere	188
Basic Sphere lesson	189
Cube	190
Basic Cube Lesson	191
Cone	192
Basic Cone Lesson	192
Cylinder	194
Basic Cylinder Lesson	194
Torus	195
Basic Torus Lesson	195
Section 25. Surface Creation	196
Surface Creation	196
Rectangular Plane	197
Circular Plane	198
Planar Surface	198
Extrude Curve	200
Extrude Surface	202
Revolved	203
3 Edge	209
4 Edge	210
Sweep	211
Cross Section	211
Skin	216
Multi-Sided Patch	218
Skin Surface & Toolpath with G-Code Training Lesson	219
Cross Section Surface Training Lesson	226
Extrude Curve Training Lesson	231
Project Curves on Surfaces	233

Surface Modifications	235
Section 26. Boolean Operations	237
Union	237
Basic Union Lesson	238
Difference	240
Basic Difference Lesson	241
Intersection	243
Basic Intersection Lesson	243
Section 27. Solids Attributes	245
Section 28. Basic CAD Design & Lesson	247
CAD Design Lesson 1 & 2	247
Section 29. Fillets & Chamfer Surfaces	263
Constant Radius Fillet	263
Variable Radius Fillet	264
Variable Radius Fillet Training Lesson	265
Independent Fillet	269
Constant Chamfer	270
Independent Chamfer	271
Section 30. Generate Toolpath	272
Planar Finish Options	272
Radial Toolpath	278
Spiral Toolpath	284
Section 31. Solid Tool Depth Settings	287
Section 32. Quick CAM Training Lesson	289
Section 33. Offset Surfaces & Solids	291
UCS Note	294
Section 34. Stitching or Un-Stitching Surfaces & Solids	295
Stitch Surfaces To Solid	295
Un-Stitch Solid To Surfaces	296
Un-Stitch Surface from Solid	296
Section 35. Extract Edges	297
Extract All Edges – Arcs and Lines	297
Extract All Edges as Lines	299
Extract All ARC Edges	299
Extract Edge Training Lesson	300
Section 36. Remove Holes	306
Driver Setup for Drilling	310
Section 37. Render	311
Section 38. Questions & Answers	315
CHAPTER 6. THE 3D MENU	317
Section 39. The 3D menu	318
Equalize	319
Skin	320
Skin Surface Box	322
Connect	327
Tip on 3D Surfacing	329
Set Surface	329
Cut Above Z=0	333
Connect at Z	333
Z From Point Y	336

Move to Z	336
Cross Section	337
Interpolate	340
Surface	340
Ruled	340
Auto Ruled	341
Follow Curve	342
Radial	343
Revolution	344
CHAPTER 7. 3D PROJECTION & SKIN SURFACE LESSONS	347
Section 40. 3D Projection Lesson	348
Section 41. 3D Skin Surface Lesson	355
CHAPTER 8. THE DIMENSION MENU	359
Section 42. Horizontal	360
Section 43. Vertical	361
Section 44. Tips on modifying a dimension	361
Section 45. Other Distance	362
Section 46. Angle	363
Section 47. Radius	363
Section 48. Auto Dimension	363
Section 49. X Ordinate	364
Section 50. Y Ordinate	364
Section 51. Move/Align	365
Section 52. Chain	365
Section 53. Reference	365
Section 54. To Lines + Text	366
Section 55. Setup	367
Section 56. Text	368
Section 57. Vectorize Text	369
Section 58. Fit Text to Path	370
Section 59. Tips on Text Along a Curve	371
Section 60. Single-Line fonts & Modifying Text	372
Modifying Text	373
Section 61. Creating Solid Fonts	374
Section 62. Working with Text Training Lesson	376
CHAPTER 9. THE SPECIAL/NC-CAM MENU	381
Section 63. The Special/NC-CAM menu, Verify & Close	382
Section 64. Generate NC & Insert NC	383
Insert NC	384
Section 65. Axis	385
Reset	387
Section 66. Selection	390
Section 67. Script & Record Script	395
CHAPTER 10. NC-CAM	397
Section 68. NC CAM Basics & Icons	398
Section 69. Customizing the CAM icons & adding Toolboxes	401
Additional NC CAM Toolboxes	402
Insert NC Object	403

Section 70. NC CAM main menus	404
Section 71. The CAM File Menu	404
Section 72. Send File	407
Section 73. Receive File	410
Section 74. The CAM Edit menu	411
Section 75. Simulate	415
Section 76. The Machine menu	417
Profile Wizard	418
Profile Training Lesson	418
Section 77. The Contour menu	424
Section 78. The Cycle menu	426
Section 79. Customizing Drill Cycles Lesson	427
Section 80. The CAM Tool menu	430
Section 81. The CAM Macro menu	432
Section 82. The CAM Setup menu	434
Configuration	434
Driver	435
ISO	438
Coordinates	439
Section 83. Standard RS 232 Setup Checklist	442
Section 84. The Coordinate Format Checklist	445
Section 85. The Conversion Function	447
Section 86. Rotary Conversion	450
Section 87. 4th Axis Indexing Lesson	452
Section 88. The CAM Help menu	461
CHAPTER 11. CNC LATHE	463
Section 89. CNC Lathe Basics	464
Tool Merge & Tool Setup	464
Merging & Setting up Lathe Tools	465
Lathe Tools Library & Customizing a Tool	467
Lathe Tool Lesson	469
Quick Lathe Lesson 1: Simple O/D Profile	477
Using Scripts	482
Quick Lathe Lesson 2: Simple I/D Profile	484
Quick Lathe Lesson 3: Simple Facing Operations	487
Quick Lathe Lesson 4: O/D Profile w/Built-in Facing	490
Using Scripts	496
Quick Lathe Lesson 5: Internal Lathe Cutting	497
CHAPTER 12. TROUBLE SHOOTING & FAQ	503
Section 90. Troubleshooting BobCAD-CAM	504
CHAPTER 13. SCRIPTING	513
INDEX	541

CHAPTER 1

INTRODUCTION

Section 1. Welcome

Section 2. Technical Support & Software maintenance

Section 3. More on Technical Support

Section 4. Advanced Training

Section 5. System Requirements & Software Installation



Section 1. Welcome

An Introduction to BobCAD-CAM Version 21 Software

Welcome to the BobCAD-CAM Version 21 software and operator manual. BobCAD-CAM software has been developed with the manufacturer in mind, offering powerful CAD *and* CAM functionality for both simple and complex part making. This operator manual includes function descriptions, explanations and special examples on how to use the software.

Section 2. Technical Support & Software Maintenance

BobCAD-CAM offers all customer technical services & software support free of charge by phone, fax and by email for your first 45-days as a new customer.



BobCAD-CAM **Technical Support** representatives are standing by to assist you in using your Version 21 CAD-CAM software. Technical phone support is free for the first 45-days once you receive your software as a new customer. Available hours are **9AM – 6PM Monday – Friday Eastern Standard Time**. Technical support will be closed on all major holidays recognized in the United States.

You can reach a technical support representative by calling **727-489-0003**.

If you are **faxing** documentation regarding a support matter, do so by dialing **727-734-8239 ATTN: Technical Support**. BobCAD-CAM also offers advanced technical support services:

BobCAD-CAM TECHNICAL SUPPORT **ON-DEMAND**

For continued support and software maintenance beyond your first 45-days, we recommend the **Technical Support ON-Demand** service program for all customers. Special benefits include:

- Extended hour telephone support.
- Saturday telephone support from 10AM – 2PM EST.
- Private technical support email service.
- CAD guidance.
- CAM, NC Configuration & Toolpath guidance.
- Same Version updates and system enhancements.

Technical Support ON-Demand is an annual 1-year membership program starting from the date that you sign up and is recommended to all customers. To sign up for the Technical Support ON-Demand program **call 877-262-2231 or 727-442-3554**.

Technical Support:

Phone: 727-489-0003

Fax: 727-734-8239

Email: support@bobcad.com

Section 3. More on Technical Support

POST PROCESSOR & CAM RELATED QUESTIONS:

1. Before calling, fax us the name of the machine controller being used, controller model and a sample NC Code/G-Code program for that specific controller model. This allows our support team to have all of the information on record when you call in and will speed up the process of answering your questions.
2. Make sure that when you call in, you have your CNC machine handbook or user guide ready.
3. Make sure that you have your BobCAD-CAM Version 21 CAD-CAM software open and ready for use.
4. Now call the BobCAD-CAM technical support number listed above.

By following the above procedure, you will spend less time on the phone and our technical support representatives will be able to service you better. **IMPORTANT:** Please follow this procedure for all **RS232 Communications questions** as well. **NOTE:** In the case that you are looking for your post processor, you may want to visit www.bobcad.com. Once there, you can click on the **SUPPORT** link and will find a list of downloadable post processors by clicking the **V18, 19 & 20 Series Post Processors** link.

TOOLPATH RELATED QUESTIONS:

If you are having trouble with generating or editing toolpath for making an NC Code/G-Code program, please follow the procedure below.

1. Before calling, email support at support@bobcad.com. Include a description of what the problem is, how you generated the toolpath and include the .cad file from the Version 21 software along as a file attachment. You may also want to include the original DXF, IGES or STEP file as necessary. This allows our support team to have all of the information on record when you call in and will speed up the process of answering your questions.
 2. Make sure that you have your BobCAD-CAM Version 21 CAD-CAM software open and ready for use.
 3. Now call the BobCAD-CAM technical support.
-

CAD FILE IMPORT/EXPORT RELATED QUESTIONS:

If you are experiencing difficulty with an external CAD file that you are attempting to import into the BobCAD-CAM Version 21 software, follow the procedure below.

1. Before calling, email support at support@bobcad.com. Include a description of what the problem is, how you generated the toolpath and include the CAD file as a file attachment. This allows our support team to review the problem file and give us all of the information that we will need to speed up the process of answering your questions.
 2. Make sure that you have your BobCAD-CAM Version 21 CAD-CAM software open and ready for use.
 3. Now call the BobCAD-CAM technical support.
-

GENERAL SOFTWARE RELATED QUESTIONS & WEB:

BobCAD-CAM now offers **Online support** at www.bobcadsupport.com. This technical support forum has been created for our customers to post questions and review related questions and answers from earlier postings. The chance is high that your technical question has already been answered and is available on this technical support forum. By selecting the REGISTER link, you will be able to submit a request for a user name and login password. Once you have submitted your request, within a few minutes you will be emailed back a password enabling you to join the forum.

Section 4. Advanced Training



BobCAD-CAM schedules special 3-day **training seminars** all over the United States. For schedule dates and locations please contact the BobCAD-CAM department of operator training at **877-262-2231** or **727-442-3554**. You can also access schedules and information by visiting our website at **www.bobcad.com**.

BobCAD-CAM now offers **one-on-one web-based training** sessions as well. This new service allows you to be trained by a trainer in the comfort of your own office. You can schedule a one-on-one training session by contacting the training coordinators at 877-262-2231 or 727-442-3554. Trainers are standing by to help you get the most out of your new BobCAD-CAM Version 21 software. Thank you.

Section 5. System Requirements & Software Installation

Minimum System Requirements:

- 900 MHz Processor.
- 256 MB Ram.
- 60 MB Available Hard Disk.
- Windows 2000, ME or XP
- 64 MB Video Card.

Recommended System Requirements for installation:

- 1.4 GHz Processor or higher.
- 512 MB RAM.
- 60 MB Hard Disk.
- Windows 2000, or XP
- 128 MB Video Graphics card or higher.

The Version 21 software is a solids and surface modeling system. If you are using a computer that has less RAM than the minimum requirements listed above, you may experience delays in executing functions that relate to rendering models, generating toolpath and G-Code programs. Understand that these delays are **not** caused by the BobCAD-CAM software. By upgrading your computer to the recommended requirements above, you will be able to execute those functions more efficiently.

Version 21 Software Installation Procedure:

The Version 21 CAD-CAM system is available on CD only.

IMPORTANT INSTALLATION NOTE: It is important that you decide which computer you will be loading the Version 21 software onto first. Version 21 is protected through a password system. Therefore, when you load the software you will be asked to contact BobCAD-CAM to register your system. Each additional site license or new computer installation requires a new password-unlocking code to register and fully use the system without limitations.

Whether you have purchased an individual site license or multiple site licenses, it is important that you decide which computer or computers you will need to load the Version 21 software onto first before registering the software.

STEP 1

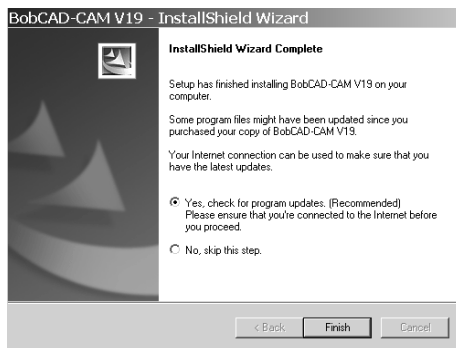
Insert the BobCAD-CAM Version 21 CD in your CD drive. The software will perform an auto-install.

STEP 2

Go through the installation wizard. Make sure that you understand the licensing agreement and click the **NEXT** button once you agree to the terms.

STEP 3

BobCAD-CAM Version 21 offers a unique **SOFTWARE UPDATE** option that will allow you to check for same version updates. These updates can include software enhancements as well as any fixes that BobCAD-CAM has performed on the software. It is recommended that you select the YES option to complete the installation process. Once you have done this and selected the FINISH button, the software will perform this action to see if there is an update already existing on the internet. If there is, the software will tell you what the revision is offer you the ability to update your existing version so that you have the newest revision of the Version 20 software.



NOTE: If you do not have internet access on the computer you will be running Version 21 on, you may visit our support site for downloading updates at www.bobcadsupport.com from another computer that *does* have internet access or simply contact BobCAD-CAM at 877-262-2231 or 727-442-3554.

If there are **NO** revisions available once you have completed the installation, a box will appear telling you so.



NOTE: This SOFTWARE UPDATE process will **ONLY** occur automatically after the initial installation process if you check the option to perform the update upon completion of the installation process. Therefore, you will **NOT** have this automatically occurring each time you open the BobCAD-CAM software after having closed out of the program.

The alternative way to check for updates is to select START from your main taskbar, PROGRAMS, BobCAD-CAM V21 folder and then the **“Check for Updates”** function.

Alternate sources for **update information**:

BobCAD-CAM Directly: 877-262-2231 or 727-442-3554.

Internet: www.bobcad.com or www.bobcadsupport.com

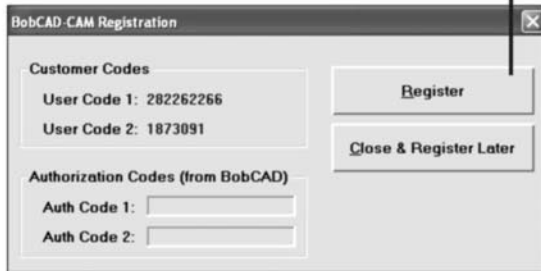
Technical Support: 727-489-0003

STEP 4

When you open the BobCAD-CAM software for the first time you will have the option of running the software in **DEMO MODE** or to **REGISTER** your system.



Run Demo allows you to evaluate the Version 21 software with limitations for 30 days. You will not be allowed to copy, paste or generate g-code programs with more than 1000 lines of code. The RS 232 communications will be disabled until you register.



REGISTER allows you to access User key codes that we will need to provide you with password-authorization codes. You will need to contact BobCAD-CAM at 877-262-2231 or 727-442-3554 to purchase the software and receive these codes during normal business hours (9AM - 5PM Eastern Standard Time.)

You can also fill out your registration form and fax it to BobCAD-CAM at **727-442-1773**. We will then contact you with the Authorization Codes. By selecting the Close & Register Later button, you will open the software in DEMO MODE.

CHAPTER 2

OPENING THE SOFTWARE & WORKING WITH FILES

Section 6. Opening the Software
Section 7. Importing External CAD Files

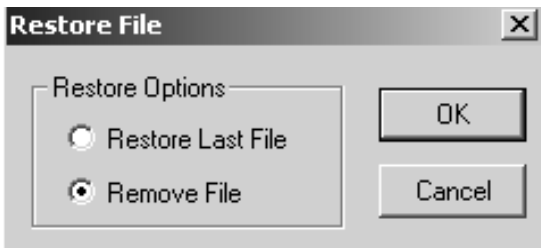


Section 6. Opening the Software.

IMPORTANT: When you open the BobCAD-CAM Version 21 software for the first time, the system will open both the CAD *and* the CAM screens together. This is the **ONLY** time that this will occur.

The **second time** and each time after that you open the BobCAD-CAM version 21, the software “splash screen” will appear and then you will see an empty screen with the FILE and HELP menus available and there will also be a “**Restore File**” box. This box offers 2 options:

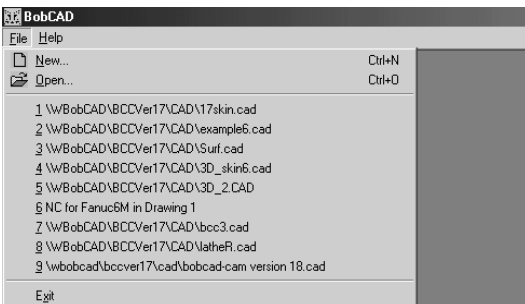
- **Restore Last File:** This allows you to restore the last file to the last point saved or auto-saved in the case that your computer goes down.
- **Remove File:** This allows you to remove the last file from your temp directory so that you do not have too many files building up there.



Once you click the OK button in the File Restore box you will have the NEW File box appear on the screen available for selection. If you click the CANCEL button you can go directly to the FILE NEW or FILE OPEN menus at the upper left corner of this screen. The NEW FILE box appears as a short-cut to opening the software with a new CAD screen or to opening one of the pre-made blue print templates that V21 has to offer. You also have an option in this box to open an NC File only rather than a CAD file.



By selecting the File menu, the **File submenu** will drop down allowing you different options.

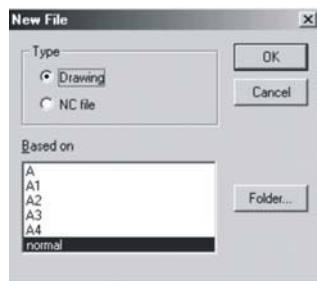


By selecting FILE you will be able to choose different options. They are:

- **New file.**
- **Open file.**
- **Select from the last 9 files opened.**
- **Exit BobCAD-CAM Version 21 software.**

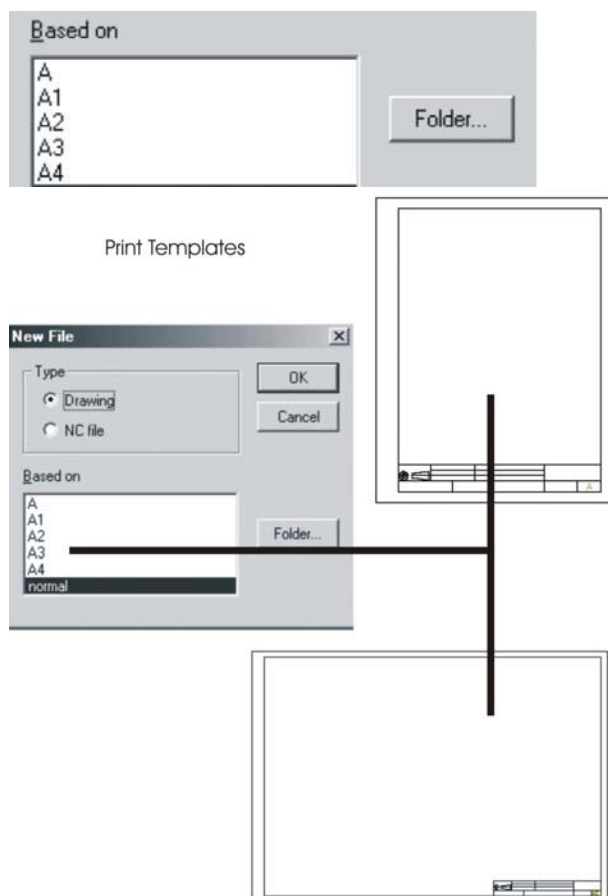
Let's take a closer look at these options.

NEW FILE: By selecting NEW, a second dialog box will appear asking you for the TYPE of file you would like to begin with.



- **Type:** This allows you to select a new “**NORMAL**” drawing screen format under “**Based on.**” By clicking the **FOLDER** button, you can browse your files for the drawing template that you want. However, by default they will be listed in this box if you saved your templates (*.cct files) to the templates folder under your main BobCAD-CAM V21 folder.

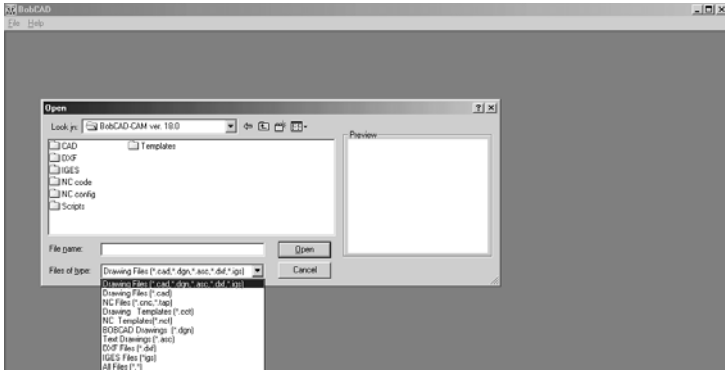
The standard 5 **Design Templates** that are included with the V21 software are: **A, A1, A2, A3 & A4**. These special templates can be used for making prints. You can modify or edit them as necessary. To open a template, select the one that you want and then select OK.



If you would like to have your own **print template**, first design it and then save the file as a **.cct file in your BobCAD-CAM V21 Templates folder**. They will then appear in the new file box as a custom template.

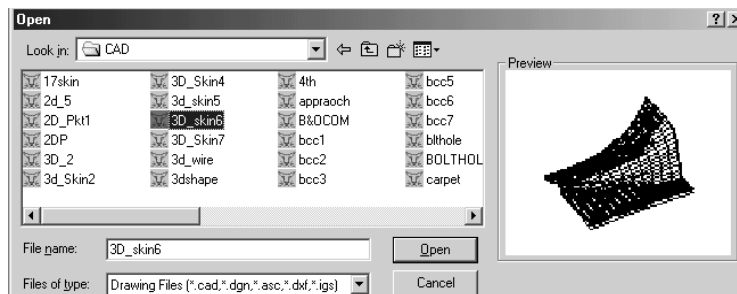
NC File: By selecting this button, the “**Based on**” section will change to list saved NC files that you have. If you are opening the software for the first time, “**NORMAL**” will be highlighted and will be the only option available. Then, by selecting OK, the NC CAM side of the software will open and the CAD side will NOT. This option has been included for fast access to your NC CAM files without having the CAD side open at the same time. As with the drawing selection, you can select the **FOLDER** button to access NC files that you may have on your computer.

OPEN FILE: By selecting OPEN from the file menu, the OPEN dialog box will appear. This will allow you to locate the file folder and files that you are looking for.



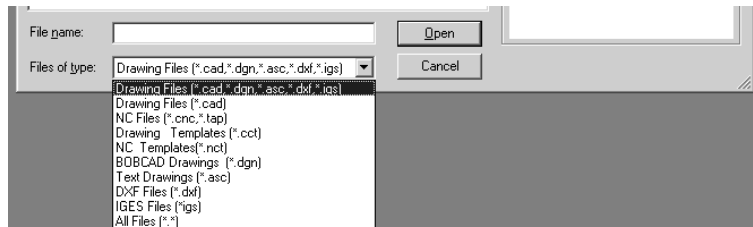
This dialog will list the default folders that come with the BobCAD-CAM Version 21 software. They are **CAD (Native BobCAD-CAM), DXF, IGES, STEP, DWG, SAT, ACIS, 3DM (Rhino CAD), SLDPRT (SolidWorks), Parasolids Files X_T & X_B, NC Code, NC Configurations, Scripts and Templates**. Double click on the folder that you want to look inside and the files will all be listed. Let’s take a closer look at each aspect of the OPEN file dialog.

- **Look In:** This is a drop down menu that will provide you with what is called a “TREE” giving you access to your directories, folders and files. This is standard Windows functionality that you should know. By selecting the folder of your choice, all of the files will be automatically listed in the window below “look in” and will be available for to open a file. For example, in the image above, you can see that the CAD file is listed under BobCAD-CAM version 21. By double clicking on that folder, all of the .cad files will be listed in the view window.



Then by clicking on the file, you can see a preview of it before opening the file altogether. **Remember**, if you change the **FILES of TYPE** to a different type of file, you will only be able to view and open those types of files.

- **File Name:** This area allows you to type in the name of an existing file to access it or when you click on a file the name will automatically appear in this area.
- **Files of Type:** This is a drop down menu listing all of the types of files that can be opened in the BobCAD Version 21 software.



There are the **drawing file extensions**, .CAD (which is native to BobCAD software as in earlier versions), .dgn, .asc, .dxf, .stp, .dwg, .sat, 3dm, .sldprt, .Parasolids X_T & X_B and .igs. There is also .cct (template file extension), .nct (NC file extension) and all files *.* at the bottom. If you are working with drawing files inside of BobCAD Version 21, it would be easy to simply use the “drawing files” type as it includes all of the basic files BobCAD can open. Remember, for example if you select .cct as the type of file you will be looking for, you will only be able to view and select those types of files etc.

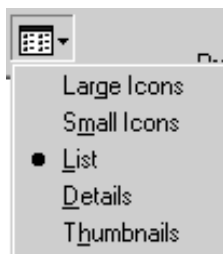
And so, if you find yourself wondering where your files are, make sure you have those types of files selected. To open the file, select the OPEN button or double click on the file with your mouse.

- **File Controls:** These controls/buttons allow you fast access in locating file folders and individual files.



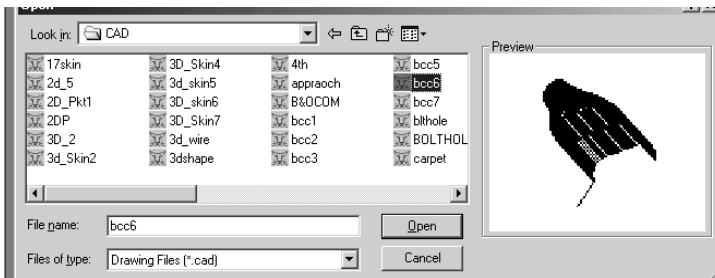
If you are already familiar with Windows technology, you will know how to use these buttons. Let's have a brief look at each one of them.

1.  This takes you back to the last folder that you visited during your search. It will not be active unless you have previously searched through your directory.
2.  This button allows you to automatically search “Up One Level.”
3.  This allows you to create a new folder.
4.  This is the “View Menu.” The selecting the drop down arrow, you will be able to choose how you would like your files to be listed in the open dialog box. See this submenu below.

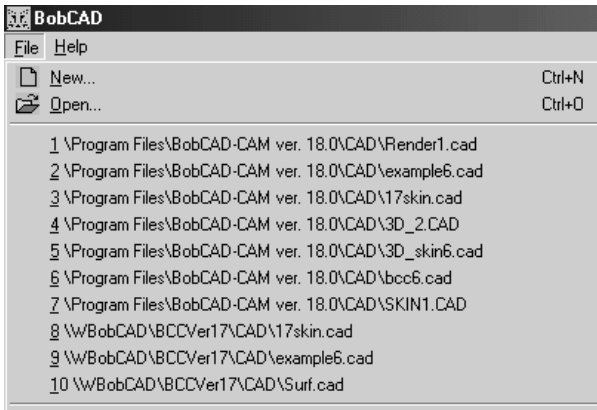


Preview Window: When you single-click on a file in one of the BobCAD drawing file folders, a preview of the drawing will appear before you actually open the file itself.

Single-click on the file to see the preview.

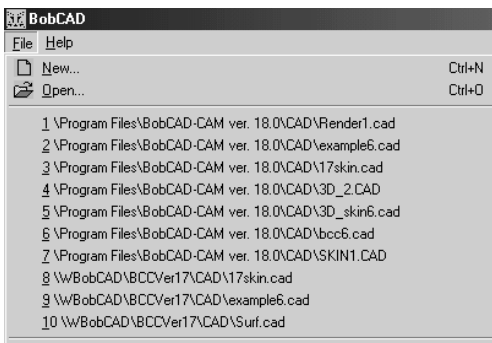


- **The last opened files:**



This allows you to open one of the last several files that you had earlier opened, faster than searching through your folders for it. The directory, program, file type and the file name will all be listed next to the number. If you are familiar with the names of the files you are working with, you can gain speed here.

- **Exit:** By selecting “exit” you will automatically close out the BobCAD Version 21 system entirely. If you actively have a drawing open in the software and select “exit,” the software will ask you if you would like to save the opened drawing first. This way, you will not lose active drawings.



IN ADDITION: **CTL+N** allows you to use your keyboard to select File>NEW. **CTL+O** allows you to use your keyboard to select File>OPEN.

Section 7. Importing External CAD Files

The Version 21 software offers you the ability to import the following files:

- **DGN Files:** This is an early BobCAD-CAM software file. Wireframe only.

DXF Files: You can import *.dxf, DXF files from external CAD systems. If you are attempting to import an AutoCAD DXF file. First explode the file into points, lines and arcs while still in AutoCAD. Then export the file as a Release 12 or 13 AutoCAD DXF file. Then open it in the BobCAD-CAM software directly by choosing **OPEN** from the V21 **FILE** menu. The file will open automatically supporting layers, colors and exploded text or exploded dimensions.

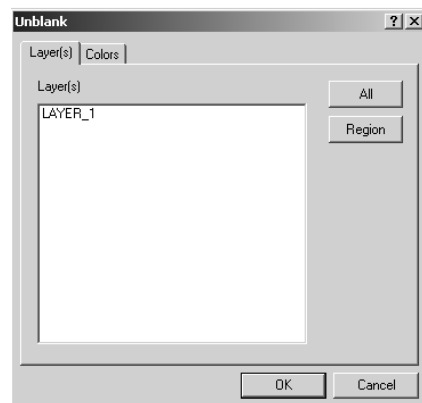
TIP regarding importing DXF geometry:

If you open the file and do not see the geometry you can click on the **VIEW ALL** button from the main CAD toolbar to position the CAD screen so that you can see everything. This is the button on the main toolbar that has the letter “A” inside of a magnifying glass.

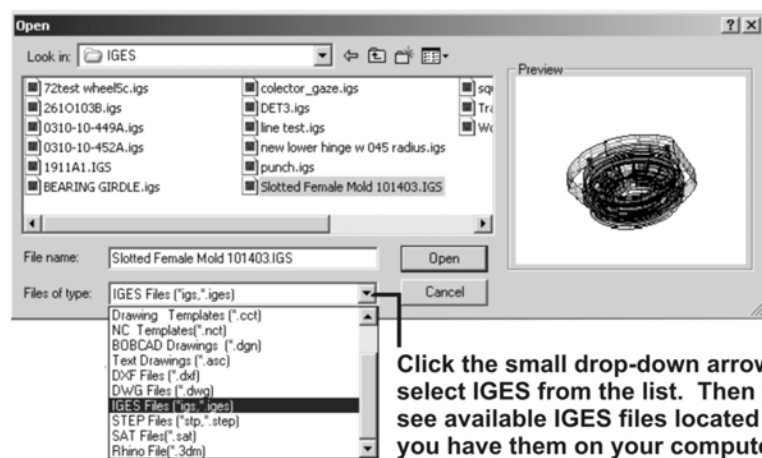
The “View ALL” button:



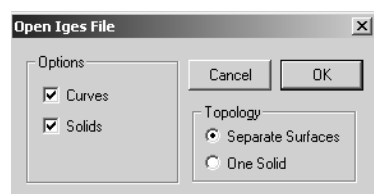
There may also be certain instances of imported geometry being blanked out. To un-blank geometry, go to the CHANGE Menu and check to see if the un-blank icon button is active (not grayed out) and click on it if it is active. If it is active and you click on it, the un-blank box will appear offering you access to blanked out geometry, layers and colors that may be blank.



- **IGES Files:** Version 21 uses modern IGES translators. Version 21 supports *.igs files in wireframe, Surface files and Solid Model formats. To import an IGES file, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, “**FILES OF TYPE**” select the IGES (*.igs) file format and then locate the IGES file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file.

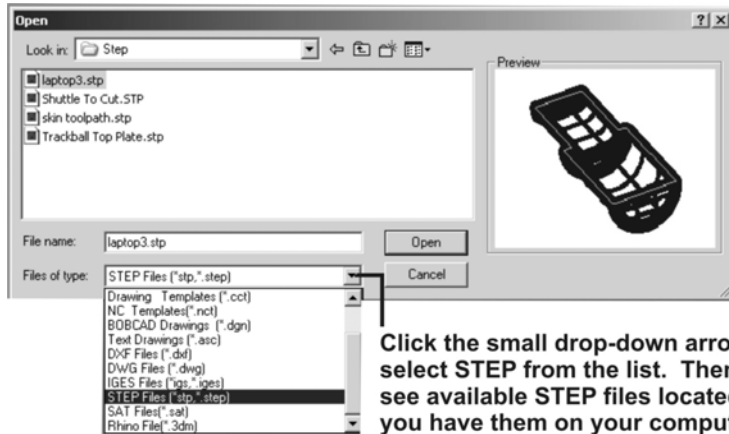


Once you select the file and then the OPEN button you will see the “Open IGES FILE” box on the screen offering you options for opening these types of files.



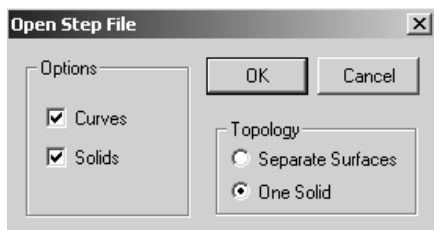
The **IGES** import features offer you four options for opening solids and surface files. Options include importing just the curves, solids or both solids and curves. You can also import the part as one solid model or separate all of the surfaces of the solid upon importation. If the IGES file contains 2D or 3D **wireframe ONLY**, select the curves option (this is not mandatory) and click OK. The wireframe geometry will open correctly even though there may not be any surfaces or solids in the file.

- **STEP Files:** Version 21 provides you with STEP, *.step import translators. To import a Step file, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the STEP (*.step) file format and then locate the STEP file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file.



Click the small drop-down arrow and select STEP from the list. Then you will only see available STEP files located wherever you have them on your computer.

Once you select the file and then the OPEN button you will see the "Open STEP FILE" box on the screen offering you options for opening these types of files.



- **DWG Files:** Version 21 provides support for the direct importation of **AutoCAD DWG *.dwg** files that contain wireframe, surface or solid model data. To import a DWG file, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the DWG (*.dwg) file format and then locate the DWG file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file. The DWG file will open up automatically in the CAD window of the software.

TIP regarding importing DWG files:

If you open the DWG file and do not see the geometry you can click on the **VIEW ALL** button from the main CAD toolbar to position the CAD screen so that you can see everything. This is the button on the main toolbar that has the letter "A" inside of a magnifying glass.

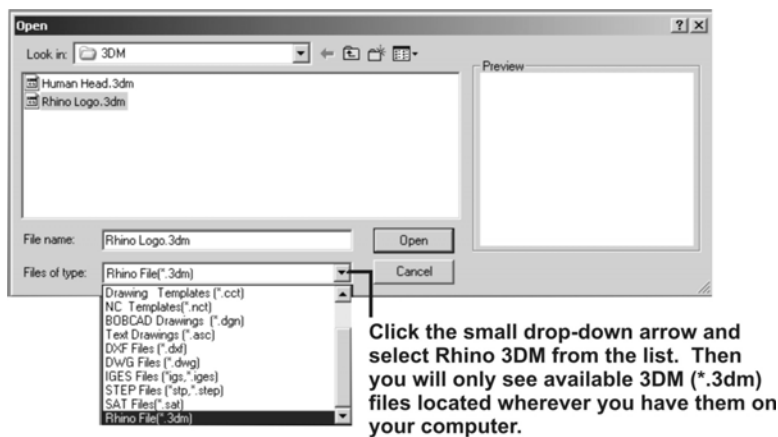
The "View ALL" button:



Supported DWG geometry/data:

Polygon Mesh
Poly Face Mesh
Face
3D Polyline
2D Polyline
Polyline
Ellipse
Spline
3D Solid
Region
Body
Radial Dimension
Diametric Dimension
Aligned Dimension
Rotated Dimension
Ordinate Dimension
2 Line Angular Dimension
3 Point Angular Dimension
Circle
M Text
Text
Line
Point
Arc
Layer table
Line type table
Dimension table

- **SAT/ACIS Files:** Version 21 software offers direct file support for *.sat. This is an ACIS file format. To open the SAT, *.sat, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the SAT (*.sat) file format and then locate the SAT file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file. The SAT/ACIS file will open up automatically in the CAD window of the V20 software with no intermediate option boxes.
- **Rhino CAD 3DM Files:** Version 21 software offers direct file support for Rhino CAD *.3dm files. These types of files are native to Rhino CAD software. To open the 3DM, *.3dm files, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the 3DM (*.3dm) file format and then locate the 3DM file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file.



- **Template Files:** BobCAD-CAM V21 offers special print **Templates** that you can use for creating your prints. These are designated as *.cct files and will be located in the TEMPLATES folder inside your main BobCAD-CAM V20 folder. To open a print template you will need to go to the CAD FILE menu and select OPEN. Then locate your main BobCAD-CAM V21 folder and open it. Then locate the Templates folder and open it. Inside of this folder you will see several files listed as A1.cct through A5.cct. These are your pre-loaded print templates. Select the one that you would like to use and then click the OPEN button or simply double-click on the file.
- **SLDPRT (SolidWorks part file):** Version 21 software offers direct file support for SolidWorks *.sldprt part files. To open this type of file, *.sldprt files, go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the SolidWorks (*.sldprt) file format and then locate the file that you wish to open. You can click on it once and then click the OPEN button or simply double-click on the file.
- **Parasolids X_T & X_B:** Version 21 software offers direct file support for Parasolids part files. To open Parasolids file go to the CAD FILE menu and select OPEN. The OPEN box will appear. In the section that says, "**FILES OF TYPE**" select the Parasolids file format and then locate the file that you wish to open. You can click on the file once and then click the OPEN button or simply double-click on the file.

CHAPTER 3

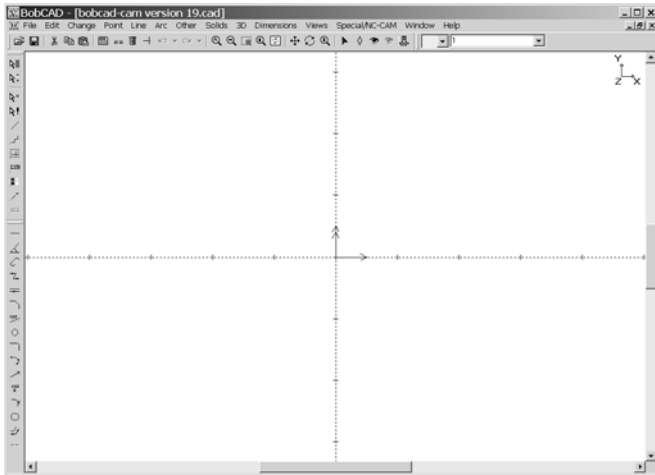
SOFTWARE INTERFACE

- Section 8. Basic Software Interface
- Section 9. Customizing the Main Toolbar
- Section 10. Selecting Entities or Solids & Surfaces
- Section 11. Modify Mode
- Section 12. The UCS – User Coordinate System
- Section 13. The Software Icons



Section 8. The Basic Software Interface.

The BobCAD-CAM Version 21 software has been designed to be user friendly for you. Let's take a look at the main **Software Interface** and how the coordinate system of Version 21 works.



Here you can see the main BobCAD-CAM Version 21 **Drawing Screen**. This is what the software should look like when you open the program for the first time.

BobCAD-CAM Version 21 was developed with modern Windows technology making the system more efficient and easier to work with. In this section we will show you some examples of this technology and how you can use it to actually customize the software interface, making it easier for you to work with for your specific needs.

You can see in the image below, there is the main menu bar consisting of the main drop down menus that contain the majority of the software functionality.



The Main Menu bar consists of:

The FILE menu: This menu will provide you with the File management functions, Print management, opening, saving and merging files as well as the software environmental settings. You will learn more about these functions in detail in the FILE section of this manual.

The EDIT menu: This menu will offer all of the functions relating to the editing of drawings or imported files. This includes the selection of entities that make up a drawing, cut, copy and paste functions as well as the ability to insert specific images, bitmaps etc into a drawing file.

The CHANGE Menu: This menu offers functionality enabling you to modify drawing entities; mirror, rotate, translation of geometry and much more.

The POINT menu: This menu offers a complete list of functions that deal directly with points.

The LINE menu: This menu offers a complete list of functions that deal directly with lines.

The ARC menu: This menu offers a complete list of functions that deal directly with arcs and circles.

The OTHER Menu: This menu offers functions for design as well as the integration of toolpath generating functions; pocketing, thread functions as well as a spline generator and more.

The SOLIDS Menu: This menu offers all of the functions that deal directly with designing models using primitive solid shapes, offsetting surfaces & solids, generating toolpath for surfaces & solids as well as stitching & un-stitching solids & surfaces and more. This is also the menu that offers solid model rendering functionality.

The 3D menu: The 3D menu allows you to create 3D toolpath through a variety of surfacing functions.

The DIMENSIONS menu: This menu gives you all of the standard dimensioning function as well as the ability to work with text and true type fonts.

The VIEWS menu: This menu gives a variety of standard viewing functions; panning, zoom, model rotation and much more.

The SPECIAL/NC CAM menu: This menu allows you to open the CAM side of the BobCAD-CAM Version 21 software. It deals with CAM related functions.

The WINDOW menu: This menu offers you multiple functions dedicated to views.

The HELP menu: This menu deals with software instructions and related “Help” documentation.

This is just a brief overview of the main menus. The software main screen will list the name of your file in the **UPPER RIGHT** corner of the main screen as you can see below.



As a note, if you are working on a drawing and have NOT saved the file, this space will list out what number the drawing is in that Version 21 session. This means that you may have more than one drawing started at a time that are open on screens behind the one that you are currently working on that are also NOT saved. Remember to save your files as you go so that you do not lose or corrupt them.

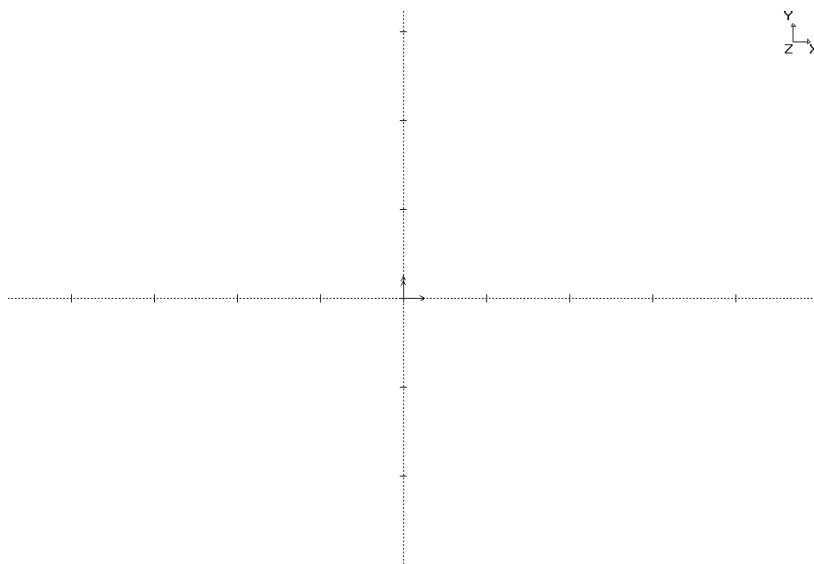
If you have multiple drawings open at the same time that are NOT saved and you attempt to close the BobCAD software, you will be asked to save those files. Then the next time you open the software, those drawings will open as well until deleted or saved. This is a fail-safe system that we have integrated into the software and it is changeable. The setting is called “**Auto Store Drawings**” and can be located under the main FILE menu, then ENVIRONMENT and the DEFAULT tab. This will also be covered in the FILE menu section of this manual. However, it is important to know at this point as well.

Remember, it is important to save your files as you go with the .cad file extension.

MAIN SCREEN AXIS:

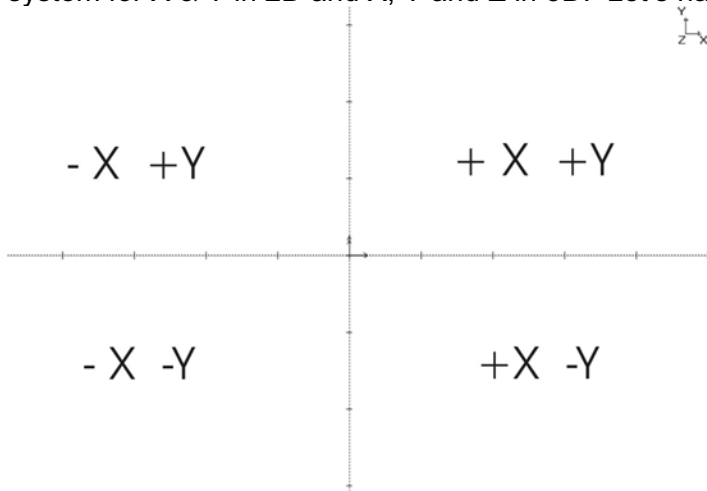
When you open the BobCAD-CAM Version 21 for the first time, you will have a main **drawing Grid Axis** on the screen. Let’s take a look at this grid axis and the basic coordinate system of the software.

The main screen grid axis:



THE BASIC COORDINATE SYSTEM:

The main drawing grid axis that you can see below has been designed for use with a specific coordinate system for X & Y in 2D and X, Y and Z in 3D. Let's have a closer look.



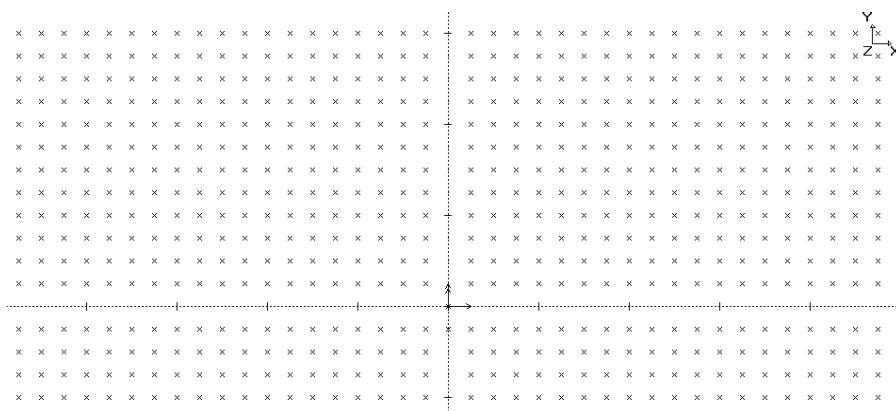
As you can see the screen is divided into four separate quadrants. When placing entities (points, lines or arcs) on the screen using the basic coordinate system, you will need to know how the screen is divided up. As you progress through this training manual, you will become more familiar with this system.

This grid axis can also be changed to make each mark represent Inches, Centimeters, Decameters or Millimeters. This value can also be changed into the desired **UNIT**. To do this you would select FILE from the main menu and then ENVIRONMENT. The environment dialog will give you various options. By selecting the COORDINATES Tab, you will see that you can choose the options or enter your own specific unit value. Once you have made the change, simply select OK and the screen will automatically adjust to the new setting.

This dialog will be covered in the FILE menu section of this manual to give you an in-depth look at how the software works and how to customize the main defaults to suit your needs.

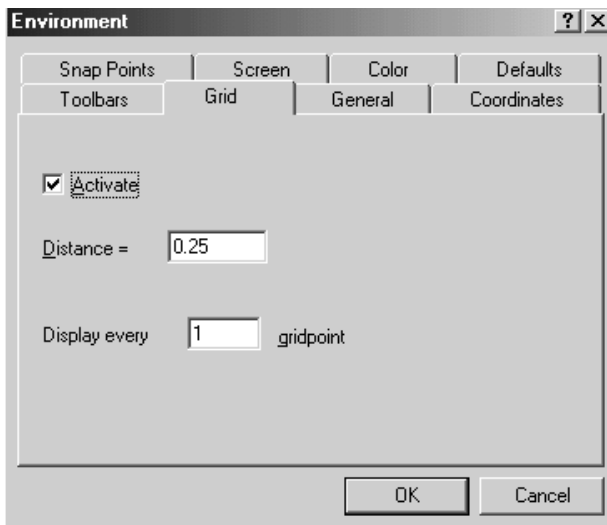
SNAP GRID:

In addition to the regular/default drawing grid axis, you have the **Snap Grid** option.



You can switch to the snap grid axis by selecting FILE, then ENVIRONMENT and then selecting the grid tab.

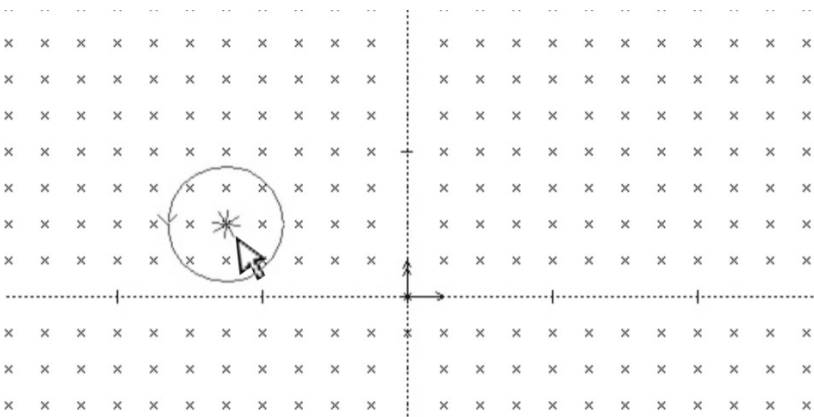
As you can see in the next image, you are given the option to activate the snap grid.



The **DISTANCE** is the actual distance between each snap grid point that makes up the grid itself.

You can also display a snap grid point based off of the distance. In the image above, the software is set to display a grid point every .25 inches on the screen, as the “display every” value is set to 1. If this were changed to 3, the grid would be broken into .25 places; yet, you would have a grid point displayed (shown on the screen) every .75 inches. This will be covered again in the FILE menu section.

The snap grid will allow you to snap entities to points easily. For example, by selecting the ARC menu and SKETCH CENTER you can snap a circle or part of a circle to any point on the screen easily.



The small star by your cursor will indicate the exact snap point that the entity will be snapped to. This also applies to drawing and sketching points or lines.

NOTE: The snap grid and axis are visible in the last example image. This can be changed to see only the snap grid. In other words, you can turn off the main axis grid and only have the snap points on the screen if you would like. You will find this setting under the FILE menu, ENVIRONMENT and then the SCREEN tab. You can then check or un-check the SCREEN AXIS option. You will not have to restart the software when making changes of this nature. The change will automatically take place once you select OK to exit the ENVIRONMENT dialog. You will learn about the point, arc and line menus and will have an opportunity to use these functions with the snap grid as you progress through this training manual.

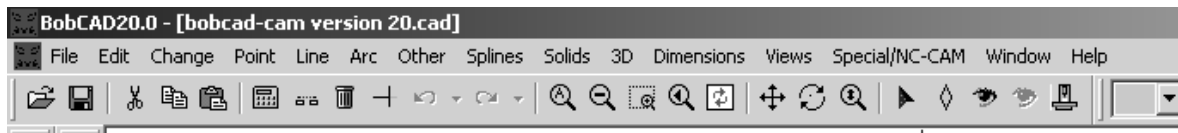
A basic option in BobCAD-CAM Version 21 is the ability to activate or deactivate the button **Tool Tips**.



By resting the cursor over a button on the screen, you will see a tool tip explaining what the function is. This can be turned on or off by selecting FILE, then ENVIRONMENTS and the SCREEN tab. There will be an option for tool tips. Check the box to turn the tool tips on and uncheck the button to turn them off. The tool tips function will help you learn what each of the function icons mean.

THE MAIN TOOLBAR:

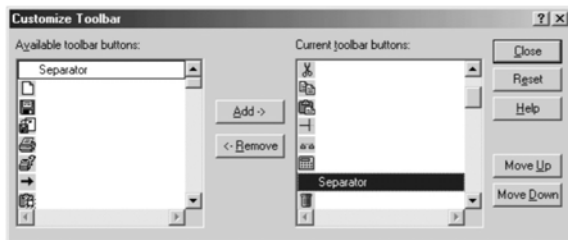
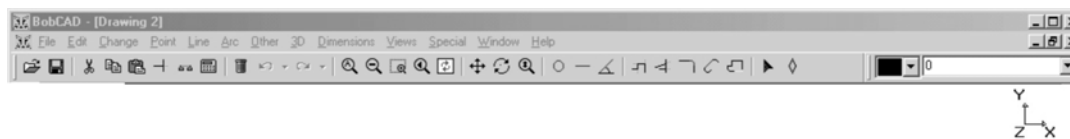
The next aspect of the basic V21 software screen is the main tool bar and each of the basic icons that are on it. Version 21 was developed with standard Windows technology, making the software flexible and easier to use and customize. Let's take a closer look at this tool bar.



When you open the BobCAD-CAM Version 21 software for the first time, it will look like the above image of the **Main Toolbar** bar. This toolbar can be customized.

Section 9. Customizing the Main Toolbar.

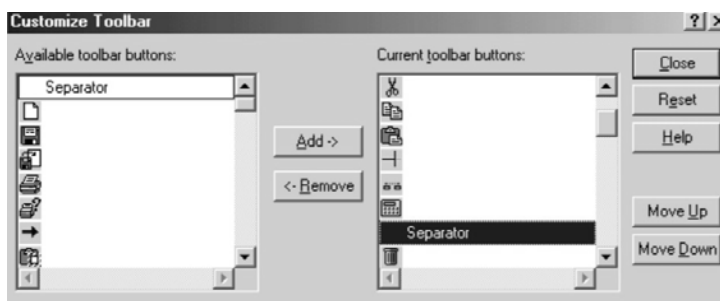
To customize the **Main Toolbar** you would take your cursor and DOUBLE CLICK on a non- button space or what is called a separator bar, on the toolbar itself. This would bring up the **CUSTOMIZE TOOLBAR** dialog. See below.



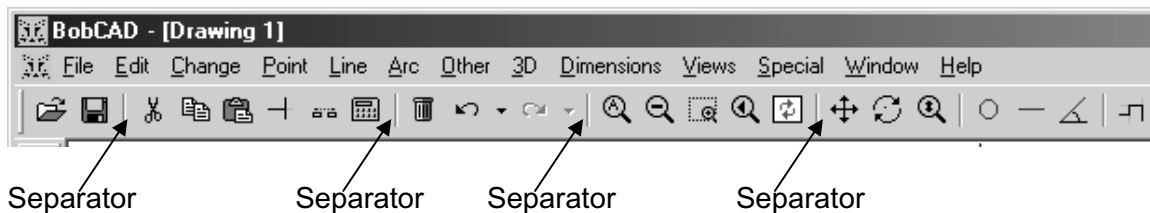
The Customize **Toolbar** dialog has two sections to it.

- The available tool bar buttons.
- The current tool bar buttons.

The "**Available Toolbar Buttons**" section lists out each and every icon that is available in the software for use. These can be used to make smaller toolboxes or they can be used on the main toolbar that runs along the top of the main screen. A **SEPARATOR** is used to separate a set of buttons from another set.



Here is a closer look at a separator.



Once again, the separator bar helps you isolate one group of buttons from another group of buttons.

To add a separator or any button to a toolbar you would:

- First double click your mouse on an existing separator to open the “Customize Toolbar” dialog.
- Select the button that you would like to add under Available Toolbar Buttons.
- Select the ADD-> button to move the selected button to the “Current Toolbar Buttons” section on the RIGHT.
- Then select the MOVE UP button or MOVE DOWN button to position the button where you need it to be.

When finished select OK.

To REMOVE an existing button from the main toolbar or a specific toolbox you would: Select the button from the “Current Toolbar Button” section and then select the REMOVE button.

To REPOSITION an existing button on the main toolbar or a specific toolbox you would:

Select the button from the “Current Toolbar Buttons” section and then select the MOVE UP button as many time as you need to or select the MOVE DOWN button as many times as you need to.

NOTE: if you decide that you do not want the added buttons, you can select the RESET button to remove any settings that you have made.

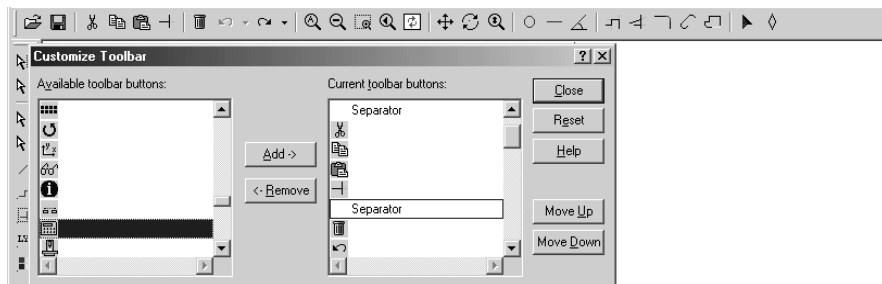
CUSTOMIZING THE TOOLBAR LESSON:

Let’s take an even closer look at how this is done in a lesson format:

STEP 1

Open the BobCAD-CAM Version 20 software to a NEW drawing screen.

Have a look at the top main TOOLBAR. Choose a separator and double click on it. This will open the Customize Toolbar dialog. See below.



NOTE: Notice that the separator bar that you double clicked on appeared in matching position in both windows of the Customize Toolbar dialog. Therefore, it makes a difference on which separator you double click on for the quick adding or taking away of buttons. However, you still have the MOVE UP and MOVE DOWN buttons to reposition the button or buttons that you add.

STEP 2

Select the **VERIFY ENTITY** Button on the “**Available Toolbar Buttons**” side which is the button that has the small calculator on it.

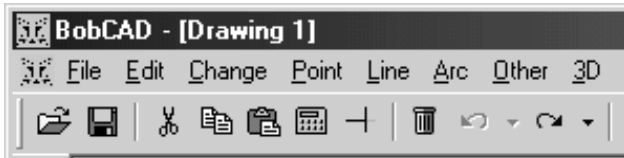
STEP 3

Now select the ADD button and you will automatically add this button to the main toolbar.

STEP 4

Now under the “**Current Toolbar Buttons**” section, select this button and then click on the MOVE UP button or MOVE DOWN button (depending on your position) until this new button is just below the PASTE ITEMS TO CLIPBOARD button (the small clipboard).

Now click CLOSE. The Verify Entity button will now be added.



This is the fundamental procedure for creating your own toolbars and toolboxes in the BobCAD-CAM Version 21 software.

Section 10. Selecting Entities & Solids or Surfaces.

The first part of this section will cover selecting basic entities and geometry. Once you have covered this, we will get into selecting surfaces and solids.

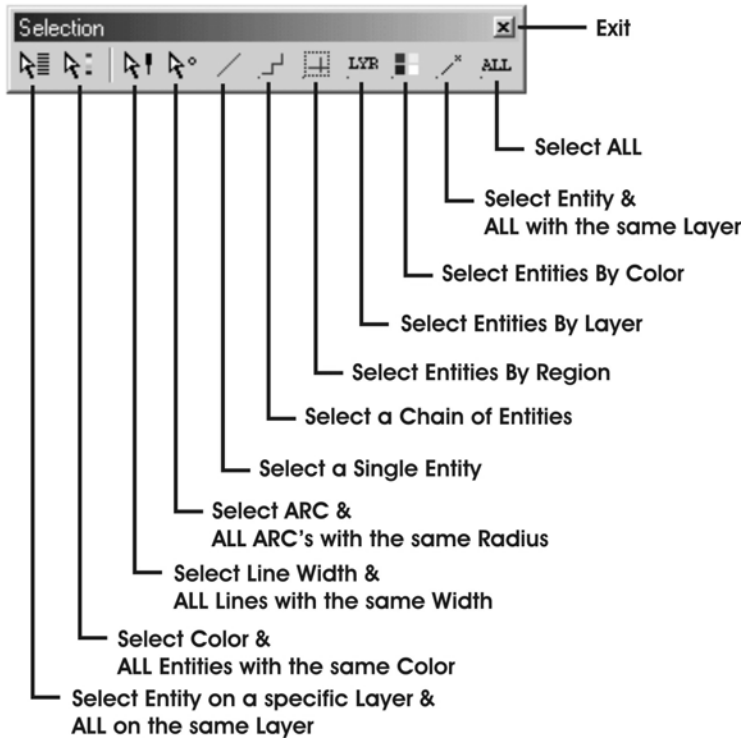
A primary basic you should know about the BobCAD-CAM Version 21 system is the **Object/Action – Action/Object** approach that the software allows you to use.

OBJECT/ACTION – ACTION/OBJECT IN SELECTION MODE

The BobCAD-CAM Version 21 software is the same as the previous Version 20 when it comes to the selection of entities. There are several ways to do this along with their corresponding icons. A fundamental of BobCAD-CAM Version 21 is the **OBJECT/ACTION – ACTION/OBJECT** approach. Many of the functions of the software can be used by FIRST, selecting your entities and then executing a function (object-action) and then the opposite, selecting the function and then the object or entities (action-object) to be modified. Typically, if you are trying to modify a group of entities at the same time, you can select them first and then choose the specific function. However, this is not always going to be the case and is why training with this manual is so important. Each function will be described to you in detail and will also inform you whether the function is Object/Action or Action/Object.

When opening the software, you will notice the function icons and toolboxes than run along the left wall of the system. One of these offers selection functions. By learning to use these, you will gain a lot of speed when drawing and editing geometry and even using functionality for toolpath creation when that function is Object/Action. Let's take a look at this **Selection Toolbox**.

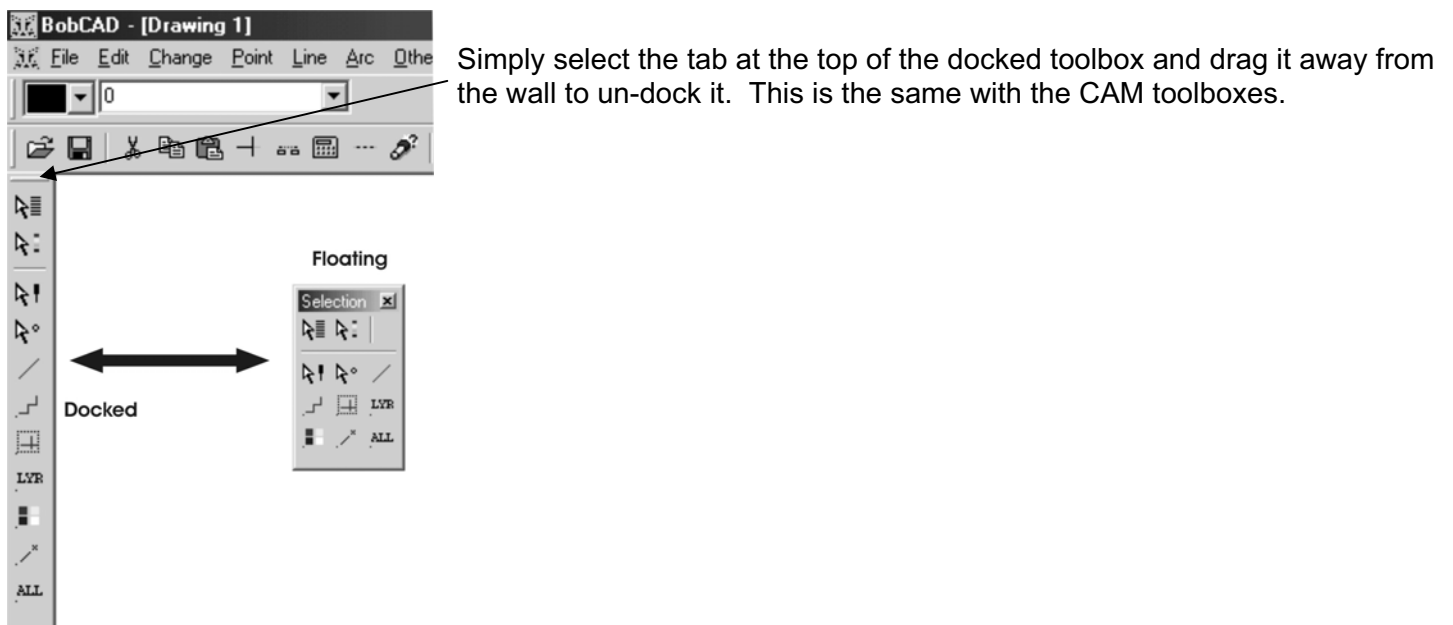
THE SELECTION TOOLBOX:



When you open the BobCAD-CAM Version 21 software for the first time, the **Selection Toolbox** will be docked on the left side wall of the software as just mentioned. As you learned in the earlier section about customizing the toolbars, this edit toolbox can be customized the same way.

All of the toolbars and toolboxes in the software are “**Docking**.” This means that you can move them around the walls of the software by clicking on the top part of the box, while holding down your mouse button, can move them to a new location or dock them on a different wall. Here is an example.

Here you can see that the toolbox can be moved and is seen floating on the drawing screen.



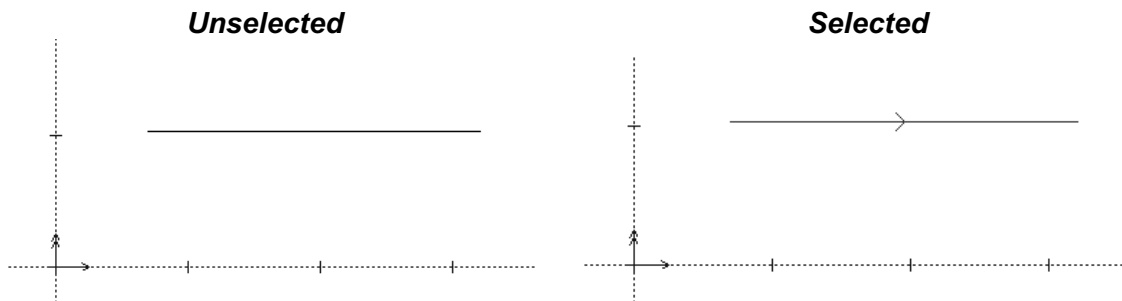
Now let's have a look at each of the selection buttons/icons and how you can use them to edit or your drawings.

SELECTION TOOLBOX ICONS:



This is the **Select Entity Single** button/icon. This allows you to select a single entity at a time to modify or edit. If you are using this function to select more than one entity at a time you **MUST** hold down the **CONTROL KEY** on your keyboard as a rule. Let's take a closer look at what happens to entities when they are selected.

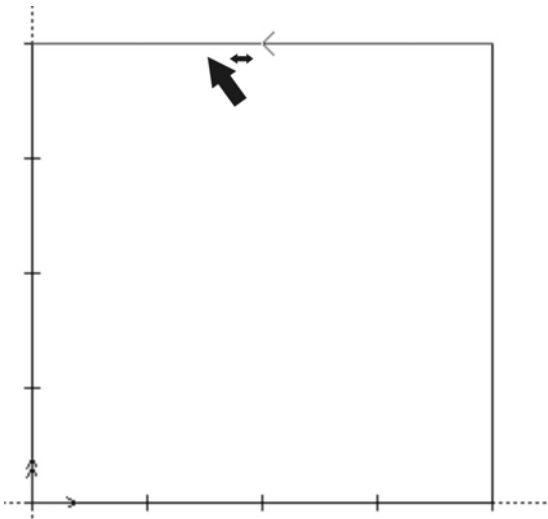
Here is an unselected line below.



The selected entity will turn RED or the designated color and you will have a selection directional arrow. You can change the direction of the selection by clicking on the entity at the closest end that you want the direction to go.



This is the Select Entities by **Chain** button/icon. This allows you to select an entire chain of entities. Take the example of a square. Click on one of the lines first, move your cursor toward the left or toward the right to choose the direction and then click your mouse button one time.



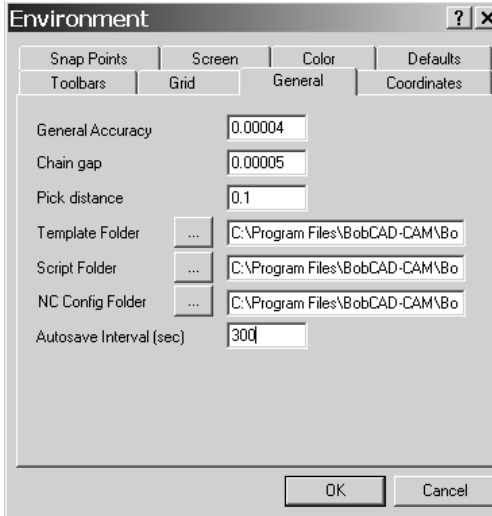
Then simply hit the F3 key on your keyboard to select the entire chain or click the last entity of the chain.

TIP on selecting a chain of entities:

Another way to select an open or closed chain of entities is to hold down your SHIFT key and click the last entity at the end of an open chain and to hold down the shift key and click an entity of a closed shape chain.

SELECTING CHAIN GEOMETRY NOTE: If you are selecting a **Chain** of entities and you have trouble getting the entire chain to select, you may need to adjust your CHAIN GAP settings. You can do this by selecting the FILE menu, then ENVIRONMENT. Once you have opened the Environment box, select the GENERAL tab.

You will see the **Chain Gap** setting listed just below the General Accuracy setting. Simply increase this value to make the software look further from the end of one entity to the beginning of another entity. This is how it works in selecting chained geometry.



This is the Select Entity by **Region** button/icon. Simply click and drag a selection window over the entities that you want to select and then click your mouse button one more time to select them.



This is the Select Entities by **Layer** button/icon. You can select this icon to bring up the active layer dialog box. This box will list all of the layers that you have saved in the drawing. Click on the desired layer or layers and then click OK to select them.



This is the Select Entities by **Color** button/icon. Select this icon and the color dialog box will appear. Simply click on the color or colors and then click OK to select them. This function also allows you to select different color surfaces and solids.



This is the **Pick & Match** by Layer button/icon. If you have a drawing that has multiple layers, you can select this icon and then click on an entity in the drawing and all of the entities, surfaces or solids on the same layer will be selected as well.



This is the Select **ALL** Entities button/icon. By selecting this icon, everything in the drawing, entities, objects and text will be selected.



This is the **Pick & Match** by **Radius** button/icon. By selecting this icon, you will be able to select a circle with a radius and all other circles with the same radius will be selected as well. This is a good function for selecting a group of arcs with the same radius for use with drilling cycles.



This is the **Pick & Match by Line Width** button/icon. By selecting this icon, you can select a specific line with a certain line width and all other lines with the same line width will be selected also. An example would be that you have extracted all of the line geometry from a surface or solid and you want to view it easier. You can assign this geometry a color once it is selected through the attributes box or by modifying the line width. Now you can use this function for selecting lines with the same width attribute.



This is the **PICK & MATCH COLOR** button/icon. By selecting this icon, you can select an entity with a certain color and all other entities with the same color will be selected. If you have disconnected the surfaces of a solid to separately machine them, you may want to also assign them a different color than the other surfaces or solid. If so, this is a great way to quickly select the surfaces that you want to machine by their colors.

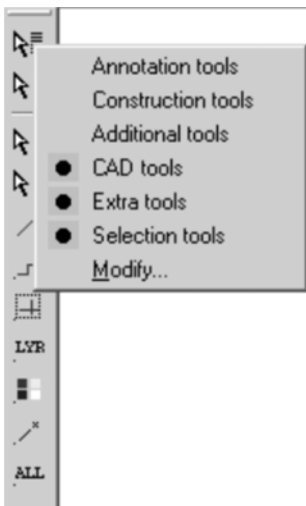


This is another button/icon for **PICK & MATCH LAYER**. Simply select an entity and all other entities, surfaces or solids on the same layer will be selected as well. This is NOT an OBJECT/ACTION function. In other words, you must select the icon first and then the entities.



This button/icon is the Select **ALL SOLIDS** function and allows you to select ALL of the solids on the drawing screen while avoiding points, lines or arc wireframe geometry.

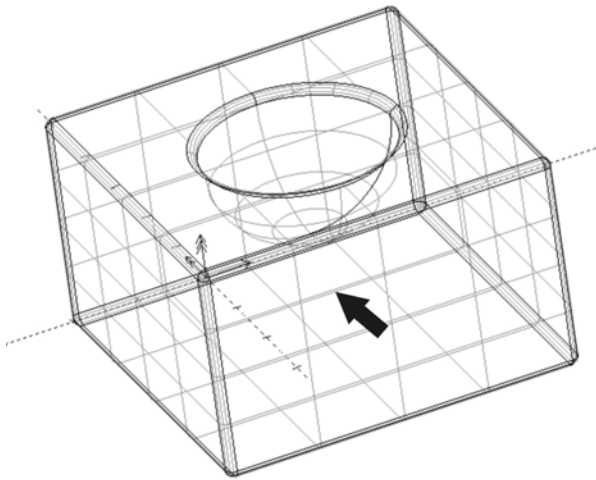
You also have the ability to access additional toolboxes by clicking on one of these icons and then RIGHT-clicking your mouse. See below.



Simply click on the desired toolbox and it will open up on the screen. We will cover the available toolboxes in the section after the available buttons. Modify, in the image above will take you to the CUSTOMIZE TOOLBAR dialog box.

SELECTING SOLIDS OR SURFACES:

The selecting of **Solids** and surfaces is easy. Have a look at the solid cube in this next image.



Whether you are attempting to select a solid model or a specific surface, you simply place your cursor on an interior or exterior edge line and click your right mouse button.

There are other ways to **select** a solid or a surface. They are:

- **By Color.** This is possible if you have selected the solid or surface and then changed the color through the Attributes box located in the CHANGE menu. Then use the Pick & Match Color function to make the color selection.
- **By Layer.** If you select a solid or surface and then assign it to a layer through the Attributes box located in the CHANGE menu, you can then select them easily through the Select Entities by LAYER function.

Section 11. Using the Modify Mode.



This is the **Modify Mode** button/icon. This allows you to change entities in different ways. When this mode is active, you can change selected as well as deselected entities. Selected entities have however, a slightly higher priority than the unselected entities in your drawing. The Modify Mode allows you to change entities like:

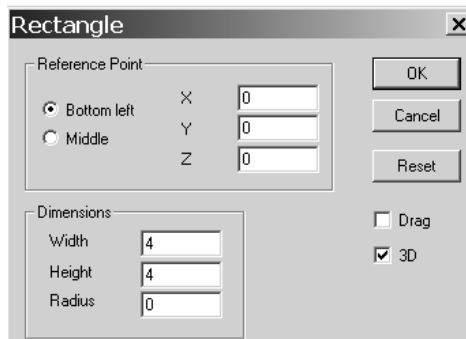
- **Extending/trimming entity(s)**
- **Moving entity(s)**
- **Copying /linking entity(s) from another drawing**
- **Rotating entity(s)**
- **Rotating and extending/trimming entity(s)**
- **Manipulating dimensions**

By using the CONTROL KEY and the SHIFT KEY on your keyboard, you can further modify entities. This is something that you will want to practice using to become more efficient with while modifying entities.

Here is an example using a square:

STEP 1

Draw a simple square by choosing the OTHER menu and then RECTANGLE.



Select the BOTTOM LEFT option first. Now enter 4 for the width, 4 for height and leave the radius value at 0. Select the 3D box and un-check the DRAG box. Now click OK.

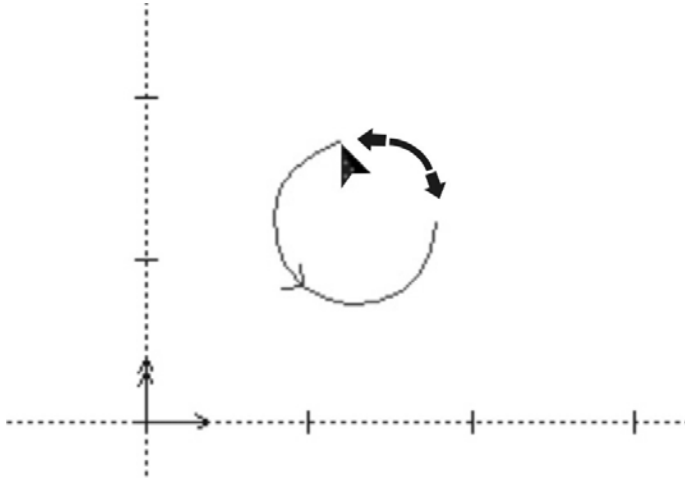
This will draw your square.

Click your mouse button again in a new location and you will have another copy and the Transform toolbox will reappear allowing you further modification to the geometry.

NOTE: You will have to **UNCHECK** the Rotate After Move dialog to exit that function. When finished with this function entirely, click your **RIGHT** mouse button to exit the function all together.

The **TRANSFORM** toolbox offers functionality that is useful for manual part nesting and many other similar type actions.

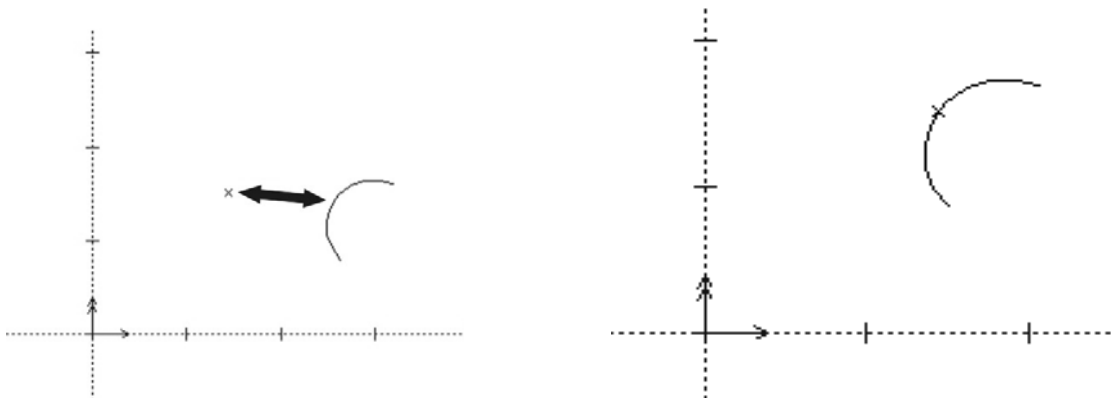
The **Extending & Trimming** of geometry can also be accomplished easily in **MODIFY MODE**. Take a look at the example below with arcs.



By holding down the **SHIFT KEY** you can modify the arc as you can see in the image above. By holding down the **CONTROL KEY**, you can modify the size/radius of the arc. By holding down **BOTH** the **SHIFT** key and the **CONTROL** key you can do both at the same time; change the angle and the radius size.

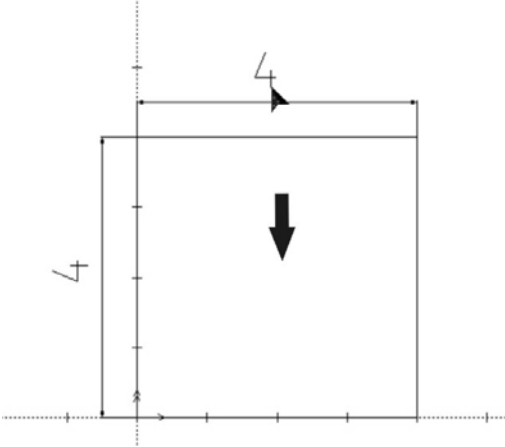
You may also click and drag the arc or entity to a different location altogether. You may also snap the arc to a point or an existing point to an arc.

Click and drag the arc to snap it to the point.

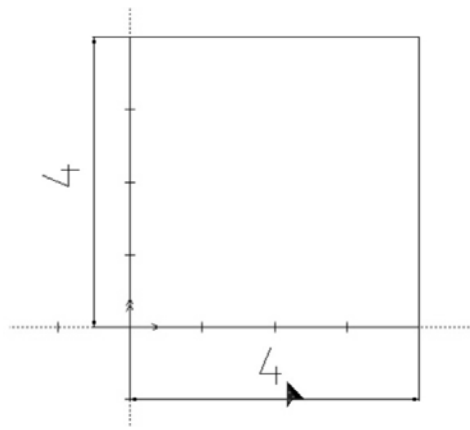


You can use the **MODIFY** function to **MODIFY DIMENSIONS** as well. Simply click on the dimension and drag it to a new location. By clicking on the directional arrow on a selected arc, you can move the arc to a different location on the screen.

First location.

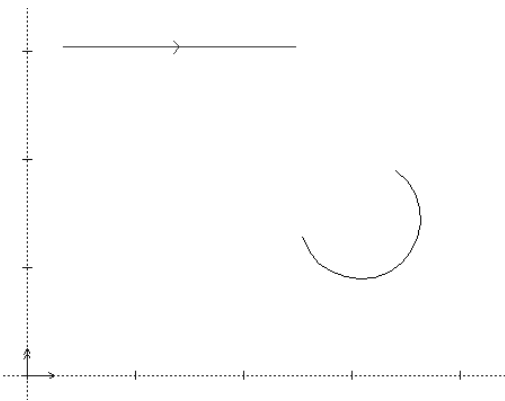


Second location.

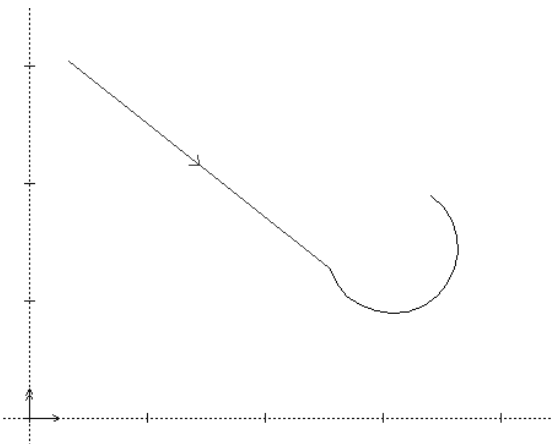


The entity that you select while in Modify Mode is given a higher priority than the other entities on the screen. This means that you can change/modify the entity selected by clicking and dragging it to a new location or performing the modify move that you want to do.

You can use the Modify Mode to change **LINES** as well. Let's take a look. In the image below, we have an arc and we have a line.



By using the Modify Mode and holding down the **CONTROL KEY**, you are able to modify the **angle** and the **length** of the line at the same time. In the next example, we did this and snapped the end of the line to the end of the arc automatically.

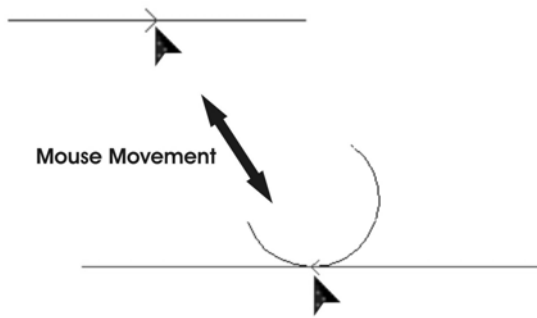


The portion of the line that we selected became free for modification. The furthest end from the part of the line that we selected became anchored to its original position.

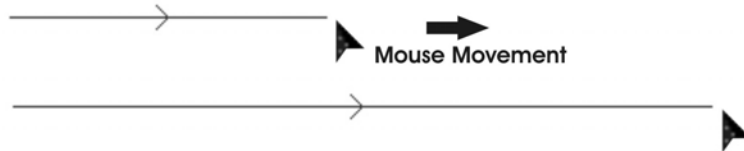
By clicking on the directional arrow on the selected line we can move the entire line to a different location on the screen, or snap the center of the line to a point or other entity.



Here is an example below of this type of modify move.



By clicking on the end of the line and holding down your mouse button, you can make the line longer or shorter by moving your mouse.

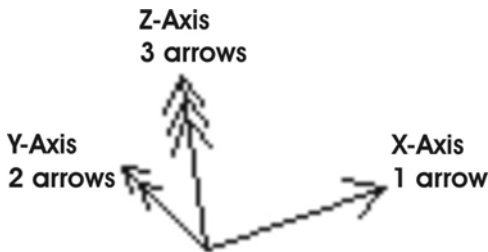


Practice using the software in **MODIFY MODE** to become more efficient in changing or modifying your drawings.

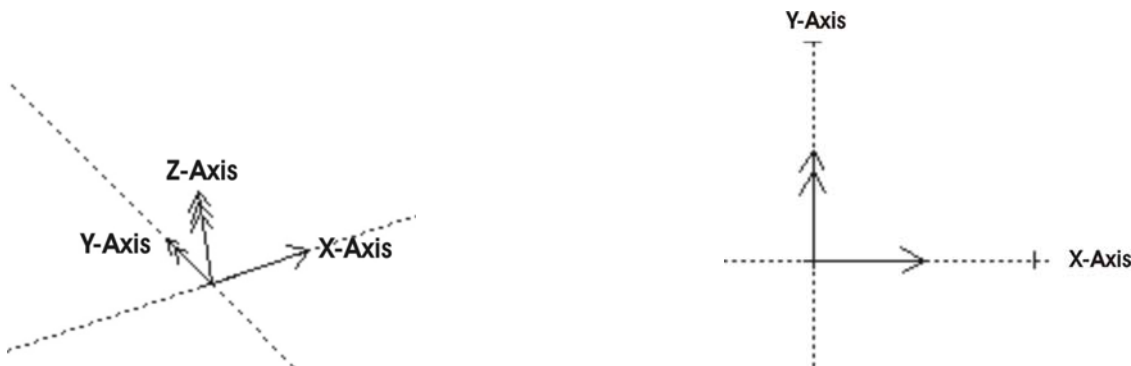
Section 12. The UCS – User Coordinate System.

This section is dedicated to what is called the “**User Coordinate System**.” This defines the drawing system when using coordinates to place entities on the screen with X, Y and Z coordinates. When we refer to “The UCS” we are speaking of the main AXIS on the BobCAD-CAM drawing screen. See below.

THE UCS:



Once again, the UCS below is shown in 2D & 3D mode.

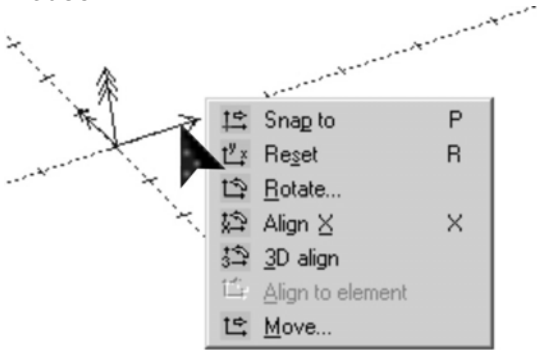


By resting your cursor over the UCS and RIGHT-clicking your mouse, a new menu will appear. Let's have a look at each of the UCS options which include function icons.

THE UCS CONTROLS:

The UCS has a full set of **controls** allowing you to change the software user coordinate system easily. By placing your cursor on one of the axis arrows and RIGHT-clicking your mouse you can automatically access them at any time.

The UCS Controls are accessed by resting your cursor over the UCS axis itself and then by right-clicking your mouse.

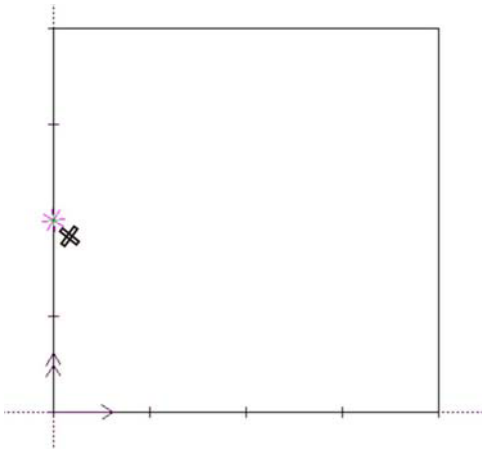


Let's take a closer look at what each of these UCS control functions are and how they work.

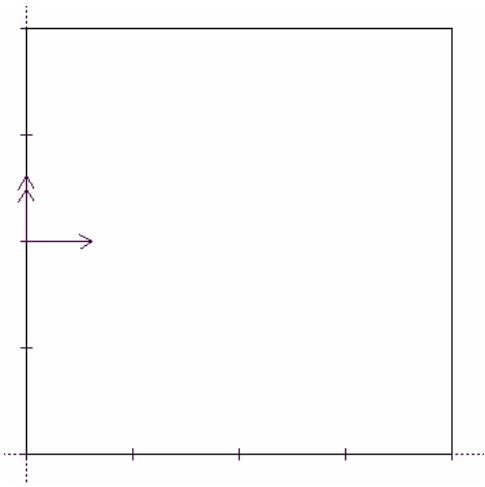


This is the **Snap To** button/icon. This allows you to snap the UCS to a specific location on your drawing. Let's take a closer look at this.

Below, we have a square and have selected the "Snap To function." The star represents one of the snap locations on our geometry.

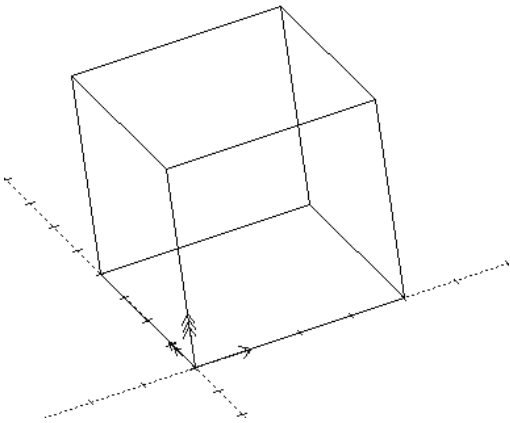


By clicking your mouse at this location, the UCS is automatically moved to its new location. Now the new location represents the ZERO position on the screen.

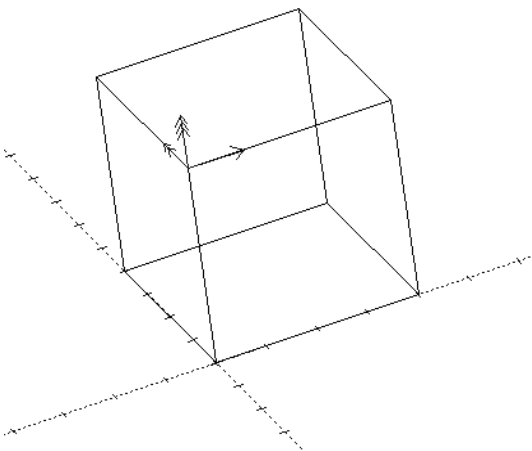


Let's take another look at this in 3D mode.

In the next image, you can see that the top of the cube is 4-inches raised in the Z-Axis.



By using the SNAP TO function, we were able to move the UCS to the upper corner of the cube. By doing this, we changed the coordinate system to make this corner the new Z-Zero. You can see the result in the next image.

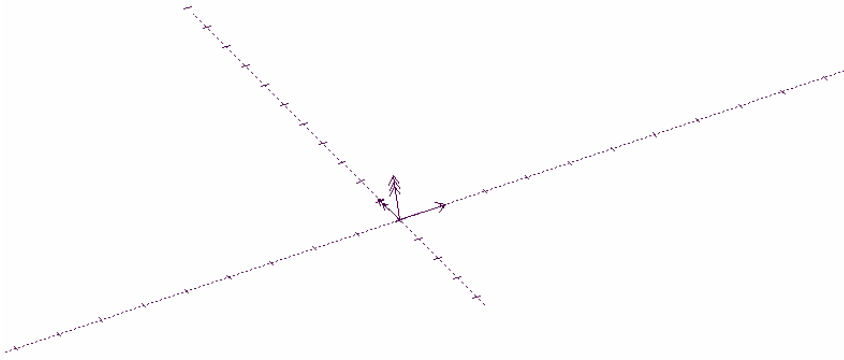


NOTE: To **RESET** the UCS, you would hover your cursor over the UCS, **RIGHT-click** your mouse and select **RESET**.

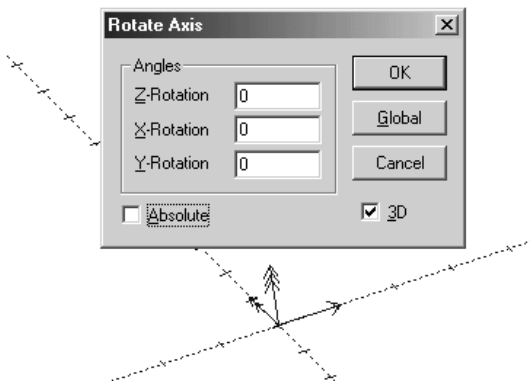


This is the **UCS Reset** button/icon. If you have changed the UCS, you can select the **RESET** function to bring the UCS back to its native home position on the grid axis.

The DEFAULT home position of the UCS is at the native Z-Zero location at the center of the grid axis as seen below.

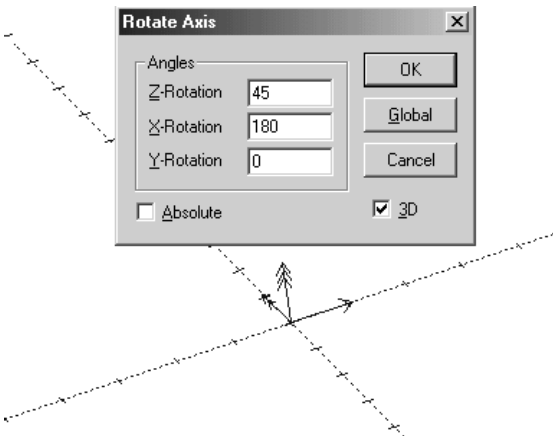


This is the **ROTATE THE UCS** button/icon. This allows you the ability to rotate the UCS around at its anchored location or at a designated coordinate location on the drawing screen. See this in the next image.

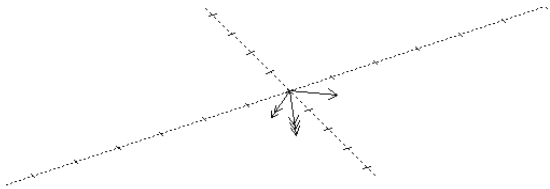


By selecting the 3D box, you can rotate in all 3-Axis at once as needed.

In the next image, you can see that we entered 45-degrees for the Z and 180-degrees for the X-Axis rotation.

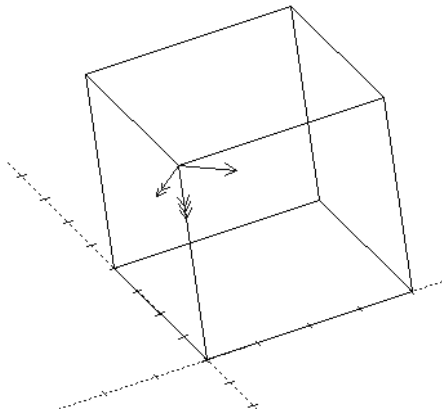


By selecting OK you can see the UCS has been changed below.

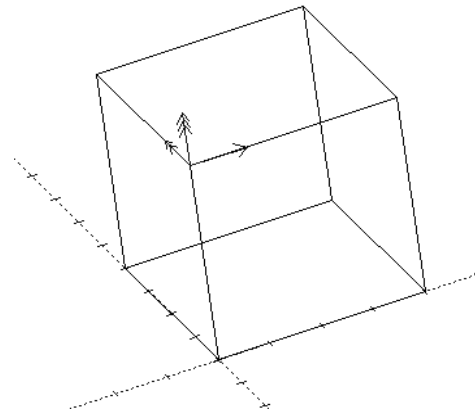


If you have snapped and applied a rotation to the UCS as seen in the image below, you can select the function and **GLOBAL** to reset the UCS at its snapped location.

The UCS snapped and rotated.



The UCS after GLOBAL was selected.



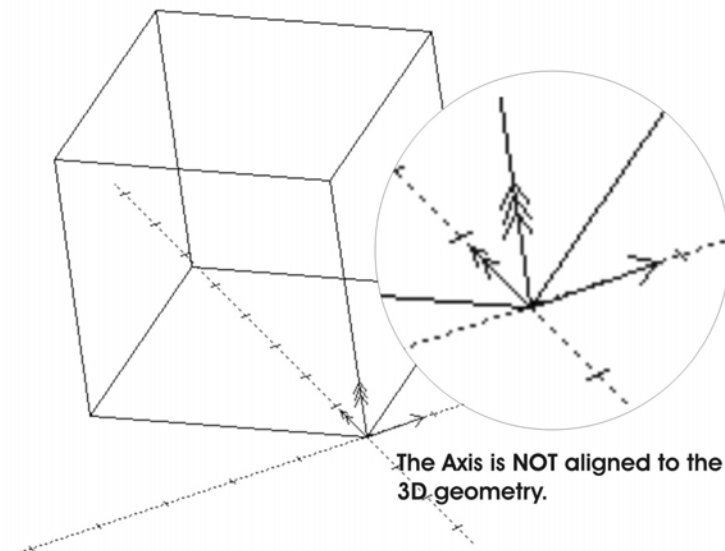
Selecting RESET would reposition the UCS at its native home-default location. **REMEMBER, the coordinate system changes when the UCS is changed.**



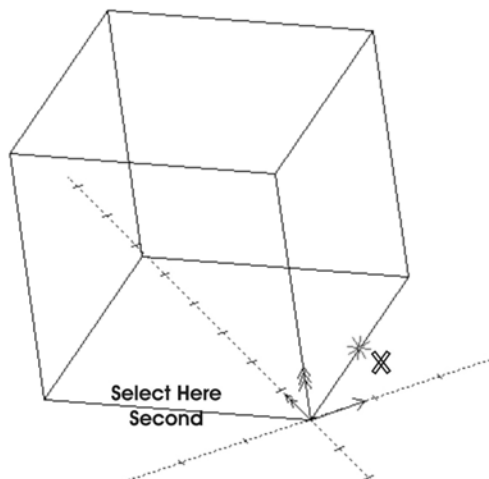
This is the **ALIGN THE UCS IN X-AXIS** button/icon. This function allows you to designate an X-Axis location to automatically change and align the UCS to. Once you select this function, you will have snap locations on the drawing. When your cursor comes near one of the designated snap locations, you will be able to click your mouse and automatically change the UCS to align to a new X-Axis.



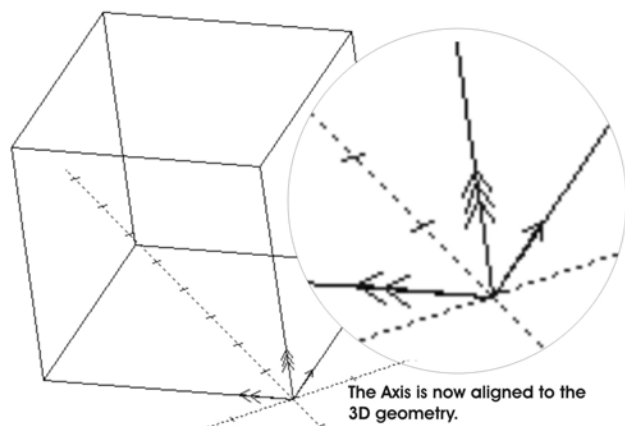
This is the **3D ALIGN** button/icon. This allows you the ability to align the UCS in 3D. In the image below, you can see that the 3D geometry is not aligned to the UCS.



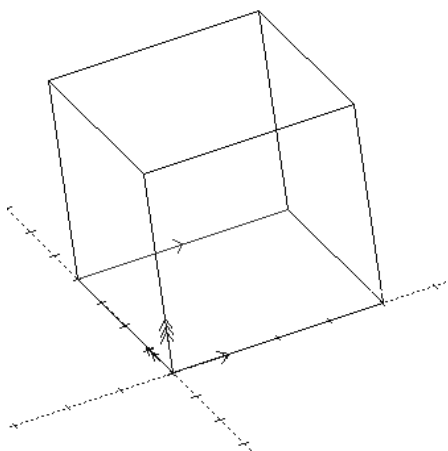
By selecting the available snap points as you can see in the next image, you can automatically align the 3D geometry.



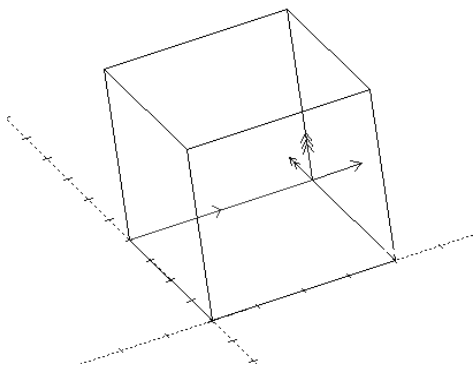
Below is the result of using the **ALIGN** function.



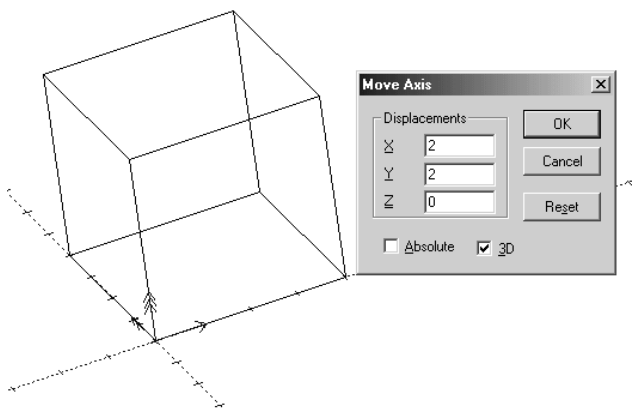
This is the **ALIGN TO AN ELEMENT-ENTITY** button/icon. This function allows you the ability to snap (align) the UCS to the end of a **PRE-SELECTED** entity. You must select an entity first before being able to activate and use this function. The direction arrow will determine which end of the entity the UCS is aligned and snapped to. See this in the next image.



You can see in the last image that the back line of the bottom square has been selected. By selecting this function you can see that the UCS was automatically aligned to the end of the selected line.

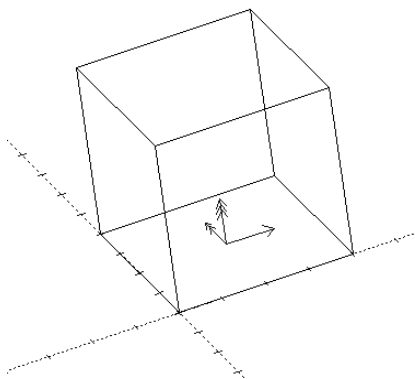


This is the **Move UCS** button/icon. When you select this function, you will open the MOVE AXIS dialog. See below.



In the image above, we have entered 2 for X and 2 for Y. By selecting OK be repositioned the UCS to its new location.

Below you can see the new UCS location. You can use X, Y and Z coordinate values to do this. The **ABSOLUTE** option pertains to the default ZERO position on the screen. Un-check the Absolute option and the software looks at the move as an **INCREMENTAL** move altogether.

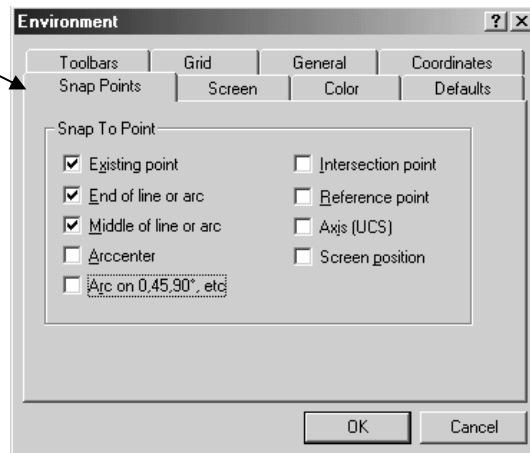


UCS TRAINING LESSON:

The purpose of this lesson is to give you a better understanding of the UCS-User Coordinate System for making drawings.

STEP 1

Select **FILE** from the main menu and choose **ENVIRONMENT** to open this dialog. Now select the **SNAP POINTS** tab as you can see below.



STEP 2

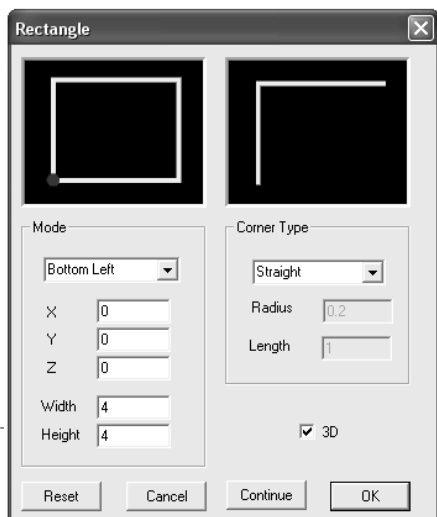
The ENVIRONMENT dialog deals directly with the main software default settings. The SNAP POINT section/tab is the focus for now in this lesson. You will want to make sure that the “Snap To Point” options are checked as instructed below.

- Check the option for **EXISTING POINT**.
- Check the option for **END OF LINE OR ARC**.
- Check the option for **MIDDLE OF LINE OR ARC**.

Now we can actually start the lesson, as these settings are important for this UCS lesson. *The ENVIRONMENT dialog is covered in detail in the FILE menu section of this manual.*

STEP 3

Select the OTHER menu and RECTANGLE. This brings up the rectangle box.



Make the following selections as seen in the image above:

- Under MODE select the “**BOTTOM LEFT**” option from the pull-down.
- Leave the X option 0.
- Leave the Y option 0.
- Leave the Z option 0 (if the 3D button is checked).
- Under Corner Type, select Straight from the pull-down.
- Enter 4 for the Width and enter 4 for the Height.
- Now select OK.

You should now have a 4x4 square on the screen. You can select the View Entire CAD Drawing icon to view all of the geometry.

STEP 4

Select the **3D menu** and **TURN 3D ON**.

NOTE: If the “**Auto Pre-select**” option is checked in the environment defaults, the square will already be selected. If not, we will select it at this time as the next step.

STEP 5

Hold down the **SHIFT** key on your keyboard and click on the geometry of the square. This will select it if not already selected.

STEP 6

Now select the **COPY** icon from the main toolbar.

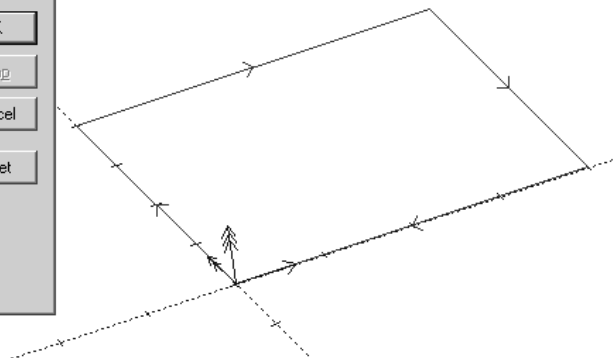
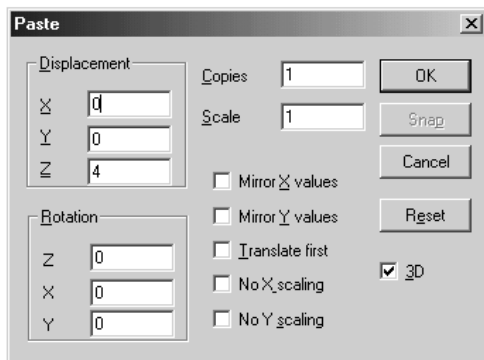


STEP 7

Now select the **PASTE** icon from the main toolbar.



This brings up the PASTE dialog as you can see in the next image.



Enter 1 for COPIES.

The scale is 1.

The DISPLACEMENT values for X and Y are both 0.

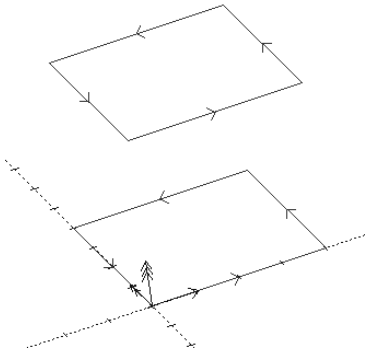
The DISPLACEMENT value for Z is 4.

The ROTATION values for X, Y and Z are all 0, as we are not going to rotate the part.

If the Z values are not available, check the 3D option.

Select OK.

There is our copied square 4 inches in the Z.

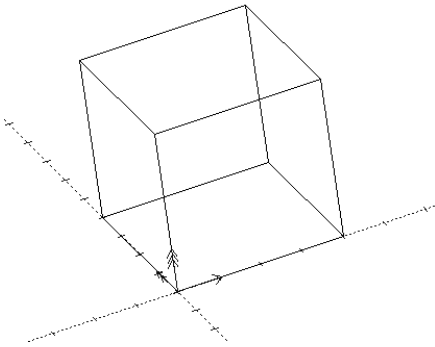


STEP 8

Deselect the geometry by clicking your mouse OFF of the part in an empty area of the drawing screen. Now we want to join all 4 corners from the top to the bottom with the LINE>JOIN command/function.

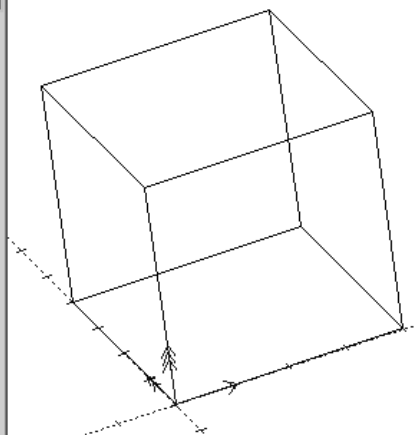
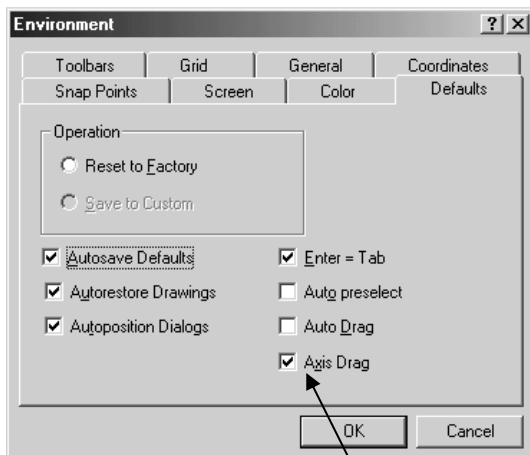
STEP 9

Select the **LINE** menu and then the **JOIN** function. This allows us to click on a corner and drag a line down to the matching corner below or above. Then click your mouse again to join the line. This is a repetitive function. Therefore, go around the geometry until you have joined all 4 corners making a cube. When finished, click your **RIGHT** mouse button to exit the function. See the result in the next image.



STEP 10

OK. Now we need to make sure that the software environment is set properly. Select FILE from the main menu and then ENVIRONMENT. Select the DEFAULTS tab.

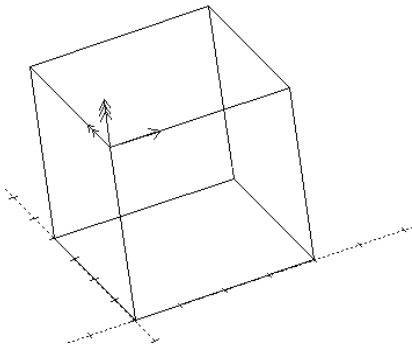


Select the option, "**AXIS DRAG.**" Now select OK.

The Axis Drag option allows us to click on one of the axis arrows and move it to a specific position or to move the entire axis itself by holding down the mouse button while executing the move.

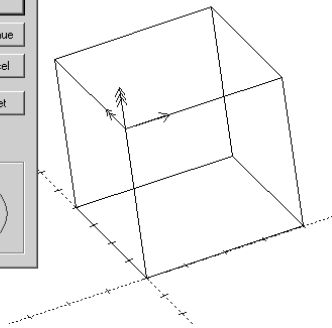
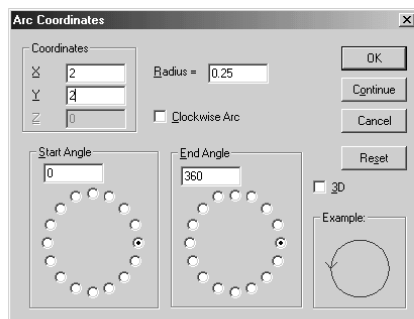
STEP 11

Now click on the UCS axis itself and drag it to the bottom LEFT corner of the topside, directly above its default location.



STEP 12

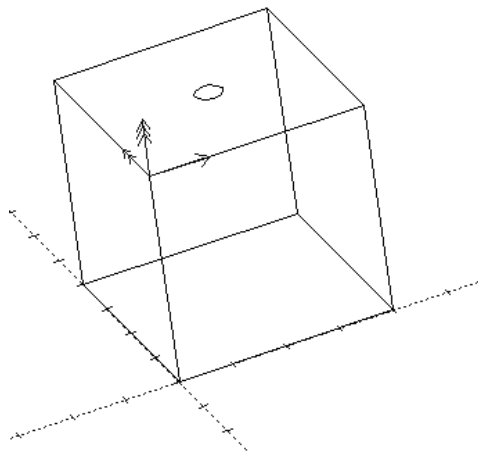
Select the **ARC** menu and **COORDINATES** to bring this dialog up.



Enter the following variables:

- A. UN-CHECK the 3D option.
- B. Enter 2 for X.
- C. Enter 2 for Y.
- D. Enter .25 for the Radius.
- E. Start Angle is 0 and End Angle is 360 to draw a full circle.
- F. Select OK.

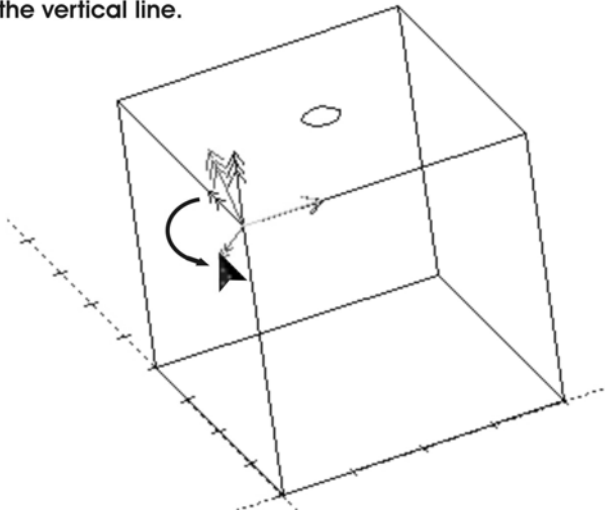
As you can see in the next image, the Z plane has now changed and is raised 4-inches to the topside of the cube.



STEP 13

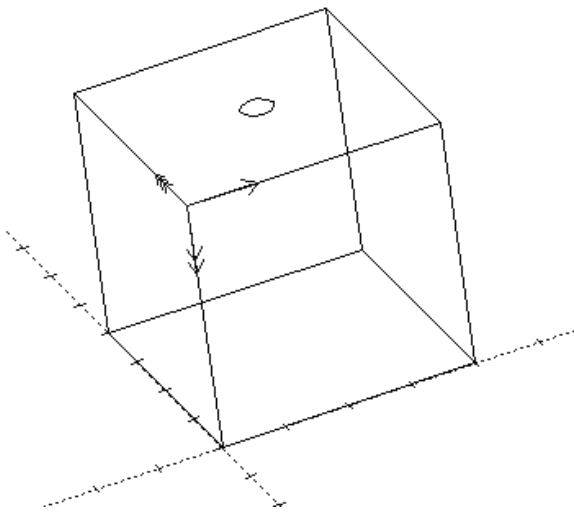
With your mouse, click on the Y-Axis (double arrow) and drag/rotate it while holding down your left mouse button so that it aligns with the front-left vertical line.

Click on and rotate the y-axis (double arrow) while holding down the left mouse button. align the y-axis with the vertical line.



Drag the cursor to the very lower left corner to fully align the UCS.

The result:



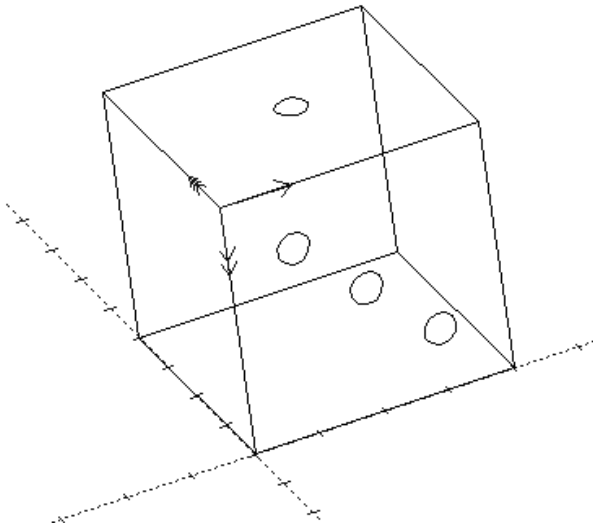
STEP 14

Now we are going to draw 3 arcs on the new Z-Plane. Select ARC from the main menu and then COORDINATE CENTER. Enter the following coordinates:

- A. Enter 1 for X.
- B. Enter 1 for Y.
- C. Enter .25 for the Radius.
- D. The Start Angle is 0.
- E. The End Angle is 360.
- F. It is not necessary to check the 3D box.
- G. Select CONTINUE.
- H. Enter 2 for X.
- I. Enter 2 for Y.
- J. Select CONTINUE.

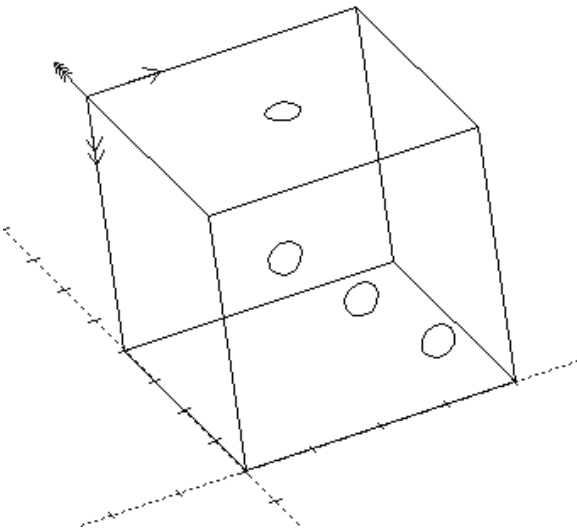
- K. Enter 3 for X.
- L. Enter 3 for Y and select OK.

The RESULT:



STEP 15

Now we are going to change the UCS one last time. Click on the center of the UCS, hold down your left mouse button and drag it to the top left corner as you can see in the image below.

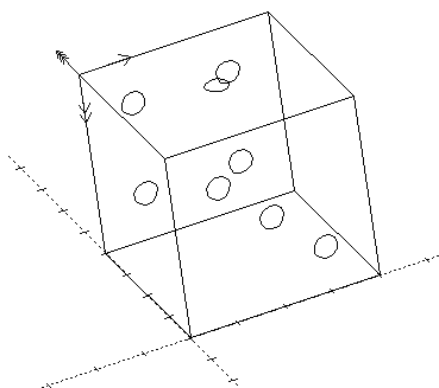


STEP 16

Go back to the ARC menu and select COORDINATE CENTER. Enter in the following coordinates:

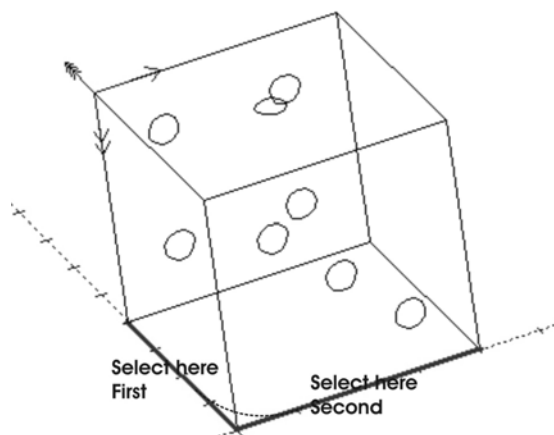
- M. Enter 1 for X and 1 for Y. Select CONTINUE.
- N. Enter 3 for X and 1 for Y. Select CONTINUE.
- O. Enter 3 for X and 3 for Y. Select CONTINUE.
- P. Enter 1 for X and 3 for Y. Select OK.

The result:

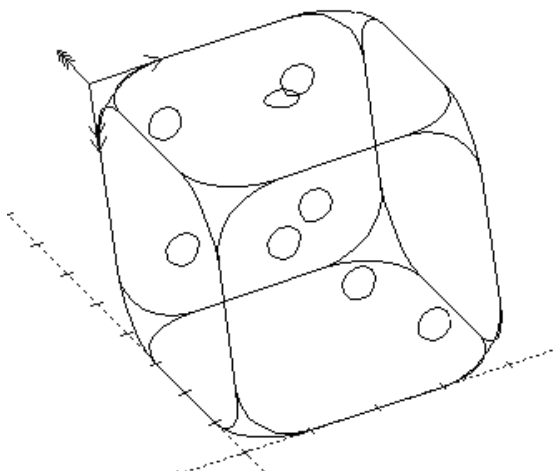


STEP 17

We could continue going around this cube drawing circles and would end up with what looks like a dice. The point here is that you can switch drawing planes fast with the new UCS. Let's go ahead and place 3D fillets on the corners of this cube. Select the **ARC** menu and **FILLET**. Enter the radius value of .5 and select OK. Follow the directions as you can see them below.



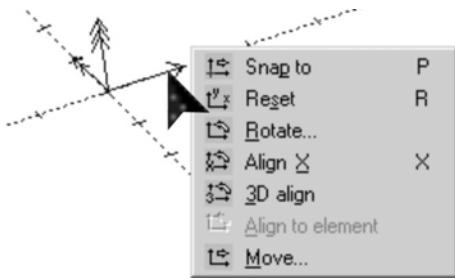
Select the first line and then hover your cursor over the second line to see a preview of the move. Then click your left mouse button again to execute the fillet move. Then do the next corner. If you make a mistake, simply select the UNDOES LAST COMMANT button and reselect the fillet function using the .5 radius value. Go around the entire cube drawing the fillets until you get the result as seen below. When finished, click your RIGHT mouse button to exit the fillet command. **REMEMBER: Make your selections of the lines going counter clockwise to draw the fillets.**



You have successfully completed this lesson.

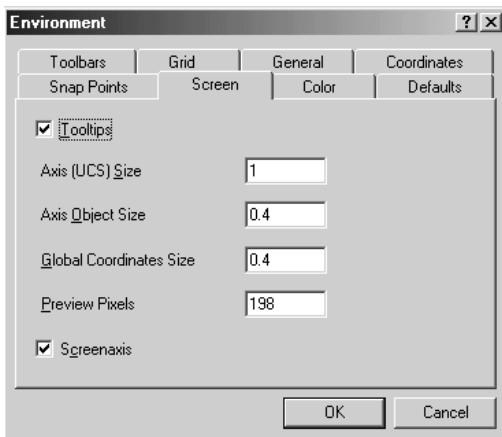
MORE ON THE UCS:

Once again, by resting your cursor on the **UCS** itself and then RIGHT-clicking your mouse, you will bring up the controls for it. See below.



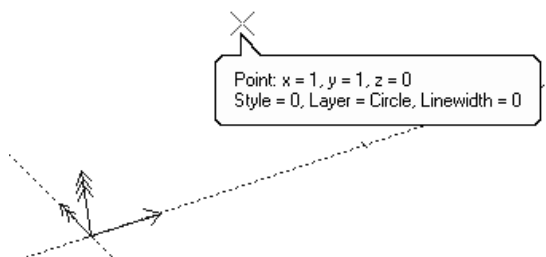
The UCS is a powerful aspect of the software that will enable you to change the entire coordinate system on the fly. You should practice using the UCS to make drawing your 2D and 3D parts faster and easier.

The **ATTRIBUTES** of the **UCS** can be changed in the **ENVIRONMENT** dialog under the **FILE** menu. Let's take a look.

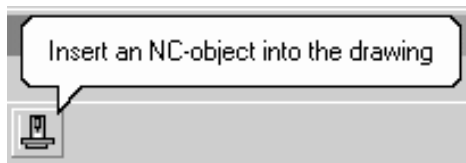


You can access this by selecting the FILE menu and then ENVIRONMENT and by then choosing the SCREEN tab. There are several options available in this dialog. Let's take a closer look at each of them.

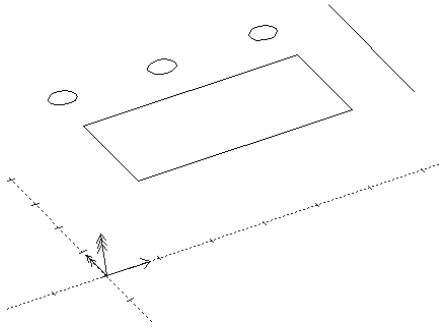
- **TOOLTIPS:** The **Tooltips** option allows you to rest your cursor over an entity or an icon and gather the information on that item. See below.



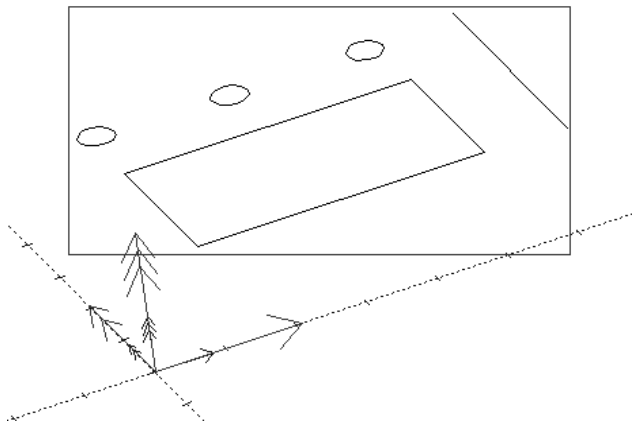
You can also verify or gain the information on what a function icon is.



- **The UCS SIZE:** This allows you to change the actual size of the UCS itself.
- **The UCS OBJECT SIZE:** This allows you to change the size of the UCS for grouped geometry. Take the example of an imported IGES or DXF file. The entities may be together in large groups. You can select these groups and then automatically change the size of the UCS. See below.

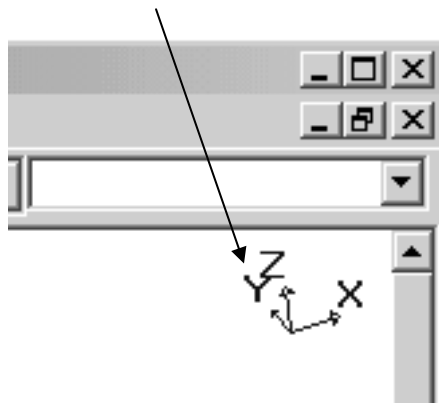


The entities in the image above are all grouped. By selecting the entities, the UCS will automatically change to the OBJECT SIZE value that is entered into this location. See below.



- **GLOBAL COORDINATE SIZE:** The Global Coordinate Size directly has to do with the Global Axis in the upper RIGHT corner of the screen. From the earlier VIEW functions, we covered this as a fast access to the various views by double-clicking on it. See below.

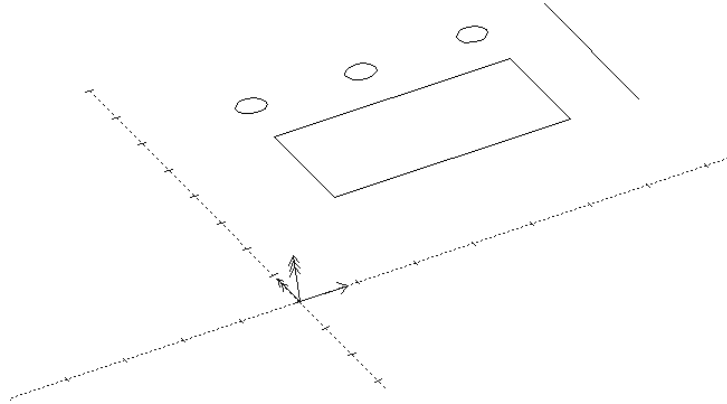
The Global Coordinate Axis.



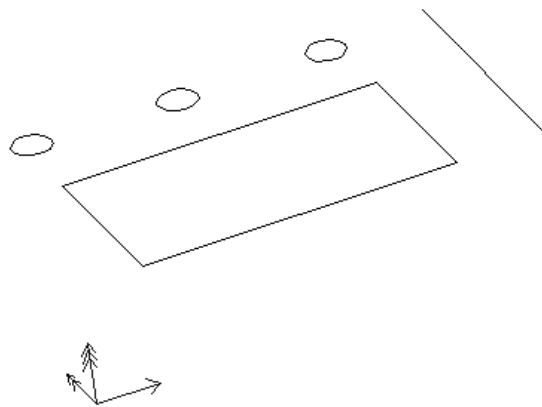
You can also click on this axis and rotate it with your mouse by holding down the left mouse button while doing so. With this function; Global Coordinate Size, you can make this axis larger or smaller.

- **PREVIEW PIXELS:** This adds or subtracts the number of pixels used to display the part in the PREVIEW window when you import or open a drawing in the software.
- **The SCREEN AXIS option:** This option allows you to turn the main drawing axis on or off. Generally, this option is selected. However, by un-checking this option, you will remove the main drawing/grid axis altogether. See below.

The Screen Axis ON.



The Screen Axis OFF.



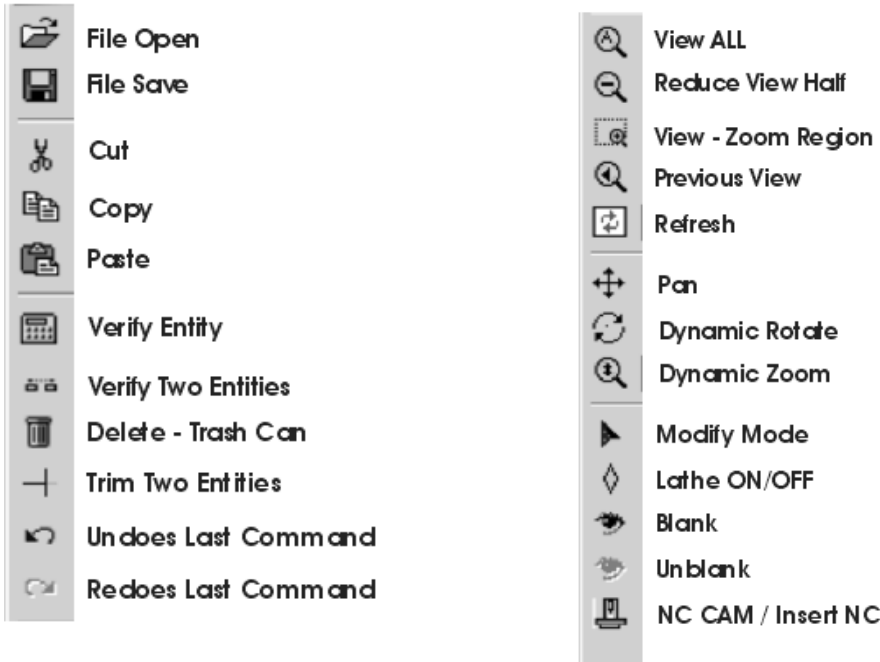
After making a change in this environment tab, you will always need to select OK to change the attributes. If you decide NOT to use the changes that you entered, simply select CANCEL. **REMEMBER:** *The AXIS DRAG option is located in the DEFAULT tab of the ENVIRONMENT dialog. As discussed earlier, this allows you to click on the UCS and drag it to another location in a drawing.*

SPECIAL UCS NOTE: In addition to what you have just learned, before generating toolpath for a solid model or surface you **MUST RESET** the **UCS**.

Section 13. The Software Icons.

The Version 21 system has a wide variety of function **Icons**. You have already learned how to add them to toolbars and toolboxes. Here is a brief look at them.

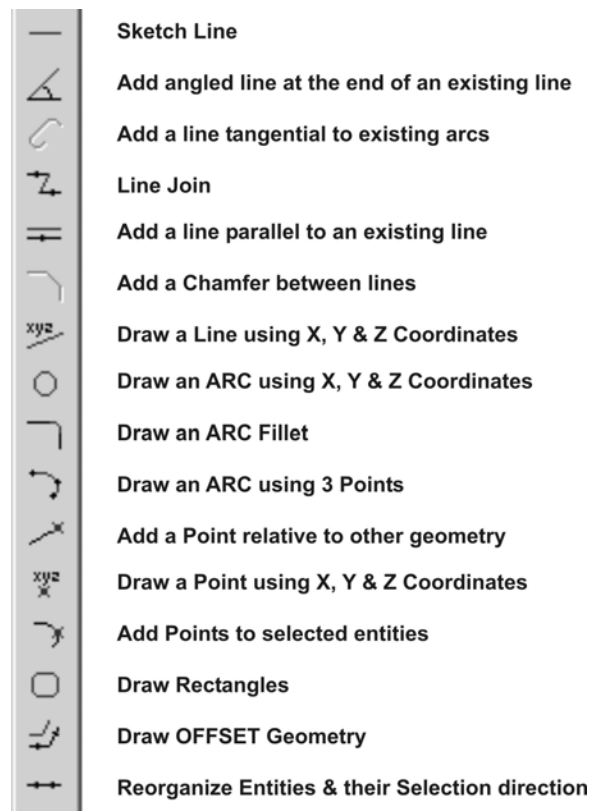
Main Screen Toolbar Icons:



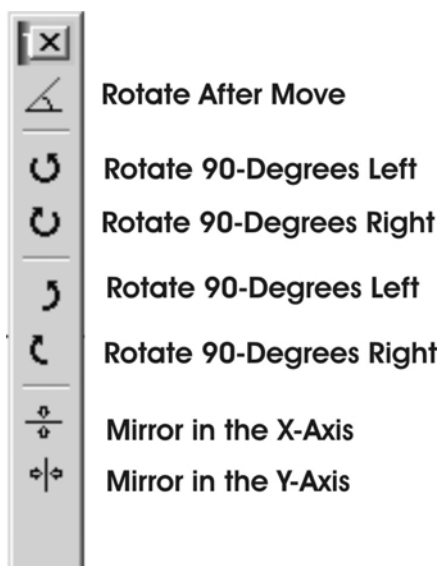
The Selection Toolbox:



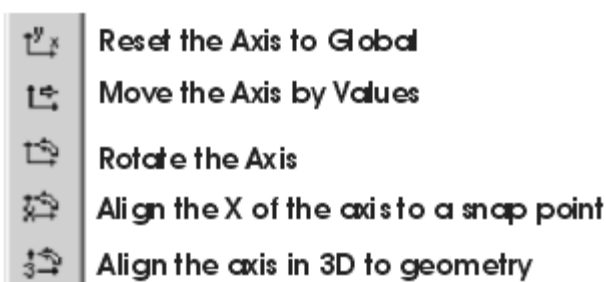
The Construction Toolbox:



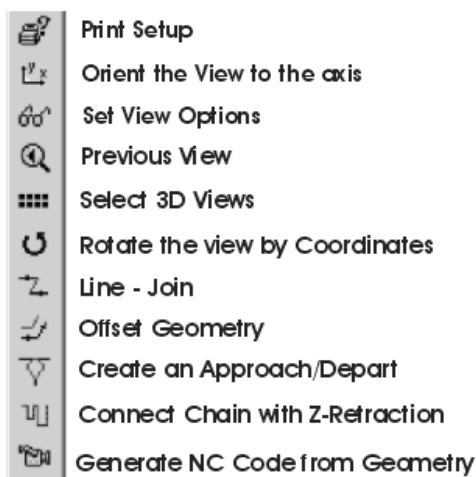
The Transform Toolbox:



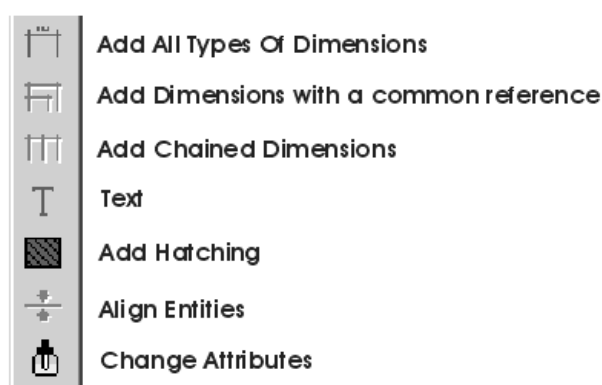
Additional Tools, Icons:



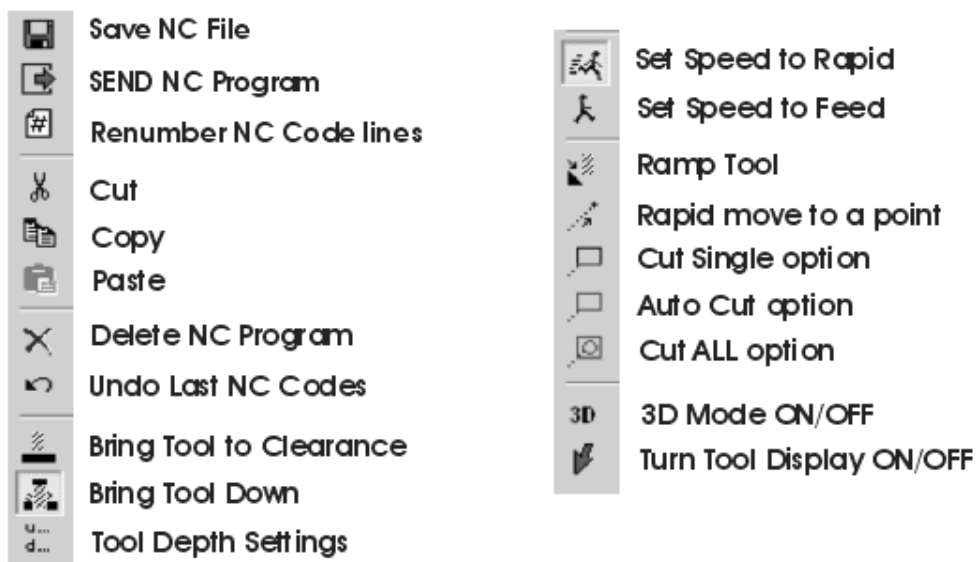
More Additional Tools, Icons:



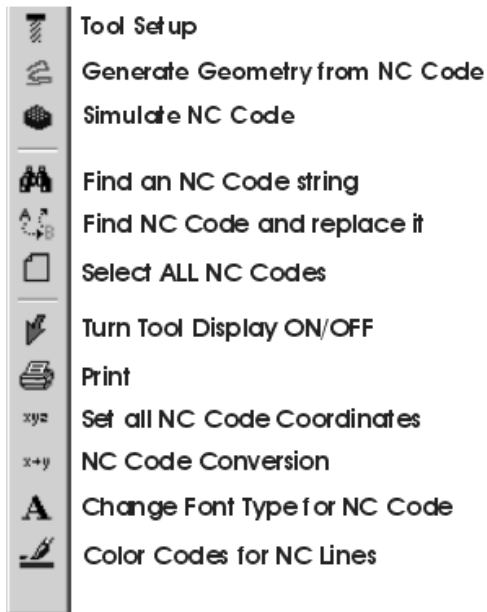
The Annotation Toolbox:



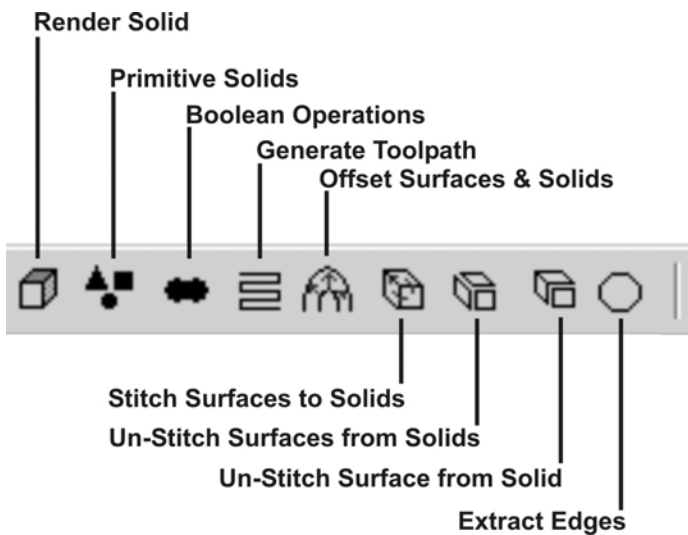
Standard CAM Toolbar:



Additional CAM Icons:



Solids & Surfaces Icons:



You can use this section for fast reference as needed while you are training with this manual.

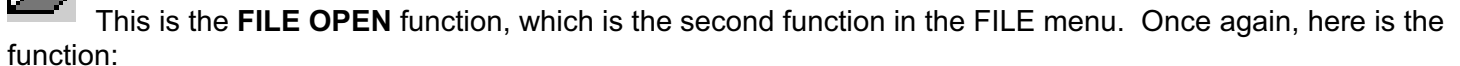
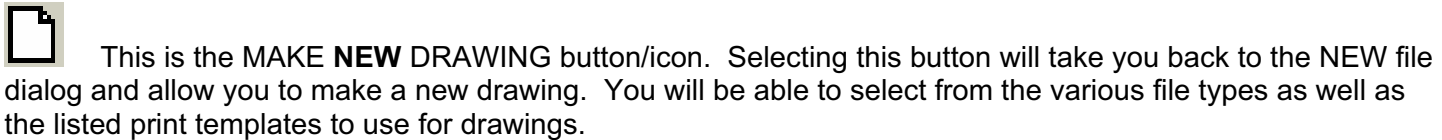
CHAPTER 4

THE FILE MENU THROUGH THE OTHER MENU

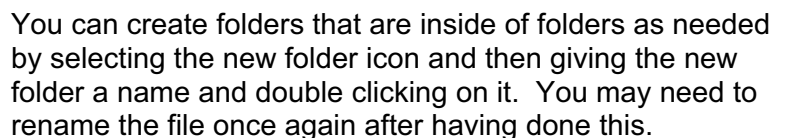
- Section 14. The File Menu
- Section 15. The Edit Menu
- Section 16. The Change Menu
- Section 17. The Point Menu
- Section 18. The Line Menu
- Section 19. The Arc Menu
- Section 20. The Spline Menu
- Section 21. The Other Menu



The **File** menu primarily deals with functions having to do with CAD files and the basic environmental software settings. Here, we are going to explain what each of the functions under FILE will do and how to use them.

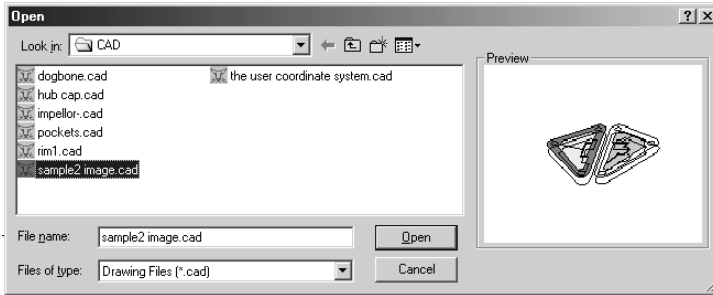


The screenshot shows the 'Open' dialog box in BobCAD-CAM version 18.0. The 'Look in:' dropdown is set to 'Templates'. The file list on the left includes CAD, DXF, IGES, NC code, NC config, and Scripts. The 'Files of type:' dropdown is open, showing a list of file types including Drawing Files (*.cad, *.dgn, *.asc, *.dtd, *.igs), NC Files (*.nc, *.tap), and Test Drawings (*.asd). The 'Open' button is highlighted.

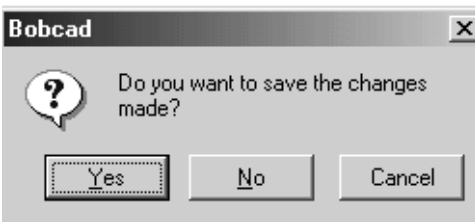
 This is the **FILE Save** function. By selecting this function, the save dialog will appear allowing you to save the drawing and direct it to a specific folder or location so that you can come back to it at a later time. If you have not saved the current drawing on the screen, the SAVE dialog will appear asking if you would like to save the file before exiting the software. REMEMBER, it is good to save an existing drawing as a .CAD file, which is native to the BobCAD-CAM software and will prevent any corruption from taking place. Simply select CAD under SAVE AS TYPE, name the file and direct it to a desired folder or the **CAD folder which should already be opened** and select OK.



This is the **FILE MERGE** function. By selecting this icon you will bring up the OPEN dialog to select a file for **Merging** with an existing file. An example of using this function would be for lathe part programming. You would merge in a specific lathe tool with the lathe profile drawing for creating the roughing toolpath.



FILE CLOSE: This function is used for exiting your drawing. If you select this function before saving an existing drawing, the software will automatically ask you if you would like to save the file.



This is the **FILE SAVE AS** function. By selecting this function you can save the same drawing for a second time with new changes or a different name. You will NOT lose the initial drawing and is a way to make multiple files of the same drawing or part. Selecting this function brings up the SAVE AS dialog. This is the same dialog used in the FILE SAVE function.

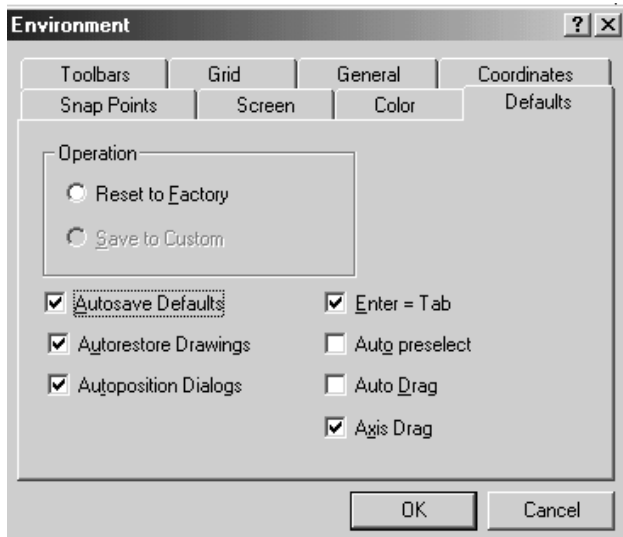


This is the **FILE PRINT** function. Selecting this function will bring up your printer dialog allowing you to print files or file templates that you have created in BobCAD. This will also allow you to switch from printer to plotter or to a different printer.



This is the **PRINT SETUP** function. This allows you to configure your printer properties through BobCAD. You may want to use a printer or a plotter for your drawings. This is also where you can switch between printers and settings.

ENVIRONMENT: The **Environment** dialog offers you the ability to change or modify some of the main software settings. Let's take a look at the ENVIRONMENT dialog below.

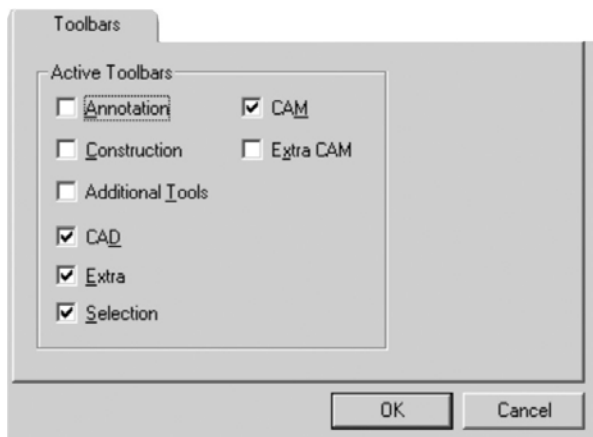


As you can see in the Environment dialog image, this box has eight tabs. Each one of these tabs represents a separate section of the software and allows you to make specific changes to the main settings. The available TABS are:

- **Toolbars.**
- **Grid.**
- **General.**
- **Coordinates.**
- **Snap Point.**
- **Screen.**
- **Color.**
- **Defaults.**

If you have made changes to the Version 21 software and would like to RESET these defaults or main settings, you can select the **RESET TO FACTORY** button under OPERATION, located on the DEFAULT TAB. **HOWEVER, REMEMBER THAT IF YOU SELECT THIS OPTION YOU WILL LOOSE ALL PREVIOUS CUSTOM SETTINGS THAT YOU MAY HAVE MADE.** Let's take a closer look at each one of these tabs individually.

TOOLBARS TAB: The **Toolbars** tab lists important toolboxes that the Version 21 system offers. You can enable or disable them in this dialog tab. Let's take a look at what each option is and what it does for you.



- **ANNOTATION:** This toolbar offers geometry, text, dimension and attributes functions. See below.



All of the functions of the ANNOTATION toolbox will only be available if both geometry and dimension of the geometry are selected together.

- **EXTRA:** This is also located on the main toolbar of the software. The two options for this box as a default setup are the **ACTIVE COLOR** menu and the **ACTIVE LAYER** pull-down menu. See below.

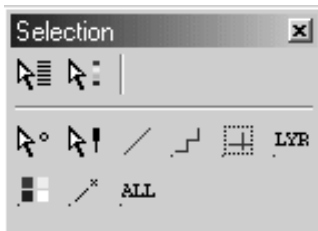


By selecting the active color drop down menu, you can switch drawing colors on the fly while using BobCAD. Its default drawing color is black. The drop down active layer menu just to the right, allows you to change the layer in which you are drawing quickly.

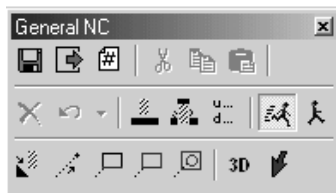
Example, if you had a drawing with the layers; circles, holes and squares and you wanted to start drawing on the circles layer, you would select this menu and circles. At that time, everything that you drew would automatically be set on that layer. By switching to another layer, everything that you drew from that point until changed would be drawn on the new layer. When there are NO layers in your drawing, you may see DEFAULT listed or nothing will be listed.

You cannot name a new layer from this box. To start a new layer, you would select the CHANGE menu, then ATTRIBUTES and name a new layer for your drawing. Then the new layer will be listed in this active layer menu.

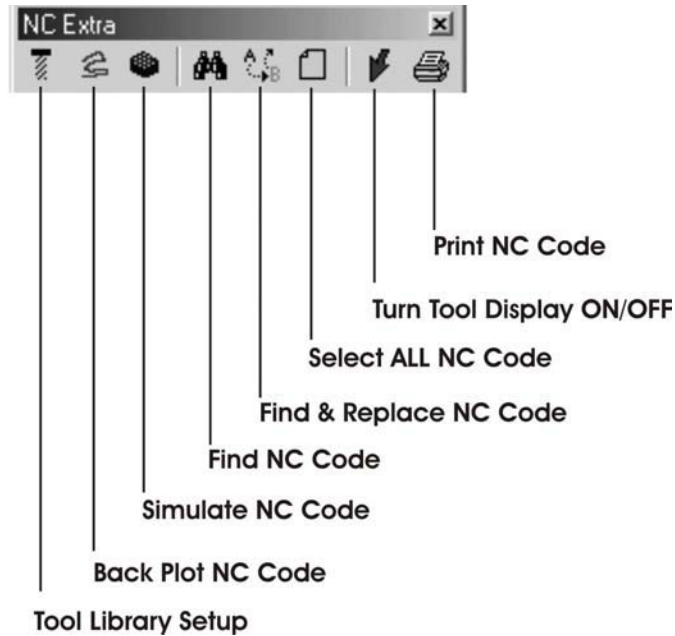
- **SELECTION:** This is the standard SELECTION toolbar/toolbox. When this is locked on a sidewall it can be referred to as a toolbar yet can be moved to float on the main drawing screen if needed.



- **CAM:** This is the main CAM toolbar that is located on the LEFT side of the NC CAM when opened. This offers tool depth settings, tooling functions, cutting options as well as file management. See below.



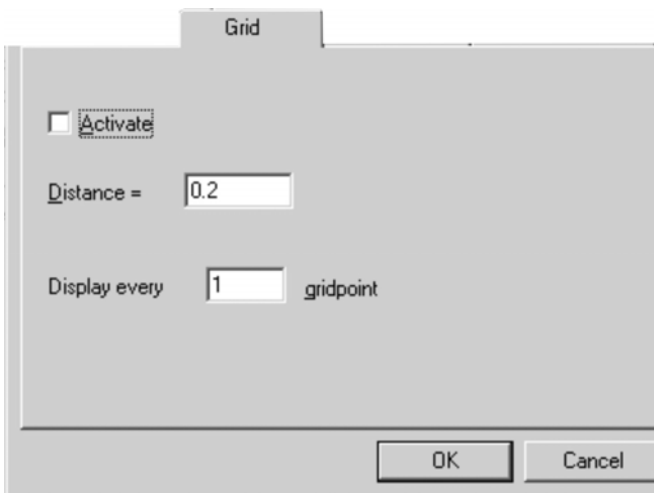
- **EXTRA CAM:** This toolbox seen below offers CAM functions for setting up tools, back plotting NC Code and more.



In order to use the Extra CAM toolbox, you must have the NC-CAM open for the generation of G-Code/NC Code. The items that are listed as default for this toolbox are located in the FILE menu of the NC-CAM and will be covered later in this training manual.

By selecting the options of the TOOLBARS tab you can turn or turn off these toolboxes and toolbars. Now let's have a look at the next TAB of the ENVIRONMENT dialog.

GRID TAB: The **Grid** tab gives you special options for the drawing Snap Grid. See below.

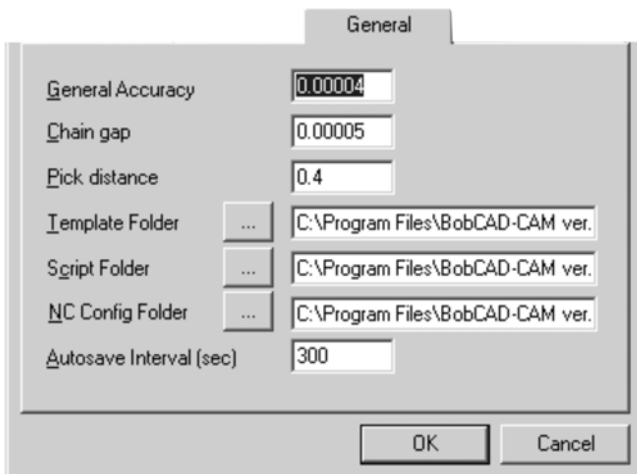


The GRID tab offers you:

- The option to activate / deactivate the snap grid itself.
- The **distance** between the grid points.
- **Display Every** refers to how many of the grid points are displayed based off of the distance value. For example, if the distance value was .5 and the display value was 1. There would be a grid point at every 1-inch location, NOT every .5 location. Here it is again, for example, if the distance was every .25-inches and the display was set at 1, there would be a grid point at every 1-inch location. If you then changed the Display Every value to .25, then a grid point would be at every .25-inch location. You see, the display simply allows you to *DISPLAY* a point. If the Display Every value is smaller than the DISTANCE value, nothing happens because the DISTANCE value controls the location of each snap point on the grid.

Therefore, the Display Every value would typically be the same or greater than the Distance value. Using the Snap Grid can speed up your drawing time once you are familiar with it. Let's take a look at the next TAB.

GENERAL TAB: The **General** TAB gives you options on adjusting the general software accuracy (how precise the software reacts to functions), the picking distance when working and selecting geometry (how close or far away your mouse must be when selecting entities), the accuracy of CHAIN GAPS (A string or chain of entities), where the destination is on your computer for the TEMPLATE folder, Where you will be directing visual basic scripts that you create on your computer, where the NC Configurations will be directed on your computer and how often the software will ask you if you would like to save your drawing while working.



GENERAL ACCURACY IN RELATION TO TOOLPATH:

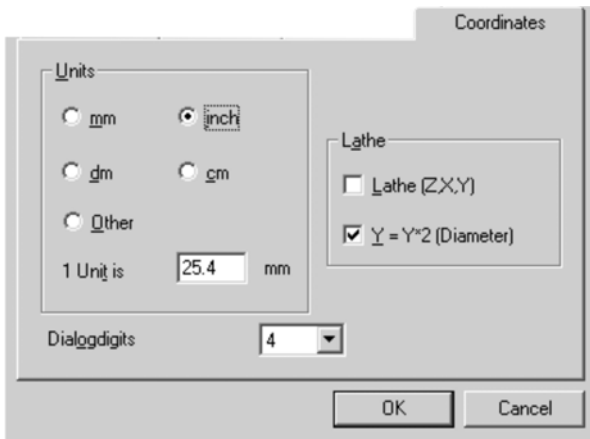
You will want to set the **General Accuracy** setting to match your machine accuracy settings. In other words, you may want to choose .0001 as your accuracy setting in Version 21 as most machines do not handle more than 4 decimal digits in accuracy for machining toolpath. If this value is higher than what your machine accepts, you may experience errors in the toolpath which will be discovered in simulation. These toolpath errors are called "crop circles."

"Crop circles" are caused by forcing the NC side of BobCAD-CAM to round the endpoints of arcs that are too small for the machine to properly cut. These arcs are smaller in length than the normal number of decimals of accuracy a machine can accept. When BobCAD-CAM's NC generation functions are used to produce code for them, they must be rounded to the maximum number of decimals the machine will read, causing the endpoints of these arcs to round down to the same coordinates as the starting points. On most machines, this leads to the machine cutting full circles in unwanted places on the part. The General Accuracy setting in BobCAD-CAM avoids this by causing the NC side to avoid these entities thereby guaranteeing good NC code.

TIPS ON CHAIN GAP SETTINGS:

Here is a tip on chain gaps. If you are using another CAD system than BobCAD-CAM and importing the files as DXF or IGES, you *may* want to increase the CHAIN GAP setting in the GENERAL tab of the environment dialog. The term, “**Chain Gap**” refers to the distance between each of the entities that make up a chain. That allows the software to *LOOK* farther for each entity of the chain.

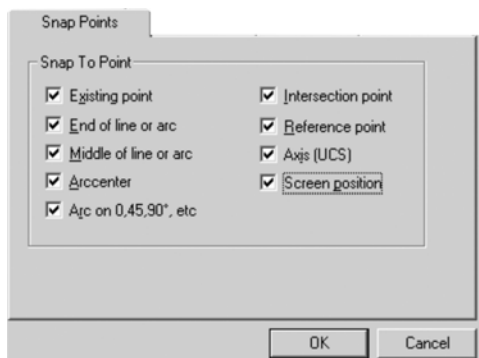
COORDINATES TAB: This tab offers the default and customizable coordinate settings/parameters for the BobCAD Version 21 software. Let’s take a look at this dialog below.



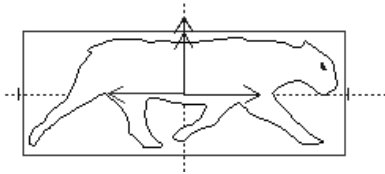
- The **UNIT** options allow you to change the coordinate value settings for the software from Inches to Millimeters, Decameters, Centimeters or OTHER which is user definable. Changing the UNIT value changes the drawing grid and the way the software deals with coordinates.
- **Dialog Digits** refers to the number of decimals to the RIGHT of the decimal point for all displayed coordinates including dimensions.
- **LATHE:** This section allows you to turn the software lathe mode ON. This changes the MILL X, Y & Z axis to X & Z for lathe.
- **Y=Y*2:** This is the diameter programming option for writing NC programs for a CNC Lathe. It basically changes the radius values to diameter values for lathe programs.

SNAP POINT TAB: This tab offers functionality that directly affects the use of **Snap** locations. As you can see in the dialog below, there are specific areas that can be activated as snap points when working with geometry or drawing. For example, snap functions would be LINE JOIN or ARC CENTER, that gives you the ability to work with snap points or snap locations on the screen.

Let’s take a look at each of the **snap** options individually.



- **EXISTING POINT:** This allows you to use an existing point on the screen as a snap location or snap point.
- **END OF A LINE OR ARC:** This allows you to use the ends of an existing line or the zero position of an arc as a snap point.
- **MIDDLE OF LINE OR ARC:** The middle of an arc is considered the 180-Degree location. This would be the 9-O'clock position on a clock.
- **ARC CENTER:** This is the very center of the complete arc or arc angle.
- **ARC ON 0, 45, 90-DEGREES etc:** This option allows you to make the 0, 45, 90, 120, 180, 225, 270, 315 and 360-degrees of an arc a snap location or snap point.
- **INTERSECTION POINT:** This option allows you to turn the intersections of all geometry on the screen into snap locations or snap points.
- **REFERENCE POINT:** This option allows you to use the reference point of an OLE object as the snap point or snap location. OLE stands for Object Linking & Embedding. For example, if you copied and pasted an object into BobCAD and then selected the object, you would see the reference point. This would become a snap point if you selected this option.

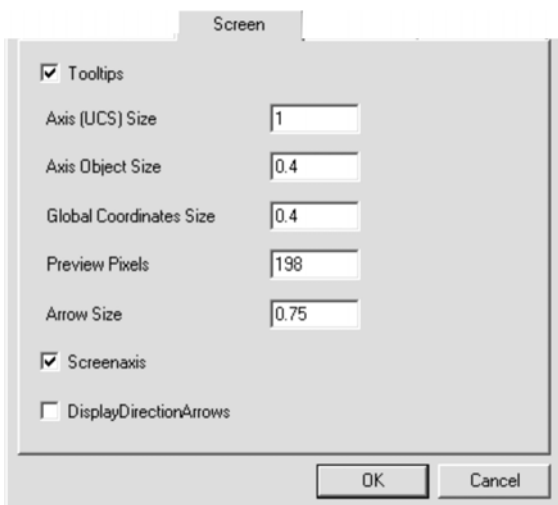


Grouped entities also have a reference point. Typically, if you group entities without moving them or changing the origin of the axis, the snap point location will be at the zero position on the drawing screen. That also goes for text that is placed on the screen and not yet smashed or interpolated into line and/or arc segments.

- **AXIS (UCS):** This allows you to make the UCS a snap location on the screen. This would be the origin of the coordinate axis. Remember, the global origin of the axis is Y 0, Y 0 and Z 0.
- **SCREEN POSITION:** This option allows you to choose the snap point or location yourself.

When you open BobCAD Version 21 for the first time, the default settings will have the first four options selected. After making a change, you will always want to select OK to activate the changes.

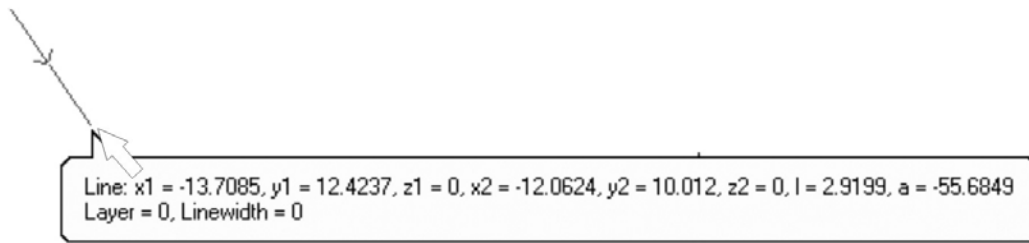
SCREEN TAB: This tab offers specific settings that relate to the main **Screen** and the UCS. Let's take a look at this tab below.



These options allow you to create an environment that suites your requirements.

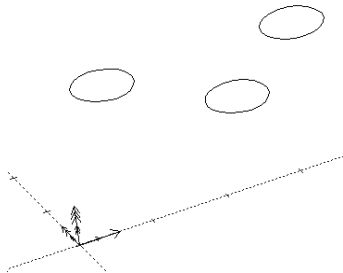
The SCREEN tab options are:

- **TOOL TIPS:** The tool tips option allows you to turn on and off the ability to gain information on whatever your cursor is hovering over. See below.

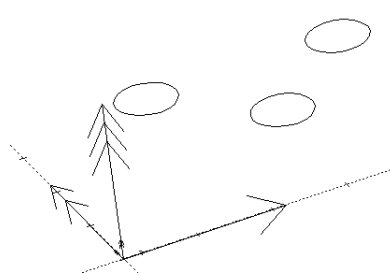


- **UCS SIZE:** This is the actual size of the UCS (User Coordinate System).
- **AXIS OBJECT SIZE:** This is the actual size of the object axis. For example, in the image below we have 3 arcs that are grouped together. You can see that the axis object size is only slightly larger than the UCS. By increasing the object axis size you can see the difference.

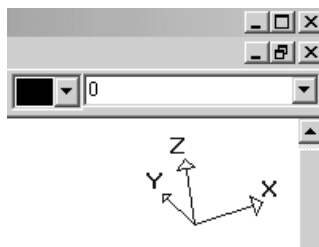
The axis size at 1-inch.



The axis size at 4-inches.



- **GLOBAL COORDINATE SIZE:** The global coordinate size is the size of the axis in the UPPER right corner of the screen. This axis allows you to change the view or rotation of the screen by clicking and while holding down your left mouse button, dragging the axis to a new location and view.

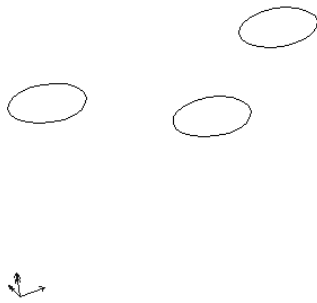


- **PREVIEW PIXELS:** This variable allows you to increase or decrease the number of actual pixels that make up a drawing in the preview window when opening or merging drawings.

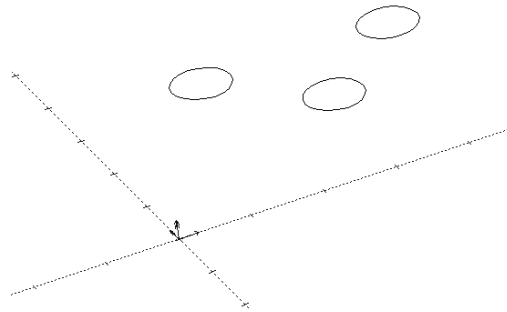


- **SCREEN AXIS:** This turns on and off the actual BobCAD screen/grid axis. See below.

Screen axis OFF.



Screen axis ON.

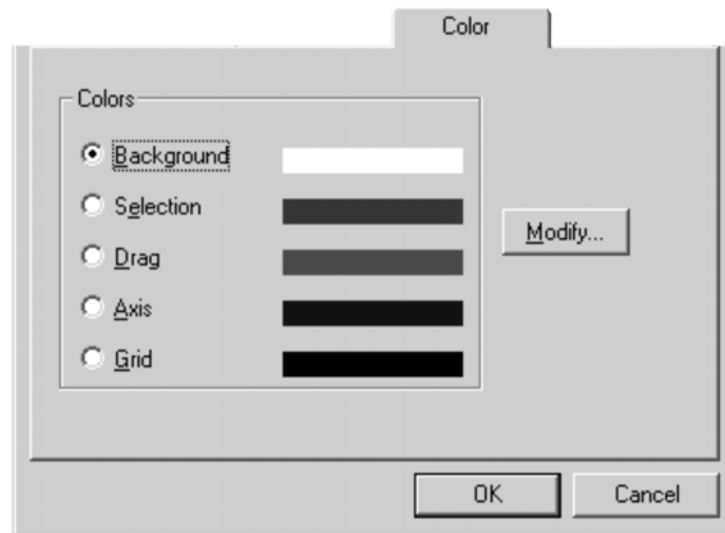


Once you make a setting change, you will need to make sure that you select OK to exit the environment dialog. If you decide to revert to the default “Factory” settings, select this option under OPTIONS in the DEFAULT tab.

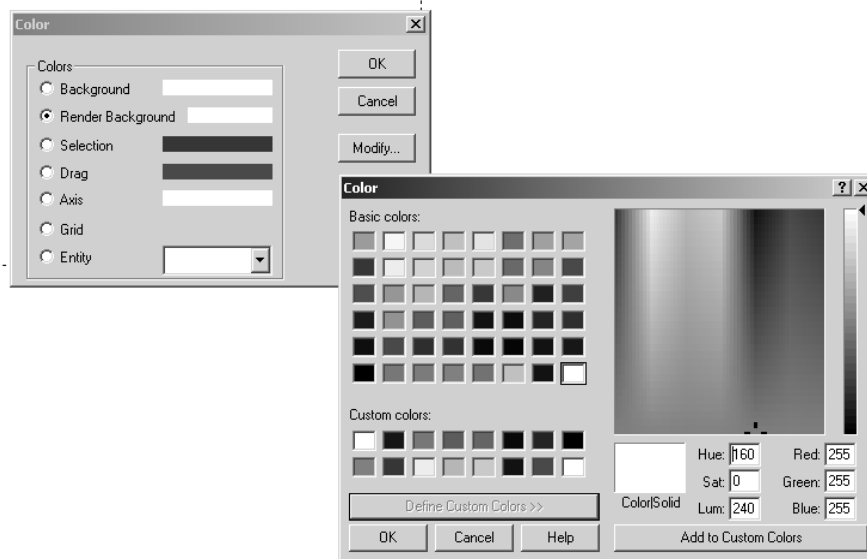
The next tab of the ENVIRONMENT dialog is the COLOR tab. Let’s take a look at this tab.

- **DISPLAY DIRECTION ARROWS:** This option allows you to turn on and off the directional arrows on selected geometry, chained geometry and selected toolpath. Check this option to turn them on and uncheck this option to turn them off.

COLOR TAB: The **Color** tab offers you the ability to modify the standard color settings for the BobCAD version 21.



- **BACKGROUND:** This is the actual background color of the screen. Typically, this color is white but can be changed by selecting the option and then MODIFY.
- **RENDER BACKGROUND:** This allows you to change the color of the background in the render window.



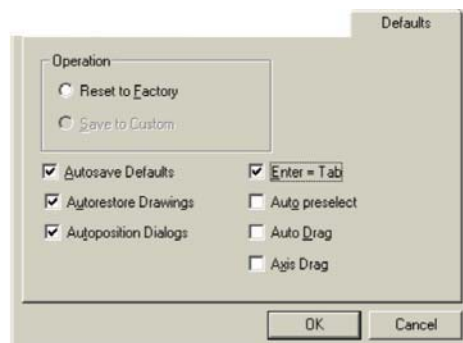
By selecting the Modify button, the custom color palette will appear as you can see in the image above. You will be offered 48 color boxes to select colors from. You can also enter your own specific RED, GREEN and BLUE (RGB) values for a specific color. After making a custom color change, select the ADD TO CUSTOM COLORS button and then select OK to use the color. You can simply make a color selection and click OK to see this color in the BACKGROUND color preview. Then, by selecting OK, the background of the main screen will automatically be changed to that selected color.

NOTE: That is the same method that would be carried out to change any of the options listed in the COLOR tab.

- **RENDER BACKGROUND:** This changes the render window background color.
- **SELECTION:** This is the actual selection color that appears each time that you click on an entity to select it. The default selection color is red.
- **DRAW:** This is the color that is used when you have a function that uses the click and drag method of selection including the VIEW REGION function.
- **AXIS:** This is the color of the main grid axis.
- **SNAP GRID:** This is the option that allows you to change the color of the snap points when you are using the snap grid for drawing.

REMEMBER to always select OK to execute the changes that you make in the environment dialog.

DEFAULTS TAB: The **Defaults** tab offers factory or standard software options for selection, file management and special settings. See this tab below.



Let's take a look at each one of these options individually. You will be offered 48 color boxes to select colors from.

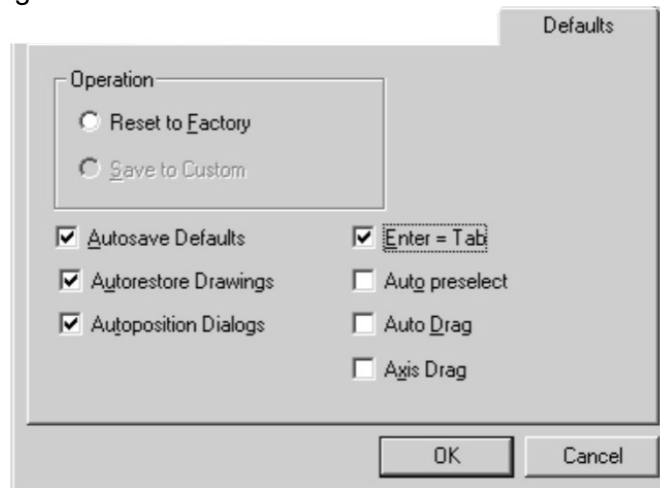
You can also enter your own specific RED, GREEN and BLUE (RGB) values for a specific color. After making a custom color change, select the ADD TO CUSTOM COLORS button and then select OK to use the color. You can simply make a color selection and click OK to see this color in the BACKGROUND color preview. Then, by selecting OK, the background of the main screen will automatically be changed to that selected color.

NOTE: *That is the same method that would be carried out to change any of the options listed in the COLOR tab.*

- **SELECTION:** This is the actual selection color that appears each time that you click on an entity to select it. The default selection color is red.
- **DRAW:** This is the color that is used when you have a function that uses the click and drag method of selection including the VIEW REGION function.
- **AXIS:** This is the color of the main grid axis.
- **SNAP GRID:** This is the option that allows you to change the color of the snap points when you are using the snap grid for drawing.

REMEMBER to always select OK to execute the changes that you make in the environment dialog.

Here is the DEFAULTS TAB again.



Let's take a look at each one of these options individually.

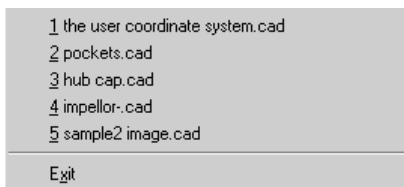
- **OPERATION:** This option allows you to completely reset the software default settings back to the way they were when you first received the software. It also allows you to SAVE CUSTOM settings. Then you can save them by selecting the SAVE TO CUSTOM option.

NOTE: *To activate the Save to Custom option you must UNCHECK the AUTO SAVE DEFAULTS option first. Then make the changes and select the SAVE TO CUSTOM option. Then RECHECK the AUTO SAVE DEFAULTS option to lock the settings into place.*

- **AUTO SAVE DEFAULTS:** This setting allows you to automatically save the current settings in the environment dialog. When you exit the software and return to the software, your settings will have remained the same.
- **AUTO RESTORE DRAWINGS:** This option allows you to automatically save and restore drawings. For example, if your computer crashes and you need to reboot, you will be able to open the BobCAD software and have your drawing still on the screen without having lost anything. This doesn't mean that you don't have to save your work as you go. It is simply an auto-backup type function of the software. **There are never any guaranties that you will not lose valuable data by not saving your work as you go.**

- **AUTO POSITION DIALOGS:** This option allows you to automatically position dialog boxes at specific locations on the screen for use. If you select this option and use the POINT COORDINATE function for example, each time you select the CONTINUE button the dialog will appear again in the same location on the drawing screen. If you unselect this option and use a dialog type function the box will continue to go back to the initial location that it was opened at. This is not a very significant function, however, can be useful when drawing with coordinate based dialog functions.
- **ENTER=TAB:** By selecting this option, you are allowing yourself to use the TAB key on your keyboard as the ENTER key on your keyboard. This will help you increase your drawing speed.
- **AUTO PRESELECT:** This option allows you to have the entities that you create in a drawing automatically be selected once drawn. For example, if you used the RECTANGLE function and entered specific coordinates and then clicked OK, the rectangle would appear *selected* rather than just appearing as a drawn rectangle.
- **AUTO DRAG:** This option allows you to click on the UCS and drag it to new locations on the screen or geometry.
- **AXIS DRAG:** Selecting this option will allow you to drag an **axis arrow** of the UCS to a new position. This is a valuable option for changing planes for drawing and machining. This also gives you the ability to click and drag geometry to new locations on the drawing screen. This includes solids.

THE LAST FILES: This is not necessarily a function so to speak, however can be used to open one of the last series of files that you had open and then closed out of. See below.

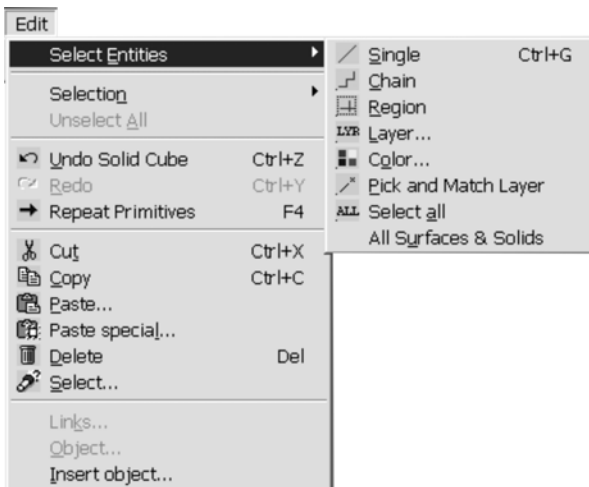


For example, the image above shows you the last 5 files that were opened on my system. By clicking on one of them, that file will automatically be opened in my Version 21 software.

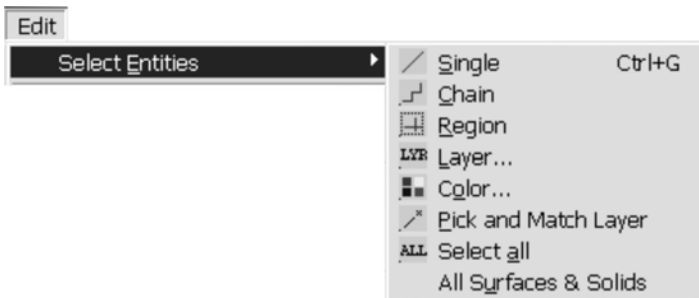
EXIT: This command will close the BobCAD-CAM Version 21 software entirely. If you have existing drawings open and unsaved, the software will ask you if you would like to save them before exiting. It is always important to save your work as you draw and/or edit them.

Section 15. The EDIT Menu.

The **EDIT** menu of the Version 21 software is very important to being able to work with drawings, solids and imported files. Below, you can see the image of the EDIT menu.



The EDIT menu has a special SUB-MENU for the first function called, “**Select Entities.**” This sub-menu can be seen below.



Let's take a look at each of these menus, their functions and their sub-menus.

EDIT – SELECT ENTITIES: As you can see the edit, select entities sub-menu offers the main selection functions.

The Version 21 CAD-CAM system deals with being able to draw and import wireframe geometry. Not everything you do will deal with solid models or surface machining. This is another reason why BobCAD-CAM is so effective. This training section will teach you how to use the edit menu for working with wireframe geometry; points, lines and arcs and how to select them. You can also use the edit, selection functions for solids. Each EDIT menu function is covered below.

- **SINGLE:** This function allows you to select a single entity for editing. If you hold down the **SHIFT** key on your keyboard you will be able to select multiple chained entities. Click the entity a second time or click your mouse button OFF of the selected entities and they will deselect. If you hold down the **CONTROL** key you can randomly select entities without selecting their chains, if the entities being selected are chained. Hold down the **CONTROL** key to randomly select entities without deselecting other selected geometry. Hold down the **SHIFT & CONTROL** keys on your keyboard and be able to select chains and other entities randomly without deselecting other entities. If you make a mistake, simply click on the entity again while holding down those keys or just the **CONTROL** key to deselect it without interfering with other selected geometry. This also works with solids.

SPECIAL NOTE: *If you are not familiar with this method of making geometry selections, you will need to practice working with geometry to become more efficient in drawing and editing. You may want to refresh your knowledge at this time on the MODIFY mode of the BobCAD Version 21 software.*

- **CHAIN:** This function is used for selecting entities that are chained together. Its basic use would be to select the function, click on the first entity of the chain, point your directional arrow in the direction of the other entities and click your mouse button again. Then hit the F3 key to select the entire chain. This is a primitive method of selecting chained geometry in BobCAD Version 21.
- **REGION:** This function allows you to click and drag a window over an area of a drawing for special selections of entities. Click your left mouse one time and drag the window over the desired geometry, then click your mouse button a second time to select all geometry in the selection window.
- **LAYER:** This function allows you to select specific layers in a drawing. You **MUST** hold down the CONTROL key to make random layer selections in the Layer Dialog. When finished making the selections, click OK to automatically select those chosen layers.
- **COLOR:** This function allows you to select colors in a drawing easily. Select the colors randomly by holding down the CONTROL key. By clicking OK, those chosen colors will automatically be selected.
- **PICK & MATCH LAYER:** This function allows you to select geometry or an entity in a drawing and automatically select ALL other entities that are saved on the same layer.
- **SELECT ALL:** This function will select **ALL** of the entities on the drawing screen.
- **SELECT ALL SURFACES & SOLIDS:** This function allows you to select all of the surfaces and/or solids on the drawing screen.

UNSELECT ALL: This is the menu command for deselecting ALL selected geometry on the drawing screen.

UNDO DELETE/PASTE: This function is basically the UNDO menu command. It will only be available in the EDIT menu as UNDO PASTE, if you have pasted an image or entities in the software. If you have and did not want to, simply select this function to undo the paste move. Otherwise, this function will be listed as UNDO DELETE. It is also a ONE STEP command meaning that it is not unlimited. The unlimited UNDO and REDO are located on the default main toolbar. They are seen as a forward and backward direction arrows. See below.

Unlimited UNDO & Unlimited REDO.



REDO DELETE: This is the function menu command for REDO and will be available if you have previously deleted entities from the drawing screen. Another way of putting it is that this undoes the last deletion of an entity. The menu command is NOT unlimited in functionality. Therefore it is available for one undo only.



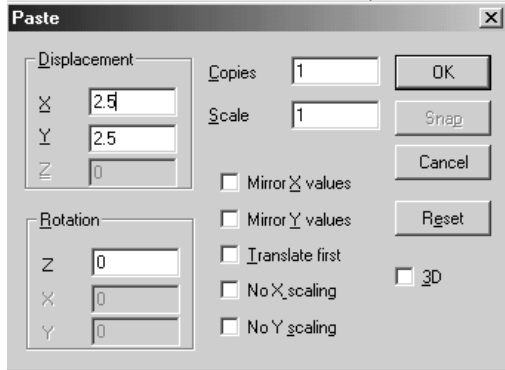
REPEATE DELETE: This function will be available for repeating the deletion of an entity. In other words, if you deleted an entity and then selected the UNDO command to put the entity back onto the drawing screen, you can select this menu command to re-delete the entity.

CUT: This menu command is also available as a main toolbar icon function and basically cuts selected geometry out of a drawing.

COPY: This menu command is also available as a main toolbar function icon and basically copies selected entities in a drawing.

PASTE: This menu command is also available as a function icon on the main toolbar and basically pastes copied or cut geometry or images into a drawing or onto the drawing screen.

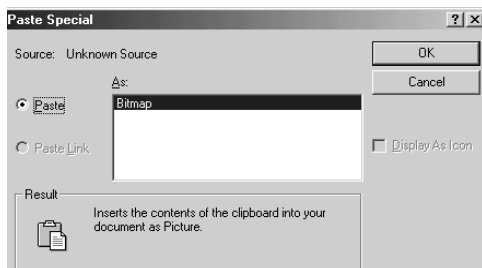
There is something important about this **paste** function: If you are pasting geometry or images into BobCAD, the PASTE dialog will appear allowing you options for placing the pasted object or entities. See below.



You can choose where you want the pasted geometry or object to be placed. By selecting the 3D option you can enter a Z value as well. When pasting an object, the reference point will be located where you have set the displacement, X, Y and/or Z. The DISPLACEMENT is relative to the origin of the axis. You can make more than one copy and mirror in the X and/or Y at the same time. You can also scale and/or rotate the object to paste before pasting. This function can be valuable when working with images for toolpath and NC Code generation.

PASTE SPECIAL: This menu command is used for importing Bitmap images into BobCAD. This also inserts the copied item as a picture.

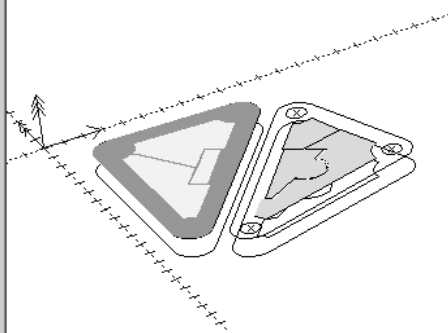
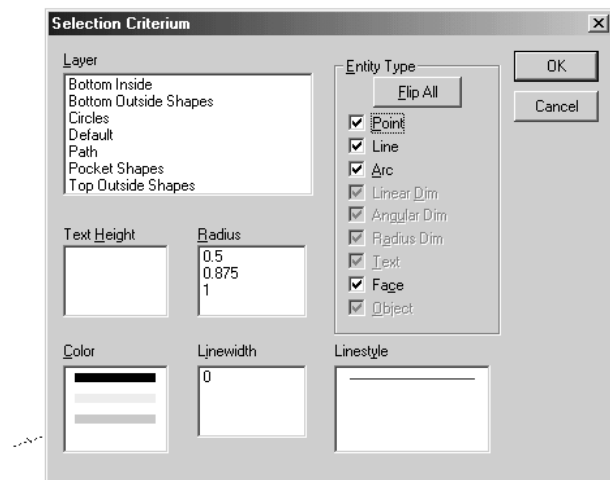
See this dialog below.



DELETE: This is the menu command for deleting selected entities from your drawing screen.



SELECT ELEMENTS IN A DIALOG: Selecting this icon brings up the Selection Criteria dialog. See below.



You can then **select elements/entities** by **layer, text height, line style, line width, color** and/or **element type**. There are nine different types of elements:

These entities are:

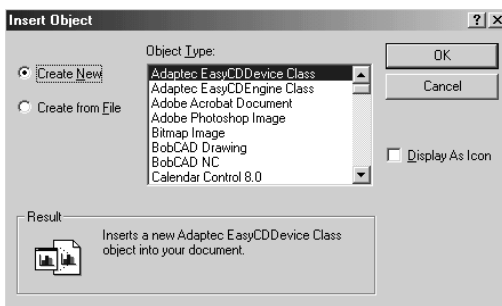
- **Construction point:** You can select all the points in a drawing.
- **Line:** You can select line styles, widths or all lines.
- **Arc:** You can select all arcs or arcs by their size radii.
- **Linear dimension:** You can select this style dimension.
- **Angular dimension:** You can select this style of dimension.
- **Radius dimension:** You can select these types of dimensions.
- **Text:** You can select the text and/or specific text heights.
- **Face:** You can select specific faces or face colors.
- **Object:** You can select objects that you may have in a drawing.
- **Layers:** You can make specific layer selections.
- **Colors:** You can select specific colors in a drawing. The matching entities will then be selected.

Remember, you MUST hold down the CONTROL key to make selections randomly. When finished, select OK to select the chosen items. This function is important for the selection of multiple drawing attributes.

LINKS: You can view and modify all of the links in the drawing. When Links is activated, a dialog box is displayed: "Links," displays the source of the links. "Update," allows you to update the links "automatically" or "manually". "Update Now," allows you the ability to update the links immediately. "Cancel Link," allows you to cancel the selected links altogether. The objects lapse into simple pictures. "Change Links," enables you to change the link to another file (e.g. when the file has been moved). All of this has to do with pasted OLE objects into the BobCAD software and allows you to work with a variety of image types for the creation of toolpath and NC code in the end.

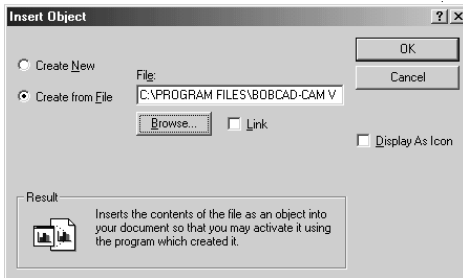
OBJECT: This is a menu function command that has to do with pasted objects; bitmaps, jpeg images etc. For example, you can paste a clip art image into BobCAD and then select this function to open the clip art image in Microsoft Paint for editing changes to the object itself. For those of you that use Corel Draw for instance, you can select the OBJECT menu command to open Corel and edit the object, bitmap, jpeg image etc. Another way to access this function is to select the object and RIGHT-CLICK your mouse. OBJECT will be listed for selection as needed. This function will only be available if you have pasted an object into BobCAD.

INSERT OBJECT: When you select this menu function/command the following dialog will appear as you can see in the next image.



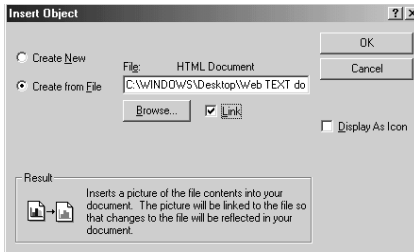
This Insert Object box will allow you to select a variety of object types for pasting into the BobCAD. You can attach and display the object as an icon or simply select OK to paste or import the object into BobCAD. The section under RESULT will provide you with details about the object that you are attempting to insert into the software.

If you select the CREATE FROM FILE option, the dialog will change as you can see in the image below.



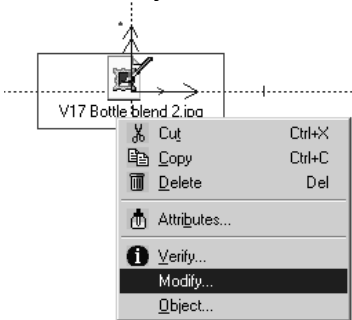
Now you can browse through your hard drive for the object that you wish to import. By selecting the LINK option, the dialog changes again and inserts a picture of the file contents into the BobCAD. The picture will then be linked to the file itself so that changes to the file will be reflected in your document.

See below.

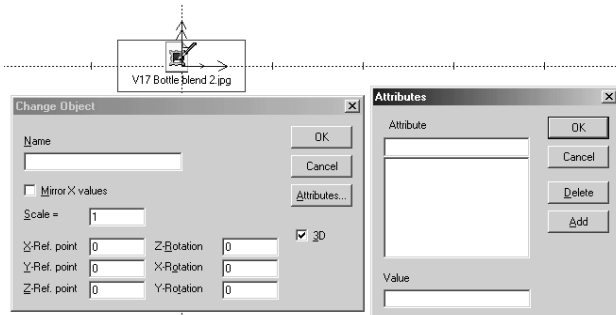


NOTE: This function is useful when you are creating a blueprint template and want to insert your company logo into the template so that when you print out the template your logo appears properly.

Another way to work with imported objects is to select the object and RIGHT-CLICK your mouse.

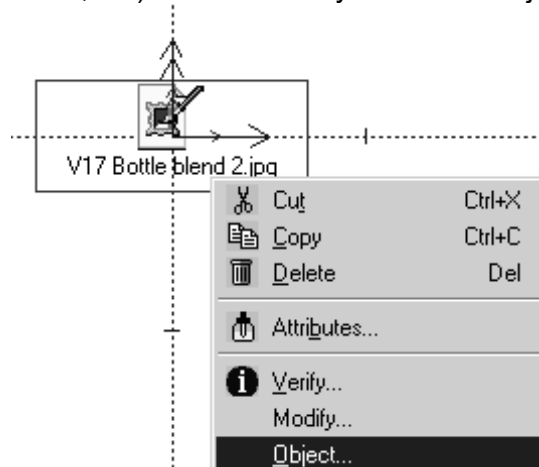


By selecting MODIFY, you will open the CHANGE OBJECT dialog. See below.



This new dialog will allow you to modify the object and even add attributes to it.

If you selected OBJECT from this menu that you can see below, you will open up the linked system (Microsoft Paint, Corel, etc) that will allow you edit the object.

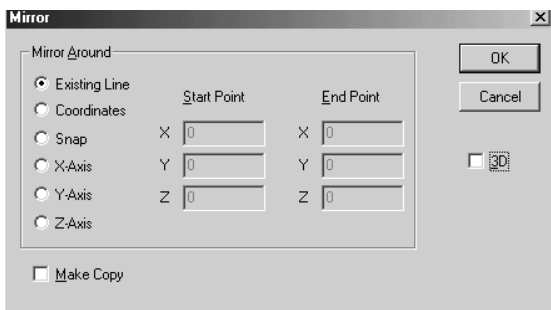


Section 16. The CHANGE Menu.

The **Change** menu offers a wide variety of geometry/entity changing functions that are important to the creation of toolpath and NC code. Below is the image of this menu. This section is going to offer you an in-depth look at each of this menus functions, what they do, how to use them and when you can gain programming speed by using this menu.



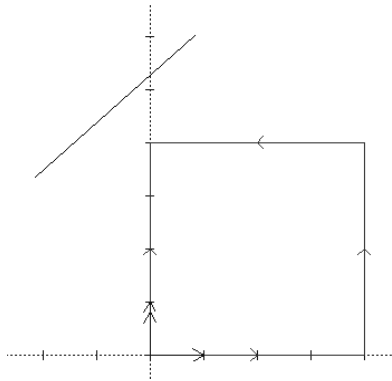
MIRROR: This function allows you to **Mirror** geometry on the screen a number of different ways. Let's take a closer look at this function below.



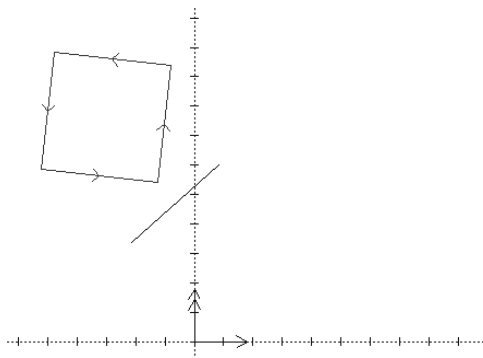
A basic fundamental about the MIRROR function is that before it can be used, you must have selected geometry/entities on the drawing screen. Only then will this function be available for use.

The mirror dialog offers:

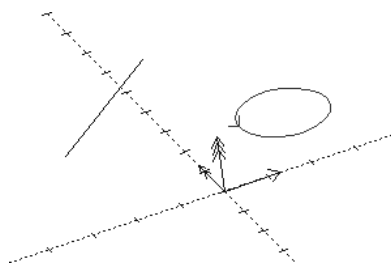
- **EXISTING LINE:** This allows you to mirror selected geometry around an existing line. Below you can see that we have a square with a line that has been sketched. The square is selected.



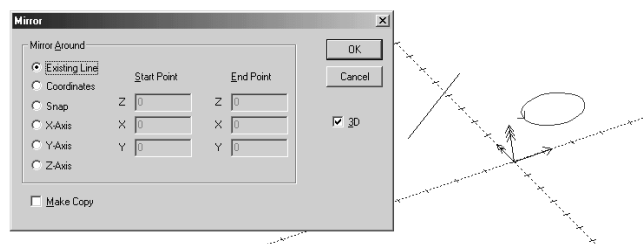
By selecting the mirror function and then EXISTING LINE, you would then select OK and click on the entity that you want to mirror the square around. See the result below.



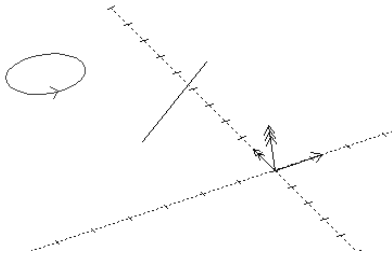
You can see that the square was automatically mirrored. Let's take a look at this option using an arc as the example. Below you can see an arc and the sketched line in 3D mode. The arc is pre-selected.



By selecting the mirror function, the dialog appears. The 3D option is already checked as we are in 3D mode already.



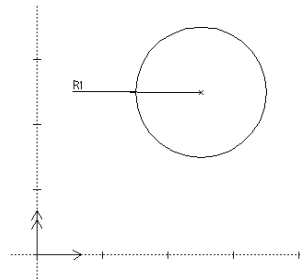
Select OK and click the existing line. You can see the result below.



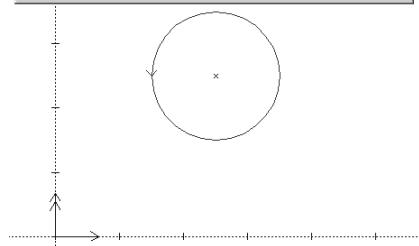
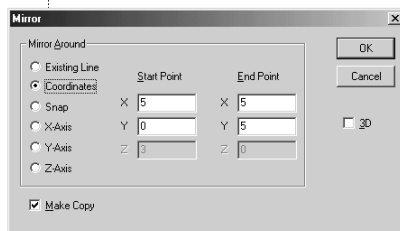
You can also make a copy by selecting the copy option. The mirror function can be used with 2D and 3D entities as well.

- **COORDINATES:** The **Coordinates** option allows you to mirror geometry using X, Y and/or Z coordinates. Basically, what you are doing is drawing an invisible line using coordinates in this dialog box. Let's take a closer look at this option as well.

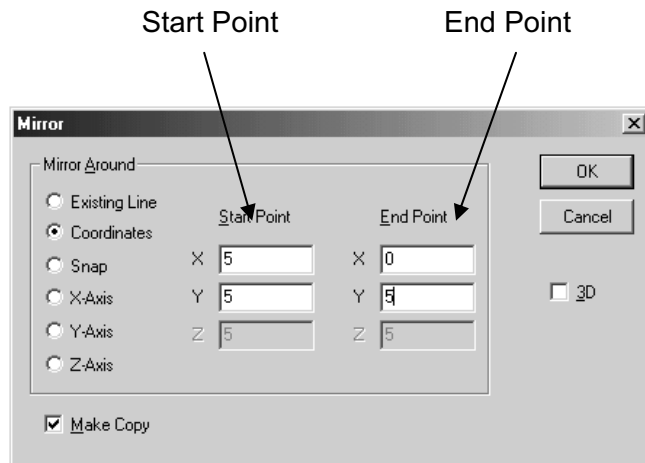
Below, you can see a 1-inch arc at X 2.5 and Y 2.5. We want to mirror this arc using the COORDINATES option.



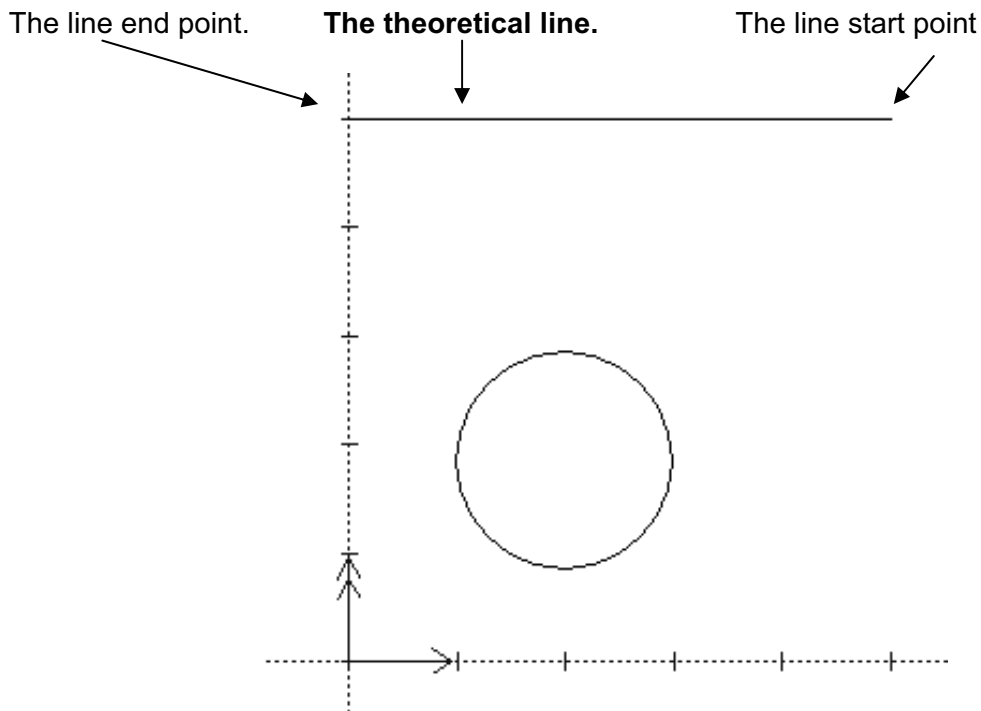
By selecting the arc and then the MIRROR, COORDINATES option (seen below) we can enter in the coordinate values that we want to use to draw an invisible line and mirror this arc. We selected the COPY option so the mirrored arc will be a copy.



When you are working with coordinates in this case, you will be dealing with a START POINT in X, Y and/or Z as well as an END POINT in X, Y and/or Z. The reason for this is that a line has a start point and also has an end point. See below.

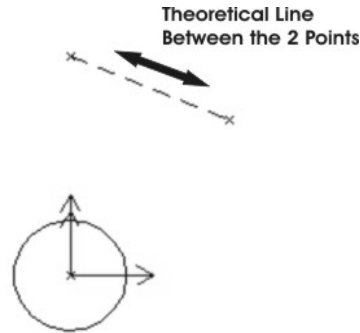


We have entered X 5 and Y 5 for the start point of the line and X 0 and Y 5 for the end point. Remember, we are drawing a theoretical line to mirror the circle.

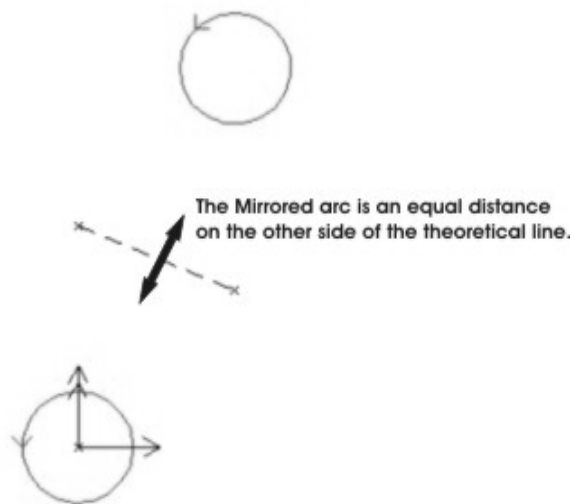


Select OK and the geometry will automatically be mirrored. REMEMBER, in this function, you are drawing a theoretical line to mirror the geometry around.

- **SNAP:** This function allows you to snap the theoretical line to two points in order to mirror geometry. Let's take a closer look below.

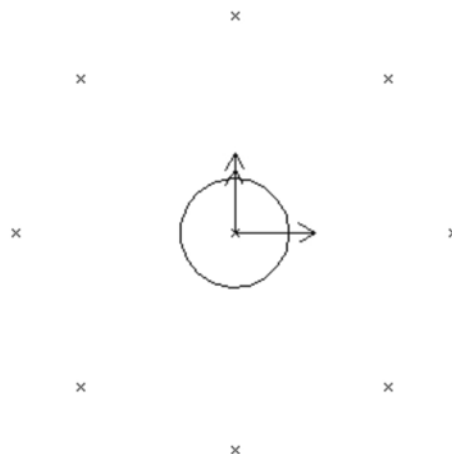


You would select the arc first to enable the use of the MIRROR function, then choose the SNAP option and select the MAKE COPY option. Select OK, click on the first point and the second point to snap a theoretical line and the arc would automatically be mirrored. See below.

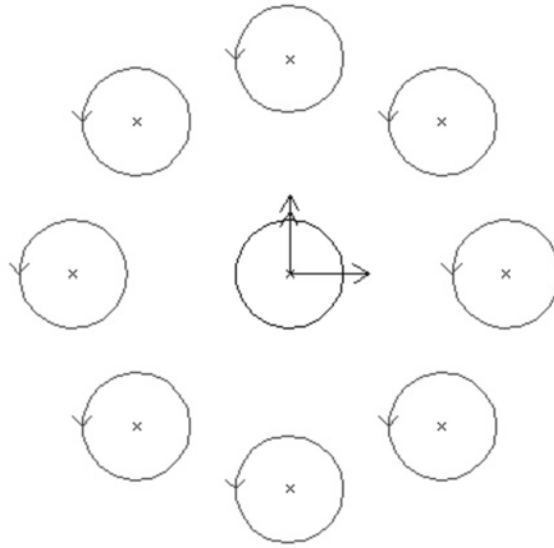


TIPS ON SNAPPING ENTITIES:

Here is a **Tip** on snapping entities. There is a fast way to snap geometry/entities to points by using the COPY and PASTE function. Here is an example below.

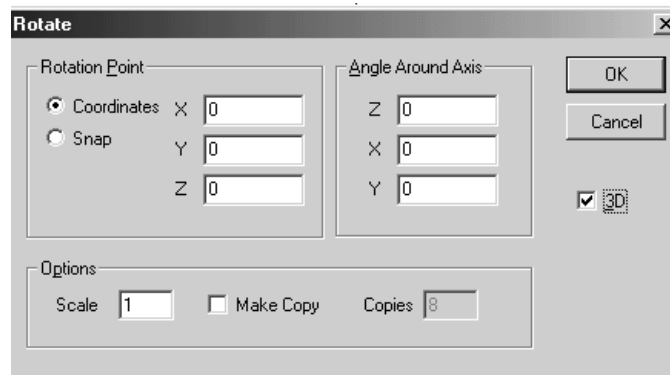


In the last image, you can see the arc with 8 points rotated around it. By selecting the arc and right-clicking your mouse you can choose the **COPY** command. Then click OFF the arc to deselect it. Then hold down the **SHIFT KEY** and select each of the points. When all the points are selected, right-click and choose **PASTE** to bring up the PASTE dialog. By selecting SNAP and selecting the **MAKE COPY** option in the PASTE dialog and then clicking OK you can snap a copy of the arc to each of the selected points.



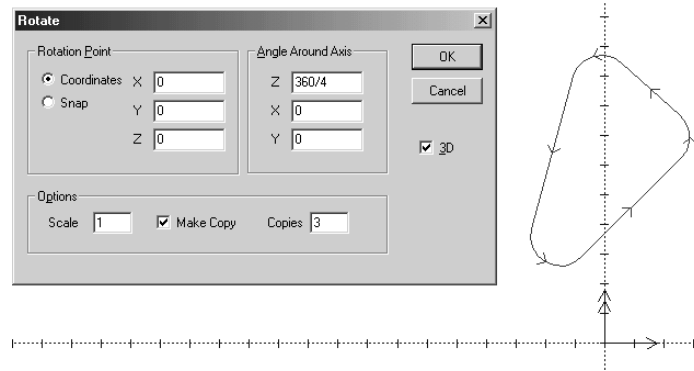
- **Y-AXIS:** This allows you to mirror geometry an equal distance from the zero/zero position in the Y-Axis.
- **X-AXIS:** This allows you to mirror geometry an equal distance from the zero/zero position in the X-Axis.
- **Z-AXIS:** This allows you to mirror geometry an equal distance from the zero/zero position in the Z-Axis. The geometry/entities to be mirrored will need to either be – or + in Z to see the mirrored change. Otherwise, the entities will be doubled up on top of each other.
- **The MAKE COPY option:** This allows you to make a single copy of the entity or entities being mirrored. Un-check this option to NOT make a copy.

ROTATE: The **Rotate** menu command allows you to rotate geometry around the screen as needed. As a basic rule, you must have entities selected before this function will be available in the CHANGE menu. Let's take a closer look at this dialog box below.

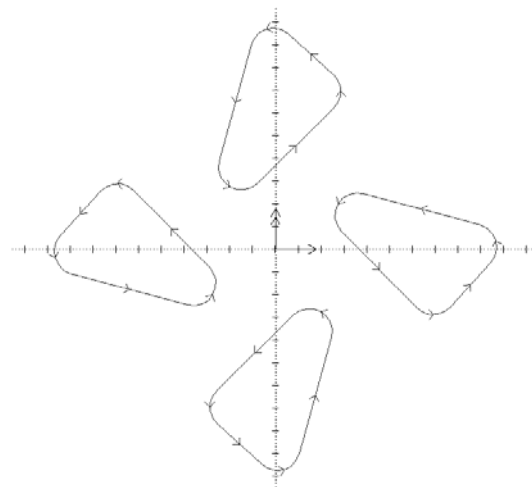


The rotate function gives you two options for rotating geometry in the X, Y or Z-Axis. Let's take a look at this below.

- **COORDINATES:** This option allows you to enter in X or Y coordinates as well as coordinates for the Angle you want to rotate around in X, Y and/or Z. To enable the Z entries, you must check the 3D option first. Let's take a look at this option.

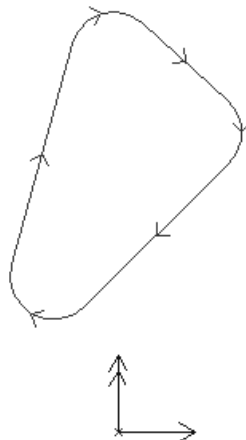


In the image above, you can see that we entered 360/4 (360 divided by 4), you would then hit the enter key, which gave us 90 as the DEGREE value to rotate around the Z-Axis. We selected the MAKE COPY option and entered 3. We entered 3 instead of 4 for copies as we already have one image on the screen. By selecting OK, you can see the result below.



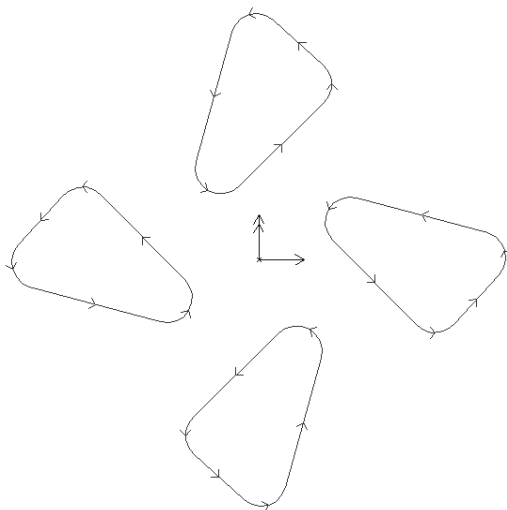
The geometry was rotated around X 0, Y 0 and Z 0, which is the center of the drawing grid axis as you can see in the image. By entering in an X and Y value under the ROTATION POINT section, you can change the location of the new copies that have been rotated.

- **SNAP:** This allows you to select a point to snap the rotation around. By selecting the SNAP option, the rotation point X, Y and/or Z entry spaces become deactivated. See below.



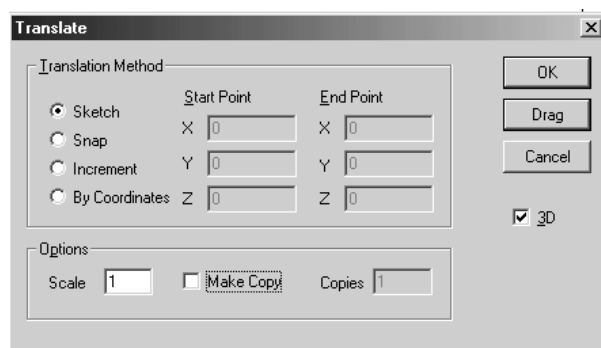
In the last image, there is a point at the zero X and Y position.

By holding down the SHIFT KEY and clicking on the geometry, you can then select the rotate function, the SNAP option and after having entered your angle variables, select OK and then click on the point to snap the rotated copies to. See below.



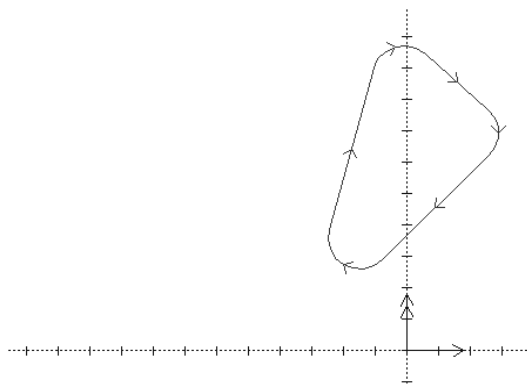
The **ANGLE AROUND** section refers to 360-degrees divided by the number of total copies, **LESS ONE** (as you already have it on the screen). The snap point that you choose on the screen will be the center of the rotation for the new copies to be rotated around. We will revisit the rotate function throughout this manual for special training lessons and for the GEAR function.

TRANSLATE: The **Translate** function is the next function down in the CHANGE menu. This function offers several options and ways to modify or translate geometry on the screen. Let's take a closer look at this function dialog below.



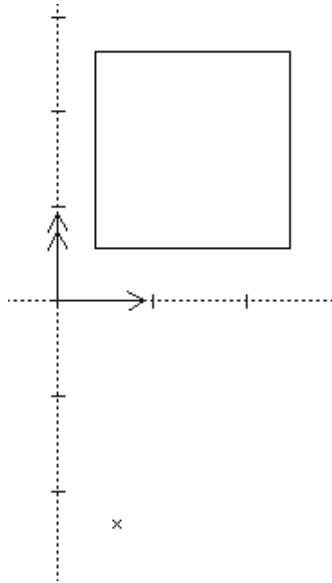
This function dialog as you can see offers 4 separate methods of translation entities and geometry. Take a closer look at each of these options below.

- **SKETCH:** The **SKETCH** method allows you to simply click your mouse on the screen to roughly sketch a new location for the entities to be moved. Let's take an even closer look.

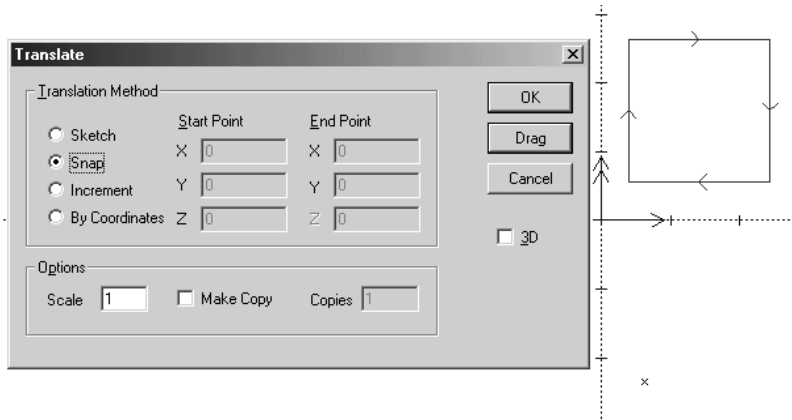


First of all, you must first select the entities that you want to translate before being able to activate this function. The SKETCH option gives you the ability to click your mouse at a FROM location on the screen and then click a second time at the desired TO location on the screen.

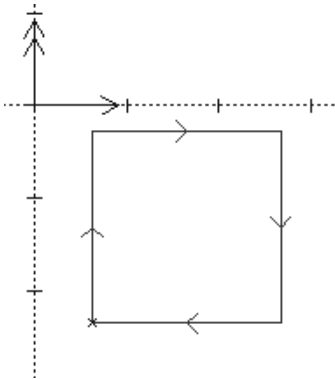
- **SNAP:** The SNAP translation method allows you to use a point for relocating geometry on the drawing screen. Let's take a quick look at this in the next image.



Here is a square and a point drawn on the screen. We want to relocate the square's lower left corner to the point that we drew. The first things that we need to do is select the square and then the TRANSLATE function in the CHANGE menu.



Select the SNAP method and then select OK. Now simply select the lower left corner of the square and then the point.

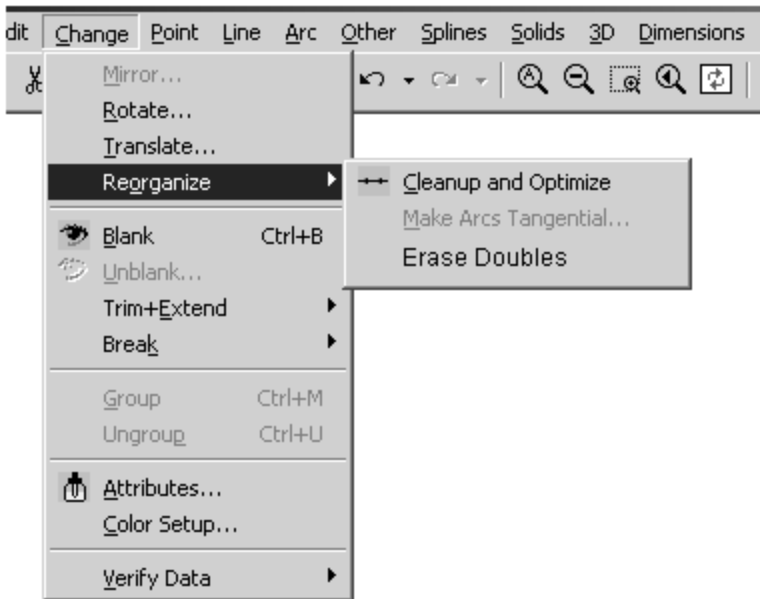


The square has been relocated. If you would like to make a copy, select the MAKE COPY option and enter the number of copies before selecting OK.

- **INCREMENTS:** This translation option allows you to move geometry around the screen incrementally. Meaning, from the geometries existing X, Y and/or Z location to a new X, Y and/or Z location.
- **BY COORDINATES:** This translation option gives you the ability to move geometry incrementally around the screen by using a START X, Y and/or Z Coordinate and an END X, Y and/or Z Coordinate. To activate a Z coordinate value, select the 3D option. You can also make copies if needed.

REMEMBER: To activate the TRANSLATE menu function you must have pre-selected geometry on the drawing screen.

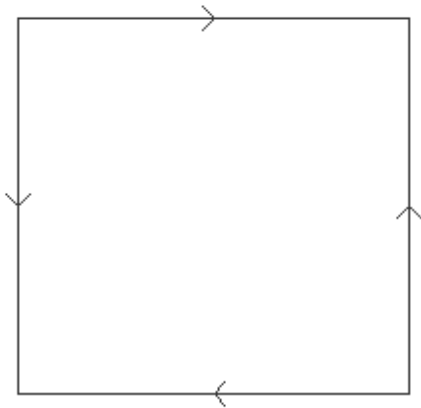
REORGANIZE: The **Reorganize** menu function has a special sub-menu with two separate functions listed.



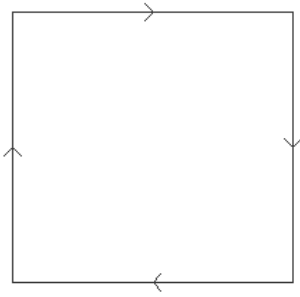
Let's take a closer look at these two REORGANIZE options.

- **CLEANUP & OPTIMIZE:** This function will automatically reorganize the selection arrows of a chain so that they are all going in the correct direction for use. For example, if you are using a function of the software that calls for the chain selection arrows to all go in the same direction and they currently are NOT, you can use this function to correct the direction of all selection arrows. See the example below.

Below are selection arrows going in un-organized directions.

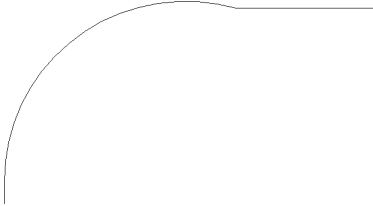


By using this function you can correct and reorganize the direction arrows.

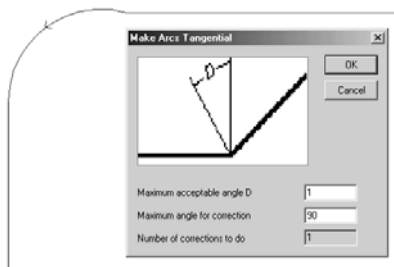


- **MAKE ARCS TANGENTIAL:** This sub-menu function allows you to modify an arc to make it tangent to a line. Let's take a closer look below.

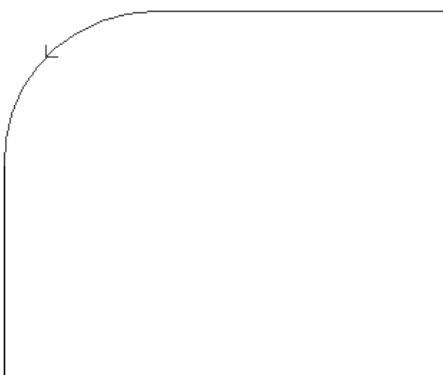
Here we have an arc with a line that is offset .01 from being correctly tangent from the arc.



By selecting this function, the Make ARCS Tangential dialog will appear as you can see below. The preview window shows what the Maximum Acceptable Angle is. In the image, you can see that for this situation we have entered 1 and 90-degrees for the Maximum Angle for Correction. There is only 1 correction to do as well. By selecting OK you get the result, also displayed below.



The RESULT: The geometry has been corrected.

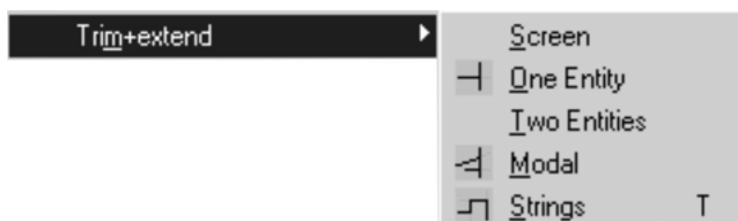


- **ERASE DOUBLES:** This function allows you to eliminate double entities that are of equal dimension and orientation from a drawing automatically.

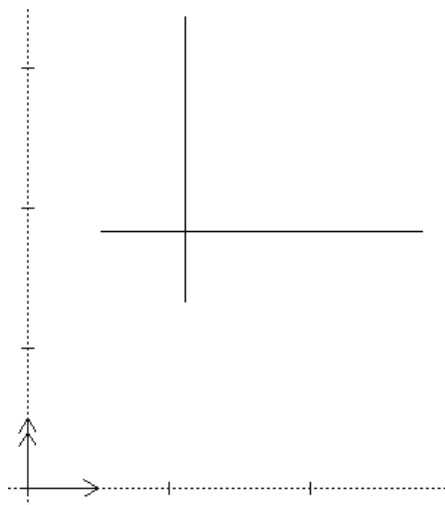
BLANK: This function is the next in order in the CHANGE menu and allows you to **Blank** out selected geometry in a drawing. It doesn't delete the selected geometry, only blanks it from view and selection. You can blank layers, colors, specific entities or all geometry on the drawing screen.

UNBLANK: This function allows you to un-blank previously blanked geometry from a drawing. If in fact, you do have blanked geometry in a drawing, the UNBLANK dialog will appear asking you for specific. You can choose between layers, colors, regions or all blanked entities.

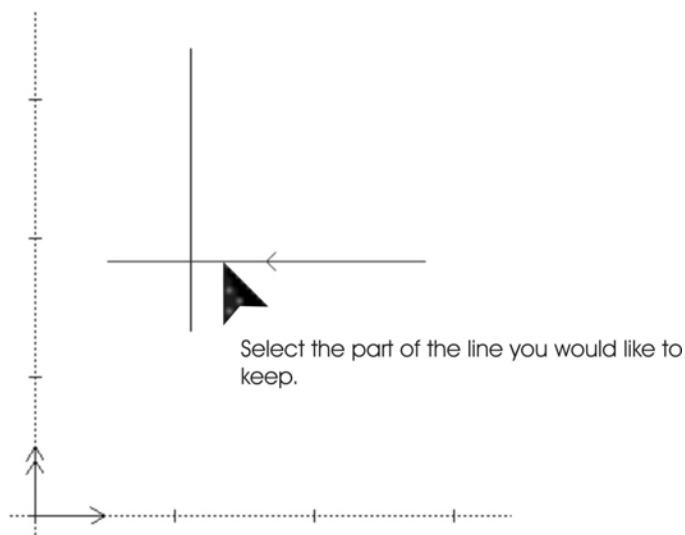
TRIM & EXTEND: This menu function has a special sub-menu offering specific trimming and extending options. See below.



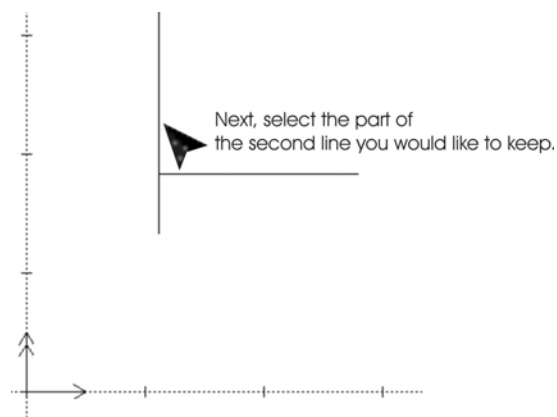
- **SCREEN:** This function allows you to pick multiple entities on the screen to trim together.
- **ONE ENTITY:** This function allows you to trim two entities together on the screen. There is a specific way to trim entities on the screen. Let's take a closer look.



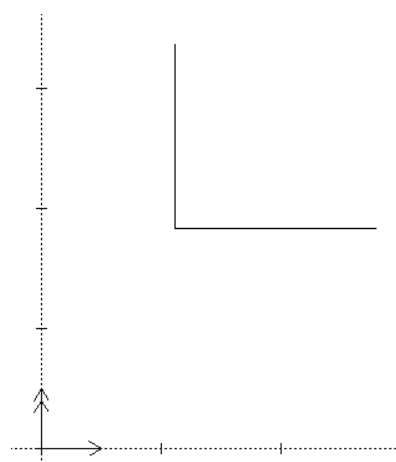
The last image shows two intersecting lines. To trim these two lines together, you must select the first line at the position that you want to keep. Then select the second line at the position you would like to keep as well.



After having selected the first line or entity, you will need to select the part of the second entity that you wish to keep.

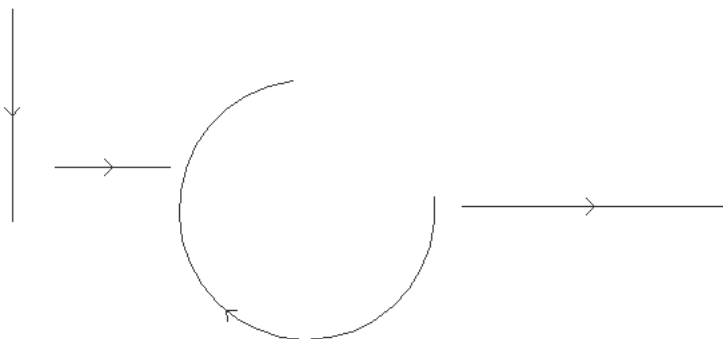


The result.

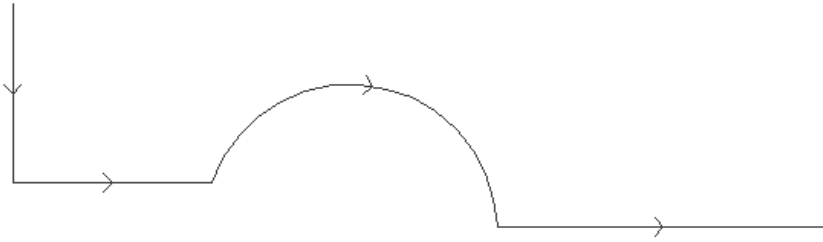


- **TRIM TWO ENTITIES:** This is a faster method of trimming more than one entity. This function allows you to trim two entities together at the same time. Select the first entity, hold down the SHIFT key to select the second entity and then select this function to trim both at the same time.
- **MODAL:** This function allows you to trim entities to a selected contour.
- **STRINGS:** This allows you to trim a group of entities together.

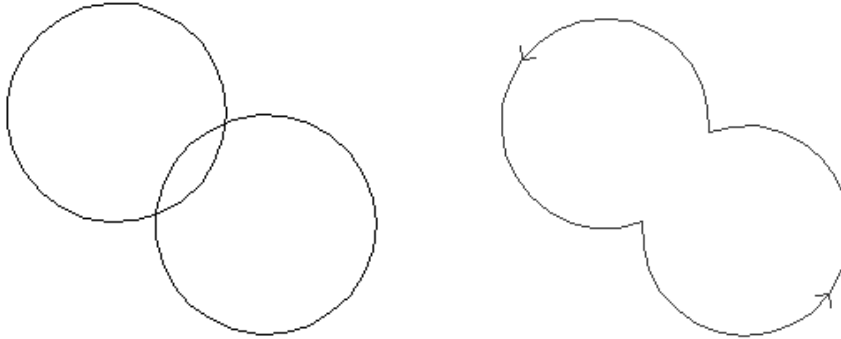
The image below shows a group of pre-selected entities.



By selecting this function, you can automatically trim the entities together to form a string or chain. See below.



Again, the image on the left shows two intersecting circles. The image on the right shows the TRIM STRINGS function once used.

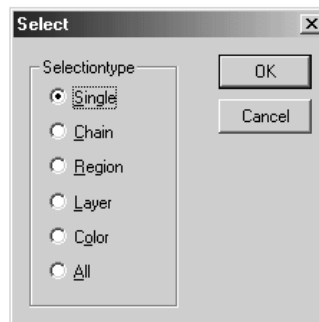


BREAK: The **Break** function offers a special sub-menu with more detailed break functions.



This function is primarily used for breaking up or dividing entities on the drawing screen. Let's take a closer look at each of the sub-menu functions starting below.

- **SCREEN:** Using this function on non-selected geometry, you can break all entities at their intersecting points, by clicking on them individually.
- **POINT:** This gives you the ability to break an entity at a specific point.
- **ANGLE:** This allows you to break an arc at a specific angle on the arc itself.
- **BREAK MANY:** This function allows you to select options from the SELECT dialog. See below.

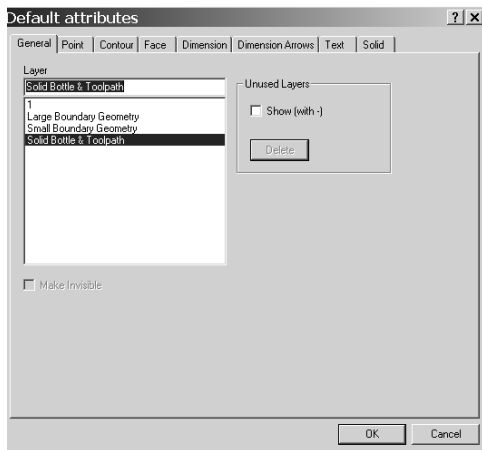


You can choose the option that you need to use for breaking up the entities on your drawing screen. Remember, this function breaks up the entities that are intersecting each other.

GROUP: This function allows you to group specific entities on the screen. First select the entities to be grouped and then select this function.

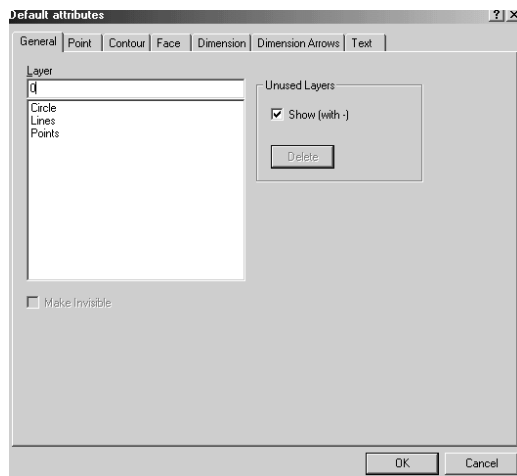
UNGROUP: This function un-groups grouped entities. Select the grouped entities first and then use this function to un-group them.

ATTRIBUTES: The **Attributes** dialog and corresponding functions provide you with many important options for modifying geometry, dimensions, points, lines, arcs, text and solids. Solids attributes also apply to surfaces as well. This dialog as seen below, offers many different options for changing or modifying geometry as well as setting up your software to meet your specific needs. Let's take a closer look at this dialog.



Fundamentally, this dialog box directly effects both pre-selected and un-selected geometry. We want to have a look at both.

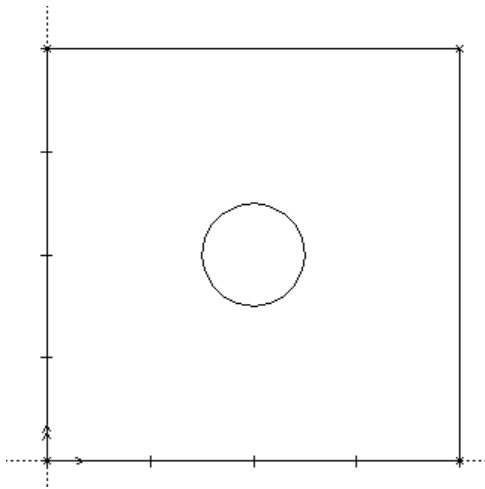
- **GENERAL:** This is the first tab that appears when opening this dialog. The General section allows you to save parts of your drawing on specified **Layers**. See below.



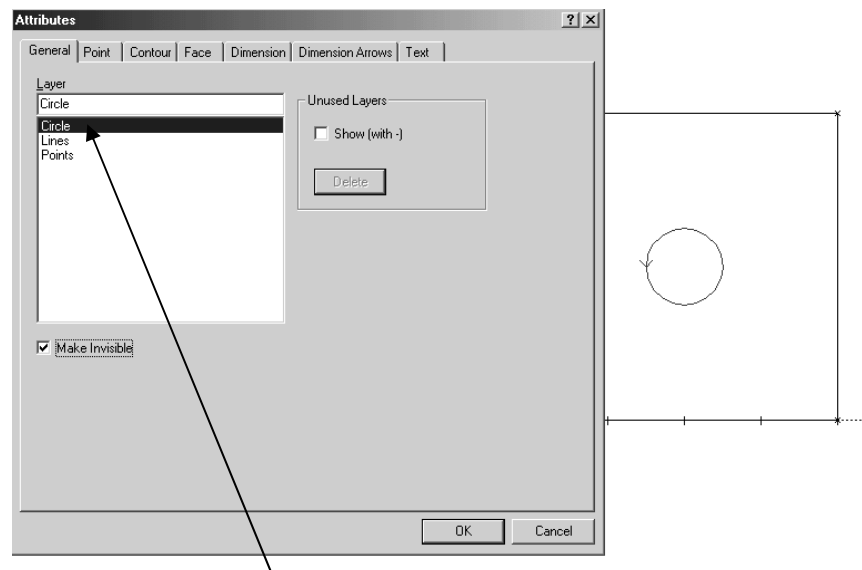
You can see in the image above, there are three different layers listed. Points, Lines and Circles are listed in this image. These would be the different layers that already exist in the drawing if you open a drawing for the first time without having changed the layer information. If there are layers that are NOT used or deleted in your drawing, you can check the box next to **SHOW (with-)**. Those layers will appear in the layer box with a – symbol next to them. Then you can select them from this box and select the DELETE button to remove them entirely. If you select an entity on the screen and then open this dialog, you can select the MAKE INVISIBLE option to blank those entities from your drawing but you will not lose them. You can un-blank them at a later time as needed.

Let's take a closer look at this.

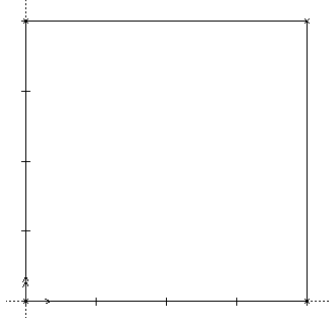
On the screen we have a square, four points and one circle, each saved on a separate layer.



By selecting the circle and then selecting the ATTRIBUTES icon, we will bring up the dialog seen in the next image.



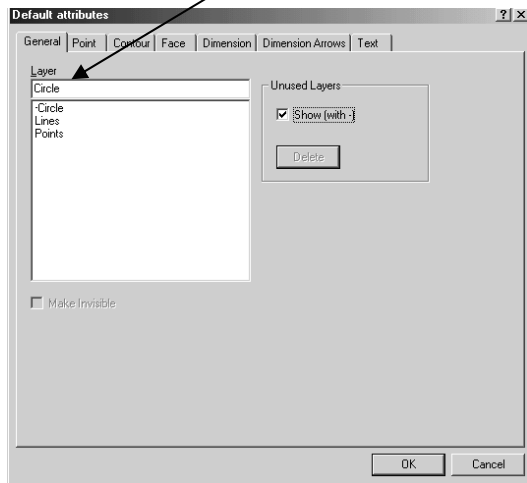
You can see that we selected **CIRCLE** and then checked the MAKE INVISIBLE option. When we selected OK, the circle was automatically blanked.



Instead of having blanked out the circle, we could have selected the MARK (with-) option, selected OK and then deleted the entity.

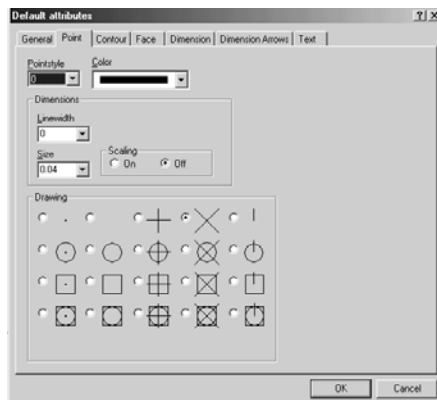
Then you could reopen the Attributes dialog and the circle would be listed with the –symbol in front of it. You can check or un-check this option to add the symbol. To delete the layer name, select the layer name itself and then the delete button.

Circle



This dialog is also the section that you would use to SAVE geometry and name it to a specific layer. To do this, you would first select the geometry that you want to save on a named layer, and then select this function and type in the name that you would like and select OK. This would automatically save the layer.

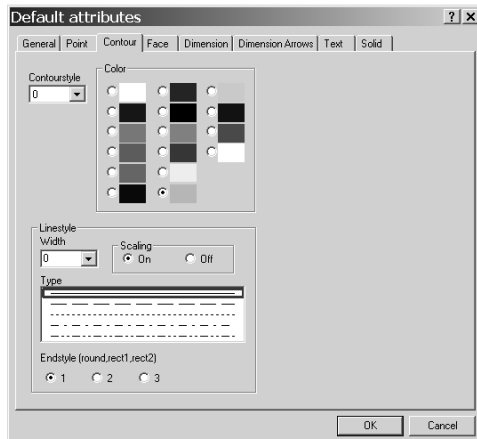
- **POINT:** The next tab of this dialog is the **Point** section. The POINT section offers you the ability to pick and choose from various point styles for CAD designs, colors and the ability to enter a value for the size of the points you need to work with. Points are made in BobCAD software with very small lines. You can adjust these point attributes in this section. SCALING refers to the ability to zoom in on a point as needed.



You can also change existing points that you may have in a drawing. Simply select the points that you would like to change and then select the attributes function, point tab and make the changes.

The ATTRIBUTES function is listed in the CHANGE menu of the software and can be accessed by RIGHT-Clicking your mouse as well.

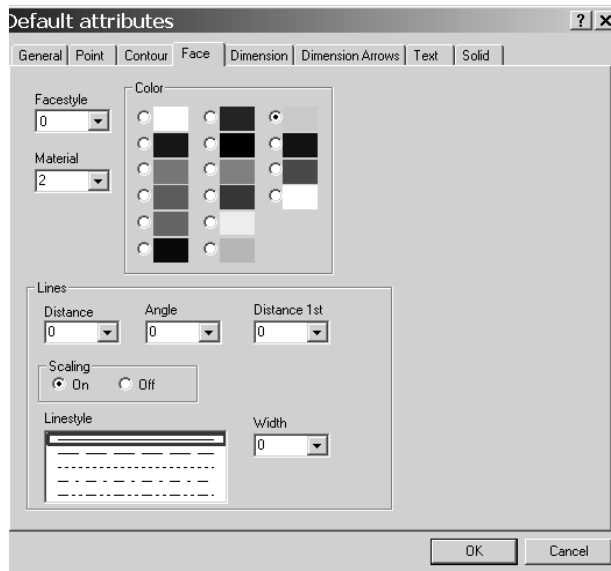
- **CONTOUR:** This tab/section offers you the ability to change the attributes of the lines in a drawing or the line attributes that you will be using to draw with. See the next image.



The CONTOUR section will allow you to change the size and the style of line that you are using as well as the drawing color. Once again, in this dialog you have the SCALING option. As in the point tab/section, you can select this option to be able to zoom in directly on the lines to see them better. Generally, you may want to have the scaling option ON during normal drawings.

You also have the ability to change existing geometry on the screen by selecting the geometry first, then making the attribute modifications in this dialog.

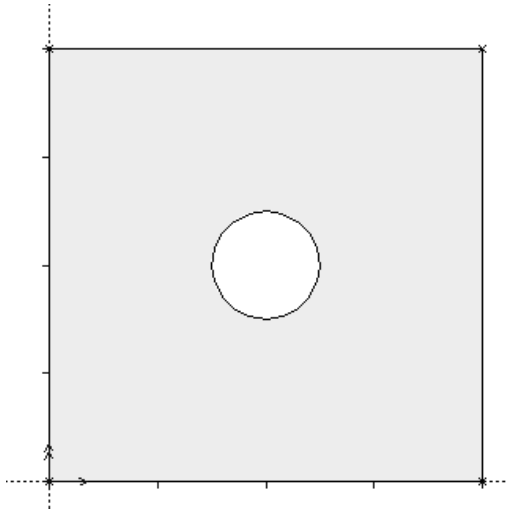
- **FACE:** The next tab/section in the attributes dialog is the FACE section. See below.



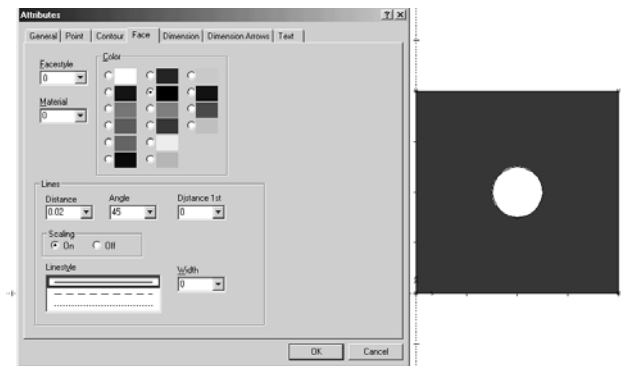
From the earlier FACE description, you now understand how to create or draw a face with existing geometry. Now, let's take a closer look at how to modify them or change the attributes for drawing/creating different and unique faces for your drawings. You can also change an existing face by selecting it first and then using this section to modify it.

Fundamentally, faces are made up of lines with chosen colors. Therefore, you can change the attributes of single or multiple faces in a drawing.

Let's take a closer look at this.

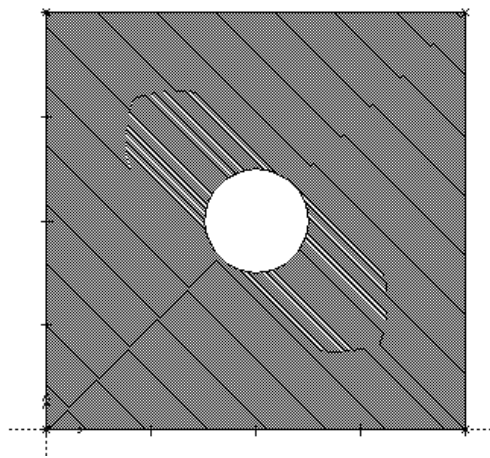


Above, you can see that we have a basic square that has been faced leaving the circle in the middle un-faced. By selecting the attributes function, we can change this face in a number of ways.

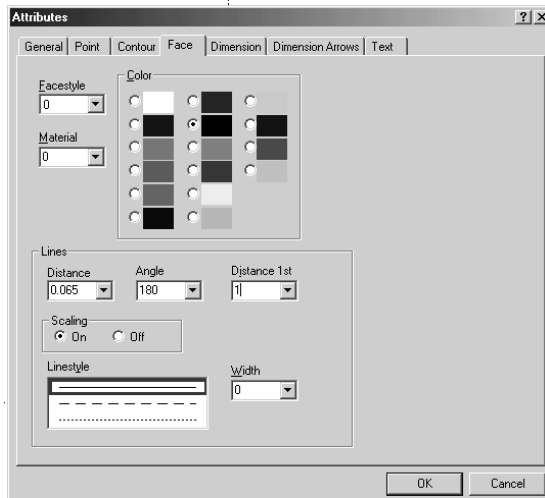


In the last dialog, we entered a line distance of .02 and an angle of 45-degrees. We did not change the line style or width. We did change the color to black from gray. We left the scaling option on so that we can zoom in on the face if needed. We did not change the Material option or decrease or increase the size of the face as we want the boundaries of the face to remain inside the original four lines of the square.

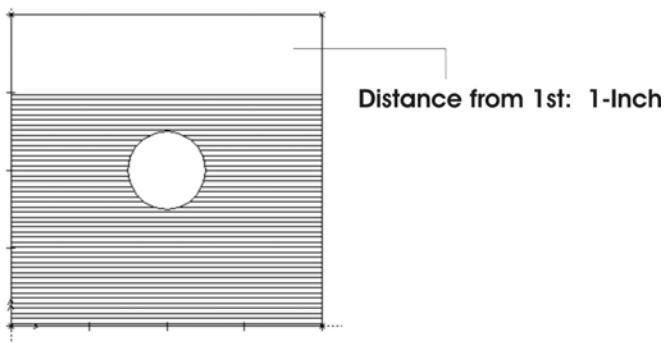
By selecting OK we can see the result.



The face was automatically changed to meet our new settings. To change them again, we would RIGHT-click our mouse button and select Attributes again.

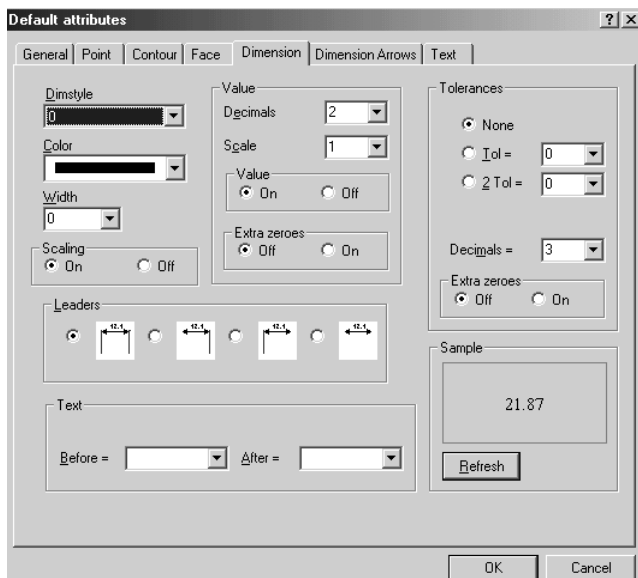


By changing the attributes again, we get the following result.

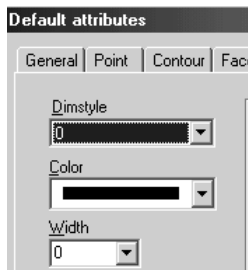


Remember, you can change the attributes of an existing face or draw/create faces with specific attributes.

- **DIMENSION:** This tab/section of the attributes dialog deals directly with dimensions. Let's take a look.

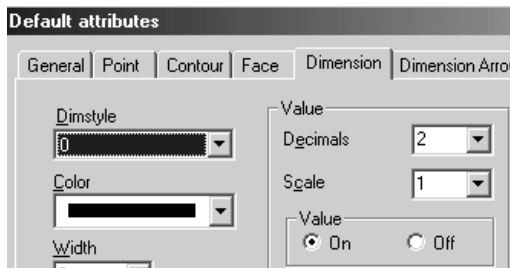


There are some important options available for the dimensioning of drawings that exist in this dialog. You have the ability to change the color of dimensions and the dimension lines.

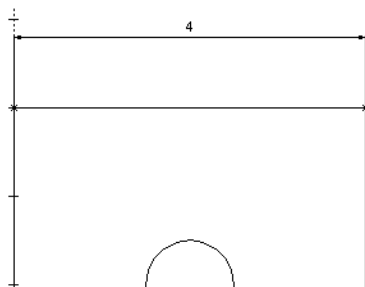


You may also change the line width by entering in a larger value than 0 (which will be the software default).

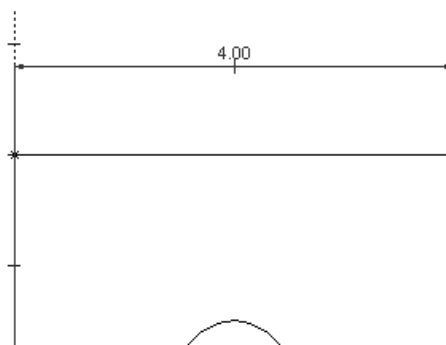
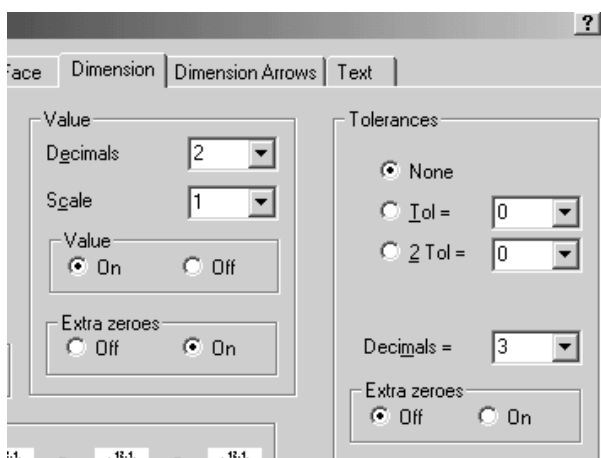
Under the VALUE section, you can change the DECIMALS option so that you dimensions display a specific number of digits to the RIGHT of the decimal point. You may also change the viewing scale of the dimensions.



The horizontal dimension that you can see below is 4-inches.

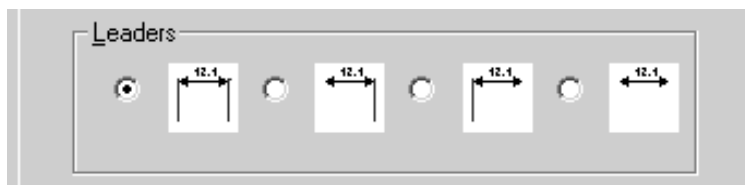


By changing the attribute; EXTRA ZEROES, you can see the result.

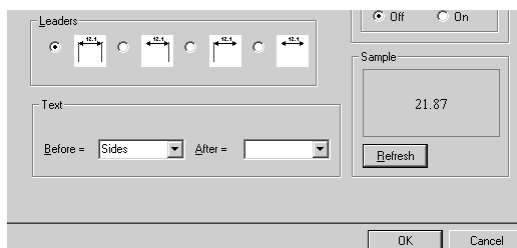


You can also change/modify the number of overall decimals in the dimension.

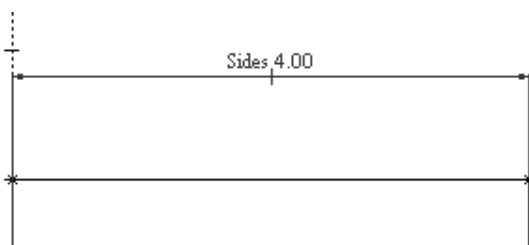
In the image below, you can see that there are also different leader styles that you can choose from.



You can clearly see the different styles directly from the selection area.



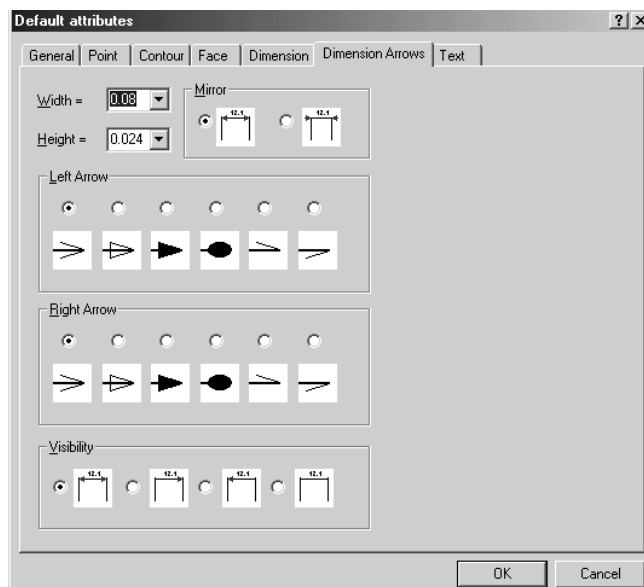
You also have the option to add text to the dimensions, either before or after.



The dimensions tab/section also gives you the ability to see a preview of the modifications before selecting OK. You can do this by selecting the REFRESH button.

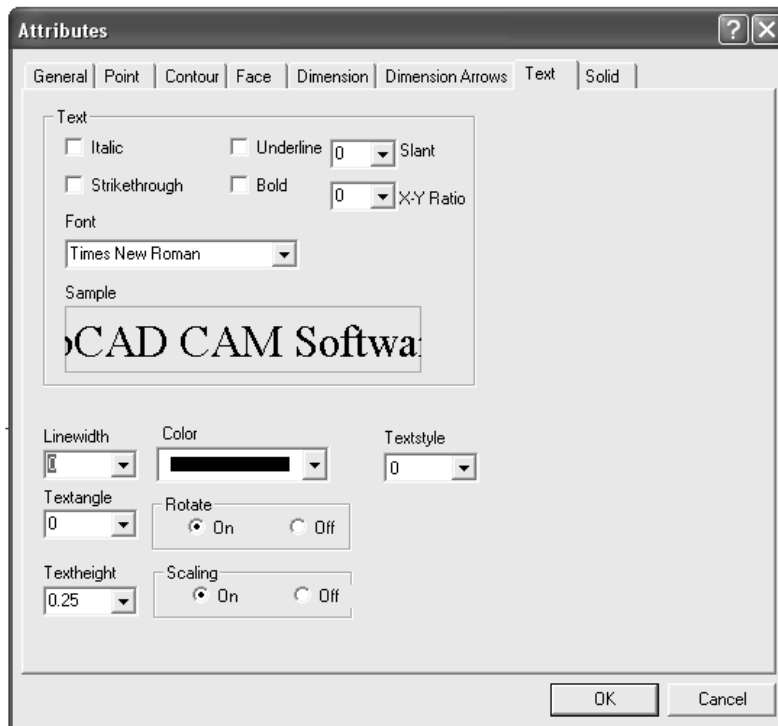
You can change the attributes of existing geometry by selecting the dimension first, RIGHT-clicking your mouse and then selecting the ATTRIBUTES function to change your settings.

- **DIMENSION ARROWS:** This TAB/Section gives you options for your **Dimensions**.



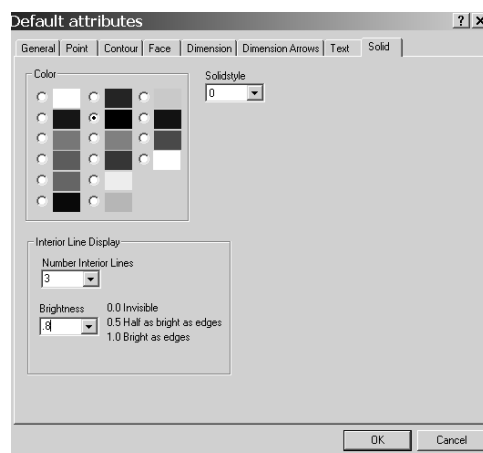
This section allows you to define the WIDTH and HEIGHT of the dimension lines as well as the style of the arrows. You can also use the mirror option change the way the arrows face. You can change an existing dimension by selecting it and then using this attributes section to apply a change as needed.

- **TEXT:** This tab/section of the ATTRIBUTES dialog directly effects text as well as allowing you to change the attributes of existing **Text** on your drawing screen.



The TEXT section offers you the ability to change text from one style to the next and effects the software in a global manner. You can also select some existing text on your screen and then use this dialog to change or modify it. However, there is already a Modify Text function that you can access through selecting your un-vectorized text and then right-clicking your mouse to use the “Modify Text” option from the pop-up menu. This includes font type, color, width and height as well the angle. Version 21 also offers you the ability to define spacing. Font spacing is not available as a global setting, yet it is available in the Text box in the Dimensions menu as well as in the Modify Text functions. The Slant & X, Y Ration are used for the 9 stroke fonts that are listed in the Font TYPE pull-down.

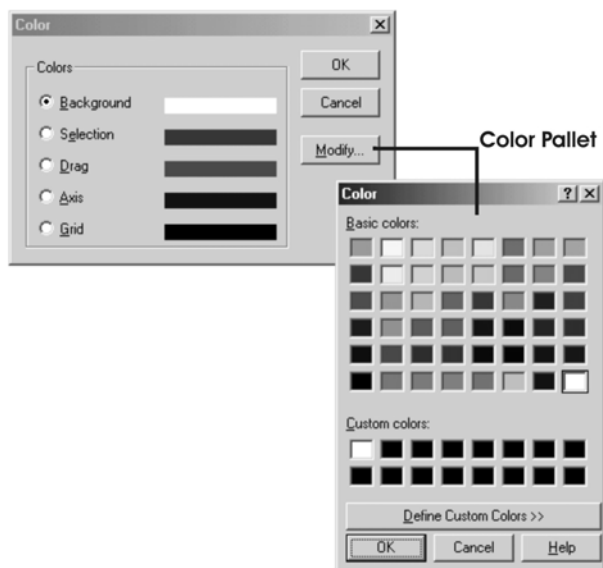
- **SOLIDS:** This tab/section allows you to modify the attributes of **Solids** as well as surfaces.



By selecting a solid or surface and then right-clicking your mouse you can access the ATTRIBUTES box and Solids tab. Pick a color or modify the number of interior lines that make up the surface or solid and then select OK to execute the change. You can also change the level of how bright the solid or surface is.

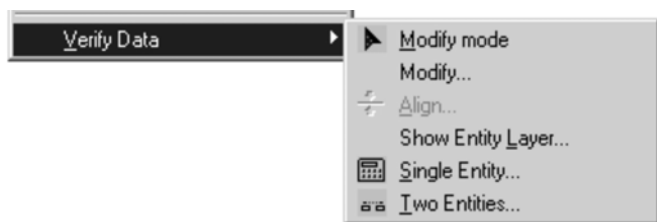
The **ATTRIBUTES** dialog has many functions that you will use when drawing or changing existing drawings. When you select geometry on the screen and then RIGHT-click to select the **ATTRIBUTES** function, the information displayed in this dialog will describe the geometry that you had selected. This is another way to use the **ATTRIBUTES** dialog.

COLOR SETUP: This function allows you to modify or setup the **Colors** for the software drawing screen as well as other items. Let's take a look below.



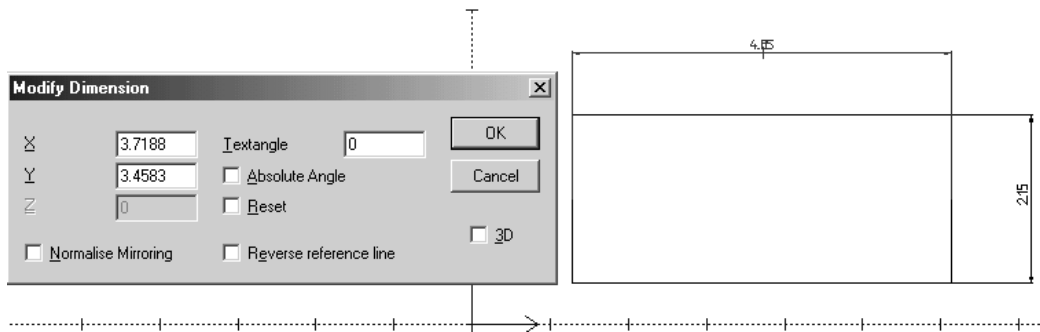
You can change the Background color, the color of selected entities, the drag box window for click and drag functions, the axis color and the color of the drawing grid. By selecting the Define Custom Colors button, you can change the colors to become more specific to what you need. You may also enter the RGB decimal values needed (Red, Green & Blue).

VERIFY DATA: This function offers a special sub-menu with more functions that are available. See below.



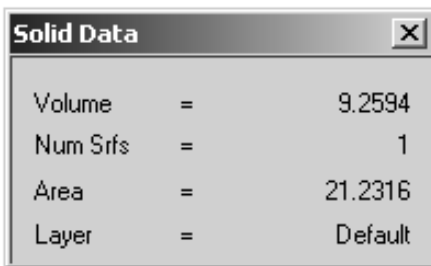
This sub-menu offers:

- **MODIFY MODE:** This automatically allows you to switch into the Modify Mode.
- **MODIFY...:** This allows you to modify selected entities or dimensions, text, etc. For example, you can select a dimension in a drawing and then this function, you can modify the dimension itself.



The dimensions X, Y and/or Z location will appear as well as special options for modifying the dimension. This will be covered again in the section that discusses dimensions.

- **ALIGN:** This allows you to align the dimension text itself. This is because the dimension is considered text.
- **SHOW ENTITY LAYER:** Selecting an entity and then this function will allow you to see what layer the selected entity is on.
- **SINGLE ENTITY:** You can gather all of the information regarding a single entity through using this function. This includes a surface or a solid. Because this function is action/object oriented, you will first select this function and then click on the surface or solid edge or interior lines and the box will appear giving you the information you need.

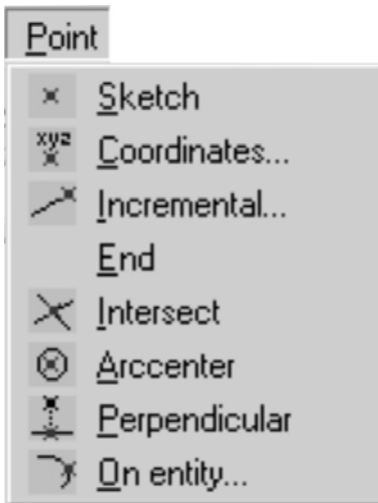


- **TWO ENTITIES:** This allows you to view the distance between two entities. Select the function, select one entity and then the second to bring up the verify dialog. This function can be used after extracting edge geometry from solids or surfaces as well.

This concludes the CHANGE menu.

Section 17. The POINT Menu.

The **Point** menu of the Version 21 software is primarily CAD wireframe design. However, there is usefulness for CAM as well in terms of working with drilling cycles. This section is dedicated to explaining and showing the CAD aspect of this menu and how to use its menu functions for drawing.

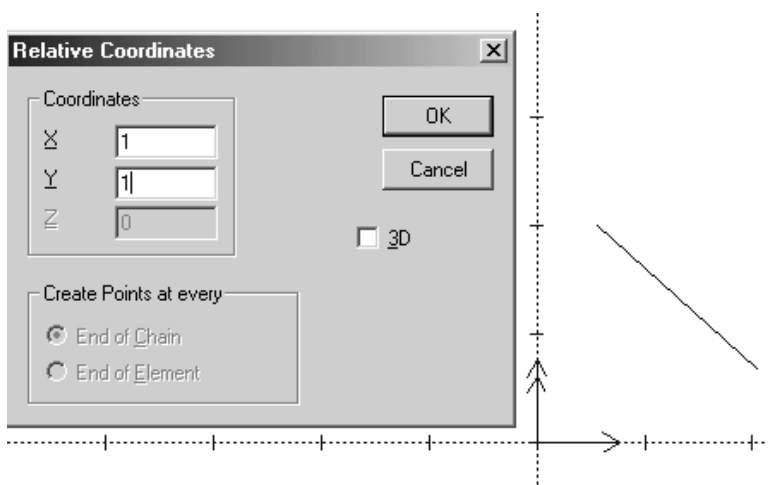


Let's take a look at each one of these menu functions.

SKETCH: This function allows you to freehand sketch points anywhere on the drawing screen. Simply click your mouse in one location and then drag your cursor to another location and then click your right mouse button again to draw the line. After you do this there will be a box that appears asking if you would like to add a coordinate position. If you do not wish to do this, just click CANCEL.

COORDINATES: This function allows you to draw points on the screen by entering in an X, Y and/or Z coordinate value. You will want to be familiar with the coordinate system of the software to use this function.

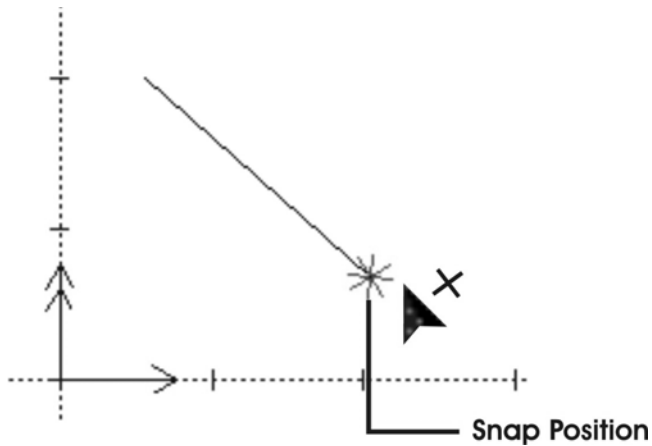
INCREMENTAL: This function allows you to place points on the screen incrementally to existing geometry.



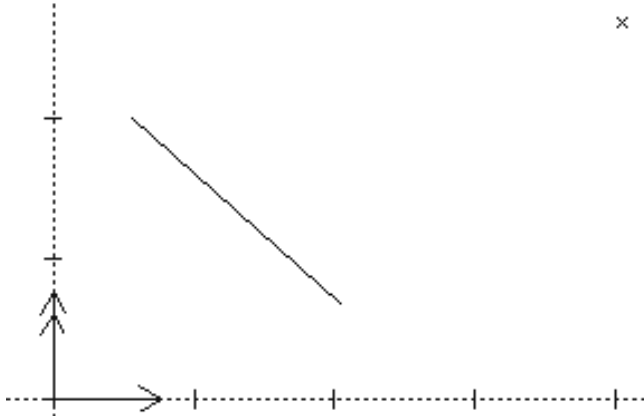
In the last image, you can see that we selected the function and entered 1 for both the X and Y boxes.

The new points that we draw will be relative to the existing line. By selecting OK, you can then choose a position of the line to base the new point off of. Hover the cursor over the line and you will see a star appear at the snap locations.

Choose a snap location and click your mouse button to draw the point incrementally lines snap location. See below.



By choosing the snap location and then clicking your mouse button, you will automatically draw the new point. See below.



This function can also be used with arcs.

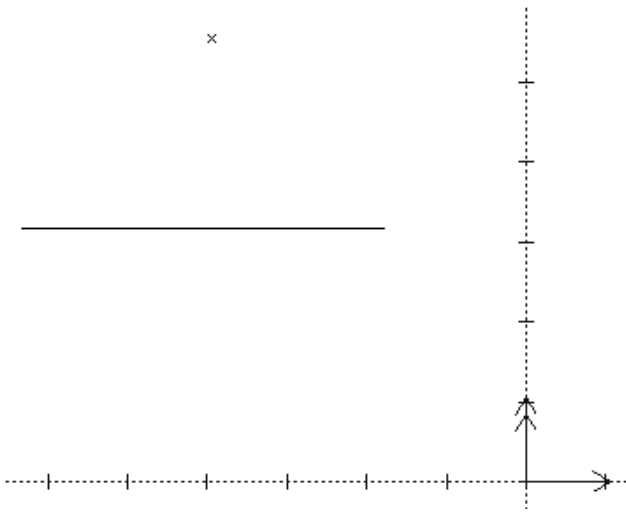
END: This function allows you to snap points to the end of entities. The end of an arc is the zero position/3 o'clock position. This is a repetitive function as well. And so, when finished, simply click your right mouse button to exit the function.

INTERSECT: This function allows you to snap points at the intersection of entities.

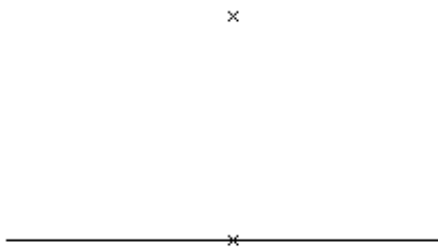
ARC CENTER: This function automatically allows you to snap points to the center of arcs or arc segments. Because this function is both object/action as well as action/object you have can multiple arcs on the screen and snap points to them all automatically. To do this you can select the arcs first and then the function. Then the arcs will no longer be selected but the points will.

As Version 21 now allows you to select arcs and use drilling operations directly, you no longer need to snap points to the arcs first.

PERPENDICULAR: This function allows you to draw points perpendicular ON GEOMETRY that is perpendicular to other points. Let's take another look at this function.



Above, you can see that we have a line and a point. By using this function to draw a point on the line that is perpendicular to the point, you would select the function, then the line and then the point above the line in this case. See the result below.



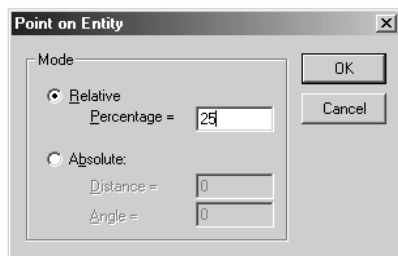
You can see that the newly drawn point is perpendicular to the first point.

ON ENTITY: This is an ACTION/OBJECT, OBJECT/ACTION function. You can select the geometry first and then place a point based off of a relative percentage. Let's take a line for example. If you selected the line first, and then selected this icon, you would be able to place a point at 50% which is right at the center of the line etc.

Let's take a closer look at this function as it pertains to percentages and the way that you select geometry.

Sketch a line using the LINE, SKETCH function. Simply click your mouse and drag a line across the screen. When finished, click your mouse button again. Now click OK to position the line closest to the actual X & Y coordinates.

Now select the line toward the RIGHT end of it. You will see a directional arrow on the line itself in selection mode, indicating that your selection is going toward the RIGHT. Now select the "Add Points To Selected Entities" button.



You can see in the last image that the selected line arrow points to the right. We have entered .25 in the “Relative – Percentage” section. Do the same. Click OK and you will have a point placed at the .25% of that line.



Here is the difference between **RELATIVE** and **ABSOLUTE**:

- **RELATIVE:** You are asked to input a percentage. If you choose 0% then all the construction points are created at the start points of the chosen geometry while for 100% they are created at the end points. You may also input an intermediate or negative value or even a value greater than 100%.
- **ABSOLUTE:** You are asked to input a distance (for lines) and/or an angle (for arcs). A value of zero results in all the construction points being created at the start points of the chosen geometry. It is also possible to input a value that is greater than the maximum value (length or angle) of the selected entity. In this case the construction point is drawn in the extension of the element.

Here it is again.

We have the same line on the screen and we select the ADD ENTITY icon. If we enter 100 as the percentage, click OK and then rest our cursor on the line, you will see the preview of the point show up at the end of whichever side our cursor is leaning toward. By clicking our mouse button the point will be drawn at the end. If you now changed that percentage value to 25%, you would see the point preview at .25 on the line itself. NOT .25 of the line based off of the coordinate system. Practice using this function until you understand how it works and feel comfortable.

Section 18. The LINE Menu.

This section is dedicated to the **Line** menu of the BobCAD-CAM Version 21 software. The line menu has a variety of functions that deal directly with the drawing of lines and line type objects. We are going to take a close look at each of these functions.



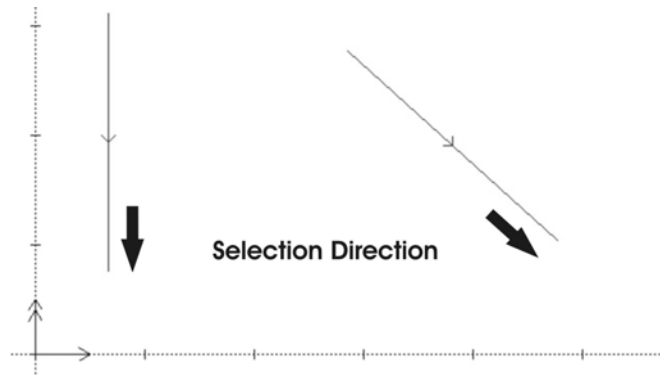
SKETCH: This function allows you to freehand sketch lines on the screen. Select the function, click your mouse button once to pick the starting location of the line and then drag the line to the desired angle and length. When satisfied, click your right mouse button again to lock the sketched line into position. Once you have done this, the LINE ALIGN WITH AXIS dialog will appear giving you coordinates. If the sketched line is more HORIZONTAL than vertical, by selecting OK the line will become perfectly horizontal to the nearest X, Y and/or Z location automatically. If the line is more VERTICAL then horizontal, by selecting OK the line will become perfectly vertical to the nearest X, Y and/or Z location automatically.

COORDINATES: This allows you to draw lines by using X, Y and/or Z coordinates. It is important to understand the coordinate system before using function. First you draw the start point of the line and then the end point of the line.

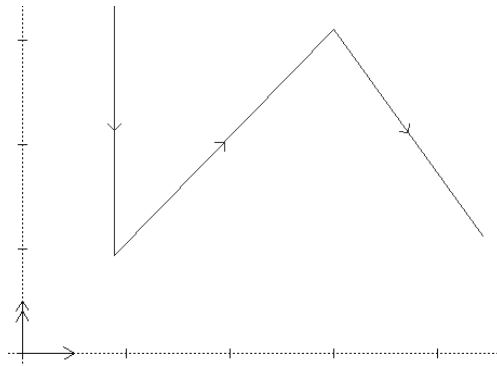
JOIN: This function allows you to join two entities with a line. Click on an end point of an entity and then click on the other end point to automatically draw the line. This is a repetitive function. When finished, you can right-click your mouse button to exit the function.

TIPS ON USING THE LINE, JOIN FUNCTION:

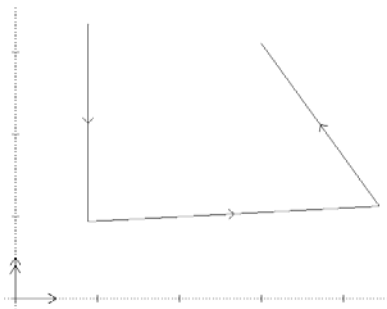
Here is a **TIP** on using the LINE-JOIN function. You can use this function by selecting two entities first and then selecting the function. However, if you do this, the direction arrows of the selected geometry make a difference in how the new joined line is drawn.



Above, we have two selected lines. The direction arrows are going in the same direction. They are both going down. By selecting the LINE JOIN function the new line is directed to be drawn as you can see below. Now you have formed a chain with the direction arrows going in the direction of a chain.

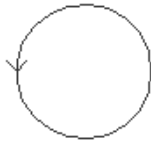


You can change the direction of a selection arrow by moving your mouse just past the middle position on the entity and then clicking your mouse to change the direction. You may have a large drawing on the screen that consists of hundreds, maybe thousands of entities. You can optimize and correct arrow directions by using the optimize function in the change menu. In the image below, you can see that the directions of the selection arrows have been changed.

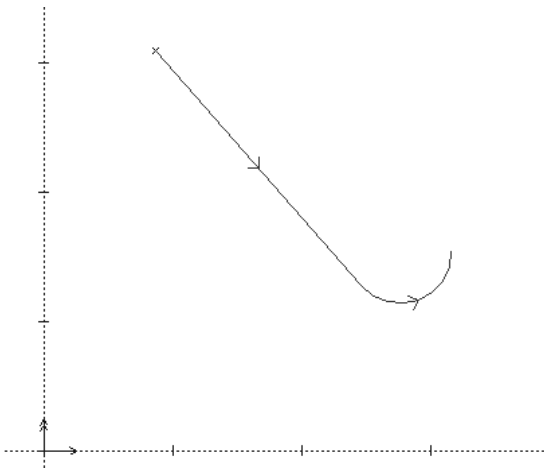


TANGENT: This function allows you to draw lines that are **Tangent** to other geometry. When you make a tangential transition between two arcs, then the transition touches both of the arcs. Curves are trimmed or extended so that they join exactly to the entities connected. This is an ACTION/OBJECT, OBJECT/ACTION function. Let's have a closer at pre-selected geometry below. See below.

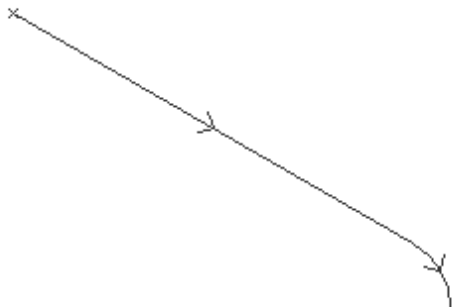
x



The arc was selected on this side. (counter-clockwise selection) Both the arc and the line in the image have been selected. Now, by choosing this function, the tangent line is then automatically drawn, trimming the arc to its 0-degree position. See below.



Have a look at the same function, yet this time selecting the arc going clockwise. The arc was trimmed to the 0-degree position again. Only this time, the tangent line was drawn on the other side. See below.

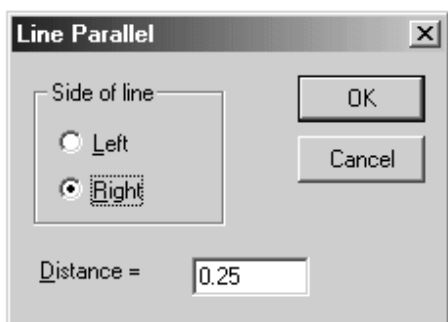


PARALLEL: This function allows you to draw a **Line** parallel to another line. This is an ACTION/OBJECT, OBJECT/ACTION function. Let's take a look.

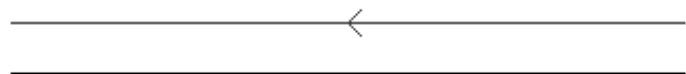
The line below was previously selected.



Now by selecting this function, the LINE PARALLEL dialog will appear. See below.

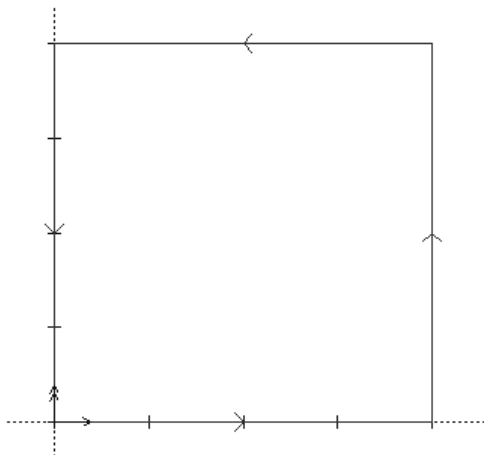


In the image, you can see that you have an option for the distance to draw the new line as well as the option for RIGHT or LEFT. We selected RIGHT and then selected OK. See the result below.

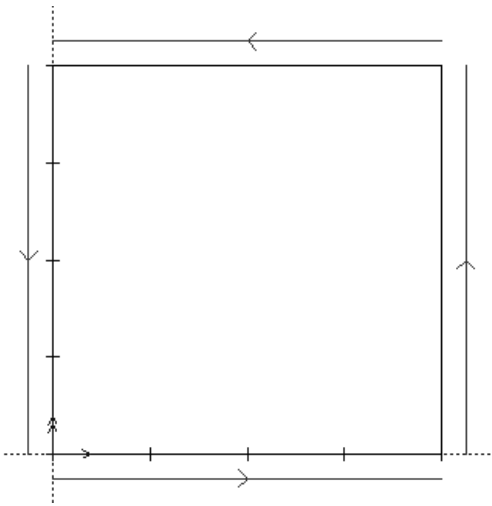


Let's take another look at this function. For example, we have a previously selected square.

The square, pre-selected.



Now select the LINE PARALLEL function. Using the same values as we did with the single line. .25 for the Distance and the SIDE was RIGHT. Then we selected OK. See the next image.



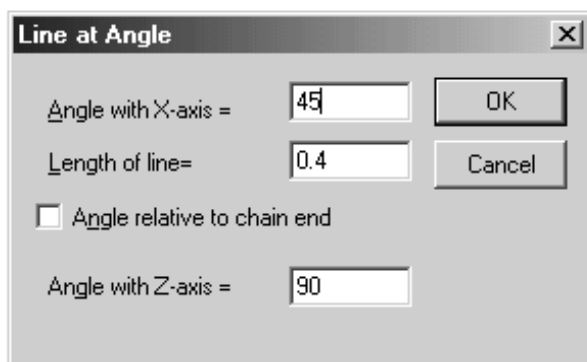
Now let's have a look at the function **WITHOUT** pre-selected entities taking the example of the single line. See below.



Now select the Line Parallel function, and then enter the distance you would like to draw the new line. Then select OK. You will then be able to click on the line and see a preview of the new parallel line on either side by moving your cursor to the right or to the left.

When satisfied with the side, simply click your mouse button to create the new line. This is a repetitive function, allowing you to continue making parallel lines until finished. When finished, click your right mouse button to exit the function entirely.

ANGLE: You can use this function to draw **Lines** at angles. Below, you can see the **Line Angle** dialog.



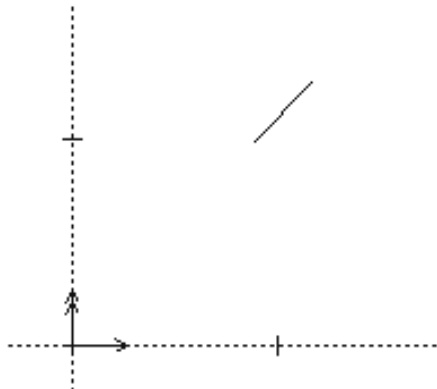
The Line at Angle dialog allows you to draw lines with a specific angle degree, allows you to specify the length of the line itself and allows you to draw angled lines that are based off of earlier angled lines with the "Angle Relative to Chain End" option.

We will start by using the same values that are displayed in the image above. By selecting the **LINE** menu and then **ANGLE**, the Line at Angle dialog will appear. We enter the following values and instructions:

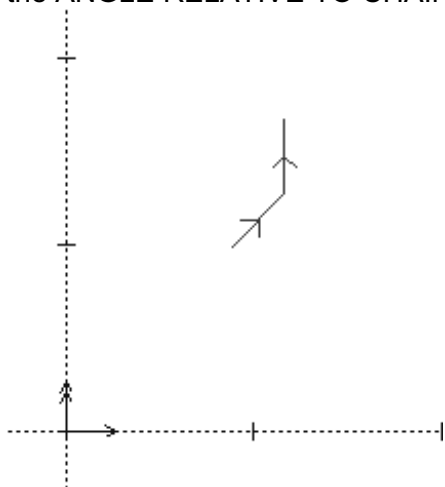
Angle with X-Axis: 45-Degrees
 Length of Line: .4
 Angle with Z-Axis: 90-Degrees
 Angle Relative To Chain NOT activated.

Now we click OK, then choose a location on the screen and click our mouse button once. Now RIGHT-click to end the function. This places our first angled line on the screen.

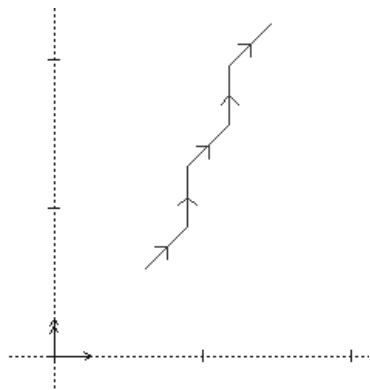
See below.



Now, selecting the line by clicking on it once, we can select the LINE menu and then ANGLE again. In the LINE at ANGLE dialog, leave the values the same, however, ACTIVATE (by placing a check mark in the box) the ANGLE RELATIVE TO CHAIN option and click OK.

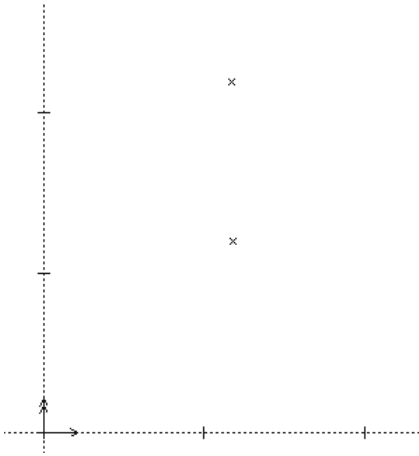


As you can see, the second 45-degree angled line is drawn 45-degrees from the original 45-degree line. By repeating this cycle 2 or 3 more times, activating and deactivating the Angle Relative to Chain option, you would get something similar to the image below.

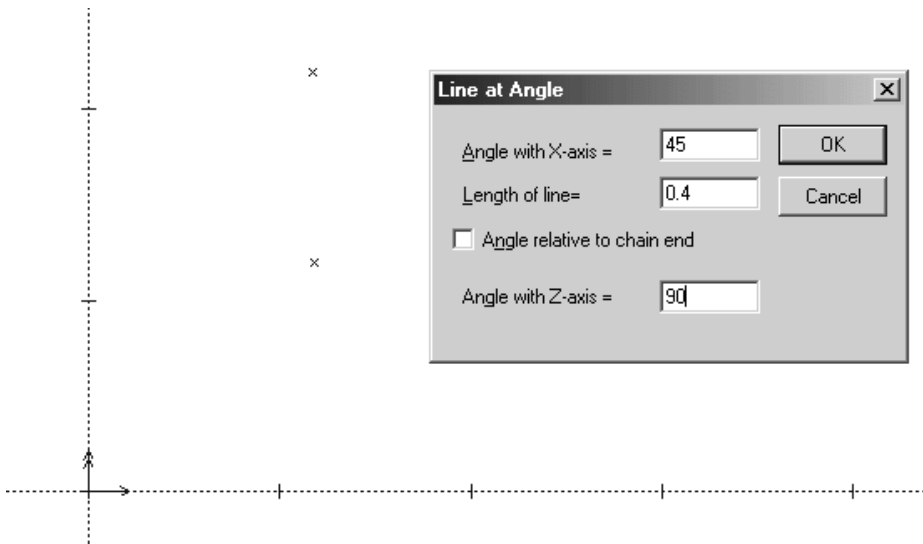


Now, another way that this function can be used is to snap the angled line to a point. Let's take a look at this.

In the image below, we have two points.

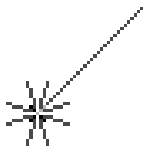


By selecting the LINE menu and then ANGLE, you will open the Line at Angle dialog.

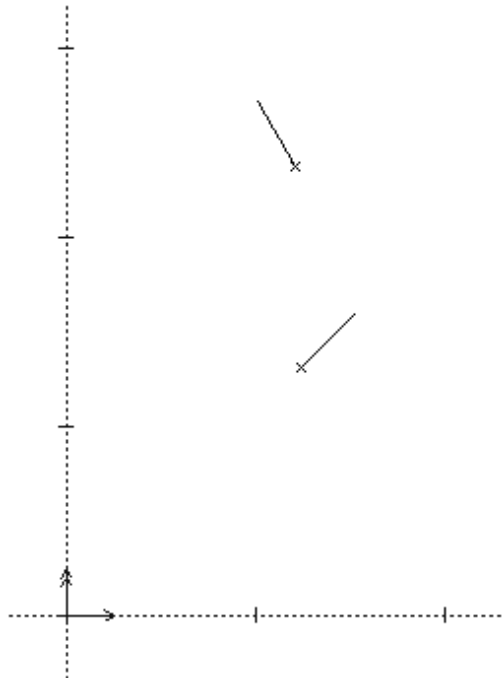


Use the same values as the image above: 45-degrees for Angle Around Axis, .4 for the length of the line and 90-degrees for Angle with Z-Axis. Do not check the Angle Relative to Chain option.

By placing your cursor near or on the point, you will see the small star next to your cursor indicating the position. Click your mouse button and snap the angled line to the point.



This is a repetitive function. Therefore, by clicking your right mouse button after the snap move, you will exit the function entirely.

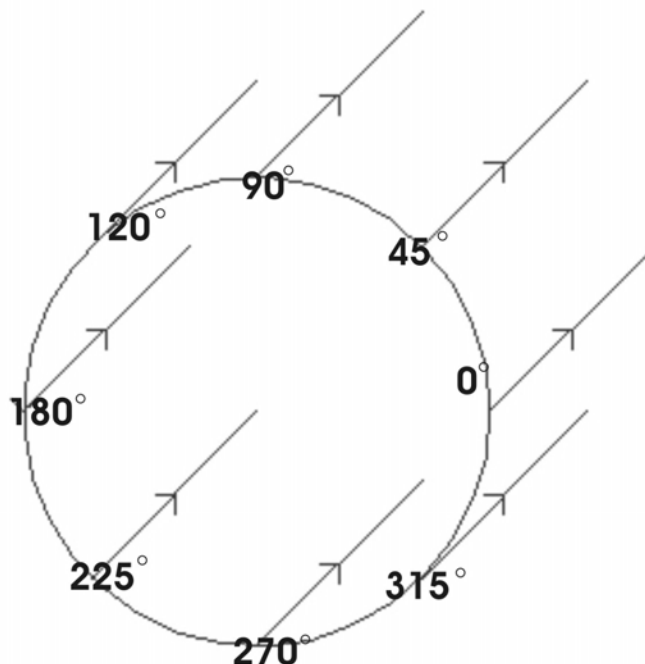


Above, you can see the two points with different line angles. One is at a 45-degree angle and the other at a 120-degree angle.

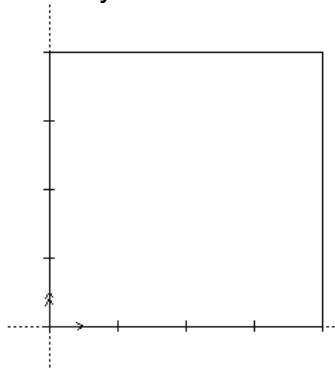
The **ANGLE WITH Z-AXIS** option will allow you to rotate the angled line in a negative and positive Z-Axis as well. You can use this function to draw lines for connecting toolpath as an example.

You can also use angled lines with arcs. The image below shows 45-degree lines snapped to each of the default degrees around an arc. You do not need to have points on a radius to snap lines or attach them at angles to a arc or radius. Just like the snap to point example earlier, you will see a small star indicating the snap position on the arc.

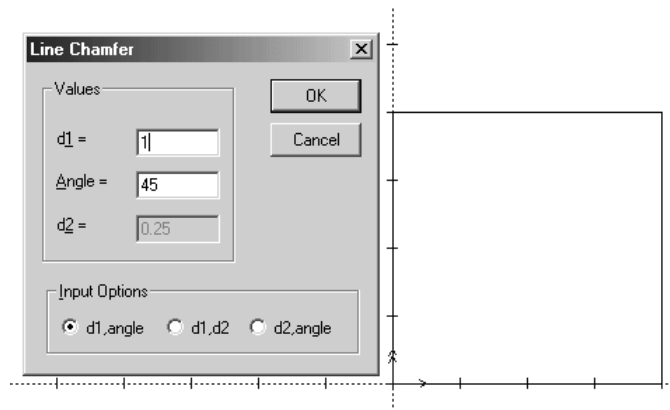
Simply click your mouse button to snap the angled lines. It is a repetitive function, allowing you to continue the function. RIGHT-click your mouse to exit the function entirely.



CHAMFER: This function allows you to draw a **Chamfer** using existing entities on the drawing screen. Let's take a look at using this function with a square as you can see below.

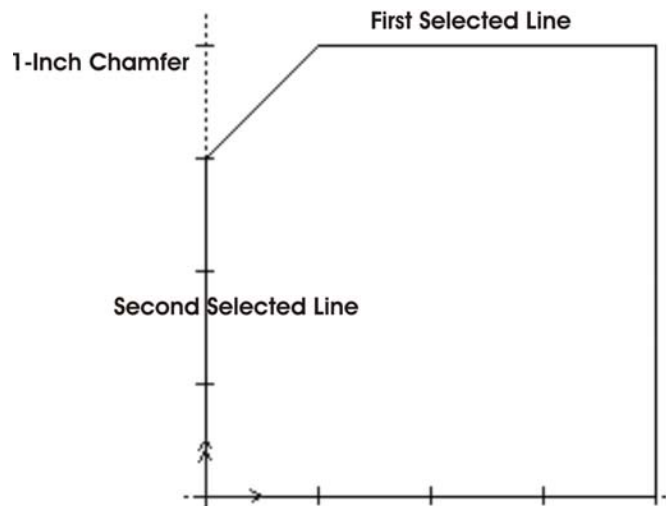


We have selected the INPUT OPTION D1-Angle. This has changed the VALUES box allowing us the ability to enter in the total distance of the chamfer line (1-inch) and make the chamfer line a 45-degree angle between the two lines that we want to use for the creation of the chamfer.

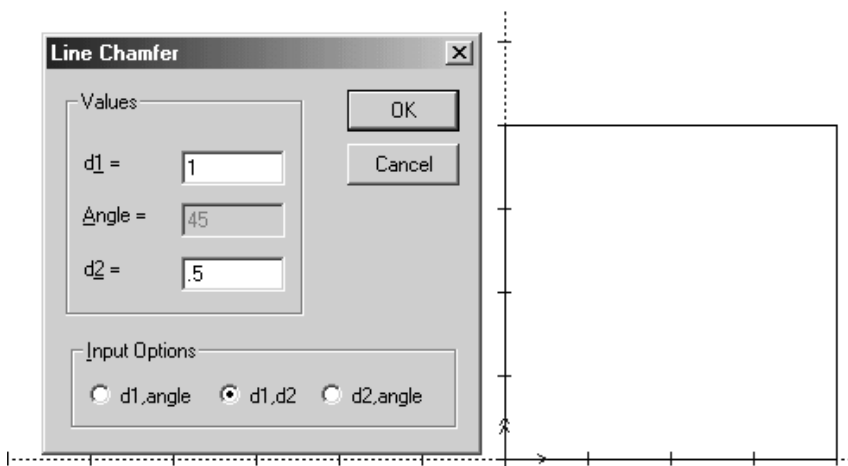


So, once again the VALUES section offers a distance for the first entity selected. Then you can enter an angle degree for the chamfer.

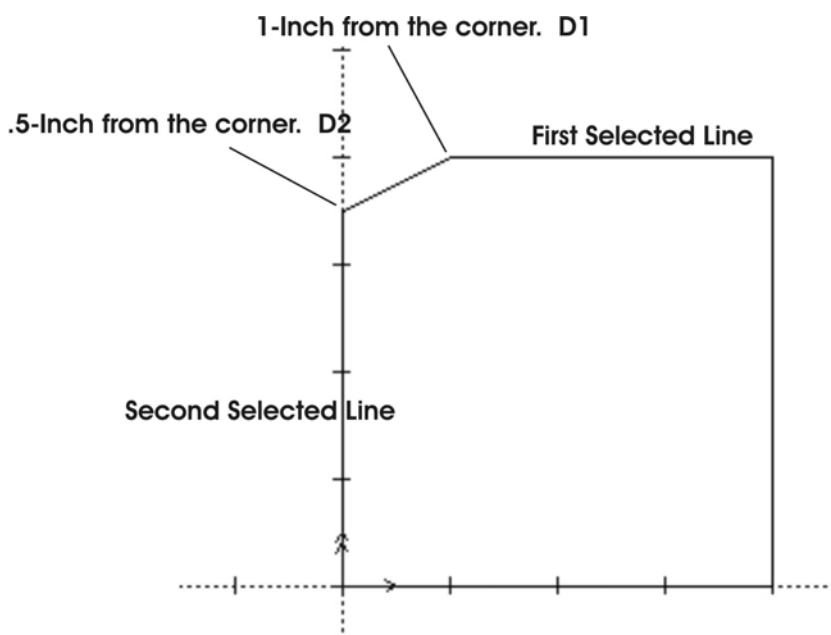
In the last image, we have selected D1- Angle, which will allow us to click on the top line, then the left line to create the 1-inch chamfer that you can see below.



In the next example image, you can see that we have selected the INPUT OPTION D1, D2 (D=distance) and have 1-inch for the first selected line and .5 for D2, the second selected line. The angle is set at 45-degrees and became unavailable for changes once we selected the Input Option D1, D2.

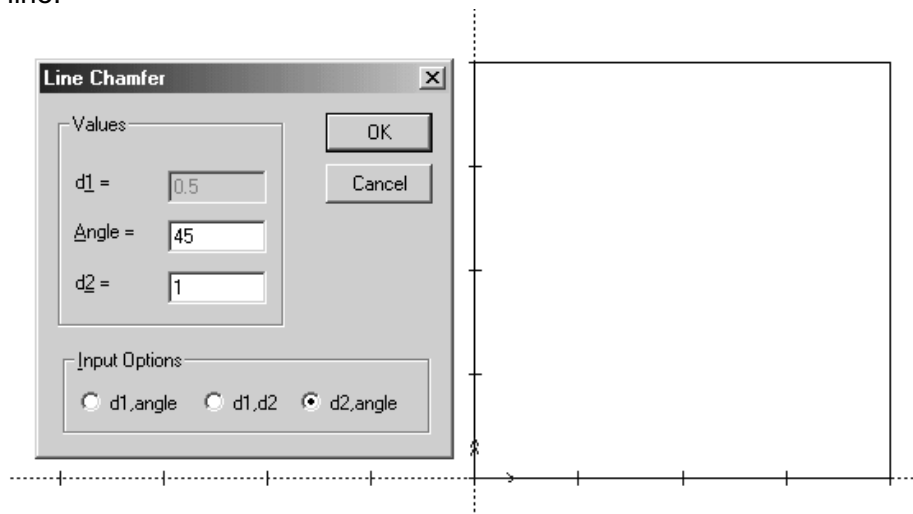


Then, by selecting OK we get the result that you can see below.



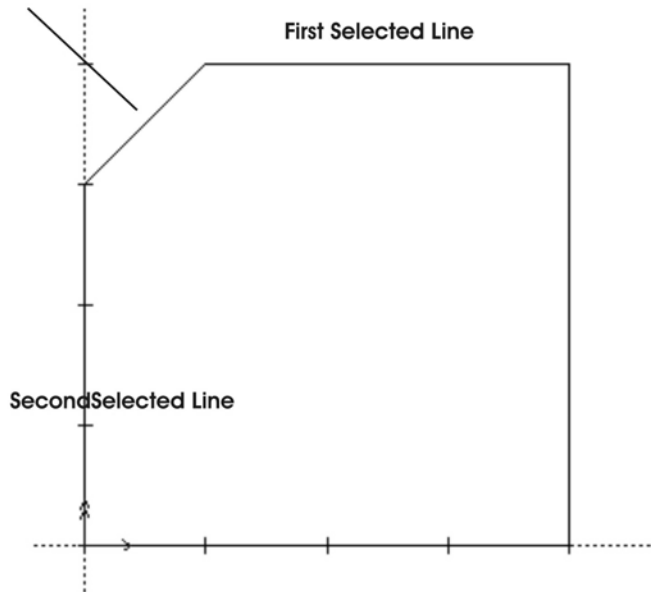
If we had selected the lines in reverse order, we would have drawn our chamfer differently as well. Now let's have a look at the other option.

In the image below, we have selected D2 under Input Options and entered a 45-degree angle for the chamfer line.



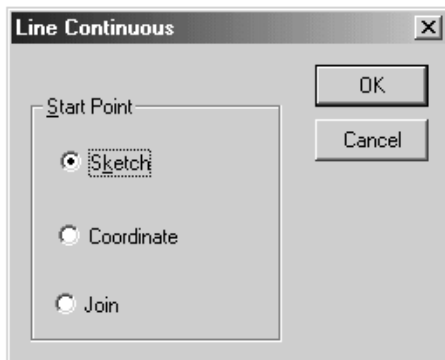
By selecting OK, we now have the chamfer that you can see in the image below.

The 1-Inch Chamfer Line

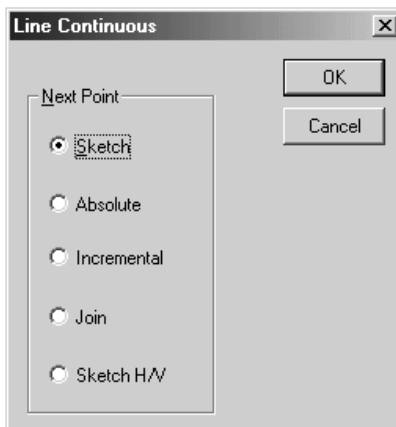


In the image above, you can see that both of the lines were trimmed to add the 1-inch chamfer line. As a side note, it does not matter which line you select first.

CONTINUOUS: This function allows you to draw continuous lines on the drawing screen. This function has various options. Let's take a closer look at the function. By selecting the **Line Continuous** function, its dialog will appear as you can see below.



The dialog above gives you three options for making the START POINT of the line. By selecting sketch, the dialog as you can see below will appear.



The second dialog gives options that relate to the NEXT point of the continuous line. By selecting SKETCH, you can then click and drag a series of lines around the screen. When finished you would select the right mouse button and then select cancel to exit the function.

The **ABSOLUTE** option will give you a coordinate dialog asking for X, Y and/or Z coordinates. This would be in ABSOLUTE mode, meaning that you are placing specific coordinates on the drawing screen NOT relative to the start point that you originally clicked.

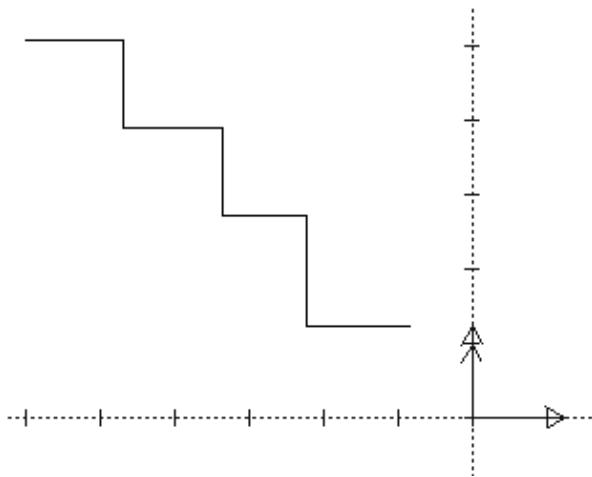
The **INCREMENTAL** option gives you the ability to enter X, Y and/or Z value coordinates that will be relative to the initial selection that you made.

The JOIN option allows you to join points or other entities around the screen.

The **SKETCH/HV** option allows you to draw continuous HORIZONTAL or VERTICAL lines ONLY.

Once again, when you are finished with any of these continuous functions, click your right mouse button and select CANCEL to exit. However, you will also be able to make another selection for drawing continuous lines if desired. This is because when you right-click, the NEXT position dialog will reappear.

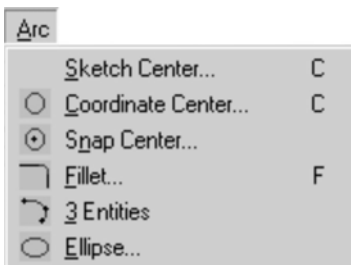
Here is an example below of the SKETCH/HV (Horizontal/Vertical) option.



This concludes the LINE menu.

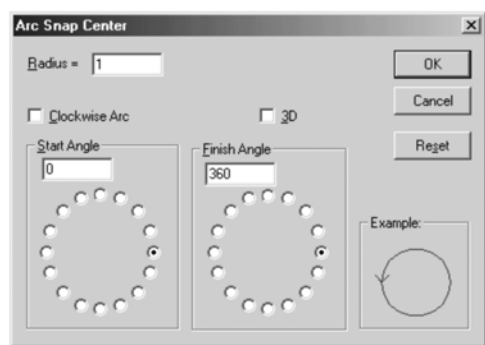
Section 19. The ARC Menu.

This section is dedicated to the **ARC** Menu of the Version 21 software and its relationship to CAD Design. Below you can see the ARC menu itself.

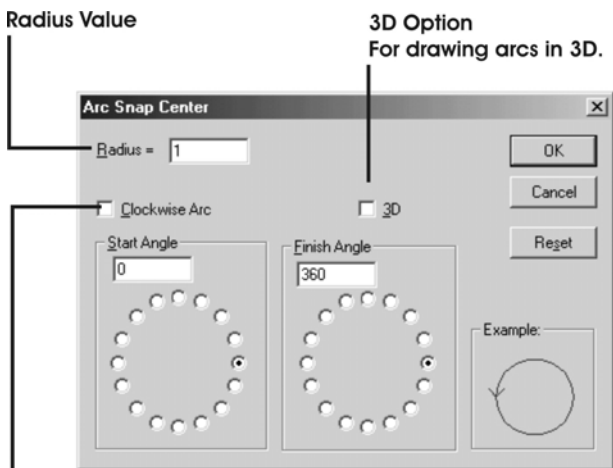


Let's take a look at each one of these menu functions in detail.

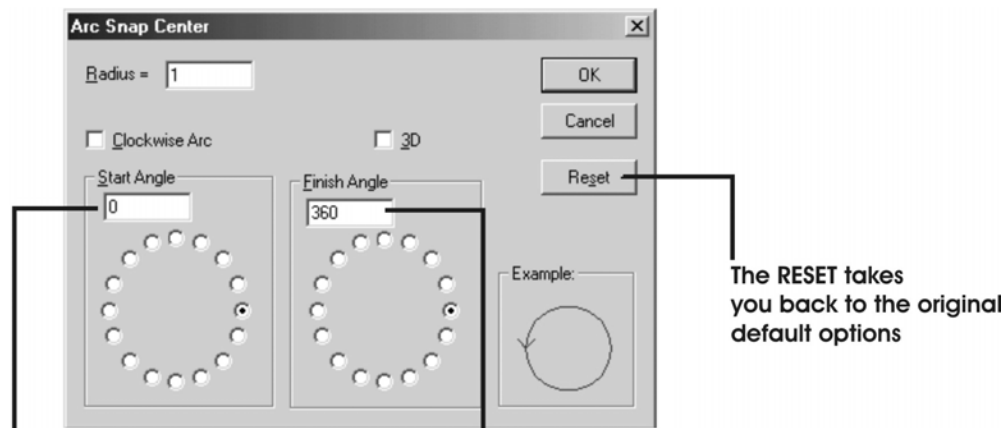
SKETCH: This function allows you to freehand sketch ARCS on the screen easily. You can also snap arcs to selected points. This is the Snap Center function. By selecting this function, the **ARC** dialog will appear as you can see in the next image.



Let's take a closer look at what this dialog consists of starting below.



Drawing arcs with an automatic Counter Clockwise selection.



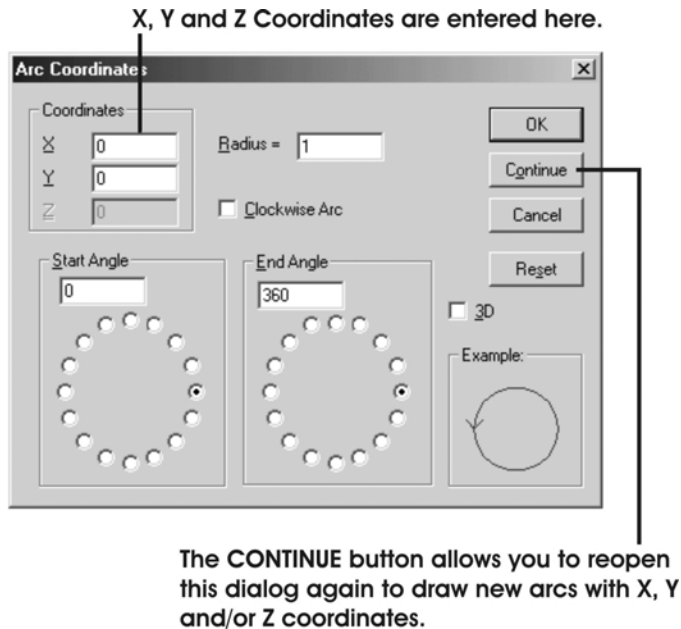
The START Angle of the ARC.
Each selection option represents the angle degree going counter-clockwise around the arc.

The END Angle of the ARC.
Each selection option represents the angle degree going counter-clockwise around the arc.

When you make a START ANGLE or END ANGLE selection, the example preview will change and show you what the arc is going to look like. This is a repetitive function, therefore, there is NO continue button as there was in earlier versions of the software.

REMEMBER: The **CLOCKWISE ARC** selection changes the way the arc will be drawn, basically reversing the direction and changing the start and end angle. Use the example window to view the changes before selecting OK. When you are ready to start sketching arcs, simply select OK.

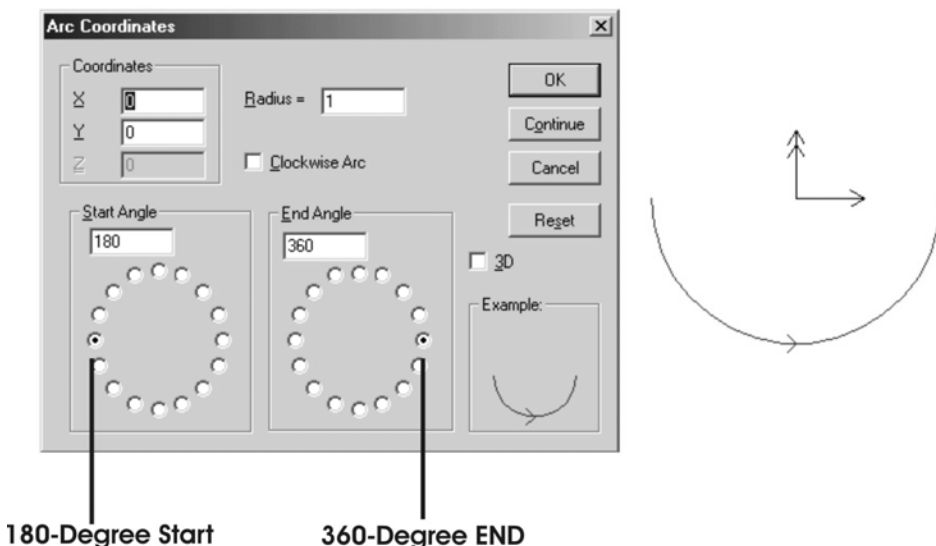
COORDINATES: This function allows you to draw arcs on the screen through the use of X, Y and/or Z coordinates. By selecting this function, you will open the **ARC COORDINATE** dialog as seen below.



The only difference between this dialog and the ARC/SKETCH dialog is that the CONTINUE BUTTON and the COORDINATES section are added.

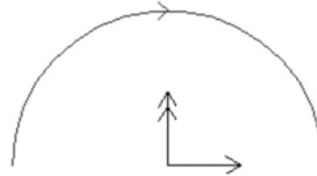
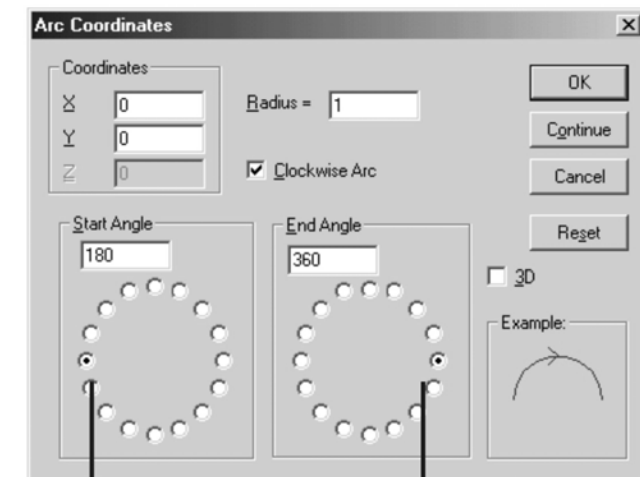
REMEMBER: The **CLOCKWISE ARC** selection changes the way the arc will be drawn, basically reversing the direction and changing the start and end angle. Use the example window to view the changes before selecting OK.

Counter Clockwise selection



The last image represents a counter clockwise arc. Let's take a look at what happens when you select the CLOCKWISE ARC option.

Clockwise selection



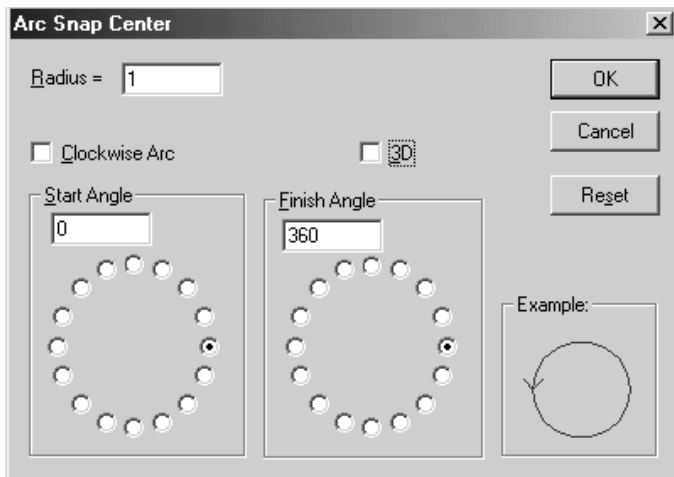
180-Degree Start

360-Degree END

As you can see the arc was drawn clockwise.

Select the 3D option to draw arcs in the Z-Axis.

SNAP CENTER: This function allows you to **Snap** arcs to the center of points on the screen. Let's take a look at this dialog below.

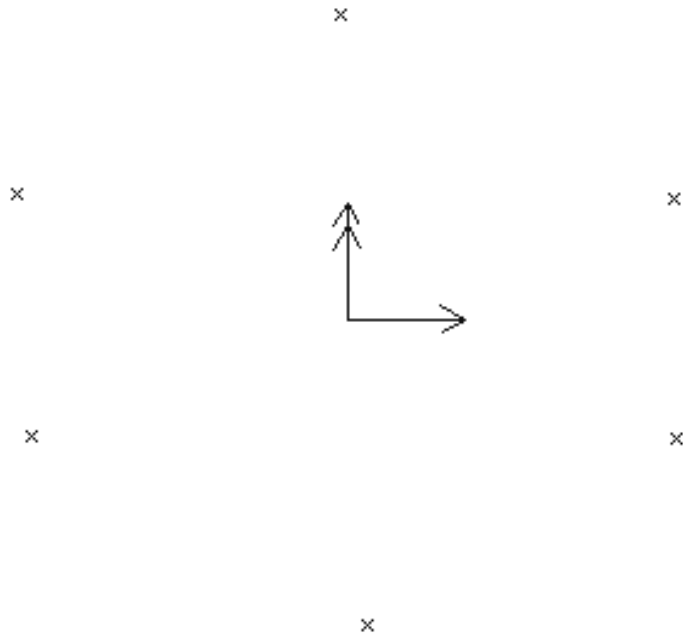


Once again, you have the ability to change between counter clockwise and clockwise as well as reset the values if needed. Simply enter the radius value and select the options needed, click OK and then click on the points that you want to snap the arcs to.

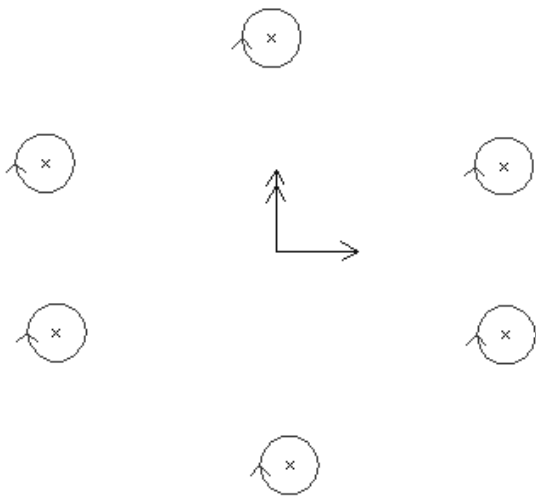
TIPS ON SNAPPING ARCS:

Here is a **TIP** on snapping arcs. If you have a series of points on the screen and wish to snap a specific size arc to all of the points in one action, you can do this by pre-selecting the points and then using the function.

For example:



Here we have 6 **pre-selected** points that we want to snap arcs to. Simply select the function and then select OK to automatically snap the arcs to all of the pre-selected points. See below.



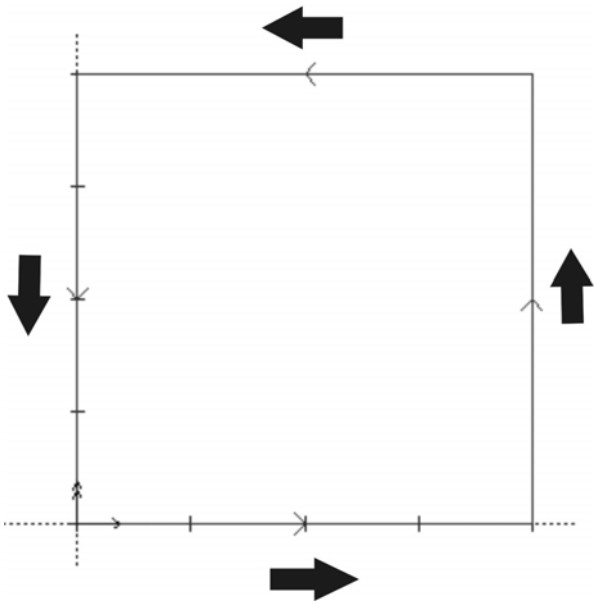
Now the points are no longer selected and the arcs are selected, all going in the same counter clockwise direction as opted in the dialog.

FILLET: By selecting the **Fillet** function, you can draw fillets or radii with existing geometry. This is an ACTION/OBJECT, OBJECT/ACTION function. Let's take a closer look at this function.

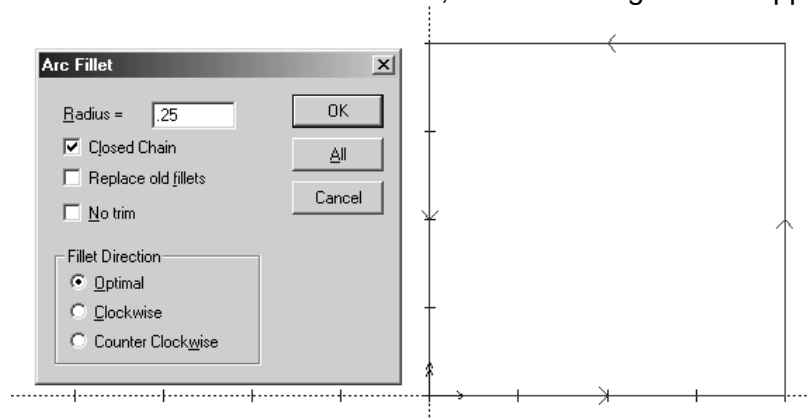
- **PRE-SELECTED GEOMETRY.**

In the image below you can see that we have a basic square that is pre-selected.

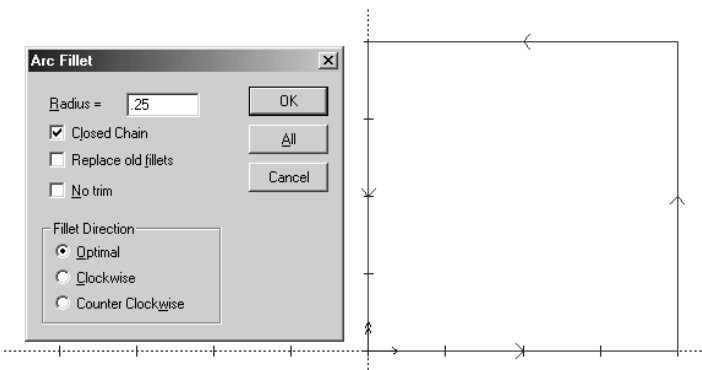
All Chain Selection Arrows going Counter
Clockwise



By selecting the ARC menu and then the FILLET function, the fillet dialog box will appear. See below.

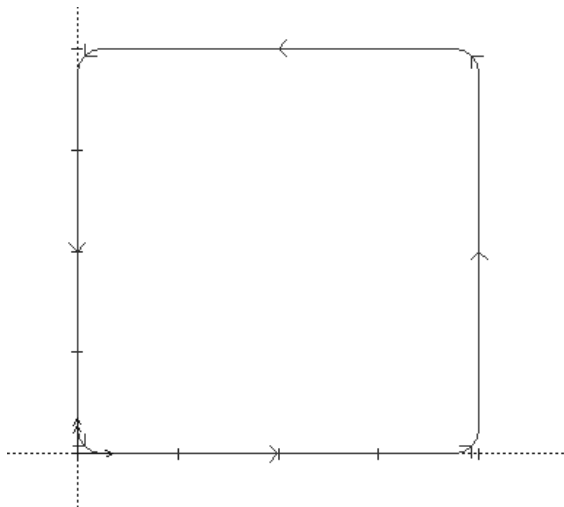


SPECIAL NOTE ON SELECTING ENTITIES: If you are selecting entities one at a time, you must hold down the SHIFT KEY on your keyboard. You can select a chain of entities by selecting the first entity of the chain, hold down the shift key and then select the last entity of the chain to select all of the entities in the same selection direction.



The ARC FILLET dialog allows you to enter the fillet radius and then choose the applicable options for a closed chain (which is what we have in the example), replace earlier fillets that you now have decided to change or to add the fillets but not remove the trimmed geometry.

Then, you also have the **FILLET DIRECTION** options. Optimal, which is basically the standard fillet, Clockwise if that is the specific direction you want the fillets to be drawn or Counter Clockwise.

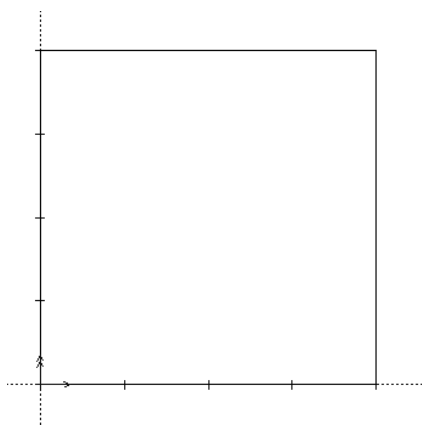


In the above example image, the directional arrows are all going counter clockwise. If your directional arrows were going clockwise, you would still have gotten the same result by pre-selecting the geometry first. However, this is not the case when making the fillets without having selected the geometry first.

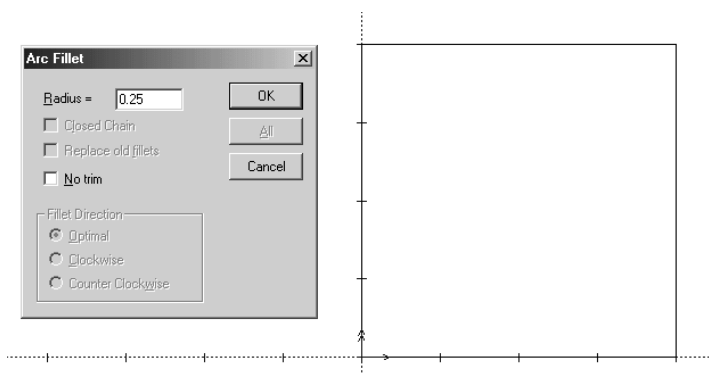
- **NON-SELECTED GEOMETRY.**

Let's take a look at applying fillets to geometry on the screen without the geometry being pre-selected.

Here we have an example of a square seen below.

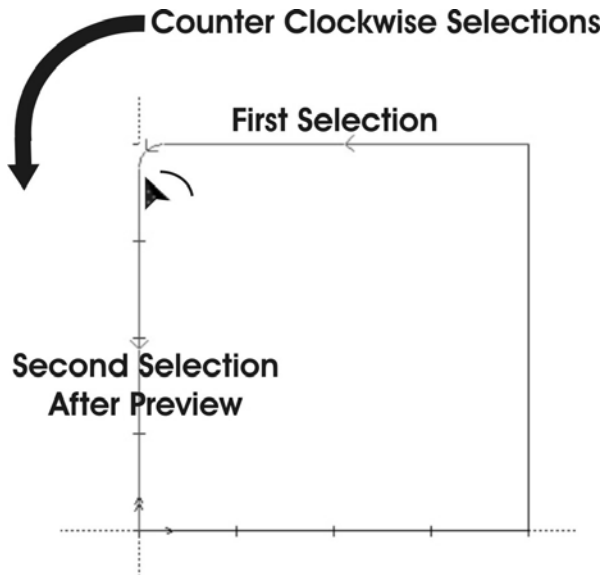


We want to draw .25-inch fillets on each of the corners of this square. To do this we'll select the ARC menu and then FILLET. See this dialog below.



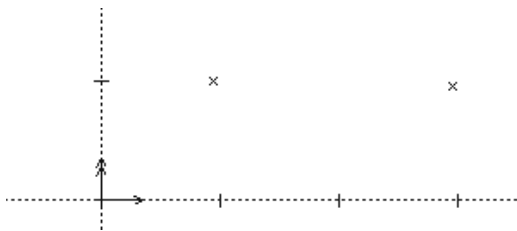
The selection options are different when working with non-selected entities than they are with pre-selected entities as you can see. You would enter the radius value of the desired fillet, and then select OK. The **NO TRIM** option allows you to create the fillet while leaving the original geometry on the screen as well.

When making fillets you will need to select your geometry in a **COUNTER CLOCKWISE** direction. After selecting OK in the fillet dialog box, you would select the first entity and then the second entity. Because this is a repetitive function, you can continue making counter clockwise entity selections to draw as many fillets with the same radius value as needed without having to go back to the fillet function. When finished, click your right mouse button to exit the function completely.



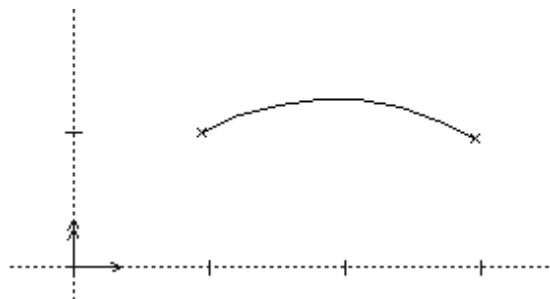
REMEMBER: 1. When drawing fillets with NON PRE-SELECTED geometry, make your selections in a COUNTER CLOCKWISE direction. 2. You can see a preview of the move before executing it by resting your cursor over the second entity to select before selecting it.

Let's take a look at another way to draw fillets. This time we will use 2 independent points on the screen. See below.

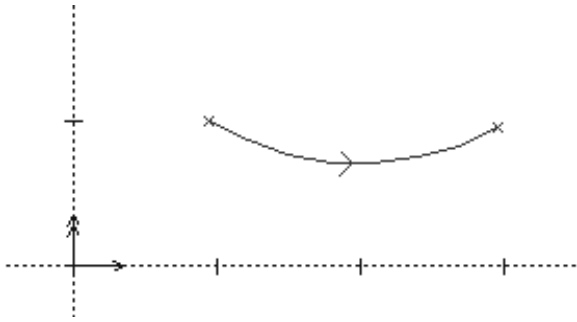


Select ARC and then FILLET. Now select the first point on the RIGHT and then the second point on the LEFT.

Here is the arc between 2 points with a counter clockwise selection.



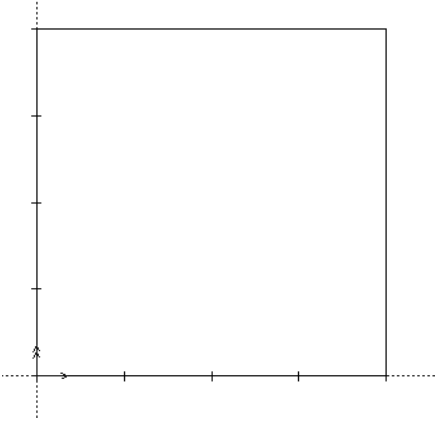
Now make the selection going clockwise. See the result below.



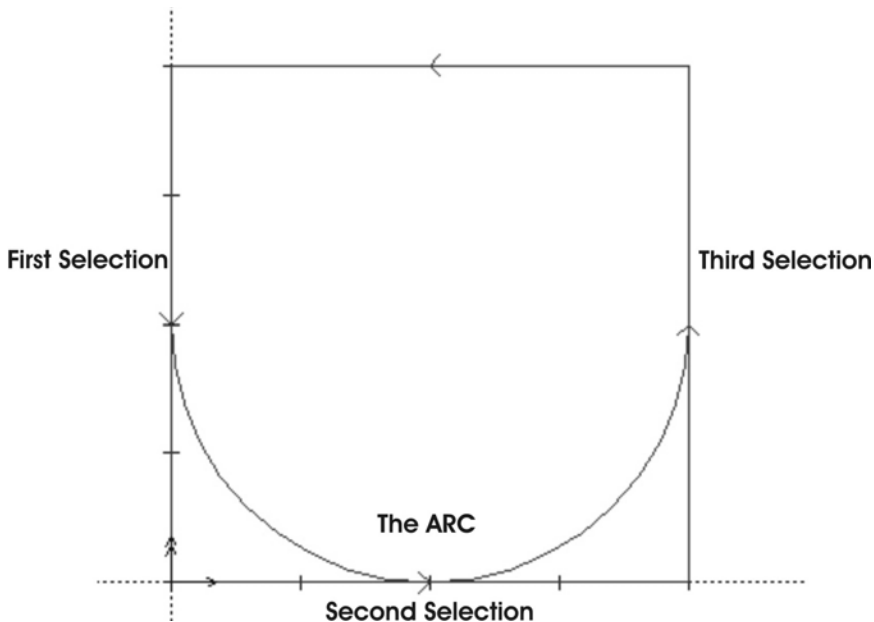
You can see the difference between the two selections. An important note is that the radius of the arc value in the fillet dialog needs to be the same or greater than **HALF** of the distance between the two points. Otherwise, the software will not draw the fillet. This will also work if you pre-select the two points and then go to the ARC/Fillet function.

3 ENTITIES: This function allows you to draw an arc or radius through 3 points or 3 entities. Let's take a closer look.

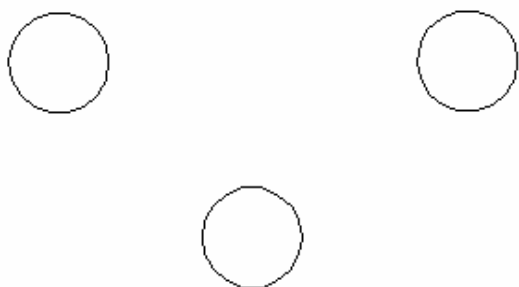
Once again we start with a square as you can see below.



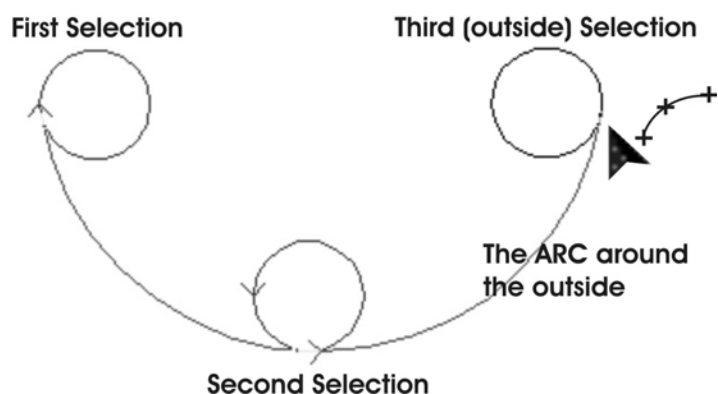
By selecting the ARC menu and then this function, you can select the 3 entities that you would like to use to create the arc between. The distance between the 3 entities will determine the radius of the arc.



To use this function again, you would need to return to the ARC menu or toolbox to select the function/icon. The ARC 3-Entity function does not require you to select your entities in a counter clockwise direction. The function depends on the fact that you are selecting 3 separate entities to perform the action. Here is the example of 3 circles below.

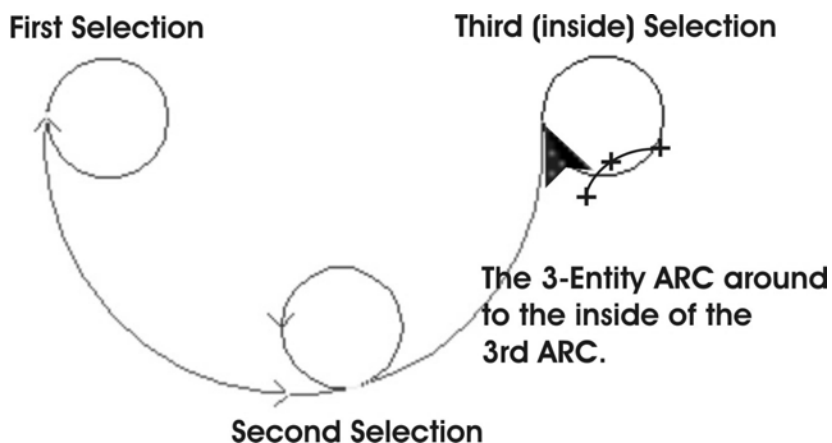


In the next image, you can see the selections.



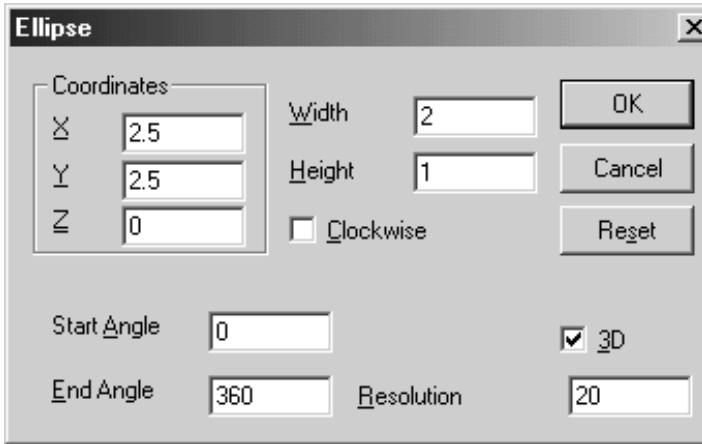
The arcs were selected on the outside. The first arc at the 180-degree position, the second arc at the 270-degree position and the last arc selected at the 0-degree position of the arc. REMEMBER, you can see a preview of the arc to be drawn by resting your cursor over the last arc that you would select, BEFORE selecting it.

Here is a different example of the same scenario below.



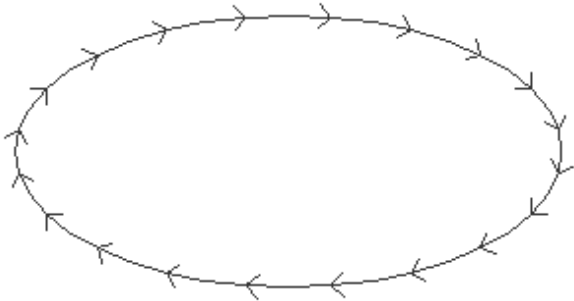
You can see the preview of the move before executing the move by resting your mouse over the 3rd entity to be selected. When satisfied, click the 3rd entity to draw the 3-Entity arc.

ELLIPSE: This icon and function will allow you to create an ellipse through entering specific values. Let's take a closer look at this function below.



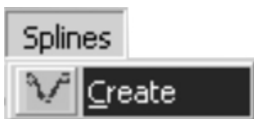
In the Ellipse dialog box above, you can see that there are several sections. Let's take an even closer look at them.

- **COORDINATES:** this allows you to position the ellipse around the screen using the BobCAD coordinate system. You can enter an X, Y and even a Z coordinate value for positioning. These coordinates will represent the CENTER of the ellipse.
- **WIDTH:** This is the X value.
- **HEIGHT:** This is the Y value.
- **START ANGLE:** This is the degree around the ellipse that you want to be the starting location.
- **END ANGLE:** This is the degree around the ellipse that you want to be the end position.
- **RESOLUTION:** This is the number of arc segments that you want to use to create the ellipse itself.
- **CLOCKWISE:** This allows you to draw an ellipse with the arc segments selected counter clockwise.
- **3D:** The 3D option simply allows you to enter a Z coordinate value.



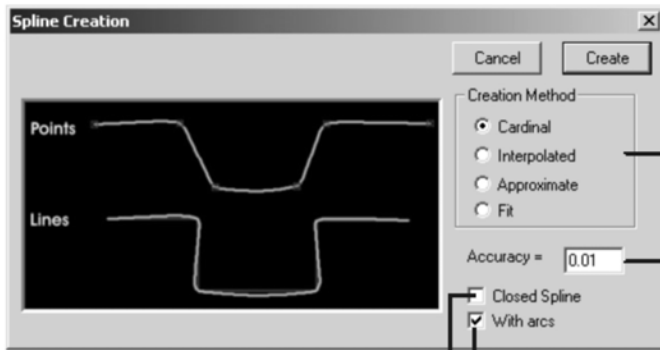
Section 20. The SPLINE Menu.

The Version 21 CAD-CAM system offers a **Spline** menu with 4 spline operations that we want to introduce to you in this section. In essence the system is allowing you to create a spline that is automatically converted to arc segments.



By selecting this function from the OTHER menu, the Spline Creation box will appear.

Spline options



The Closed Spline option creates a complete closed chain of arc and/or line segments.



The Accuracy option controls how closely the spline conforms to the geometry used to create the spline. In other words, how small the converted spline geometry is when created.

The "With Arcs" option converts the spline to add arc segments. If you un-check this option, the spline will be converted only to line segments.

The **Spline Creation** function offers 4 types of splines that can be created and automatically converted to line and arc segments. They are:

- Cardinal.
- Interpolated.
- Approximate.
- Fit.

Each of these are accompanied by an accuracy option as well as options for creating a closed spline, meaning that it will automatically create a spline without open ends and the "**With Arcs**" option that uses arc segments to generate the spline itself. By un-checking the "With Arcs" option, you will create a spline that uses lines rather than arcs.

Each of these spline functions can be used with existing points or existing line geometry on your CAD screen. If you do not want to use existing points or lines you can simply select the spline function that you want and then sketch the lines on the screen. When finished, right-click your mouse to create the spline. You see, **if you do not use selected lines or points, the spline function automatically goes into a continuous line sketch mode** allowing you to sketch the layout of the spline using continuous lines.

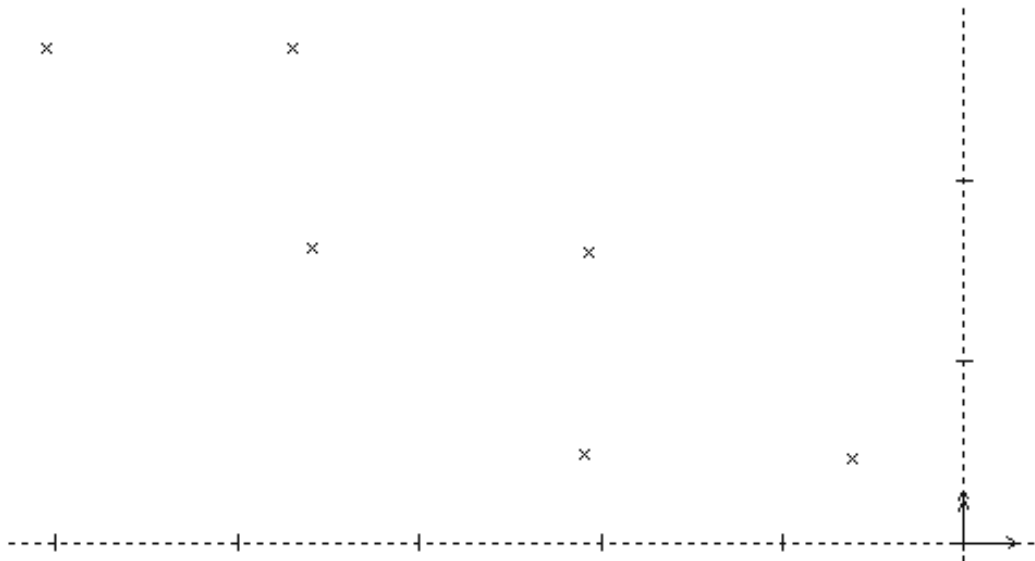
IMPORTANT: If you are using points, you must select the points **FIRST** in the order that you wish to create the spline. If you are using lines you will need to chain select the lines before selecting the spline option that you wish to use. A tip in performing a fast chain selection is to hold down your shift key and click on the last line of the chain. Once you have done this you can use the spline functions to produce the result that you need. The spline functions will not work on existing arc geometry.

Another important fact about the spline function is that when the software creates the spline for you, your original geometry will be left on the screen. Therefore, it will be important for you to change your active color to something other than the color of the original geometry for easy editing if you do not want the original geometry left on the screen.

Let's take a look at each of these separately.

CARDINAL: This type of spline drives through each of the points that are selected using longer linear geometry and arc segments through the mid-points.

Here are six points that we want to use to create a spline.



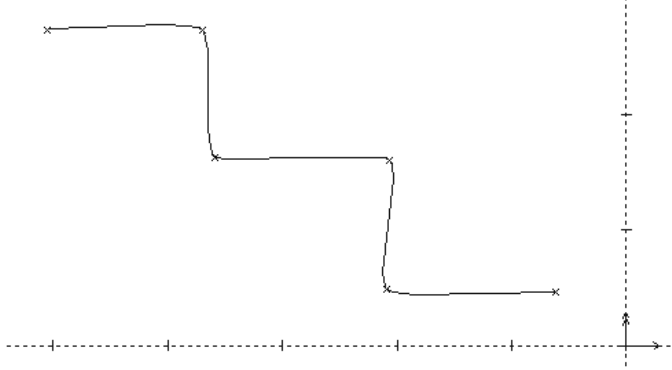
The first thing to do is to change the **ACTIVE COLOR** to something other than the color of the points.



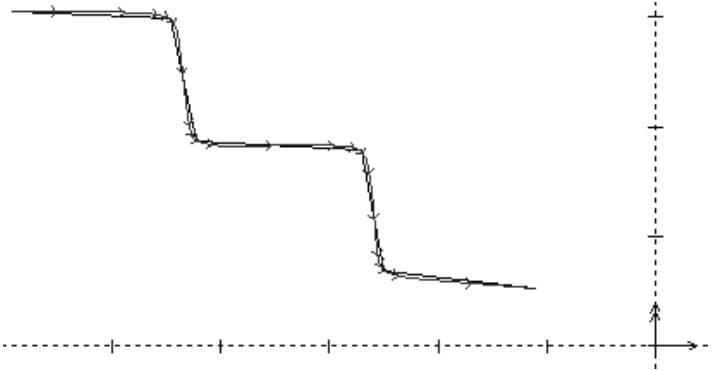
Now select the points in the sequence that you want the spline to be created and right-click your mouse when finished.

The result of the cardinal spline using points on the left and lines on the right:

Points:



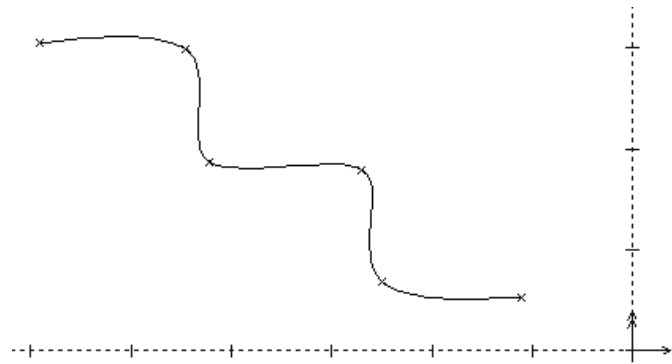
Lines:



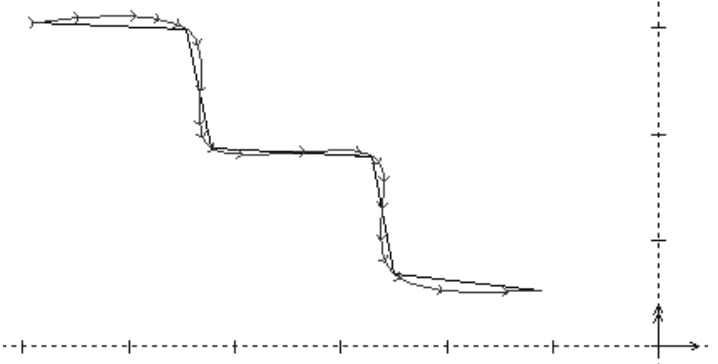
INTERPOLATED: This type of spline drives through each of the points that are selected using longer arc and linear geometry and arc segments through the mid-points.

The resulting interpolated spline using points on the left and lines on the right:

Points:



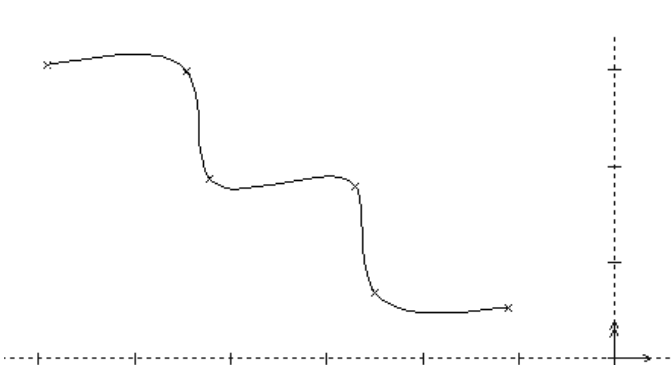
Lines:



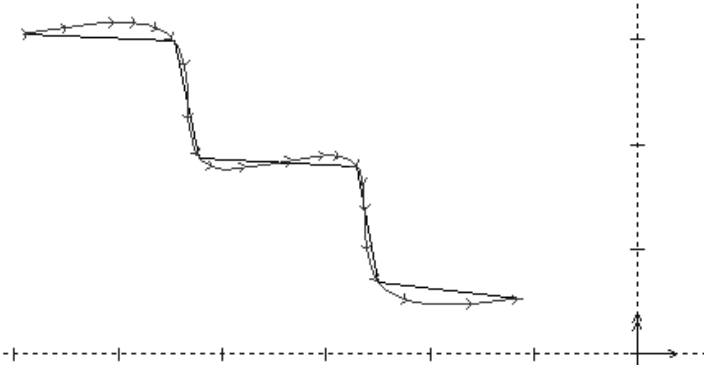
APPROXIMATE: This type of spline drives through each mid-point, also using slightly larger line and arc segments.

The resulting approximate spline using points on the left and lines on the right:

Points:



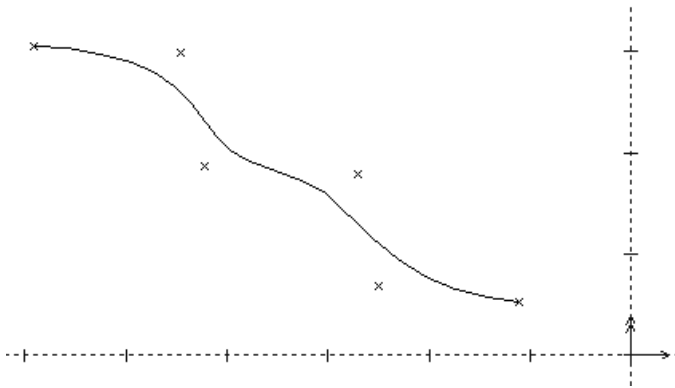
Lines:



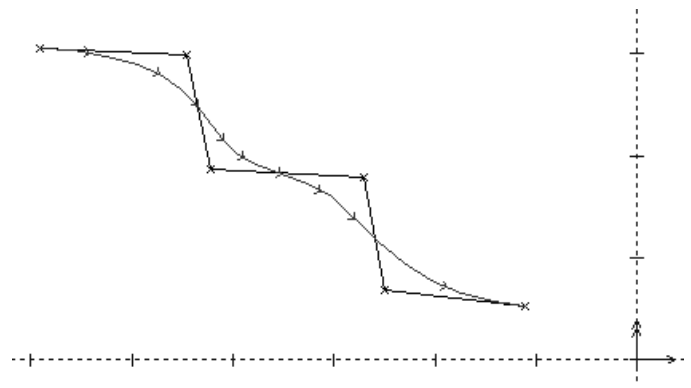
FIT: This type of spline is most similar in behavior to the Bezier Spline. As you can see in the example images, this spline drives through the approximate mid-location of the points that are selected.

The resulting “Fit” spline using points on the left and lines on the right:

Points:



Lines:



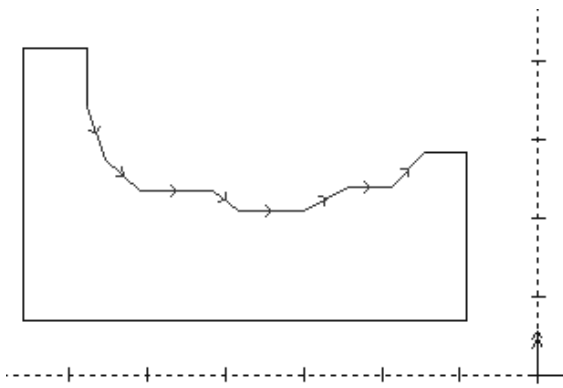
CREATING SPLINES WITH LINE GEOMETRY:

In addition to **Spline** creation with points, you can also use existing line geometry or even sketch lines on the drawing screen through the spline option box. Let's take a look at sketching continuous lines through this option.

1. With a new drawing screen, select the SPLINE menu, choose a spline option and click OK.
2. Now click your LEFT mouse button anywhere on the screen and by moving your mouse you will see that you can sketch a line. Draw the first line and click your left mouse button again to draw the second line. In actual fact you are simply sketching a continuous line. Each time you left-click your mouse you will begin the next line sketch. When you have several lines sketched on the screen, RIGHT-click your mouse and the line geometry will be replaced by the spline that you initially chose to create.

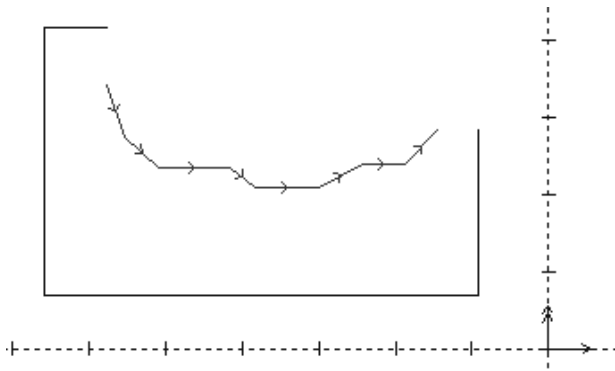
SELECTING LINE GEOMETRY FOR SPLINE CREATION:

To select an existing chain of lines you can use the “select chain” option in the CAD Edit menu. Once you select this function simply click on the first line of the chain at either end, point the first line's direction arrow toward the end of the chain, click your left mouse button once again to set the direction and then click on the last line of the chain to select the entire chain of lines. This is a good way to select a series/chain of lines that are connected to other entities that you do not want to select.



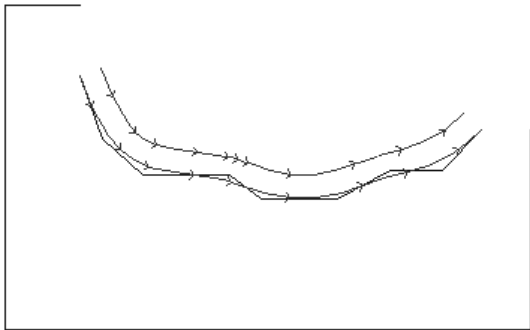
Also, if you have an independent chain of lines that is not connected to other geometry, simply hold down your shift key and click on the last line in the chain at either end.

This will select the entire chain.



USING SPLINE GEOMETRY FOR OFFSET AND TOOLPATH:

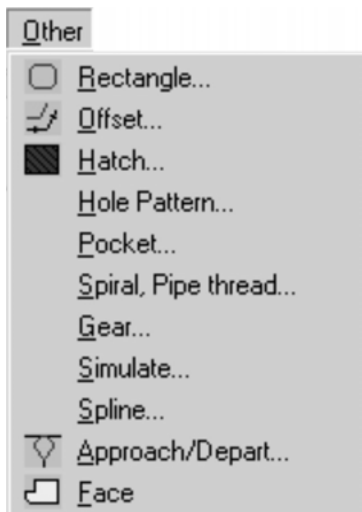
The **Spline** is automatically converted to arc and/or line segments when generated on the drawing screen. This geometry can be offset for toolpath by using the offset function in the OTHER menu of the software.



Section 21. The OTHER Menu.

This section is dedicated to the **Other** menu of the BobCAD-CAM Version 21 software. In this section, we will begin the integration of toolpath into the manual for various types of operations.

Let's take a quick look at the image below of the OTHER menu.

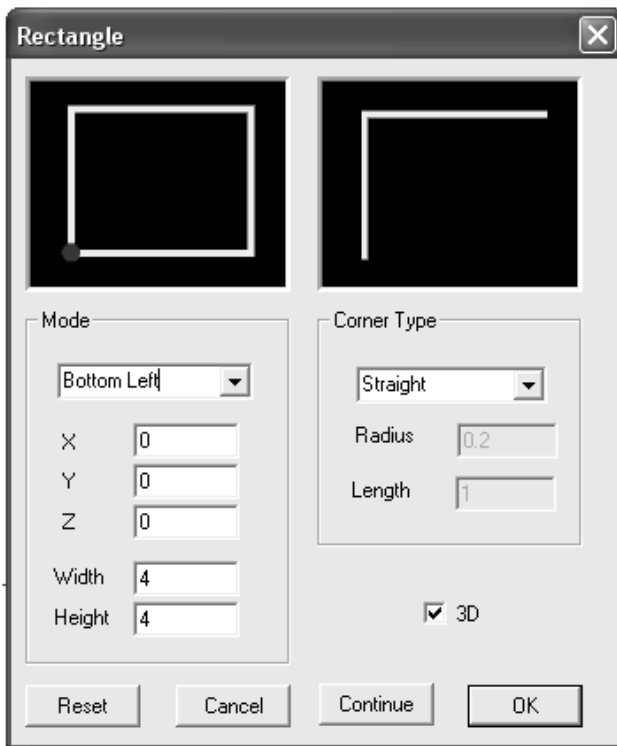


Several of the OTHER menu functions deal with CAD as well. Such as:

- The RECTANGLE function.
- The HATCH function.
- The FACE function.

Let's take a close look at each of these menu functions individually.

RECTANGLE: This function allows you to draw rectangles or squares on the screen using a coordinate system for X, Y and/or Z as well as the freehand ability to sketch these shapes. Let's take a closer look at the RECTANGLE box and each of the available drawing options.



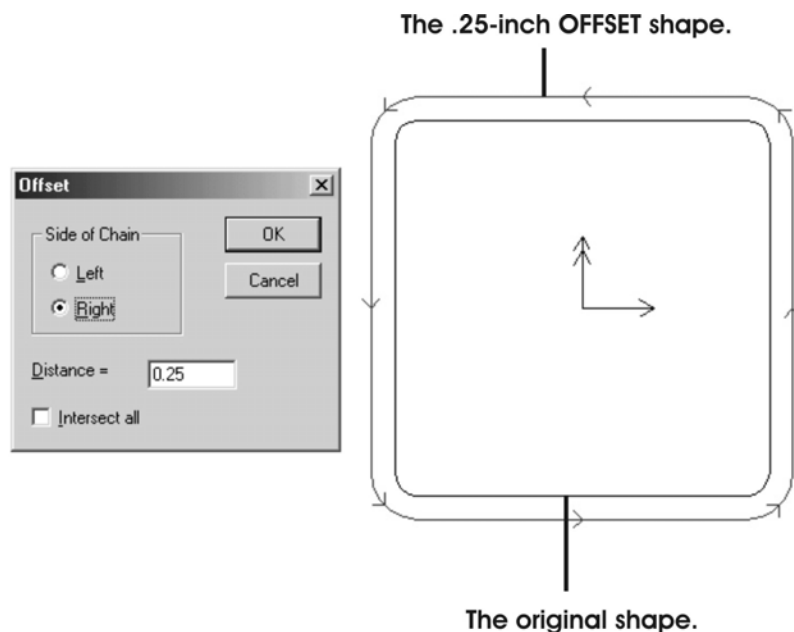
The 2 main sections in this box are:

MODE: The mode pull-down offers you a reference point for the shape. You have Bottom Left, Center and Drag. Each time you select a mode the example dialog image will change. Bottom Left places the bottom left corner of the shape at your specified X, Y and/or Z coordinate location. This would apply as well to the Center option. By choosing DRAG you have the ability to draw the shape freehand by left clicking your mouse where you want the lower left corner to be placed and then dragging and left clicking where you want the upper right corner to be placed. Even if you use the DRAG option you can still apply a corner type to the shape.

CORNER TYPE: The Corner Type section allows you to define a straight, Radius or Chamfer type corner to the shape. Simply use the Corner Type pull-down to make your selection. If you choose to create a straight corner the Width and height values command the shape unless you are freehand dragging the shape into position on the CAD screen. If you select Radius you will be able to apply a radius value as well as when you are choosing the chamfer corner. In that case you can enter a chamfer value that will be the same on all 4 corners of the shape when drawn.

The RESET button resets the values and selections to the last drawn shape. You also have a cancel button that closes the box, a continue button that allows you to continue drawing shapes and the OK button that executes the command.

OFFSET: The offset function allows you to select a single or chain of entities and create an offset geometry on the inside or the outside of it. Let's take a closer look at this function below.

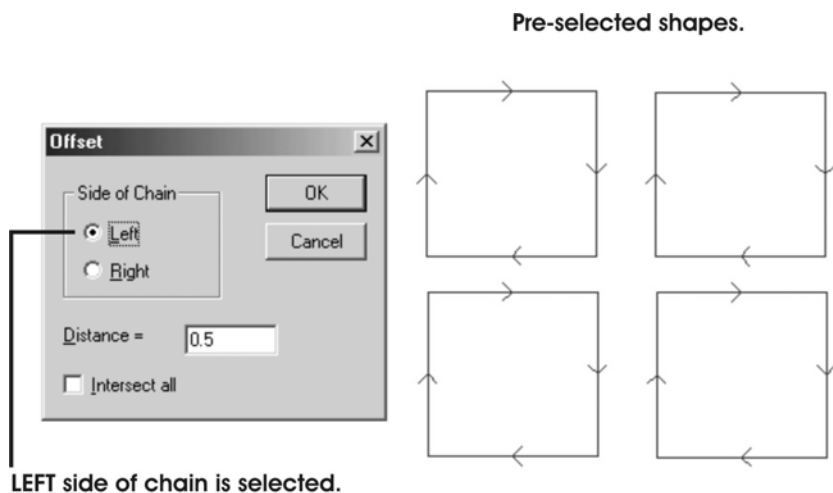


By choosing the RIGHT option, the offset is drawn on the OUTSIDE of the original geometry. By choosing LEFT, the offset would be drawn on the INSIDE of the geometry.

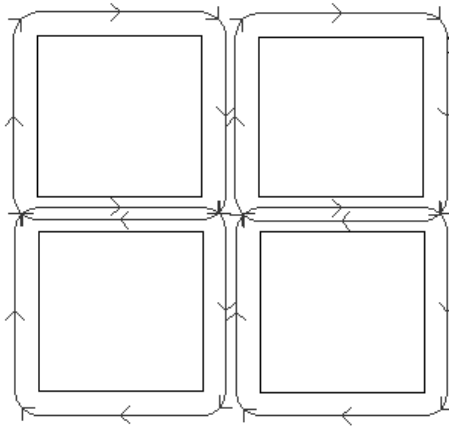
The **INTERSECT ALL** option refers to making the offsets of multiple shapes intersect each other or create a larger outside offset altogether.

Let's take a closer look at this.

You can see below that we have multiple shapes pre-selected, we have chosen LEFT as the offset side and did NOT select the INTERSECT ALL option.



By selecting OK, you can see the result below.

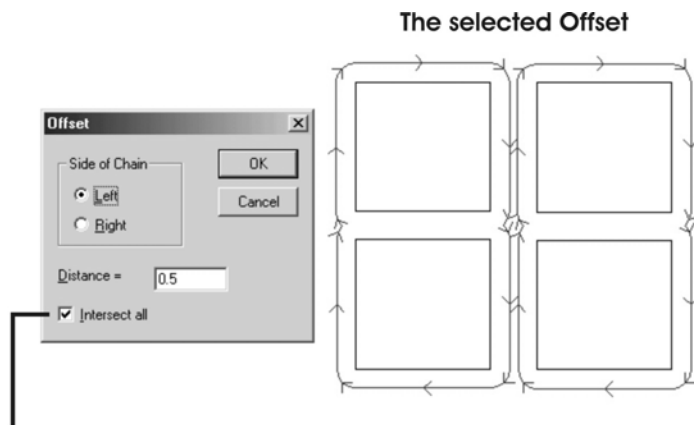


Each of the offsets intersect and cross over each other. If you were using the offset function as toolpath, half the diameter distance of the tool you were going to use, it would not work. You would need to use a smaller tool because the shapes are too close together to use the larger tool here.

USING OFFSET GEOMETRY AS TOOLPATH:

It is important to remember, that when cutting a single line **Offset** or entity, the line used as toolpath will be representing the center of the cutting tool. V21 offers a complete **Profile Wizard** in the MACHINE menu of the CAM that incorporates everything you will need for 2D profile milling. However, because the OFFSET function is available in the “**Other**” menu we are covering it here.

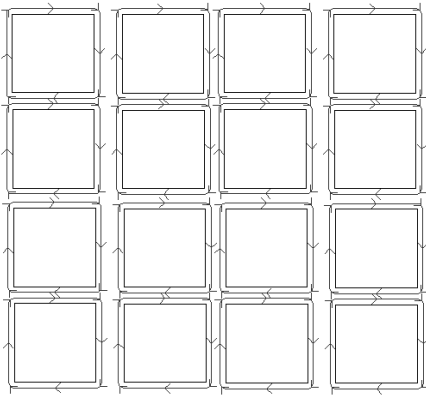
Here you can see that by selecting the Intersect ALL option, the offset doesn't cross over. This is a good option to use for shapes that you want to cut out. Now you could basically connect both of the offsets and use it as toolpath if you could not use a smaller tool.



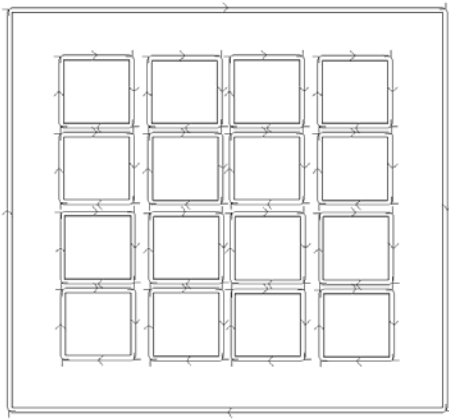
The INTERSECT ALL option was selected here.

OFFSETTING MULTIPLE SHAPES:

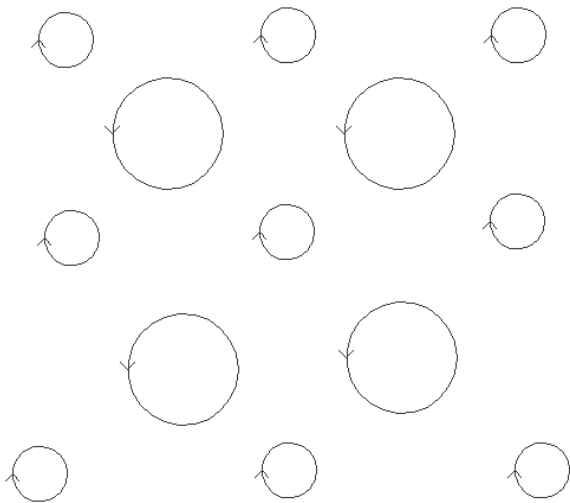
Here is a **TIP** on offsetting multiple shapes at once. If you would like to offset a series of shapes, you would select all of the shapes first. It is important to make sure that the arrows are all going in the same direction, and then use the offset function to create multiple offsets at once. This is a fast way to generate toolpath for many shapes all at once for use with the same size tool when cutting.



Here is an example that you can see below of multiple shapes inside of a larger shape, all offset automatically.

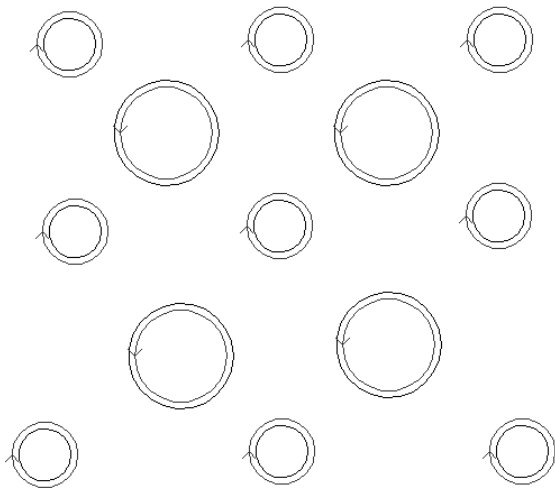


Here you can see that we have two sizes of circles. The larger circles have been selected COUNTER Clockwise. Let's see what happens when we use the INSIDE offset function.



All of the circles were offset automatically. However, the larger circles that we selected counter clockwise were reversed and the offset is now on the inside. The circles that we selected CLOCKWISE are all offset on the OUTSIDE. Remember this when using the offset function. The direction in which you select the shapes and arcs will determine the offset side whether you check INSIDE or OUTSIDE.

The circle with the selection arrow is the offset circle below.



This would be a good time to practice using the offset function with clockwise and counter clockwise selected shapes.

OFFSET CIRCLE LESSON:

You will need to generate offsets for the following:

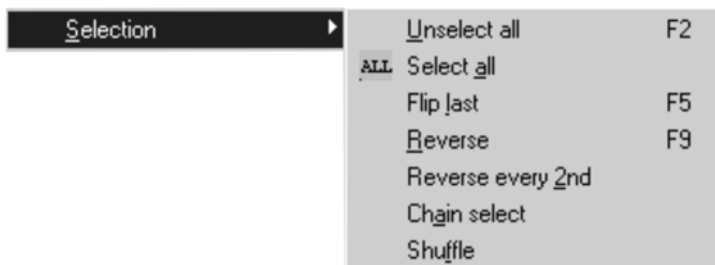
- INSIDE offsets for arcs that are selected COUNTER CLOCKWISE.
- OUTSIDE offsets for arcs that are selected CLOCKWISE.
- INSIDE offsets for arcs that are selected CLOCKWISE.
- OUTSIDE offsets for arcs that are selected COUNTERCLOCKWISE.

Do this exercise by freehand sketching 1-inch arcs around the screen first.

TRAINING TIP:

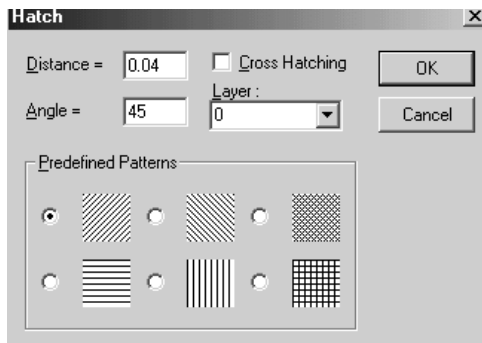
When you are creating offsets as toolpath for multiple shapes, you can use a fast selection function of the software to reverse the selection arrows on the screen.

At this time, go to the SPECIAL menu of the software and choose SELECTION. You will notice the SELECTION sub-menu that can be seen below.



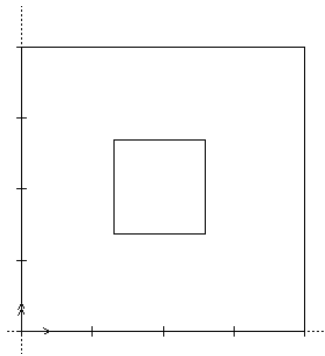
You can use the REVERSE function to reverse selection directions fast. You can also use the REVERSE EVERY 2nd function for the lesson you are doing. The REORGANIZE function in the CHANGE menu will automatically make all of the selection arrows go in the same direction on the screen. We will cover this SPECIAL menu function and sub-menu later in this manual again.

HATCH: A “Hatch” refers to a section of a drawing that is etched out to represent a filled area or stock.

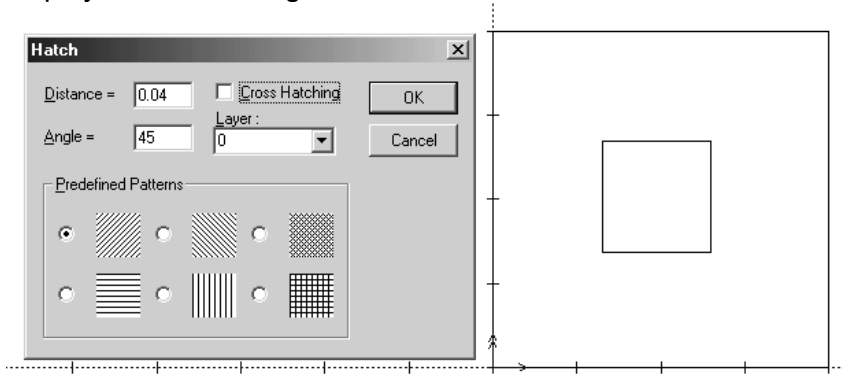


The Hatch dialog offers you the ability to choose from different hatch styles, define the angle of the hatch lines, choose a layer to place the hatch on as well as draw a criss-cross hatch line style. Let's take a closer look at this function.

Here we have two squares, one inside of the other.



By selecting the HATCH function, you will see the hatch dialog. We are going to use the same values that are displayed in this dialog.

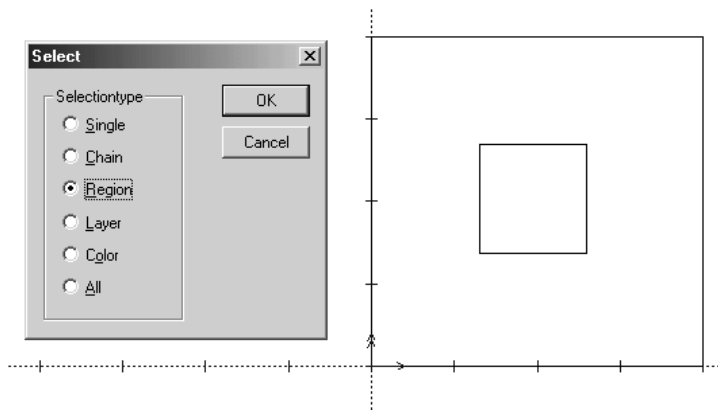


We are going to use the following values:

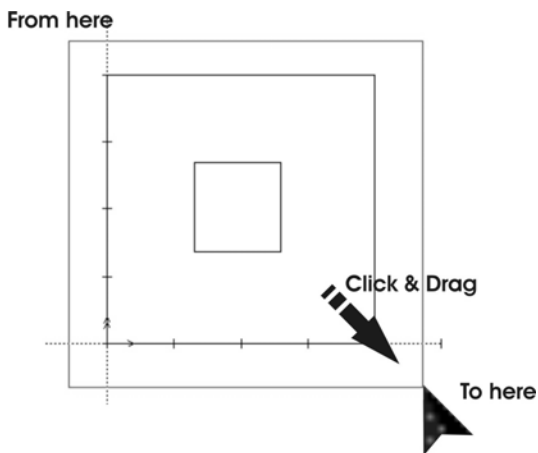
- **Distance:** We want the hatch lines to be .04-inches apart.
- **Angle:** The angle of the hatch lines will be 45-degrees.
- **Layer:** The layer will remain “0” as we do not have any layers in this example to save the hatched geometry on.
- **Predefined Patterns:** We have the first of these patterns selected to lean at a 45-degree angle toward the left.
- **Cross Hatching:** We are not going to select this option. If we did, the hatch pattern “criss-cross” style would automatically be selected for us.

By selecting OK, we are ready to make our selection.

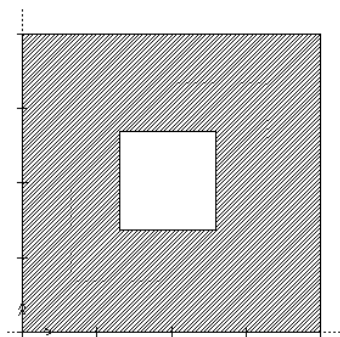
We are going to check the REGION option in the selection dialog and click OK. See below.



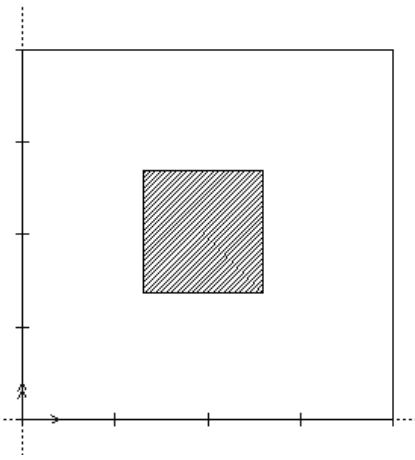
Once you have selected OK, you can click and drag a selection box over the entire drawing. Once you have done this, click OK.



The drawing will automatically be hatched. The middle box was avoided. All closed shapes inside of an outside shape will be avoided with this method.

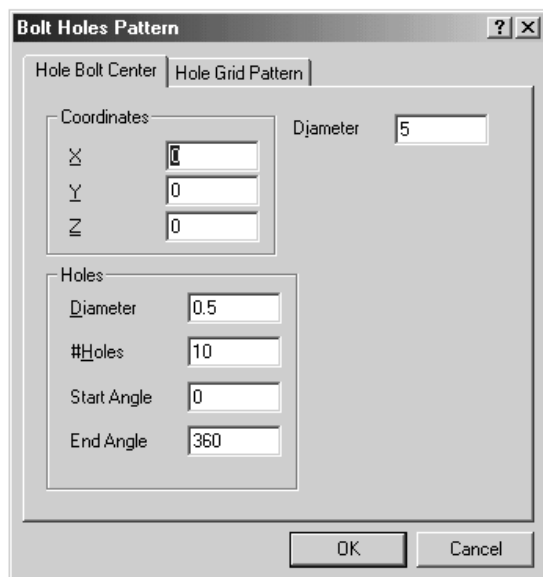


If we wanted the inside square to be hatched we would have clicked and dragged our selection box over the inside square only. See this in the next image.



The CHAIN selection option can also be used for creating hatched geometry. This function can also be used with **PRE-SELECTED GEOMETRY**. Simply select the parts that you would like to be hatched and then select the function. To hatch everything leaving the inside closed shapes un-hatched, use the SELECT ALL function first.

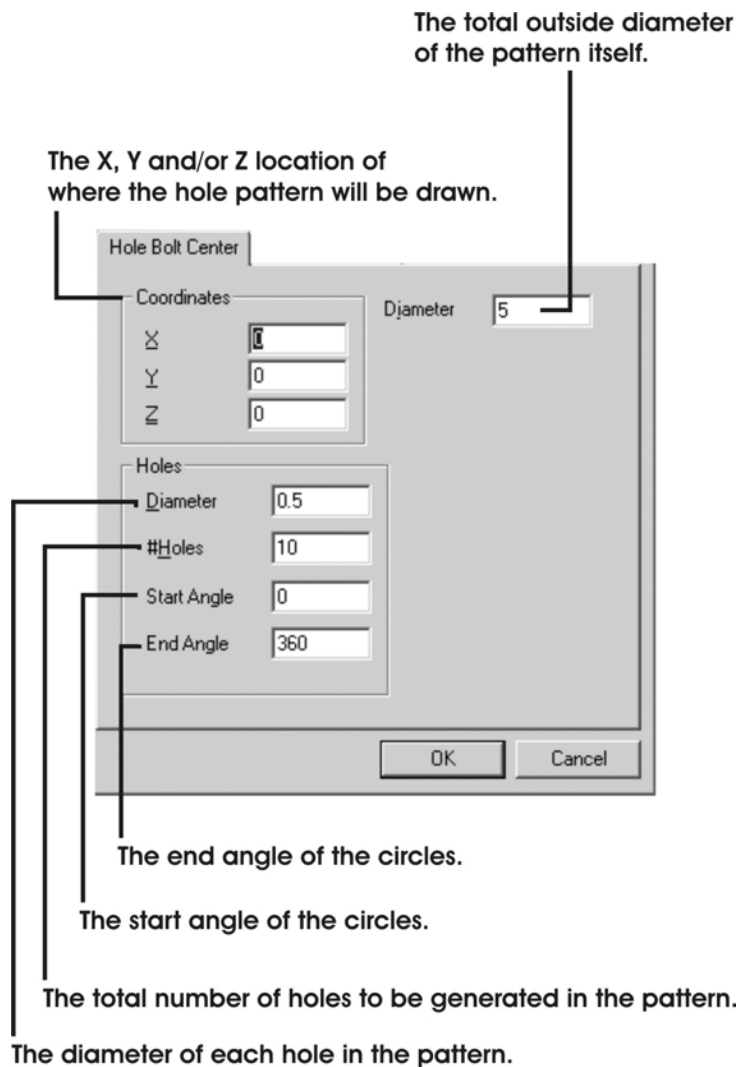
HOLE PATTERN: This function offers special bolt-hole style patterns. Let's take a closer look at this dialog.



The HOLE PATTERN dialog offers two different methods of generating them.

They are:

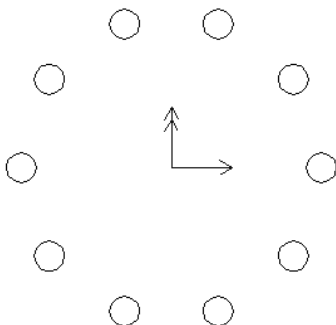
- **HOLE BOLT CENTER:** This offers you the ability to create circular style hole patterns by defining the overall outside radius of the pattern, The diameter of each hole, the total number of holes as well as the ability to define the start and end angles of each hole. You can place the pattern on the screen using X, Y and/or Z coordinates.



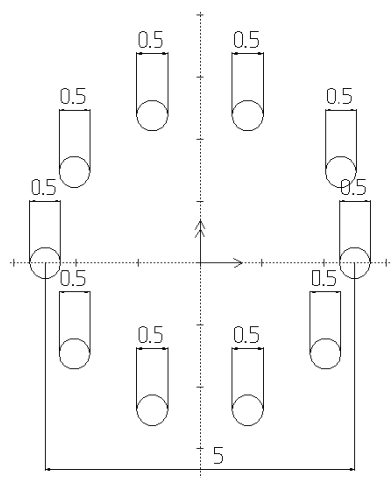
In the last diagram we had entered certain values:

- X 0, Y 0 and Z 0 for the coordinates, basically placing the circular pattern right in the center of the drawing grid axis.
- The overall DIAMETER is set at 5-inches total.
- Each hole diameter is .5-inches.
- The total number of holes is 10.
- The START angle is 0 and the END angle is 360-degrees to make complete circles.

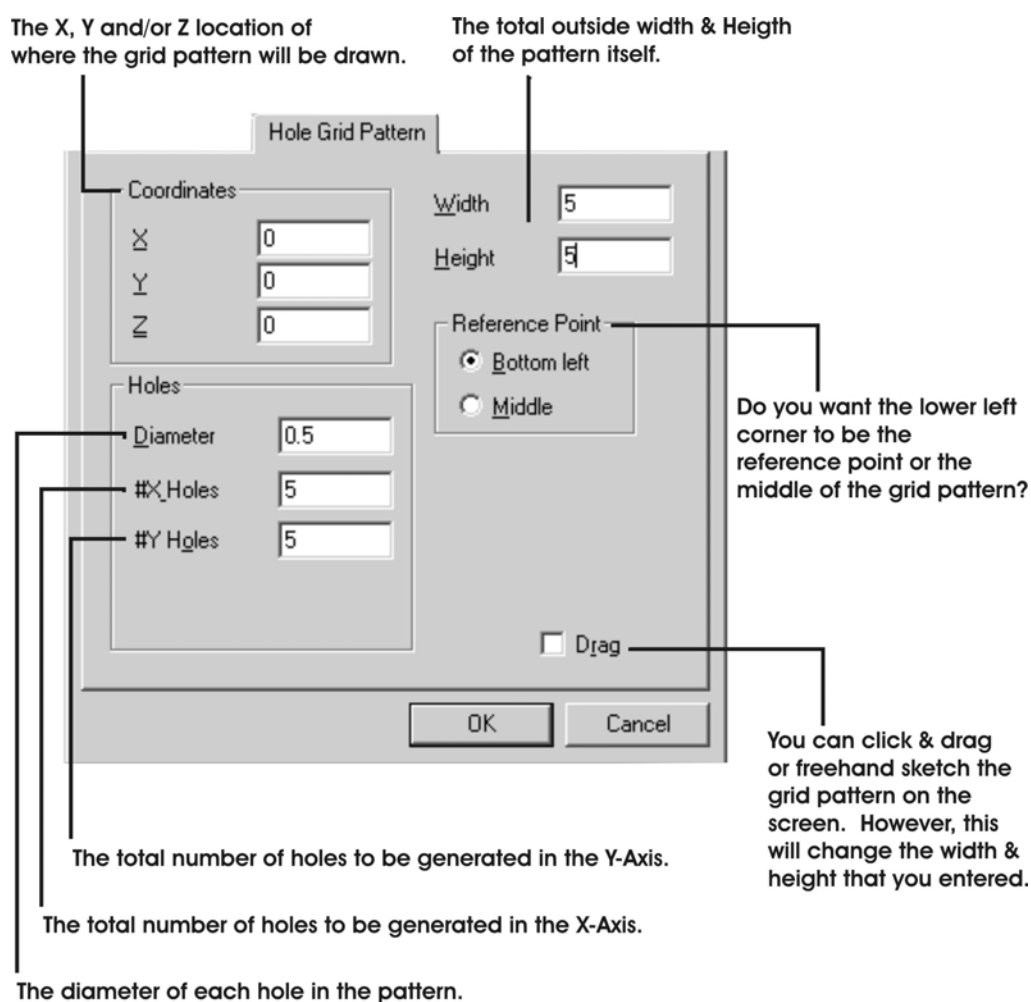
By selecting OK you can see the result below.



Below, you can see the same bolt-hole pattern with dimensions.



- **HOLE GRID PATTERN:** This function allows you to create grid type patterns rather than circular style patterns. Let's take a look at this dialog below.

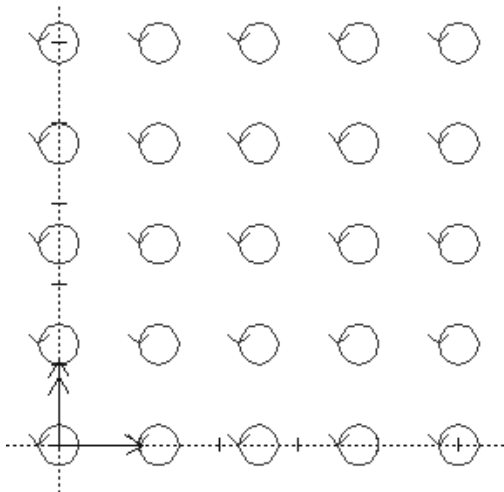


We have entered specific values in the diagram that you see above. They are:

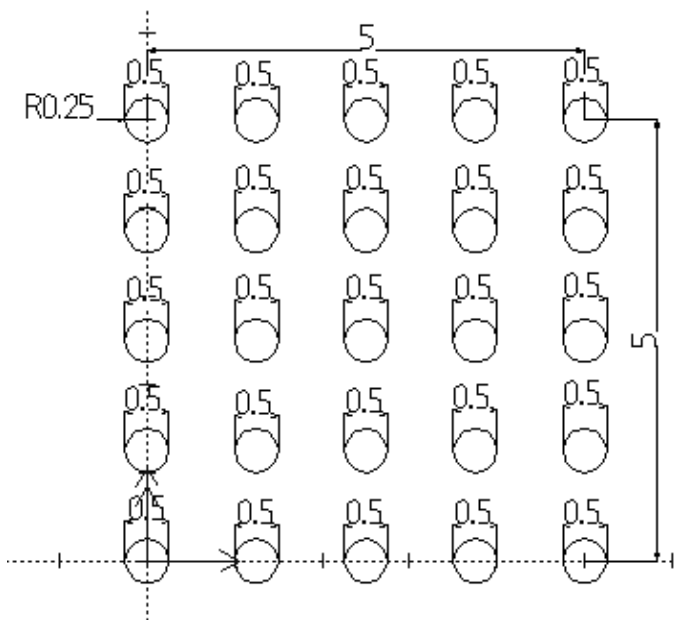
- The overall width is 5-inches.
- The overall height is 5-inches.

- We left the positioning coordinates at X 0, Y 0, Z 0 which will place the reference point of the grid pattern at this location.
- Each hole diameter is .5-inches.
- There will be 5 grid circles in the X-Axis.
- There will be 5 grid circles in the Y-Axis.
- The reference point is set to be the BOTTOM LEFT of the grid pattern itself.

Now by selecting OK you would get the following result.



The result with dimensions:



With this function you will be able to create bolt patterns for a wide variety of machining functions. Continue on for tips on machining hole patterns.

MACHINING HOLE PATTERNS:

Now that you have learned about creating bolt type patterns, we want to show you a fast way to machine them with the BobCAD Version 21 software. You can follow along with this example if you would like.

Below, we have a bolt **grid pattern** that was produced with the following variables:

COORDINATES:

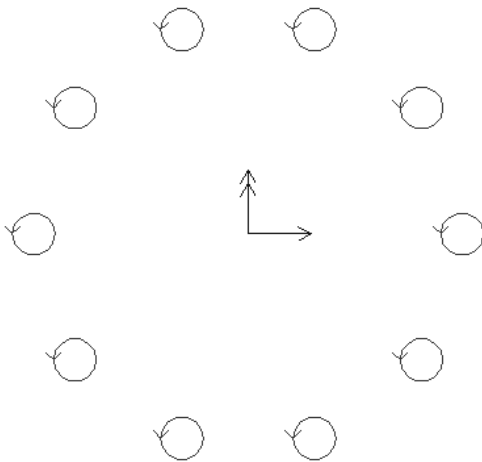
- X0
- Y0
- Z0

DIAMETER: 5

HOLES:

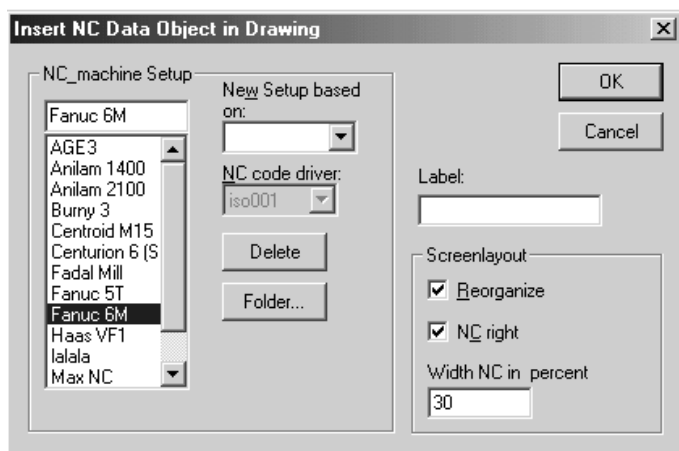
- Diameter of .5
- Number of holes is 10
- The Start Angle is 0
- The End Angle is 360

Select OK to see the result below.



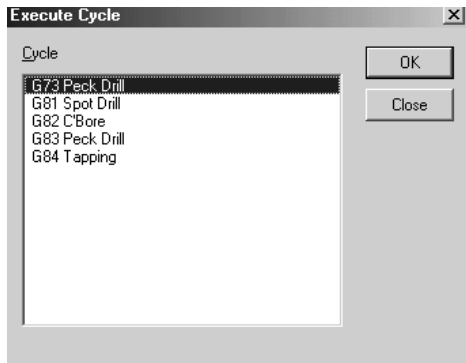
In earlier versions of BobCAD-CAM software you would now have to snap points for drilling. You NO LONGER NEED TO DO THIS as you can select the arcs and drill them directly in Version 21. Therefore, simply select the arcs you wish to drill and move on to select a post processor and open the CAM side of the software.

By selecting the SPECIAL/NC CAM menu and INSERT NC you will bring up the INSERT NC DATA dialog. SELECT **FANUC 6M** for the example.



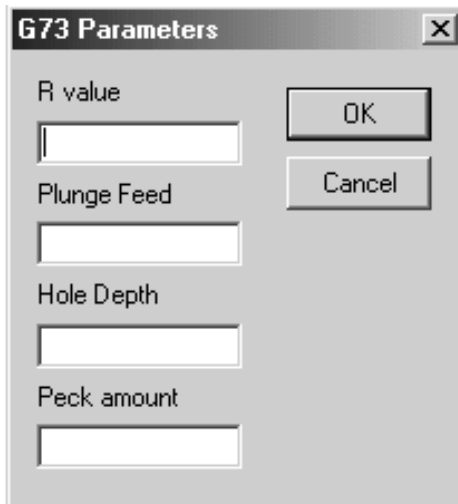
This dialog is covered in depth in the SPECIAL/NC CAM MENU section of this manual. Once you have selected the FANUC 6M post processor, select OK to open the NC CAM side of the software.

Now that the NC CAM side is open, select the MACHINE MENU and then DRILL & CYCLES to open the pre-stored drill cycles. See below.



Select the G73 Peck Drill cycle and click OK.

The toolpath will automatically be drawn and the wizard dialog will appear asking for drill variables.



G73 PARAMETERS:

- R is the retract value for clearance.
- Plunge Feed can be entered and does not need the letter F in front of the value.
- Hole Depth speaks for itself. Enter a – symbol in front of the value.
- Peck Amount will stop the drill and retract the tool at this depth for material removal before returning to the hole for the next depth. If the total depth is –1 and the PECK depth is .24, the tool will retract 4 times for material removal.

If you are still following along, enter the following values:

- R Value of .2
- Plunge Feed of 5
- Hole Depth of .75
- Peck Amount of .25
- Select OK.

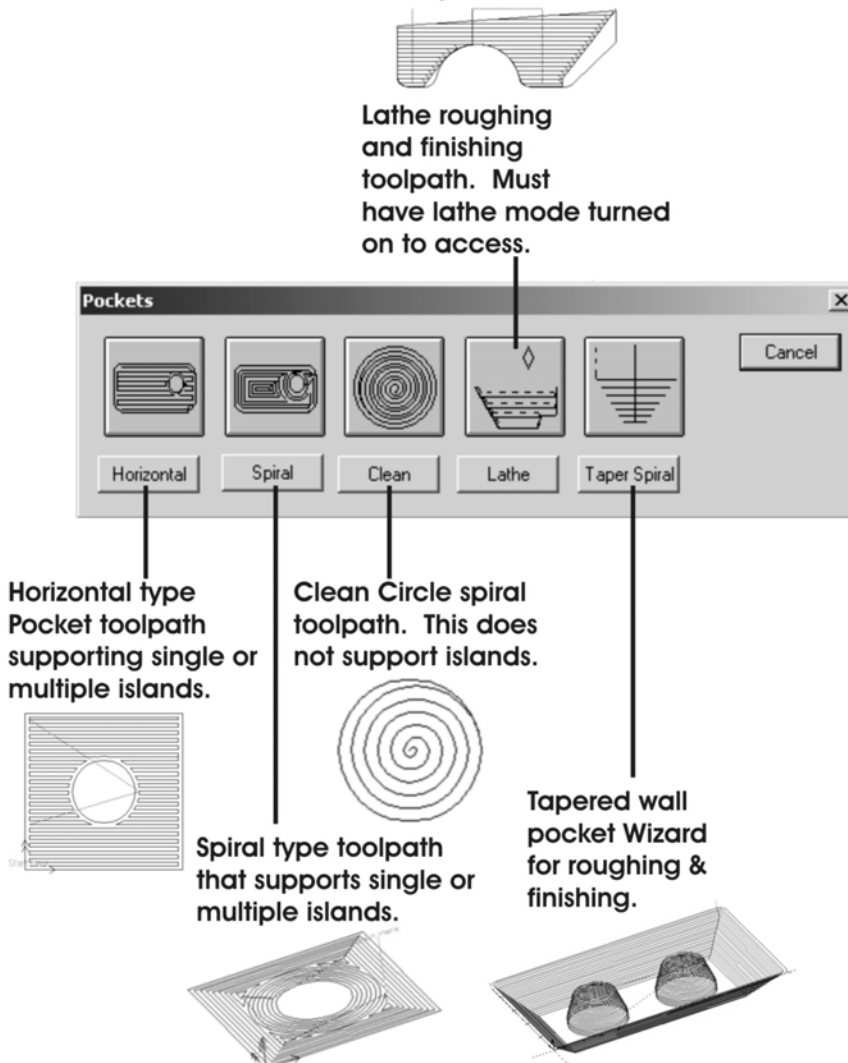
This will have generated the code for the Bolthole pattern automatically for you.

YOU DO NOT NEED TO SNAP POINTS TO THE CENTER OF ARCS FOR HOLE DRILLING IN VERSION 21. SIMPLY SELECT THE ARC AND USE THE DRILL CYCLE OF YOUR CHOICE.

POCKET: The **Pocket** function offers both Milling and Turning functionality depending on what mode you are in and the style of pocket toolpath that you want to generate. In this section of the manual we are going to cover the Mill pocketing operations. You can study about the Lathe toolpath generation in the Lathe section of this manual. By selecting the OTHER menu and then POCKET, the Pockets Dialog will appear offering different style pocketing toolpath options. See below.



Each one of these buttons will provide you with a different style toolpath result. As a note, when using the Lathe toolpath option, you will need to have Lathe Mode turned on which is the small blue diamond button located on your main CAD toolbar that runs across the top portion of your drawing screen. Let's take another brief look at this dialog in the next example image.



HORIZONTAL: This operation allows you to create a straight lace cut type pocket toolpath that basically runs back and forth across the part, inside the outer boundary profile while avoiding any islands inside. To use this function you do not need to have anything selected on the CAD screen in terms of your profile or inside island geometry. However, because this function is Action/Object-Object/Action, you can pre-select your geometry first and then execute the operation. If you select your geometry first you must select the outer profile so that your selection arrows are going clockwise and any island geometry selected with the arrows going counter-clockwise. Therefore, if you find that your toolpath is not correct, you may want to check and make sure that your arrows are correct. To avoid this, simply deselect everything, choose this pocketing operation and then make your geometry selections, following instructions from the lower instruction bar on your CAD screen to get the desired result.

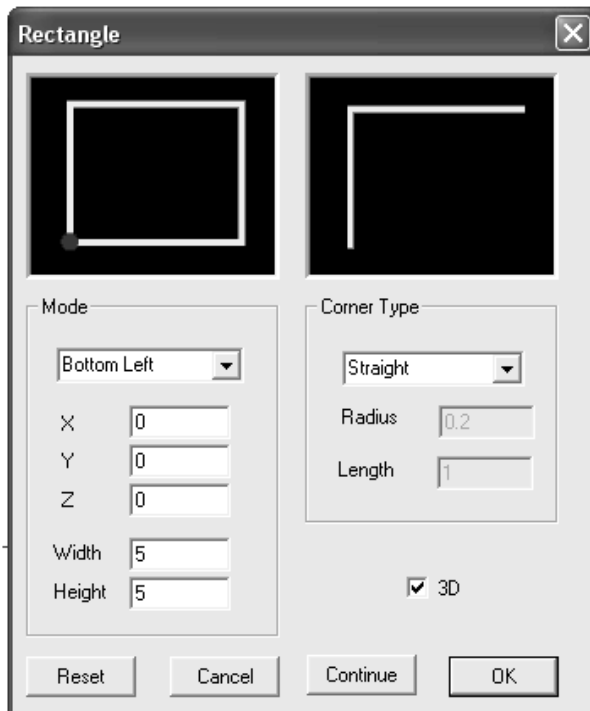
It would be best to do the quick lesson on the next page to learn more about this operation.

QUICK HORIZONTAL POCKET LESSON:

IMPORTANT: V21 has global environmental option that is called, “Auto Pre-select.” This allows you to draw geometry whether freehand or by using coordinate values and have the geometry appear on the CAD screen “selected” and ready to be used for functions that are considered Object/Action. This simply allows you to draw shapes and not have to select them first to use a function. For advanced operators, this can help save time. In this lesson, we would like you to turn this option OFF to prevent any discrepancies executing the steps of this lesson. To do this, go to the main FILE menu and select ENVIRONMENT. Choose the DEFAULTS TAB and UN-CHECK the Auto Pre-select button if it is checked. Now click OK to exit the dialog.

STEP 1

Use the Rectangle function from the OTHER menu to draw a 5x5 square box first. To do this, go to the OTHER menu and choose Rectangle.



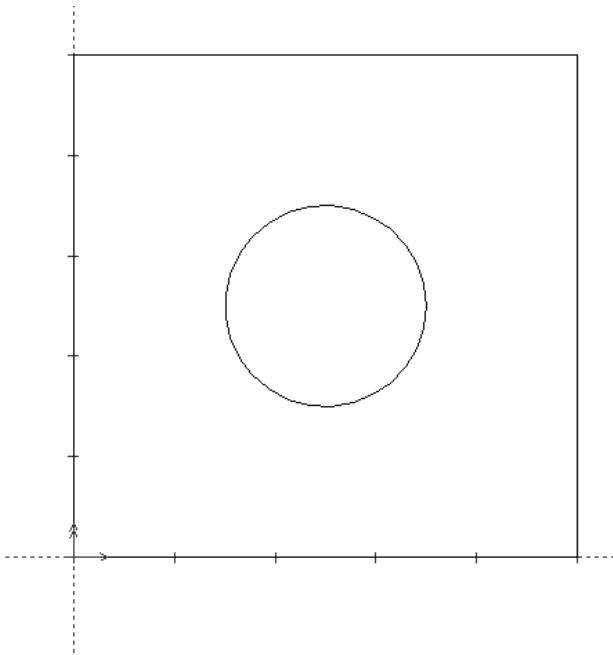
- Choose Bottom Left as the Mode. Use the pull-down to make the selection if needed.
- Leave the X, Y and Z values at 0.
- Enter 5 for the width and the height.
- Choose Straight as the Corner Type. Use the pull-down to make the selection if needed.
- Click OK to draw the shape.

STEP 2

Go to the ARC menu and choose Coordinate Center. This box allows you to define the arc you intend to draw as well as the radius and command its coordinate position on the CAD screen.

- In the Coordinates section enter 2.5 for X, 2.5 for Y and leave the Z value 0. If the outer and inner profiles were on different planes or levels in the Z axis you would not be able to produce the toolpath correctly. If the Z entry is grayed out, check the small 3D button and change the value to 0.
- Enter a Radius of 1.
- We are not concerned with the Clockwise arc check box as this is used once you become more proficient in using the auto pre-select option for faster drawing and using object/action operations. Remember, this is both an object/action-action object function. Therefore, when drawing shapes for pocketing quickly and using the auto pre-select option you must know what direction your selection arrows need to go. Outside profiles should be selected with direction arrows going counter clockwise with any islands going clockwise.
- The Starting Angle should be 0 and the End Angle should be 360 to draw a full circle. Click OK to draw the circle.

The result:



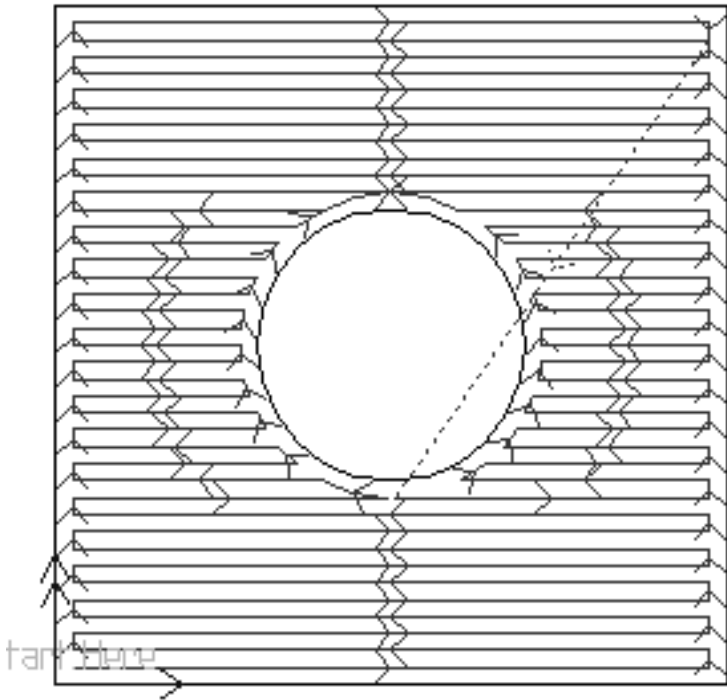
STEP 3

Now, with nothing selected, go to the OTHER menu and choose POCKET. Click on the Horizontal button at the far left end of the Pockets dialog.

Enter the following values in the HORIZONTAL POCKETING dialog:

- Tool diameter: .25-inch
- Distance Between Lines: .125-Inch. This is the tool step-over.
- Stock Distance: 0. We are leaving no extra stock around the profiles
- Contour Is First Path: Un-Check. This uses your original profile geometry as a part of the toolpath.
- Mark Start: Check. This gives you a written starting position on the screen and creates a green line indicating where the tool will begin plunging down into the cut.
- Click OK.
- Place your cursor on any part of the outer profile and left-click on it. It will all select. Now place your cursor on the circle and left click on it. Now right click your mouse to generate the toolpath.

The toolpath result:



The toolpath that is created, goes all the way against the walls of the pocket and is horizontal in style. The dashed line represents a rapid move in which the tool will retract to the clearance value set in the CAM and then will start its next cut.

This toolpath is based off of the tool diameter and the step-over for that tool. This function offers automatic cutter compensation for the tool that is used because you are telling the software that information. If you would like to leave stock for a finish pass around the outside walls of your part, you will want to add some STOCK DISTANCE for this.

IMPORTANT NOTE FOR MACHINING POCKET TOOLPATH:

When machining pocket toolpath and generating the NC Program you will want to turn the COMP option OFF in the CAM side of the software BEFORE creating any g-code. You have already used a form of tool comp when entering a step-over distance for the tool in the horizontal pocket dialog and you can see that amount in the "Step %" value. To do this you would go to the CONTOUR menu in the CAM side of the software and select NO COMP before generating the NC program.

SPIRAL: The SPIRAL **Pocket** style toolpath allows you to generate toolpath that spirals around islands creating a smoother path for the cutter. If you are attempting to create pocket toolpath with islands that have radii on the corners or are circular islands, it is recommended that you use this toolpath for a smoother cut and finish.

The Spiral Pocket dialog offers separate roughing and finishing operation settings that you can see in the next example image.

This option finishes the contour after roughing.

Tool Diameter for Finishing.

Auto Roughing Option.

Auto Finishing Option.

The image shows a 'Spiral Pocketing' dialog box with several sections and annotations. The 'Operation' section has 'Roughing' and 'Finishing' radio buttons, with 'Finishing' selected. The 'Roughing parameters' section includes 'Tool diameter' (0.5), 'Distance between lines' (0.25), 'Stock distance' (0), and a 'Contour is first path' checkbox. The 'Finishing parameters' section includes 'Tool diameter' (0.25), 'Step %' (50), and a checked 'Finish contour afterwards' checkbox. The 'Advanced connection of islands' section has three radio buttons: 'No connection', 'Connect', and 'Connect and close path' (selected). There are also checkboxes for 'Connect rest with dotted lines' and 'Mark Start'. The 'Maximum tool diameter information' section has input fields for 'For finishing' (7874015) and 'For roughing' (0), with a 'Calculate' button. Annotations with lines point to various elements: 'Auto Roughing Option.' points to the 'Roughing' radio button; 'Auto Finishing Option.' points to the 'Finishing' radio button; 'Tool Diameter for Finishing.' points to the 'Tool diameter' field in the finishing parameters; 'This automatically provides a START HERE line.' points to the 'Mark Start' checkbox; 'This option lets you connect multiple pockets with a dashed line representing rapid moves for the CAM.' points to the 'Connect and close path' radio button; 'This gives you maximum tool information for roughing.' points to the 'For roughing' field; and 'This gives you maximum tool information for finishing.' points to the 'For finishing' field.

This automatically provides a START HERE line.

This option lets you connect multiple pockets with a dashed line representing rapid moves for the CAM.

This gives you maximum tool information for roughing.

This gives you maximum tool information for finishing.

This offers connection settings for Islands:

1. No Connection.
2. Connect (the islands)
3. Connect all islands & close the entire toolpath altogether for machining.

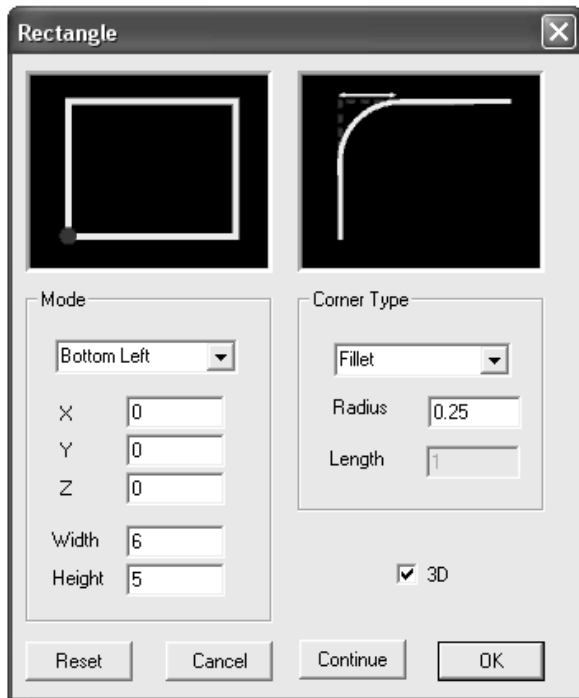
The roughing and finishing operations are used separately. In BobCAD you would generate the NC program for roughing and then go back to your geometry and use the finishing operation for creating that portion of the NC program after making the appropriate tool change in the program.

QUICK SPIRAL POCKET LESSON:

IMPORTANT: V21 has global environmental option that is called, “Auto Pre-select.” This allows you to draw geometry whether freehand or by using coordinate values and have the geometry appear on the CAD screen “selected” and ready to be used for functions that are considered Object/Action. This simply allows you to draw shapes and not have to select them first to use a function. For advanced operators, this can help save time. In this lesson, we would like you to turn this option OFF to prevent any discrepancies executing the steps of this lesson. To do this, go to the main FILE menu and select ENVIRONMENT. Choose the DEFAULTS TAB and UN-CHECK the Auto Pre-select button if it is checked. Now click OK to exit the dialog.

STEP1

Select the OTHER menu and then RECTANGLE. This will bring up the rectangle dialog.



- Select Bottom Left under Mode. Use the pull-down as needed for your selection.
- Enter 0 for X, Y and the Z values. If the Z entry is grayed out, check the 3D option button as needed to access Z.
- Enter 6 for the Width and 5 for the Height.
- Under Corner Type choose Fillet from the pull-down and enter .25 as the radius value.
- Click OK to draw the shape.

STEP 2

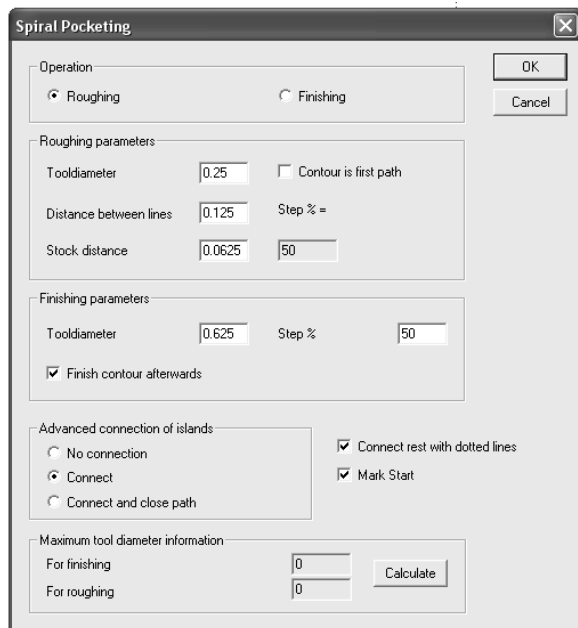
Go to the ARC menu and choose Coordinate Center. Enter the following values:

- Enter 3 for X and 2.5 for Y. Leave Z at 0.
- Enter 1 for the radius.
- The Start Angle should be 0 and the End Angle should be 360 to draw a full circle.
- Click OK to draw the circle.

Now that we have created our profiles we are ready to create our toolpath.

STEP 3

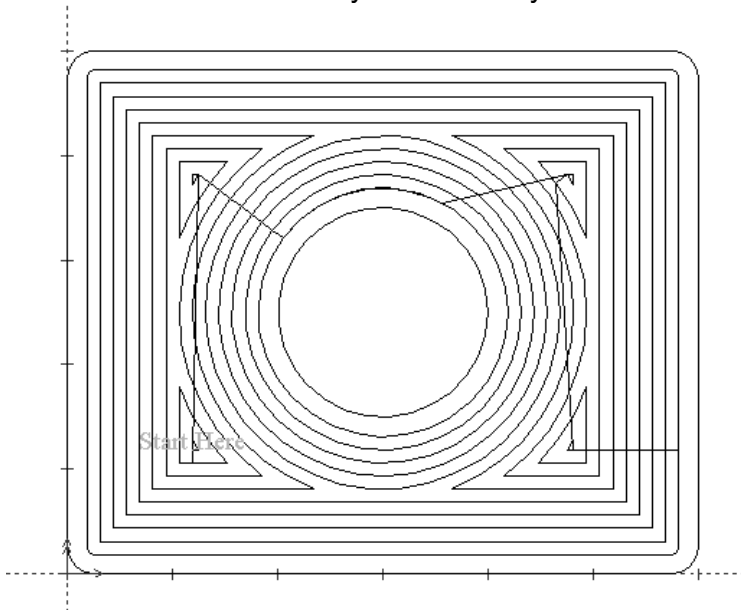
Go to the OTHER menu and choose POCKET. Now click the Spiral operation button. This opens the Spiral Pocket dialog box.



- In the Operation section, select Roughing.
- In the Roughing Operations section enter .25 for the tool diameter.
- Un-check the “Contour is first Path” option box if it is checked.
- Enter .125 as the Distance Between Lines. This is our tool step-over.
- Enter .0625 for the Stock Distance value. This will be the amount left around our profile walls allowing a finishing operation.
- Under the section, “Advanced Connection of Islands” check the Connect option.
- Check the Connect Rest with Dotted Lines as well as the Mark Start options.
- Click OK.

STEP 4

Place your cursor on the outside profile and click your left mouse button to select it. Now click on both circles one at a time the same way. Then click your RIGHT mouse button to create the toolpath.

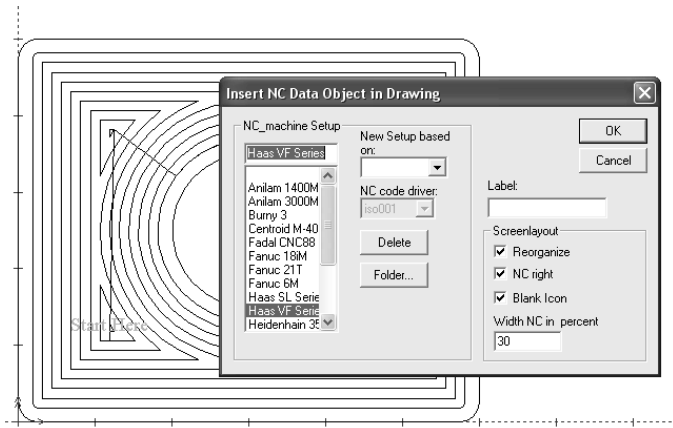


The toolpath that we created is basically editable line and arc geometry at this point. Now let's machine it.

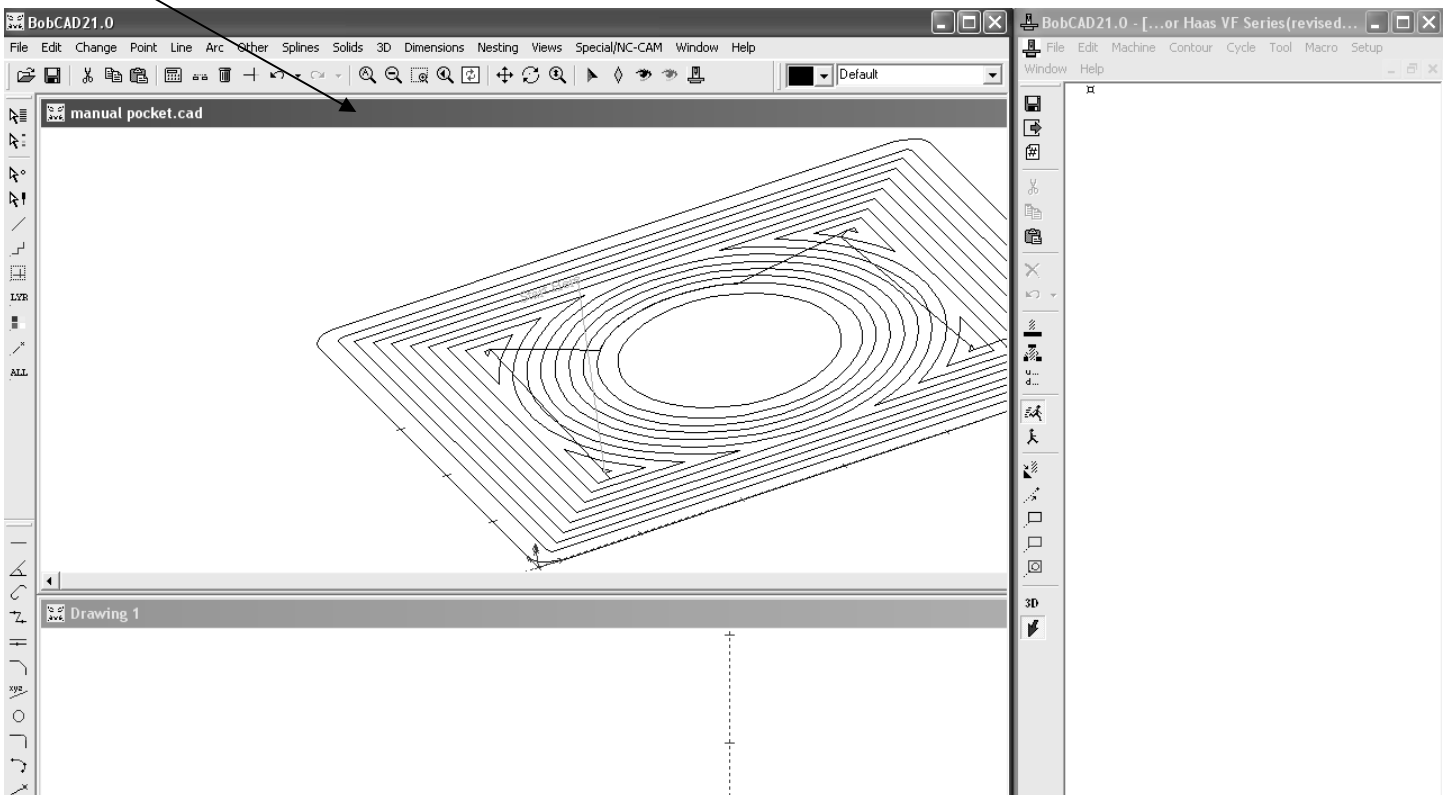
STEP 5

Go to the Special NC CAM main menu and choose Insert NC.

Choose the HAAS VF Series post processor from the list of configurations in the box.



Click OK to open the CAM in the right hand side of the software. When the CAM opens your CAD screen may split into multiple screens if you have multiple drawing screens open. You can place your cursor on the **window bar** and double click your mouse.

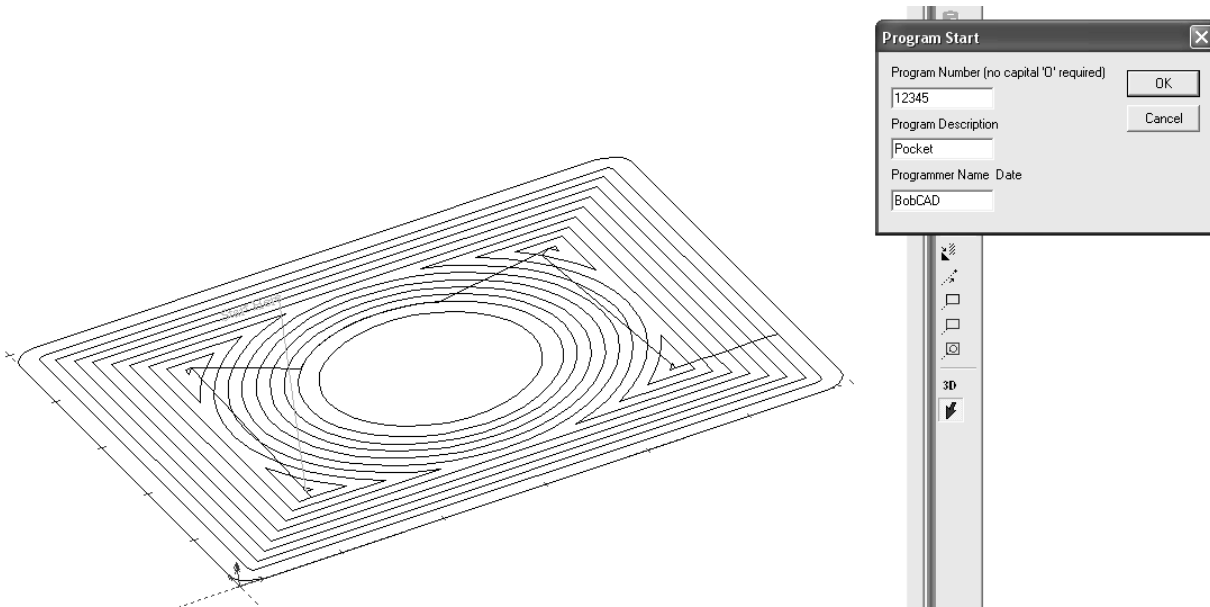


Before moving on, go to the CAD 3D main menu and turn 3D ON. This will be your first option in the 3D menu. Now select the View ALL button from the main CAD toolbar so that the part zooms into view better. This is the small function icon that has a magnifying glass with the letter A in it.



STEP 6

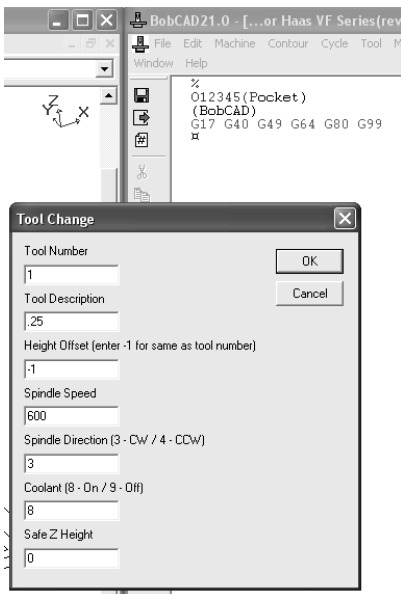
At this stage in the process you will want to build the header or start block of the program. Do this by going to the CAM Macro menu and choosing Program and then Start.



The current HAAS VF Series script will then ask for a program number, description and allow for a programmer name and date as needed. Enter 12345 for the number (HAAS Requirement), call the description: Pocket and click OK to advance.

STEP 7

Now the script will ask for more information.



- Enter 1 for the tool number.
- Enter .25 for the tool diameter.
- The height offset can be set at -1 because this will make the H & T numbers match. The height and tool numbers need to match for HAAS.
- Spindle Speed can be 600 as this is a roughing cycle.

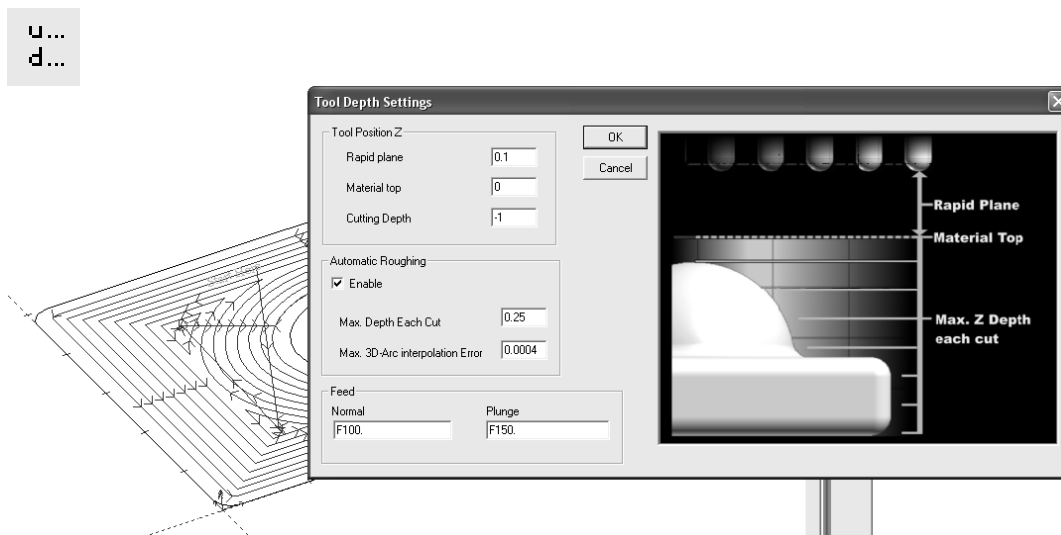
- Initial Spindle Direction is 3. This is clockwise.
- Coolant is 8. This means ON.
- Safe Z height refers to the amount of height you want the tool to retract to during rapid moves so that you can avoid clamps or fixtures. Leaving this at 0 would make the tool rapid to the start point of the pocket toolpath at the height used in the tool depth settings dialog called, "Rapid Plane." This value in the script does not overwrite the value entered in the Tool Depth Settings Parameters box under Rapid Plane. You can enter .1 as the safe Z-Height so that the tool rapids just above the part.
- Click OK. This brings up a Rapid Location box. Leave both values at 0 and click OK. Once again, this is a HAAS script being used for machining this example. Your machine control may be different and thus, requiring a custom script for you.
- Click OK to insert the start block into the program.

STEP 8

Now you want to select the toolpath. Go to the EDIT menu on the CAD side, choose Select Entities and then CHAIN. Now place your cursor on the green vertical starting line of the toolpath and click your left mouse button to select it. Move your cursor lower than the mid-point of this line to make the directional arrow point DOWN. When you have done this click your left mouse button again to secure the direction and then hit the F3 key on your keyboard to select the rest of the chain of toolpath.

STEP 9

With your toolpath selected go to the CAM side and select the Tool Depth Settings icon from the main CAM toolbar. This is the function icon that has a small U & D on it.



- Enter .1 for the Rapid Plane. This is your clearance.
- Set the material top to 0.
- Enter -1 for the Cutting Depth.
- Check the Enable button under Automatic Roughing.
- Enter .25 for the Max Depth Each Cut.

We will basically be doing 4 roughing passes here. Now click OK.

STEP 10

With the toolpath selected, go to the Machine menu on the CAM side and choose AUTO. This will automatically create the roughing toolpath on the screen as well as your G-Code program for it.

This is the procedure for creating spiral pocketing toolpath. The Finishing of this pocket would be basically the same type of procedure, only making the finish selection options in the Spiral Pocket dialog. Here are some tips for completing the program for finishing starting from where you left off:

- A. Perform a tool change in the program by going to the Tool menu in the CAM, selecting Change Tool and following the script to insert the tool change information into the code.
- B. Select the roughing toolpath and blanking it out or saving it on its own layer and then blanking it out.
- C. Deselecting all geometry and going back to the spiral pocket operation through the Other Menu.
- D. Selecting the Finishing operation and making the entries needed for finishing in the Spiral Pocket dialog box.
- E. Selecting the profiles the same way, outside first and then any islands and right-clicking to generate the finish toolpath.
- F. Setting the depths properly in the Tool Depth Settings dialog in the CAM for finishing and using the Auto Cut operation for generating the finishing g-code.
- G. Going to the Macro menu in the CAM, selecting Program and then END.

Let's take a look at how you would generate **pocketing** toolpath for multiple shapes with islands.

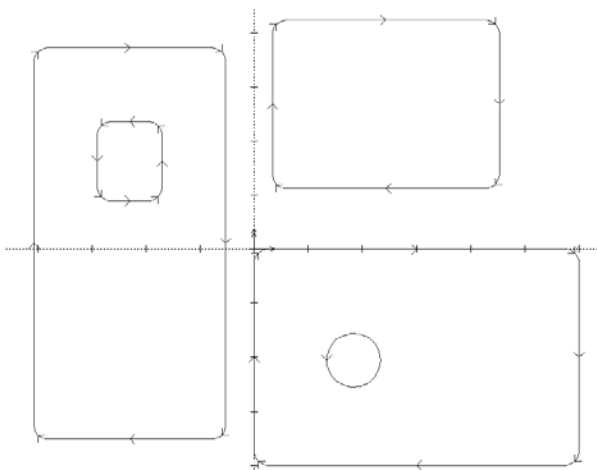
As in earlier pocketing lessons, you would start out by:

1. Selecting the function.
2. Enter in the tool diameter of the tool that you will be cutting with.
3. Enter the step over, which is half the distance of the size tool being used for compensation.
4. Enter a Stock Distance as needed as this will be leaving material around the walls for a finish pass.

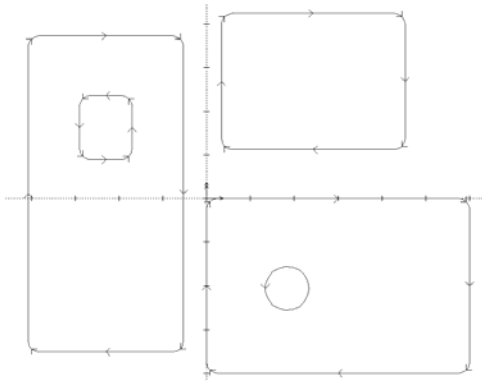
The additions made now offer:

1. The ability for you to add a finishing tool diameter as needed.
2. The options for connecting multiple pockets in one program.
3. Connecting multiple pocket toolpath with dashed lines to represent rapid moves.
4. The ability to mark the start or not.
5. Or, a completely separate finishing function. After roughing you can return to the spiral pocketing function and select the Finishing option for finishing off around the walls of the pocket.
6. Or, roughing and finishing together.

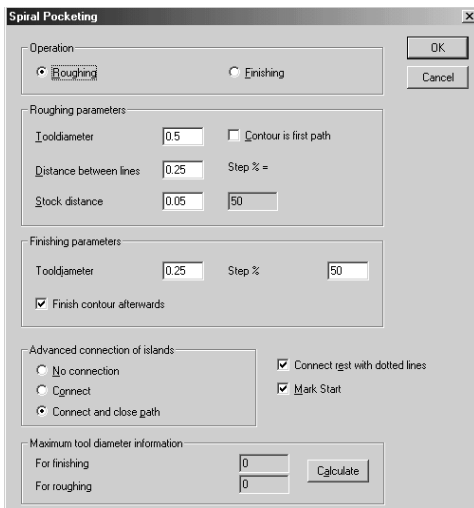
Remember, if you want to **climb-mill** the separate finish toolpath, you would NOT enter a tool diameter in the FINISH section, use the select all function and then REVERSE the direction of the selection arrows by using the REVERSE function in the SPECIAL menu under SELECTION before going back to the spiral pocket function for the finish. This is because the new spiral pocketing function does not have an option for climb milling that is built-in. Here is an example:



You can see that all of the geometry has been selected correctly in these images

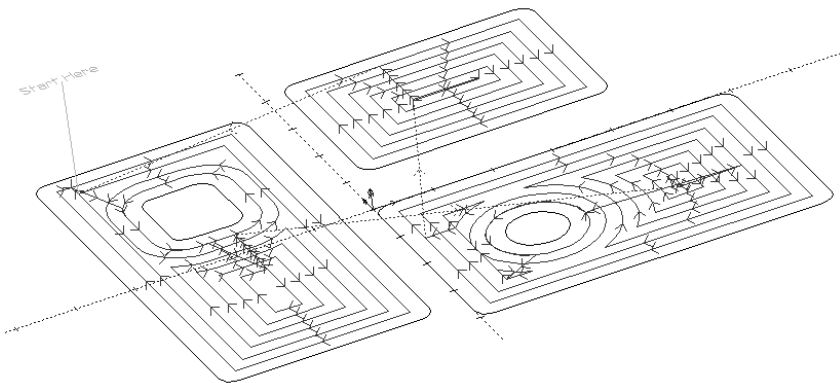


By selecting the spiral pocket function we bring up the dialog below.

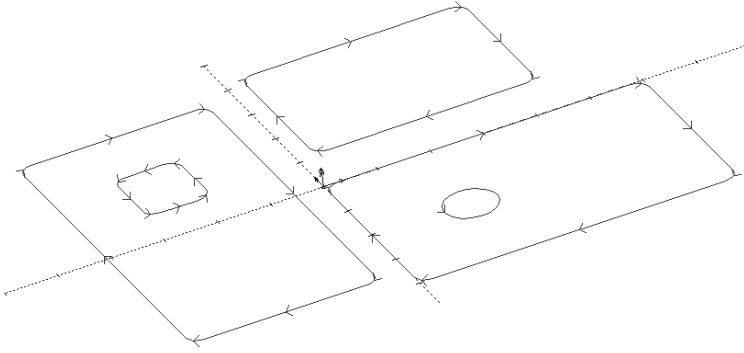


We have selected ROUGHING as the operation option as we want to rough out the pocket first. We are using a .5-inch diameter tool and a .25 step over (half the distance of the size of the tool). We have also left some stock distance. Now, because we know that we will be using a .25-inch tool for finishing the part, we then entered .25 for the tool diameter. This gives us a step over of 50%. In the example we have multiple pockets with islands, so we selected the Connect and Close Path option. We wanted all of the connecting lines to be dashed so we selected that option as well.

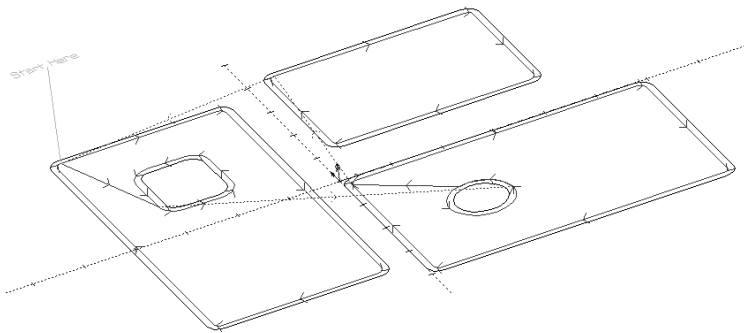
At last, because we are working with pre-selected geometry, by selecting OK the complete toolpath was generated. See below.



Now if we take the same selected geometry as seen below, this time selecting the FINISH option with the same tool diameter, we can just do a finish toolpath.



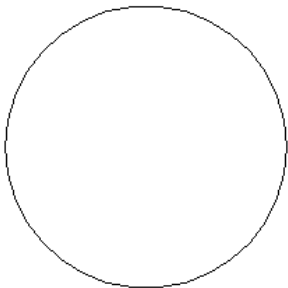
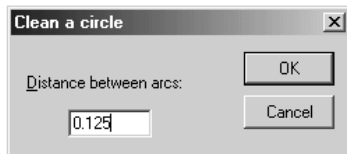
The complete finish toolpath:



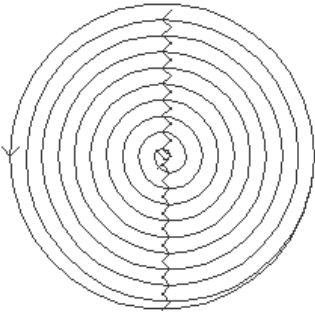
The software finished the walls of the outside shape as well as each of the islands in one function. The machining of this part in the NC CAM has not been changed. Only the spiral pocket dialog has been enhanced.

We recommend that you practice using this new spiral pocketing function until you are fast and efficient with it.

CLEAN CIRCLE: This function allows you to clean out the material inside of a circle. You will be able to generate a spiral toolpath to do this. See the example below.

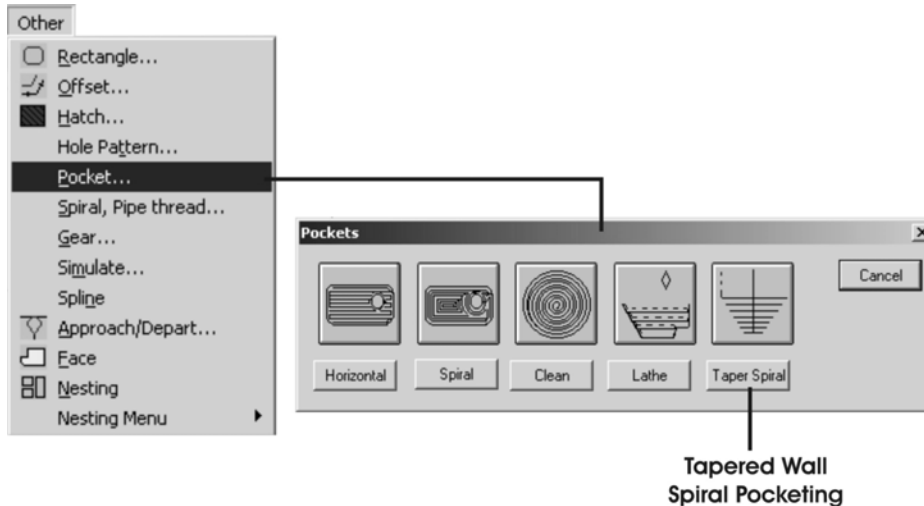


By selecting this option, you will bring up the Clean a Circle dialog as you can see above. The DISTANCE BETWEEN ARCS refers to the step over of the tool diameter that you would be using. This value will be HALF the size of the tool diameter being used to cut this toolpath. Once you have entered in the value, select OK and click on the circle to generate the toolpath.

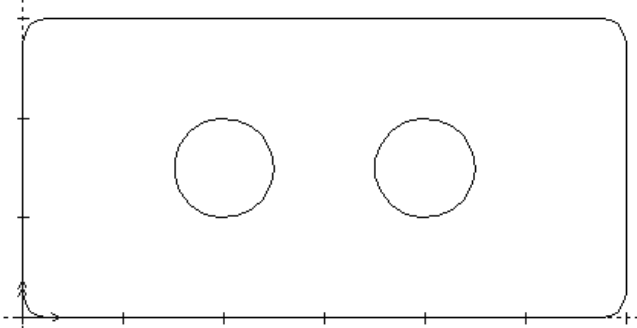


You can also use the spiral pocketing toolpath function to do the same thing. To do this you would select the circle as the outside shape and then click your right mouse button to create the toolpath. It is important to remember that the Clean Circle function takes the original ARC geometry and uses it as toolpath as well. Therefore, if you do NOT want to use the outside circle geometry, you would use the spiral pocketing function to clean out the circle. OR, offset the circle and use the clean circle function for that. However, if you do this you still need to offset HALF the diameter of your cutting tool.

TAPER WALL POCKET: Version 21 offers a tapered wall spiral **Pocket** wizard allowing you to create toolpath with tapered angles for outer walls as well as variable Z-Level tapered angles for inside islands. This function is located in the OTHER menu under POCKET.

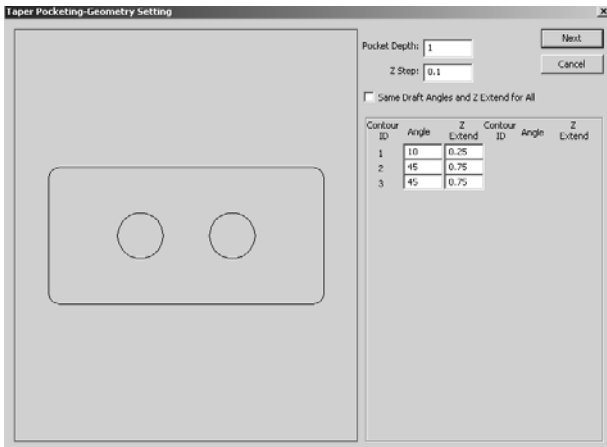


Here is an example of some basic pocket geometry including an outside contour with two inside islands.



By selecting this function you will be prompted to select the outer contour chain and then any islands that you may have inside.

Once you have made the selections you **MUST** Right-click your mouse to open the Taper Pocketing-Geometry Setting box.



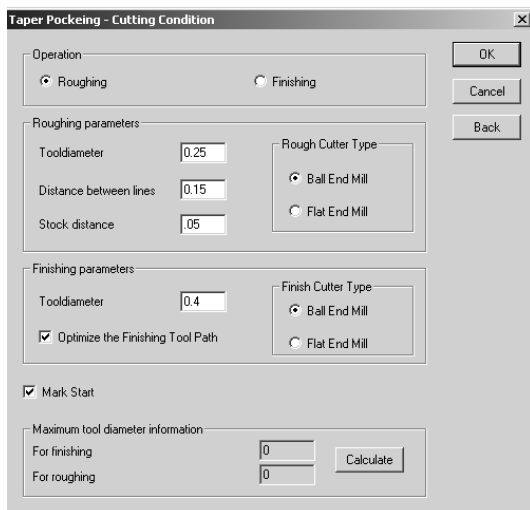
This new box is going to ask you for the following information:

- The pocket depth. You DO NOT need to type in a – symbol.
- The Z-Step for your toolpath that you will base off of the tool diameter that you will be using for roughing or finishing. You DO NOT need to type in a – symbol.
- An option you can check if the outside taper is going to be the same as any island tapers. If all tapered angles are the same you will want to select this option. If the tapers for outside and inside geometry are going to be different you can indicate the angle of the taper as well as where you want the taper to begin under the Z Extend section.

Contour ID	Angle	Z Extend	Contour ID	Angle	Z Extend
1	10	0.25			
2	45	0.75			
3	45	0.75			

Each piece of geometry is assigned a number in the order that you selected it. When you click your mouse in the Angle box next to its number, the corresponding geometry will highlight in the preview window so that you can see which chain or entity you are assigning tapers to. The Contour ID section will list out all geometry selected in the selection sequence.

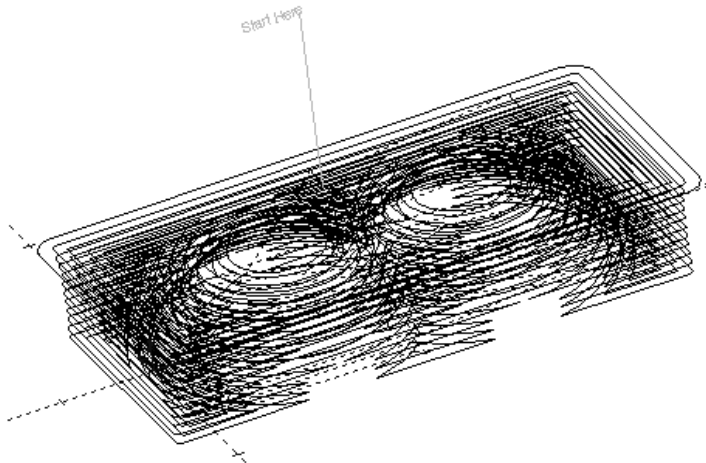
Remember, you need to select the outside contour first. When you have made your selections and entries click the NEXT button.



This new Cutting Conditions box will ask you for the pertinent information necessary to generate the actual toolpath that you will be machining.

- Roughing or finishing operation selections.
- Tool Diameter and option for Ball End Mill or Flat tool.
- Your step over (Distance between lines).
- Any stock distance you want to add, leaving stock for a finish pass.
- An option for optimizing finishing toolpath.
- A "Mark Start" option so that the software will add a line showing you where the start of the toolpath is. (recommended)
- A calculator for calculating the maximum tool diameter for both roughing and finishing as an aid to you.

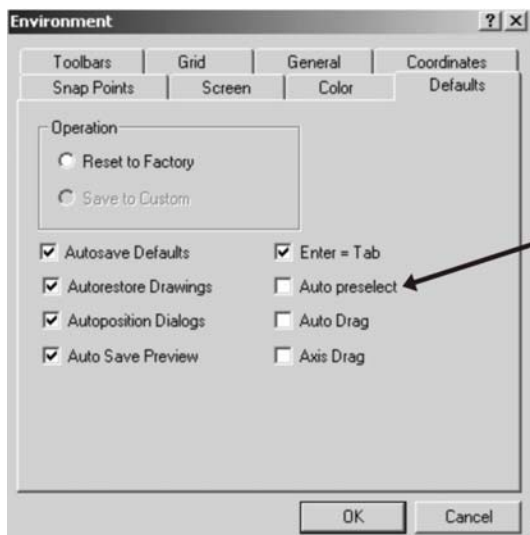
When you are finished entering in your information you will want to click the OK button. There is also a BACK button if you need to make a change to the taper information in the earlier box.



From here you would open the CAM, turn on the 3D button for 3D machining on the CAM side, set your depths and machine the part using the Auto-Cut option.

TAPERED WALL POCKET LESSON:

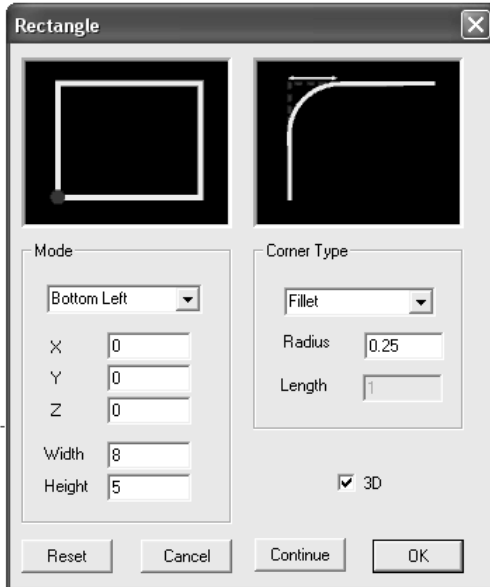
NOTE: BEFORE DOING THIS LESSON, PLEASE TURN THE "**AUTO-Preselect**" option OFF. Do this by selecting the FILE menu and then ENVIRONMENT. Choose the DEFUALTS tab and un-check this option. Thank you.



By un-checking the Auto Preselect option, the geometry You draw will NOT appear selected.

STEP 1

Start this lesson with a NEW (blank) drawing screen. Go directly to the OTHER menu and select RECTANGLE which is the first item of the OTHER menu.



Make the following selections and entries:

- Select BOTTOM LEFT as the reference point from the pull-down under Mode.
- Leave X, Y and Z all set at 0.
- Enter 8 for the WIDTH (this is along the X axis), enter 5 for the HEIGHT (this is along the Y axis).
- Select Fillet from the Corner Type pull-down. Enter .25 as the radius.

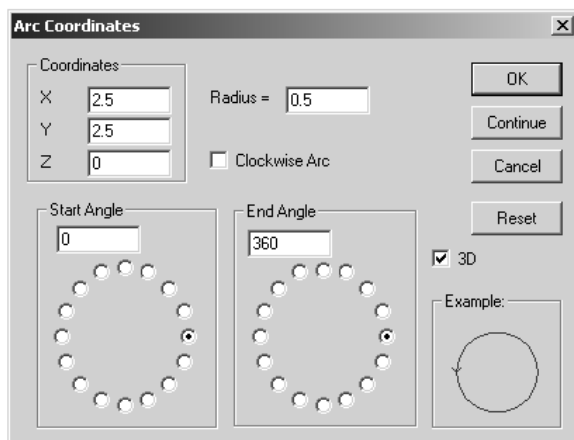
Click OK to exit the rectangle box and draw this geometry on the screen.

Now click on the VIEW ALL icon which is located along the TOP toolbar of the screen. This is the icon that features a magnifying glass with the letter “A” in it.



STEP 2

Now we are going to draw 2 circles that we will be using as our islands. Go to the ARC menu and select COORDINATE CENTER.



Do the following:

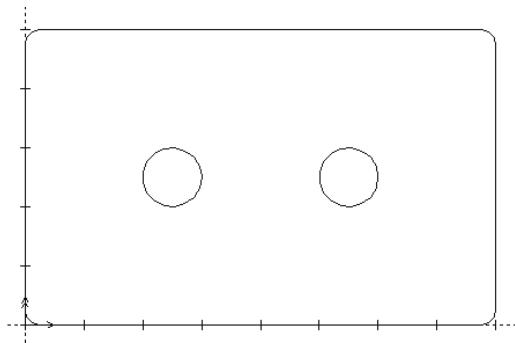
- Enter 2.5 for X.
- Enter 2.5 for Y.
- Z should be 0.
- Enter .5 for the Radius.
- Enter a Start Angle of 0.
- Enter an End Angle of 360.

Click CONTINUE.

- Enter 5.5 for X.
- Leave Y at 2.5.
- Z should still be 0.
- Leave the Radius at .5.
- The Start and End Angles should be left the same.

Click OK.

The result:

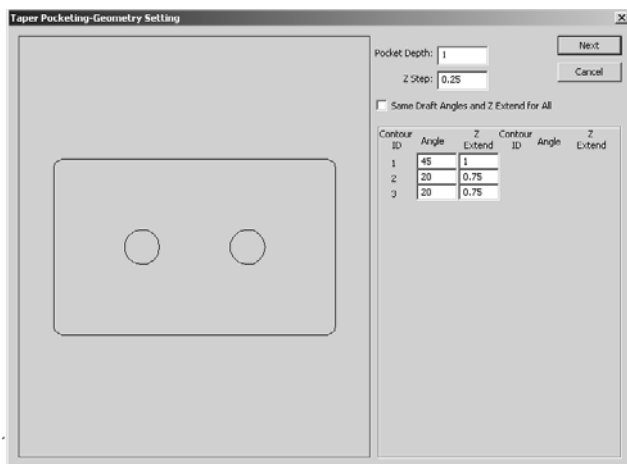


STEP 3

Go to the OTHER menu and select POCKET. The Pocket options box will appear offering 5 different styles of pocket toolpath. Select **TAPER SPIRAL** by clicking on the icon/button.

You will be immediately prompted by the software to select the pocket contour and any islands. Click OK to close this box and then click on the **OUTSIDE** contour **FIRST**. Simply place your cursor anywhere on the outside geometry and left-click your mouse. Then click on both circles as they are the islands.

When you have done this, **RIGHT-CLICK** your mouse to open the Tapered Pocketing box.



This new box will allow you to control the following:

- How deep the pocket will be entirely.
- The size of each toolpath step going down toward the bottom.
- The tapered angle of the outside walls as well as the depth that the taper will begin and/or end which is the “Z Extend” option.
- The tapered angle of each island that you may have and at what depth the taper will begin and/or end which is the “Z Extend” option.

You can see in the image above that the geometry that you have on the CAD screen will be shown. The Angle and Z Extend options in this box will reflect the sequence in which you selected them originally. In other words, the outside contour will always be listed as “1” in the angle/z extend section and so on.

NOTE: You do NOT need to put a minus symbol (-) next to the POCKET DEPTH value nor do you need to do this for the Z STEP value. If the draft/tapered angles are going to be the same for the outside contour and for any islands, check this option.

☐ Same Draft Angles and Z Extend for All

STEP 4

Make the following entries:

- The POCKET DEPTH is 1.
- The Z STEP is .25.
- Under Contour ID, the 1st angle will be 45, the 2nd angle will be 20 and the 3rd angle will be 20 as well.

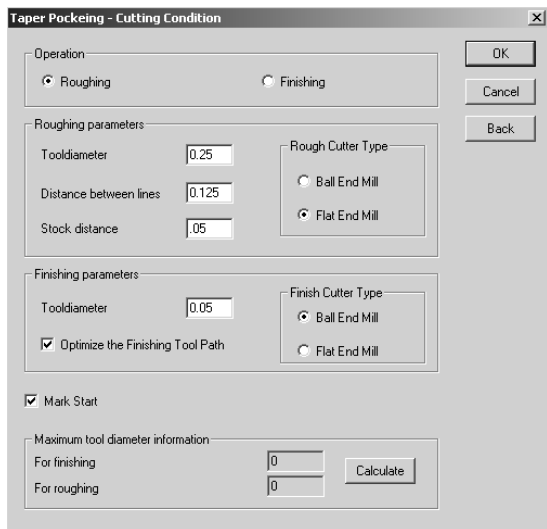
Contour ID	Angle	Z Extend	Contour ID	Angle	Z Extend
1	45	1			
2	20	0.75			
3	20	0.75			

- The 1st Z Extend will be 1. This will apply a 45 degree taper for the full depth of the part.
- The 2nd Z Extend will be .75. This means that we will pocket down .75 with a taper and then the rest of the island wall will be straight. A negative value would reverse this.
- The 3rd Z Extend will also be .75. Once again, this means that we will pocket down .75 with a taper and then the rest of the island wall will be straight. A negative value would reverse this.

Now click NEXT.

STEP 5

Now you will have the Taper Pocketing box on your screen. First off, select the **ROUGHING** option under “Operation” as we will be roughing out this pocket first. Under ROUGH CUTTER TYPE, select Flat End Mill.

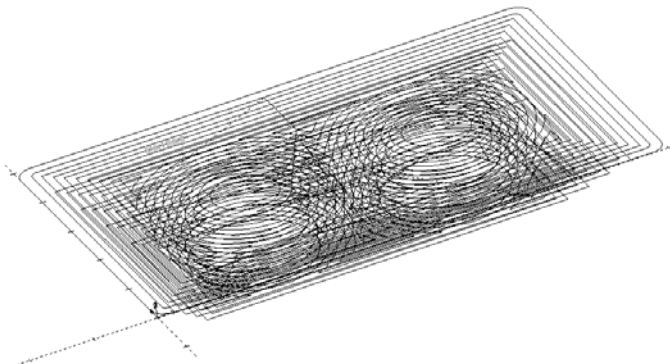


Under Roughing Parameters enter the following:

- Enter .25 for the Tool Diameter.
- Enter .125 for the Distance Between Lines as this is our tool step-over. This compensates for the tool.
- Enter .05 for the Stock Distance as we want to leave some stock around the walls for our finish pass.

Now click the OK button to generate the toolpath for this program. Now go to the 3D menu and **Turn 3D ON**.

The result:



STEP 6

Go to the Special/NC-CAM menu and choose INSERT NC. The Insert NC Object box will appear offering available post processor configurations. Click on the post called, "Fanuc 6M" and then click OK to open the CAM side of the software. In BobCAD-CAM software, the CAD is always on the left and the CAM will always open on the right side.

On the CAM side, you will see a tool/icon bar that runs down the wall. Locate the 3D button and click on it so that this button is selected. We're doing this as this is now designated as 3D toolpath in the software.

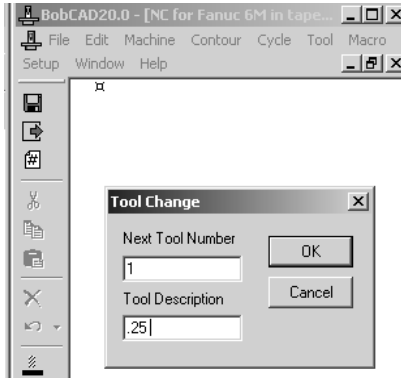


OK, let's take a look at this toolpath on your screen. Notice the green vertical line that also says "start here." We now need to perform a chain selection starting at the top of this green line. Do not click on "Start Here." You will actually be clicking on the green line itself.

Do this now by going to the EDIT menu, then Select Entities and choose CHAIN. Now place your cursor right on that green vertical starting line and click on it **ONCE**. Now move your cursor towards the BOTTOM of that line and you will see that the directional arrow on the line itself will be pointing downward. Once you have that directional arrow, at the midpoint of the line pointing down, click your LEFT mouse button ONE more time and then hit the F3 key on your keyboard to select the entire toolpath properly. You see, the toolpath is also a chain and this is why we have chosen to use the chain selection feature.

STEP 7

Now that the toolpath is selected and ready to be machined, double check that you have clicked on the 3D button on the **CAM** side wall. Good, now go to the TOOL menu on the CAM side and choose TOOL CHANGE.



In the Next Tool Number space, enter 1. For the Tool Description enter .25 and then click OK. This will add the codes to NC Editor. Now select the **U/D** button from the CAM wall.



This opened the TOOL DEPTH SETTINGS box. All we are concerned with for this operation is the RAPID PLANE and Material Top. Enter .25 for the RAPID PLANE, which is our clearance. Enter 0 for the Material Top and leave the cutting depth at 0 as well as we have already entered this in the wizard. Make sure the Enable check-box for Automatic Roughing is not checked and click OK.

Now go directly to the MACHINE menu on the CAM side and select AUTO. This will produce the complete G-Code program for the roughing.

We're not done. Move your cursor to the CAD screen where you see the words, "Start Here" and while holding down your CONTROL key (Ctrl), click on "Start Here." Now the toolpath and "Start Here" will be selected in red (unless you have clicked your mouse anywhere else). With these entities and text selected, go to the CAD side CHANGE menu and choose ATTRIBUTES.

STEP 8

Select the GENERAL tab and under "Layer" type in Roughing Toolpath and click OK. Now, before clicking anywhere else, go to the CHANGE menu again and select BLANK to blank this layer out and off of the drawing screen.

You should now have the original geometry on the screen.

STEP 9

Now go back to the OTHER menu and select POCKET. Click on the TAPER SPIRAL button. You will be prompted to select the OUTSIDE contour first and then the islands. Click OK in this box and then simply click on the outer contour geometry and then click on both circle islands.

When you have completed your selections, RIGHT-CLICK your mouse to open the next box.

In the new box, change the **Z STEP** value to **.01**. Basically, we are going to use a .25 ball cutter with a small step down for finishing.

The Pocket Depth is still 1 and each of the taper angles and z extents are going to be the same.

Click NEXT.

Now you need to select the **FINISHING** option at the top of this box. You do not need to be concerned about the roughing section now. Locate the Finishing Parameters section and enter a tool diameter of .25 and select the **BALL END MILL** cutter type.

Taper Pocketing - Cutting Condition

Operation:
☐ Roughing ☒ Finishing

Roughing parameters:
Tool diameter: 0.25
Distance between lines: 0.125
Stock distance: 0.05
Rough Cutter Type:
☒ Ball End Mill
☐ Flat End Mill

Finishing parameters:
Tool diameter: 0.25
☒ Optimize the Finishing Tool Path
Finish Cutter Type:
☒ Ball End Mill
☐ Flat End Mill

☒ Mark Start

Maximum tool diameter information:
For finishing: 0
For roughing: 0
Calculate

Buttons: OK, Cancel, Back

We used a flat tool to rough out the material and are switching to the same sized tool except this time we will be using ball cutter with a much smaller Z-Step to finish this part as just mentioned. Make sure the Mark Start option is checked and then go ahead and click OK to create the toolpath on the screen.

STEP 10

Now we want to check a few things. First, make sure that the 3D button on the CAM wall is selected. Second, click on the U/D button and ensure that your RAPID PLANE is set at .25 still and that Material Top is 0 as well as the Cutting Depth. We are not concerned with these as we already told the software what our depth is and we have already told the software what our Z-Step is. Click OK to exit the Tool Depth Settings Box.



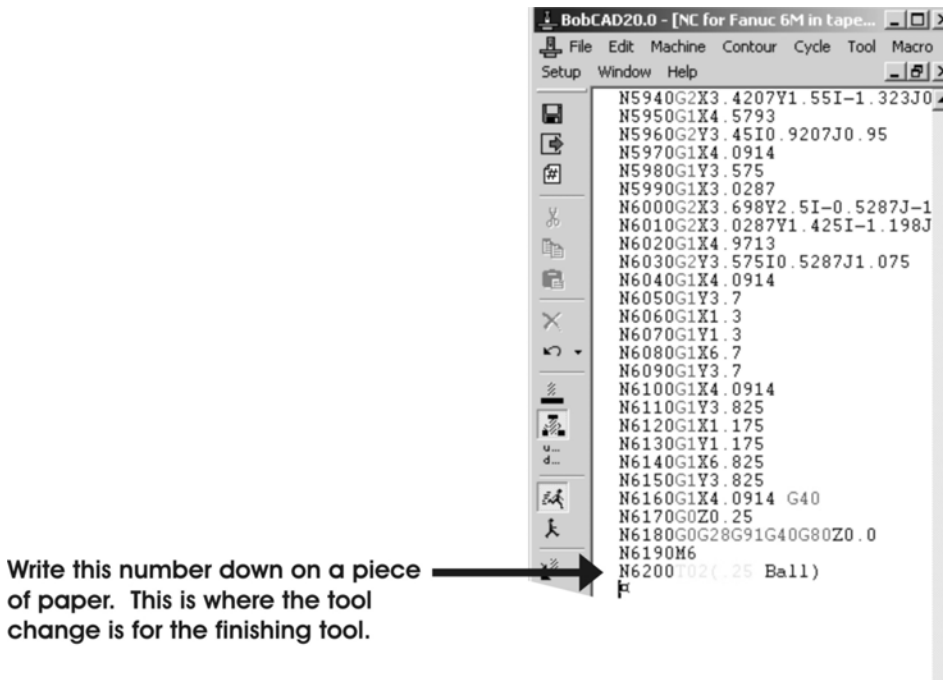
Now go to the Tool menu on the CAM side and select TOOL CHANGE. Now enter 2 for the NEXT TOOL NUMBER and .25 BALL for the Description. Then click OK to insert the tool change into the code.

STEP 11

OK. Now we are ready to select this toolpath and proceed with the finish. Do this now by going to the EDIT menu, then Select Entities and choose CHAIN. Now place your cursor right on that green vertical starting line and click on it **ONCE**. Now move your cursor towards the BOTTOM of that line and you will see that the directional arrow on the line itself will be pointing downward.

Once you have that directional arrow, at the midpoint of the line pointing down, click your LEFT mouse button ONE more time and then hit the F3 key on your keyboard to select the entire toolpath properly. You see, the toolpath is also a chain and this is why we have chosen to use the chain selection feature.

IMPORTANT: At this time look to see what the number is for the last line of G-Code and note this on a piece of paper. This is important as we will be using this information when setting up the tools for solid simulation at the end of this program.



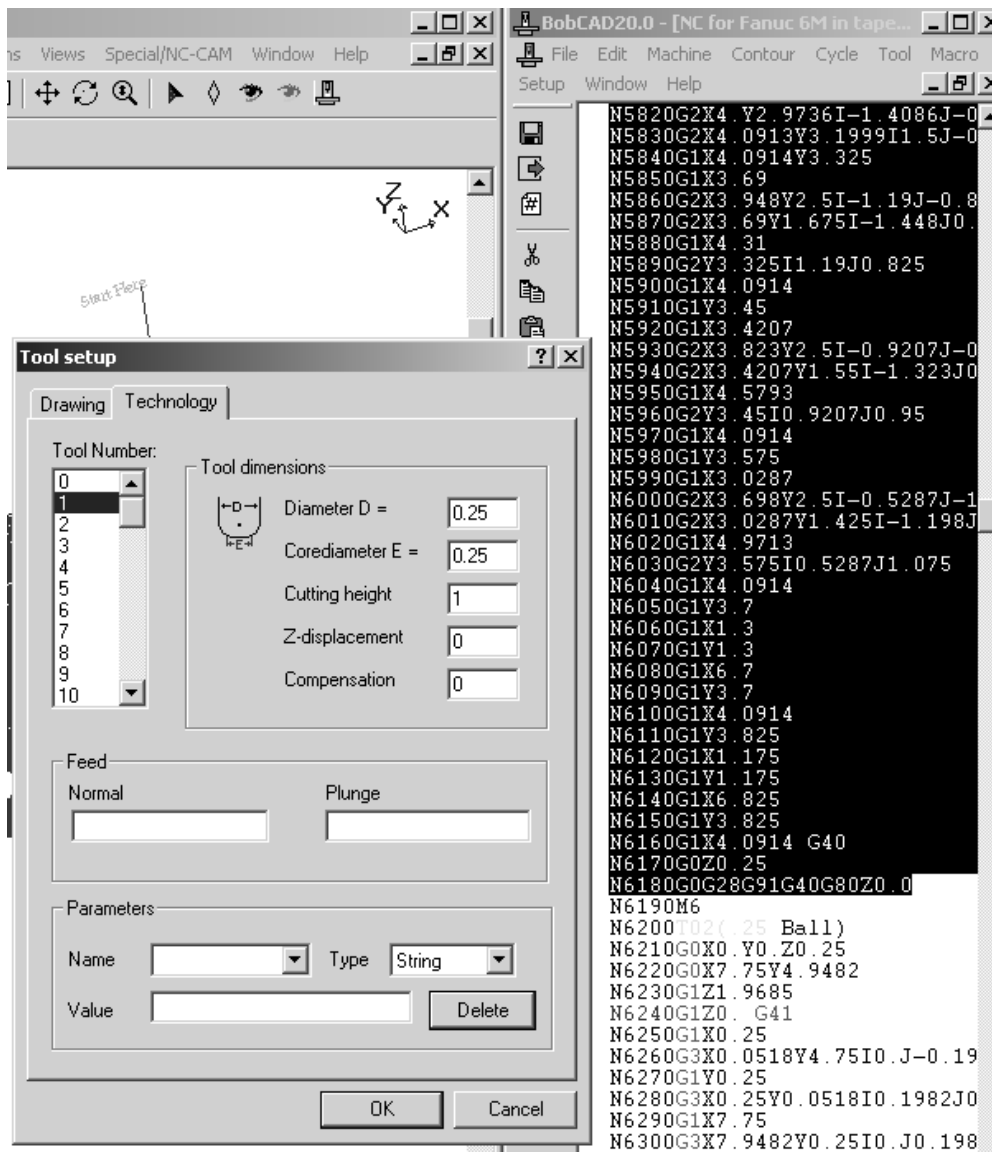
OK. With the toolpath selected, go ahead over to the CAM side and select the MACHINE menu at the top and then click on AUTO. This will generate the g-code program for the finish. Good Job!

STEP 12

Now let's go ahead and setup the tools for the solid simulation. To do this you need to scroll up to the first line of G-Code. Now click your mouse just on the left side of the first letter N. Hold down your SHIFT key and scroll down through the code (while holding down your shift key) to the number that you wrote down on that piece of paper. You can use the down arrow key on your key-pad, your mouse or by clicking and dragging the scroll bar on the right side wall of the CAM-G-Code Editor.

We are selecting all of the code that we used the first tool to create.

When you have reached this line of code with the same number that you wrote down, **STOP** scrolling downward and release your shift key. Go directly to the EDIT menu on the CAM side and select TOOL.



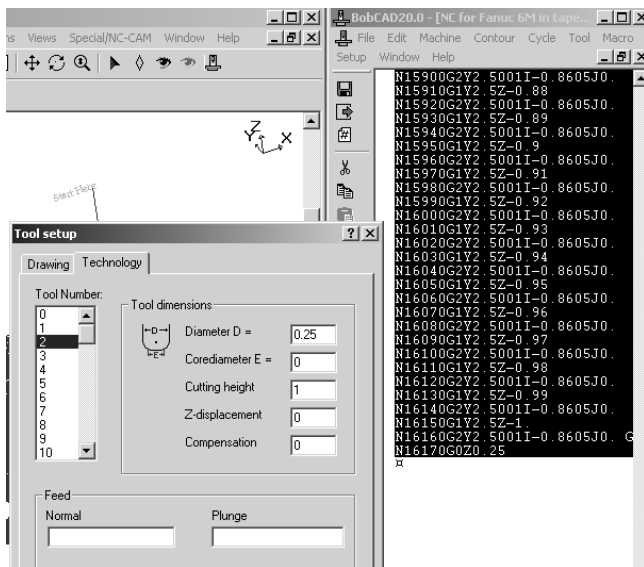
Select the TECHNOLOGY tab from the Tool Setup box. Under, “Tool Number” click on 1. For the DIAMETER enter .25.

Because we used a flat tool for the roughing we need to enter .25 for the CORE DIAMETER. Enter 1 for the CUTTING HEIGHT. Click OK at the bottom.

STEP 13

Now we need to do the same action starting from the next un-selected line of code. First click next to the letter N (which stands for “Number.”) and while holding down your SHIFT key, scroll down to the last line of code. Make sure that you have highlighted the last line of code fully. When finished, go to the CAM EDIT menu and select TOOL again.

Select the TECHNOLOGY tab and this time click on tool 2 under Tool Number.

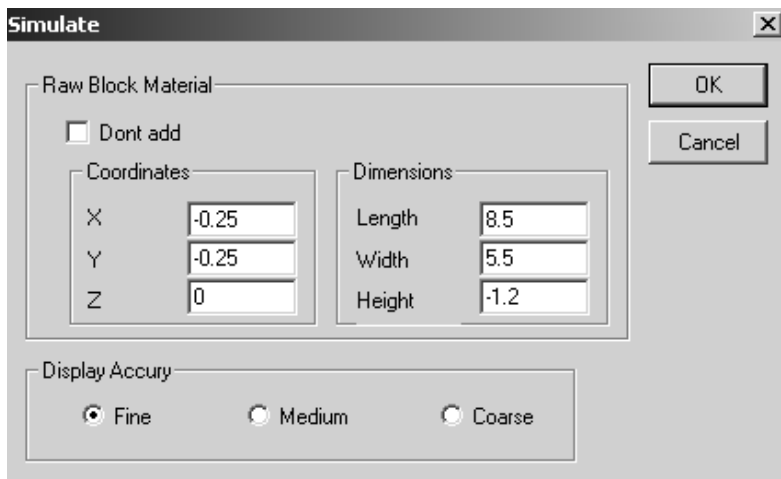


Enter .25 for the Diameter. Enter 0 for the Core Diameter and 1 for the Cutting Height. Now click OK at the bottom.

Alright. Now that the tools for simulation are setup you can go to the CAM EDIT menu and choose SELECT ALL. All of the code will be selected.

STEP 14

Go back to the CAM EDIT menu and select SIMULATE.

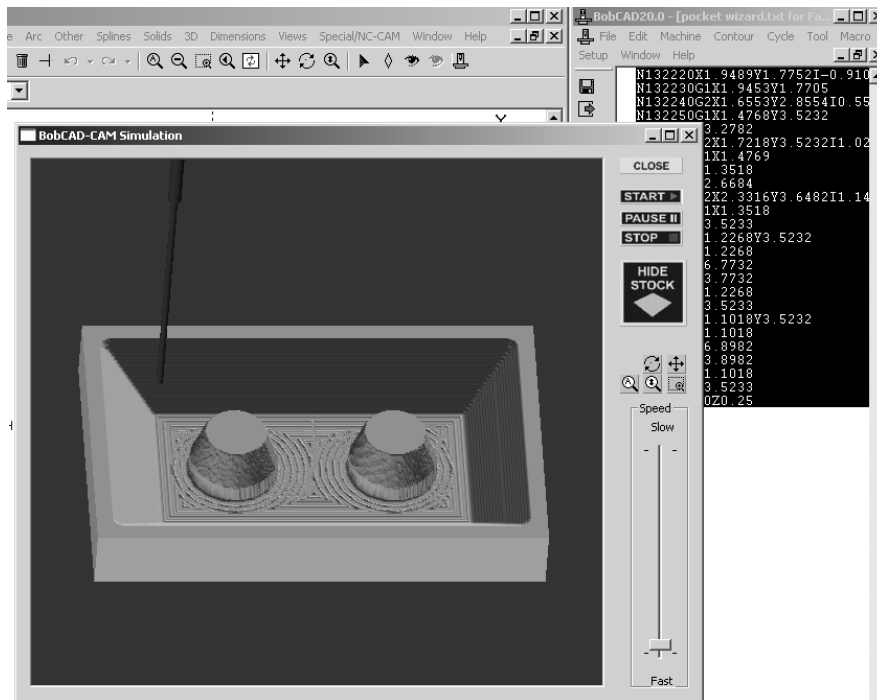


Under COORDINATES enter -.25 for X, -.25 for Y and leave Z at 0.

Under DIMENSIONS enter 8.5 for the Length, 5.5 for the Width and -1.2 for the Height. Check the FINE option for the display accuracy and click OK.

This will begin the simulation on the CAD side of the screen. You will see the roughing cycle first and then you will see the finishing occur.

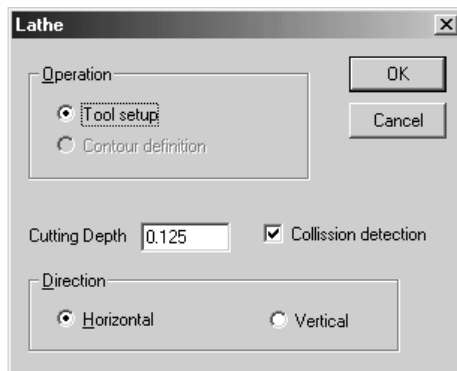
The result after simulation:



The Version 21 simulation function provides you with complete simulation for roughing and finishing with flat end or ball end cutters.

You have completed this lesson.

LATHE: The lathe option in the pocketing box allows you to perform tool setups for merged lathe tools as well as create roughing and finishing toolpath. In order to access lathe functions, lathe mode must be turned on first.

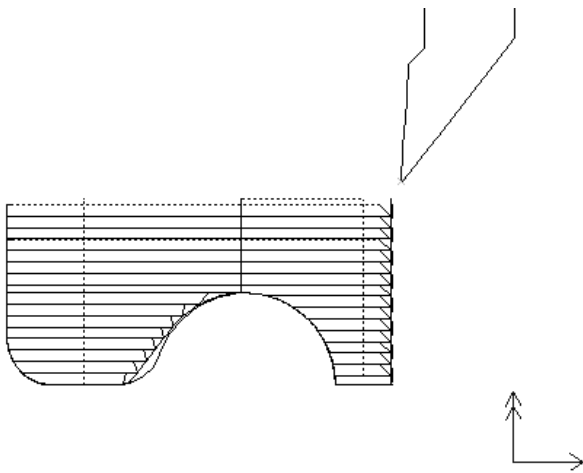


In the above image, you can see that you have a **tool setup option**, the depth of each roughing pass, automatic collision detection and the horizontal and vertical toolpath options. By entering in the values that you need and selecting OK, you will be able to select the OD (outside diameter) or ID (inside diameter) as a chain and generate the complete roughing toolpath.

The **CUTTING DEPTH** will be determined by the size of the cutting tool and the type of material that you will be cutting. This is decided by you and how much material you want to remove at a time. It's that simple when determining the cutting depth. This IS NOT necessarily going to need to be half the size of the cutting tool. The procedure for setting up your lathe tool and generating roughing toolpath:

- The first thing that you will do is turn lathe mode on.

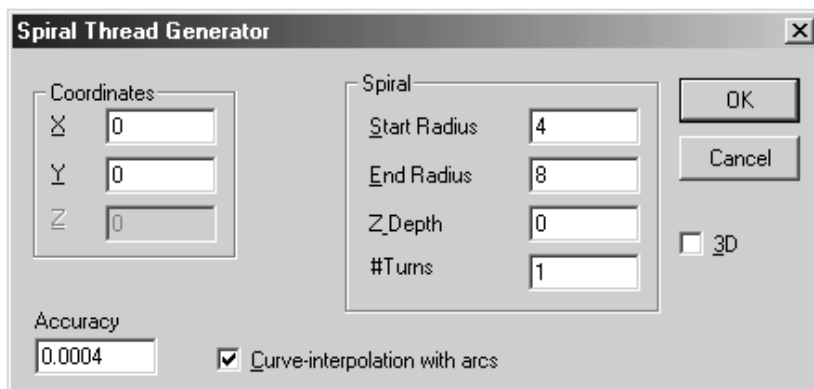
- Once you have the OD created or imported, you will then need to merge in a lathe tool from the tool folder in your BobCAD Version 21 directory.
- You would do this by selecting FILE and then MERGE. You will need to decide on the correct lathe tool for your profile, select it and click OK to bring it into the existing drawing.
- Now you can select the OTHER menu and POCKET to select the lathe option.
- The next thing that we will need to do is perform a tool setup. Do this by entering the cutting depth, set collision detection by checking the box and then select the type of toolpath you will be creating. Then click OK.
- Because the tool is actually made up of lines that make a chain, you can select it as a chain, which is what you must do at this point. Click on a line of the chain, point the direction arrow in either direction, click your mouse button again to set the direction and hit the F3 key on your keyboard.
- Now you will need to select the cutting tip of the tool. You will notice that there is a point at the end of the tool nose. Click on it with your mouse. Now you are ready to go ahead and create the toolpath.
- You would select the OTHER menu again, POCKET and the Lathe option. Double check the values that you had entered and select OK.
- Now you will be in chain selection mode. Simply place your cursor at the start of the profile, click your mouse button and point the directional arrow in the direction of the chain. Click your mouse button again to set the direction and hit the F3 key on your keyboard. This will automatically generate the complete lathe toolpath for the part.



This concludes the pocketing function of the OTHER menu.

SPIRAL, PIPE THREAD: This is the next menu option in the OTHER menu and will allow you to create spiral toolpath with variable attributes.

Let's take a closer look at this function below.

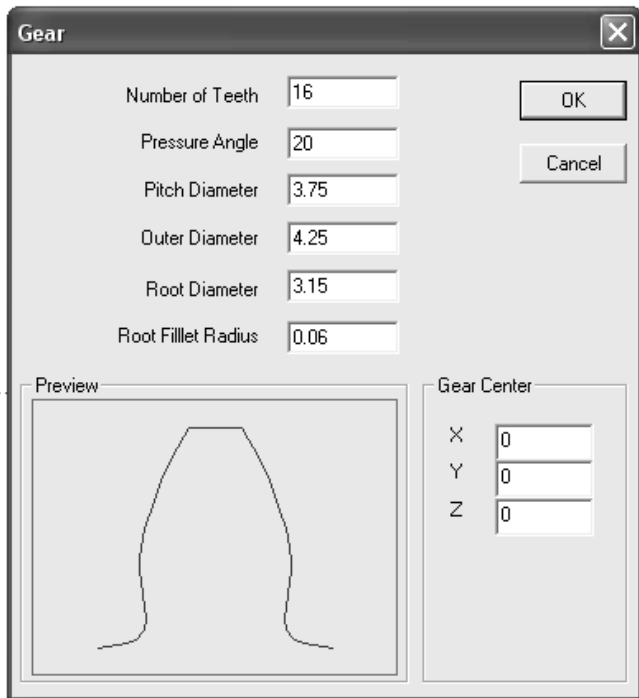


The thread generator will give you the ability to enter X, Y and/or Z coordinates for positioning. The thread is defined by:

- A Start Radius.
- An End Radius.
- A total Z depth, positive or negative.
- A value for the number of turns that you want to make up the thread toolpath.

By selecting the “Curve Interpolation with ARCS” option, you will be creating arc segments for the thread and there will be a lot less overall entities than there would be with line segments. The ACCURACY setting will determine how long or short those line segments would be if you did not use the curve interpolation option.

GEAR: The Involute **Gear** is drawn with small arc segments that can be machined in the NC CAM (G01, G02, G03). Here is the gear box below.

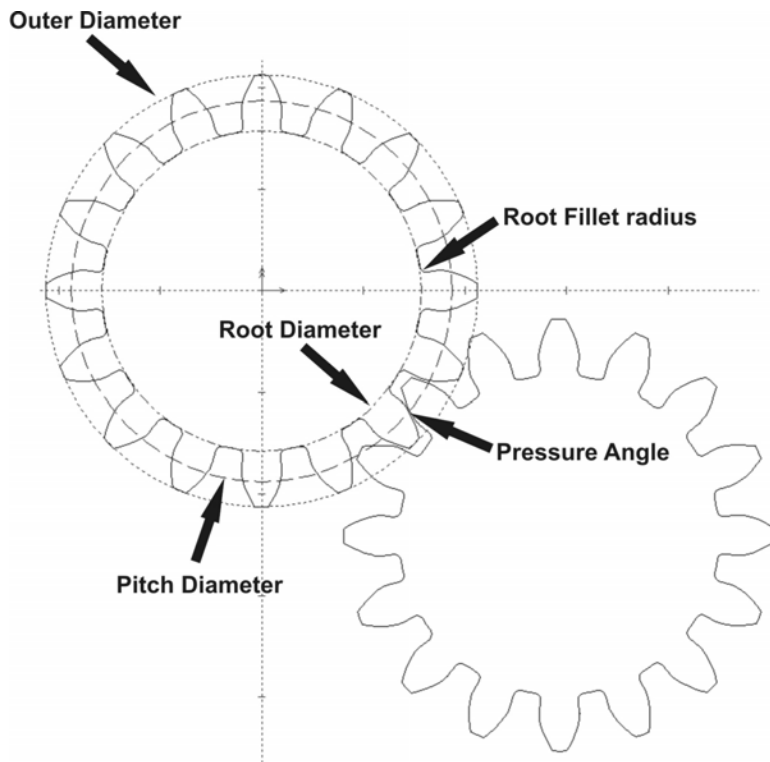


These attributes include:

- **Number of teeth.** This is basically the number of teeth that you need the gear to have.
- **Pressure Angle.** This is the angle between the tooth profile and the radial line at its pitch point. Also described as the angle between the line of action and the line tangent to the pitch circle. More information on this can be attained through your Machinist Handbook.
- **Pitch Diameter.** This is the diameter of the pitch circle. The operating pitch diameter is the pitch diameter at which the gear operates.
- **Outside Diameter.** This is the diameter of the circle that contains the tops of the teeth of external gears.
- **Root Diameter.** This is the diameter of the circle that contains the roots or bottoms of the tooth spaces.
- **Root Fillet Radius.** This is the radius of the concave portion of the tooth profile where it joins the bottom of the tooth space.

These definitions can be seen in the GEARING section of the Machinist/Machinery's Handbook. You simply enter in the values needed to create your gear and click the OK button to draw the complete gear. You can use the X, Y and/or Z values as needed for placing the gear on the CAD screen.

Here it is again in a diagram:



QUICK GEAR LESSON:

STEP 1

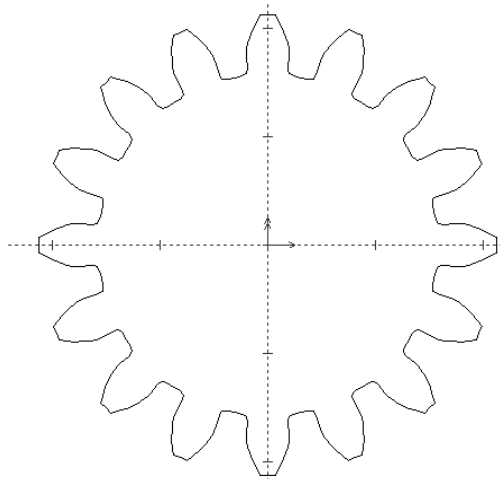
Open the software and select GEAR from the OTHER main menu.

STEP 2

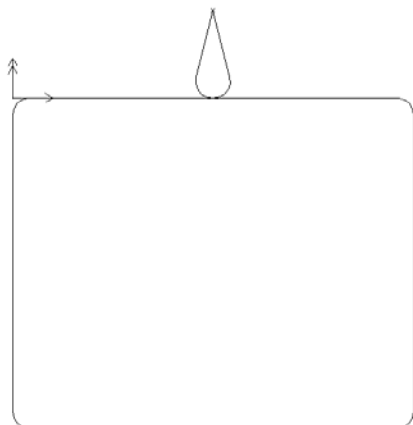
Enter the following:

- 16 for the number of teeth.
- 20 for the Pressure Angle.
- 3.75 for the Pitch Diameter.
- 4.25 for the Outer Diameter.
- 3.15 for the Root Diameter.
- .06 for the Root Fillet Radius.

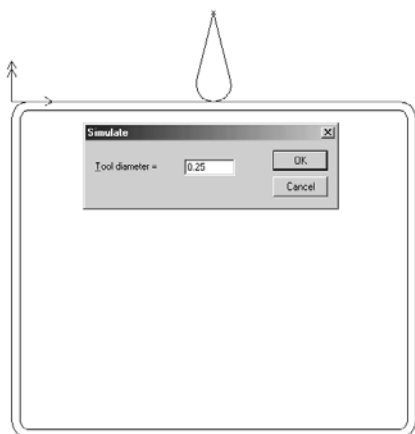
Click OK to see the result.



SIMULATE: This is the next function in the OTHER menu. The **Simulate** functions allows you to apply a tool diameter to an open or closed shape profile as well as pocketing toolpath so that you can see what will happen with the tool. The tools travel will be enlarged o the toolpath showing the result. Let's take a look at this with the example profile toolpath below.

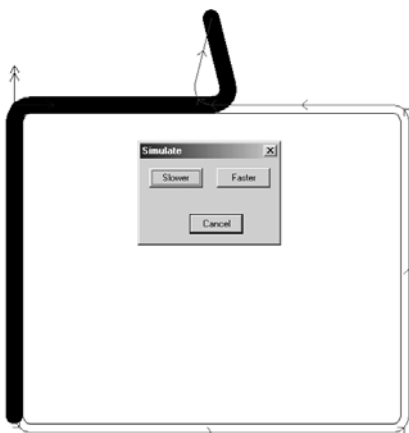


The toolpath and the approach/depart is based off of HALF the Distance of the size of the tool. By selecting the SIMULATE function from the OTHER menu you will be asked for a tool diameter. Enter the ACTUAL tool diameter that you will be using. Then select OK.

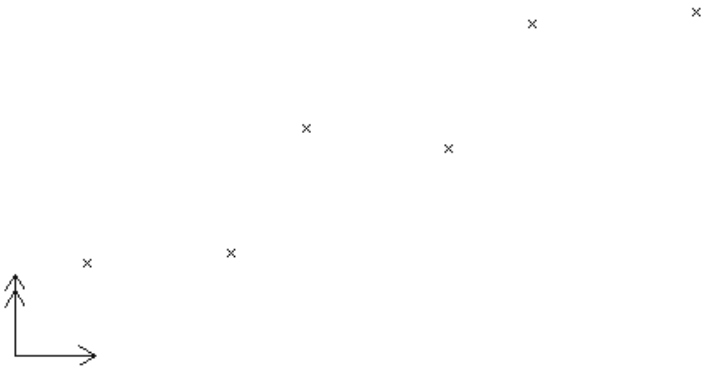


Select the START line of the approach, point your directional arrow in the direction of the chain, click your mouse button one time and hit F3 on your keyboard to select the entire chain and bring up the simulate control box. You can see this in the image below.

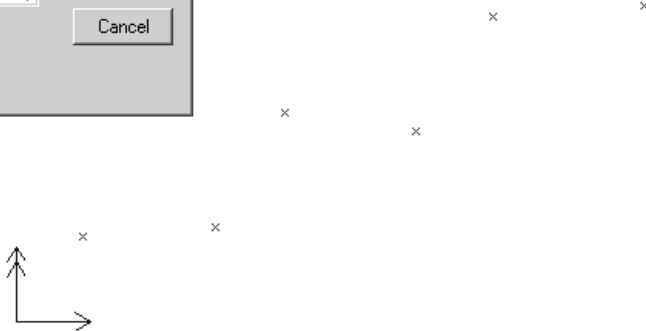
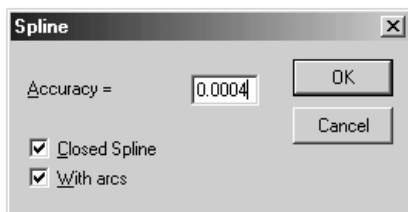
The control box will allow you to speed up and slow down the simulation cycle.



SPLINE: The **Spline** function will allow you to create a chain of arc segments through points on the screen. Let's take a closer look at this below.



You can see in the image above that there are random points on the screen. By selecting the OTHER menu and SPLINE, you can open the spline dialog box. See below.

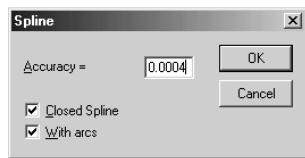


This dialog gives you several options. They are:

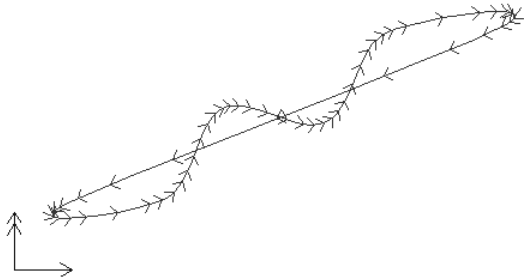
- **ACCURACY:** This is the tolerance; how large or small you want the arc segments to be.
- **CLOSED SPLINE:** This setting allows you create an entire chain of arcs through points that is closed as one chain.
- **WITH ARCS:** this lets you use arcs instead of line segments. If you uncheck this option to create a complete spline chain and then machine it, you will have many more lines of code than you would if you used arc segments to make up the spline.

Let's take a look at the example of a **CLOSED SPLINE**. In the next image you can see that there are a series of points laid out on the drawing screen. You would select all of the points and then the **SPLINE** function from the **OTHER** menu.

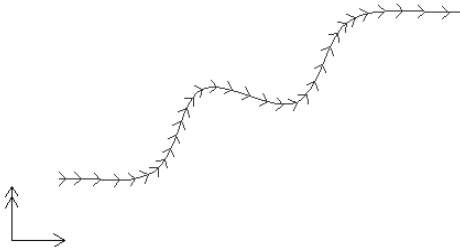
This would bring up the box you see below.



By selecting OK in the dialog example above you will get the result as seen below.

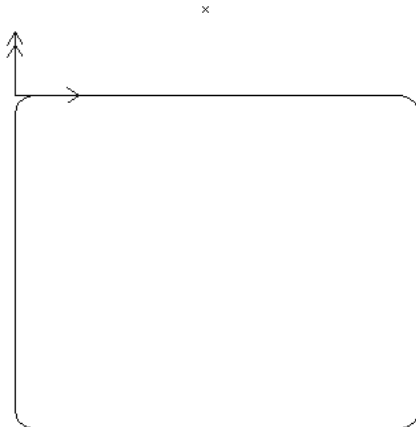


Here is an example below of the same settings and points. However, this time we UNCHECKED the CLOSED SPLINE option.

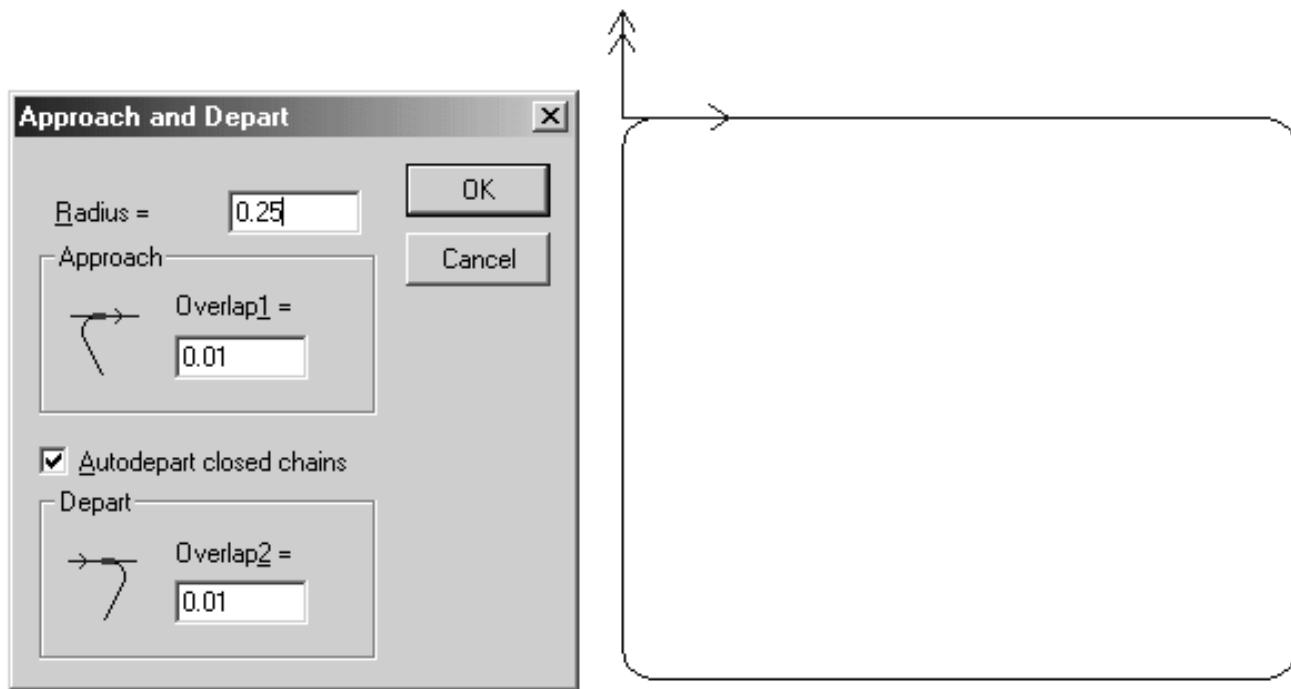


APPROACH / DEPART: This function allows you to create approach and departure toolpath for profiling. Let's take a closer look at this function.

In this example you can see that we have a profile that we want to cut out. To use this function, you must first draw a point off of the profile that will be the start position of the approach and will also be the end position of the departure of the tool. See below.



By selecting the OTHER menu and Approach/Depart we will open the dialog as seen below.



First you will want to enter in the approach and depart radius.

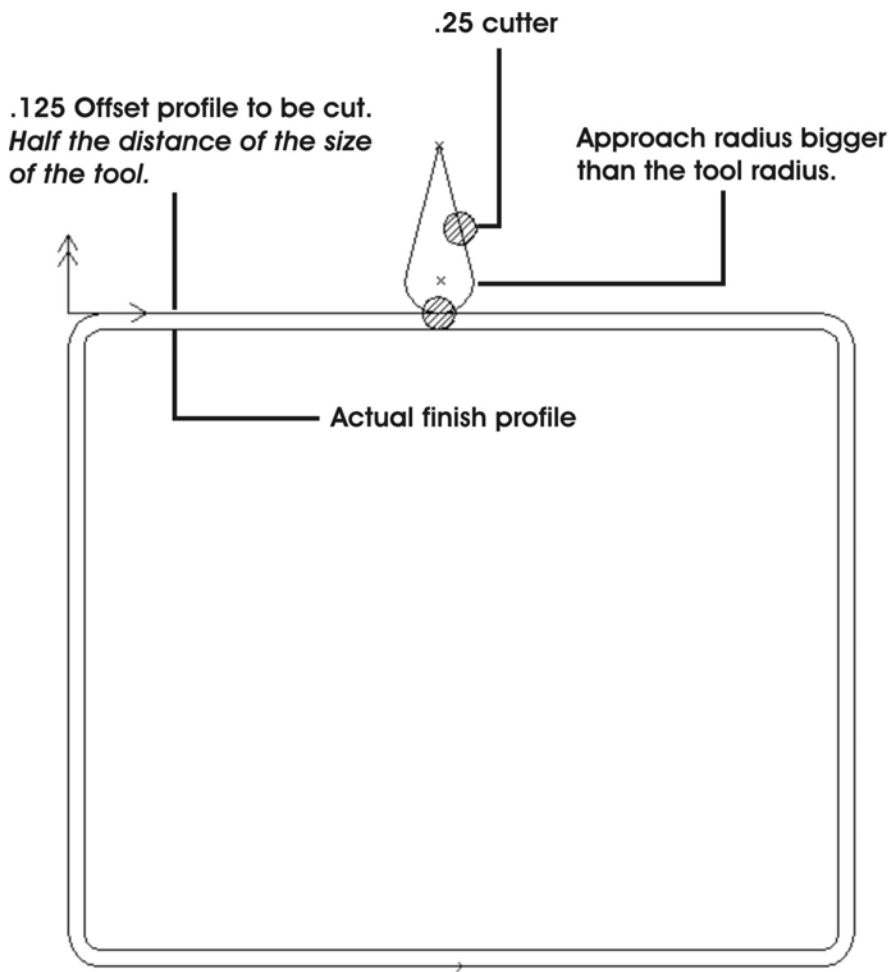
THE APPROACH & DEPARTURE RULE:

Never place a tool into an approach radius that is smaller than the radius of the tool when using cutter compensation at the controller.

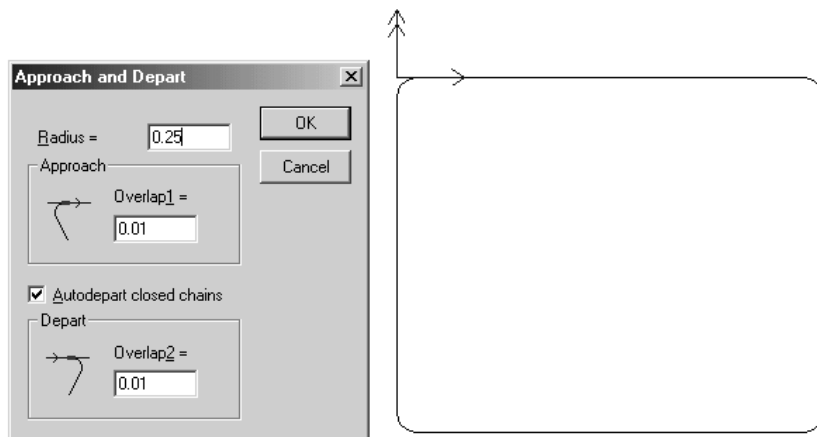
Therefore as an example, if your approach radius is .125 and you have a .375 tool, you cannot use cutter compensation that matches the radius of the tool. This is because the tool radius is .1875, which is larger than the approach radius.

Here it is again. A half-inch tool would need a quarter inch radius if you were using cutter comp at the controller. If you are NOT using cutter compensation at the controller, you may use any radius that is smaller than the size of the tool.

The starting point MUST be at least the same distance off of the profile geometry to be cut as the size of your tool. In other words, if you are using a .5-inch tool and have already created a .25-inch offset for the profile itself (half the distance of the tool because you are not using cutter comp at the control), you must create an approach with a .25-inch radius and a starting point at least .5-inches away from the profile.



Now we will enter an overlap amount to reduce the scallop.



In the example, we entered .01 for the approach overlap. We selected the AUTO PART CLOSED CHAIN option because we want a depart toolpath as well. The overlap amount is .01 for the depart as well.

Now, you would click OK, select the start point first and then the first profile entity. Point your directional arrow in the direction of the chain and click your mouse button again.

FACE: This function allows you to add solid color to geometry on the screen. Let's take a closer look at the images below.

A coordinate plane with a horizontal dashed line and a vertical dashed line intersecting at the origin. A horizontal solid line with arrows at both ends is located below the horizontal dashed line. A slanted solid line with arrows at both ends is located in the upper-left quadrant, parallel to the horizontal dashed line.

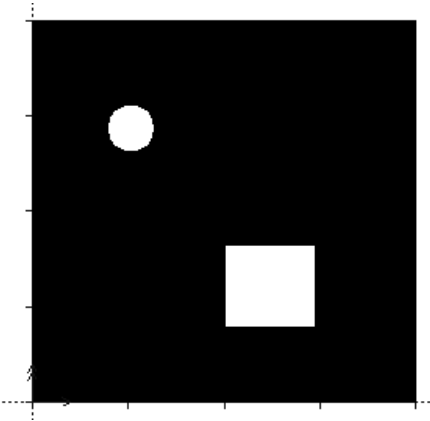
However, in order to face the circle, we will first need to interpolate or break it up into more than 1 entity. Therefore, we must do the following:

Select the CHANGE menu, then BREAK and then DIVIDE. Enter 2 into the Divide dialog and click OK. Now select SINGLE in the selection dialog. Good, now click on the CIRCLE once and then a second time. Now the circle is divided into 2 entities and can be faced.

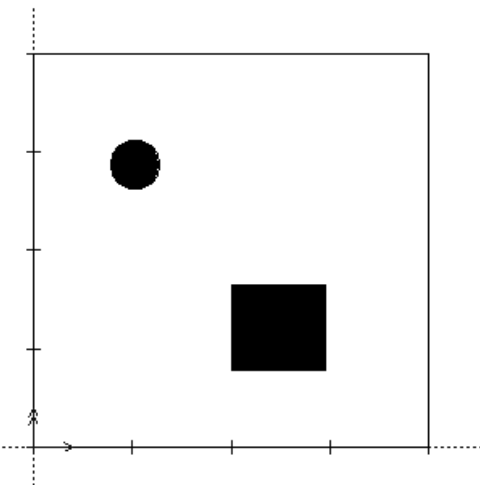
We will start with the outside shape first. HOLD DOWN THE SHIFT KEY and then select the outside square first, then rest the cursor just inside the circle and click your mouse button and at last, click the small square. Now everything will be selected red.

Select the FACE icon and you will face the part.

See below.



To face the circle and the small square you would hold down the shift key on your keyboard and only select the circle and the small square, then the FACE icon. See below.



You can create different color faces by selecting the main drop down color bar and choosing a color before creating the face. You can also change the color of an existing face by selecting the face, selecting CHANGE, then attributes and choosing a face color from the face tab.

CHAPTER 5

SOLIDS & SURFACES

- Section 22. Primitives Defined
- Section 23. The Solids Menu
- Section 24. Primitives
- Section 25. Surface Creation, Projection & Modification
- Section 26. Boolean Operations
- Section 27. Solids Attributes
- Section 28. Basic CAD Design
- Section 29. Fillets & Chamfered Surfaces
- Section 30. Generate Toolpath
- Section 31. Solid Tool Depth Settings
- Section 32. Quick CAM Training Lesson
- Section 33. Offset Surfaces & Solids
- Section 34. Stitching or Un-Stitching Surfaces & Solids
- Section 35. Extract Edges
- Section 36. Remove Holes
- Section 37. Render
- Section 38. Questions & Answers



Section 22.

Primitives Defined.

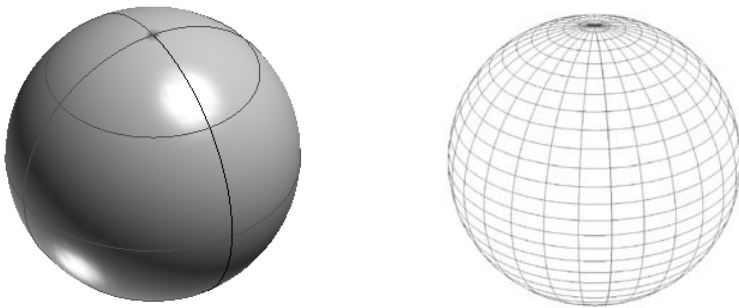
The **Solids** Menu offers all of the functionality regarding working with and creating models in the software as well as other prime functions that you will be using to create surface toolpath and more.

The primitive shapes are:

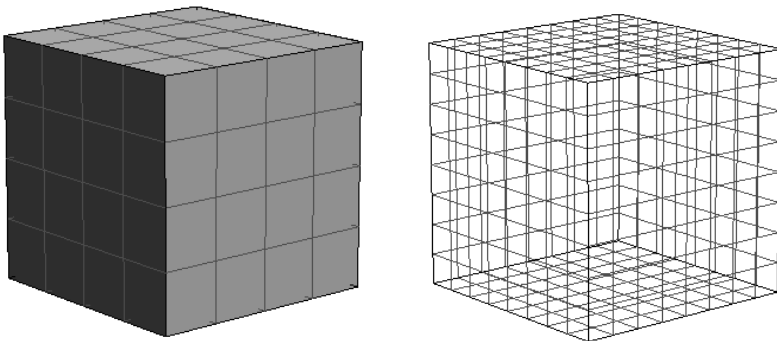
- Sphere
- Cube
- Cone
- Cylinder
- Torus

Here they are below:

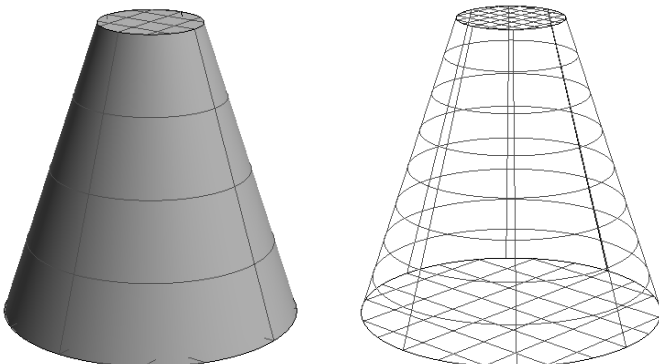
Sphere: The sphere is defined by its radius.



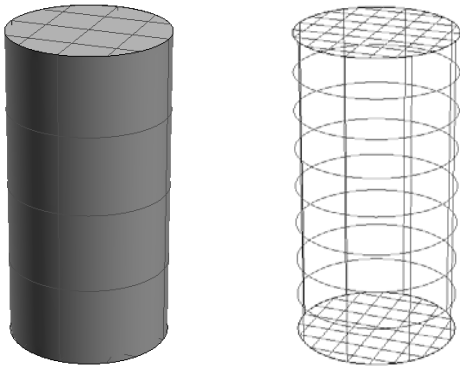
Cube: The cube is defined by its height, width and length.



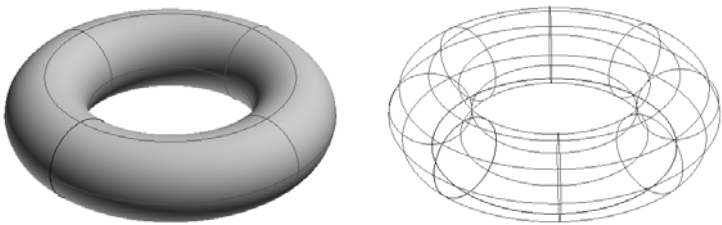
Cone: The cone is defined by its base radius, top radius and height.



Cylinder: The cylinder is defined by its total radius and height.

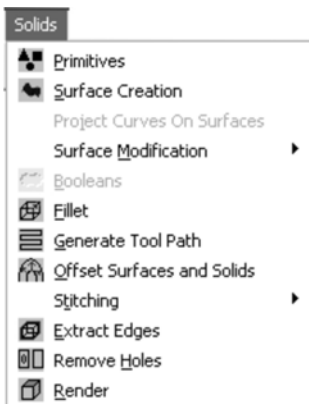


Torus: The torus is defined by its major radius and minor radius. The Major radius is the larger, total radius and the minor radius is the one that makes up how thick or large the inside will be.



Section 23. The Solids Menu

The **Solids** Menu.



The **Solids** menu offers separate functions and their corresponding sub-menus. They are:

- **Primitives:** These are the five basic solid shapes that Version 21 offers.
- **Surface Creation:** This offers a sub-menu containing functions for constructing surfaces. These also include extrusion functions for both surfaces as well as curves (2D geometry or chained geometry).
- **Project Curves On Surfaces:** This allows you to project text or curves (geometry) onto a surface.
- **Surface Modification:** This allows you to reverse a surface normal/the direction of a surface as well as extend a surface.
- **Booleans:** These are special operations that will assist you when manipulating the solid shapes for making your models.
- **Fillet:** This allows you to create surface variable radius or constant radius fillets on edges as well as chamfers.

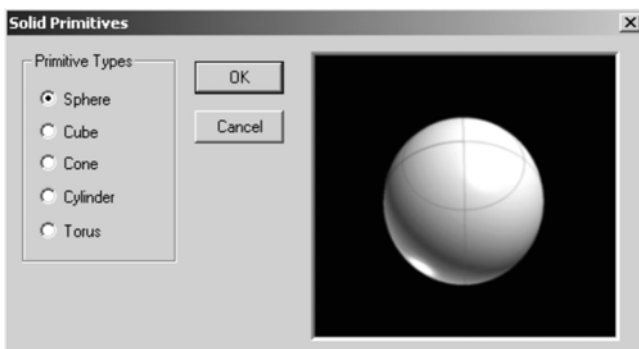
- **Generate Toolpath:** This is a special toolpath wizard that offers various automatic toolpath generating functions that include Planar, Radial & Constant Spiral for surfaces or solid parts only.
- **Offset Surfaces & Solids:** This function allows you to enter an offset value and automatically generate offsets for surfaces and solids.
- **Stitch Surfaces to Solid:** By selecting surfaces and then using this function you can automatically sew or stitch your surfaces into a water tight solid.
- **Remove Holes:** This function will allow you to automatically remove holes that are represented with cylinders from your model for generating toolpath. If you have a file that will need both toolpath and hole drilling, you will want to perform a “Save-AS” for these separate operations. This is because when you remove the cylinder holes, you cannot undo this after surface toolpath creation.
- **Stitching:** This menu has a separate sub-menu offering functions for un-stitching and stitching surfaces. You can also pick one surface to un-stitch from a solid.
- **Extract Edges:** This allows you to extract arc and line geometry from a solid for use in drilling operations and boundary machining.
- **Render:** This function allows you to render selected surfaces and/or solids to view them in a variety of ways.

Section 24.

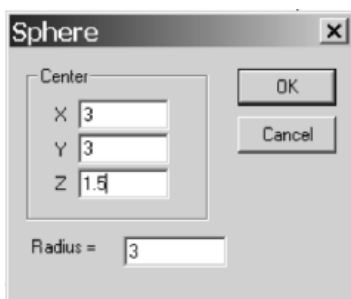
Primitives

PRIMITIVES: Version 21 provides a number of Primitives which represent basic solid forms. These are typical in solid modeling software packages. With the Boolean functions that you find in the Solids menu you can combine the primitives to make more complex shapes. You can also combine them with imported solids and surfaces or use them to make modifications.

Sphere: By selecting the primitives function you will open up its dialog box as seen below. This command box allows you to select the primitive solid that you would like to use.



Then, by selecting OK you will open the next command box.

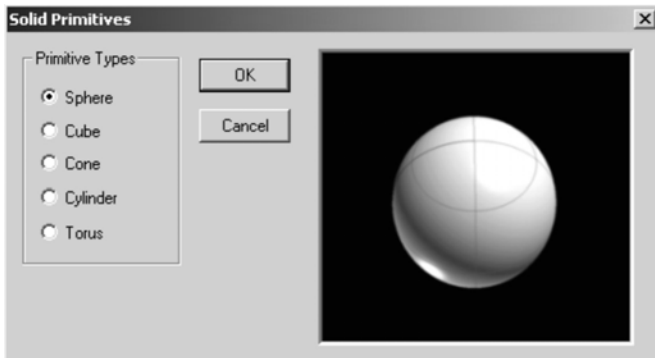


The sphere box allows you to position the shape using the X, Y and Z coordinates as well as the radius that you want.

BASIC SPHERE LESSON:

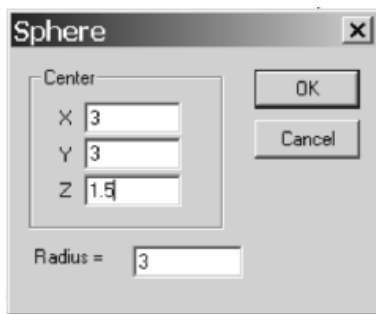
STEP 1

Select the SOLID main menu and then Primitives. Now select Sphere and click OK.



STEP 2

When the Sphere box appears, enter 3 for X, 3 for Y and 1.5 for Z. Enter 3 for the radius and then select OK.

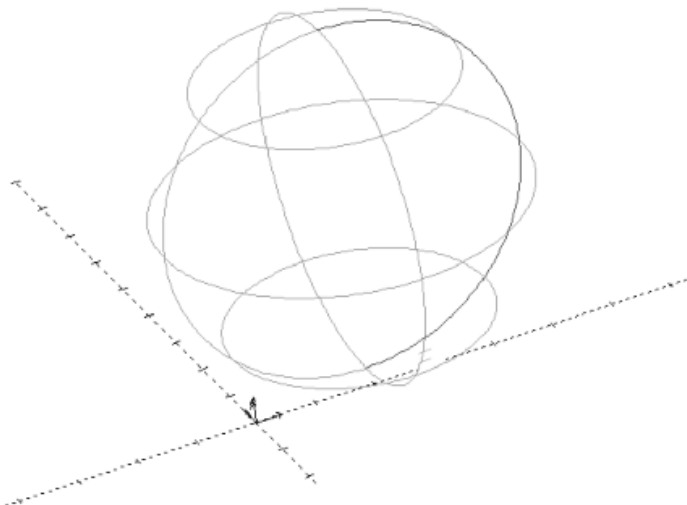


STEP 3

Now select the 3D menu and Turn 3D ON. Now select the View All icon from the main toolbar.



The result:



STEP 4

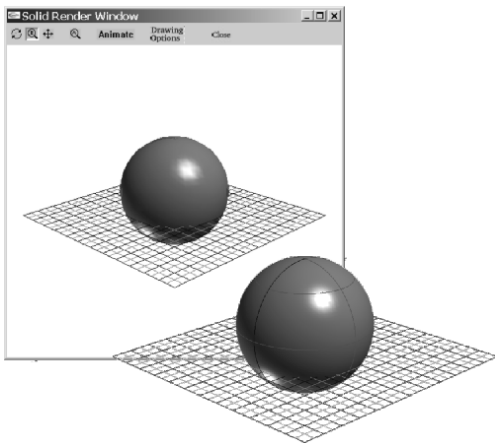
If your Auto Pre-select option is selected in the environment dialog the sphere will appear selected. If your Auto Pre-select option is turned OFF and is not selected, the sphere will appear in the color that you have chosen from the Active Color bar in the upper right corner of the main BobCAD screen.

If the sphere is not selected, rest your cursor on one of the interior boundary arcs and click your mouse button to select the shape. Once you have done this the sphere should be selected and will turn red.

STEP 5

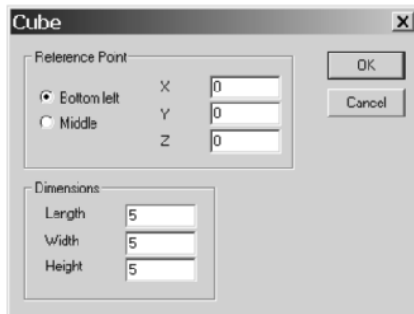
Now go back to the Solids menu and choose Render. This will open the render box.

The render box with solid image:



The render function is covered completely later in this training guide. This concludes the basic sphere lesson.

Cube: By selecting Cube from the solid primitives box, you will notice the image change. By selecting OK, the cube dialog will appear.



The cube box offers different options on coordinate positioning as well as the actual dimensions of the cube itself.

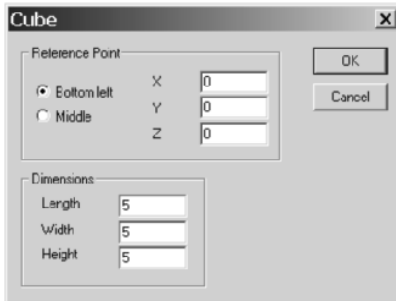
The two options are:

- **Reference Point:** This is exactly like the Rectangle function in the Other menu. The **Bottom Left** option allows you to place the bottom left corner of the shape at the default 0-position on the screen. The **Middle** option makes the coordinate values that you enter the center of the shape when it is drawn on the screen.
- **Cube Dimensions:** This area allows you to define the cubes length, width and height. You can draw these solid shapes in 2D or in 3D mode.

BASIC CUBE LESSON:

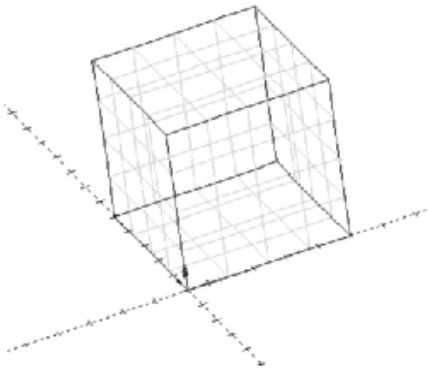
STEP 1

Select the Primitives menu and then Cube. This will open the Cube Command box as seen below.



STEP 2

Select Bottom Left under Reference Point. We are going to leave the coordinate values all at 0. Under Dimensions enter 5 for length, 5 for width and 5 for height. Now select OK and this will draw the cube.

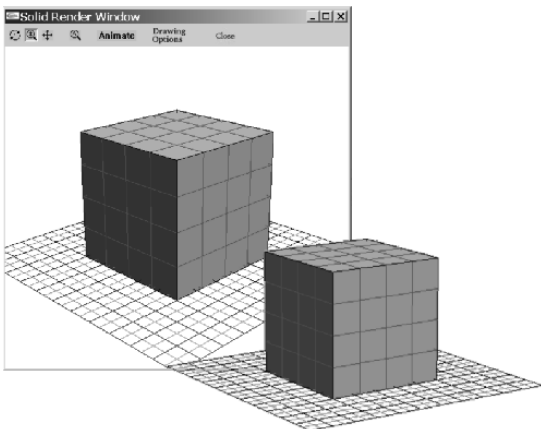


STEP 3

You will notice that the solid cube has a main edge frame as well as interior lines. To select this solid you will need to rest your cursor on one of these lines or edges and then click your mouse button. Go ahead and do this now. The solid will be selected and will turn red or your designated selection color.

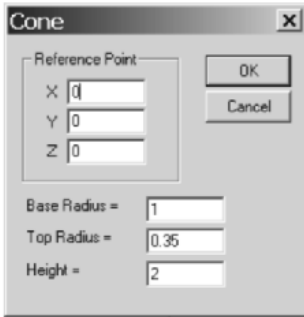
STEP 4

Now go back to the Solids menu and choose Render. This will open the render box and you will be able to preview the solid. You can see this in the next image.



Once again, the Render function will be covered in detail later in this manual. This concludes the basic cube lesson.

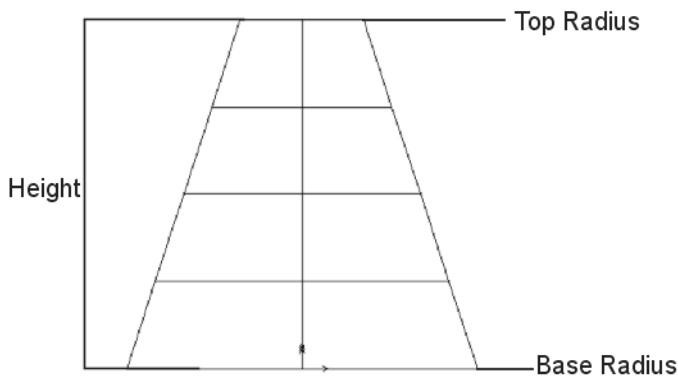
Cone: By selecting cone from the solid primitive's box, you will notice the image change. By selecting OK, the cone dialog will appear.



The cone is defined by 3 variables. They are:

- Base Radius
- Top Radius
- Height

Take a look at this below.

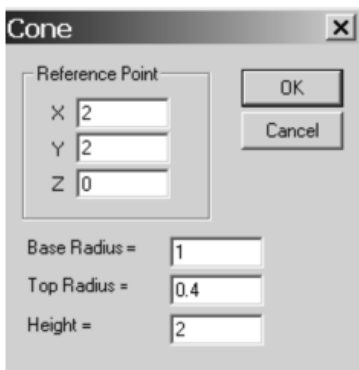


The cone dialog box also has a **Reference Point** section. This defines the base center coordinate location. In other words, if you leave the reference point values 0 for X, Y and Z the center point of the cones base will be placed at the default 0 location.

BASIC CONE LESSON:

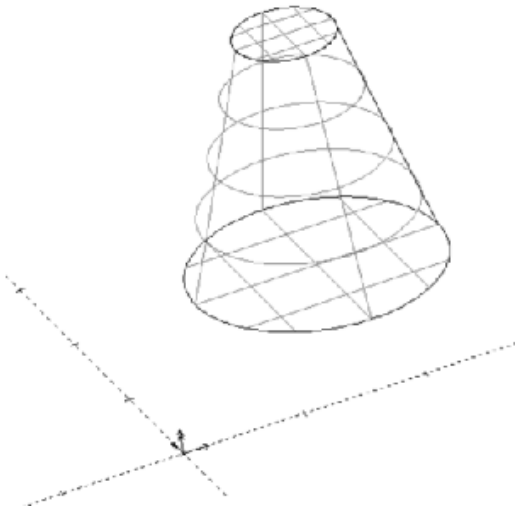
STEP 1

Select the Solids menu and then Primitives. Select the Cone option and then select OK. This will open the cone command box as you can see below.



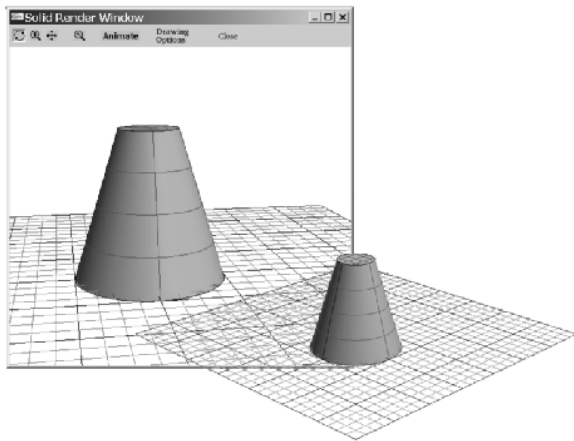
STEP 2

Under Reference Point enter 2 for X, 2 for Y and leave Z at 0. For the Base Radius enter 1, for the Top Radius enter .4 and for the Height enter 2. Now select OK. This will draw the solid cone as you can see below. You may need to select the View All icon from the main toolbar and select 3D and Turn the 3D on.



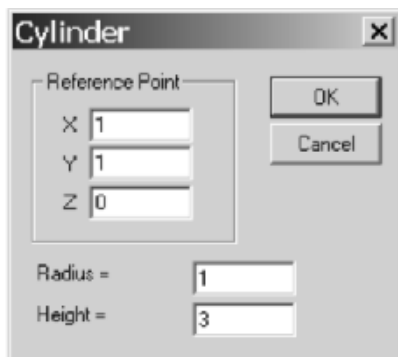
STEP 3

Rest your cursor one of the edge or interior lines and select the solid cone. It will highlight and turn red or the set selection color. Once you have done this select the Solid menu and then Render. This will open the render window so that you can view the solid cone that you just drew.



This concludes the basic cone lesson.

Cylinder: By selecting cylinder from the solid primitive's box, you will notice the image change. By selecting OK, the cylinder dialog will appear. You can see this below.



The cylinder is defined by:

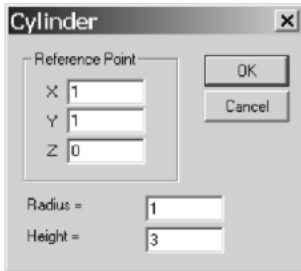
- Radius
- Height

The **reference point** indicates where the center of the cylinders base will be located.

BASIC CYLINDER LESSON:

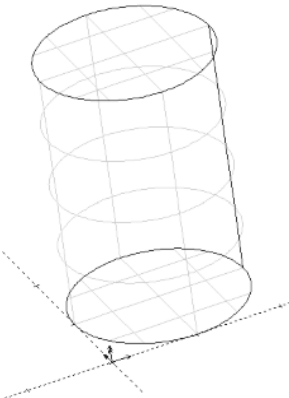
STEP 1

Select the Solids menu and then Primitives. In the Solid Primitives box select Cylinder and then click OK. This will open the new box.



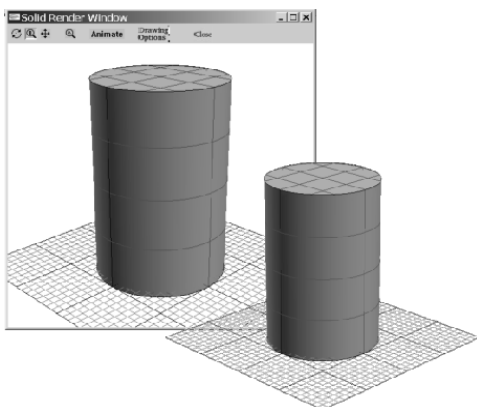
STEP 2

Under Reference Point enter 1 for X, 1 for Y and leave the Z value at 0. Enter 1 for the Radius and 3 for the height. Now select OK and the cylinder will be drawn at the entered coordinates.

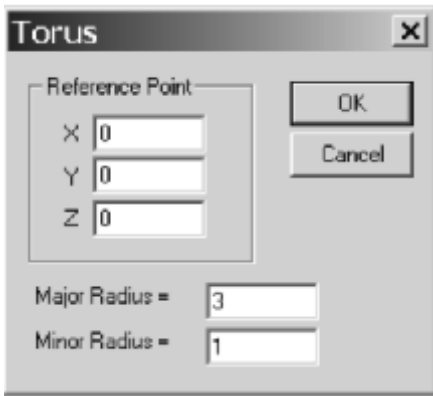


STEP 3

Place your cursor on one of the edge or interior lines and select the cylinder. It will highlight or turn the designated selection color. Now select the Solid menu and then choose Render. This will open the render window so that you can view the solid cylinder.



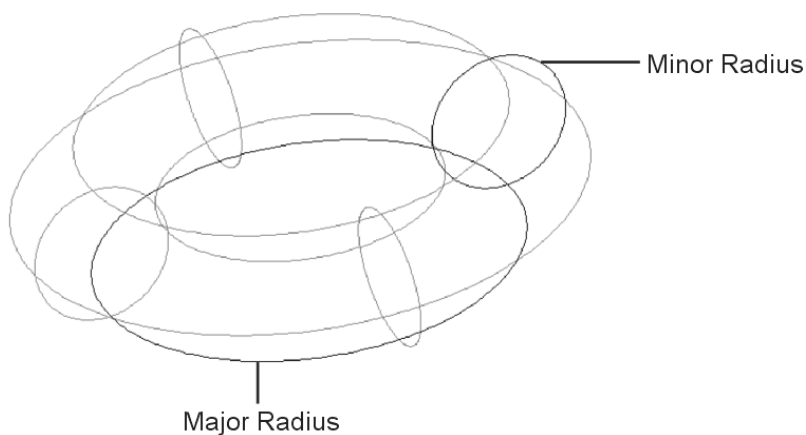
Torus: By selecting torus from the solid primitive's box, you will notice the image change. By selecting OK, the torus dialog will appear. You can see this below.



The torus is defined by:

- The Major Radius.
- The Minor Radius.

Let's take a closer look at these two attributes in the next diagram.

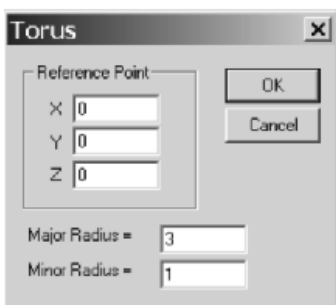


The **reference point** area of the torus command box allows you to use the coordinate system to position the **center of the major radius** where you need it to be.

BASIC TORUS LESSON:

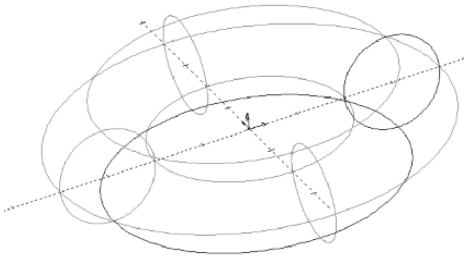
STEP 1

Select the Solids menu and then Primitives. In the Solid Primitives box select Torus and then click OK. This will open the new box as you can see in the next image.



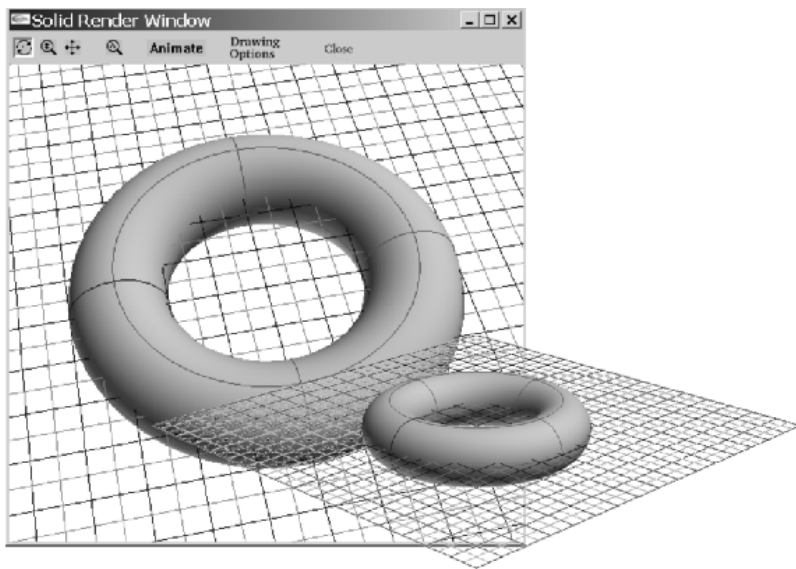
STEP 2

Under Reference point we are going to leave the values 0 for X, Y and Z. Enter 3 for the major radius and then enter 1 for the minor radius. Now select OK. The torus should now be on the screen as you can see below.



STEP 3

Now place your cursor on the edge geometry or the interior geometry and click your mouse button to select the torus. Once you have done this, select the Solids menu and then Render to view the solid torus.



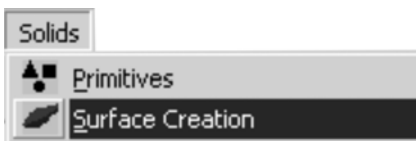
This concludes the basic torus lesson.

Section 25. Surface Creation, Projection & Modification

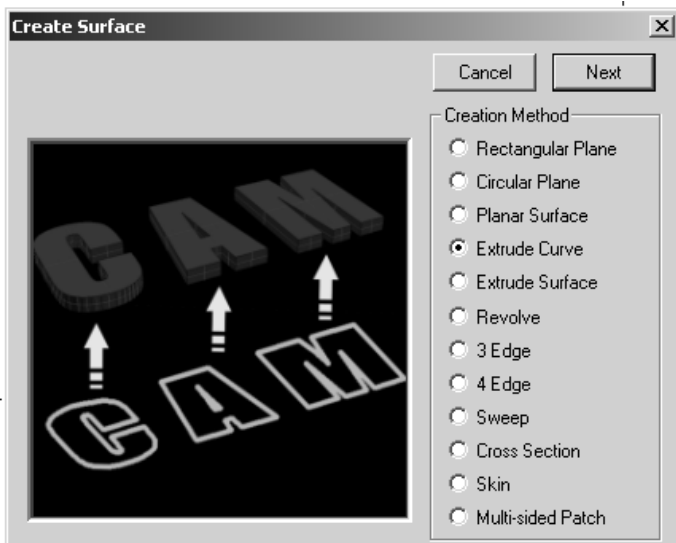
The Version 21 software offers a variety of **Surface** generating functions that can be used for creating complex surfaces for toolpath and machining.

SURFACE CREATION:

Surface Creation is located in the SOLIDS menu. By selecting the “Surface Creation” function a new box will appear offering each of these new features.

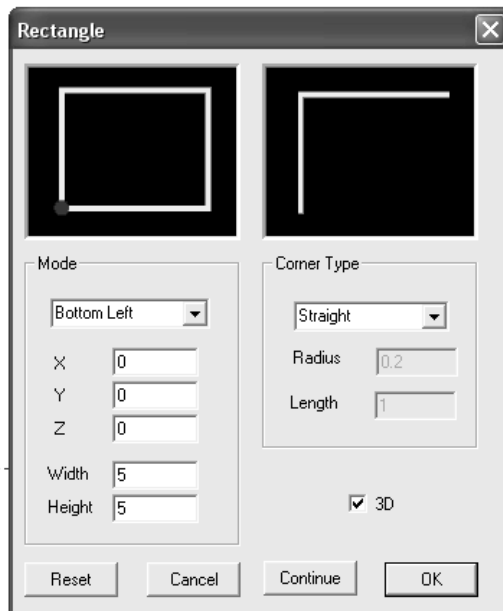


The **Surface Creation** option box:



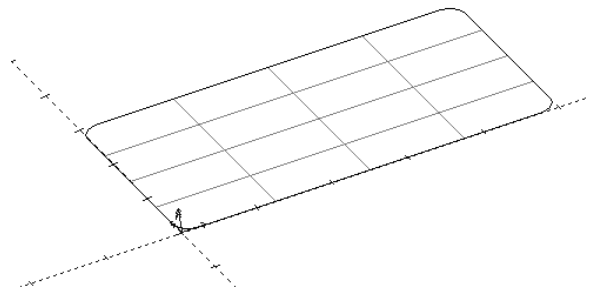
Let's take an in-depth look at each one of these new surface functions separately.

RECTANGULAR PLANE: By selecting this option and then the NEXT button, the standard rectangle box will appear.



This allows you to draw a rectangle or box type 2D surface by using coordinate values for placement and size. If you select the SKETCH option, you will be able to select OK and then click your left mouse button to drag and drop the rectangular planar surface.

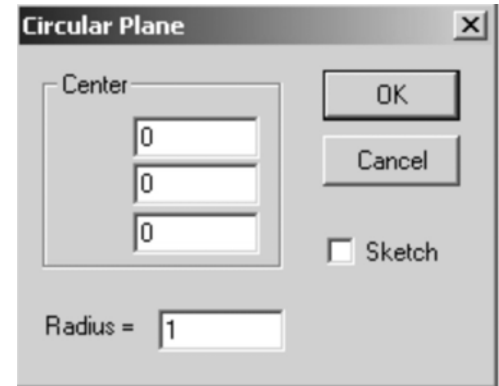
This function would be used for drawing a specific size rectangular surface in the X-Y plane. It is not a repetitive function meaning that if you need to draw a second rectangular surface you will need to go back to the SOLIDS menu, Surface Creation and choose Rectangular Plane again.



CIRCULAR PLANE: By selecting this option and then the NEXT button you can perform the following:

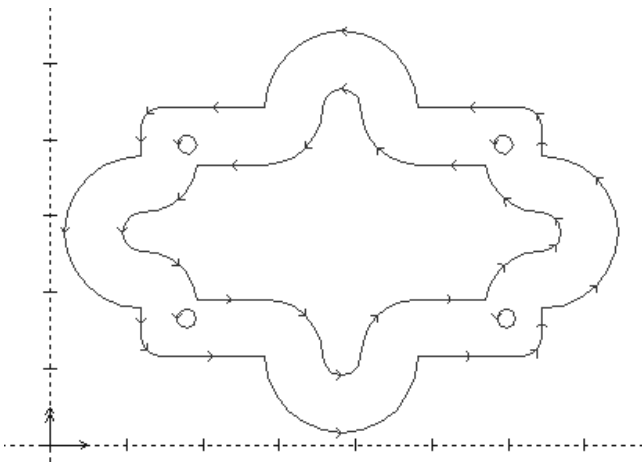
- A. Use the coordinate system to draw a circular surface. You may enter in an X, Y or Z coordinate value as well as the radius of the circular surface you are creating in this box.
- B. You can bypass the coordinate section by selecting the SKETCH option. Select OK and then click your mouse to draw the circular surface.

This is a continuous function meaning that if you are sketching circular surfaces, you may click your left mouse button drawing as many of the same circular surfaces as you need on the screen.

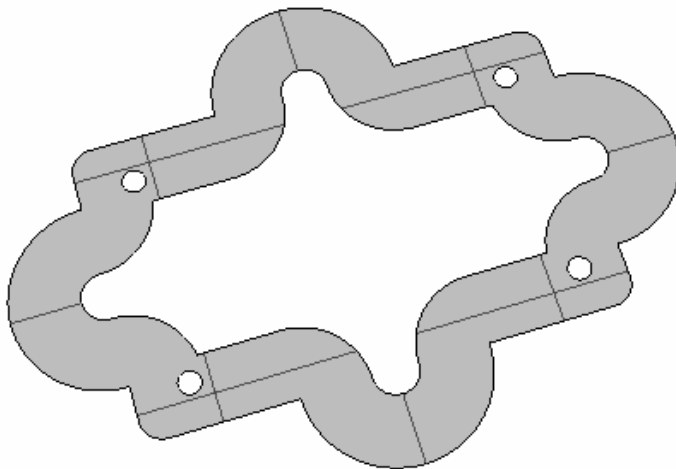


PLANAR SURFACE: This function allows you to create a surface in ANY drawing plane based off of wireframe geometry with or without island wireframe geometry. Let's take a closer look.

In this image you can see the 2D geometry with an outside contour, 4 small circles and then an inside contour chain. We want to create a surface that avoids the small circles as well as the inside contour.



To do this we simply click and drag a selection box over the geometry and release the left mouse button to select everything. Then we go to the Surface Creation function in the Solids menu and select the **Planar** option and click NEXT. This automatically generates the surface for the part.

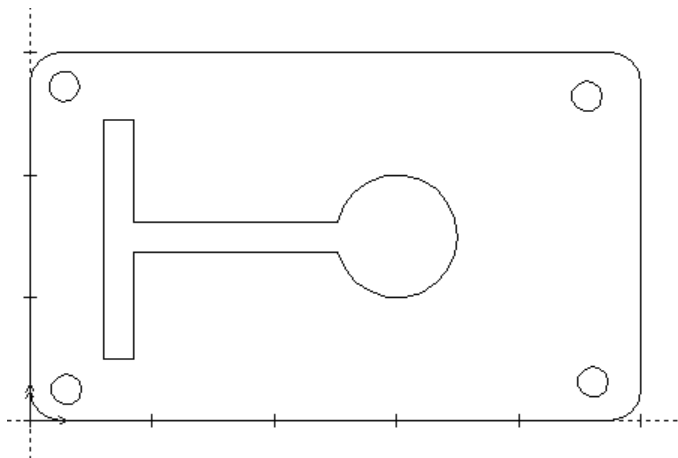


IMPORTANT: This is an **OBJECT-ACTION only** function meaning that you **must** have pre-selected geometry at the time you attempt to perform this operation.

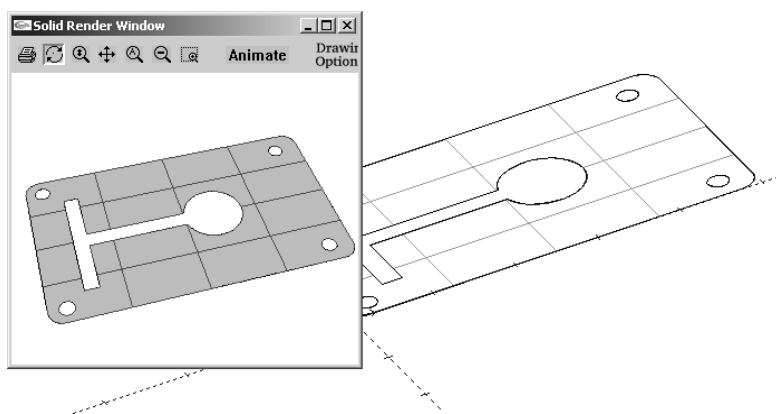
The **procedure** for creating a planar surface:

1. To do this you would *first* select the geometry that you are working with. Then go to the **SOLIDS** menu, select **SURFACE CREATION**, then the **Planar Surface option** and the NEXT button.
2. The surface is generated automatically.

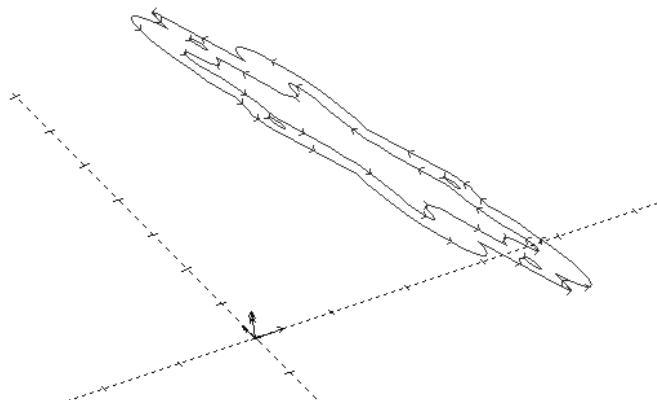
Here is an example of a wireframe part with multiple islands.



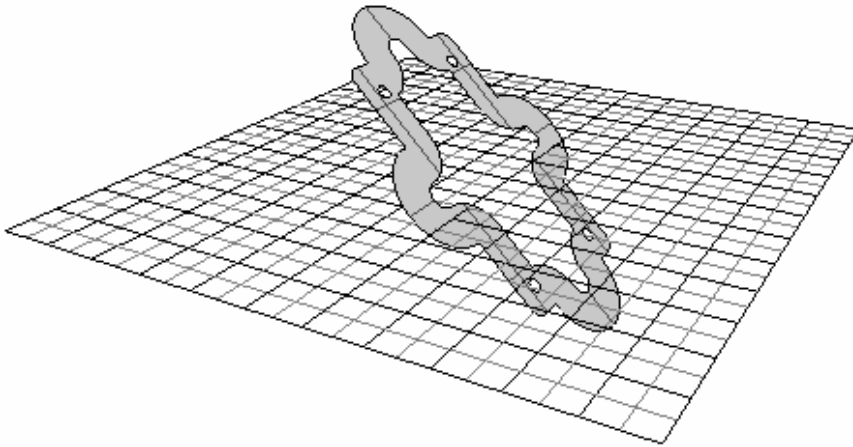
By using the planar surface function you can create a planar surface as seen here:



Here is an example of geometry that has been rotated 45 degrees in the Y axis.



By selecting the geometry and then this function, the surface is automatically generated.

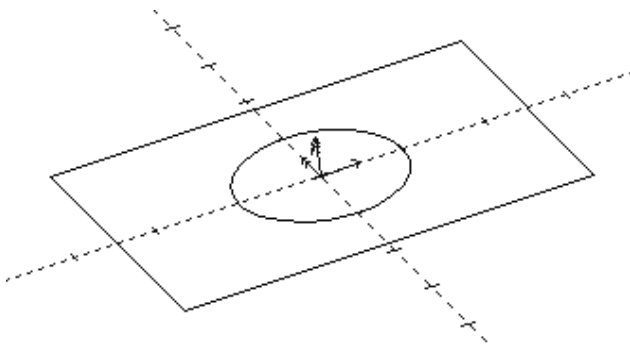


SELECTING GEOMETRY FOR CREATING PLANAR SURFACES: There are several ways to select geometry for planar surface creation.

- Click and drag a selection window over the geometry.
- You can select open or closed chains of entities by using the chain select option in the EDIT, Select Entities menu.
- You can select multiple entities (points, lines or arcs) by holding down the control key and simply clicking on the geometry.
- You can use the select ALL option if the only geometry on your drawing screen will be used for creating the surface.

Practice using different techniques of selecting geometry for creating planar surfaces to become more proficient.

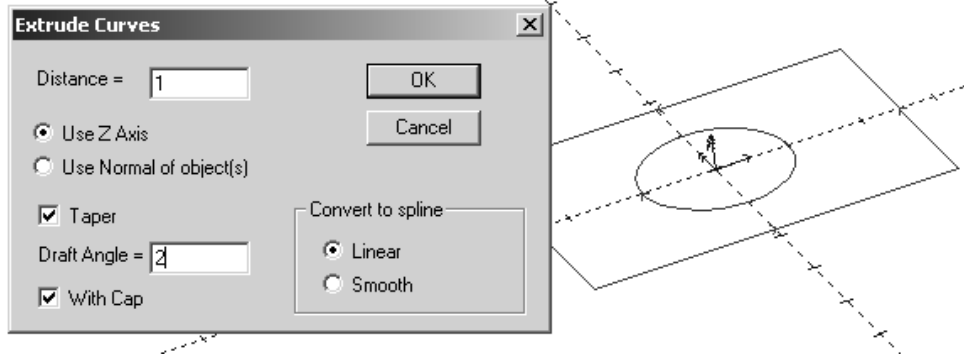
EXTRUDE CURVE: The **Extrude Curve** option allows you to create wireframe geometry with or without island geometry and create extruded surfaces that can also include tapers in any plane.



Here we have a simple rectangle with a circle island. This is an **ACTION-OBJECT** ONLY function (*selecting the function first and then the geometry*). Let's take a quick look at this function.

The **procedure** for extruding a curve:

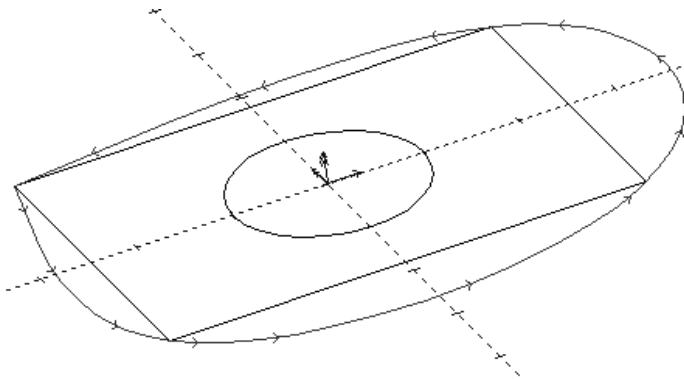
1. Go to the Solids menu, select Surface Creation, the Extrude Curve option and click NEXT.
2. Perform the chain selection operation on the square and select the circle
3. Right-Click your mouse to open the Extrude Curve dialog box.
4. Make your entries and settings and click OK to execute the operation.



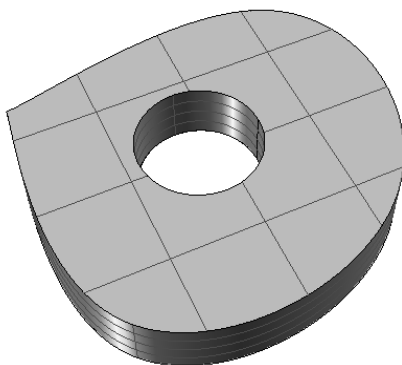
This new box offers you:

- An extrude distance.
- Ability to use the standard UCS (*User Coordinate System*) Z-Axis.
- Use the “Normal” of the objects/geometry, meaning that the surface created will be perpendicular to its origin in any plane.
- Apply a draft angle to both outside and any inside contour geometry.
- Apply caps, turning this geometry into a solid.
- Apply a smooth surface extrusion based off of a spline conversion of the geometry being used. In the example of the square, if you were to select it and create interpolated spline geometry from it, you would create a spline based off of the 4 corner control points of the square itself.

Here is an example of this:

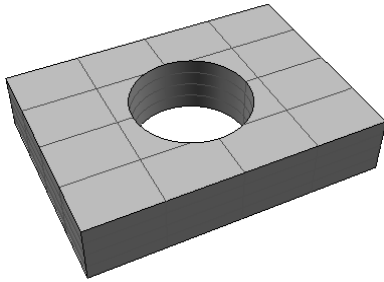


Therefore, if you created the extrusion using the “Smooth” option under “**Convert to Spline**” you would get the following result:



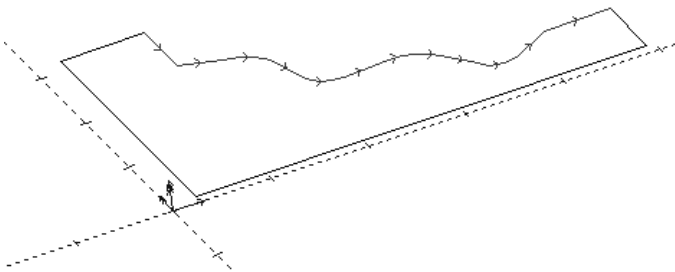
It is important to understand this option. When working with smooth geometry for curve extrusions you will get separate results.

Here is the result using the same example geometry and the “Linear” option:

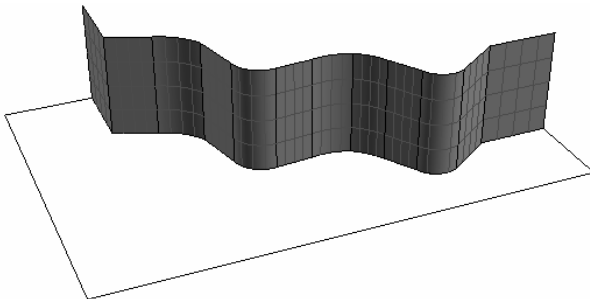


You can create extruded curves from both smooth and linear geometry as you can see in this example below.

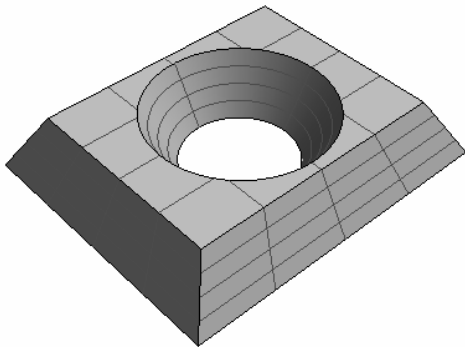
Chain-select the geometry that you need:



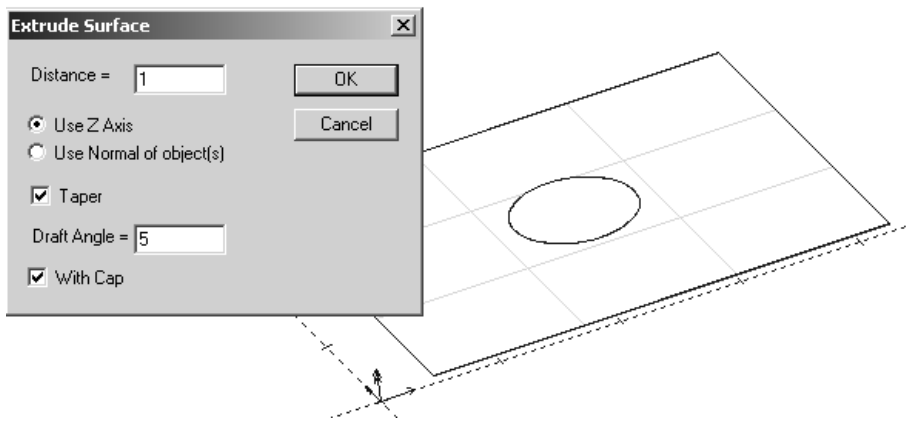
After the Extrude Curve function is used:



Extruded curves after tapered angles are applied:

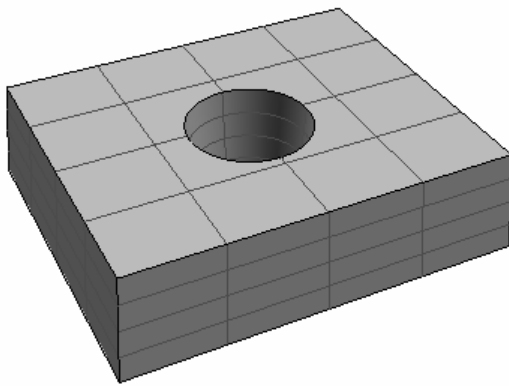


EXTRUDE SURFACE: This option allows you to **Extrude Surfaces** in any plane with or without draft angles. This is an **OBJECT-ACTION** function meaning that you **MUST** select the surface first and then the function. By doing this, the Extrude Surface box will appear offering several options.



This new box offers you:

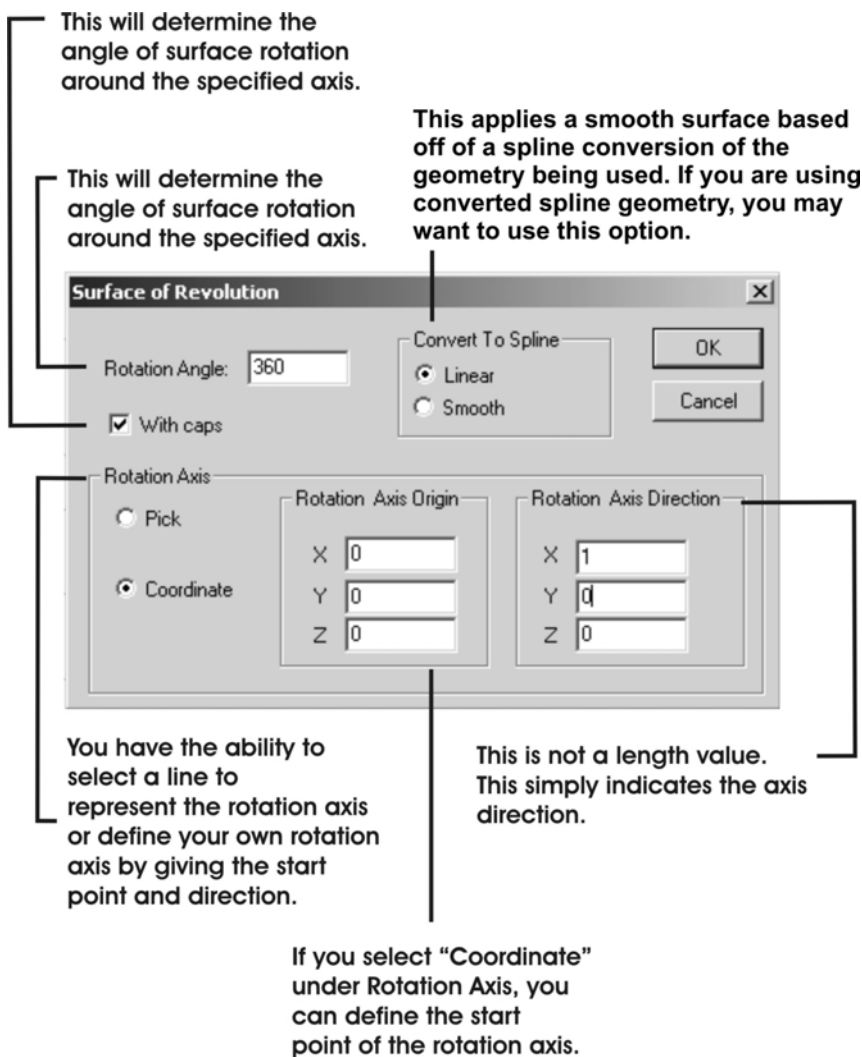
- An extrude distance.
- Ability to use the standard UCS (*User Coordinate System*) Z-Axis.
- Use the “Normal” of the objects/geometry, meaning that the surface created will be perpendicular to its origin in any plane.
- Apply a draft angle to both outside and any inside contour geometry.
- Apply caps, turning this geometry into a solid.



The **procedure** for extruding a surface:

1. Select the surface that you want to extrude.
2. Go to the Solids menu, choose Surface Creation and the Extrude Surface option.
3. Make your entries and selections in the Extrude Surface dialog box and click OK to execute the operation.

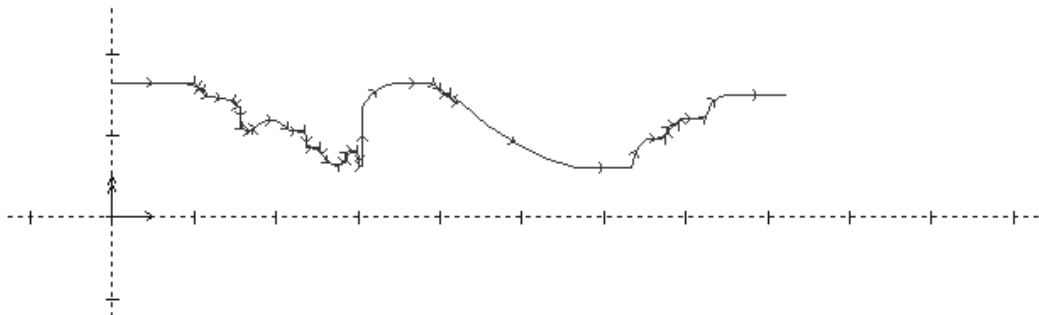
REVOLVED: The **Revolved Surface** option allows you to use existing open chain geometry and create a surface that is based off of the axis you choose to rotate the surface and a degree around that axis. The Revolved Surface function is both **Action-Object** as well as **Object-Action**.



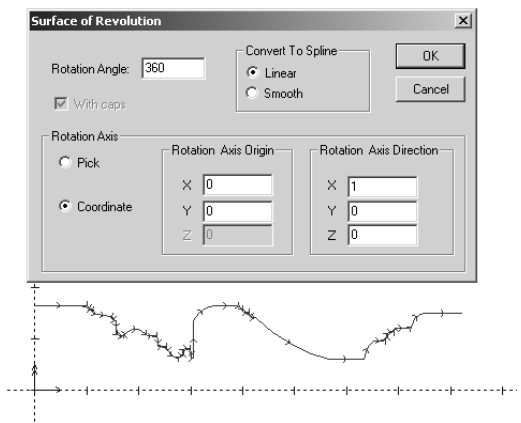
Let's take a closer look at the **ROTATION AXIS** section of this box.

Example 1

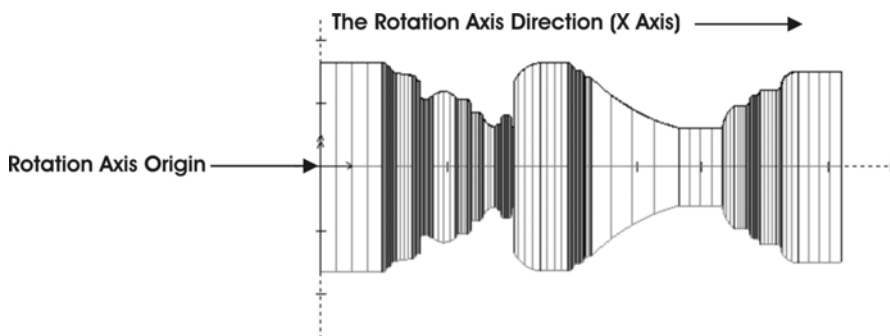
Here we have an open shape profile that we want to use to create a surface for. This contour runs lengthwise along the X axis as you can see.



With the geometry chain selected, we would go to the SOLIDS menu, select Surface Creation, Revolve and then the NEXT button to open the Revolved Surface box.

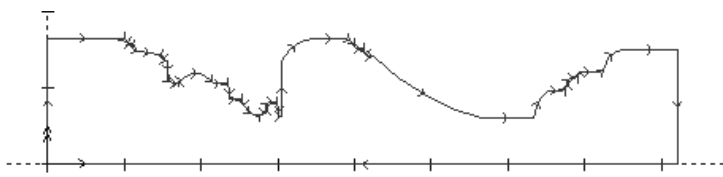


We have entered 360 degrees as the rotation angle and we are *not* going to convert the existing contour into approximated spline geometry for the surface creation so we checked “Linear.” We want to use a coordinate of X0 and Y0 as the origin of rotation and have indicated that the axis direction is in X by putting a 1 in the X box under **Rotation Axis Direction**. By clicking OK we get the following result.

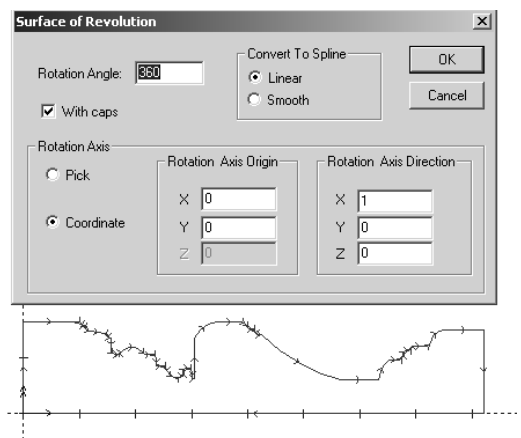


Example 2

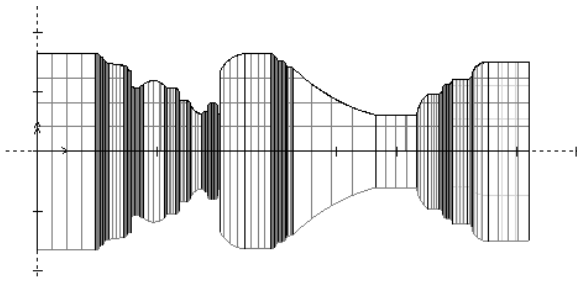
In this example we have a closed shape profile that we want to create a revolved surface for that is similar to the geometry in example 1.



With the closed chain geometry selected, we go to the SOLIDS menu, select Surface Creation, pick the Revolved option and click the NEXT button to open the box.

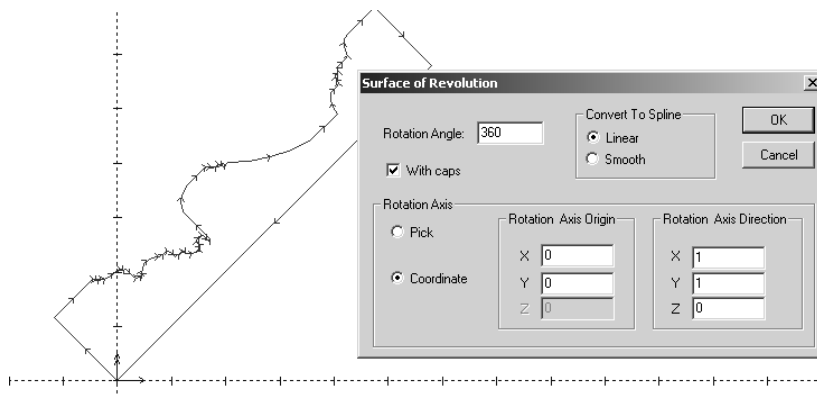


Because this is a closed chain the “With Caps” option is available for selection. We have checked it to indicate that we want this surface to have capped ends, automatically creating a solid. By indicating the Axis origin and the Axis Direction we are able to click OK and create the surface we wanted.

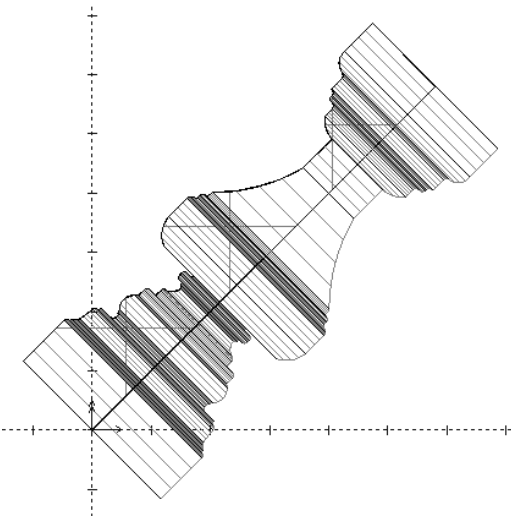


Example 3

Here you can see that we have rotated the closed chain 45 degrees around the Z axis. With selected chain geometry we can now go to the SOLIDS menu, select Surface Creation, pick the Revolve option and click NEXT to open the box.

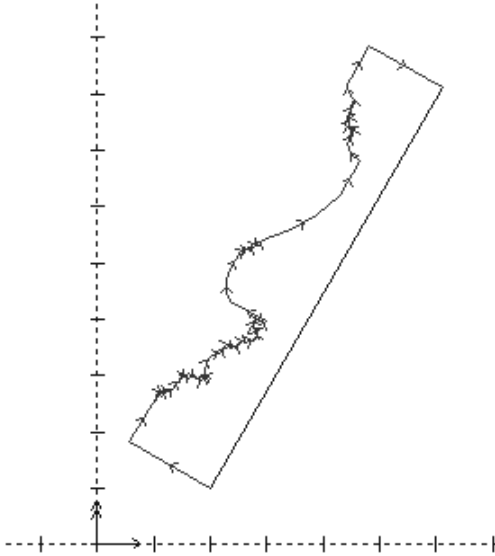


Because we know that the Axis Direction is equal in X and Y, we entered 1 for X and 1 for Y. By clicking OK we got the correct surface.



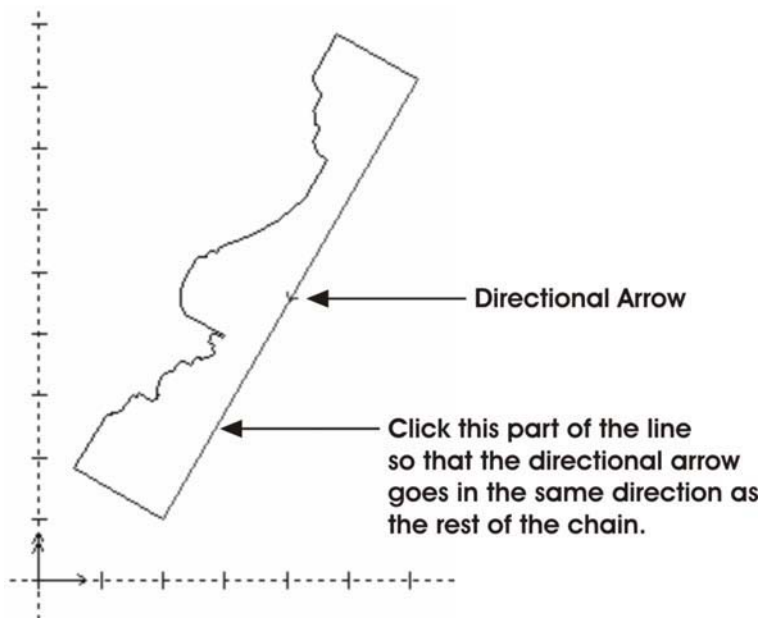
Example 4

There may be cases where you are not certain what the Axis Direction should be set to. Let's take a look at the next example image.



In this example we are unsure of what the correct Axis Origin or Axis Direction would be. Therefore, we select the chain geometry leaving the bottom line as we need to be able to pick it for our rotation. With the chain geometry selected, we go to the SOLIDS menu, choose Surface Creation, Revolved and click NEXT.

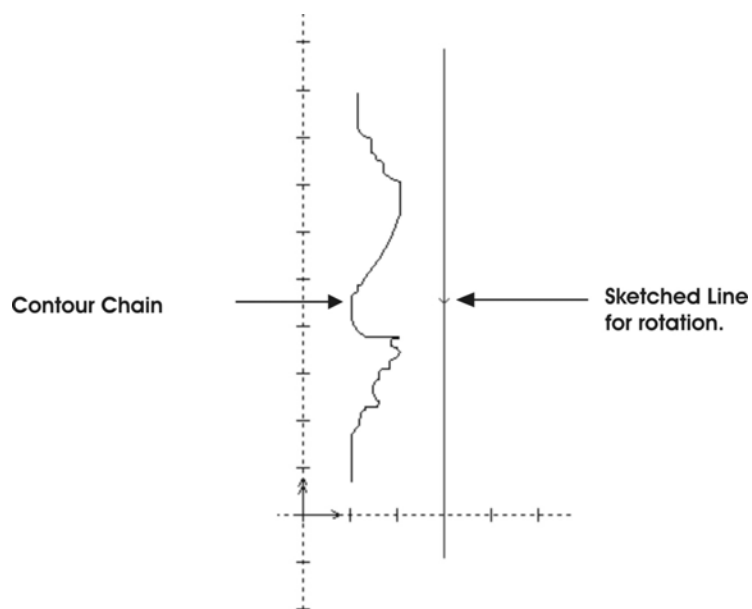
Choose the "PICK" option in the Revolution box and click OK. Now we would need to select the lower part of the line that we are using as the axis, by clicking on it, so that its directional arrow on that line points in the same direction as the rest of the directional arrows.



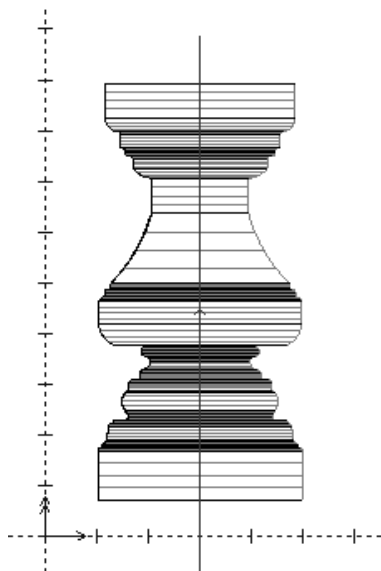
By doing this, the surface is automatically created.

Example 5

Here is the example of sketching the axis line in which you want to use to create the revolved surface.

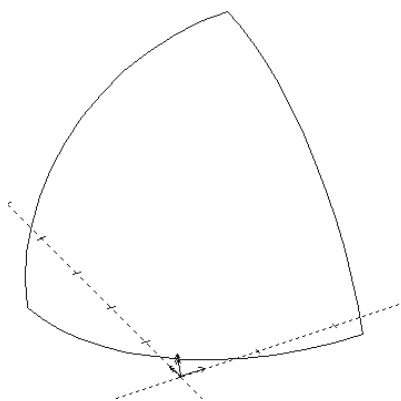


The result:

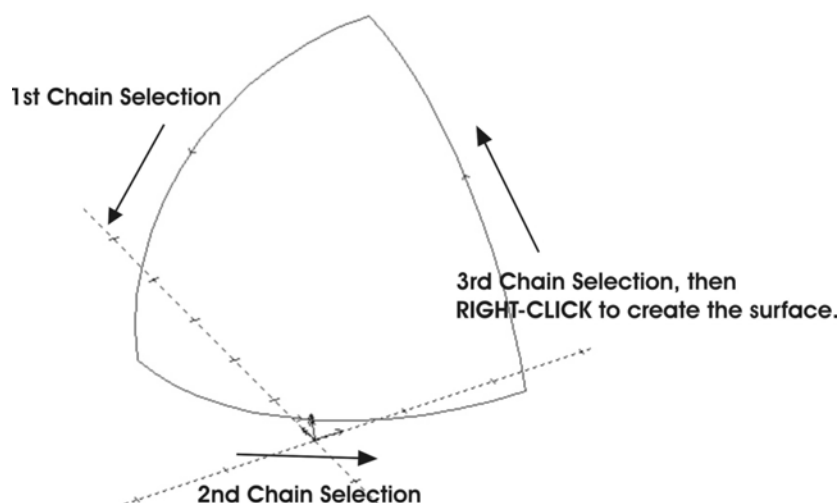


The **Revolved Surface** function is an Action-Object, Object-Action operation meaning that you can access the function with pre-selected geometry (object-Action) or you can select the function first and then select the geometry second (action-object).

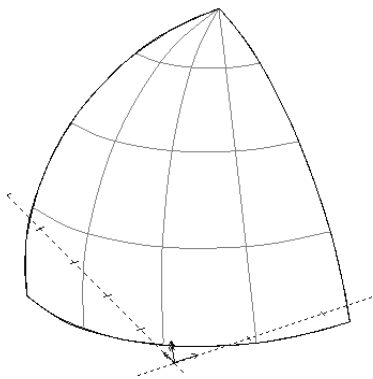
3 EDGE: The **3 Edge** surface function allows you to create surfaces for 3 edge wireframe geometry. This is an **Action-Object** function only. Let's take a look.



Here is an example of 3 arcs that we want to use for creating a surface. To do this we would first go to the Solids menu, choose Surface Creation and select the 3 Edge option. Have a look at the example below.



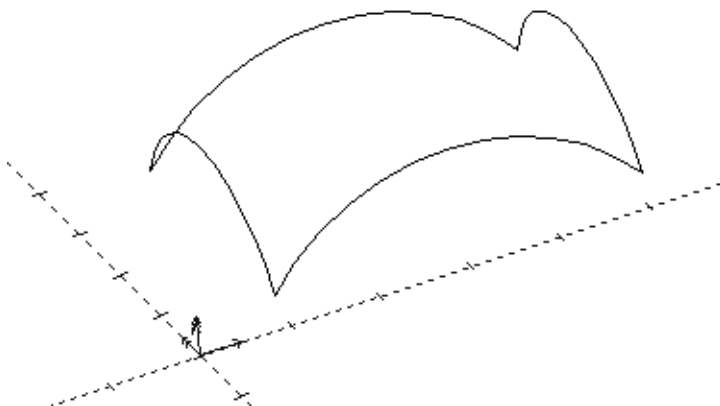
After you select the 3 Edge option and click the NEXT button, the software will be ready for you to make the first chain selection. Start off by clicking on the first arc entity (*this can be any of the 3 entities*), point the direction arrow going down and click your left mouse button again to set the direction. Now right-click or hit the F3 button on your keyboard. Now perform a chain selection the same way on the second of the 3 entities the EXACT same way. After you have chain selected the 3rd and last entity the software will automatically create the surface.



The key here is to make sure that the three chain selection directions are the same.

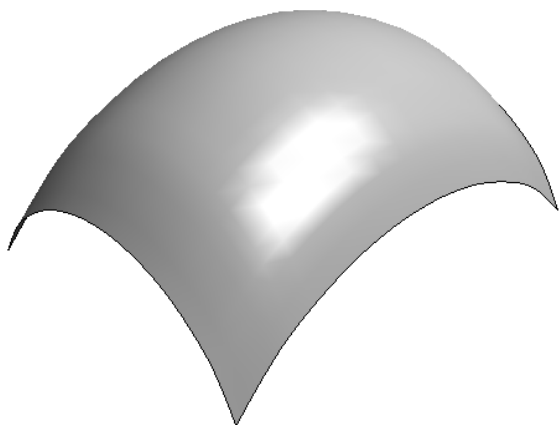
4 EDGE: The **4 Edge** surface function allows you to create surfaces for 4 edge wireframe geometry and is very similar to the method in which you create the 3 Edge surface in terms of making chain selections that all go in the same direction. This is an **Action-Object** function only. Let's take a look.

Here we have an example of 4 arcs that we want to create a surface from.



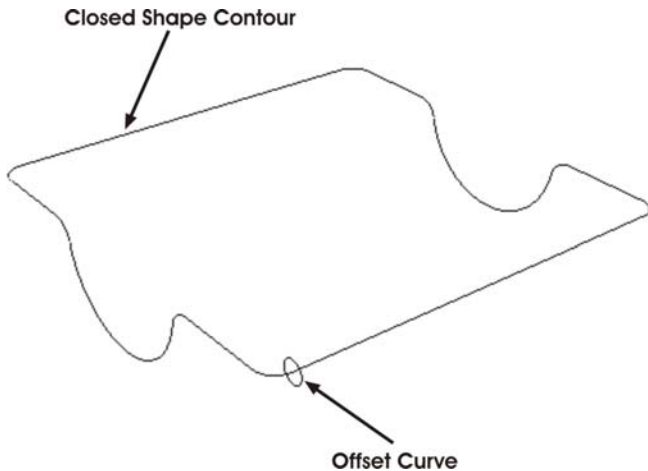
Because this is an Action-Object function, the procedure for creating this type of surface would go like this:

1. Go to the SOLIDS menu and choose Surface Creation.
2. Select the 4 Edge option and click NEXT.
3. Place your cursor on any one of the arc segments and click on it one time.
4. By moving your mouse you would point the direction arrow and left-click your mouse one more time to indicate the direction.
5. Right-click to secure the direction or hit the F3 key.
6. Place your cursor on the next arc segment going in the SAME direction as the first one and click on it one time.
7. Now make sure that the directional arrow on the next arc is going in the same direction as the first one and click your left mouse button again to set the direction and right-click to end that selection.
8. Perform chain selections on the remaining two arc segments making sure that the selection directions are going in the SAME direction.
9. On your last arc selection, when you right-click your mouse, the surface will automatically be created.



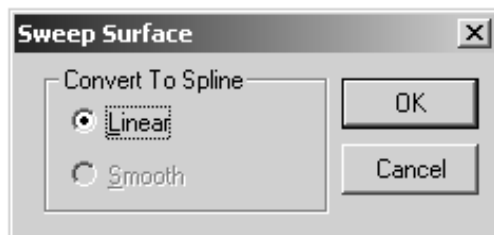
SWEEP: The **Sweep Surface** function allows you to use an offset curve to create a surface that sweeps along a specified open or closed shape contour. This is an **Action-Object** function only.

Here is an example of a closed shape contour that we want to create a sweep surface for. In order for us to do this successfully we need to have an offset curve which you can see in the image.



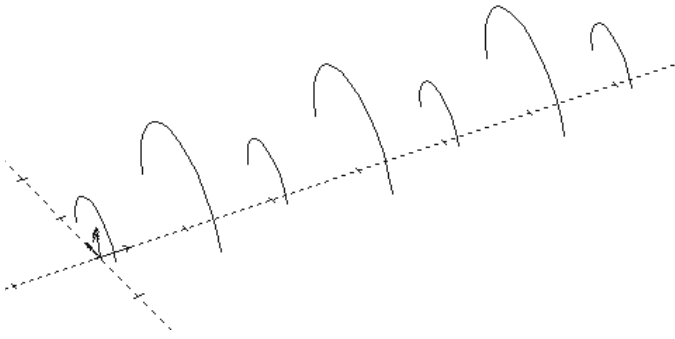
To create the sweep surface for the above example you would do the following:

1. Go to the Solids menu and select Surface Creation.
2. Choose the Sweep option and click the NEXT button.
3. Place your cursor on the offset curve and click on it one time.
4. By moving your cursor you can change the selection direction arrow, making it go in either direction. Choose a direction and click your left mouse button one more time to indicate the direction.
5. Now RIGHT-click your mouse to complete the chain selection. You can also hit F3 on your keyboard.
6. Now place your cursor on the first entity to the front or back of the offset curve and click on it one time. Make sure that the directional arrow of this entity is going AWAY from the offset curve and click your mouse button one more time to set the direction and then either click on the LAST entity of the chain or hit the F3 button on your keyboard.
7. This brings up the Sweep Surface box. This box gives you 2 options: Linear or Smooth. If you are working with converted spline geometry, the smooth option will be available. It is recommended that you choose linear for all other geometry. When you click OK in this box the surface will be created.



CROSS SECTION: The **Cross Section Surface** function allows you to create surfaces from cross sectional geometry. This is an **Action-Object / Object-Action** function. Let's take a look at some examples of this function.

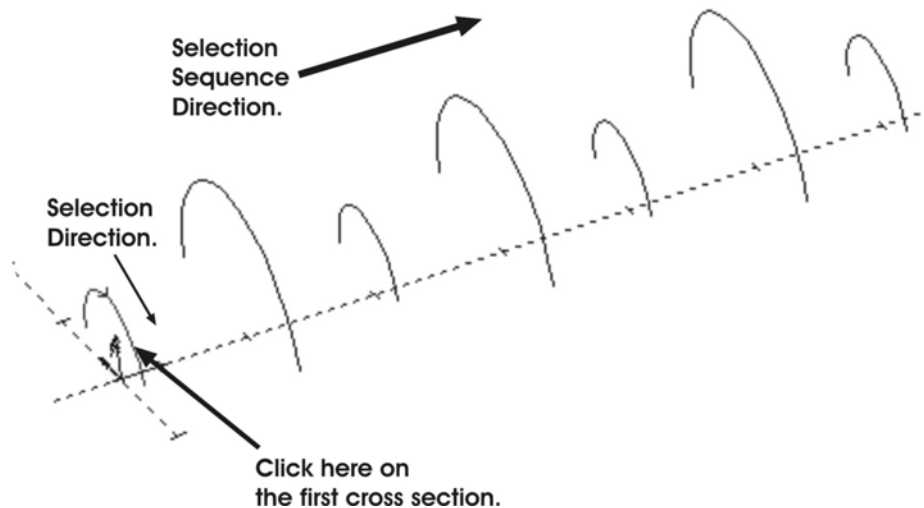
Have a look at the next image which shows cross sectional geometry.



Because this function is both action-object and object-action we can pre-select each entity or chain of entities first. To do this:

STEP 1

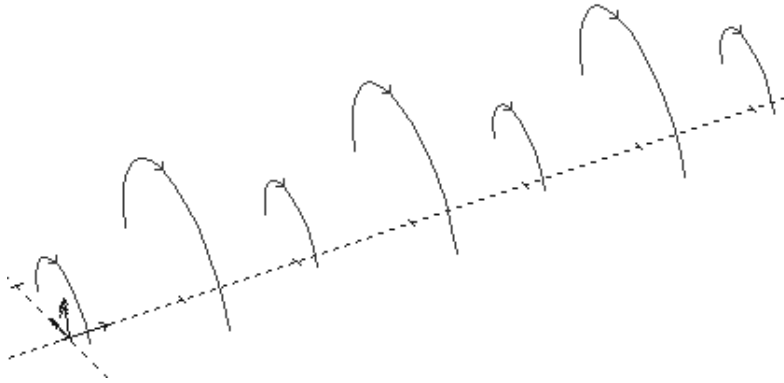
You need to make sure that your selections are in selection order, meaning that you select these cross sections going from one end to the other. In this case, the cross sections are single arcs. Therefore you would start with the far left arc, placing your cursor toward the front, bottom and click on it to select it with the directional arrow pointing toward you.



NOTE: If you select the function first (**Action-Object**) you are going to perform a chain selection on each cross section **even if the cross sectional geometry is a single entity**. Therefore, you will first select the entity or first entity of a chain, point the selection directional arrow toward the last entity of the chain (if there are multiple entities) or point your arrow in a direction, click your left mouse button one more time to set the direction and then either click on the last entity of the chain, hit F3 on your keyboard to complete the chain selection or, in the case of a single entity, right click your mouse.

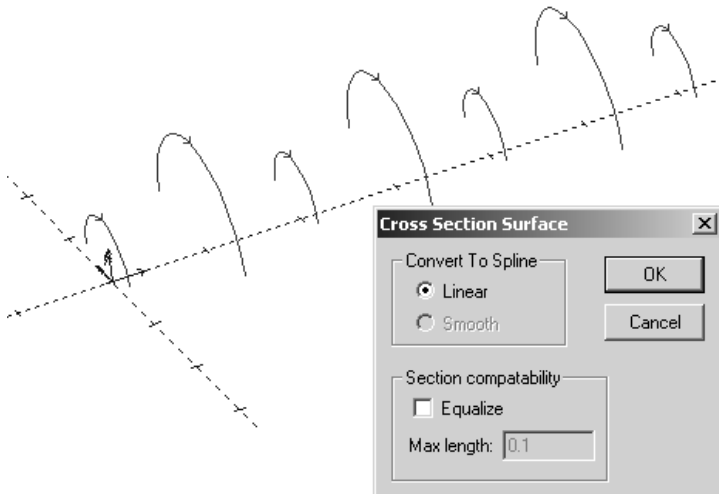
IMPORTANT to remember: *When selecting the function first (action-object) and then making your chain selections, when you chain select the last cross section and right-click your mouse to end the selection you **MUST RIGHT-click your mouse one more time** to open the Cross Section Surface box. You do not have to do this with pre-selected cross sections (object-action).*

To move forward now and select the cross sections, hold down your control key and click on each of the arc entities going from left to right until they are all selected as in the example image below.



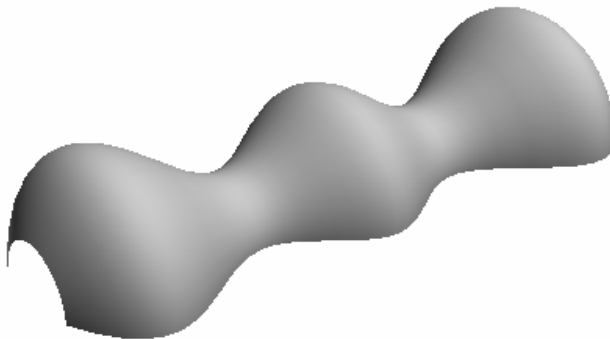
STEP 2

Now go to the Solids menu, choose Cross Section and click the NEXT button to open the Cross Section Surface box.



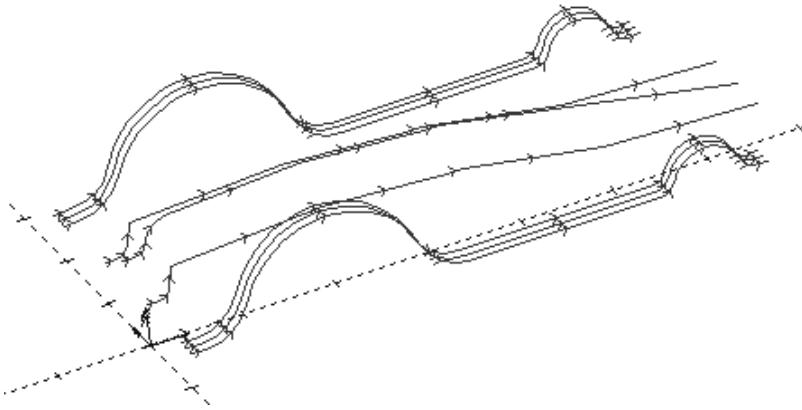
This box offers a Linear and Smooth option depending on the geometry that you are using to create the cross section. In this case we are dealing with single arcs and the smooth option is not available for selection. If we were using converted spline chain geometry the Smooth option would be available for selection and may be a good choice to try. In that case you could also equalize the chain entities to get the desired result by selecting the Equalize option and entering in a value that you would like to convert each entity to in the process.

By clicking the OK button the surface will appear.

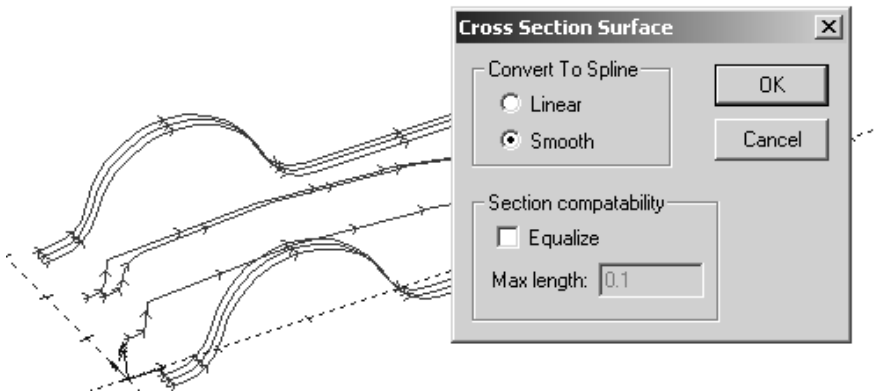


Another Cross Section Example:

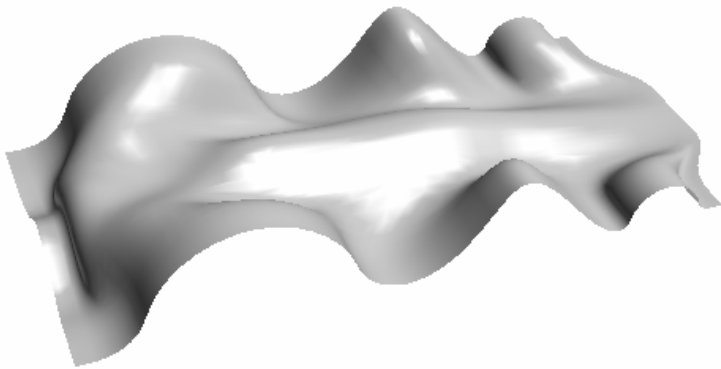
Here are a series of cross sectional pre-selected chains all going in the same direction.



By selecting the chain geometry first and then the function (object-action), we have the cross section surface box automatically appear offering both Linear and Smooth options.

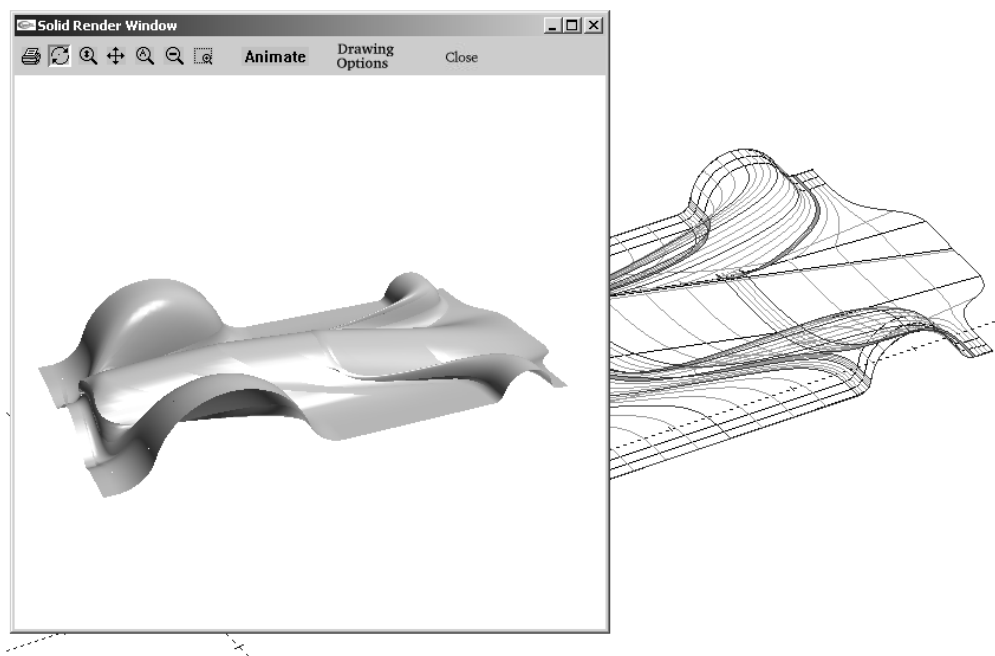


You can see from the image that we chose “Smooth” and did not check equalize. By clicking OK, we got a slightly off result from what we wanted below. This is because we selected “Smooth” and the surface was created based off of the geometry being converted to spline curves. This is also because we didn’t equalize the geometry and apply a Max Length.

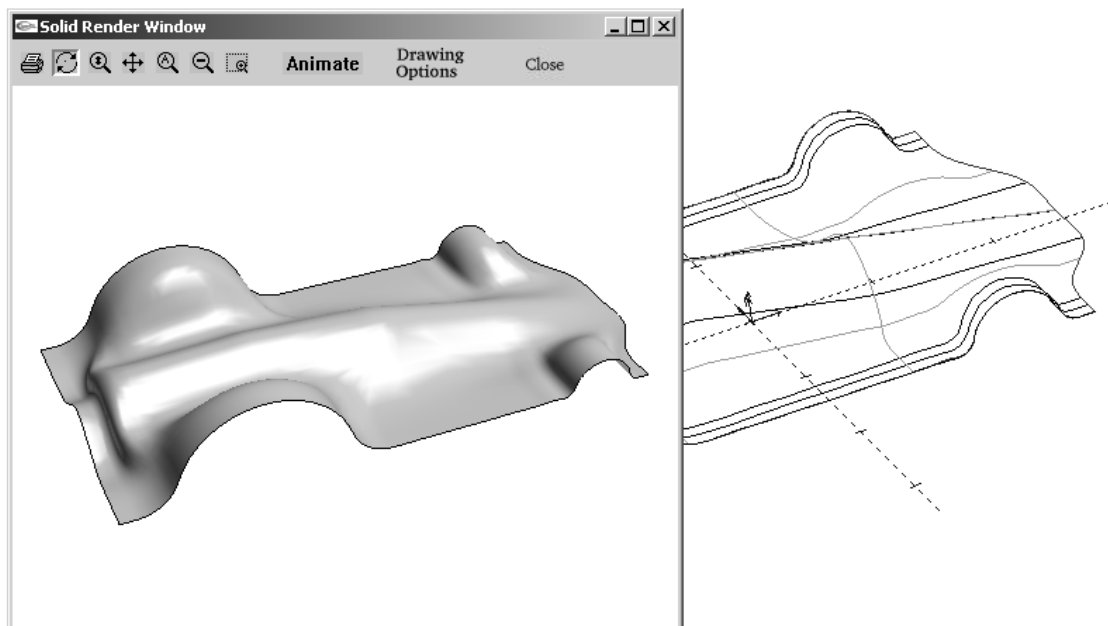


If this happens, click the UNDO button and start over by pre-selecting the cross section chains in correct sequence and all direction arrows going in the same direction. Then go to the solids menu, choose Cross Section and then click the NEXT button to open the Cross Section Surface box again.

This time we chose “**Linear**” instead of “**Smooth**” and you can see the improved result:



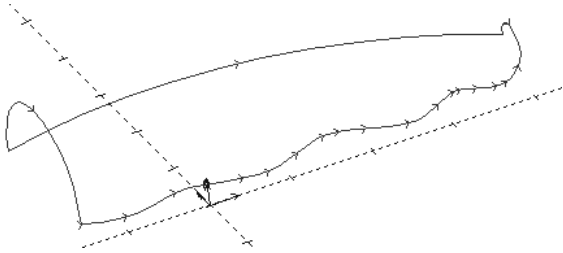
This time we selected “**Smooth**”, checked “**Equalize**” and entered .25 as our **Max Length** and got an even better result:



As you can see, your surface results can be improved depending on the cross sectional chain geometry being used.

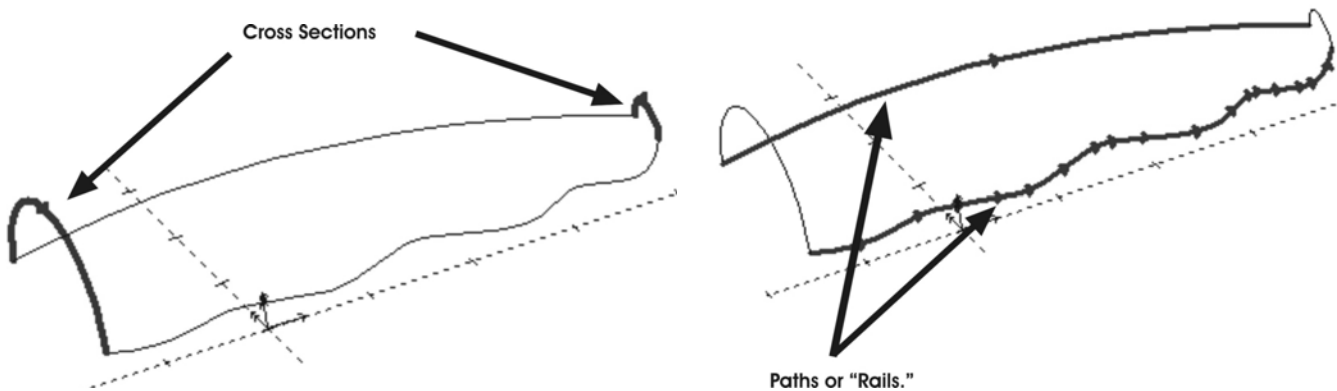
SKIN: The **Skin Surface** function creates a surface that is based off of **cross sectional geometry** as well as a path or what are called “**rails**.” This is an **Action-Object** function only. Let’s take a look.

Here we have a 3D wireframe drawing that consists of 2 end arcs that represent the construction of the surface that we want to create as well as one long arc in the back and a chain of converted spline geometry in the front, representing the path of the surface that we want to create.

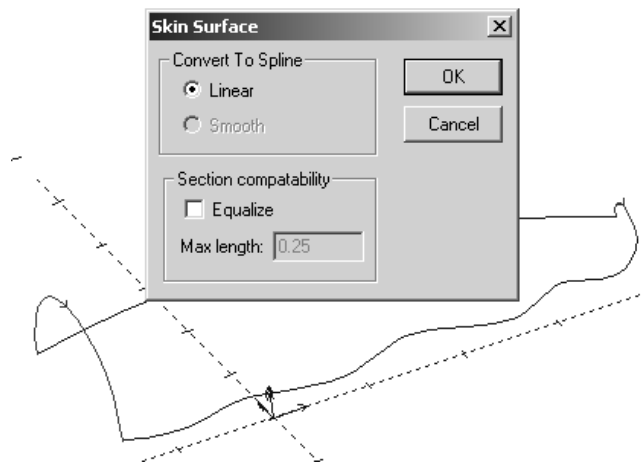


Because this is an Action-Object only function you must go to the SOLIDS menu, choose Surface Creation, Skin and then click NEXT to begin the chain selection on the cross section or construction geometry first.

You need to select the two (end) cross sectional geometry first. Then the second part will be to select the path (rails) geometry.

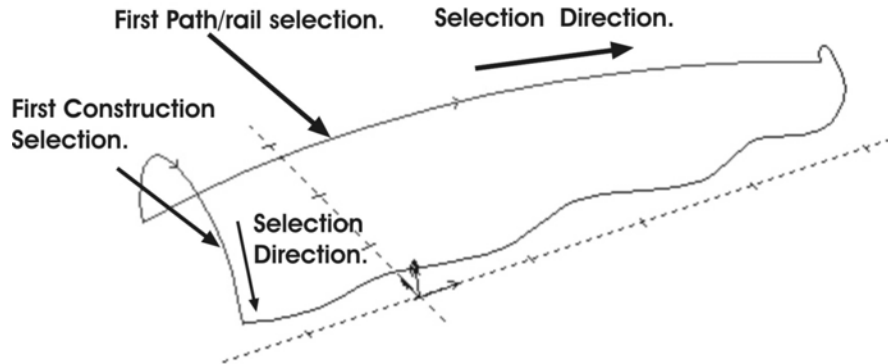


1. After selecting the skin option and the NEXT button from the Surface Creation box, you need to **chain-select** the left end first by clicking on it one time, pointing the directional arrow toward the front end of the arc and then clicking your left mouse button one more time to set the direction and then Right-click (or hit F3) to complete the selection.
2. Now chain select the second cross section in the same way with the selection directional arrow in the same direction as the first side. After right-clicking to set the direction and complete the second chain selection, RIGHT-click your mouse once more to open the Skin Surface box.

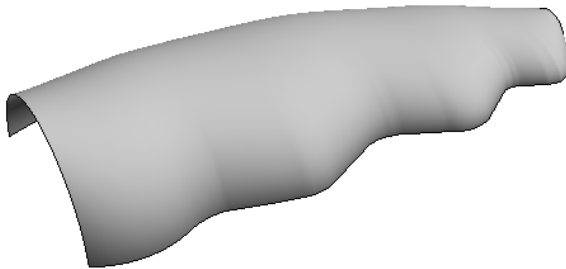


3. You will now have 2 options: Convert to Spline allows you to create a "Linear" surface or "Smooth" which creates a surface based off of converted spline geometry. In most cases "Smooth" will not be available unless you are working with converted spline cross sections. By clicking OK you will be able to move on with the path/rail selections.

It is important to select these in the correct sequence. You need to select the path that connects to the first construction cross section with the path selection direction going AWAY from the start of that construction entity or chain.



4. After clicking OK in the Skin Surface box, place your cursor on the first path entity or first entity starting that chain. In the above example you would click on the back "Path" arc, point your directional arrow going away from the left construction arc and click your left mouse button again and then right-click your mouse to complete that chain selection.
5. Now place your cursor on the far LEFT end of the FRONT chain and click on it one time. Point your directional arrow going toward the Right end of the chain and then take your cursor over to the last entity of that chain and click on it. Make sure that you select the last entity of that chain or hit your F3 key on the keyboard. This automatically generates the surface.

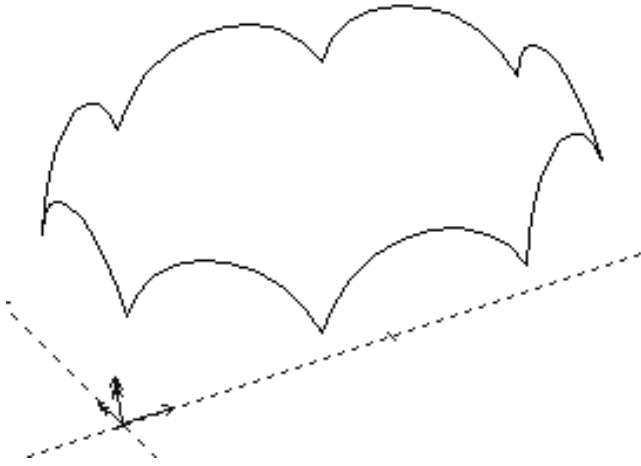


Skin Surface procedure:

1. Go to the Solids menu, choose Surface Creation, Skin and then click NEXT.
2. Chain-Select the first construct entity or chain that goes away from the first Path or rail selection entity or chain.
3. Chain-Select the next cross sectional construction entity or chain.
4. When finished chain selecting all of the cross sectional geometry entities or chains, RIGHT-click your mouse to access the skin surface box.
5. Make your selections and/or entries and click OK in the skin surface box.
6. Chain-Select the first path entity or chain that connects to the first cross sectional construction entity or chain with the directional arrow going away from that entity or chain.
7. Make the next path/rail chain selections and when finished the skin surface will automatically be created.

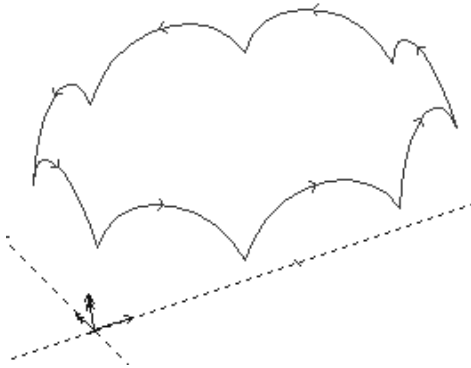
MULTI-SIDED PATCH: The **Multi-Sided Patch** Surface function allows you to create surfaces from closed chain geometry. This is an Action-Object function only and requires chain selection methods on the geometry being used. Let's take a look.

Here we have an example of 8 connected arcs that we want to create a surface for. Because this is a Action-Object function you will need to make sure that the geometry is not selected and then go to the Solids menu, choose Surface Creation, choose the Multi-Sided Patch option and then click NEXT.

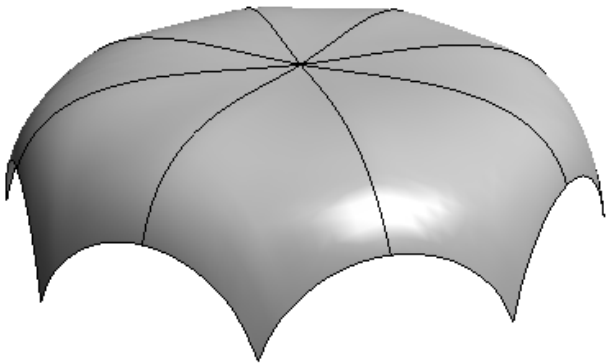


Once you click NEXT in the Surface Creation box the software function will be active and ready for you to make your series of chain selections. It is important to remember that when making the chain selections, you must do so with your directional arrows all going in the same direction and that you are making the right selection sequence without skipping entities.

Selection direction:



IMPORTANT: *When you have made your last chain selection you **MUST RIGHT-CLICK** your mouse a second time to create the surface.*



ACTION-OBJECT / OBJECT ACTION SURFACE CREATION FUNCTIONS:

Action Object: *This refers to selecting the function first and then performing the steps to carry out the operation.*

Object Action: *This refers to selecting geometry and then selecting a function to carry out the operation.*

The following is a list of each surface creation function and whether or not it is **Action-Object** or **Object-Action**.

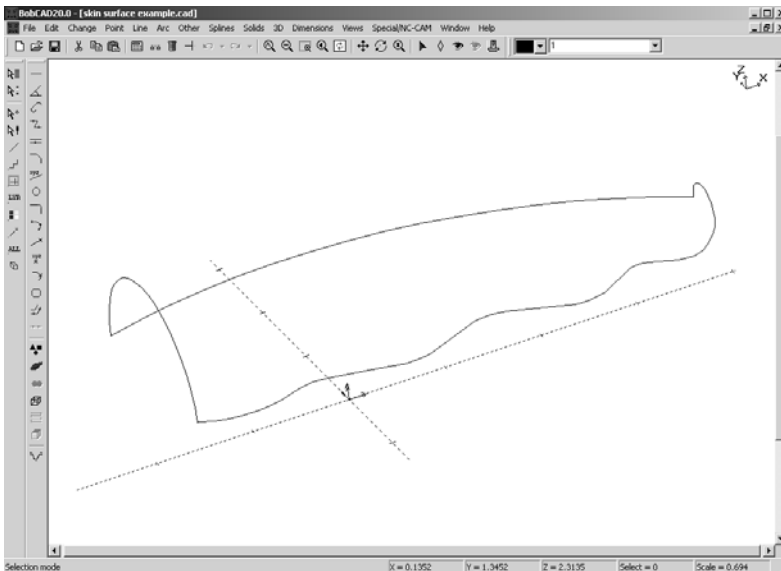
- **Rectangular Plane: Action-Object**
- **Circular Plane: Action-Object**
- **Planar: Object-Action**
- **Extrude Curve: Action-Object**
- **Extrude Surface: Object-Action**
- **Revolve: Action-Object/Object-Action**
- **3 Edge: Action-Object**
- **4 Edge: Action-Object**
- **Cross Section: Action-Object/Object-Action**
- **Skin: Action-Object**
- **Multi-Sided Patch: Action-Object**

SKIN SURFACE & TOOLPATH WITH G-CODE TRAINING LESSON:

Version 21 offers a series of surface creation functions. One of them is the SKIN function. This lesson will teach you how to use this function using an existing 3D wireframe file that is located in the Version 21 Samples Folder.

STEP 1

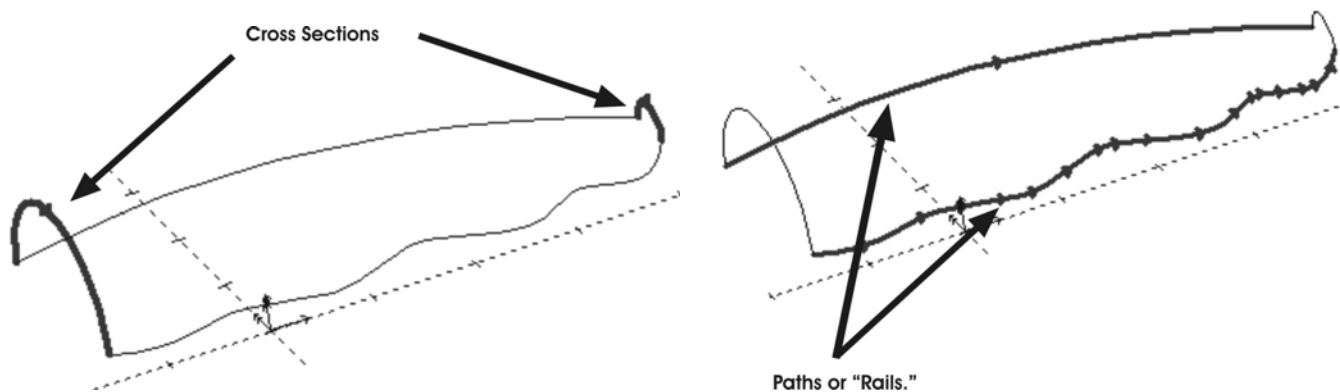
With a NEW drawing screen go to the FILE menu of the CAD side and select OPEN. Locate the BobCAD-CAM Version 21 Folder and double-click on it. Now locate the Samples folder and double click on it as well. Locate the file called, "**Skin surface example.cad**" and double click on it to open it in the CAD window.



Notice that the 2 cross sectional ends are black and the 2 paths are blue. The paths are also called "rails."

STEP 2

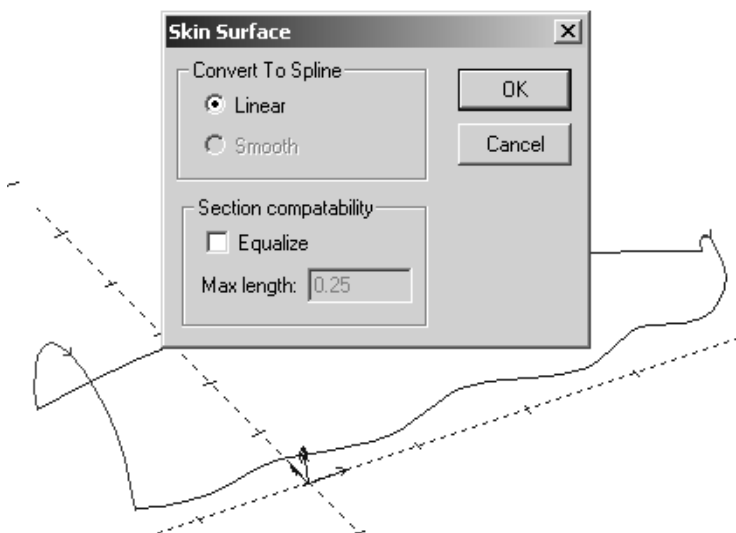
Because this is an Action-Object only function you must go to the SOLIDS menu, choose Surface Creation, Skin and then click NEXT to begin the chain selection on the cross section or construction geometry first. You need to select the two (end) cross sectional geometry first. Then the second part will be to select the path (rails) geometry.



STEP 3

After selecting the skin option and the NEXT button from the Surface Creation box, you need to **chain-select** the left end first by clicking on it one time, pointing the directional arrow toward the front end of the arc and then clicking your left mouse button one more time to set the direction and then **Right-click** your mouse to complete the selection.

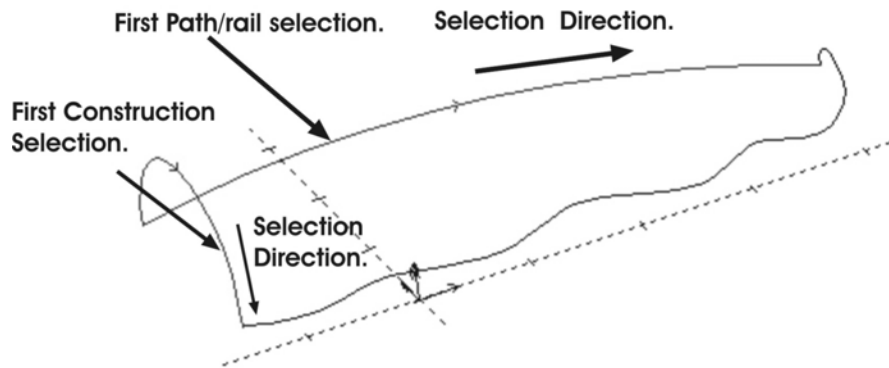
Now chain-select the second cross section in the same way with the selection directional arrow in the same direction as the first side. After right-clicking to set the direction and complete the second chain selection, **RIGHT-click your mouse once more** to open the Skin Surface box.



STEP 4

You will now have 2 options: Convert to Spline allows you to create a "Linear" surface or "Smooth" which creates a surface based off of converted spline geometry. In most cases "Smooth" will not be available unless you are working with converted spline cross sections. By clicking OK you will be able to move on with the path/rail selections.

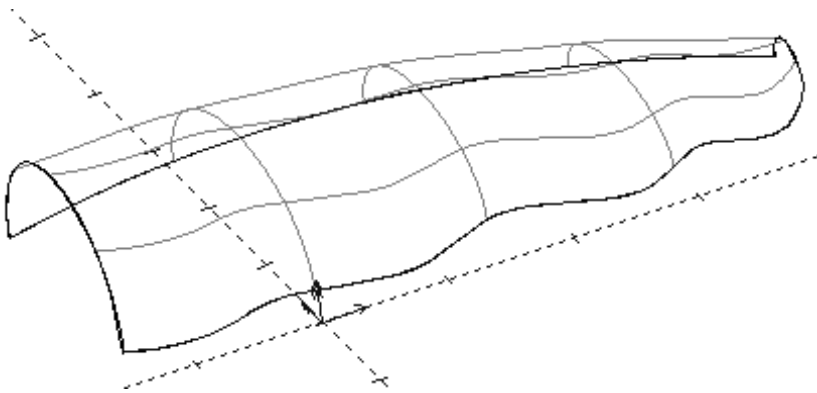
It is important to select these in the correct sequence. You need to select the path that connects to the first construction cross section with the path selection direction going AWAY from the start of that construction entity or chain.



STEP 5

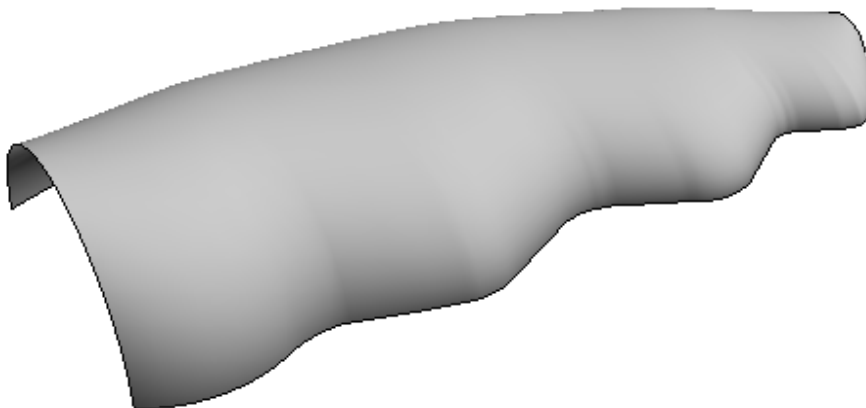
After clicking OK in the Skin Surface box, place your cursor on the first path entity or first entity starting that chain. In the above example you would click on the back "Path" arc, point your directional arrow going away from the left construction arc and click your left mouse button again and then right-click your mouse to complete that chain selection.

Now place your cursor on the far LEFT end of the FRONT chain and left-click on it one time. Point your directional arrow going toward the Right end of the chain and then take your cursor over to the last entity of that chain and click on it. Make sure that you select the last entity of that chain or hit your F3 key on the keyboard. This automatically generates the surface.



STEP 6

Now place your cursor on an interior line of the surface and click on it to select it (*it may already be selected if your auto Preselect option is checked in the environment settings of the CAD File menu*). Now go to the Solids menu and select RENDER.



STEP 7

Now, select the CLOSE button in the render window. If the CAM side of the software is open, this is fine. If not, go to the Special/NC CAM menu at the top menu bar and select INSERT NC. Locate the **HAAS VF Series** post processor file and double click on it to open the CAM. Now go to the CAD side of the screen and select the View All icon from the main icon bar at the top of the software.



Now, have a look at the CAM side. You will see the CAM icon bar that runs down the LEFT wall of the CAM. Toward the bottom of this CAM icon bar you will see a button that has 3D on it. Click that 3D button to turn the CAM 3D mode on for machining.



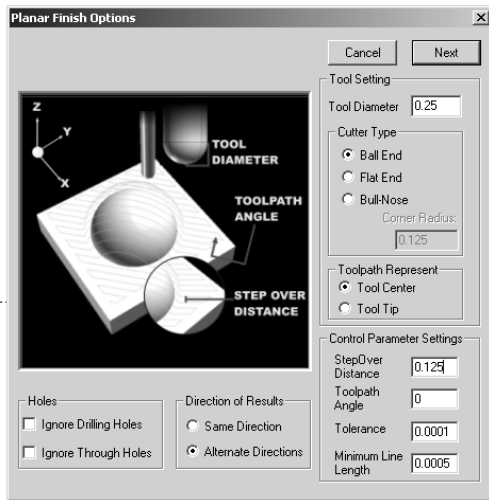
Now go to the TOOL menu on the CAM side and select TOOL CHANGE. In the tool change box enter 1 under Tool Number and enter .25 under Tool Diameter. Enter 2000 under Spindle Speed, 3 under Spindle Direction which represents Clockwise and 8 under Coolant.

A screenshot of the 'Tool Change' dialog box. It contains several input fields: 'Tool Number' with '1', 'Tool Diameter' with '.25', 'Operation Info' (empty), 'Spindle Speed' with '2000', 'Spindle Direction' with '3', and 'Coolant (8 - On / 9 - Off)' with '8'. There are 'OK' and 'Cancel' buttons on the right side.

NOTE: The HAAS VF Series post configuration will be different from other posts. If you have a HAAS CNC machine you can use this post configuration. If your post is not listed in the V20 demo, we either have it or can create it at no additional cost to the customer.

STEP 8

With the surface selected (if it isn't selected, select it now) go to the SOLIDS menu and select GENERATE TOOLPATH. (If it is not available for selection, then you didn't select the surface and need to select the surface first.) This will open the Solid Machining box. Select the Planar option in the Solid Machining box and then click NEXT to open the Planar Cut Options box.



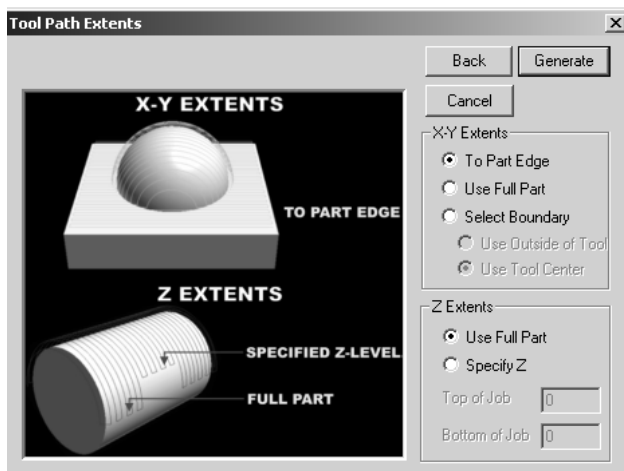
Enter the following:

- .25 for the Tool Diameter.
- Select the Ball End cutter type.
- Check the Tool Center option.
- Enter .125 for the Step Over distance.
- Enter 90 for the toolpath angle. This will set the toolpath going from right to left across the surface.
- Enter .001 for the Tolerance.
- Enter .001 for the Max Line Length so that we get a fine toolpath that follows the surface contour closely.
- DO NOT check either IGNORE Hole options even though there are none.
- Select the Alternate Direction option as we do not want to climb mill this surface.

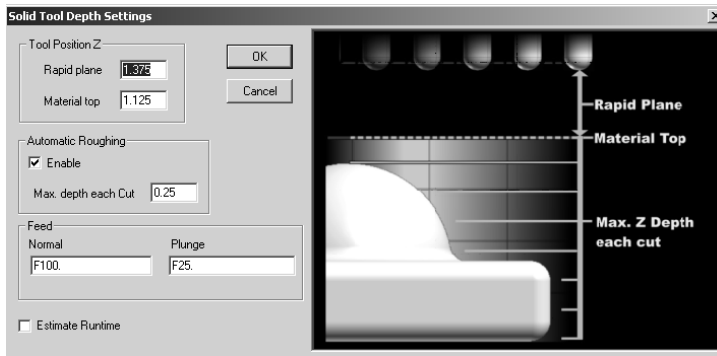
Click the NEXT button to advance in the wizard.

STEP 9

This opens the Toolpath Extents box.



Select the “To Part Edge” cutting option as well as “Use Full Part” under the Z Extents section and click GENERATE to produce the toolpath across the surface as directed and open the Solid Tool Depth Settings box when finished.



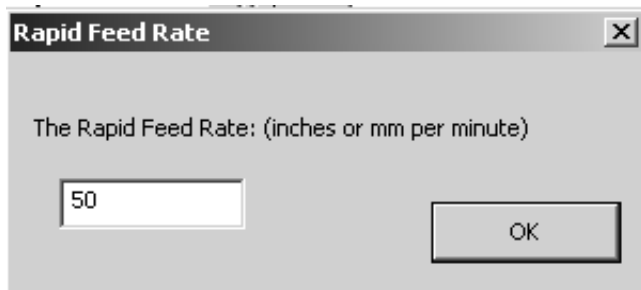
The Rapid Plane is the clearance and this box will have a value in it as the top of this surface is not at Z0. You will also have 1.125 as the top of material. This is fine as this is the material top for this example.

Check the **ENABLE** button under Automatic Roughing and enter .25 as the max depth of each cut. Go ahead and click OK. This will automatically generate the G-Code program for a HAAS VF Series control. Other post processors are available as needed.

STEP 10

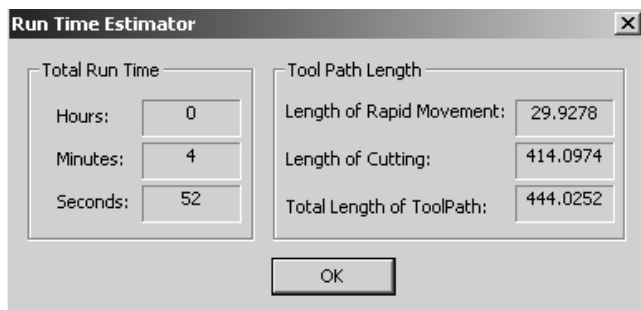
Go to the CAM EDIT menu and choose SELECT ALL. This will highlight all of the code. Now go back to the CAM EDIT menu and click on **Run Time Estimate**.

The new box will prompt you for a feed rate. You can enter in the feed rate that you will be using.



This will be in inches or millimeters per minute depending on what you have the software set to in the software environment settings (*located in the CAD side edit menu under environment and coordinates tab*). Click OK.

This will produce your run time.

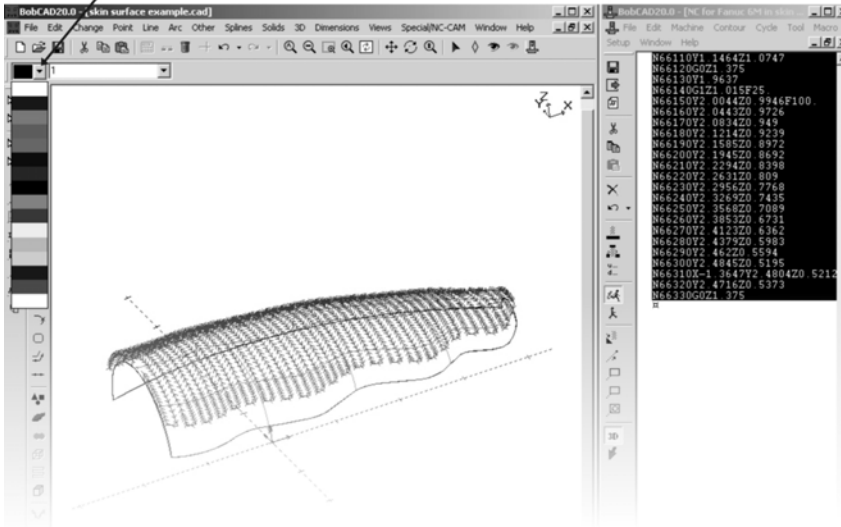


Click OK to exit the run time box.

STEP 11

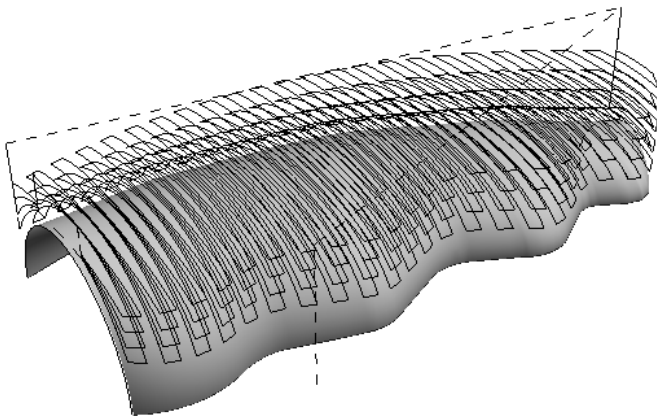
Now go to the CAD side of the screen with the g-code still selected and change the active color by selecting the active color drop-down arrow and choosing a different color than black.

Click the small drop-down arrow to access the color bar.
When you do this, everything you draw from that point forward
Until changed again will be in the new color.



STEP 12

Now go back to the CAM EDIT menu and select GEOMETRY FROM NC. This will backplot the g-code toolpath over the surface in the new color you chose. Now go to the **CAD** EDIT menu and choose Select Entities and then SELECT ALL. Everything will highlight. Now go to the SOLIDS menu (also on the CAD side) and choose RENDER.



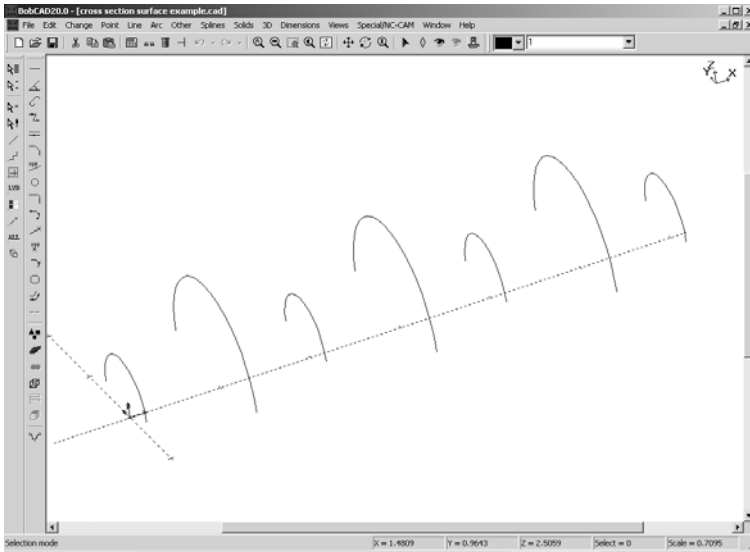
This is basically the roughing cycle. You would now perform a tool change and go back to generate your finishing toolpath and g-code for finishing in the same program if this is how you wanted to program the part. As you become familiar with how to create parts and machine them you will find BobCAD-CAM Version 21 to be both easy to use and fast in 2D or 3D part programming!

CROSS SECTION SURFACE TRAINING LESSON:

The Version 21 software offers a large variety of surface creation functions. The Cross Section surface is one of them and this lesson is going to show you how to use this function with an existing example file.

STEP 1

With a NEW drawing screen open go to the CAD FILE menu and select OPEN. Locate the BobCAD-CAM Version 21 folder and double click on the Samples folder to open it. Now locate the file called, “**Cross section surface example.cad**” and double click on it to open it in the CAD window.



This is a series of cross sectional geometry (arcs) that we want to create a surface for so that we can machine it in the CAM.

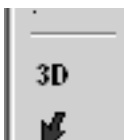
STEP 2

Let's go ahead and open the CAM side of the software by going to the Special/NC CAM menu and selecting INSERT NC. Locate the HAAS VF Series post processor and double click on it to fully open the CAM on the right side of the screen.

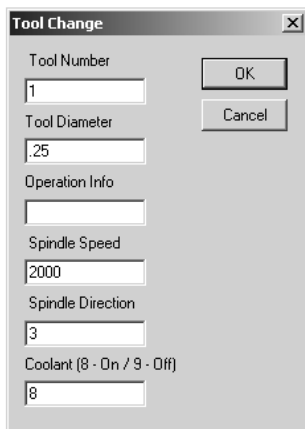
Now with the CAM open, go to the CAD side of the screen and click the View All icon from the main icon bar at the top of the software.



Now, have a look at the CAM side. You will see the CAM icon bar that runs down the LEFT wall of the CAM. Toward the bottom of this CAM icon bar you will see a button that has 3D on it. Click that 3D button to turn the CAM 3D mode on for machining.



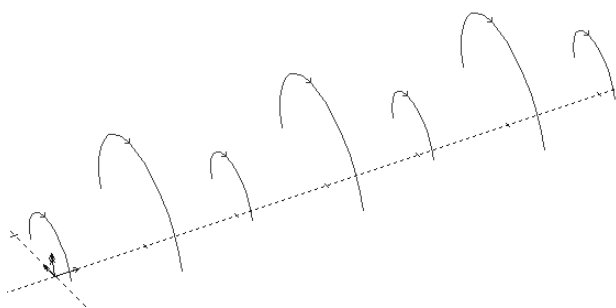
Now go to the TOOL menu on the CAM side and select TOOL CHANGE. In the tool change box enter 1 under Tool Number and enter .25 under Tool Diameter. Enter 2000 for the Spindle Speed, 3 for the Spindle Direction which represents clockwise motion and 8 for Coolant and click OK.



NOTE: The HAAS VF Series post configuration will be different from other posts. If you have a HAAS CNC machine you can use this post configuration.

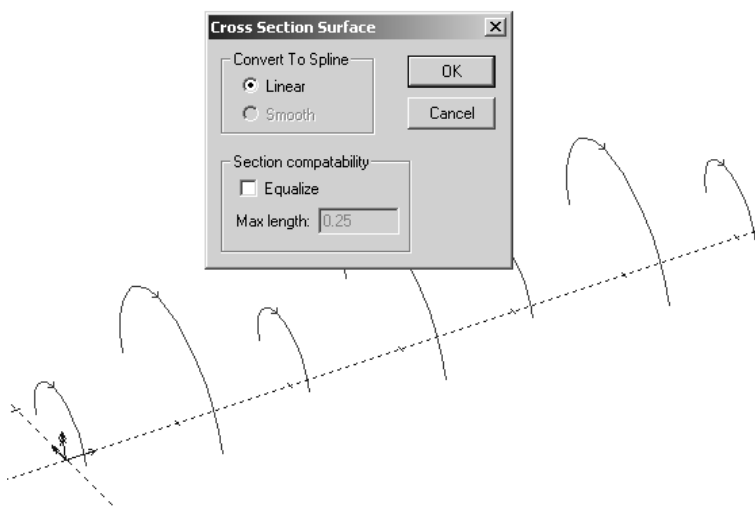
STEP 3

Go to the CAD side of the software and position your cursor at the bottom FRONT position of the far LEFT arc and click on it. The direction arrow of the arc will be pointing toward you. Now hold down the CONTROL KEY on your keyboard and click the same location of each arc going from the next across to the right until each arc is selected with all directional arrows going in the same direction.

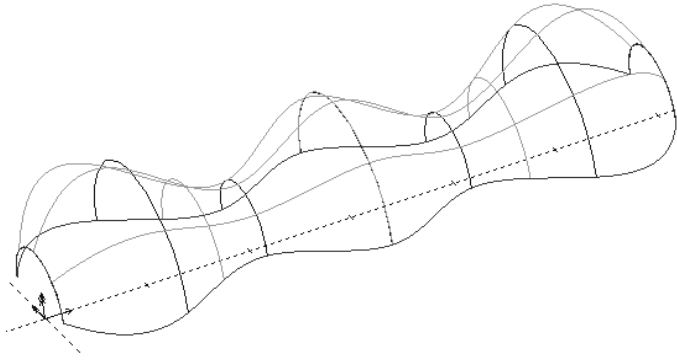


When this is done go to the Solids menu, choose Surface Creation, Cross Section and then click the NEXT button.

This brings up the Cross Section box.



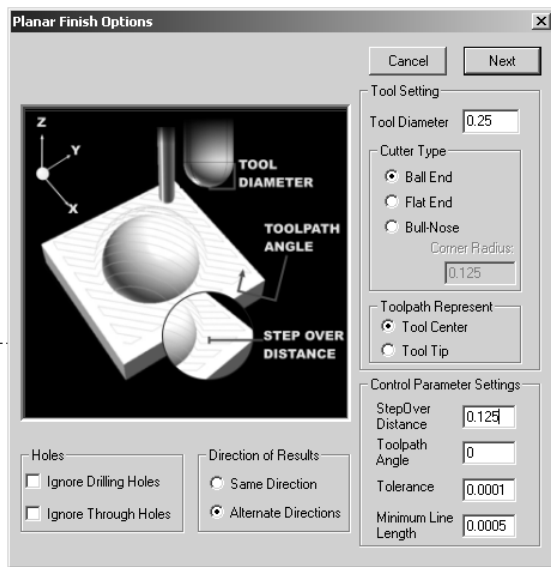
We do not need to “Equalize” these entities. Therefore, do not check this box. Simply click the OK button to generate the surface.



STEP 4

At this time the surface may be selected (highlighted in red). If it isn't, go ahead and click on one of the surfaces edge lines or an interior line of the surface. The interior lines of this surface run lengthwise from end to end.

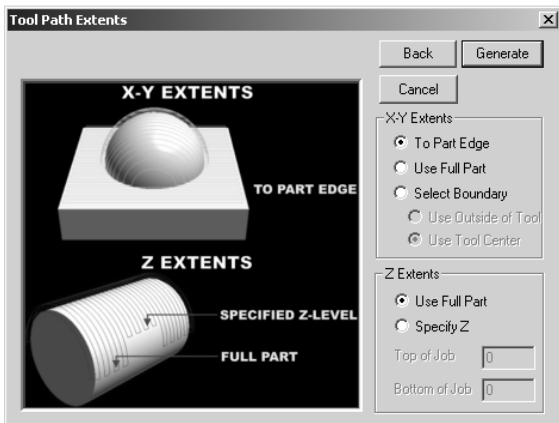
Now go to the Solids menu and select Generate Toolpath. Select the Planar option and click NEXT to open the Planar Finish options box.



Enter the following:

- .25 as the Tool Diameter.
- Select the Ball cutter type.
- Select the Tool Center option.
- .125 as the Step over Distance.
- 90 as the Toolpath Angle.
- .001 as the Tolerance as we want a nice and smooth toolpath.
- .001 as the Minimum line length as we want the toolpath to conform closely to the surface contour
- DO NOT check either IGNORE Hole options even though there aren't any.
- Select the Alternate Directions option as we are not climb milling this surface.

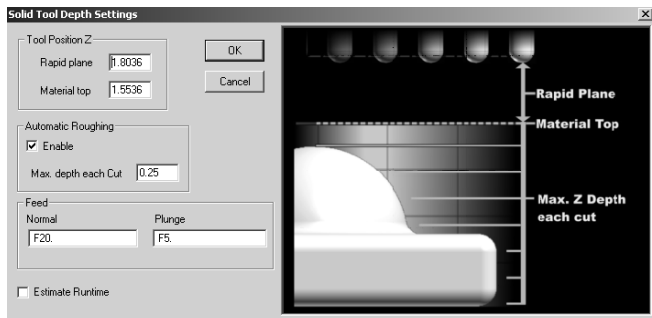
Click the NEXT button.



Select the “To Part Edge” option and the “Use Full Part” option under the Z extents section and click the GENERATE button to create the toolpath and, when finished, open the Solid Tool Depth Settings” box.

STEP 5

The Rapid Plane (clearance) will provide a clearance value based off of the Material Top of where this part is located on the CAD screen.



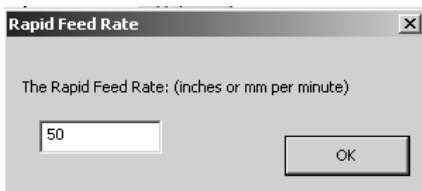
Check the ENABLE button under Automatic Roughing and enter .25 as the maximum depth (step down) for each cut.

Now click the OK button to generate the g-code program.

STEP 6

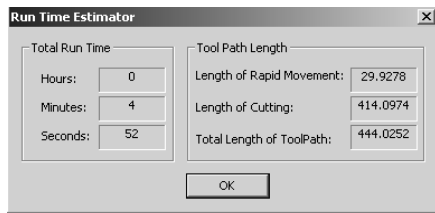
Go to the EDIT menu of the CAM and click on the SELECT ALL function which will highlight all of the G-Code. Now go back to the CAM EDIT menu and click on **Run Time Estimate**.

The new box will prompt you for a feed rate. You can enter in the feed rate that you will be using.



This will be in inches or millimeters per minute depending on what you have the software set to in the software environment settings (*located in the CAD side edit menu under environment and coordinates tab*). Click OK.

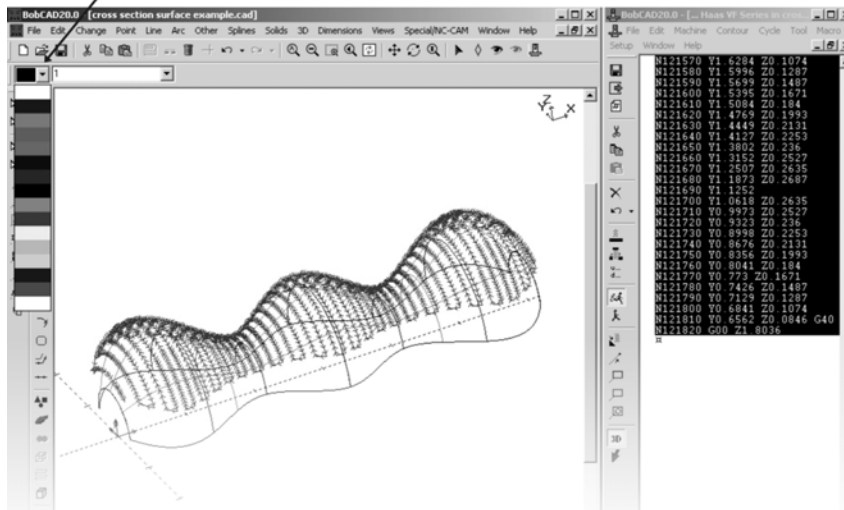
This will produce your run time.



Click OK to exit the run time box.

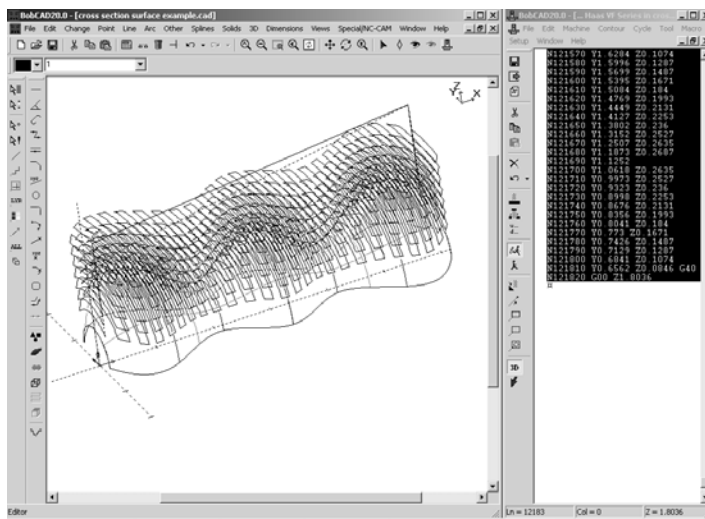
Now go directly to the active color bar on the CAD side, click the small drop down arrow and click a different color than black.

Click the small drop-down arrow to access the color bar.
When you do this, everything you draw from that point forward
Until changed again will be in the new color.



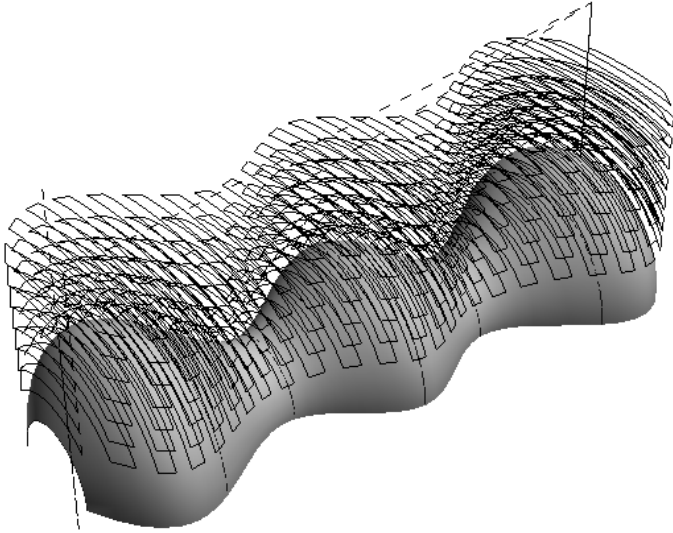
STEP 7

Now that you have changed the active color, go to the CAM EDIT menu and select GEOMETRY FROM NC to backplot the toolpath from the g-code program.



Now go to the **CAD** EDIT menu and choose Select Entities and then **SELECT ALL**. Everything will highlight.

Now go to the **SOLIDS** menu (also on the CAD side) and choose **RENDER**.



This is basically the roughing cycle. You would now perform a tool change and go back to generate your finishing toolpath and g-code for finishing in the same program if this is how you wanted to program the part. As you become familiar with how to create parts and machine them you will find BobCAD-CAM Version 21 to be both easy to use and fast in 2D or 3D part programming!

EXTRUDE CURVE TRAINING LESSON:

Version 21 offers extrusion options for open and closed chain geometry. This lesson is going to teach you how to use the Extrude Curve function.

STEP 1

With a NW drawing screen go to the **POINT** menu and select **Coordinates**. Perform the following actions.

- Enter X0, Y3, Z0 and click the **CONTINUE** button.
- Enter X1, Y2.5, Z0 and click the **CONTINUE** button.
- Enter X2, Y2.5, Z0 and click the **CONTINUE** button.
- Enter X3, Y2, Z0 and click the **CONTINUE** button.
- Enter X4, Y1.5, Z0 and click the **CONTINUE** button.
- Enter X5, Y1.5, Z0 and click the **OK** button.

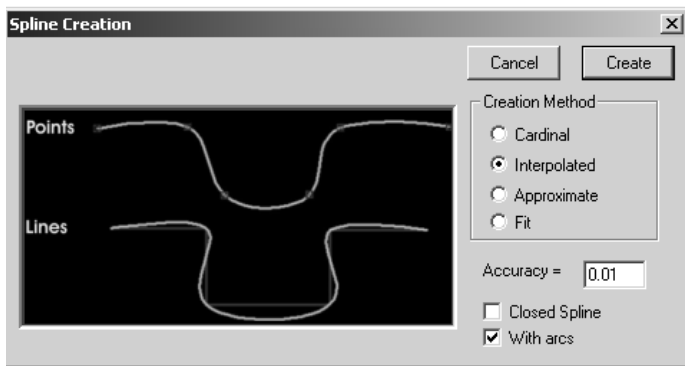
STEP 2

Place your cursor in an empty area of the drawing screen and click your mouse to make sure that the points are deselected. Now click the **VIEW ALL** button from the top toolbar.



STEP 3

Starting on the first point that you created, hold down your **CONTROL** key on your keyboard and click on each of the 6 points going from left to right. When they are all selected, go directly to the **SPLINES** menu and select **CREATE** to open the spline box.



Choose the INTERPOLATED option, enter an accuracy value of .01, check the “With Arcs” option, make sure that the “Closed Spline” option is NOT selected and click the CREATE button.

This will draw the spline geometry.

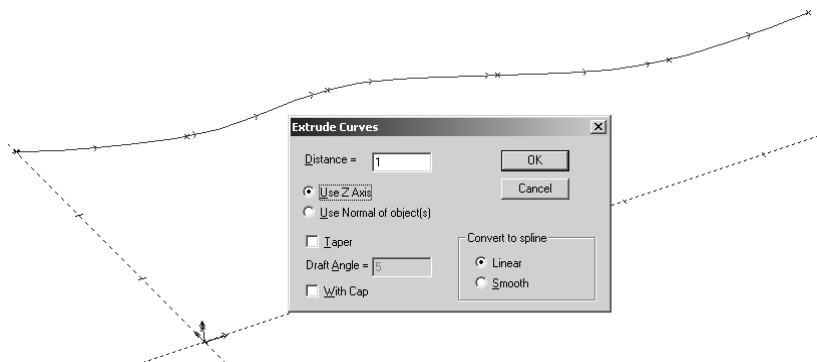
STEP 4

Place your cursor in an empty area of the drawing screen and click your left mouse button to make sure that everything is deselected. Now go to the 3D menu and turn your 3D ON. Select the View All icon from the main toolbar.



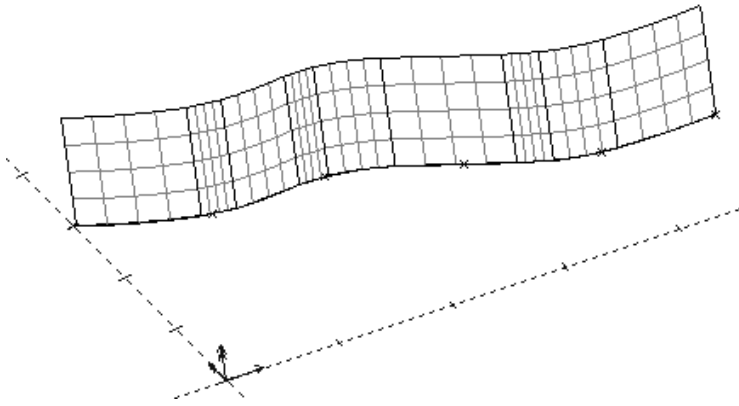
Now go to the SOLIDS menu and select SURFACE CREATION. Select the Extrude Curve option and click the NEXT button in the upper right corner of this box.

Now you are in the Chain Selection mode. Go to the left side of the spline, place your cursor on the first arc segment of the spline and click on it. It will turn purple and you will see a directional arrow at the mid-point of the arc segment. Point this arrow so that it is going toward the right and click your left mouse button one time to set the direction. Now hit your F3 key on your keyboard to select the entire chain and bring up the Extrude Curve box.



STEP 5

Enter a distance of 1, representing 1 inch in positive Z. Select the “Use Z Axis” option as the spline is actually 2D (X-Y plane). Taper and “With Caps” should NOT be checked. Select the **Linear** option as we are working with converted spline geometry already and then click OK to create the surface.



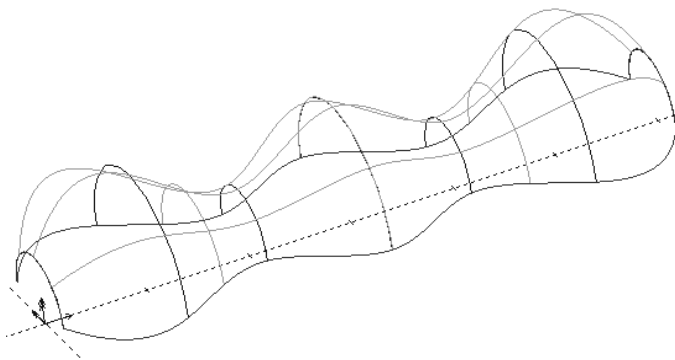
This is how you use the Extrude Curve option for an open ended chain. You can also use this for closed chained geometry.

PROJECT CURVES ON SURFACES:

Version 21 offers you the ability to project vectorized text or 2D geometry onto a surface for engraving and 3-Axis machining. Here's how you do it:

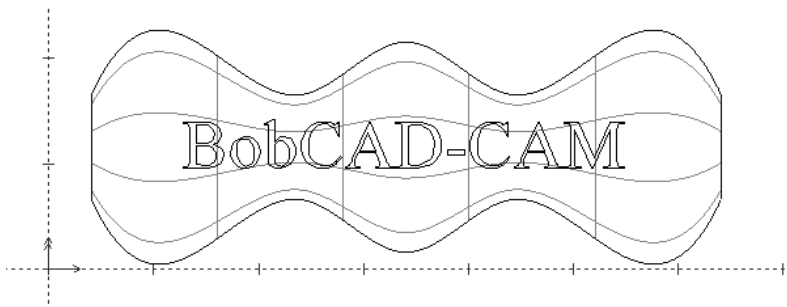
STEP 1

Create a surface or import a surface or solid. Below, we are using the cross section surface.



STEP 2

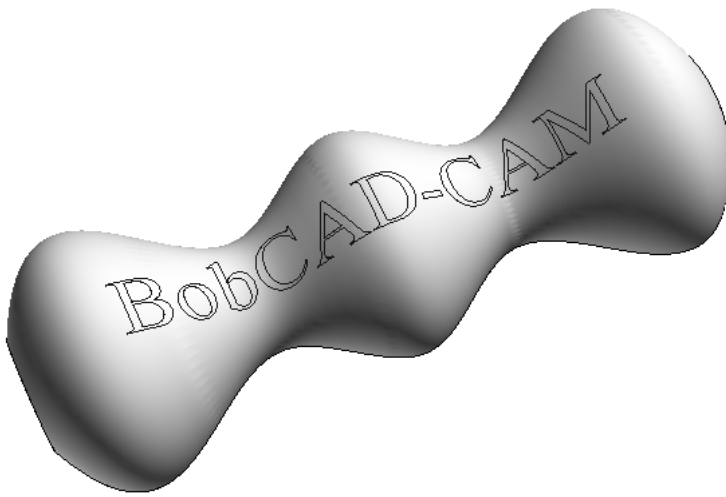
Create your text, placing it within the boundaries of the surface itself and vectorize it.



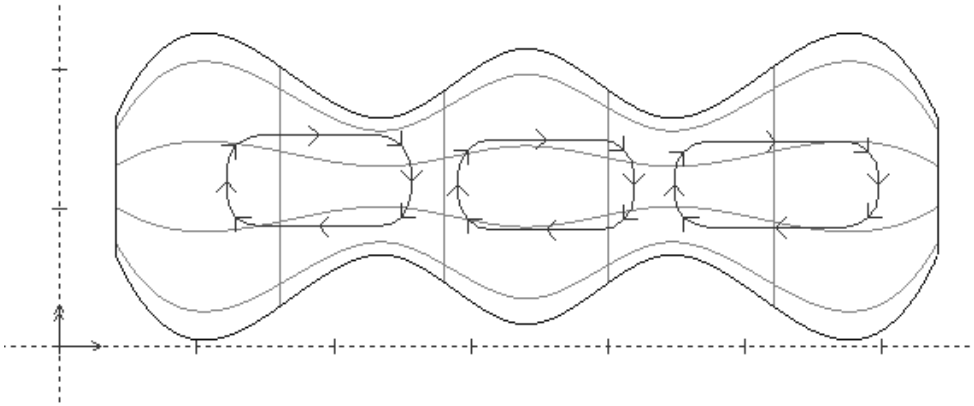
STEP 3

Select the text ONLY and nothing else. Then go to the Solids menu and choose Project Curves on Surface. Then click on one of the interior lines that represents the surface. This will project the curves/text or profile geometry onto the surface.

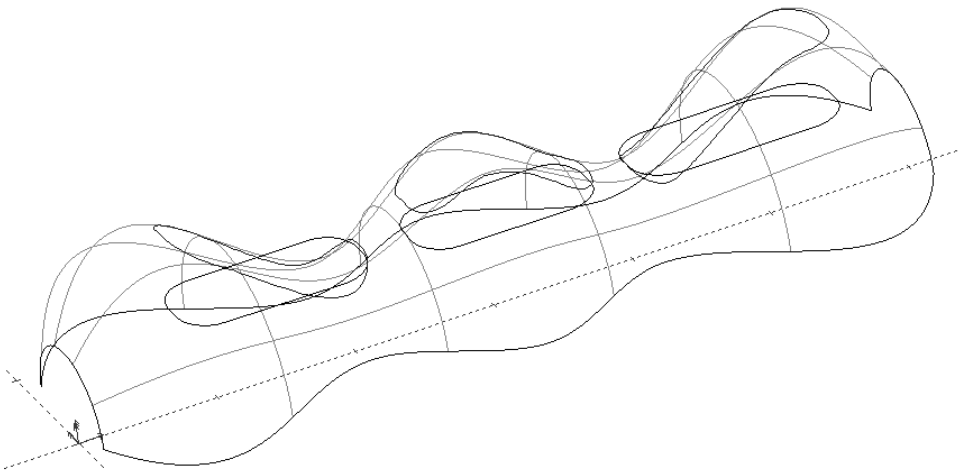
The result:



You can also select multiple entities or chains for projection at once.



In this example you can see that we selected 3 closed chain profiles and projected them.



Do note that when you project geometry, the original geometry is left on the screen. Therefore, you may want to change to a different active drawing color before using the projection function so that you can select by color and delete or blank out the original geometry after projection.

SURFACE MODIFICATIONS:

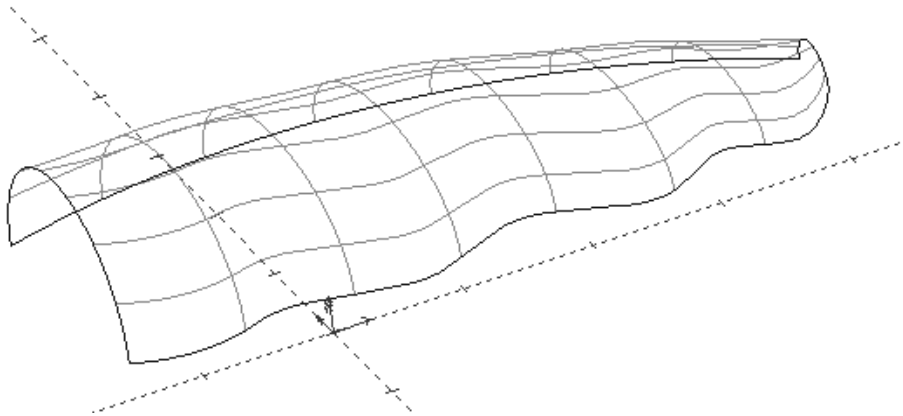
Version 21 provides functionality for limited surface modification.



The two options in the sub-menu are:

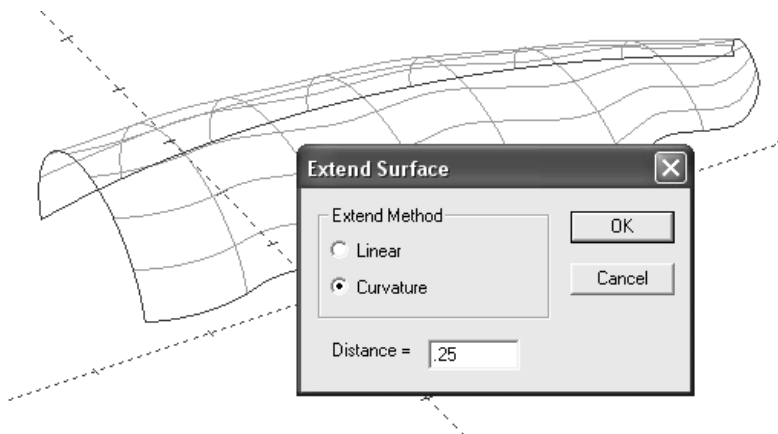
- **Reverse Surface.** This allows you to reverse a surface normal, the direction of the surface. If you have performed stitching and un-stitching on a model and then have difficulty performing Booleans or filleting operations you may use this function to reset the surface normals. First, select the model or surfaces and then use the RENDER function at the bottom of the Solids main menu and render the surface normals to see which one(s) need to be reversed.
- **Extend Surface.** This function allows you to extend a surface. Results may vary depending on the surface topology. If the result seems to warp the surface, you must reduce the extend distance value. Let's take a closer look.

Here we have a surface:

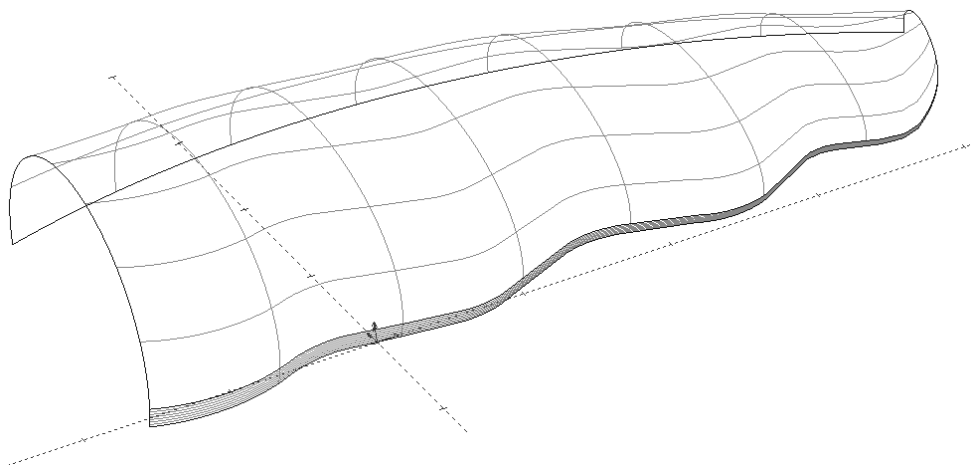


Let's say that we wanted to extend the front curved edge. First you would go and select the function and then click exactly on the front edge.

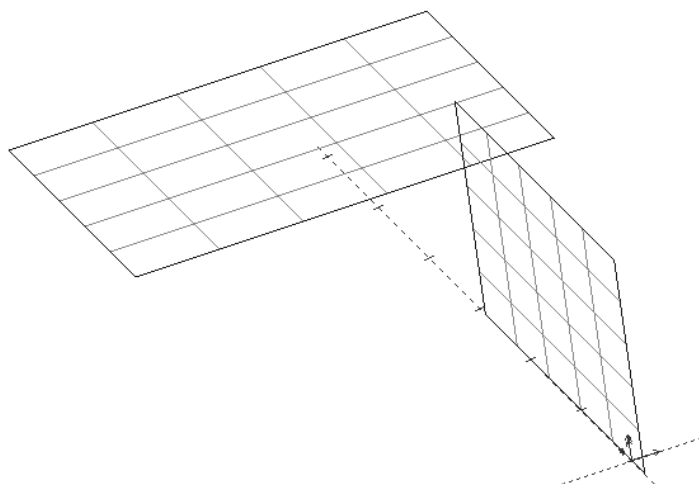
Once you click on the edge, the Extend Surface box appears with 2 options.



LINEAR is used for flat surfaces that are more 2D. **CURVATURE** is used for more 3D type surfaces. In this example we chose "Curvature" and entered a distance of .25. By clicking OK you can see the result below.

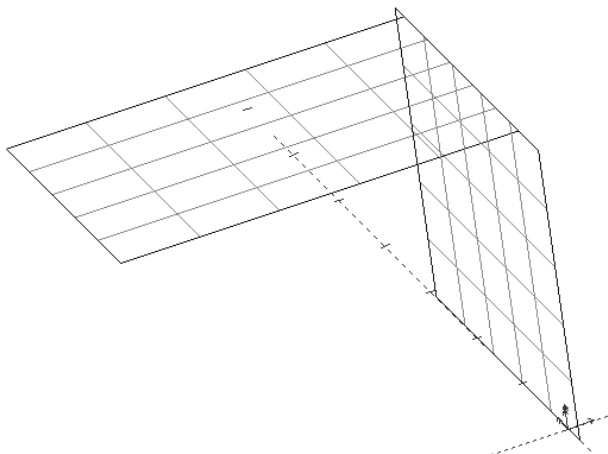


You can see that the surface was extended. In the next image you can see 2 surfaces.



The vertical surface needs to be extended. To do this we simply select the Extend Surface function and then click right on the top edge of the vertical surface and the box will appear.

By entering our distance value, selecting the Linear option and clicking OK we get the desired result.



Section 26. Boolean Operations

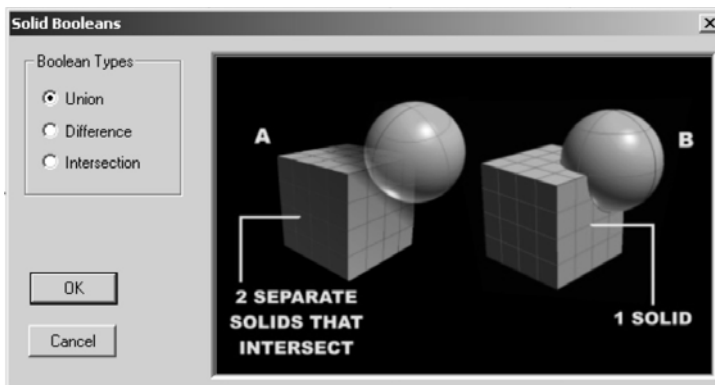
Boolean operations allow you to combine 2 solids or surfaces into a single solid, or subtract one from another or create the union of 2 solids.

When creating a toolpath from a solid or surface the toolpath function accepts only one solid or surface as input. Thus you have to use the Boolean function in order to create a single solid from multiples first. Version 21 offers 3 main Boolean operations. They are:

- **Union**
- **Difference**
- **Intersection**

A basic rule with Boolean operations is that you must have at least 2 surfaces or solids selected in order for this function to be active.

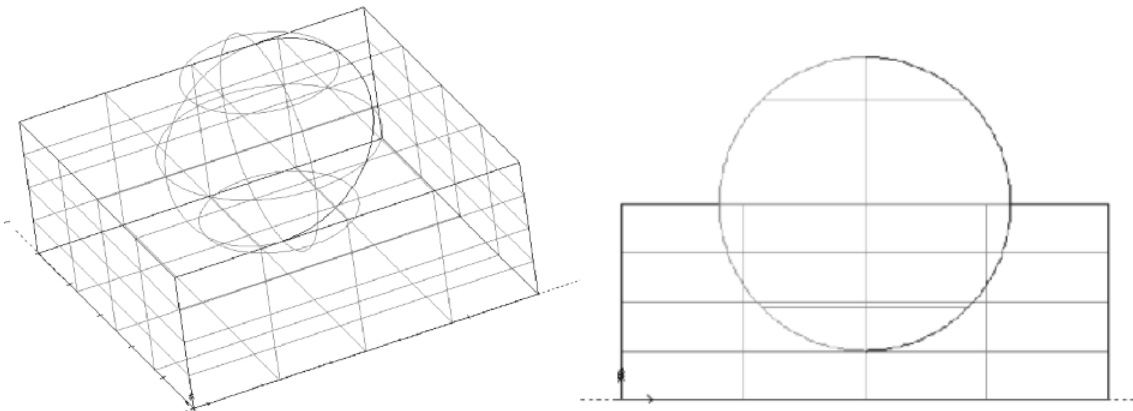
Let's move on and have a look at the Solid Boolean dialog box.



Each one of the Boolean functions will show an image that describes the function. In the image above, you can see the Union function and its corresponding image description. Let's take a look at each of these Boolean Operations individually.

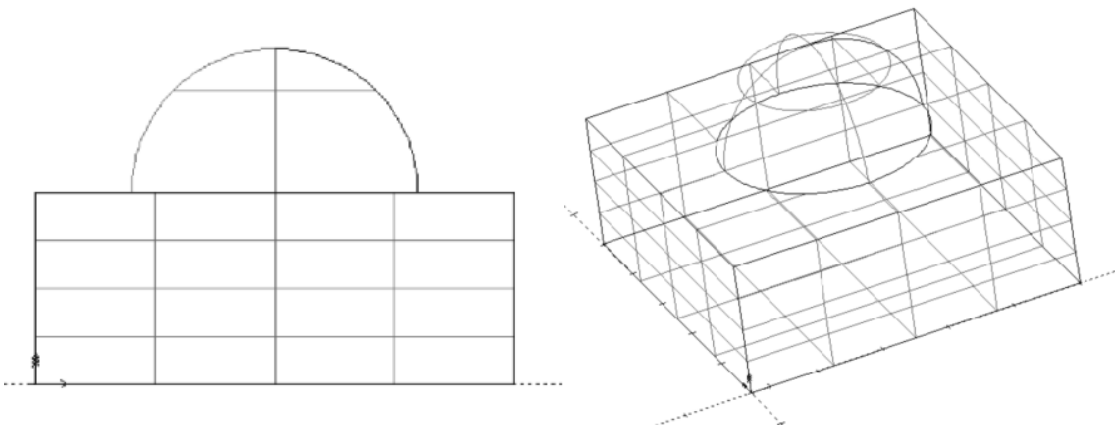
UNION: This allows you to merge two solids together to create one water tight solid. Have a look at the two solids below.

2 Solids:



As you can see, the two solids intersect. By using the Boolean Union function on them, you will notice the change in the next images. Both solids were merged to make one solid.

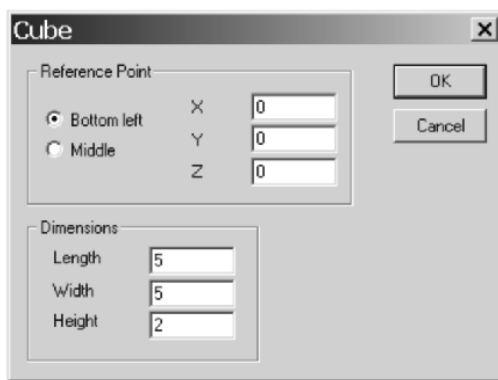
After the Union function was used:



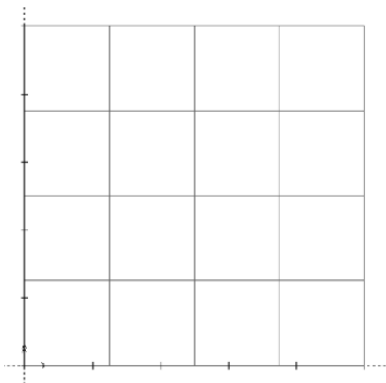
BASIC UNION LESSON:

STEP 1

With a new screen, select the Solids menu and choose cube. Under Reference Point select Bottom Left. In the Dimensions section, enter 5 for the length, 5 for the width and 2 for the height. Then go ahead and select OK.

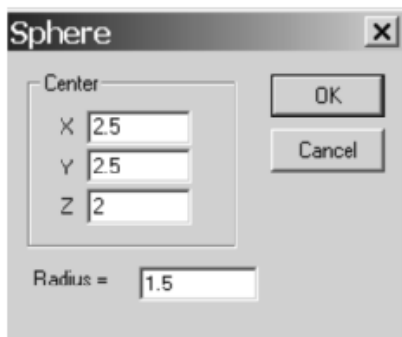


Result in a 2D view:

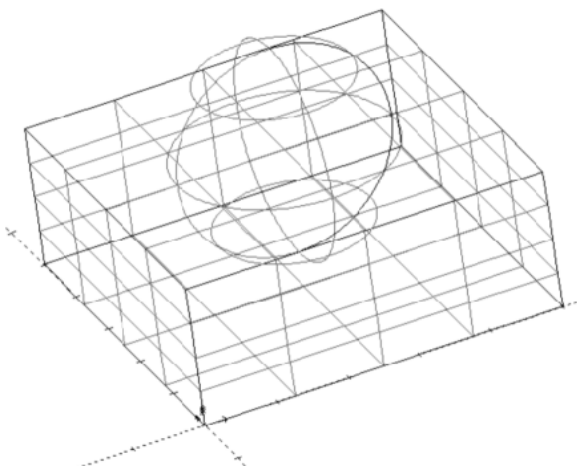


STEP 2

At this time, select the 3D menu and turn the 3D mode on so that you have a 3D view of the cube. Once you have done this select the solids menu and then Sphere. In the Sphere box enter 2.5 for X, 2.5 for Y and 2 for Z. Enter 1.5 for the Radius and then select OK to draw the sphere.



The result in a 3D view:

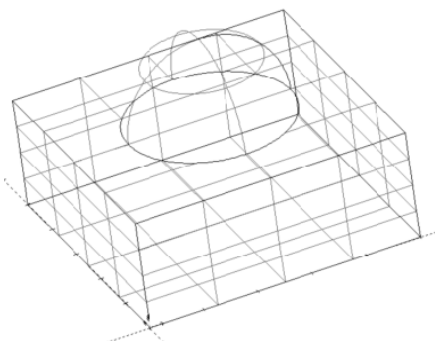


STEP 3

Now we need to select the two solids. If there were other solids or entities on the screen you would need to click on the edge or interior lines of one solid, then by holding down the shift key and clicking the edge or interior lines of the second solid they would both highlight and be selected. In this case, because there are only two solids on the screen, you can click on your Select ALL icon from the default selection toolbar. Both solids will highlight.

Now select the Solids menu and Booleans. When the new dialog box appears, select Union and then OK.

You will get the result as seen in the image below.



The two solids have been merged together and the intersecting part is now gone.

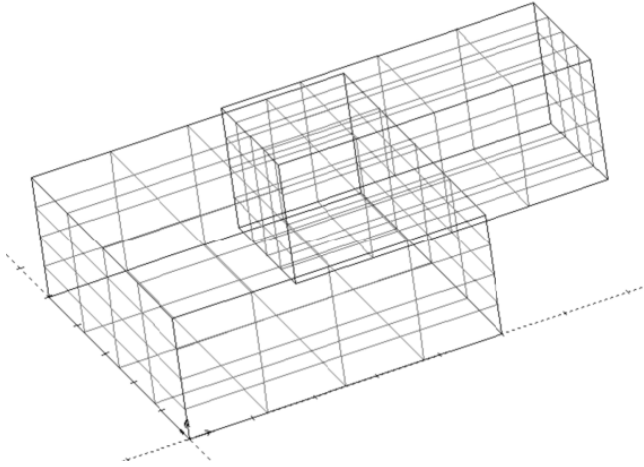
This concludes the basic union lesson.

Basic Rules for using the Boolean Union function:

1. The two solids used must intersect in some way.
 2. You must have two separate solids selected before this function will be available.
 3. You can only perform the union function on two solids at a time. In other words, if you have five solids that you want to union, you will need to perform this function on each one at a time.
-

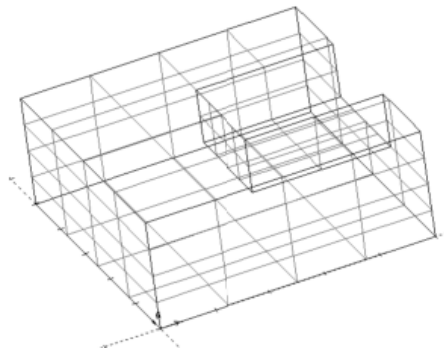
DIFFERENCE: The Boolean Difference function will allow you to use one solid against another to create a difference in the first one that you selected. You see, the order in which you select the two solids will determine the end result.

Here are two solids that intersect:



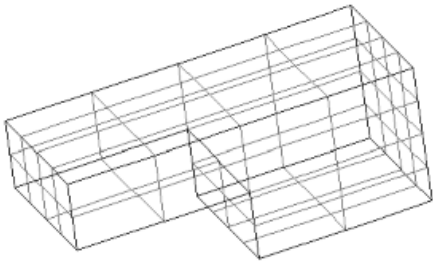
These solids are independent from each other in the image above. Now let's say that we wanted the bottom solid to be changed by using the top solid and the Boolean Difference function. Since we want the bottom solid to be changed, we would select that shape first. Once this was done, we would hold down the shift key and select the second, top solid. When they are both selected we would choose the Solids menu, Booleans and then select the Difference function. This would automatically create the difference and the shape would then be made into one solid.

The result of the Boolean Difference operation:



Now, let's say that we had selected the top solid first and then the bottom solid second before executing the difference operation.

We would have ended up with the result as seen below.



Therefore as a rule, remember that the result of the Boolean Difference function will be determined by the order in which you made the initial selection of your solids.

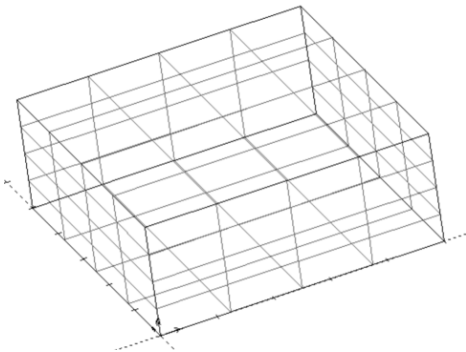
Basic Rules for using the Boolean Difference function:

1. The two solids used must intersect in some way.
2. You must have two separate solids selected before this function will be available.
3. You can only perform the Difference function on two solids at a time. In other words, if you have five solids that you want to modify, you will need to perform this function on each solid at a time, keeping in mind that the solid that you select first will be the solid changed by the second solid that you select.

BASIC DIFFERENCE LESSON: Follow this lesson and learn to use the Boolean Difference function.

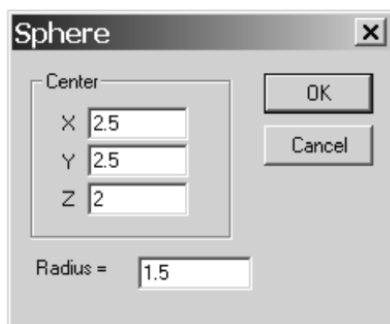
STEP 1

With a new drawing screen select the Solids menu and then choose Cube and select OK. When the cube box appears make sure that Bottom Left is selected as the reference point. The X, Y and Z coordinates should all be 0. Enter 5 for length, 5 for width and 2 for height. Now select OK to draw the first solid. Now select the 3D menu and turn your 3D ON. You may want to also select the View All Icon from the main toolbar so that you can see the shape correctly.

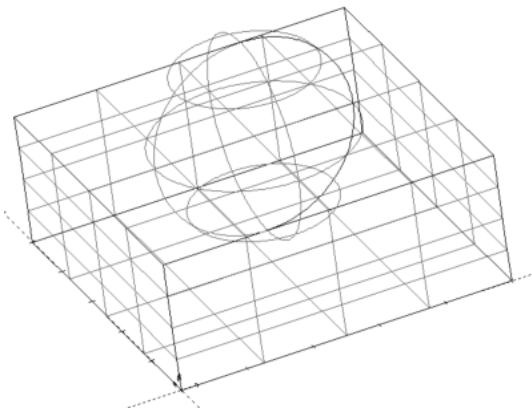


STEP 2

Select the Solids menu and then Choose Sphere and click OK to open the sphere box. Enter 2.5 for X, 2.5 for Y and 2 for Z. Enter 1.5 for Radius and then select OK to draw the sphere.



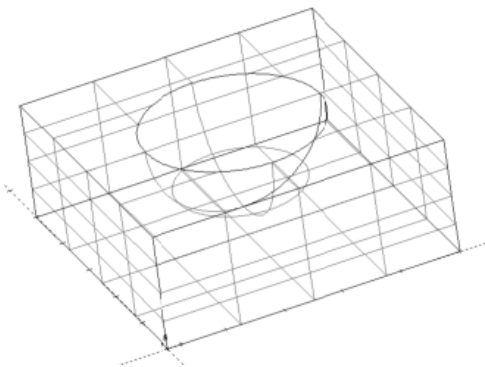
The result after drawing the solid sphere:



STEP 3

Now we are going to select the square first. Place your cursor on either the edge lines or interior lines of the square and click your mouse button to select it. Now hold down the SHIFT key and select the sphere. When they are both selected let go of the shift key and select the Solids menu, then Booleans. Select the Difference option and then click OK to execute the operation.

The result:



STEP 4

Now select the UNDO icon from the main toolbar so that the sphere comes back. Both solids should be selected. If they are, click your mouse off of the geometry to deselect the two solids.

This time select the sphere FIRST and then the cube. Once you have done this select the Solids menu and Booleans. Choose difference and then select OK to see how this function works when making your selections in this sequence.

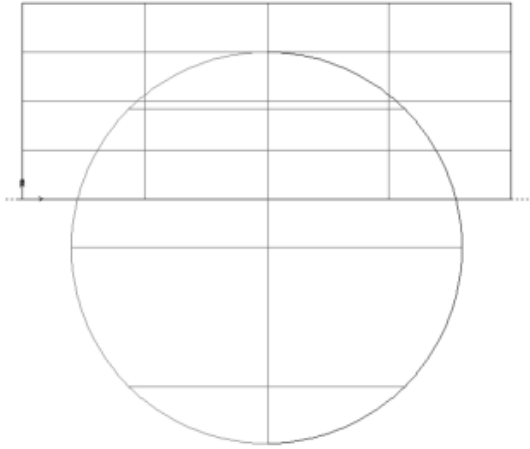
The result:



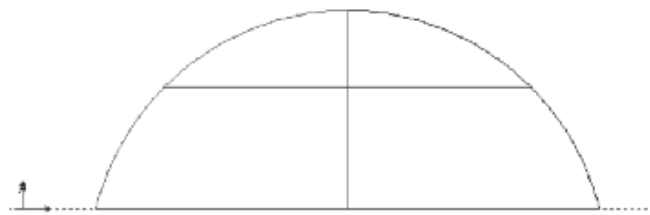
So, now you see how this function works. Remember that the outcome of this operation will be determined by the order in which your two solids are selected. It's that simple. Practice using this Difference function so that you can become more familiar with it for drawing with solids in BobCAD.

This concludes the basic difference lesson.

INTERSECTION: The Boolean Intersection operation allows you to subtract everything from two intersecting solids except for the intersecting part itself, leaving you with one solid in the end. In the image below you see a rectangular solid shape and a sphere. Both intersect.



By selecting both solids in any order, choosing the Solids menu, Booleans, then the Intersection option and clicking OK you would get the result as seen below.



As you can see, what is left over is the intersection that is now a solid.

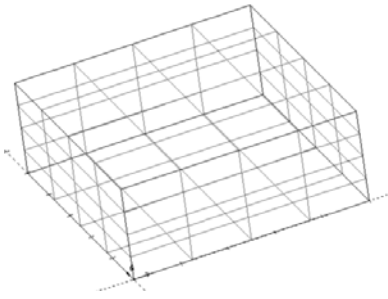
Basic Rules for using the Boolean Intersection function:

1. The two solids used must intersect in some way.
2. You must have the two solids selected before this function will be available. The selection sequence is not important as it is with the Difference operation.
3. You can only perform the Intersection function on two solids at a time. In other words, if you have five solids that you want to modify, you will need to perform this function on each solid at a time.

BASIC INTERSECTION LESSON: Follow this basic lesson and learn how to use the Boolean Intersection function on solids.

STEP 1

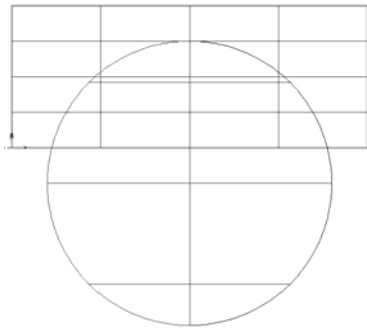
With a new drawing screen select the Solids menu, Primitives and then select Cube. In the cube box leave the X, Y and Z coordinate values all at 0. Enter a length of 5, width of 5 and height of 2 and select OK to draw the solid. Select the 3D menu and Turn the 3D ON. You may now want to select the View ALL icon from the main toolbar.



STEP 2

Select the Solids menu, Primitives and then Sphere. In the sphere box enter 2.5 for X, 2.5 for Y and -.5 for Z. Enter a Radius of 2 and select OK to draw the solid sphere.

See the side view image below.

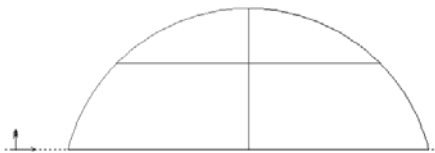


STEP 3

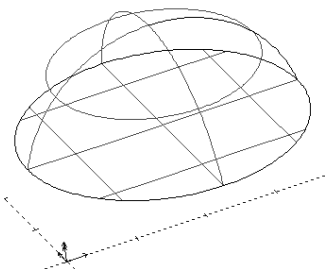
Now select both solids. You can do this by selecting the edge lines or interior lines of one solid and then hold down the shift key and select the next shape. You can also simply click the select All icon from the selection toolbox. Now select the Solids menu, Booleans and then click on the Intersection option. Now click OK to perform the operation.

You should have exactly the same on your screen as you can see in the image below from a side view and a 3D view.

Side view:



3D View:



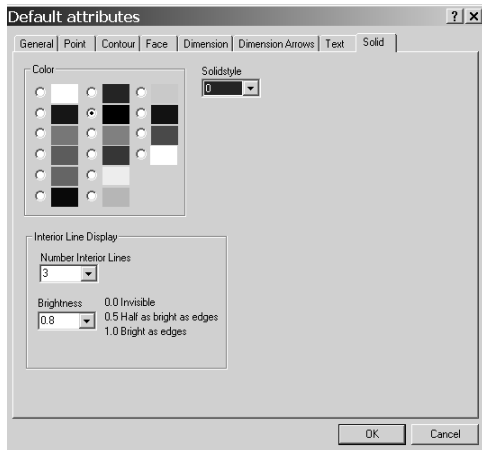
As you can see, the solid shape was automatically changed leaving a new solid from the prior intersection. *This concludes the basic intersection lesson.*

Section 27. Solids Attributes

The primitive solids that are available in BobCAD Version 21 have **Attributes** as mentioned earlier in this manual. They are:

- The brightness of the Interior Lines
- The number of Interior Lines

These lines will also be seen in surfaces that you import from an external CAD system. You may want to increase or decrease the number of interior lines that define the way the surfaces or solids look for selection and viewing. Let's take a look at the new addition to the software Attributes dialog as you can see below.



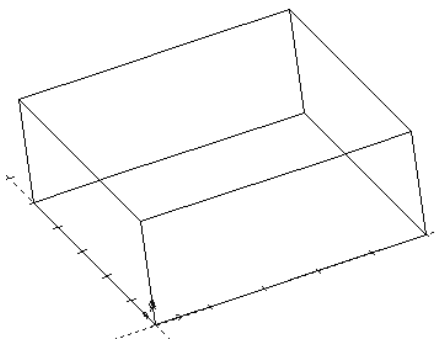
The solids tab offers 3 important attributes:

- 16 style Color pallet.
- The number of Interior lines that are seen in a surface or solid.
- The Brightness of the solid or surface EDGE LINES.

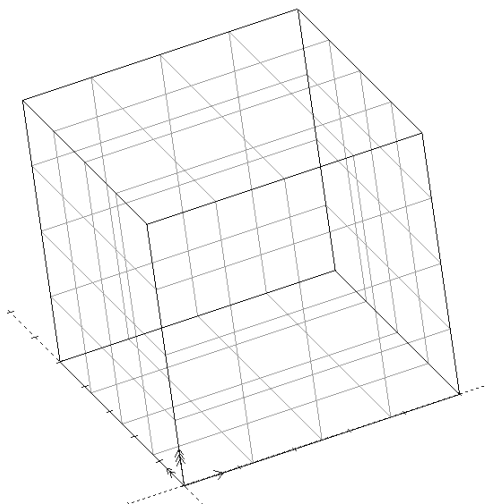
COLOR PALLET: This attribute function allows you to change the color of a solid. To change the solid color you will need to click on it first and then RIGHT-CLICK your mouse to select ATTRIBUTES. Then choose the Attribute function to open the dialog. Next you can select the desired color and click OK. Your solid will appear selected. Deselect the solid by clicking in an empty area of the screen or by selecting the EDIT menu and then choosing Unselect ALL.

You are not able to make the **Edge Lines** one color and the Interior Lines another color.

NUMBER OF INTERIOR LINES: When you draw or import solids and surfaces you will notice that there is edge geometry that defines the shape or boundary of the surface or solid. Take for example the cube image below.

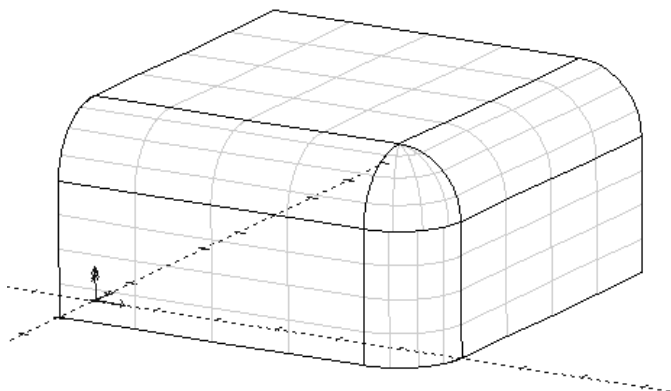


This is actually a solid with the NUMBER OF INTERIOR LINES set to 0. Here is a cube seen below with the interior Lines set to 3.

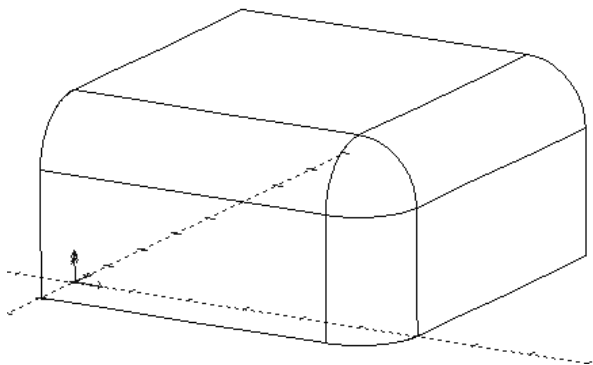


THE BRIGHTNESS OF INTERIOR LINES : This option allows you to make the interior lines more or less bright. This is more for viewing the surface or solid.

Here is a part with multiple surfaces. As you can see, the Interior Lines are set to .8.



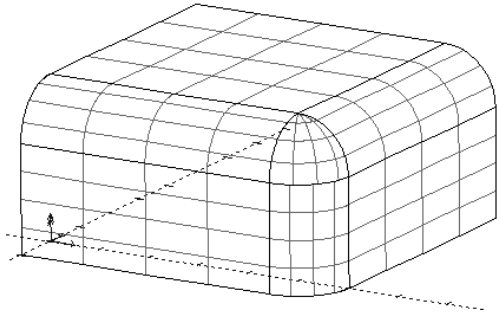
Here is the same surface file with the Brightness of the Interior Lines set to 0.



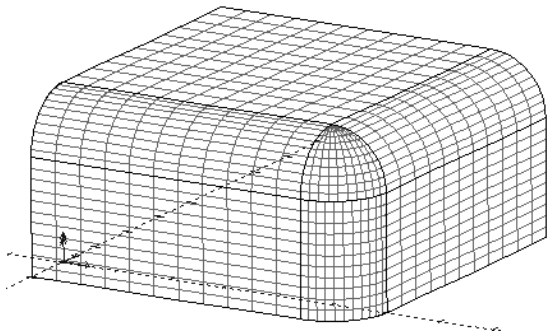
Simply select the surface, surfaces or solid, RIGHT-CLICK your mouse and choose ATTRIBUTES. After you check the option that you want, click OK and the changes will automatically take place.

MORE ON THE NUMBER OF INTERIOR LINES: This option allows you to increase or decrease the number of interior lines that make up the surface or solid. Simply select the solid or surfaces, RIGHT-CLICK your mouse button and choose ATTRIBUTES. The typical number of interior lines will be 3. However this can be changed by entering the number that you want and then by selecting OK.

Here is an example of a file that is set to **three** interior lines per surface:



Here is an example of the same surfaces with **ten** interior lines per surface:



MORE ON THE BRIGHTNESS OF INTERIOR LINES: This option allows you to increase or decrease the brightness level of the interior lines in your surfaces or solids. The software default setting for this option is .3. To change this, select the surface or solid, RIGHT-CLICK your mouse and choose ATTRIBUTES. Select the solids tab and increase or decrease the value and click OK. That will automatically make the desired change.

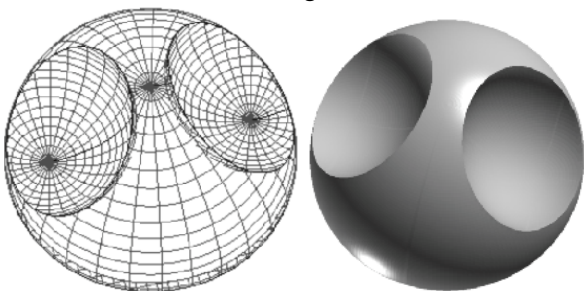
The ATTRIBUTES dialog is also located in the CHANGE menu.

Section 28. Basic CAD Design

This section is dedicated to teaching you how to draw and edit with solids in the BobCAD Version 21 software. Here are three basic lessons that use each of the Boolean Operations that you have studied thus far in this training guide.

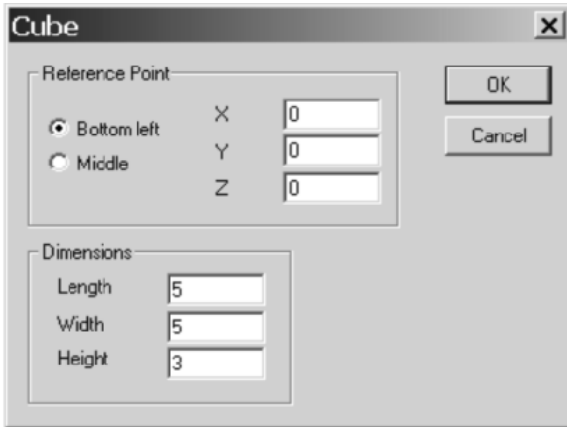
CAD DESIGN LESSON ONE:

Follow this basic training lesson and learn how to design a part in the Version 21 software.



STEP 1

With a fresh drawing screen, select the Solids menu and then Primitives. Choose Cube and click OK.



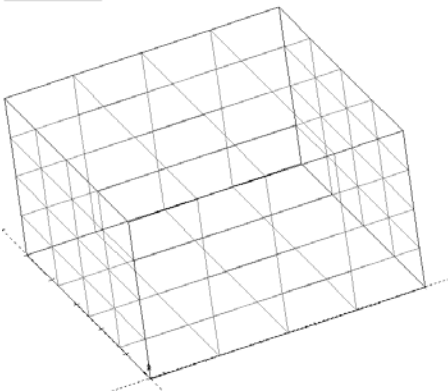
Make sure that Bottom Left is selected as the reference point and that the X, Y and Z coordinates are all 0.

Enter 5 for the Length.

Enter 5 for the Width.

Enter 3 for the height.

Now select OK. Once you have done this, select the 3D menu and Turn 3D ON. You may want to select the View All icon from the main toolbar.



STEP 2

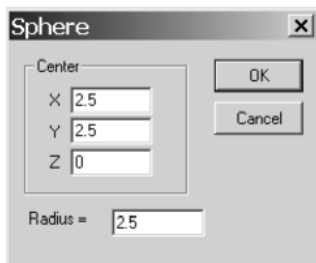
Select the Solids menu, then Primitives and choose Sphere. In the sphere box:

Enter 2.5 for X.

Enter 2.5 for Y.

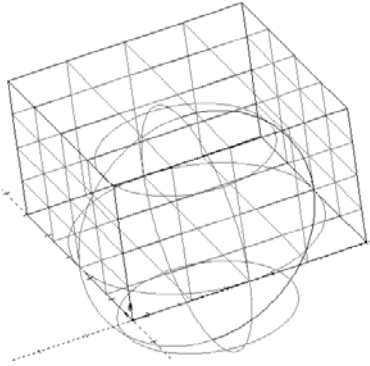
Leave the Z coordinate at 0.

Enter a Radius of 2.5.

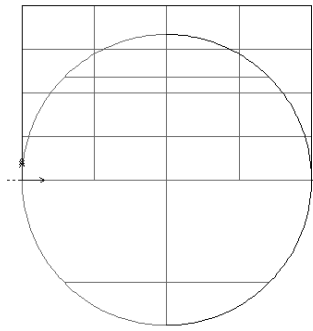


Now click OK. See the image below as this is exactly what you should have on your screen now.

The 3D View:



The Side View:



STEP 3

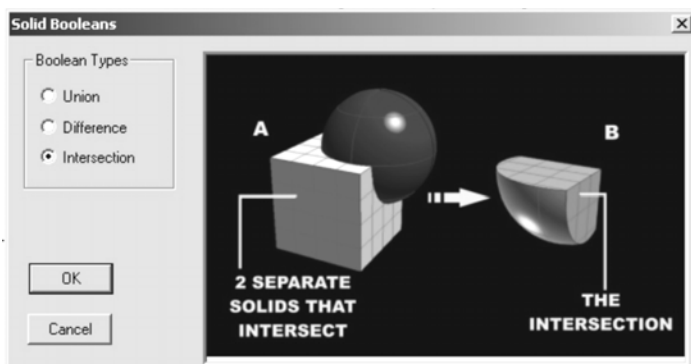
Now click on the Select ALL icon from the selection toolbox that should be on the left wall of the software.



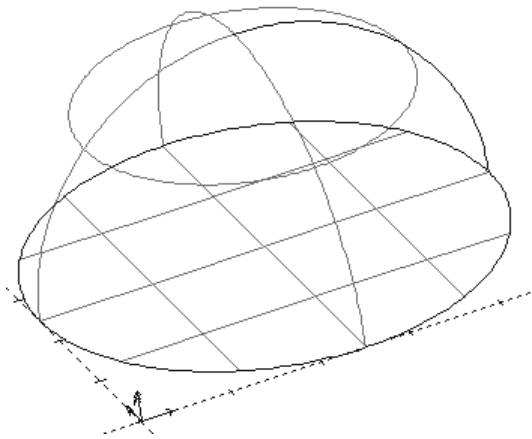
You can also do this by selecting the Edit menu, Select Entities and then Select All. This will select both solids.

STEP 4

Select the Solids menu, Booleans. In the Solid Booleans box choose INTERSECTION and click OK. This will perform the intersection operation.



See the result in the image below.



STEP 5

We are going to go ahead and draw a new sphere. And so, select the Solids menu, Primitives and then choose Sphere.

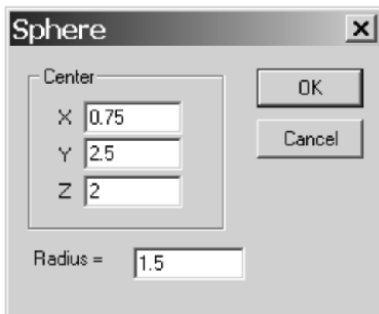
When the new sphere box appears:

Enter .75 for X.

Enter 2.5 for Y.

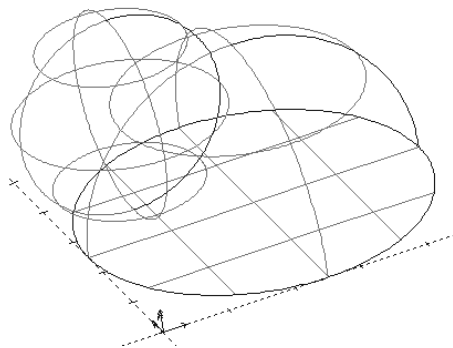
Enter 2 for Z.

Enter 1.5 for the Radius.



Now select OK to draw the new solid sphere.

Have a look at the image below as this is what you should have on your screen now.



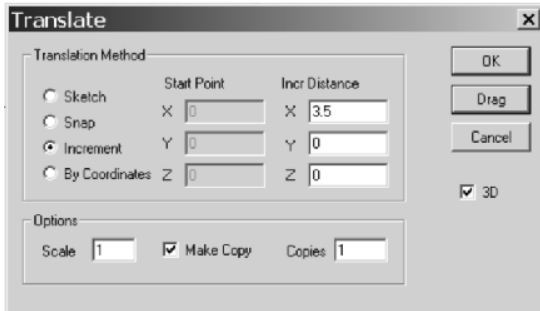
OK. Now we are going to use the Translate function to make a copy of the sphere you just created.

STEP 6

Place your cursor on one of the interior or edges of the sphere that you just created and click on it to select it.

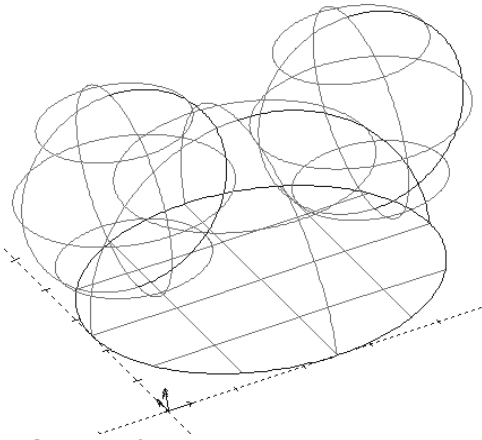
NOTE: *If both solids are still selected you will have to deselect them first before trying to make the new selection of the sphere. To do this you simply place your cursor in an empty area of the drawing screen and click your mouse button. This will deselect everything. This method is faster than going to the edit menu and choosing Unselect All.*

Once you have selected the sphere, select the Change menu and Translate. In the Translate box as seen here below, click on the INCREMENTAL option and check the 3D box if it is not already selected for you.



Enter 3.5 for the X axis, Y & Z leave at 0 with a scale factor of 1. Check the Make Copy option and enter 1 for the number of copies to be made. Now select OK.

You should now have the translated sphere as seen in the image below.



NOTE: *After the translation, both spheres will be selected. **Deselect them** as we don't want anything to be selected prior to our next action. Remember, you can do this by placing your cursor in an empty area of the drawing screen and clicking your mouse button once.*

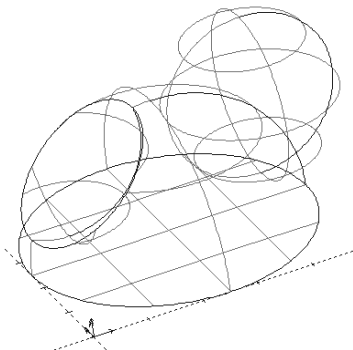
STEP 7

Now you want to select the bottom solid first. This was the first solid that you created. Place your cursor on one of the edge or interior lines and select it. Make sure that this is the one selected.

Once you have selected the bottom solid, hold down your shift key and select the top LEFT solid. When you have selected the top left solid, let go of the shift key.

STEP 8

Select the Solid menu, Booleans. In the Solid Booleans box, select Difference and click OK. This will perform the operation and change the bottom solid shape. Leave the changed solid selected. The image below shows what you should have on your screen at this time.

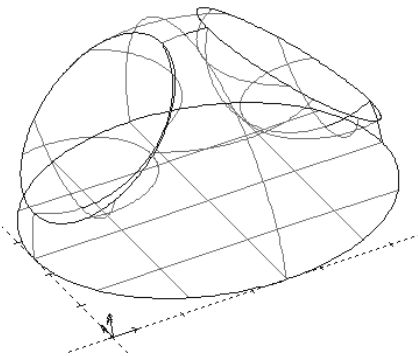


At this time, the bottom solid should be selected. If it is, that is good. If not, select it now.

STEP 9

Make sure that the bottom solid is selected. Select the last sphere next so that both solids are selected. Then choose the Solids menu and Booleans. Now select the Difference option. It may already be selected as you just performed this operation. Click OK and the bottom solid will now be changed again.

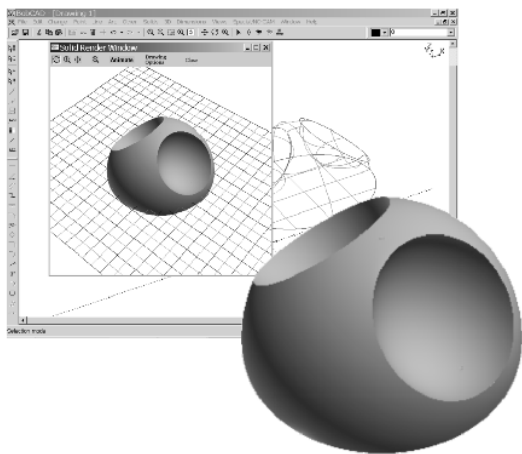
Have a look at the image below. This is what you should have on your screen at this time.



OK. In this lesson you learned how to draw with solid shapes as well as use the Boolean Operations Intersection and Difference functions. Let's go ahead and render this solid.

STEP 10

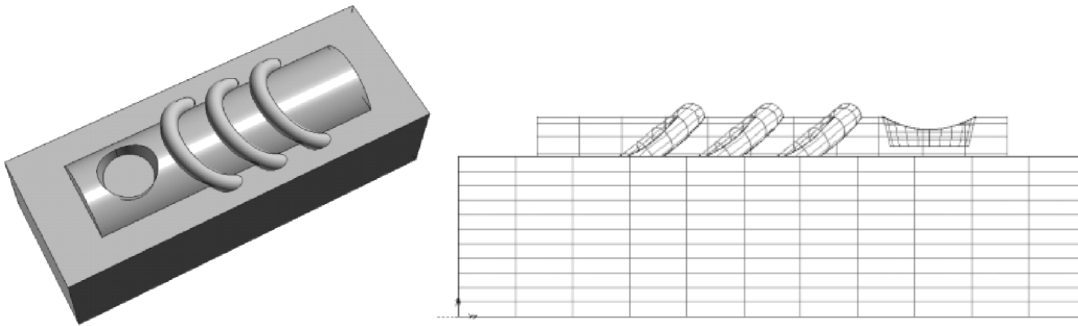
After performing the last Boolean operation, your solid will be selected. If it is not, select it now. Once you have done this, select the Solids menu and then Render. This will open the render window and give you a look at the solid part that you just made.



This concludes the CAD Design Lesson One.

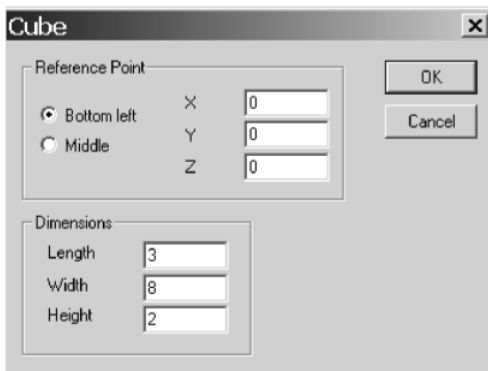
CAD DESIGN LESSON TWO:

Follow this lesson and learn more about using the Version 21 to design parts with solids.



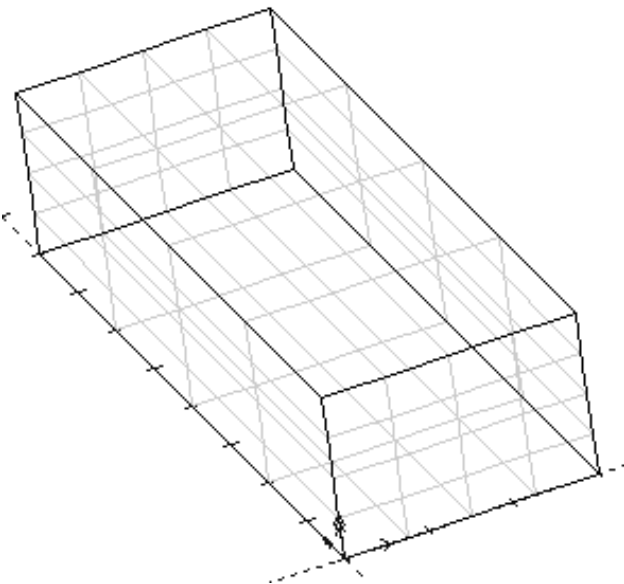
STEP 1

With a new drawing screen, select the Solids menu and Primitives. Now select Cube. In the cube box, select Bottom Left as the reference point of the cube. Leave the X, Y and Z coordinates at 0.



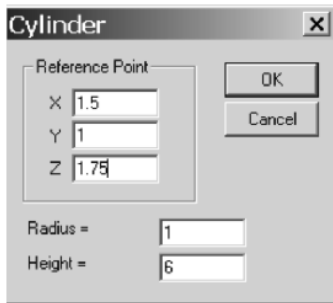
For the Dimensions, enter 3 for Length, 8 for the Width and 2 for the Height. Now select OK. You may want to Turn 3D ON from the 3D menu and then use the View ALL icon from the main toolbar.

The result:

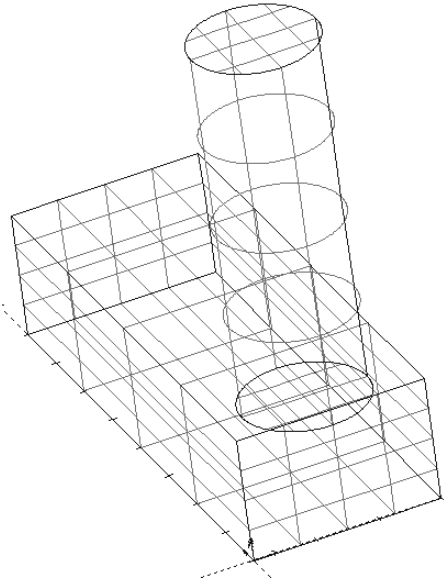


STEP 2

Now we are going to draw a solid cylinder. Select the Solids menu, Primitives and then choose Cylinder and click OK. Under the Reference Point section, enter 1.5 for X. Enter 1 for Y and enter 1.75 for Z.



Enter 1 for the Radius.
Enter 6 for the Height. Now click OK to draw the solid cylinder.



STEP 3

Now we are going to rotate the cylinder and position it for our next action. Make sure that the cylinder is the **ONLY** solid selected at this time. If you need to, place your cursor in an empty area of the screen and click your mouse button to unselect everything. Then select the cylinder by clicking on one of the edge or interior lines. Once you have the cylinder selected, go to the Change menu and choose ROTATE.

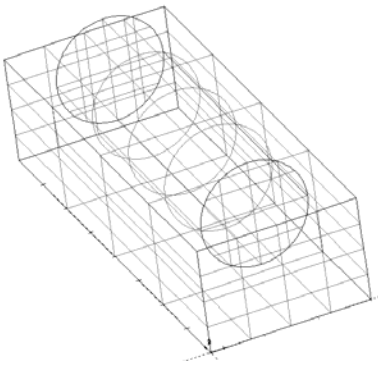
STEP 4

In the Rotate dialog, you will need to select Coordinates under the Rotation Point section. Enter 1.5 for X, 1 for Y and 1.75 for the Z coordinates. We are using the base-center of the cylinder as the rotation point.



Under the Angle Around Axis section enter -90 for X. Z and Y should be 0. The scale should be 1 and we are not going to make any copies. To enter all of the coordinates, you may need to check the 3D box.

Once you have entered everything in, select OK to rotate the cylinder. Have a look at the end result below.



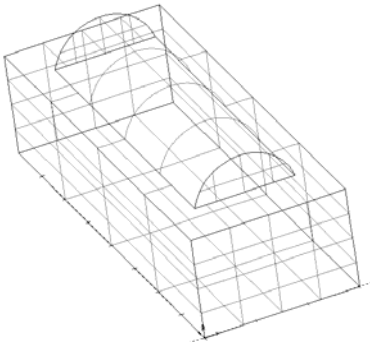
STEP 5

Now we are going to use the Union Operation on these two solids. Click on the Select ALL icon from the selection toolbox. You can also select one solid, hold down your shift key and select the other.

NOTE: Remember, we are using the **Union Operation** on these two solids. Therefore, we do not need to select them in any particular order.

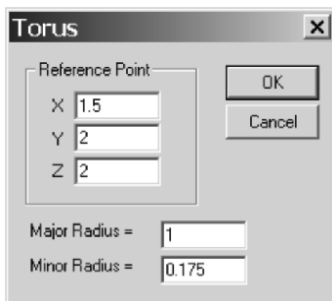
Once you have both solids selected, choose the Solids menu, Booleans and then the Union operation. Now click OK to perform the union.

You can see the result below.



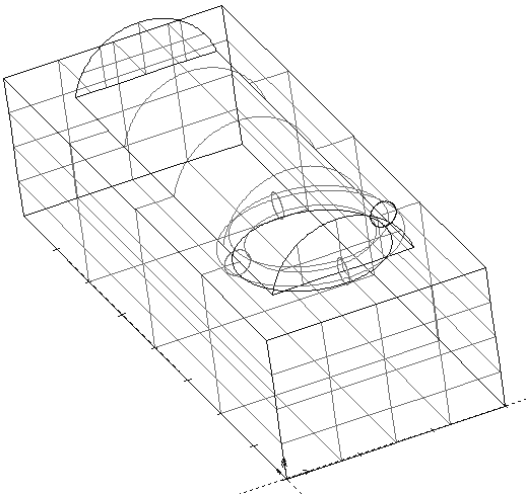
STEP 6

Now we are going to draw the first of three solid torus shapes. Select the Solids menu and then Primitives. Now choose Torus and click OK to open this box. For the Reference Point, enter 1.5 for X, 2 for Y and 2 for Z. Enter 1 for the Major Radius and enter .175 for the Minor Radius. Use the box below for direction if needed.



Now select OK to draw the torus.

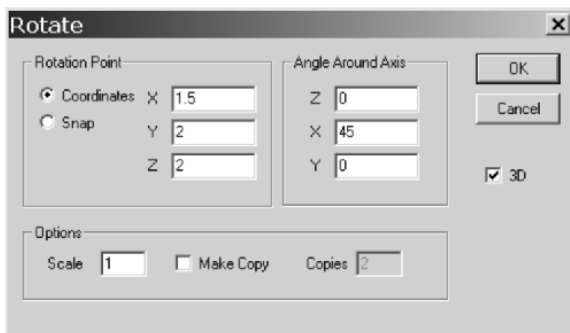
The result:



IMPORTANT: At this time, make sure that *ONLY* the torus is selected on your screen. This is important for the next step.

STEP 7

With *ONLY* the torus selected, choose the Change menu and ROTATE.

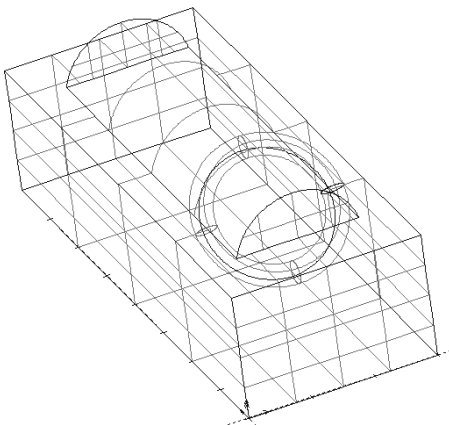


Now we are going to rotate this torus by the following coordinates and angles:

Under the Rotation Point section select Coordinates and enter 1.5 for X, 2 for Y and 2 for Z. For the Angle Around Axis, Z is 0, X is 45-degrees and Y is 0. The scale is 1 and do not make copies. Select OK.

The result of the rotation:

3D View

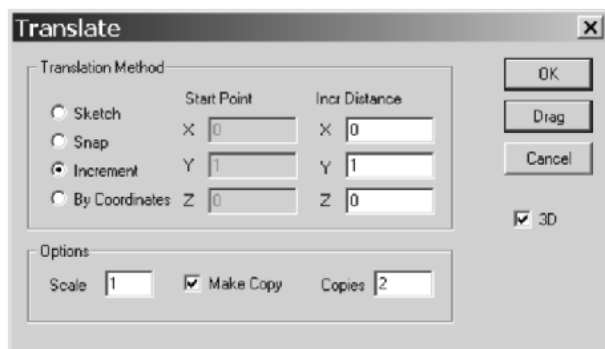


Side View



STEP 8

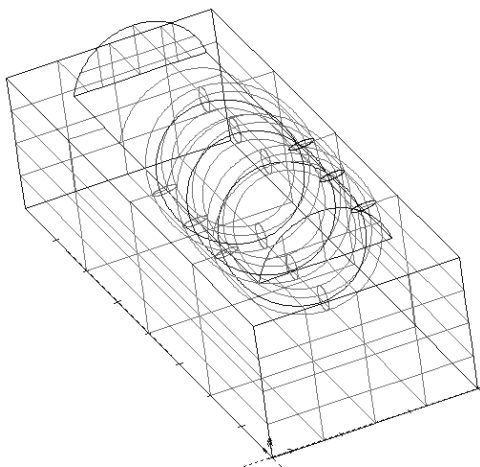
OK. Make sure that the torus is the **ONLY** solid selected at this time. Now select the Change menu and **TRANSLATE** to open this box.



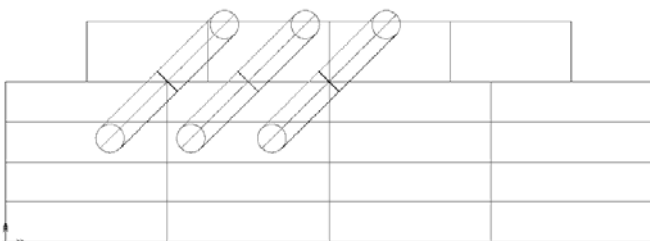
Choose **INCREMENT** as the translation method. Check the 3D box and enter 1 for Y under the Incremental Distance section. Leave the X & Z values at 0. The scale should be set to 1, select the Make Copy option and enter 2 as the number of copies to be made. Now select OK.

The result:

3D View



Side View



STEP 9

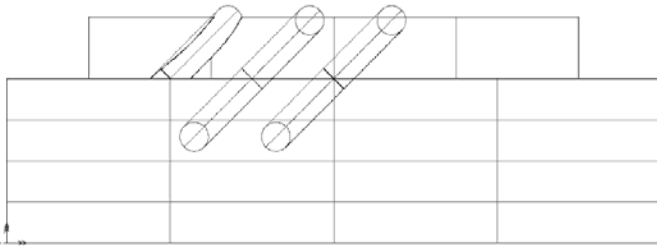
Unselect all solids at this time. OK. Now you will want to have your 3D View turned ON. The next step is to perform a Union Boolean operation on the torus solids that we just drew and the rectangular solid. Now, as mentioned earlier in this lesson, you do not have to select solids in any particular order to use the union operation. However, you can only have two solids selected at a time for any of the Boolean Operations in BobCAD. Therefore, at this time, select the rectangle solid and the first torus that you created. Do this by selecting one of the edge or interior lines of the rectangular solid, hold down your CONTROL key and select the first torus.

Only select these two solids. When you have done this, select the Solids menu and Booleans.

STEP 10

In the Solid Booleans box, select Union and then click OK.

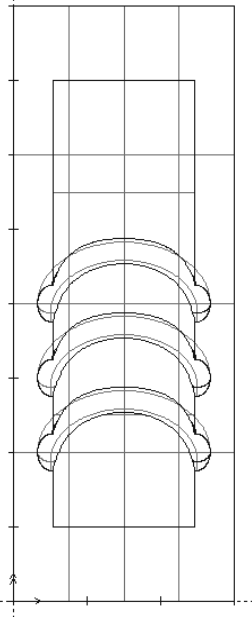
This will perform the first union. You can see a side view of the result in the image below.



After the union operation is finished, the new solid may be pre-selected and ready for the next selection. If not, select it now and then the next torus solid. Remember to hold down your Control key when making 2 selections so that you don't lose your sequence (*if you are performing a Difference Operation*) or so that you don't un-select the first solid. OK, when both solids are selected, choose the Solids menu and then Booleans. Select the UNION operation and click OK to union the second torus.

Now do this one last time for the third torus. Select both solids, then the Solids menu and Booleans. Select Union and click OK.

The result:

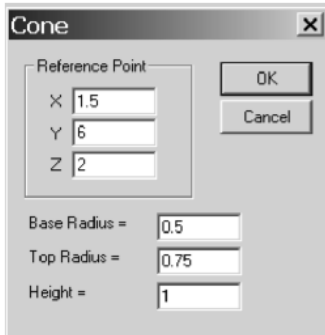


STEP 11

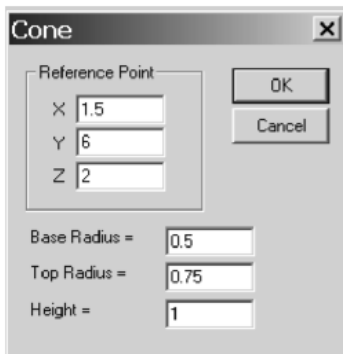
Select the 3D menu and Turn 3D OFF. Then select the View ALL icon from the main toolbar.



OK. Now we are going to draw a solid cone for this part and then finish it off. Select the Solids menu, Primitives and then Cone. In the cone box, there are two parts.



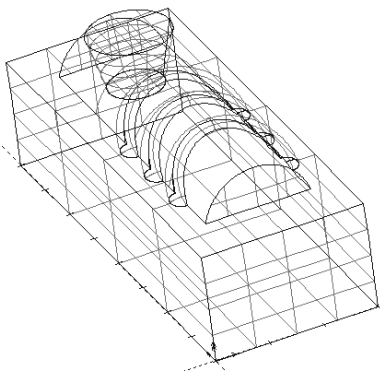
For the Reference Point enter 1.5 for X, 6 for Y and 2 for the Z coordinates. Now you will define the cone itself.



Enter .5 for the Base Radius.
Enter .75 for the Top Radius.
Enter 1 for the Height.

Click OK.

The result:

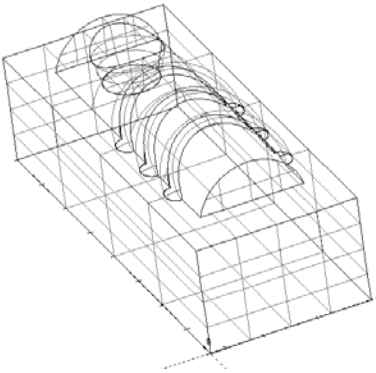


STEP 12

Make sure that there is nothing selected on your screen at this time. As you learned earlier, when performing the Boolean Difference function in BobCAD, you need to take the sequence of selected solids into consideration. Go ahead and select the Base Solid shape at this time. Then hold down your Control (ctrl) key and select the cone second. We are making our selection sequence this way because we want to change the base solid and extract the difference of the code.

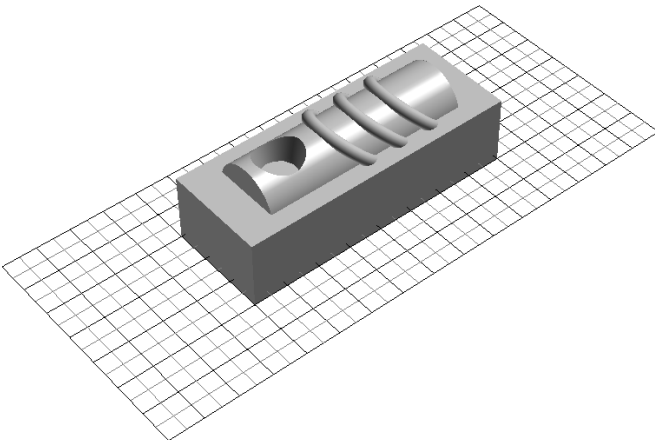
STEP 13

Select the Solids menu and then Booleans. Select the Difference option in the Solid Booleans dialog and click OK to perform the operation. Have a look at the image below as this is what you should now have on your screen.



STEP 14

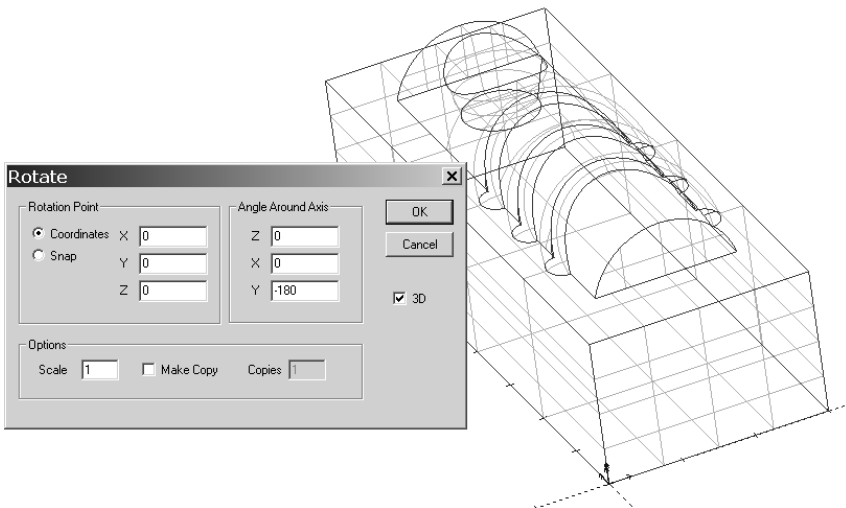
Select the solid and then the Solids menu and RENDER. This will open the render window so that you can see the part you just finished as a solid.



STEP 15

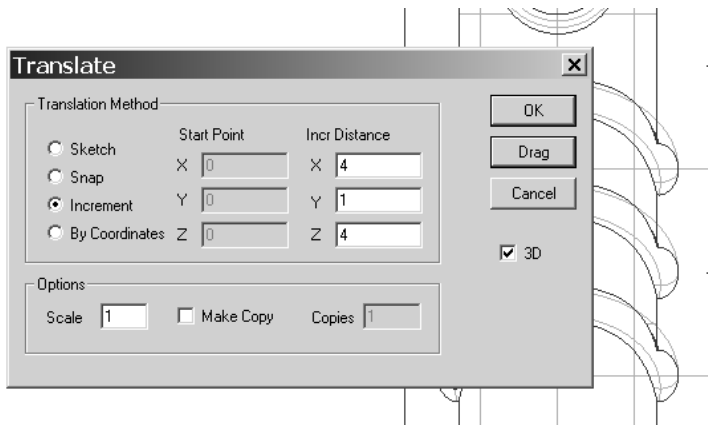
Go ahead and close out of the render window. Now, we are going to take this one step farther. We are going to use this solid to create a negative model of the same part.

At this time, select the part. Now select the Change menu and ROTATE. Select Coordinate under the rotation point. Under Angle Around Axis enter -180 for the Y axis. Scale should be 1 and we are NOT going to make any copies. Select OK to rotate the solid.



STEP 16

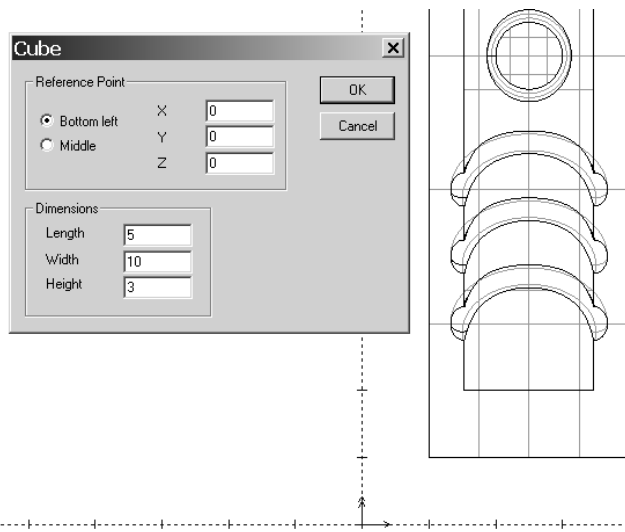
Now we are going to use the TRANSLATE function in the CHANGE menu to position the solid. Select the Change menu and then Translate.



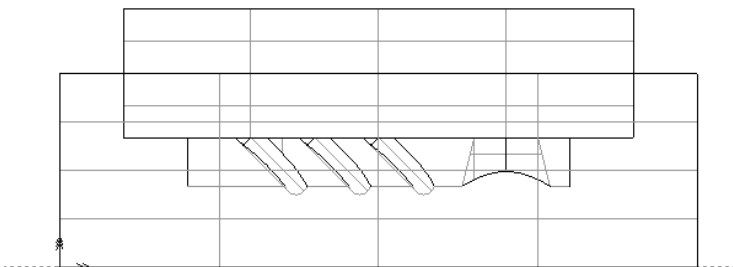
Select the INCREMENTAL option under Translation Method. Enter 4 for X, 1 for Y and 4 for Z. Scale should be set at 1 and we will not be making any copies. Go ahead and select OK now to make the change.

STEP 17

Now we are going to draw a second solid shape that we will be using to create the negative. First go to the 3D menu and Turn 3D OFF. This way you will only see the top view (2D View) of the part. Now go to the Solids menu, select Primitive and then choose CUBE. In the cube box select Bottom Left for the reference point and leave the X, Y and Z values all at 0. Under Dimensions enter 5 for the length, 10 for the width and 3 for the height. Now select OK to draw this new solid.



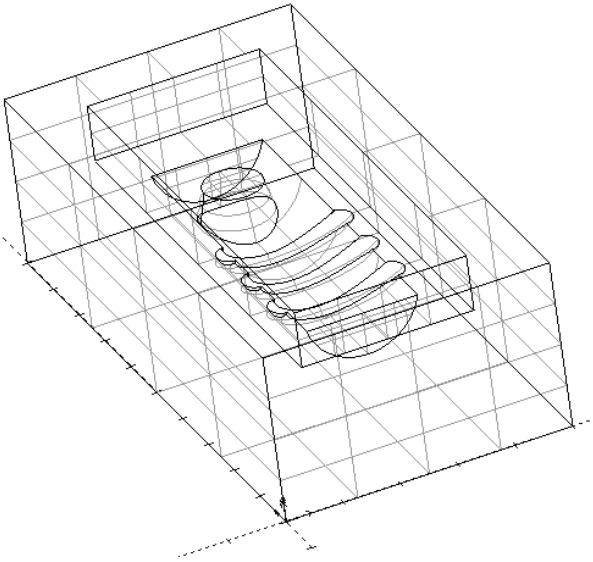
The result seen from a side view:



As you can see, the two solids intersect. Now we are going to perform a Boolean Difference operation, using the top solid to modify the bottom solid. Make sure that nothing is selected at this time. A quick way to deselect everything is to simply place your cursor in an empty area on the drawing screen and click your mouse button once. As a rule, when using the Boolean Difference operation, you will first need to select the solid that you want to change or keep, then the solid that you want to use to make the change without deselecting the first solid. To do this, click on the bottom solid first. Now hold down the SHIFT key and select the top solid second. Both will be selected.

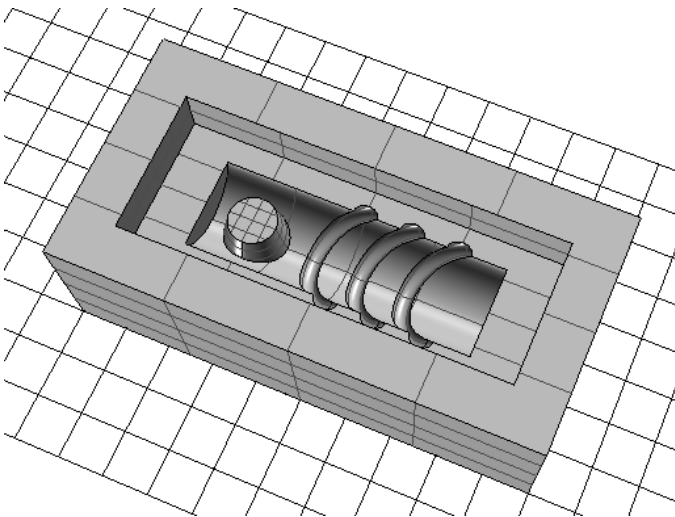
Now go to the Solids menu and select Boolean and then choose DIFFERENCE and click OK. This will perform the operation and make the change that we want.

The result:



STEP 18

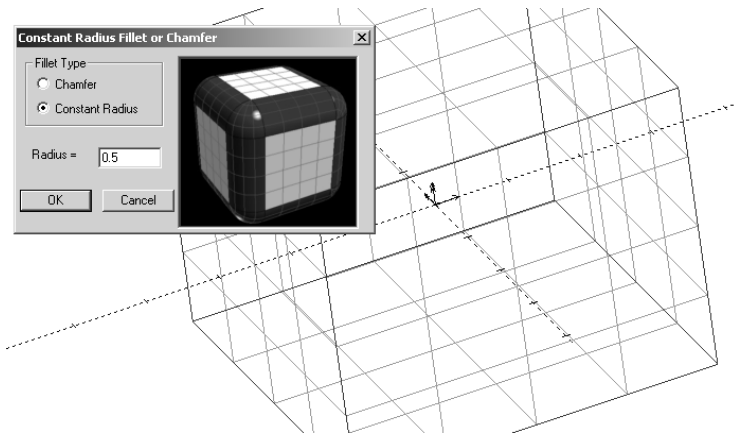
Now select the solid if it is not already selected. Go to the solids menu and select RENDER to view the part.



Section 29. Fillets & Chamfer Surfaces

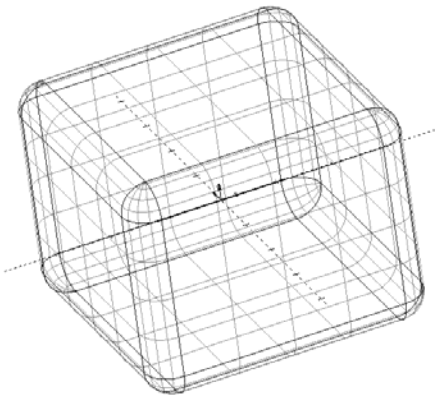
Version 21 offers you 2 different types of **Fillet** options for Surfaces and Solids as well as a **Chamfer** option. Here they are:

CONSTANT RADIUS FILLET: To create a constant radius fillet you **MUST** pre-select the solid shape first. Only then, go to the Solids menu and select Fillet. You will see the following type box:



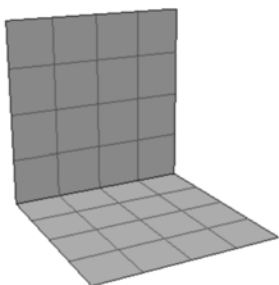
You will notice that this box does not include the Variable Radius Fillet option. In any case, when you select the model first and then the function, you will only have a constant chamfer and constant fillet option. This means that all edges will be chamfered or filleted.

Simply enter the radius or chamfer amount and click OK.

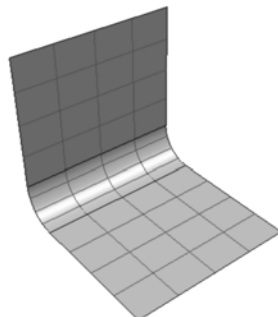


You can also perform the fillet function on 2 individual surfaces. Take the example of the 2 surfaces below.

2 surfaces:



3 surfaces:



If you want to apply a fillet or chamfer surface between these two surfaces you must have them stitched together first. Then select the solid and go to the Solids menu, choose Fillet, enter in the radius or chamfer value and select OK.

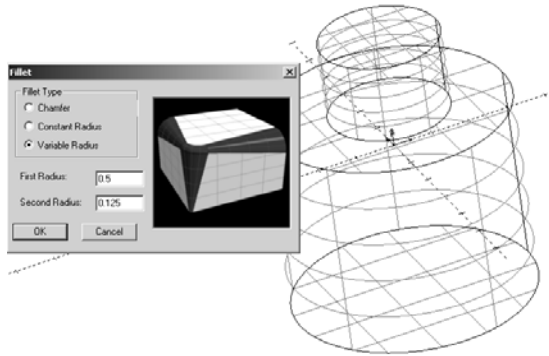
Variable Radius Fillet procedure:

STEP 1

Make sure that the solid or surfaces are not selected.

STEP 2

Select the SOLIDS menu and then FILLET.

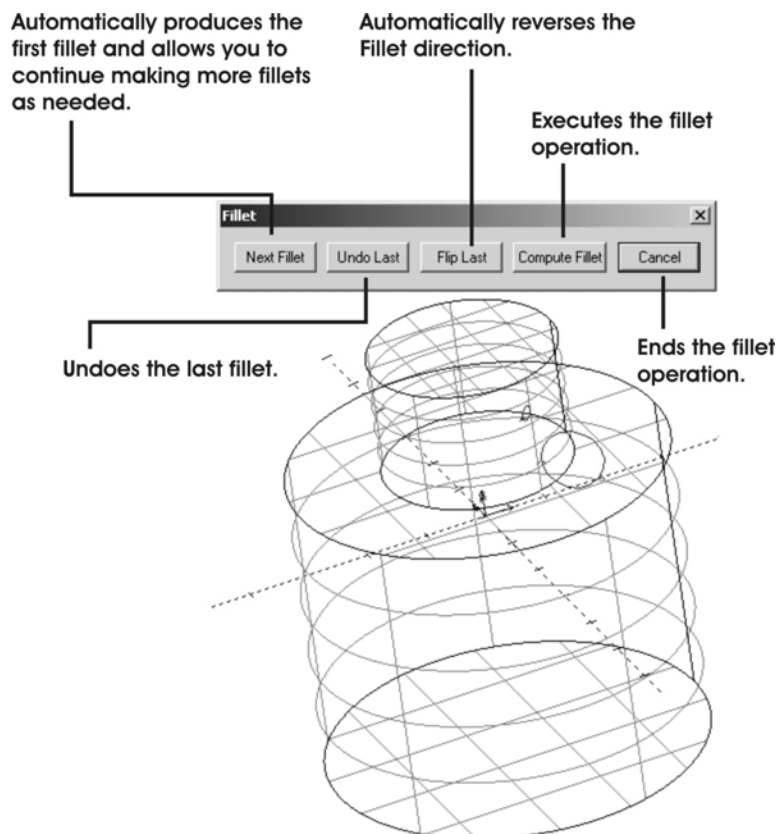


STEP 3

Select the Variable Radius option and enter in the First Radius and then the Second Radius value. When finished, click OK.

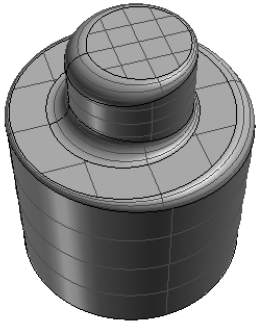
STEP 4

Select the edge that you would like to create the fillet. Once the edge is selected, you will see two arcs representing where each size fillet will be placed along with the command box.



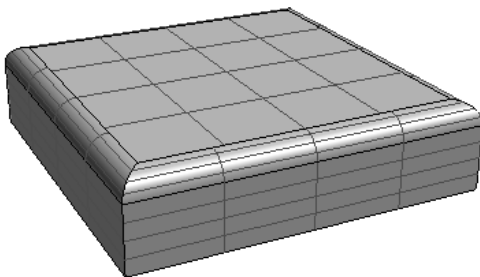
STEP 5

Make your selection in the command box to execute the fillet or continue making more variable radius fillets. If you select the Next Fillet option, the original fillet box will reappear offering you the ability to change the first and second radii.



Again, this is an Action-Object function meaning that you select the function first before selecting the model or surfaces. In order to access the variable radius fillet option you need to make sure that the surfaces or solid is not selected at the time of choosing the fillet function.

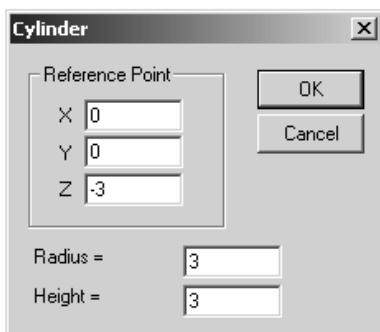
NOTE: You can use this function with the same values for the first and second radius to perform **fillet** surfaces easily rather than un-stitch and re-stitch surfaces for independent fillets. For example, below we have a cube. In the earlier Version 19 you would have to un-stitch all of the surfaces, then select the side and top surfaces and then re-stitch just to produce the fillet around the top edges. Now all you have to do is use the variable radius fillet option, using the same value for the first and second radii and select the top edges, selecting the next button until finished and then select the Compute Fillet button to execute the operation. You have to select OK in the fillet box to continue picking your edges. This gives you the flexibility of creating as many equal or variable radius fillets as you need when drawing. Remember to NOT have the surfaces or solid selected first when going to the Solids menu and attempting to use the Variable Radius Fillet function.



VARIABLE RADIUS FILLET TRAINING LESSON:

STEP 1

With a new drawing screen go to the SOLIDS menu and select PRIMITIVE, Cylinder and then click OK.



In the cylinder box enter:

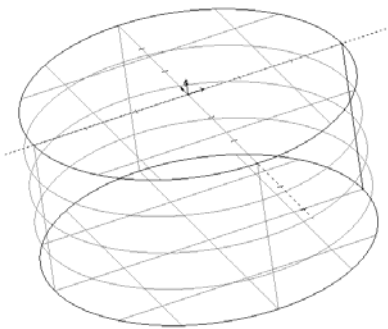
- X 0
- Y 0
- Z -3
- Radius 3
- Height 3

Now click OK to draw the first cylinder. By giving the cylinder a height of 3 and a Z -3 reference point, the top of the cylinder will be located at Z0.

Now go to the 3D menu and TURN 3D on, then click on your VIEW ALL button from the top toolbar.



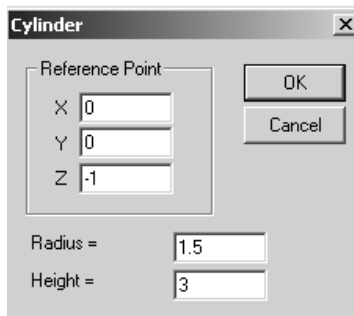
The result:



STEP 2

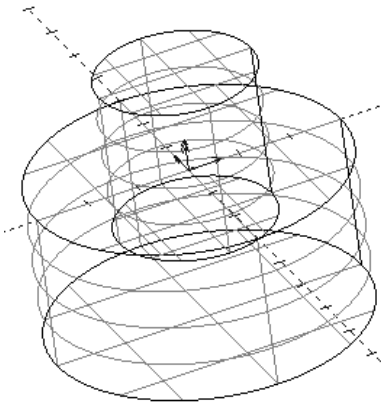
Now go back to the Solids menu, select Primitives, choose Cylinder and click OK.

Now draw another cylinder by entering in the following coordinates:



- X 0
- Y 0
- Z -1
- Radius 1.5
- Height 3

Click OK to draw the second cylinder.



STEP 3

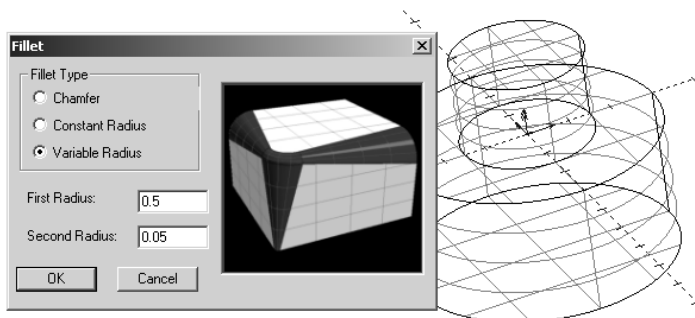
Now we want to select both cylinders. Place your cursor in an empty area of the drawing screen and click your mouse button to make sure everything is deselected first. Now place your cursor on an edge line of either solid and click your left mouse button to select it. Hold down your CONTROL key and click the other cylinder so that both are now selected.

With both cylinders selected go to the SOLIDS menu and select BOOLEANS. Now choose UNION in the Booleans box and click OK to create one solid.

Now place your cursor in an empty area of the drawing screen and click your mouse to deselect the solid.

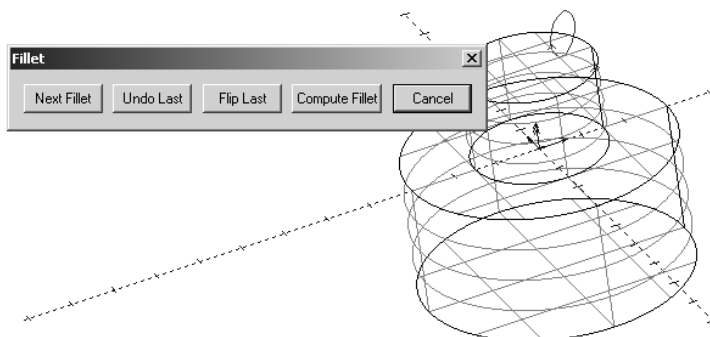
STEP 4

With the solid deselected, go to the Solids menu and select Fillet.



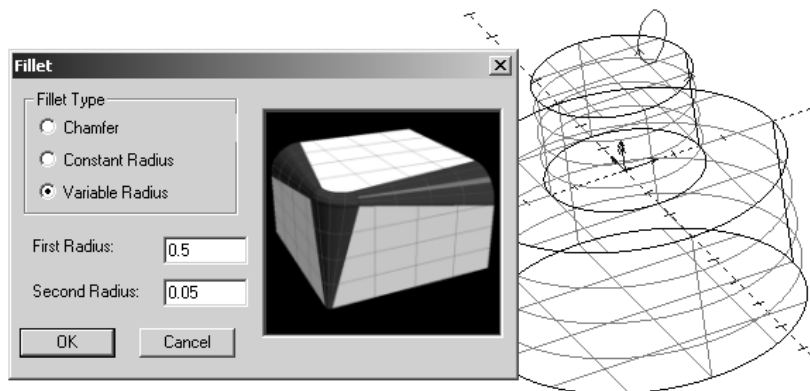
Select the Variable Radius fillet option first.

Enter .5 for the First Radius and .05 for the Second Radius. Now click OK. Place your cursor on the very top edge and click your mouse button.



You will see 2 small arcs, one larger than the other representing where the fillets will be located.

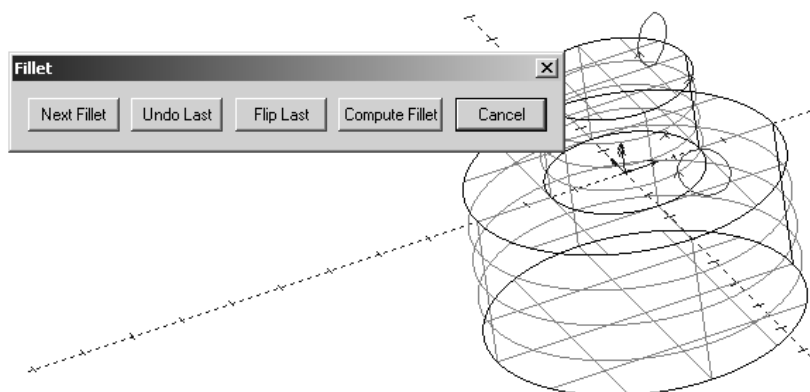
Now click on the **NEXT FILLET** button. This opens the fillet box again.



Leave the values the same and click OK.

STEP 5

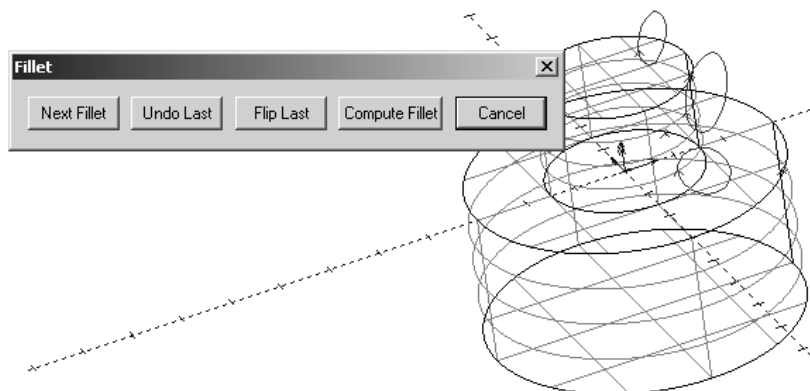
Now click on the second radius down from the top on the left edge. The command box will re-appear.



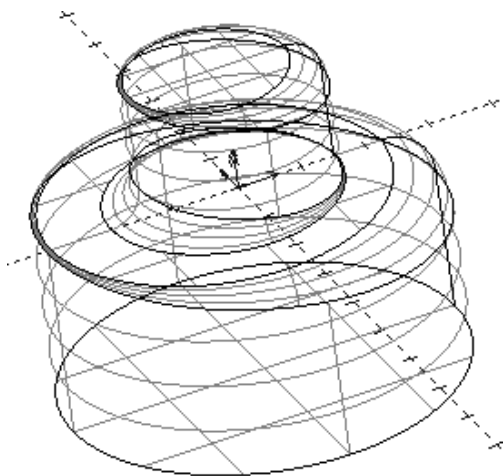
Now click on the FLIP LAST button and then click the NEXT FILLET button.

STEP 6

In the Fillet box, change the First Radius value to .75 and leave the Second Radius value at .05. Click OK. Now click the top larger radius at the left side edge. The command box will re-appear.

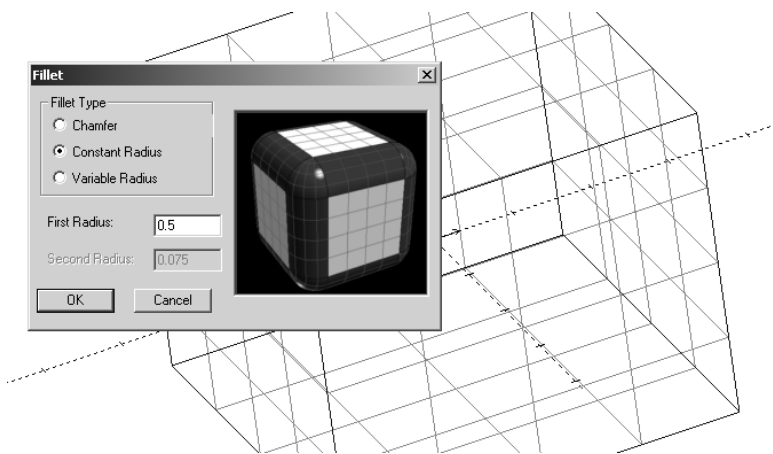


Now select the COMPUTE FILLET button to create the variable Radius fillets.

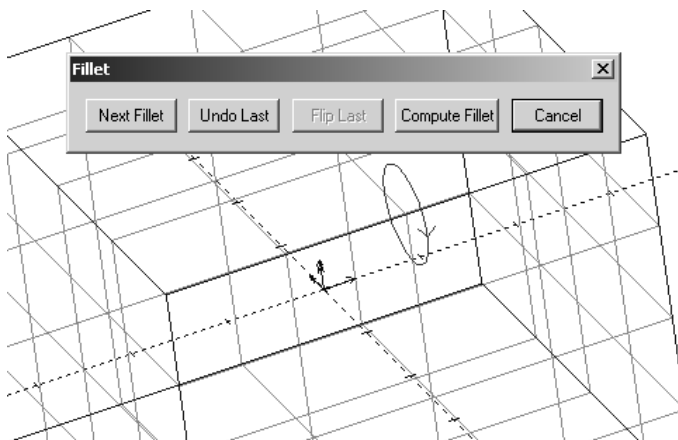


This is how you can create multiple variable radius fillets easily.

INDEPENDENT FILLET SURFACES: In order to create independent **Fillet** edges you would NOT select the solid first. You would go to the Solids menu and select Fillet first.

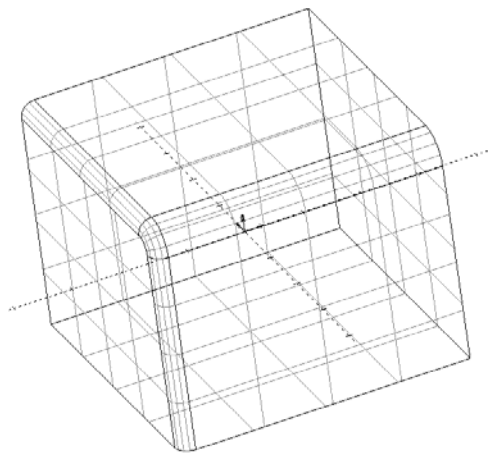


Then you would select the Constant Radius option, enter the radius value and click OK. Then click on the edge that you would like to create a fillet surface for.



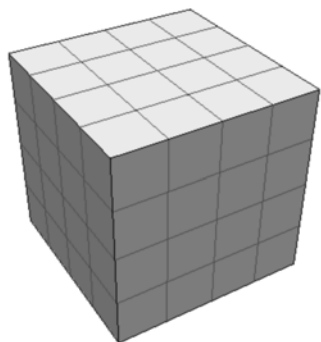
You will notice that an arc will appear on that edge that you selected. If you only want to create that fillet you would click the “Compute Fillet” button to execute the operation. If you wanted to create a second fillet or more you would repeat the operation by selecting the NEXT FILLET button and then the COMPUTE FILLET button when finished.

When you select the NEXT FILLET button the box reappears offering you the ability to change the radius value for the next fillet if needed.



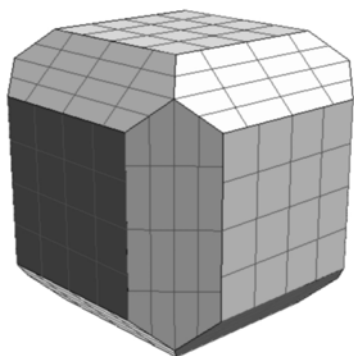
CONSTANT CHAMFER: A chamfer is just like a beveled edge. This function is just like the constant fillet function in terms of performing the action on all edges of the solid as well as independent edge chamfers.

Take the example of the cube below.



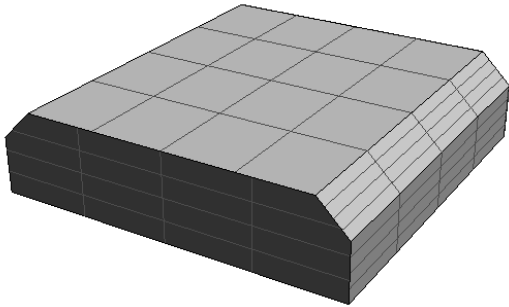
If you would like to apply a chamfer to all of the edges, select the solid, go to the Solids menu and select FILLET. When the new box appears, select Chamfer and enter the value of the chamfer that you would like to create and then select OK.

The result:



As you can see in the last image, the software applied chamfers on all of the cubes edges. There may be cases where you want to create a chamfer but do not want all of the edges changed. To chamfer ONLY specific edges you would make sure that the part is NOT selected and then go to the Chamfer function.

INDEPENDENT CHAMFER: Similar to the Variable Radius Fillet function, you have the Independent Surface **Chamfer** option, allowing you to create chamfered surfaces that are not constant. In the example image below you can see that we have created chamfered surfaces on only 2 sides.



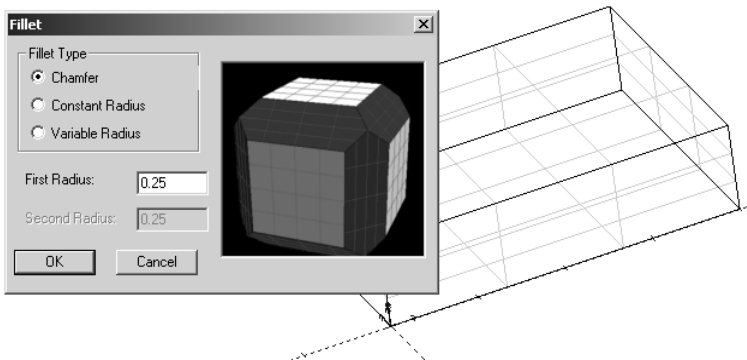
The procedure for using this function:

STEP 1

Deselect the solid.

STEP 2

Select the Solids menu and then Fillet. Select the chamfer option and enter a chamfer value. You will only have the first fillet value available. Enter a value and click OK.

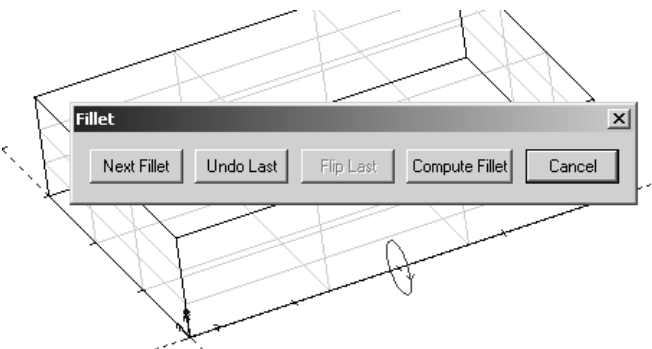


STEP 3

Now click on the edge that you would like to chamfer.

STEP 4

Use the command box to execute the operation or to continue using creating other independent chamfers for the model.



STEP 5

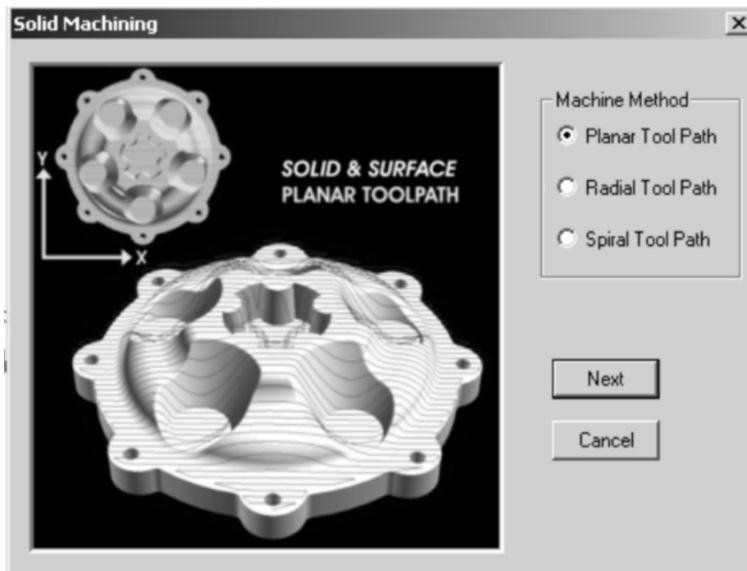
When finished, select Compute Fillet. If you click the “Next Fillet” button you will be returned to the fillet box to enter a new chamfer value if needed. If not, simply click OK in the fillet box and continue using the fillet command box this way until done.

This is how to create independent chamfered edges.

Section 30. Generate Toolpath

This section is going to teach you the fundamentals of the **Generate Toolpath** function and wizard in the Solids menu. Let’s go ahead and have a look at the first box in the Generate Toolpath Wizard. After selecting the model or surface file to create toolpath for, you must go to the solids menu and select Generate Toolpath to open the Solid Machining box as seen below.

The **Solid Machining** box:



The solid machining box offers three types of toolpath.

- Planar Toolpath
- Radial Toolpath
- Spiral Toolpath

Each of these have separate wizards that, once selected, you will click the NEXT button to access that options wizard. Let’s go ahead and take a look at each one of these starting with the Planar Toolpath option.

PLANAR: The planar toolpath option creates roughing and finishing toolpath that runs horizontally back and forth across the part. You can apply an angle such as 45-degrees, 90-degrees etc as well as other options for ignoring drilling and through holes and more. By selecting this option from the Solid Machining box and then clicking the NEXT button, you will open the first Planar box of the wizard. This is called the **Planar Finish Options** box.

This is the **Planer Finish Options** box.

This is the area where you will enter in the actual tool diameter that you will be creating the toolpath for.

You can select from Ball, Flat or Bull Nose cutters. When you select the Bull Nose you will need to enter this cutters Corner Radius.

Here you can select to create toolpath based off of the tool tip or tool center.

You will enter the step over for your tool here. The step over will be roughly half the size of the tool diameter that you will be using.

The Toolpath Angle option allows you to apply an angled type path. An example would be 45-degrees, 90-degrees etc.

The Tolerance option pertains to how small you want line and arc segments in the toolpath to be when following along the solid or surface contour.

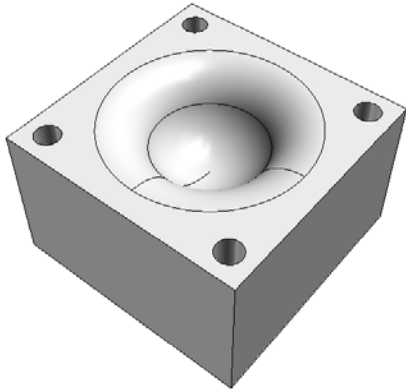
The Minimum Line Length allows you to control the minimum length of a resulting toolpath line. In other words, the toolpath is made up of lots of little lines. If any one of them is shorter than this value then the V20 combines it with an adjacent line. This means that no line in the toolpath should be shorter than this value. However, it also means that the toolpath may be a little less accurate.

The **IGNORE HOLES** option allows you to create toolpath that does not take the tool down inside the hole represented by a cylinder. This is used for avoiding drill holes that are represented by cylinders that do not go through the entire model. **IGNORE THROUGH HOLES** would create toolpath that would NOT go down inside of any type of opening that goes through the entire model.

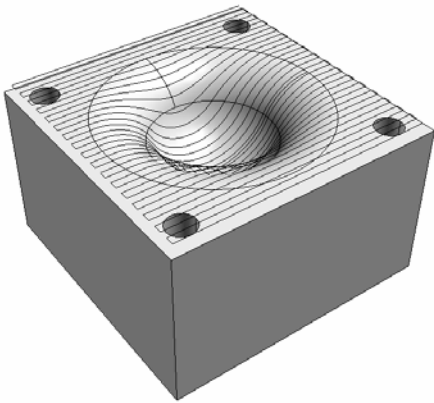
The Direction of Results section allows you to further define the toolpath. "Same Direction" creates toolpath that will only travel in one direction for cutting as you can see in the image below. The Alternate Directions option allows you to create regular Planar type toolpath.

Remember, the Planar Finish Options box offer the options to ignore drilling holes in a model that are represented by cylinders that **DO NOT** go all the way through the model as well as what are considered **Through Holes**. Through Holes would be any shape that cuts all the way through a surface or solid model.

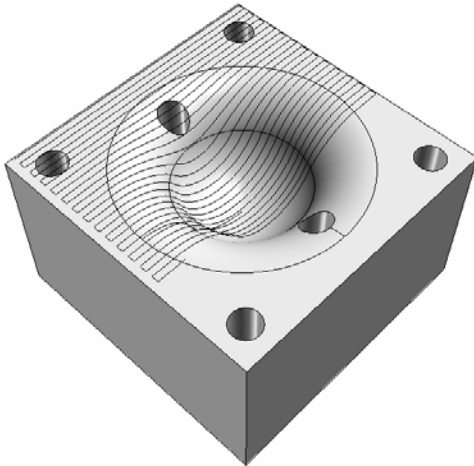
Take the example here where we have a part that you want to create toolpath for, leaving the drill holes separate.



By selecting the IGNORE DRILLING HOLES option you can generate toolpath that doesn't try to go down inside of cylinders. In this example the top of the holes are in a flat 2D plane.



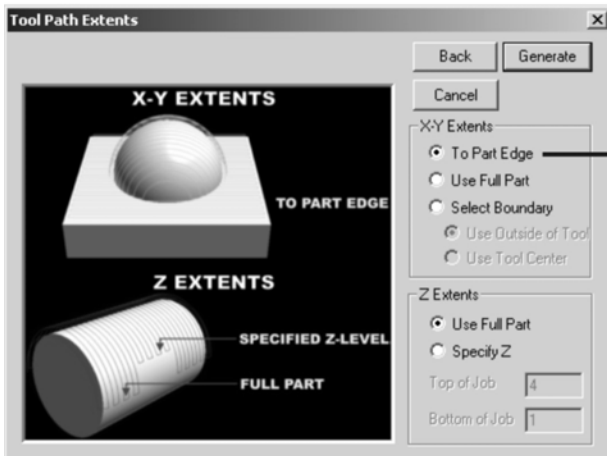
Here is an example of a drill hole on a 3D surface. The toolpath can be generated to avoid these as well by selecting the IGNORE DRILLING HOLES option.



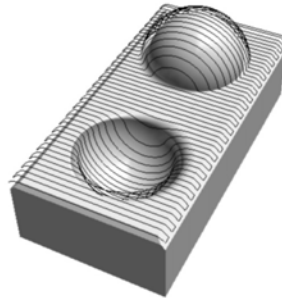
If you **Un-check** the IGNORE DRILLING HOLES option, the software will create toolpath that goes down inside of the cylinder **if the tool will fit**. This would also apply for the IGNORE THROUGH HOLES option. Once you have made your entries in the **Planer Cut options** box you will select the NEXT button to move forward in the toolpath wizard. The next box is called the **Toolpath Extents box**. This box offers several options for changing the type of toolpath that you want to create for your part. Let's take a look at the different aspects of this box starting next.

Here we have the **USE FULL PART** and **TO PART EDGE** options.

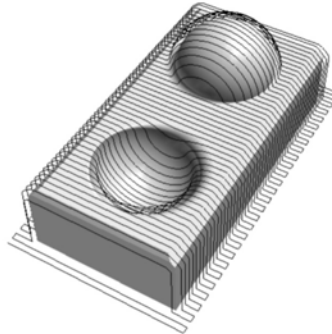
The **TO PART EDGE** option takes the toolpath to the edge of the models top edges and stops.



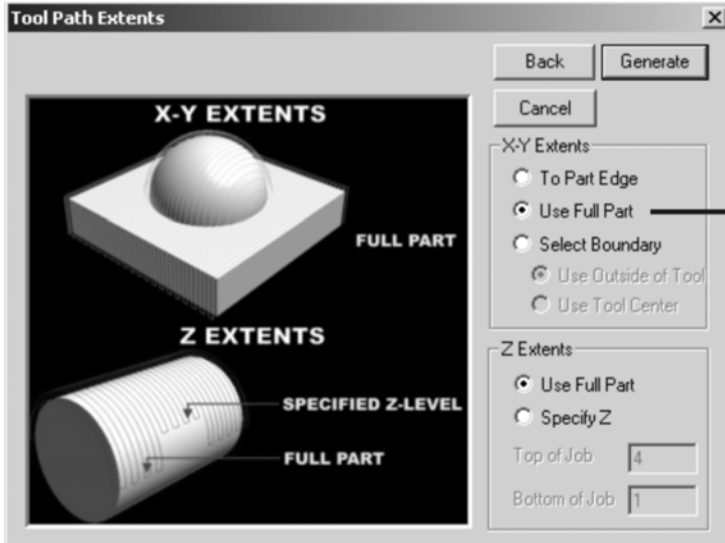
To Part Edge



Use Full Part



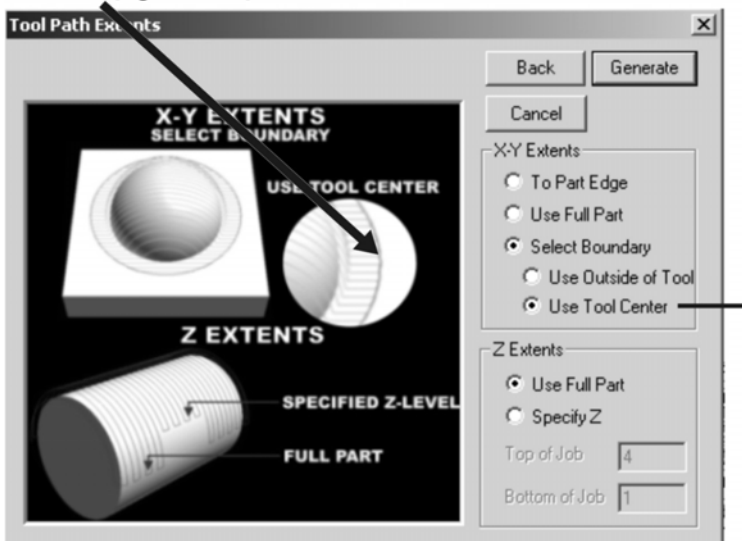
The **USE FULL PART** takes the toolpath all the way out to the very edge of the boundaries. The toolpath will waterfall over the edges of the part.



Version 21 also offers the ability to create machine toolpath within specified X & Y boundaries.

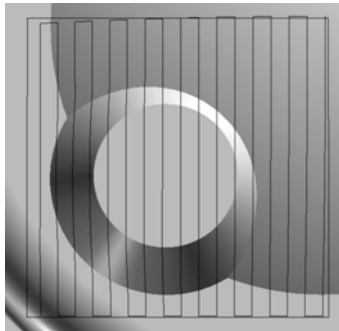
Here you can see the “Use Tool Center” option.

In this image you can see the “Use Tool Center” option has been selected. This creates toolpath used with a boundary. However the tool center goes all the way to the boundary. This means that you would not want to use the tool center option if you did not want the tool to go outside of the boundary geometry.



You can see these differences below.

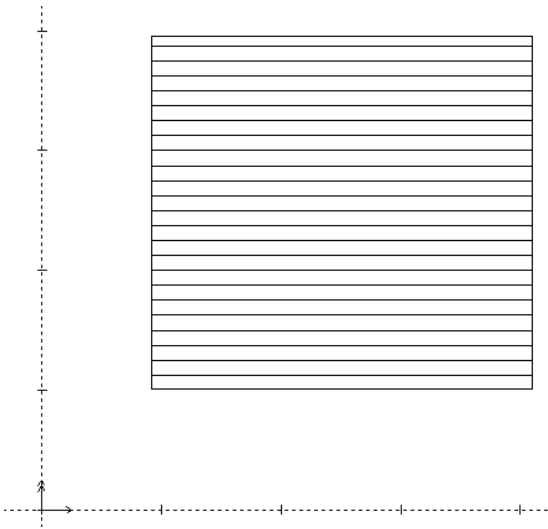
“Use Tool Center” type toolpath. “Use Outside of Tool” type toolpath.



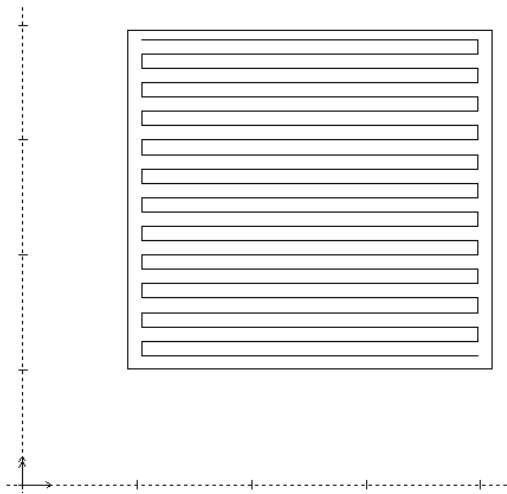
By using the TOOL CENTER option you will notice that the toolpath will extend all the way to the boundary geometry. By using the OUTSIDE OF TOOL option, you will automatically apply compensation for the tool so that the tool does not exceed the boundary at all. This is a very useful function in some cases.

Here it is one more time.

The next image shows a toolpath created inside of a square boundary using the center of the cutter. The toolpath goes directly to the edges of the square geometry.

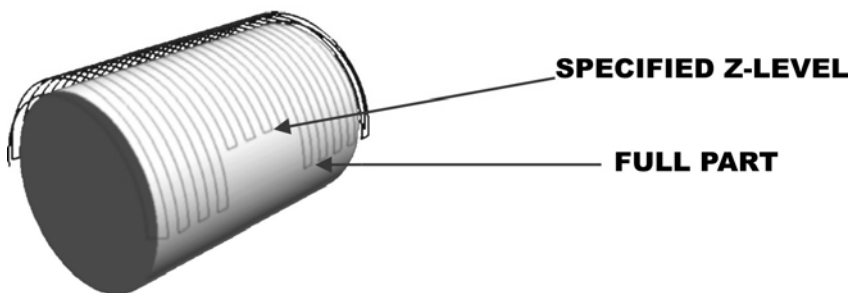


Here we have the same square boundary geometry with the Outside of TOOL option used to create the toolpath. Notice that the toolpath that is created prevents the tool from exceeding the boundary geometry.

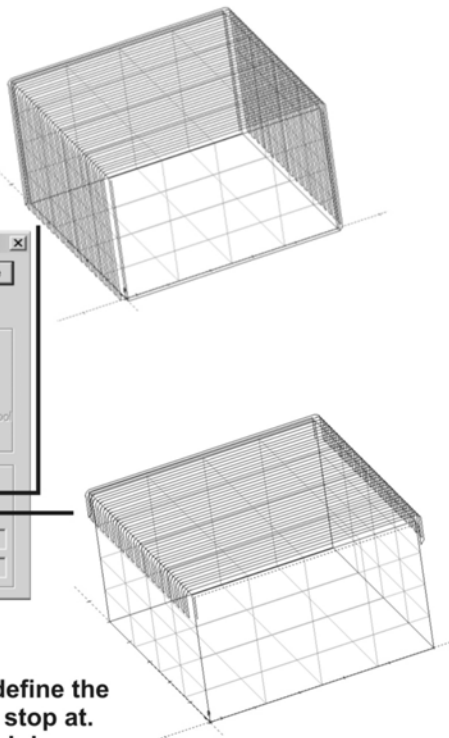
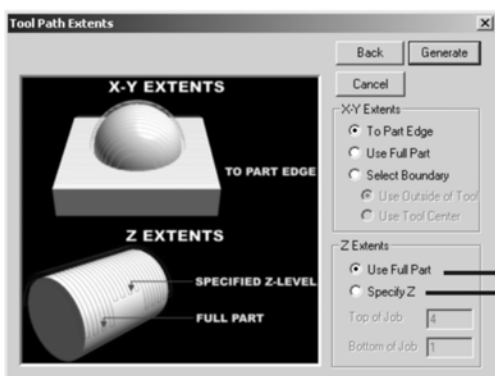


When using Version 21 to create toolpath for surfaces or solids, you can either draw the boundary geometry for toolpath creation or merge in the original wireframe geometry that was used for creating the surfaces that you have. You will need this to get an accurate boundary for creating the toolpath that you need. The boundary geometry that you use for creating toolpath can be flat 2D or 3D wireframe geometry with or without arcs or curves. The boundary geometry however must be a connected chain.

Now, let's take a look at the **Z-EXTENTS** section of this box starting below.



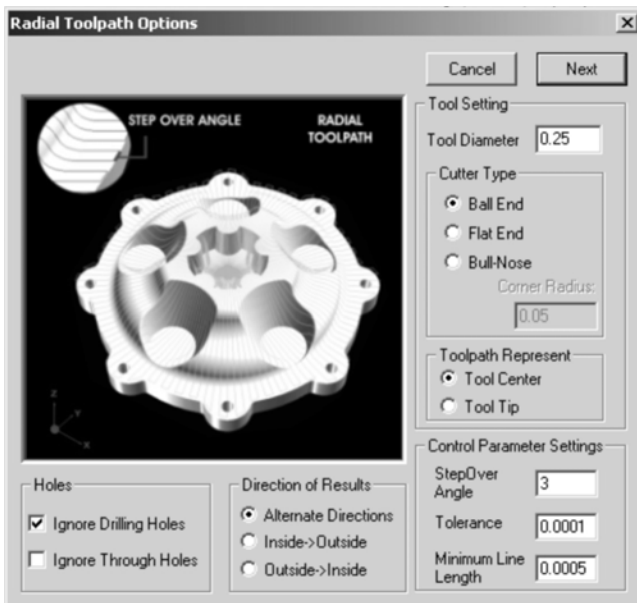
The **USE FULL PART** option allows you to create toolpath that waterfalls over the edges down to the bottom of the part.



The **SPECIFY Z** option allows you to define the -Z value that you want the toolpath to stop at. You will need to indicate the Z-Top of Job and the Z-Bottom of Job.

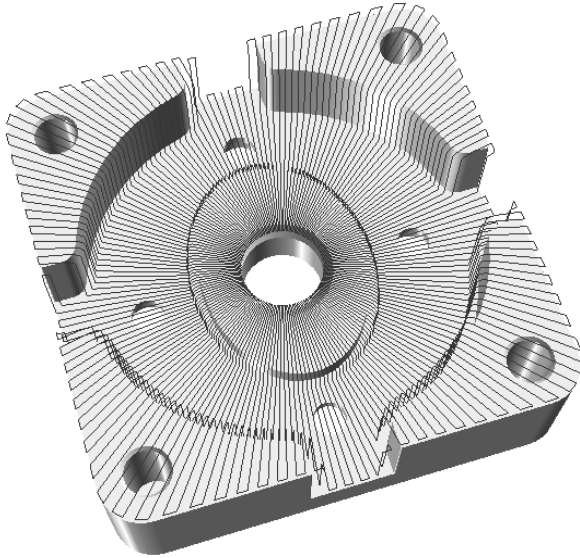
Once again, the Z-Bottom of Job is going to be the Z-Level that you want the toolpath to stop at. Once you have made the necessary entries, simply select the **GENERATE** button and the software will create the toolpath for you automatically.

RADIAL: The **Radial** toolpath option allows you to create toolpath that starts from the center of the model going outward and sweeping around the part in a 360-degree fan. This option also offers you control for cutting in one direction from the inside-out or the outside-in. In order to access this option you must first click on the solid or surface file to select it and then select **Generate Toolpath** from the Solids menu. This will open the **Solid Machining** box. At that point, click the Radial option and then the **NEXT** button to open the first box of the Radial Toolpath wizard. This is the **Radial Cut Options** box.

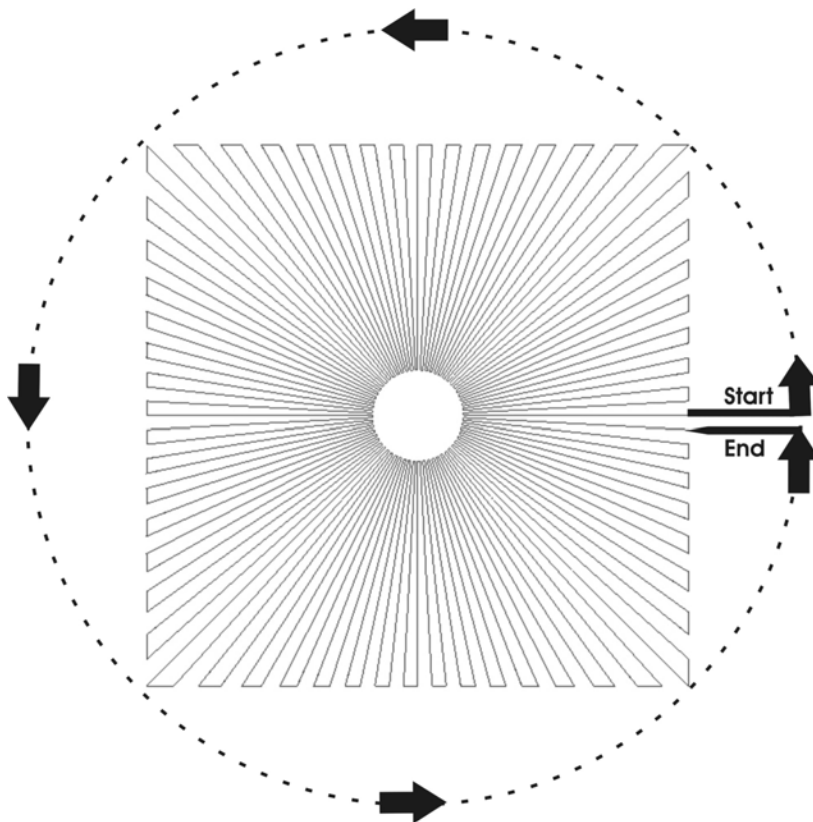


Similar to the Planar Finish options box, you have control options for ignoring drilling holes or through holes as well as 3 separate cutting direction options. Let's take a look at each of these cutting directions so that can use them.

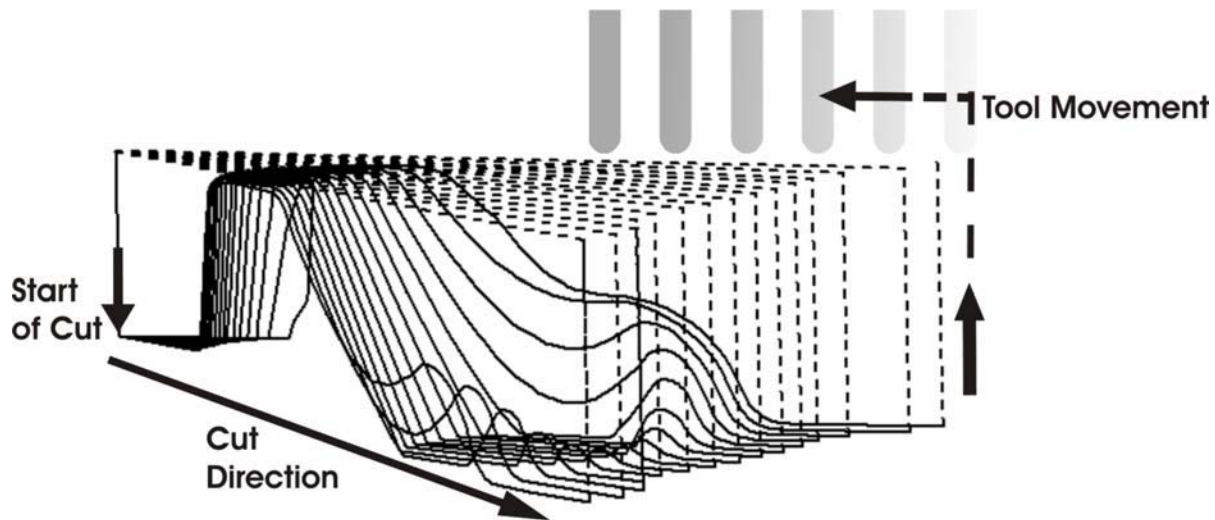
- **Radial Alternate Direction:** This basically creates toolpath that starts at the center of the part or boundary and goes outward to the 0 position of a theoretical arc and sweeps counter clockwise around the surface of the part or model until the toolpath reaches the starting position.



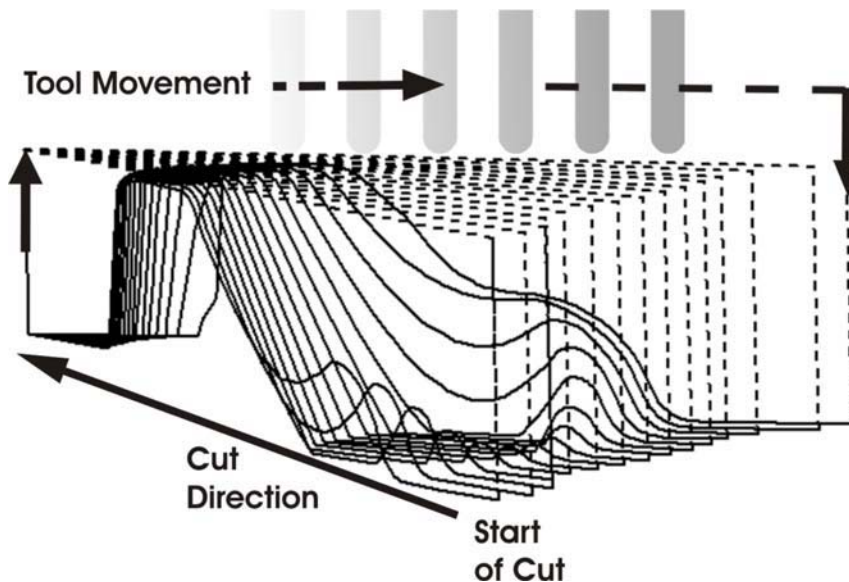
Here you can see that the toolpath starts at the 0 position of an arc and sweeps 360-degrees around the part based off of the specified step over angle that you enter.



- **Radial Inside>Outside:** This option allows you cut in one direction starting from the inside and cutting to the outside. Then the tool retracts back to the clearance position and rapids back to the next cut from the inside going out and so on.

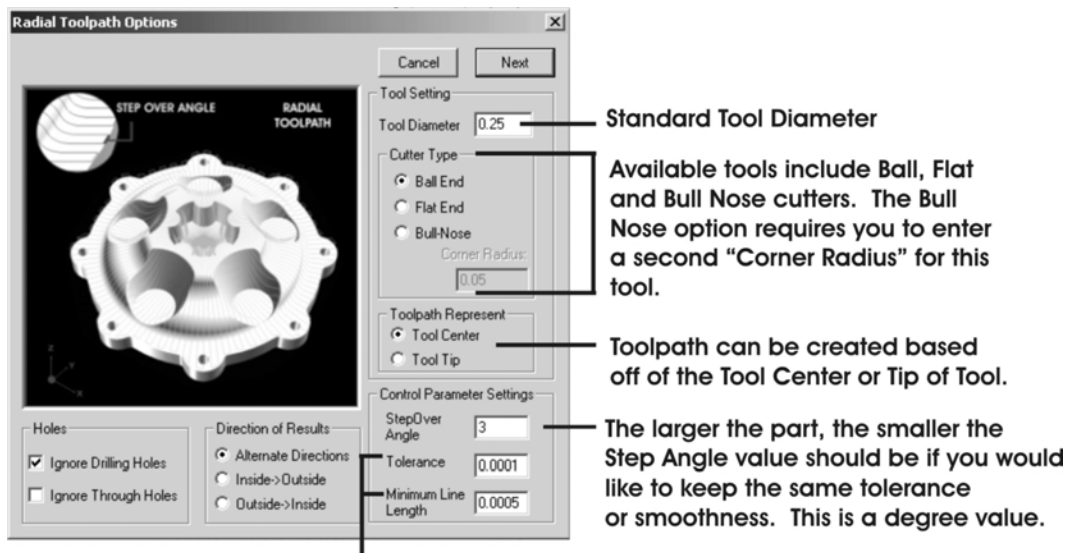


- **Radial Outside>Inside:** This option allows you cut in one direction starting from the outside and cutting toward the inside. Then the tool retracts back to the clearance position and rapids back to the (outside) next cut from the outside going in and so on.



Let's go ahead and take a closer look at the Radial Toolpath options box.

The Radial Toolpath Options box:

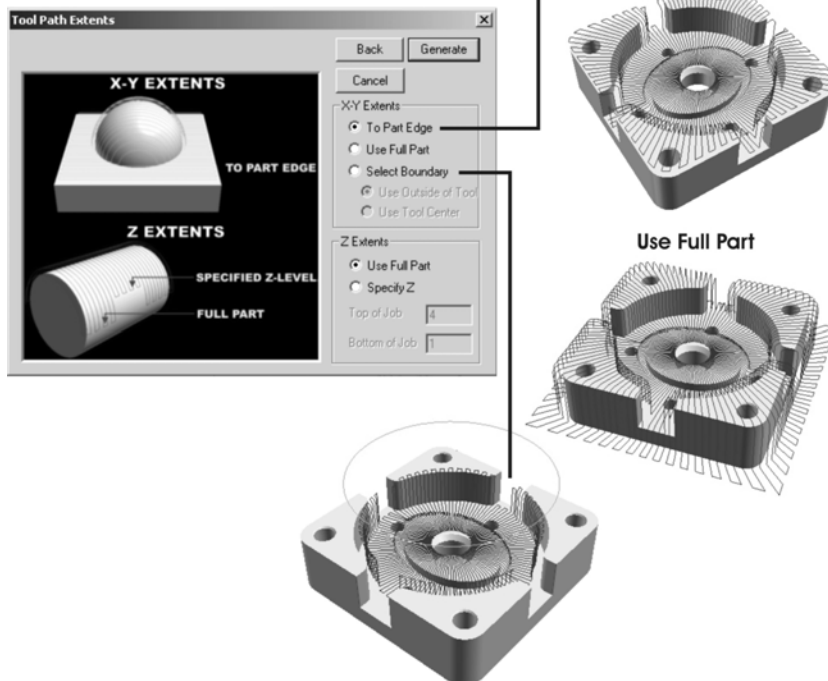


The **TOLERANCE** option pertains to how small you want line and arc segments in the toolpath to be when following along the solid or surface contour.

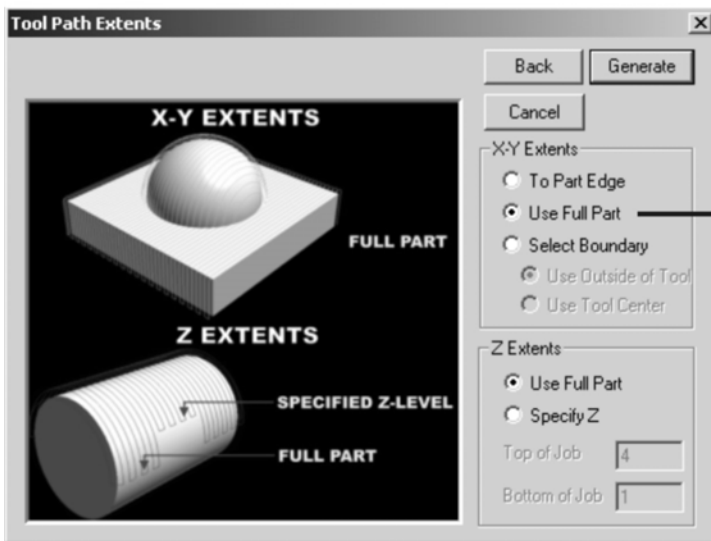
The **MINIMUM LINE LENGTH** allows you to control the minimum length of a resulting toolpath line. In other words, the toolpath is made up of lots of little lines. If any one of them is shorter than this value then the V20 combines it with an adjacent line. This means that no line in the toolpath should be shorter than this value. However, it also means that the toolpath may be a little less accurate.

Once you have selected and made your entries, click the **NEXT** button to move to the next box of the wizard. Here we have the **USE FULL PART** and **TO PART EDGE** options.

The **TO PART EDGE** option takes the toolpath to the edge of the models top edges and stops.



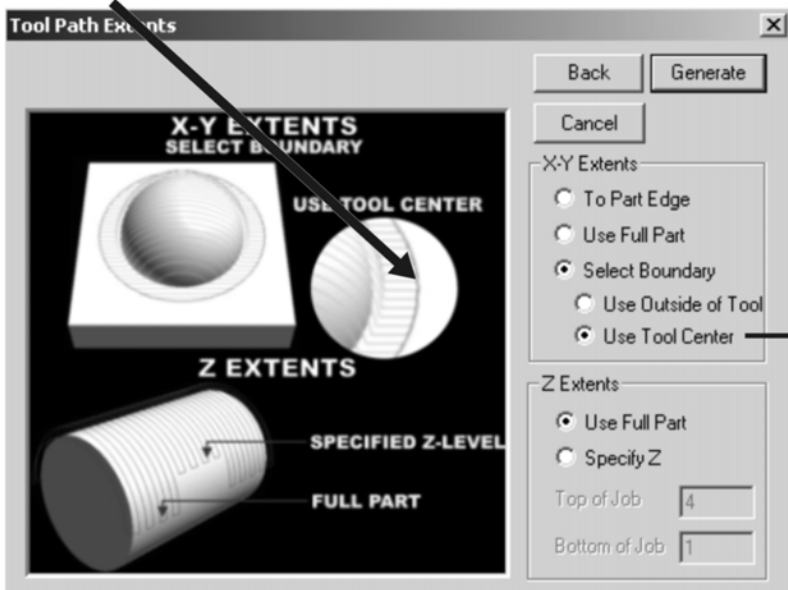
The **USE FULL PART** takes the toolpath all the way out to the very edge of the boundaries. The toolpath will waterfall over the edges of the part.



Version 21 also offers the ability to create machine toolpath within specified X & Y boundaries.

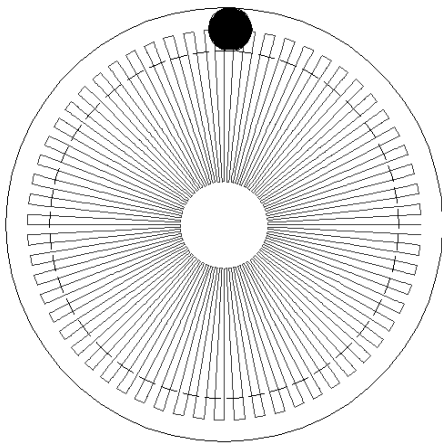
Here you can see the “Use Tool Center” option.

In this image you can see the “Use Tool Center” option has been selected. This creates toolpath used with a boundary. However the tool center goes all the way to the boundary. This means that you would not want to use the tool center option if you did not want the tool to go outside of the boundary geometry.

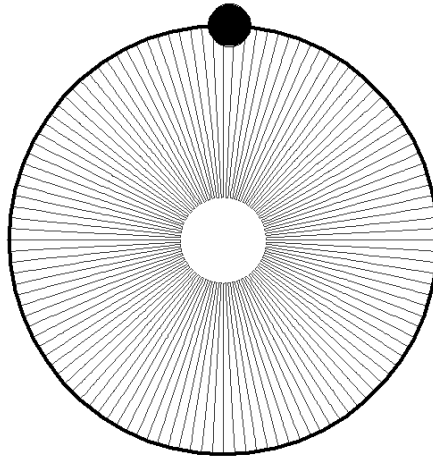


You can see these differences below.

“Use Outside of Tool” type toolpath.

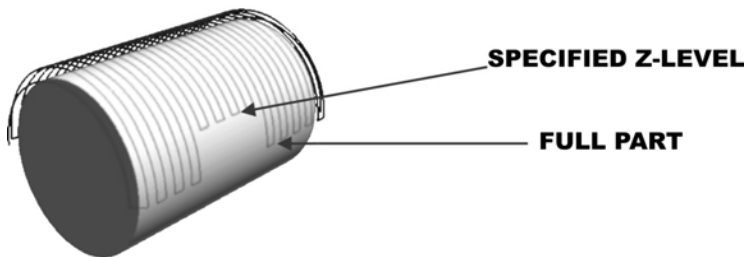


“Use Outside of Tool” type toolpath.

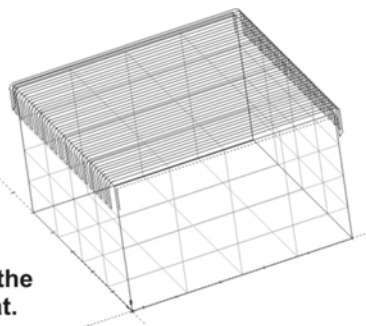
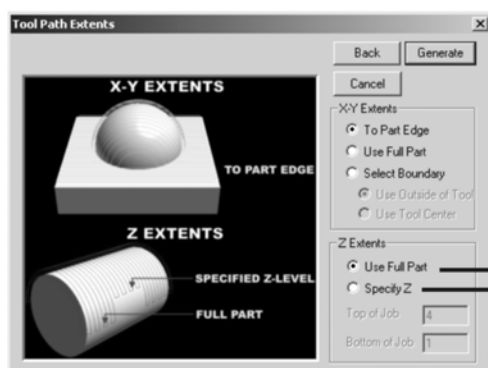
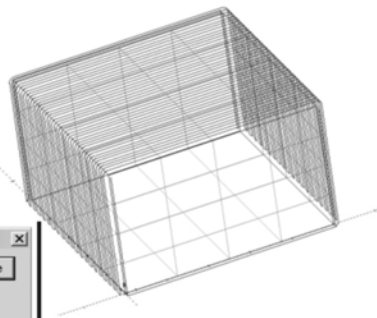


By using the TOOL CENTER option you will notice that the toolpath will extend all the way to the boundary geometry. By using the OUTSIDE OF TOOL option, you will automatically apply compensation for the tool so that the tool does not exceed the boundary at all. This is a very useful function in some cases.

Similar to the Planar Extents box you have the **Z-EXTENTS** section.



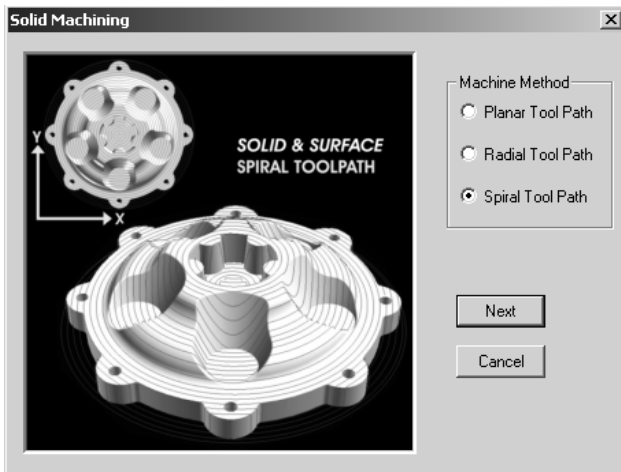
The **USE FULL PART** option allows you to create toolpath that waterfalls over the edges down to the bottom of the part.



The **SPECIFY Z** option allows you to define the -Z value that you want the toolpath to stop at. You will need to indicate the Z-Top of Job and the Z-Bottom of Job.

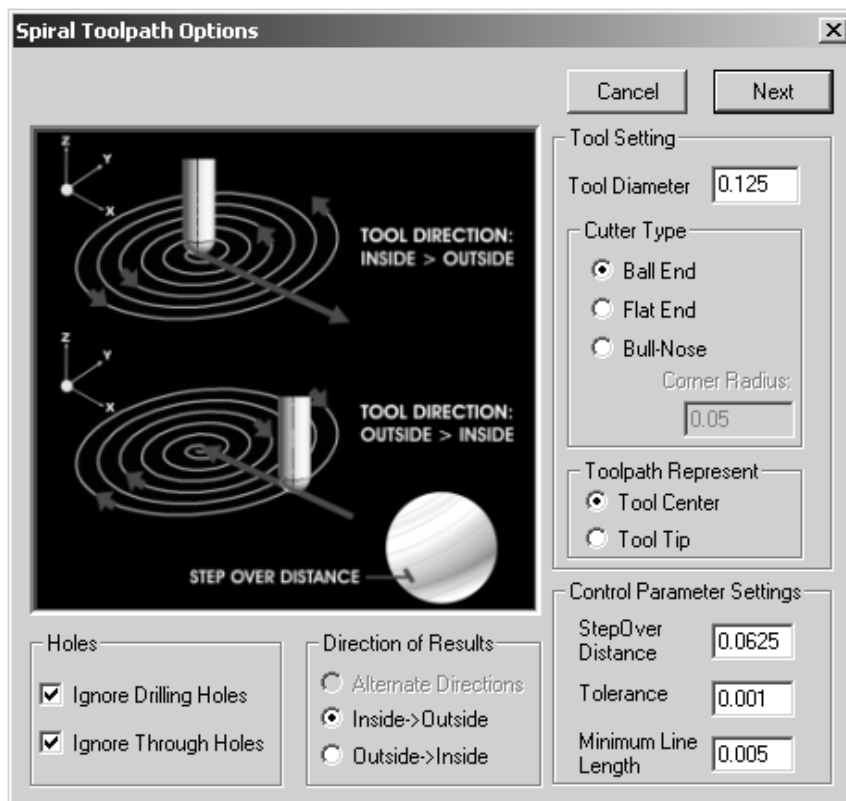
Once again, the Z-Bottom of Job is going to be the Z-Level that you want the toolpath to stop at. Once you have made the necessary entries, simply select the **GENERATE** button and the software will automatically create the toolpath for you.

SPIRAL: The **Spiral** toolpath option provides you with options for beginning the spiral from the outside of the part toward the inside or the inside spiraling towards the outside. There are the standard 3 X-Y extent options as well for machining the full part, machining to the edge of the part or machining within a specified 2D boundary. First select the model or surface file, go to the Solids menu and select **Generate Toolpath**. The Solid Machining box will appear.

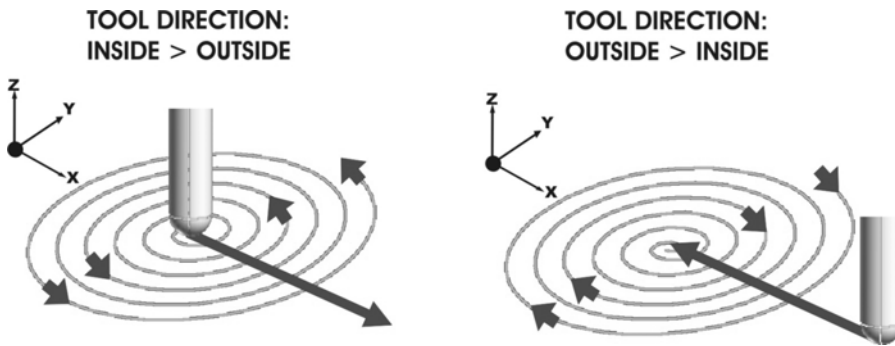


Select the Spiral option and then click the NEXT button to open the first box of the spiral toolpath wizard. The first box is called the “**Spiral Toolpath Options**” box.

The “**Spiral Toolpath Options**” box:



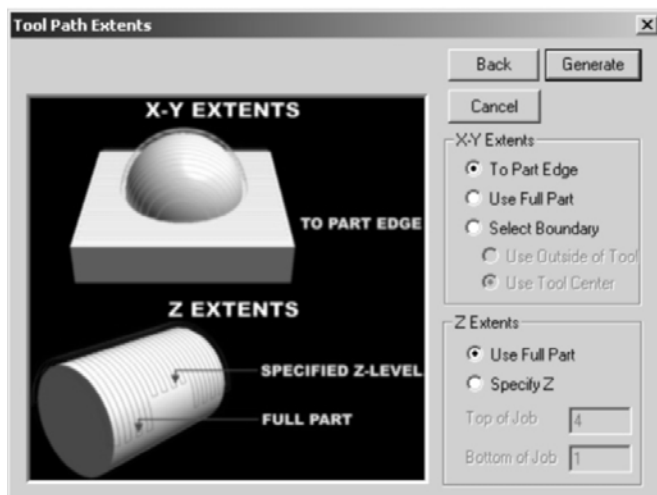
This box provides you with control options for starting the toolpath on the outside, working its way toward the center of the part or from the inside outward as you can see in the example image below.



Other aspects of this box include:

- **Tool Selection:** You can create toolpath for Ball, Flat and Bull Nose type cutters.
- **Tool Diameter.**
- **Hole avoidance.** The two options provide you the ability to create toolpath that avoids drilling holes or through holes. The spiral toolpath that is created produces a dashed line over the holes or through hole area rather than a solid toolpath line over them. If these options are un-checked the tool will attempt to go down inside holes if the tool can fit. This is normal. If the tool is too large the toolpath will be created over the top of the hole.
- **Tool Tip or Tool Center.** This allows you to program using either of these options.
- **Tool Step Over Distance.** This is a constant step over distance for the continuous spiral toolpath that is created.
- **Tolerance & Minimum Line Length.** This determines how tight the toolpath conforms to the surface. The toolpath is created with tiny line segments. The higher the minimum line length is set, the more lines of g-code you will produce when writing the actual program.

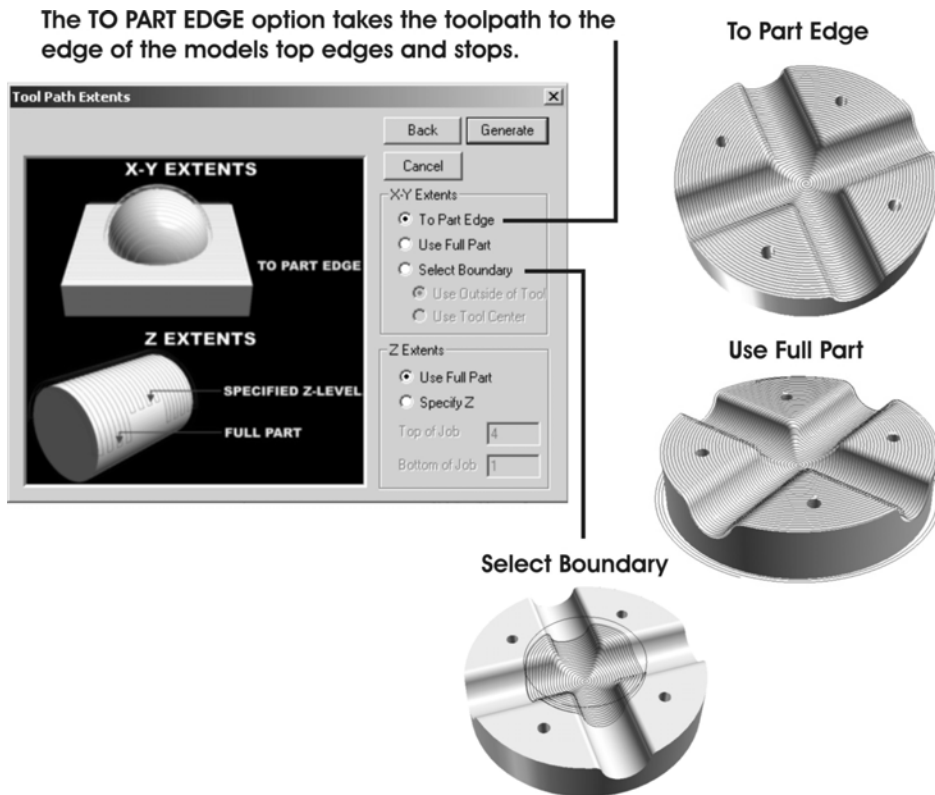
Once you have made your selections and entries, click the NEXT button to continue to the next box. This is the Toolpath Extents Box.



The Toolpath Extents box offers special control options.

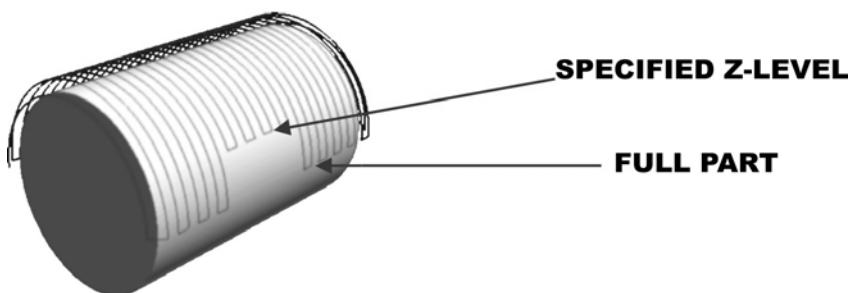
The Spiral X-Y Extents:

- **To Part Edge** allows you to drive toolpath out to the edge of the selected part surface and no further. Any avoided drill or through holes (if those options are chosen) will be avoided with a dashed line. The dashed line will bring the tool to the specified clearance and then a rapid move over the hole and back down.
- **Use Full Part** allows you to cover the complete part and part area boundary.
- **Select Boundary** allows you to generate toolpath within a specific 2D boundary.



The Spiral Z-Extents:

- **Use Full Part** allows you to generate toolpath for all Z level surfaces.
- **Specify Z** allows you to stop or start the toolpath a various Z levels. This option allows you to enter a Top of Job and a Bottom of Job value.



The procedure for using the GENERATE TOOLPATH functions: This would apply for all 3 toolpath types.

1. Open the CAM end of the software by selecting INSERT NC from the Special/NC CAM menu, choose a post processor and click OK. This opens the CAM.
2. Insert your program header from the CAM macro menu.
3. Insert a tool from the CAM tool menu.
4. Select the surface or solid model.
5. Go to the Solids menu and select GENERATE TOOLPATH.
6. Make the toolpath selection in the Solid Machining box and click the NEXT button.
7. Make the selections and entries in the first wizard box and click NEXT.
8. Make the selections in the Extents box and click the Generate button. If you chose the boundary option, click the boundary geometry.
9. The toolpath is created and the Tool Depth Settings box appears.
10. Enter your clearance, cutting depth and roughing option if you are roughing and click OK.
11. The G-Code program is created.
12. Select the Program end option from the CAM Macro menu.
13. You are done.

Section 31. Solid Tool Depth Settings

The following is the way to generate toolpath and the G-Code program with BobCAD Version 21 when machining solids and /or surface type files. This will also explain why the CAD and CAM windows open together the first time you open the V21 software after installation.

NOTE: The following is just a step by step example of how to generate G-Code through using the Solid Tool Depth Settings Box for 3D programming.

STEP 1

Import the part that you want to generate machine toolpath for. Once you have positioned the part where you want it to be on the screen, go to the SPECIAL/NC CAM menu and select INSERT NC. This will open the Insert NC box. Select a post processor by clicking on it one time and then select OK to open the CAM. You may need to select the **VIEW ALL** button from the CAD main toolbar.

The View ALL button:



STEP 2

You will now need to set up your start block in the CAM. Select the CAM MACRO menu, PROGRAM and then Program Start. Then any fixture offsets, G41 or G42 if using compensation, G43 height, Tool number and any M-Codes that you need to use. This is just for example. Basically, build your header or start block for the G-Code program.

The CAM G-Code editor is similar to a text editor. Therefore, you can edit and type in items that you need to have in the code. If you are manually setting up your header/start block, when finished just tab down to a new line or click your cursor at the beginning of the next line after your manual entries.

STEP 3

Now you will go back to the CAD side to generate the toolpath while the CAM is open. First select your solid model so that it highlights. Next, go to the CHANGE menu and choose the TRANSLATE function.

Incrementally move the part so that the TOP of the part is at Z0. The top of your part may already be at Z 0. If so, ok.

STEP 4

Decide if you want to machine the entire part or create a boundary to only machine inside of. If you want to create boundary geometry, make sure you are in 2D mode and draw a rectangle or circle over an area that you want to machine. When you have done this, deselect everything by clicking your mouse off of the model.

STEP 5

Now select **ONLY** the solid model so that it highlights. Next, go to the **SOLIDS** menu and choose **Generate Toolpath**, select the **Planar** option and click **NEXT** to open the first box of the Planar toolpath wizard. This is the Planer Cut Options box. Each of the 3 toolpath options have their own wizards to go through that are similar to each other.

STEP 6

Enter an appropriate tool diameter for your part. Pick one that you can use to cut this test part on your machine. Now enter a Step Over that is **HALF** the size of that tool. An example would be if you were using a .5-inch tool, your step over would need to be .25.

Leave the Toolpath Angle at 0, select **ALTERNATE DIRECTION** and then select the **NEXT** button.

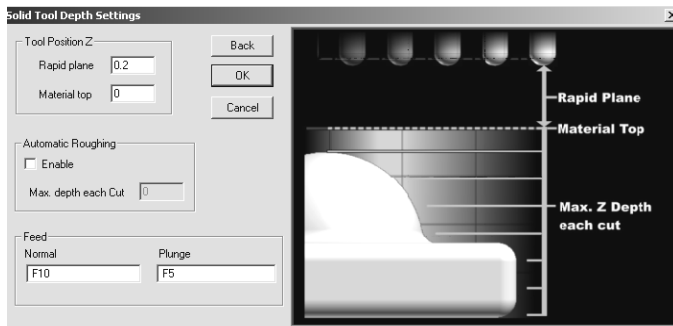
STEP 7

This is the **Z-EXTENTS** box. If you are using boundary geometry, select **Boundary** under **X-Y Extents**. Then choose a tool option for using the center or outside of the cutter/tool.

Under the **Z-Extents** section select **USE FULL PART** or enter in a negative **Z** level amount that you want the toolpath to stop at. Now select **GENERATE**. If you are using boundary geometry, select it to generate the toolpath. If you are creating toolpath for the full part, the software will automatically begin drawing the toolpath.

STEP 8

Once the toolpath is generated, the Solid Tool Depth Settings box will automatically appear.



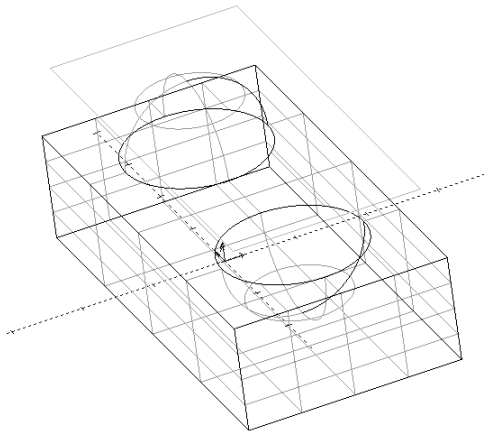
The Solid Tool Depth Settings box asks you for:

- Rapid Z Plane/Clearance
- Material Top
- Allows for Auto Roughing
- If roughing, you can enter a maximum Z-depth for each roughing pass.

You can also manually enter in speeds and feeds as needed. Once you click **OK** the G-Code program will be created based off of this information. If you are roughing, you will want to click on the **ENABLE** check box and enter your maximum depth in accordance with the part and the size tool that you have earlier decided to use for this part. That's basically it. Do the next lesson to learn more about creating toolpath and the G-Code programs that you need.

Section 32. Quick CAM Training Lesson

NOTE: This quick CAM lesson uses a pre-loaded cad file that is saved in the SAMPLES folder that loaded with V21. Select the OPEN function in the FILE menu and locate your **SAMPLES** folder, double click on it to open the folder and then select the file called, “**Quick CAM1.cad**” and open it in the software.



STEP 1

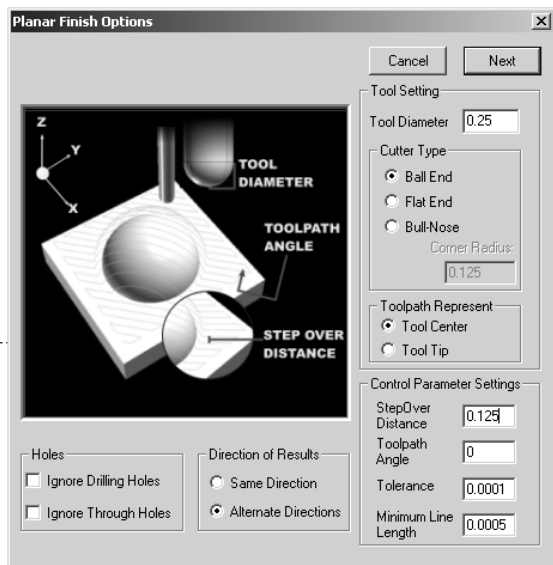
Both the CAD *and* the CAM screens should be open at this time. If they are not, select the SPECIAL/NC CAM menu and select INSERT NC. In the Insert NC Object box that appears, select the **FANUC 6M** post processor and then click OK to open the CAM. Now click on the VIEW ALL button on the main CAD toolbar.



NOTE: *This is the time that you would normally go to the CAM side of the software and build your start block or header for the program. You would select the MACRO menu, PROGRAM and then Program Start, add fixture offsets, compensation, add your tool and any M-Codes needed etc.*

STEP 2

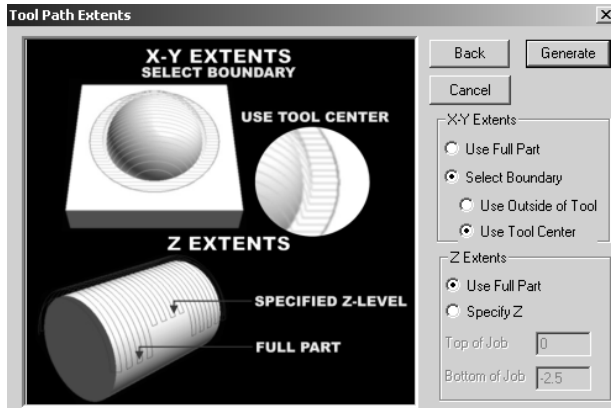
Now click on the solid part to select it. Now go to the solids menu and select **GENERATE TOOLPATH**. Select the Planar option and click NEXT. This will open the Planer Finish Options box.



Enter the following values:

- .25 for Tool Diameter
- Select Ball End as the cutter
- Select Tool Tip.
- .125 for the Step Over
- 0 for Toolpath Angle
- .0001 for Tolerance
- .0005 for the Minimum Line Length.
- DO NOT check the IGNORE DRILL or THROUGH HOLES options under holes.
- Select the ALTERNATE Directions option.

Now select NEXT to advance to the TOOLPATH EXTENTS box.



Under the X-Y Extents section, select **BOUNDARY** and then click the **USE TOOL CENTER** option.

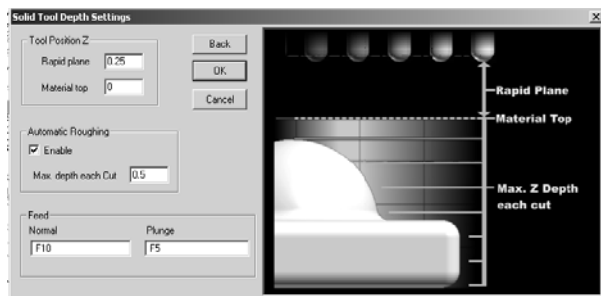
Under Z-EXTENTS select **USE FULL PART**.

Now select GENERATE. You will then need to click on the wireframe rectangle geometry that you see above the solid. Just click on it once.

Once you have clicked on the rectangle geometry the toolpath will begin to be created. This may take a moment depending on the power of your computer.

STEP 3

Now that the toolpath is completed, you should have the Solid Tool Depth Settings box on the screen.



Enter .25 for the Rapid Plane. The material top is 0. Check the ENABLE box and enter .5 for the Maximum Z-Depth for each pass. Then select the OK button to generate the G-Code program.

NOTE: This may take a moment depending on how powerful your computer is.

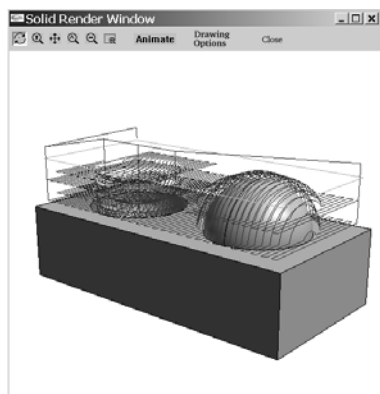
You should now have the complete G-Code program.

STEP 4

Now select the EDIT menu in the CAM and choose SELECT ENTITIES and then "Select ALL." All of the G-Code program will highlight. Once the code is selected, go to the EDIT menu of the CAM again and select GEOMETRY FROM NC. This is the Backplot function. Now the software will redraw the roughing toolpath on the CAD side of the software. When this is done, go to the EDIT menu of the CAD side and choose SELECT ALL. Everything on the CAD side will highlight.

STEP 5

Now go to the SOLID menu and select RENDER. The solid render box has various options as well for viewing, panning, dynamic rotation and zoom. You will learn more about this as you go through this training guide.

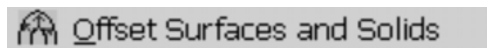


This is an excellent way to inspect the toolpath to make sure that you have the right G-Code program.

IMPORTANT TOOLPATH NOTE: ONLY select the Toolpath when generating the G-Code in the CAM. You will notice that once the toolpath is generated, the surface or solid will deselect and only the toolpath will be selected. Another way to select (only) the toolpath is to change the active color before entering the toolpath wizard so that the new toolpath is generated in a different color. Then, simply use the color selection feature, choose the toolpath color and click ok to only select the toolpath by color. The type of toolpath generated for solids and surfaces may end up giving you a large amount of G-Code lines. The faster speed and the larger the hard drive you have in your computer will determine the speed in which the G-Code program is created.

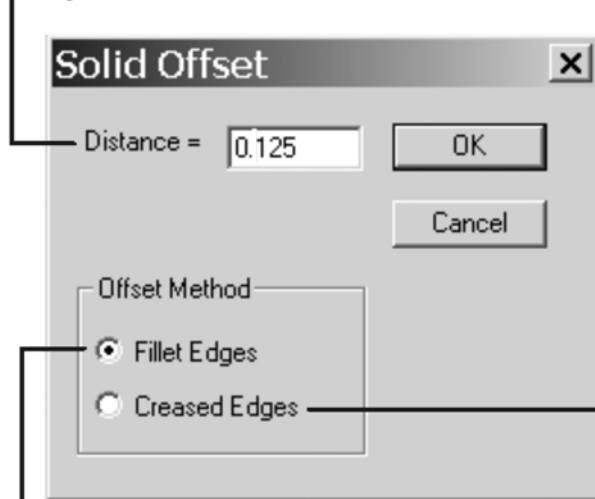
Section 33. Offset Surfaces & Solids

The next item in the Solids menu is the **Offset** Surfaces & Solids function.



By selecting your model and then the Solids menu, then the Offset function, the new offset box will appear. Let's have a look below.

Here, you will enter in the offset distance that you want.



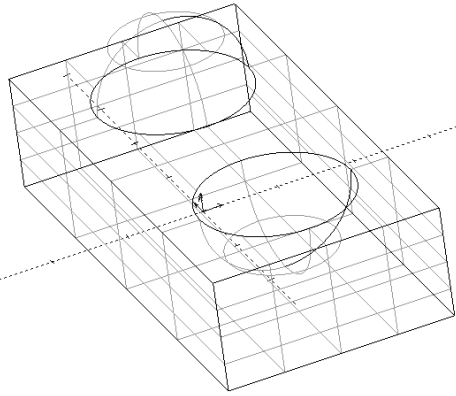
By selecting the Fillet Edges option you will create an offset that has fillet surfaces connecting all of the edges.



By selecting the Creased Edges option you will create an offset surface that does not have fillet surfaces. Instead the offset surfaces will be connected with straight edges.

After the offset is drawn, it will be selected and the original solid or surface will not be selected. This makes it possible to quickly move forward and create toolpath as needed. Let's take a look at how this function works.

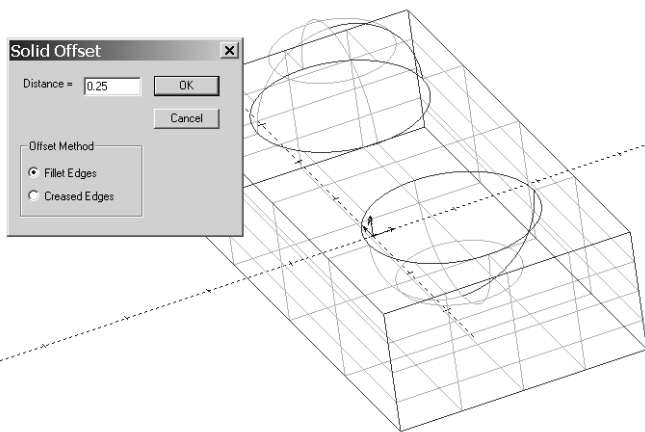
Example 1. The solid model.



Here is a basic solid part just to the left. You may want to offset this solid with surfaces that are all sewed or stitched together.

To do this you would first select the solid and then go to the Solids menu and select the **Offset** Surfaces & Solids function.

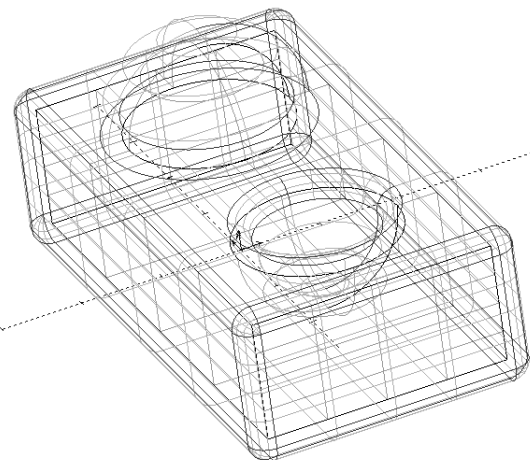
Example 2. Using the Offset function.



After selecting the solid part and the offset function, the Solid Offset box will appear as you can see in the example 2 image to the left. In this box, you will enter the offset distance that you need and then select the style of offset that you want to produce. There are two options.

There is the Fillet Edges option and the Creased Edges option. In this example we selected the FILLET EDGES option. This option is going to produce fillet surfaces on all edges. Have a look at the next example.

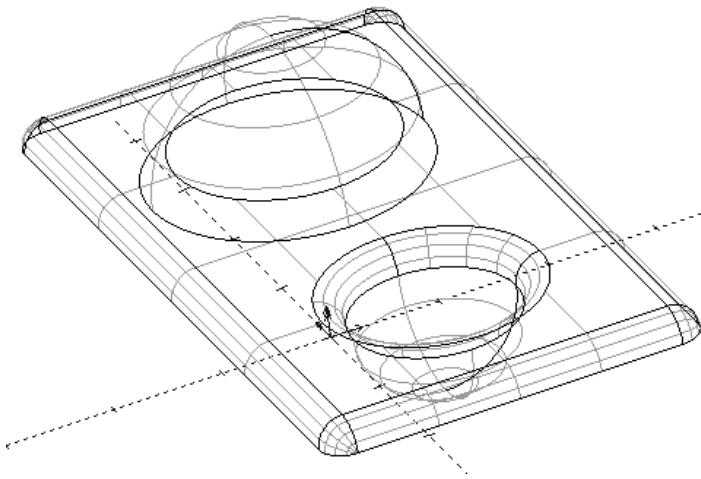
Example 3. The offset.



Once the offset surface/surfaces are created, the original solid will deselect, leaving the offset selected and ready to use the Generate Toolpath function on.

If you need to, you can now use the Unstitch Solid to Surfaces function in the **STITCHING** sub-menu to disconnect all of the offset surfaces. Let's go ahead and have a look at this as well in Example 4.

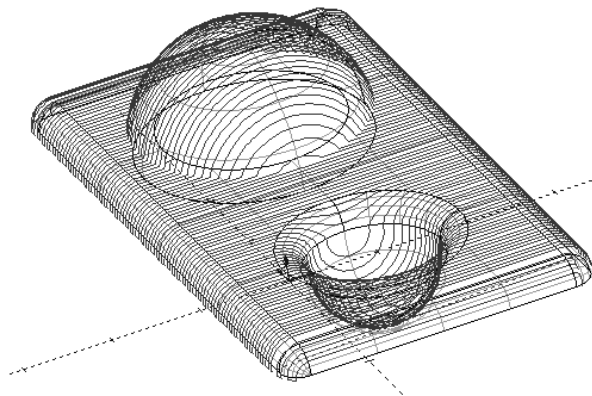
Example 4. Editing the offset.



After unstitching the solid to surfaces you can hold down the shift key and click on all of the surfaces that you would like to remove from the part. In the example image to the left, we decided to keep what you can see. Once you remove the surfaces that you want to, you can use the EDIT select ALL function to select all of the remaining surfaces and then select the **STITCH SURFACES TO SOLID** function from the **STITCHING** sub-menu to sew all of the surfaces back together for generating toolpath.

See this in the next Example 5 image.

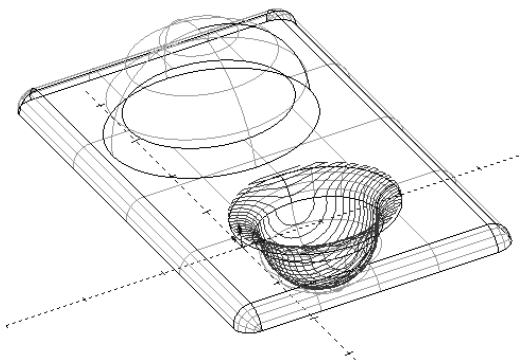
Example 5. Generating toolpath for offset surfaces.



In this example image to the left, you can see that the surface was selected and then we used the **GENERATE TOOLPATH** function to create the toolpath that we wanted.

You can also create surface toolpath for individual surfaces that you select. Have a look at Example 6.

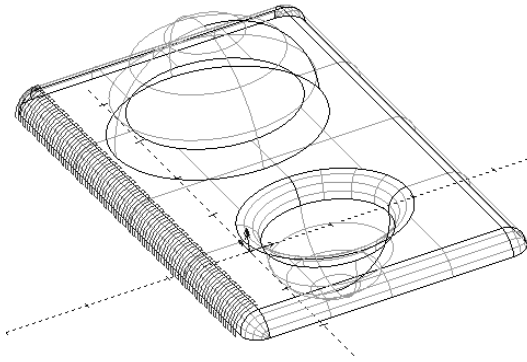
Example 6. Creating toolpath for individual surfaces. Here, you can see that we selected a specific area to create toolpath for. You must first **UNSTITCH SOLID TO SURFACES**, pick the surfaces that you want to create toolpath for and then use the **STITCH SURFACES TO SOLID** function to sew them together.



You must do this or the **GENERATE TOOLPATH** function will not be available for selection.

SURFACE/GENERATE TOOLPATH RULE: *If you are selecting more than 1 surface to create toolpath on, you MUST stitch them together first. Do this by selecting multiple surfaces first and then using the **STITCH SURFACES TO SOLID** function in the **STITCHING** sub-menu of the main **SOLIDS** menu. You can machine an individual surface by itself. Just select the surface by itself and then use the **Generate Toolpath** function in the **SOLIDS** main menu.*

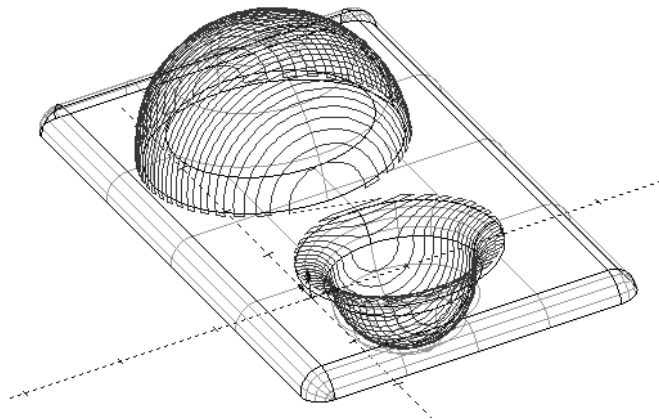
Example 7. The individual offset surface for creating toolpath.



In the example image to the left you can see that only one offset surface was selected to create toolpath.

You can also pick separate surfaces that do not connect to create toolpath. Have a look at the Example 8.

Example 8. Non-connected offset surfaces for creating toolpath.



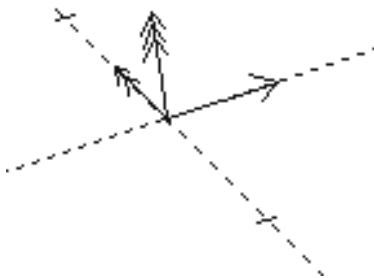
In the example image to the left you can see that toolpath was created for the recessed surface, then made a rapid move and proceeded to create the toolpath for the other section.

When there is a dashed line in the toolpath the CAM will bring the tool to clearance and rapid to the next cut or series of toolpath.

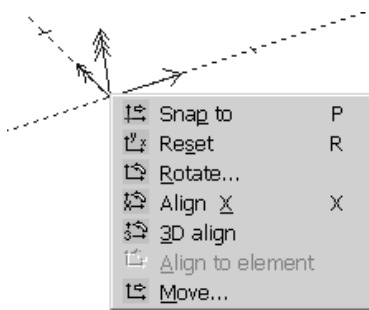
And so, if you are creating toolpath for several different offset surfaces or regular surfaces you now know that you have to use the STITCHING function on them first before being able to continue creating toolpath.

UCS NOTE: Before attempting to generate toolpath and a **G-Code program** for a surface or solid, you **MUST** reset the UCS. Do this by placing your cursor on the UCS itself, right-click your mouse and select the **RESET** option.

The UCS:



After Right-clicking your mouse.



When attempting to generate toolpath for surfaces and solids, use the TRANSLATE function in the CHANGE menu to incrementally move your part so that the TOP of the part is at the Z0 location. The UCS **CAN** be used for designing parts with primitive solids in the Version 21 software. NOTE: If you are using boundary geometry for machining inside a boundary you can relocate the boundary to Z0. Do this by selecting the 3D Menu, select MORE and then MOVE TO Z. UNCHECK RELATIVE MODE as you want to be in Absolute mode and enter 0 for the Z-Value after making your selection type based off of the boundary geometry. In other words if the boundary geometry is an arc, use the select SINGLE choice as it is one entity.

If you are using a square (4 entities) as the boundary geometry then you would choose CHAIN as the selection type. Then make your geometry selection and right-click your mouse to move the boundary geometry to Z-0. Now you will not need to reset the UCS.

Section 34. Stitching or Un-Stitching Surfaces & Solids

BobCAD Version 21 offers stitching functionality for working with surfaces or solid models. STITCHING is located in the new Solids Menu and has a special sub-menu with the following options:

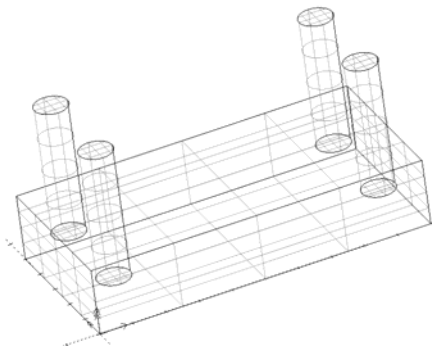
- Stitch Surfaces to Solid.
- Un-Stitch Solid to Surfaces.
- Un-Stitch Surface from Solid.

Each one of these offer a different result and have to do with creating toolpath for stitched surfaces or unstitched solids. Let's take a close look at each of these options and how they effect toolpath generation.

STITCH SURFACES TO SOLID:

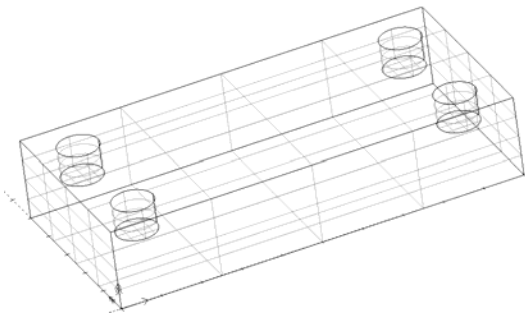
This function allows you to automatically sew two or more selected surfaces together to create a solid. Even though this function is labeled for surfaces, you can also use it on solids. Here's an example below.

Example Image 1.



In the image to the left you can see a solid rectangle and 4 non-intersecting cylinders. By holding down your shift key and selecting each of the four cylinders, you can select this function (Stitch Surfaces to Solid) and stitch them together into a group. This is a fast way to perform Boolean Operations on more than two solids at a time.

Example Image 2.

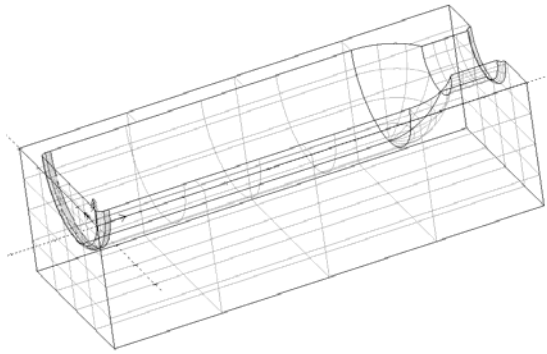


By using the Stitch Surfaces to Solid function on all four cylinders at once, we were able to use the Boolean Difference function against the rectangular solid to get the result image on the left.

NOTE: The STITCH SURFACES TO SOLID option will NOT be available for selection unless you have more than one surface that has been selected in the drawing screen.

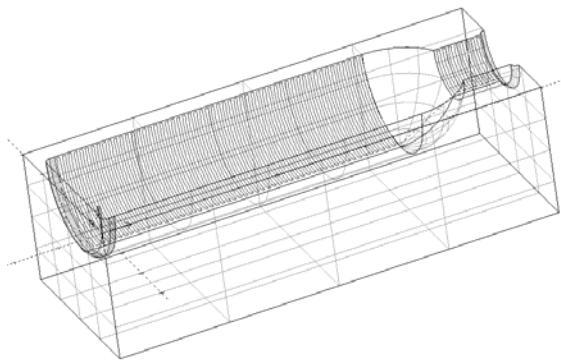
Now let's take the example of a bottle that has been un-stitched into individual surfaces.

Example 1. Individual Surfaces.



The image to the left was a solid that was un-stitched to individual surfaces. You can select an individual surface to generate toolpath on, but if you want to generate toolpath for two surface you **MUST** use the STITCH SURFACES TO SOLID option first before you will be able to generate toolpath. This is a rule.

Example 2. Toolpath after using the Stitch Surfaces to Solid option.



You can select as many surfaces as you want to. Then use this option. Simply hold down your SHIFT key while clicking on the surfaces that you want to stitch together to make them a solid. This is also a quick way to select surfaces and stitch them together, to later save them on specific layers as needed.

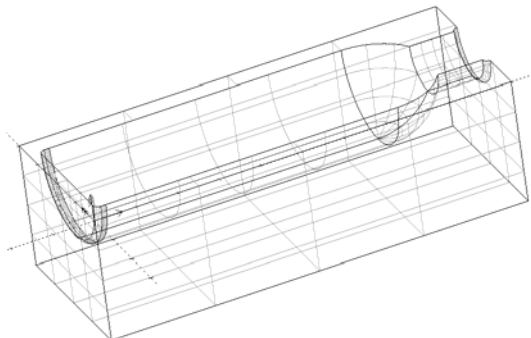
You could take an imported solid model, use the Un-Stitch surfaces from solid function and then reselect surfaces, change their colors and save various aspects of the part on different layers to use later. This can be very useful when working with complex models that require different sets of toolpath.

UN-STITCH SOLID TO SURFACES:

This function is actually very simple. By selecting a solid you can then use this function to break it up into its individual surfaces. You **MUST** first select the solid for this option to be available. Use this function to un-stitch multiple surfaces that have previously been stitched together. This function doesn't always have to be used on solid parts or primitive solids.

UN-STITCH SURFACE FROM SOLID:

This function allows you to pick an individual surface from a solid and un-stitch it by itself. Let's take a closer look.

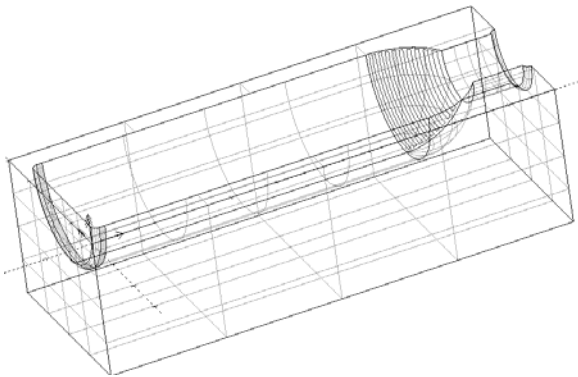


Example 1. An Imported or designed solid model.

Let's say that you designed this bottle in the Version 21 software. Now you want to generate toolpath for only one surface. **Make sure that the solid is NOT selected on the screen first.** Then go to the Solids menu, select STITCHING and then choose the Un-Stitch SURFACE FROM SOLID option. It will be the only item available as nothing was pre-selected.

Then simply click on the individual surface that you want to disconnect/un-stitch from the rest of the solid. The V21 system will automatically perform the action. Now click your mouse off of the drawing to deselect everything. Lastly, select the individual surface by clicking on it and you will find that the GENERATE TOOLPATH function will be available for selection now.

Example 2. Individual surface toolpath.



In the image to the left, you can see the end result of this procedure. You can only use this function on one surface at a time. Nothing on the screen can be selected if you want to use this function.

Section 35. Extract Edges

The **Extract Edges** function has been added for two reasons.

1. You may only want to use Version 21 for drilling holes and all you have is a solid model. Thus, extracting arcs from cylinders for snapping points and using drilling cycles.
2. You may want to extract the edges for use in creating a boundary for creating toolpath based off of your imported or created model or a 2D surface.

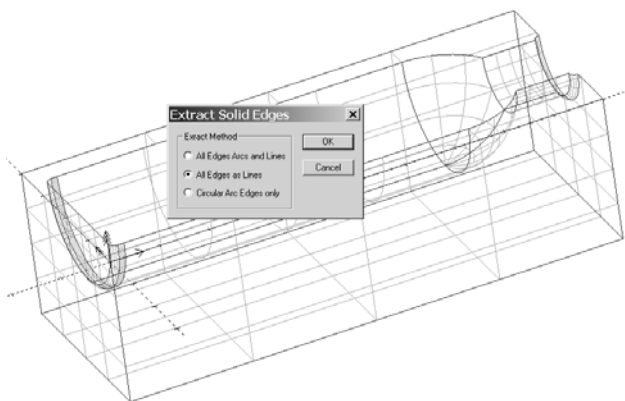
The three extract options are:

- Extract ALL Edges ARCS and Lines. *This will extract all of the edge lines and edge arcs in the solid or surface.*
- Extract ALL Edges as Lines. *This will only extract the edge lines in the solid or surface.*
- Extract Circular ARC Edges ONLY. *This will ONLY extract the ARC edges in the solid or surface.*

Let's take a look at how these EXTRACT options work.

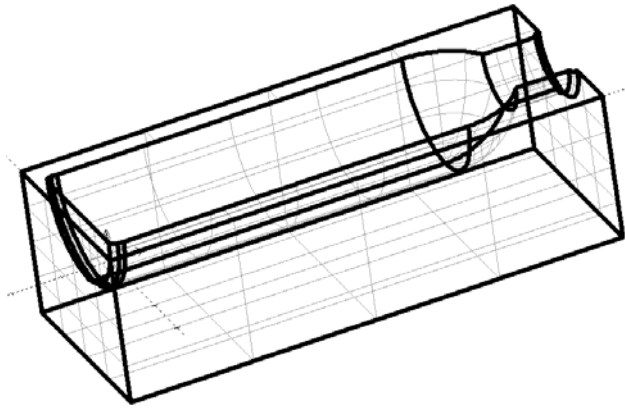
EXTRACT ALL EDGES ARCS & LINES:

Example 1. Using the Extract all Edges ARCS & Lines on a solid.



In the example to the left you can see the Extract Solid Edges box. You must first, select the solid or surface and then select this function.

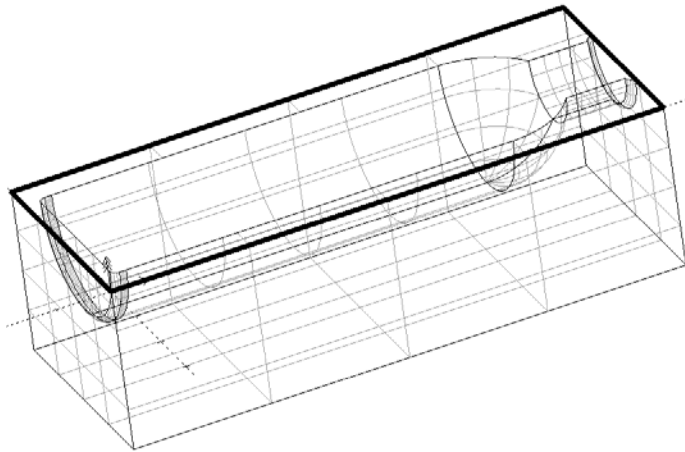
Example 2. *The result of this extract option.*



Here in the image to the left you can see that all of the extracted geometry has been enlarged.

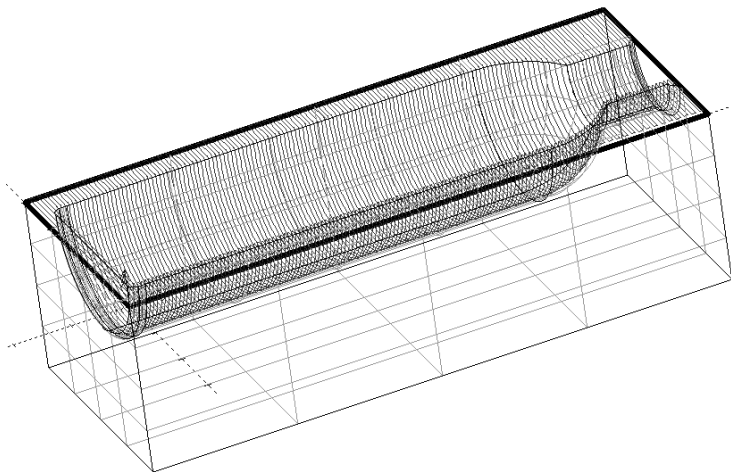
TIP: Change the ACTIVE drawing color before using this feature so that the extracted geometry is NOT the same color as the solid or surface. This makes it easy to select this geometry for saving it on a separate layer or other purposes.

Example 3. *Editing the extracted geometry.*



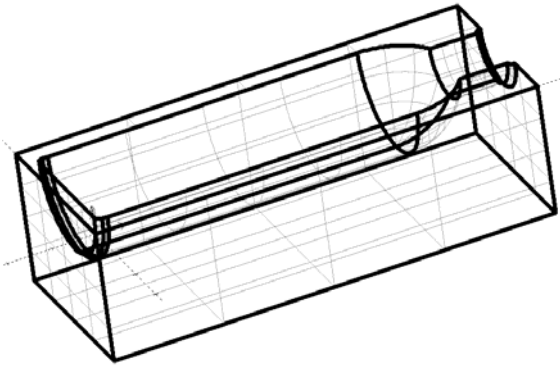
In this image you can see that we edited out the extracted edges that we did not want to use and kept a top boundary to use for creating the toolpath.

Example 4. *Using extracted geometry for Boundary Machining.*



The image here shows the toolpath created using the extracted and edited boundary geometry.

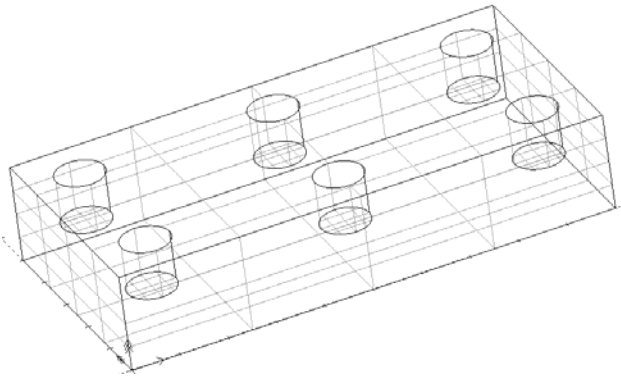
EXTRACT ALL EDGES AS LINES:



Once again, this extracts all of the edges on the solid as lines rather than lines and arcs.

EXTRACT ALL ARC EDGES: This function will extract the arcs only.

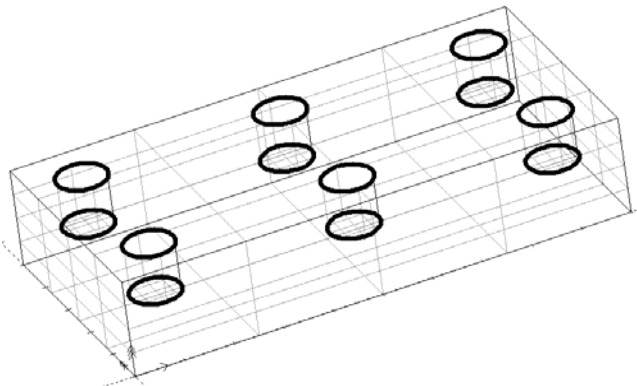
Example 1. A solid part with cylinders to be used for drilling cycles.



Here is an example of a plate that has six cylinders that we want to use for drilling holes.

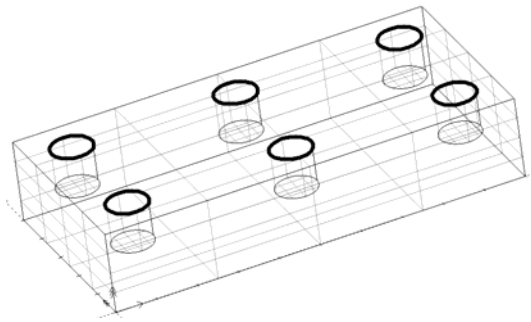
We will first change the active color to something other than the color of the solid part.

Example 2. The extracted arcs.



In this image you can see that the top and bottom arcs of each cylinder have been extracted. So now we need to edit out the arcs drawn at the bottom of the cylinders. We will do this by holding down the SHIFT key and clicking on the bottom arcs. When they are all selected we hit the delete key and delete them.

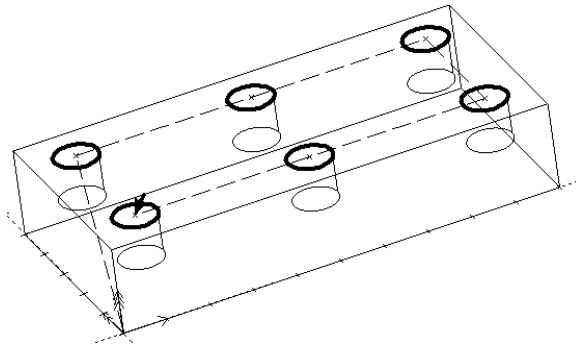
Example 3. The edited part.



Now, with the un-needed arcs edited out you can either select their color or selected them by holding down the SHIFT key and clicking on each one of them. When each of them are selected we would go to the POINT menu and choose ARC CENTER to deselect the arcs and snap selected points to the center of each arc.

Example 4. Using extracted arcs for drilling.

Here you can see that by selecting the extracted arcs, you can easily use the Version 21 software for drilling cycles as well. Remember, in Version 21 you no longer have to snap points to arcs first for drilling.



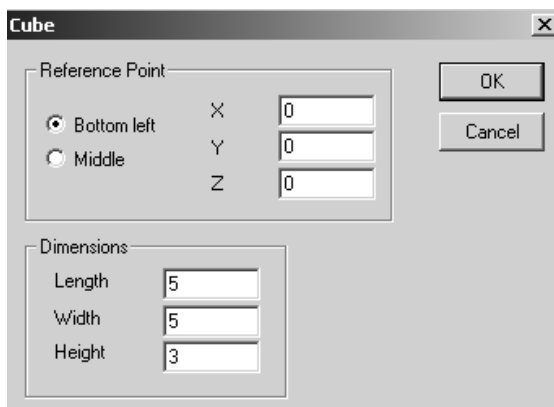
IMPORTANT NOTE: If you have a model with cylinders and the cylinders do not go all the way through the model itself, Version 21 will generate toolpath that will take the tool to the bottom of the cylinder. As the Version 21 was developed primarily for generating toolpath on solid and surface type parts, this toolpath action IS NOT a bug. This is what the software is supposed to do. If needed, you can also import the wireframe information and use this for locating or creating points for use with drilling cycles.

Extracted edge geometry can also be used for verifying geometry and for dimensioning as well.

EXTRACT EDGE TRAINING LESSON:

STEP 1

Go to the Solids menu and select PRIMITIVES and CUBE. This will open the Cube box.

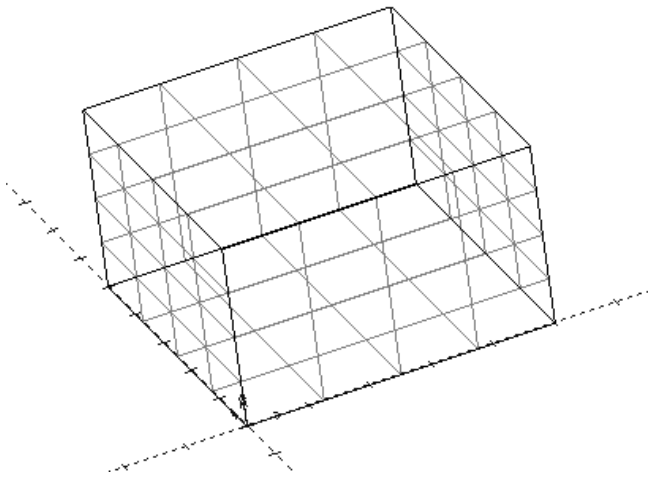


Select the BOTTOM LEFT option under Reference Point and leave X, Y and Z at 0.

Enter .5 for Length, .5 for Width and 3 for Height.

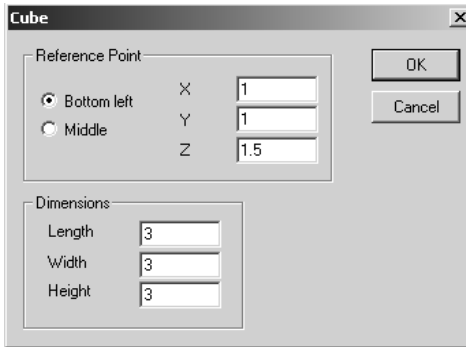
Now select OK to draw the cube.

The result:



STEP 2

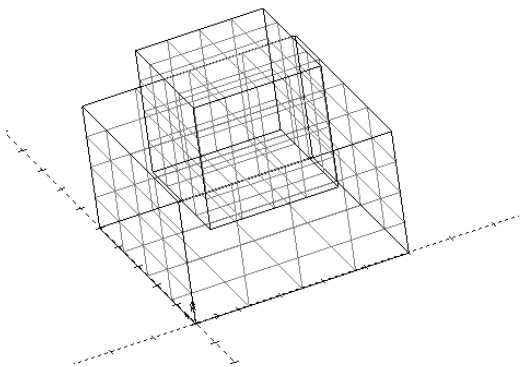
Now go back to the Solids menu, select PRIMITIVES and Cube again.



Select the BOTTOM LEFT option for the reference point and enter 1 for X, 1 for Y and 1.5 for Z.

Now under the dimensions section enter 3 for length, 3 for the width and 3 for the height. Now select OK.

The result:



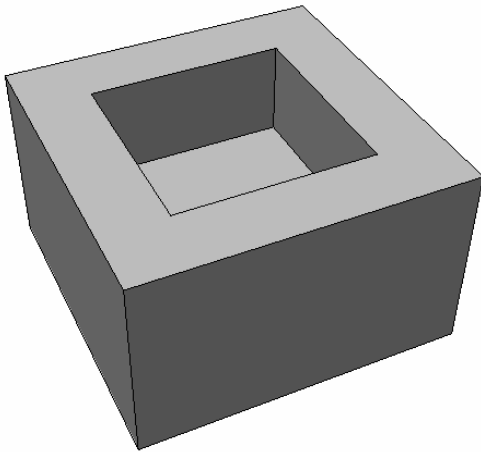
Now click your left mouse button with your cursor OFF of the geometry in an empty area of the screen to deselect everything.

STEP 3

Now, we are going to select the bottom solid first. Place your cursor on an edge line of the bottom cube and click your mouse button to select it. Now, hold down the SHIFT key on your keyboard and click on the top cube second. Both will be selected now.

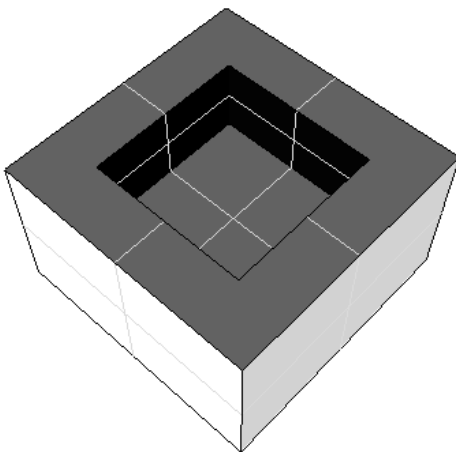
Go to the SOLIDS menu and select **BOOLEANS**. Now choose the DIFFERENCE function and select OK to extract the top cube from the bottom.

The result in a solid render view:



STEP 4

Go ahead and click on the solid to select it if it is not selected at this time. Now go back to the SOLIDS Menu and select STITCHING, then **UN-STITCH SOLID TO SURFACES**. This will automatically un-stitch all of the surfaces from the solid.



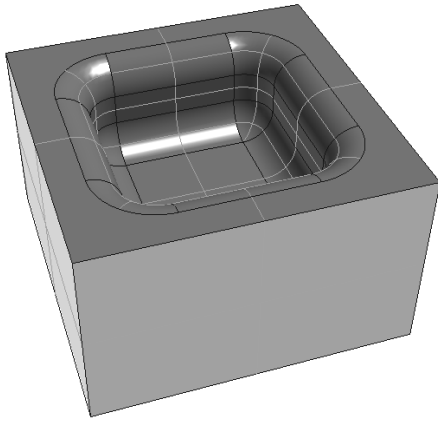
In the example image above you can see the parts that are shaded are the parts that you want to select.

This will be the top surface as well as all 4 inside walls and the bottom of the cavity. Hold down your shift key and select the top surface, the four walls of the recessed cube and the bottom of the inside cube as well.

Basically, you do NOT want to select the four outer walls or the very bottom surface of the model.

Once you have made these selections, go back to the SOLIDS menu, select STITCHING and STITCH SURFACES TO SOLID. Now click your left mouse button off of the part to deselect everything and then click on an edge line of the top surface to select this newly stitched solid.

Now you can go to the SOLIDS menu again and select FILLET. Enter **.5** in the fillet box and click OK.



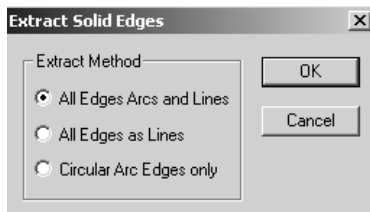
You will now have fillet surfaces on the bottom and top of the recessed cube as seen in the above image.

If the Auto Pre-Select option is turned on in your ENVIRONMENT settings, the new changes will take place and be selected. If not, click on the top part of the solid on an edge line to select it.

STEP 5

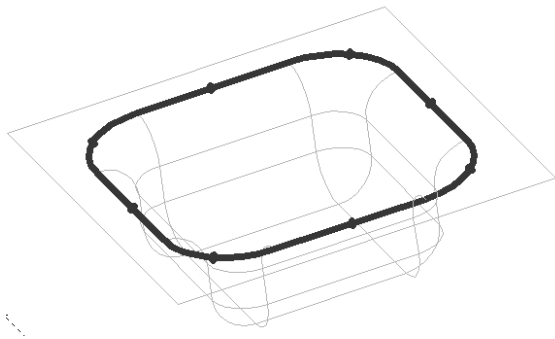
Now, with the top selected, go to the CHANGE Menu and select ATTRIBUTES. Click on the CONTOUR tab and select the color GREEN. Your ACTIVE drawing color will now be green. We are doing this because we are going to extract the edges of this solid for boundary toolpath.

Now go directly to the solids menu and select **EXTRACT EDGES**.



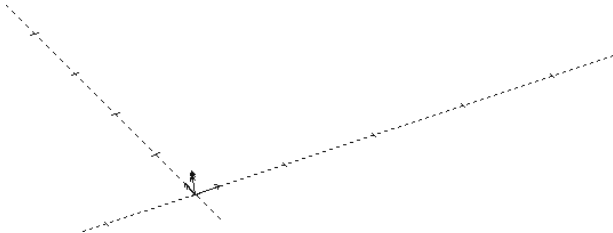
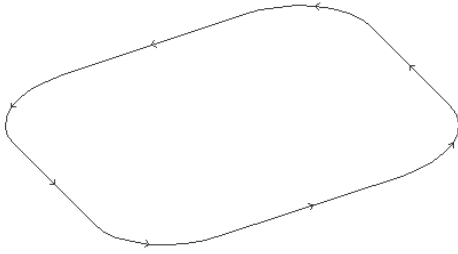
Now select the “ALL Edges Arcs and Lines” option and click OK. This will automatically extract all of the edge geometry. Click your left mouse button with your cursor off of the geometry to deselect everything.

Now go to the EDIT menu, SELECT ENTITIES and then choose COLOR. Click on the color BLACK and then select OK. Now go to the CHANGE menu and select BLANK. This will leave ONLY GREEN geometry on the screen.



All we want to keep is the top rectangular geometry. Notice in the image to the left that this geometry has been enlarged. Hold down your CONTROL key on your keyboard and select all geometry EXCEPT for this top enlarged geometry. When finished, hit the delete key on your keyboard.

The result:

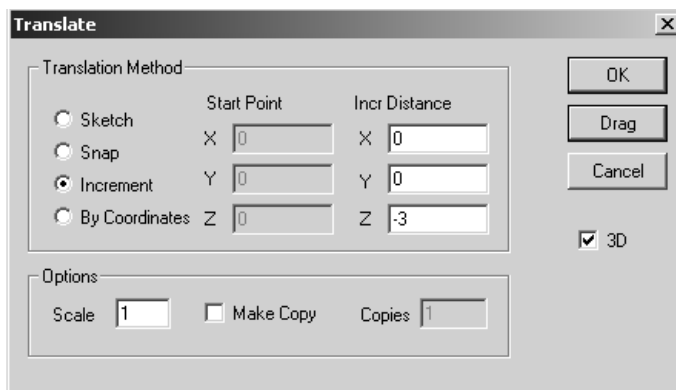


Only the top rectangular geometry should be remaining.

STEP 6

Now go back to the **CHANGE** menu and select **UN-BLANK**. When the Un-Blank box appears, select **ALL** and then click **OK**. Now click your mouse in an empty area of the drawing screen to deselect everything. Now, go to the **EDIT** menu, choose **SELECT ENTITIES** and then select **COLOR**. Select the color **BLACK** in the color box by clicking on it and then click **OK**. All of the surfaces will be selected. Go directly to the **SOLIDS** menu and choose **STITCHING** and **STITCH SURFACES TO SOLID**.

This will turn the part into a solid. Now go to the **EDIT** menu, **SELECT ENTITIES** and select **ALL**. Go to the **CHANGE** menu and select **TRANSLATE** to open this box.



Select the **INCREMENT** option and enter -3 for Z. The **SCALE** is 1 and do not check the “Make Copy” box. Select **OK** to move the model so that the top is at Z=0.

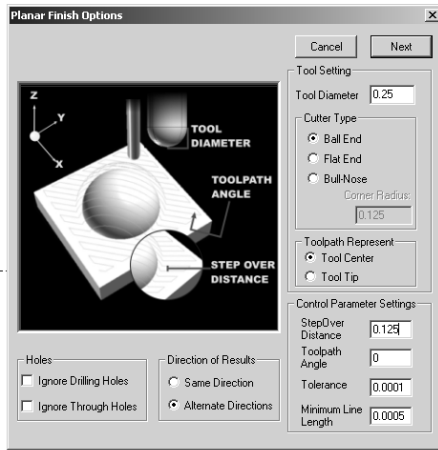
Now place your cursor off of the model and click your mouse button to deselect everything.

STEP 7

Go to the **SPECIAL/NC CAM** menu and select **INSERT NC** to open the Insert NC Object box. Double click on the **FANUC 6M** post processor and the NC CAM will open on the right side of the screen. Place your cursor in an empty area of the screen and click your left mouse button to deselect everything.

You may want to select the **VIEW ALL** icon from the main toolbar now to reposition the model on the CAD side so that you can see everything. Remember, this is the small icon with a magnifying glass and the letter “A” in it.

Now click on the solid **ONLY**, leaving the wireframe un-selected. Now go to the SOLIDS menu and choose **GENERATE TOOLPATH**. Select Planar and click NEXT. This will open the **Planar Cut Options** box.

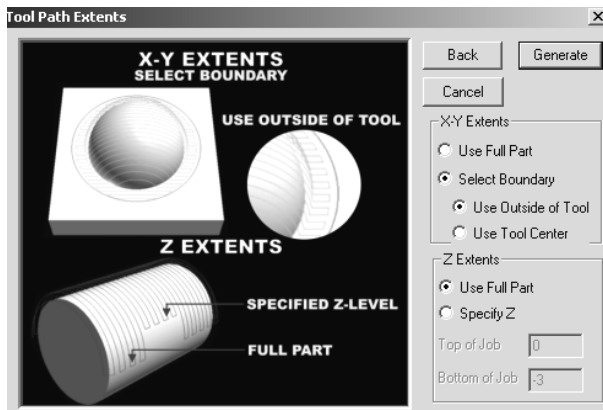


- Enter a Tool Diameter of .25
- Select Ball End as our cutter.
- Select the Tool Center option.
- Enter a Step Distance of .125
- Enter a Toolpath Angle of 45-degrees.
- Set the Tolerance at .0005
- Set the Maximum Line Length at .0005.
- Do not check either ignore hole options.
- Select Alternate Directions as the Direction of Results

Now select the **NEXT** button. This will open the Toolpath Extents box.

STEP 8

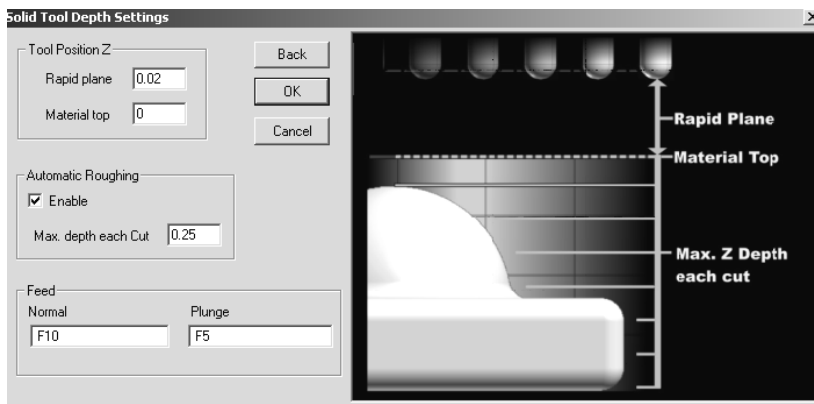
In the new box, choose the “SELECT BOUNDARY” option and then **USE OUTSIDE OF TOOL**.



Select **USE FULL PART** under the Z-Extents section. Now select OK and click your mouse on the GREEN wireframe geometry to generate the toolpath.

STEP 9

Once the toolpath has been generated, the **Solid Tool Depth Settings** box will appear.



Enter .02 for the Rapid Z-Plane. The Material Top is 0. Check the ENABLE box under Automatic Roughing and enter .25 for the Max. Z-Depth Each Cut and click the OK button.

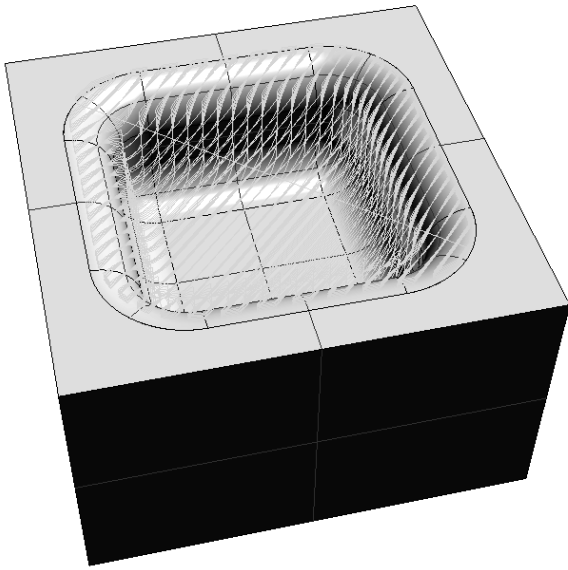
This will automatically generate all of the G-Code program and the toolpath. It may take a moment depending on your computer resources.

Once the entire G-Code program is complete, go to the EDIT menu on the CAM side and choose SELECT ALL. All of the G-Code will highlight. Now go back to the EDIT menu on the CAM side and select GEOMETRY FROM NC. This will **backplot** the toolpath geometry on the CAD side of the screen.

When this is finished, go to the EDIT menu on the CAD side and choose SELECT ENTITIES and then ALL. Everything on the CAD side of the screen will be selected.

STEP 10

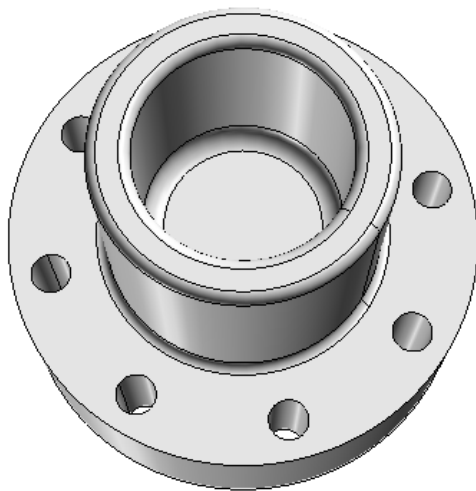
Go to the SOLIDS menu and choose RENDER.



Congratulations! You have completed this lesson.

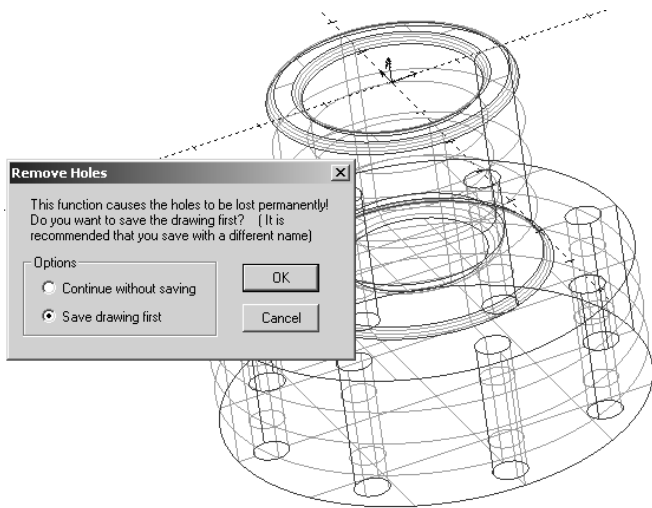
Section 36. Remove Holes

The “**Remove Holes**” function is used for models that require drilling operations.



Take for example the part to the left. You want to machine this part but do not want the tool to go inside the main cylinder or each of the smaller cylinders. All you will need to do is select the model by clicking on it and then by selecting the REMOVE HOLES function from the Solids menu.

When you do this a special box will appear for your protection, as once you remove the cylindrical holes you will NOT be able to put them back.

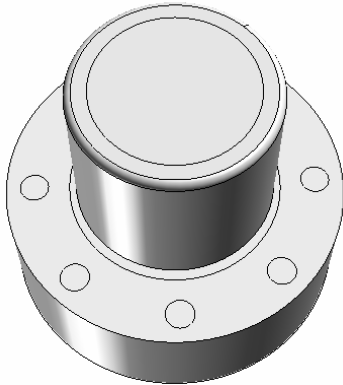


Therefore, you will want to save your drawing with another name so that you can go back and machine the holes later. Select the “Save Drawing First” option.

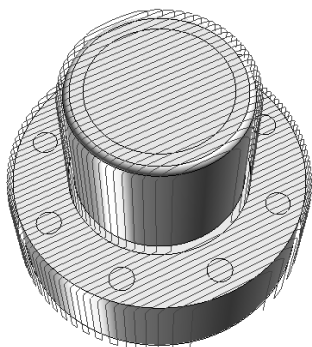
When you perform the SAVE option the “Save As” box will appear. Simply name the file that you want to save for drilling operations and click OK. This will save the drawing to the designated folder that you chose and then seal the holes for you. When this happens the surfaces that make up the cylinders will not be deleted, they will be blanked out onto layer 0.

The drawing you saved will be available for you later. You also have the option to continue working without saving the file if you want to.

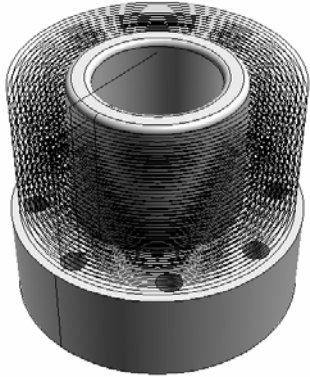
Take a look at the same example model after removing the holes:



Now, to create the toolpath for this, you can do it THREE ways. The first way would be to select the model and use the GENERATE TOOLPATH option from the Solids menu to create the toolpath.

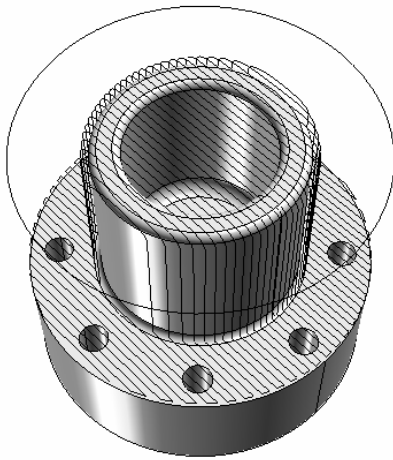


The next way would be to use the EXTRACT EDGES function to create wireframe geometry for use with the pocketing function in the software's OTHER menu. Once you extracted the edges you wanted to use for pocketing you could use the verify 2-entities button to gather your depth information for use in the CAM tool depth settings (small icon with the U & D on it). Then select the outside geometry for your pocket and, in the case of this example, the island arc edge geometry and move them to Z-0 using the Move to Z function. Then use the spiral pocketing function for your toolpath. Once you generate the toolpath you would enter your depth settings in the depth settings box, roughing parameters and generate the toolpath and G-Code.

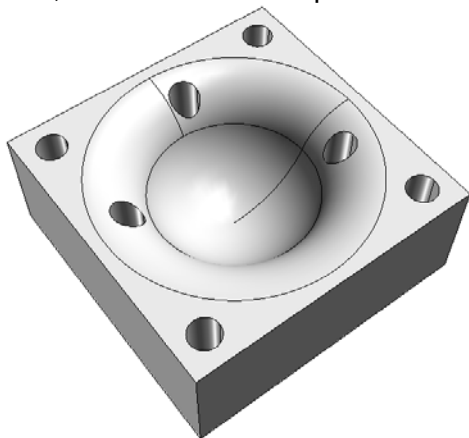


The third way to machine this type of part example would be to use the IGNORE HOLES or IGNORE THROUGH HOLES options in each of the 3 toolpath finish dialog boxes.

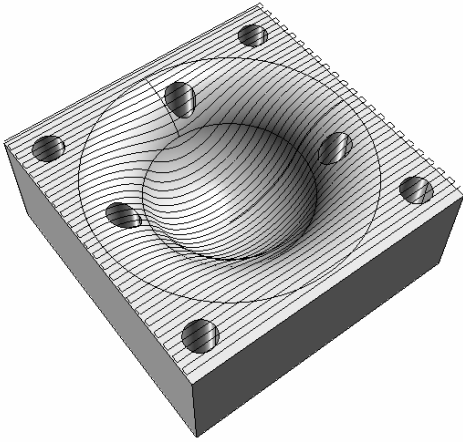
This will cause BobCAD's toolpath generation function to skim over the holes in the surfaces as seen in this example.



Now, let's take the example of a 3D model that has cylindrical drill holes that are not flat 2D.

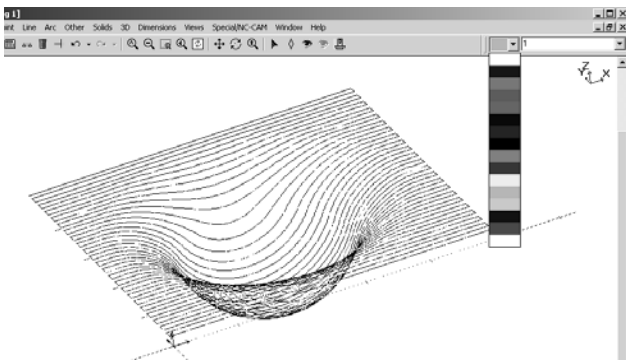


All you would have to do once you have selected a toolpath type and clicked the NEXT button in the Solid Machining box, is select the IGNORE HOLES option when creating the surface toolpath.



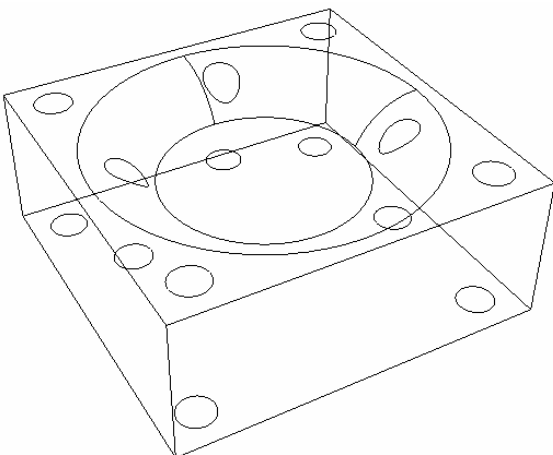
As you can see, the toolpath ignored both 2D and 3D cylindrical holes. **Now to go back and drill out the holes** after you have created the toolpath, you would select the roughing or finishing toolpath and save it to a layer first, then blank it out using the CHANGE menu's BLANK function.

Once you have done this, you will use the EXTRACT EDGES and then EXTRACT ALL ARCS function from the Solids menu. It is wise to use a different color before doing this by selecting the ACTIVE COLOR bar from the main toolbar and picking a different color, then executing this operation. This will make the extracted arcs easier to distinguish from the edge curves of the top surface.

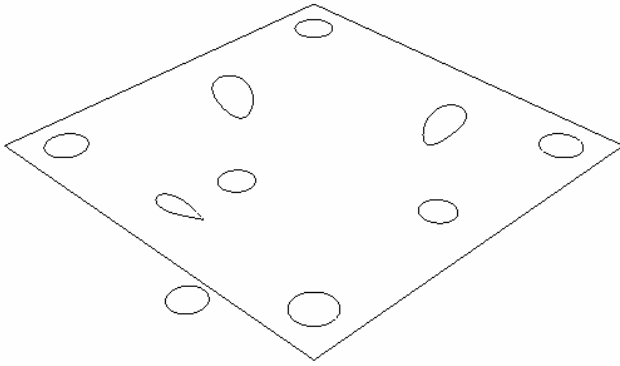


After changing the ACTIVE COLOR, extract the arcs and lines from the model.

Here is the extracted geometry.



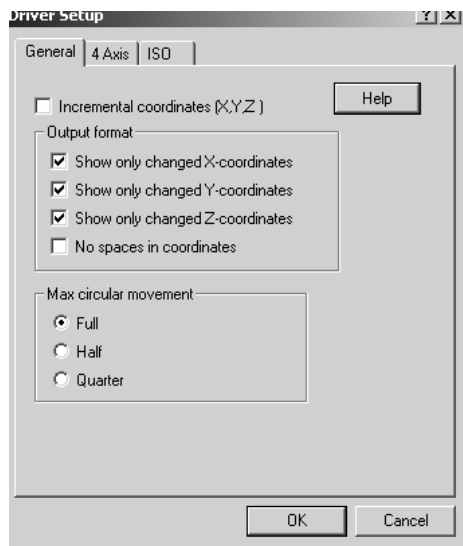
Now you can delete the entities that you don't actually need. Below is an example.



As you can see there are the tops and bottom geometry of the holes in the 3D section of this model and then there are the 2D flat hole arcs. You can actually delete the 3D geometry if you would like as we are going to select the actual 2D arcs and use the drilling cycles to machine them. BobCAD software uses scripts for drilling where you enter in depths and the information needed anyways. Therefore, the Z position on the screen isn't relevant to the code that will be generated.

The script dialog boxes used for each drilling cycle will ask for the correct cycle information including the Z-depth.

Therefore select the arcs that you will be drilling for, let's say a G81 cycle. Then execute that script after you have entered in the right parameters. Then let's say you will be tapping as well. Therefore you would perform a tool change and select the arcs for that cycle, then select the cycle for that operation and execute it after entering that operations variables in the script box that will pop-up.



NOTE: If your machine requires the X and Y coordinates both to be present in every drill cycle, be sure to go to the **SETUP** menu on the CAM side and then **DRIVER** and check:

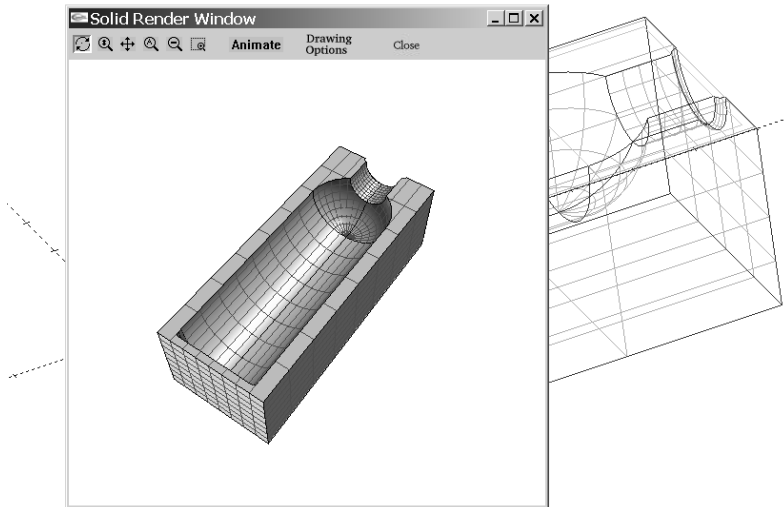
- Show ONLY changed X Coordinates.
- Show ONLY changed Y Coordinates.
- The show ONLY changed Z Coordinates does NOT matter for drilling cycles due to the scripts that you will be using for various drilling operations.

You MUST know what your machine expects to see in the Drill Cycles/G-Code program before attempting to use BobCAD drilling cycles to avoid ANY errors that may occur.

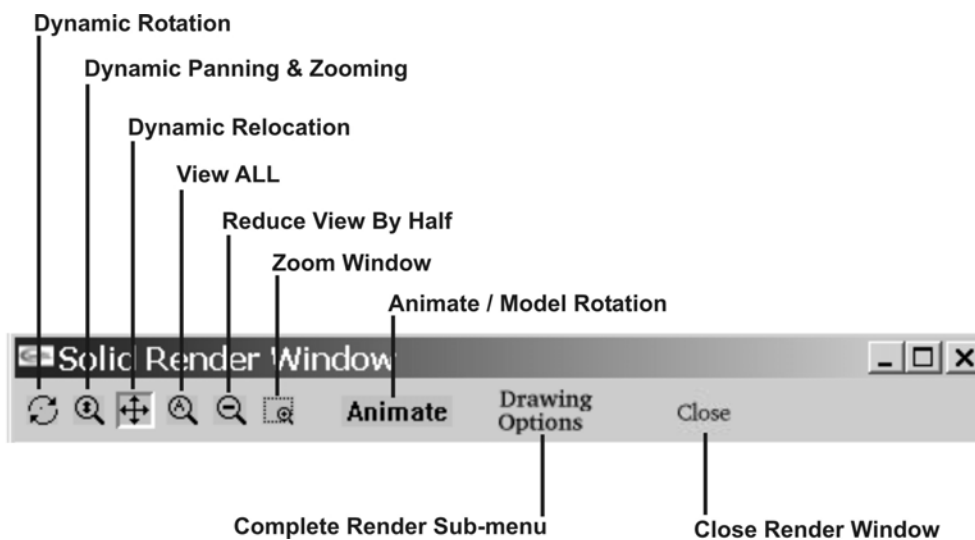
If the machine accepts MODAL coordinates and you do NOT move your X or Y axis during your tool change (some machines simply pull straight up in Z for the tool change position) then you do not need to re-enter the X or Y coordinates for the next drill cycle as you are already in position.

Section 37. Render

Version 21 offers you the ability to view your model in a separate **Render** window. You may use this function at any time whether you are working with surfaces, solids or wireframe geometry. You must always select the parts that you want to view first, and then select the render function located at the bottom of the Solids menu.



Let's take a look at the Render Window Toolbar.



Now, take a quick look at the description for each of these functions.

DYNAMIC ROTATION:



By selecting this button you can click on the rendered model and by holding down your left mouse button, dynamically rotate it around the screen in any direction of rotation.

DYNAMIC PANNING & ZOOM:



By selecting this button you can **zoom out** by pressing your left mouse button and moving your cursor in a downward direction. By pressing your left mouse button and moving your cursor in an **upward direction**, you can **zoom in**.

DYNAMIC POSITIONING:



By selecting this button you can reposition or move the part to a new viewing location by holding down your left mouse button and then moving the part.

VIEW ALL:



By clicking this button you will be able to reposition your rendered part so that you can view everything.

REDUCE VIEW BY HALF:



By selecting this button you will automatically reduce the current view of your rendered part by half the current viewing scale.

ZOOM WINDOW:



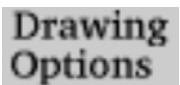
By selecting this button you can click and drag a view window over an area of your rendered part to zoom in to view only that area. This is also good for zooming in on rendered solids and back-plotted toolpath.

ANIMATE:



By selecting the **animate** button, your rendered part/geometry will begin to rotate by itself until you select the button again to stop rotation.

DRAWING OPTIONS:

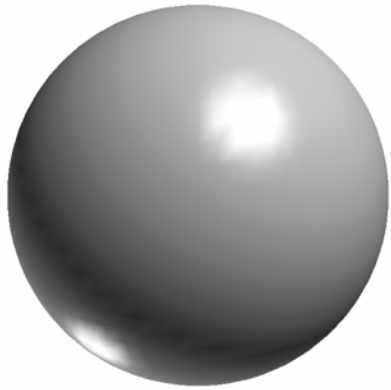


The Drawing Options menu offers functionality pertaining to the model and how it is being viewed in the render window. This functionality includes:

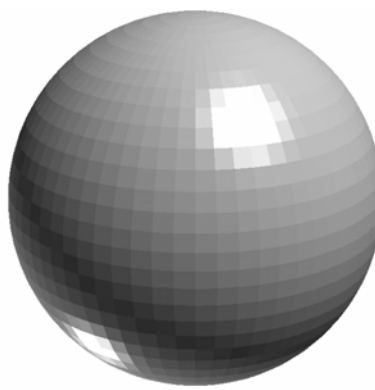
- **SURFACES:** This function allows you to adjust how you view the surfaces of your model. Let's take a look at what each of these options will do to a basic sphere starting below.



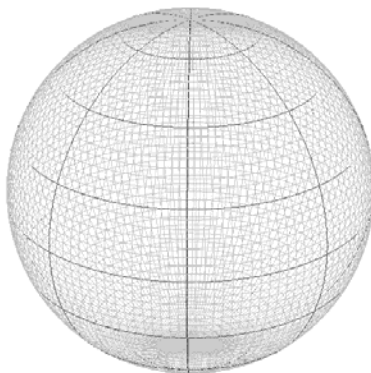
Smooth Shading



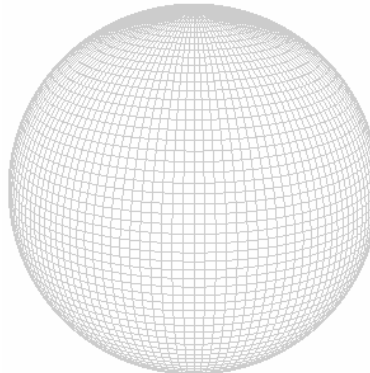
Flat Shading



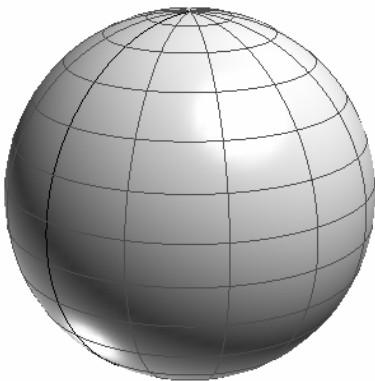
Outline Patch



Outline Polygon



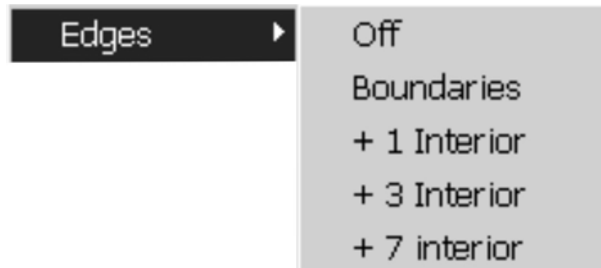
Hidden Lines



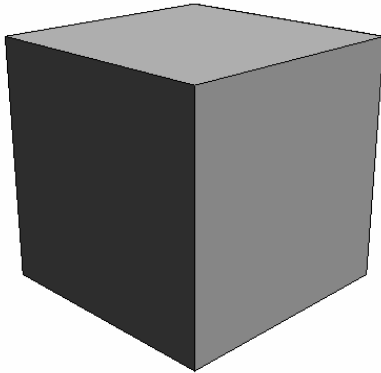
These are the various options in the Surface Sub-menu.

Let's move on and have a look at the next function under the Drawing Options menu.

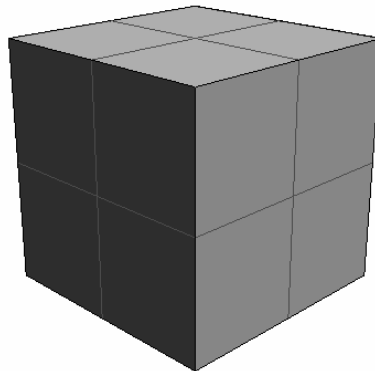
- **Edges:** The Edges sub-menu offers a variety of ways that you can view the edges of a solid. The OFF option turns off the edges altogether. Let's take a look at the other four edge options using a basic cube in the next image.



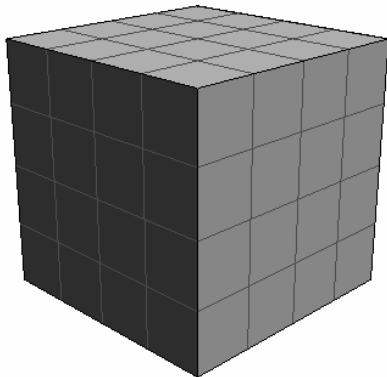
Boundaries



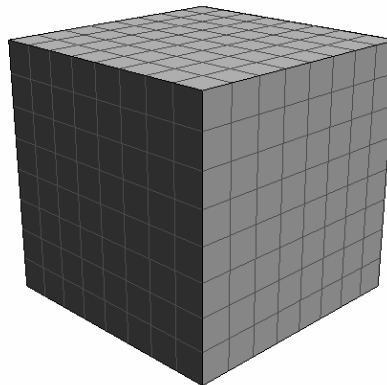
+1 Interior



+ 3 Interior



+ 7 Interior

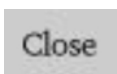


- **Resolution:** The Resolution sub-menu offers three levels that you can choose from. They are High, Medium and Low. The render function uses memory resources and may slow down your computer. If you find that your solid render doesn't look at perfect as you would like, you can use this to increase the resolution of the rendered model and make the image view sharper.



- **Toggle Grid:** This is the last item in the Drawing Options sub-menu. You can turn the render grid on and off by selecting this option.

CLOSE:



By selecting this button you will completely close the Render window.

Section 38. Questions & Answers

Question: *Can I apply an angle to my toolpath?*

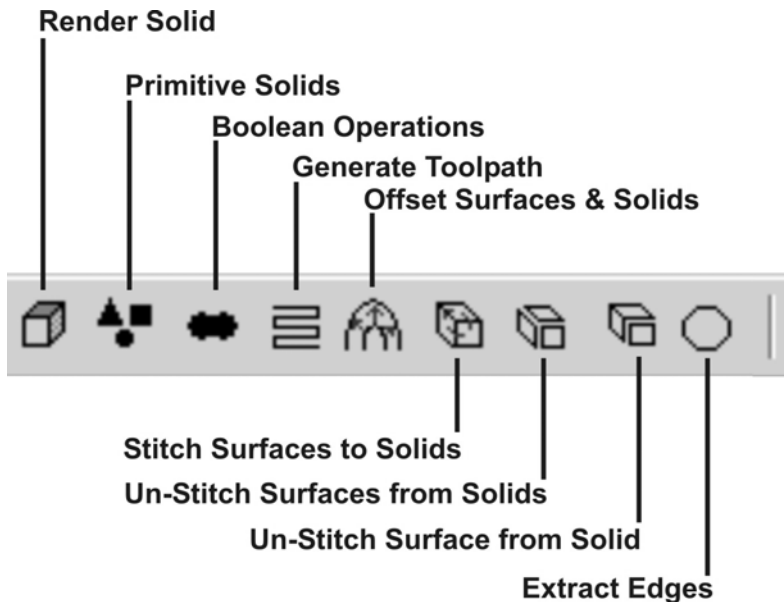
Answer: YES. This is an option in the Planer Cut Options box. You can use this to apply an angle to the toolpath.

Question: *What is the default tolerance for generating toolpath?*

Answer: The default tolerance should be set to .001 or .0001 in the Planer Cut Options box. You may increase the tolerance as needed.

Question: *Are there **Solids & Boolean Operation icons** that I can add to a toolbar or toolbox?*

Answer: YES. Double click on a spacer on the main toolbar to open the Customize Toolbar box. Have a look at the Solids icons in the next image.



Question: *Is it optimum that I use the translate function and lower the top of my model to Z0?*

Answer: Yes. This would be a good idea when generating the toolpath and G-Code program.

Question: *Can I verify my model?*

Answer: Yes. By right-clicking your mouse after selecting your model or surface you will open a command menu. Here you will notice a **VERIFY** option. Select it and you can gather important information regarding the TOP and BOTTOM coordinate values.

This concludes the Questions and Answers section.

CHAPTER 6

THE 3D MENU

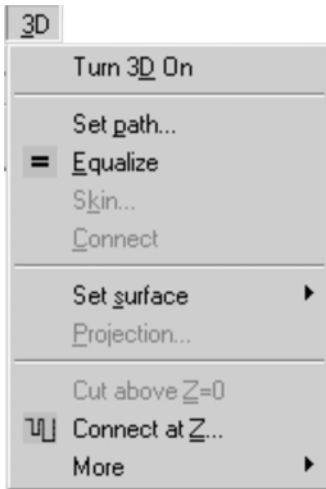
Section 39. The 3D Menu



Section 39. The 3D Menu

The Version 21 software deals with Wireframe, surfaces and solids. Therefore, V21 allows you to create surface toolpath directly from wireframe geometry. If you have an interest or applicable need for 3D drawing and part programming through wireframe geometry, then you will need to study this chapter before attempting to program complex 3D parts with your CNC machine.

Let's take a look at the **3D** menu as seen below.



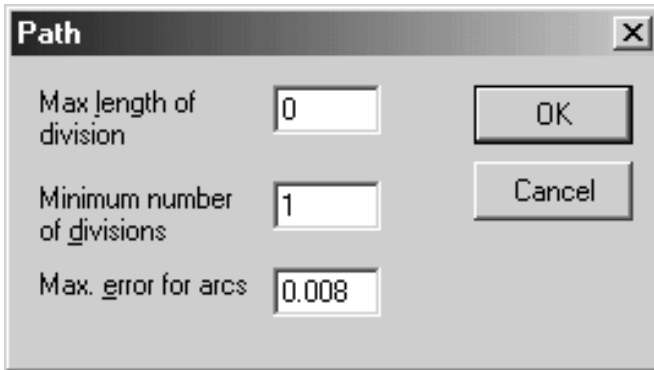
As you can see in the menu image above, there are functions that are turned off or not activated. These functions require that previous actions have taken place. We will cover these functions and details in this chapter for you so that you can become an expert in BobCAD Version 21 3D part programming. Let's take a look at each one of these functions and their special sub-menus.

TURN 3D ON: This function is used to change the screen from a 2D view to a 3D view. This menu changes in each mode. If you are in 3D mode, the menu will say: TURN 3D OFF. This is made this way because if you have modified the view, rotated the part etc. you can switch back to a 2D view automatically by using this function.

SET PATH: This function defines a path for interpolation between different cross-sections. This function is used for setting up the geometry to generate a skin surface. There are a number of ways in which you can define a path. They are:

- **LINEAR:** You can define a linear path so that a skin surface can be generated.
- **SINGLE:** You can define a single path for creating a skin surface.
- **MULTIPLE:** you can define multiple paths/geometry to proceed with generating a skin surface.
- **DASHED:** Dashed geometry can be defined for creating skin surfaces.

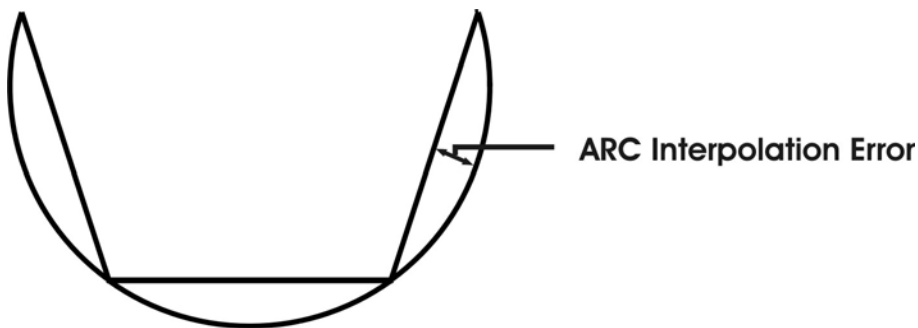
The SET PATH function is primarily used for surfacing to set the path in which the skin surface will be defined or created from. This is different from earlier versions of BobCAD software. In Version 21, you set the path first before generating the skin surface. The PATH is the part or parts of the geometry that the surface toolpath will follow or conform to. Let's take a closer look at this functions dialog.



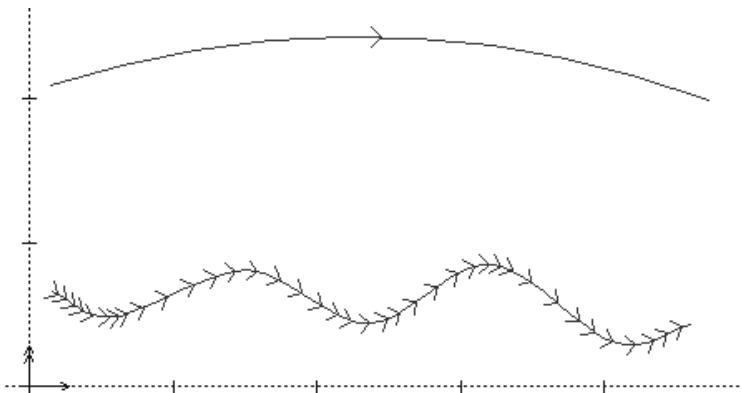
There are three variable settings in the PATH dialog. They are:

- **Max Length of Division:** This is the maximum permissible length of an entity that will be considered the path.
- **Minimum Number of Divisions:** Every element in a chain is divided into a number of entities. This represents that value.
- **Max Error for ARCS:** This is the error that occurs by the approximation of an arc by a number of straight lines. You do not machine arcs in 3D. These arcs are actually made up of line segments in a 3D contour. This will define how tight the tolerance is for the contour of line segments.

In the example below, you can see an example of ARC Interpolation error.

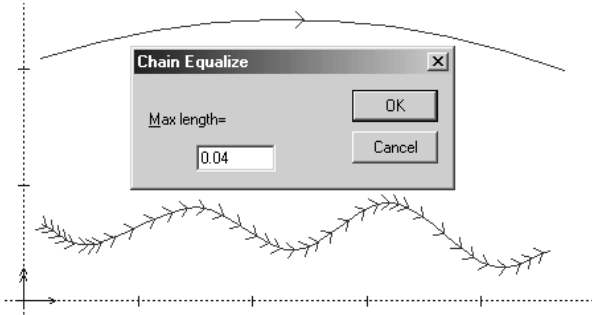


EQUALIZE: This function allows you to take two chains, one with more entities than the other and automatically make them both the same number of entities. Let's take a closer look.

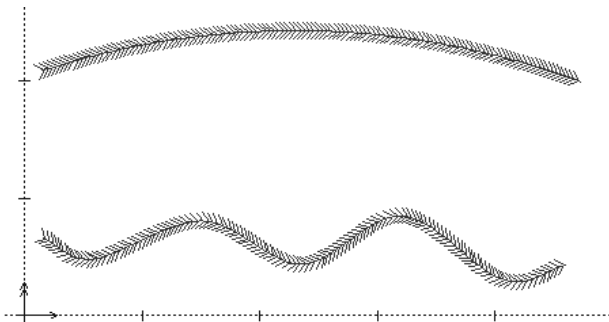


In the image above you can see that the top arc is only 1 entity. The bottom chain has many more entities in it. First, we selected all entities and then selected the EQUALIZE icon.

Once we selected the EQUALIZE icon, we are able to enter a value representing the length of each segment. See below.

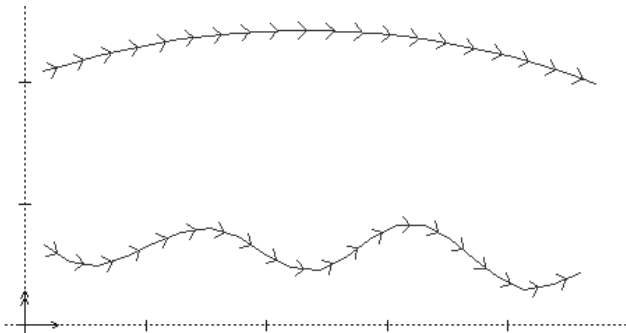


We entered in .04 and then selected OK. See the result below.



Then we selected the EQUALIZE function again and changed the value to .25 and selected OK.

This increased the size of each entity segment. See below.

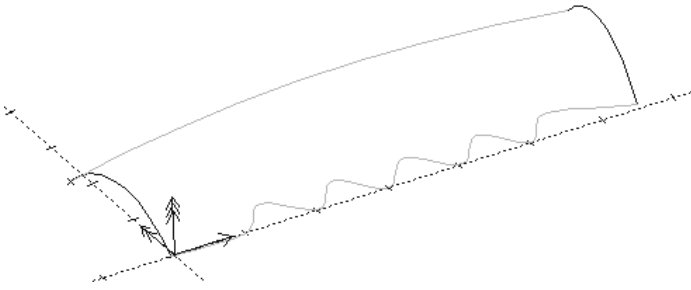


SKIN: You can generate a skin surface toolpath over previously selected sections, using the parameters that you input in the SET PATH function. You must ensure that all sections contain the same number of elements. This is done by using the EQUALIZE function. You can set a number of options in the dialog box that is displayed on the screen. The SKIN dialog allows you to set various options. They are:

- Maximum distance between the different tool paths.
- Tool radius.
- Tool core diameter.
- Machine internally or externally.

Let's take a closer look at the skin function.

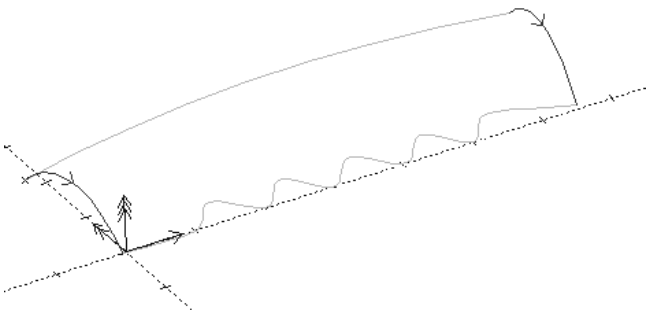
Here you can see that we have wireframe geometry ready for a skin surface.



The first thing that we will need to do is to **SET THE PATH**. By selecting the two end radii we can do this. Follow along below.

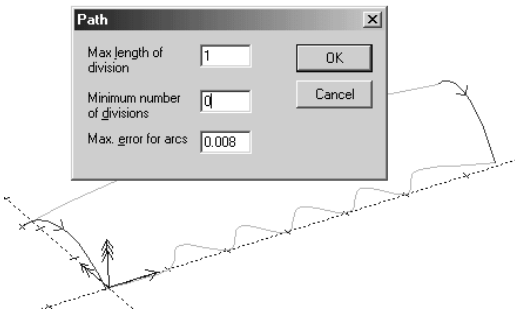
STEP 1

We will hold down the CONTROL KEY and select both end radii so that the directional arrows are both going toward us. Select the LEFT arc FIRST and then the right. This is important for creating the skin.



STEP 2

Now we will select the 3D menu and SET PATH and then select OK. See below.



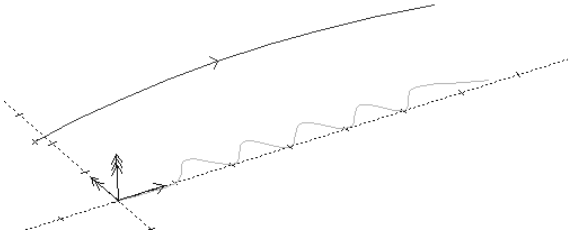
STEP 3

While those two ends are selected and the path is set, we can select the BLANK button from the main toolbar.



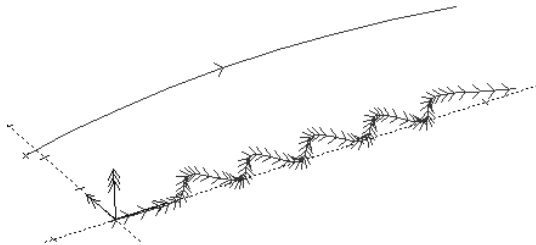
STEP 4

Remember, we used the two arcs at the ends as the path of the skin surface. We did not need to use the equalize function as the arcs were both the same number of entities. If they are different shapes we would have used the equalize function to make them both the same number of entities first and then set the path with them. Now we need to chain select both sides. The way that we do this is important. We selected the first arc on the left earlier and now we are going to select the backside geometry so that the directional arrow is pointing away from the left arc.



STEP 5

Now hold down the SHIFT KEY and click your mouse with the cursor at the RIGHT END of the front side geometry and the entire chain will select with all of the directional arrows going in the same direction as the arrow for the backside geometry. See below.



STEP 6

Now we are ready to generate the skin surface. Go to the 3D menu and select SKIN, as it will now be active and available for selection. By selecting SKIN, the dialog will appear. Let's take a look at the 3D **SKIN BOX** below.

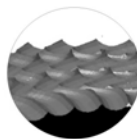
Dialog Image 1.

Enter the tools core diameter if you are using a flat tool for cutting.

Enter the tool diameter that you will be using for the surface.

The software will tell you the exact number of lines in the toolpath.

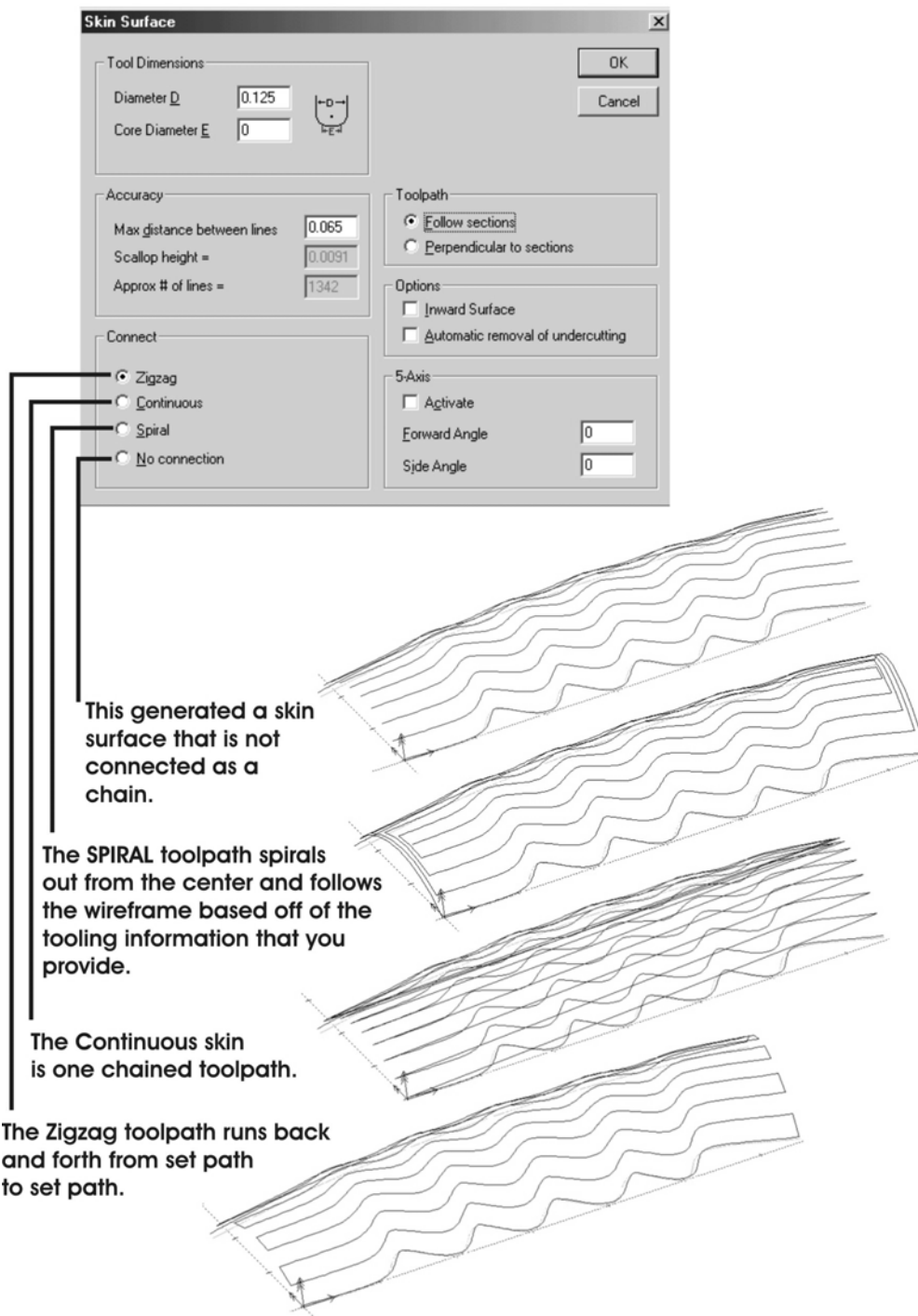
This is what is left from the tool stepping over and will be cleaned up on the finish pass when machining.



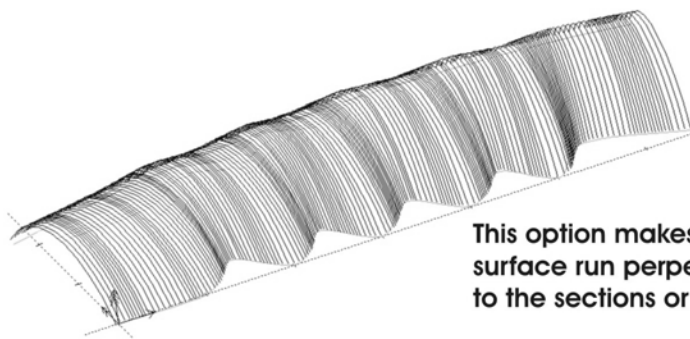
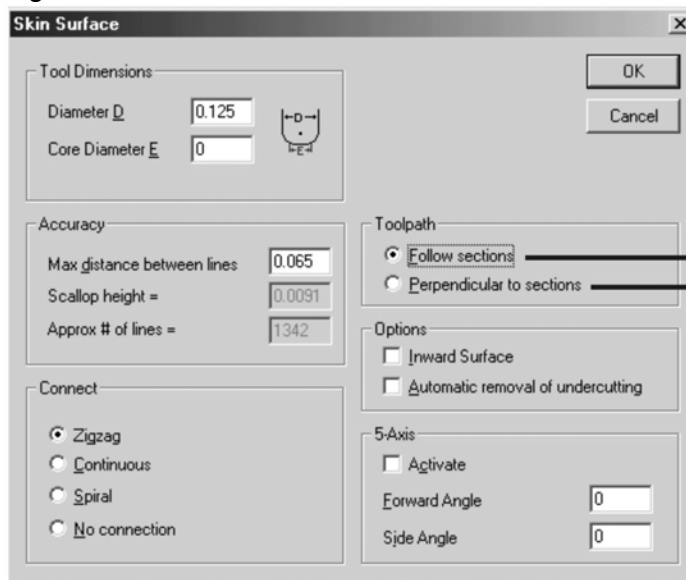
This is basically HALF the distance of the tools radius. This adds compensation automatically for you when generating the toolpath. The Scallop Height just below this value will automatically change based off of this distance between lines of toolpath.



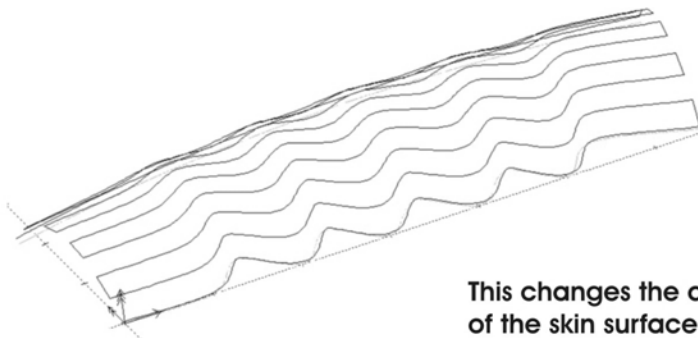
Dialog Image 2.



Dialog Image 3.

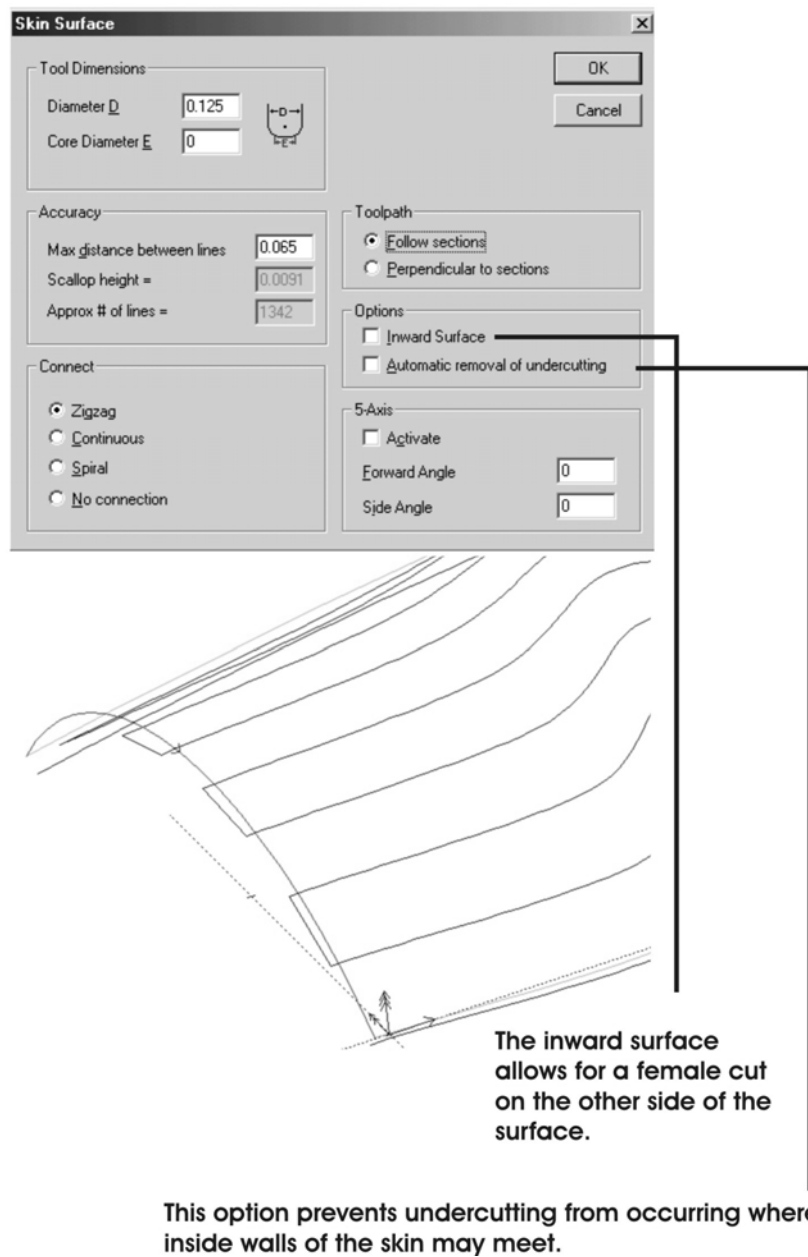


This option makes the skin surface run perpendicular to the sections or sides.

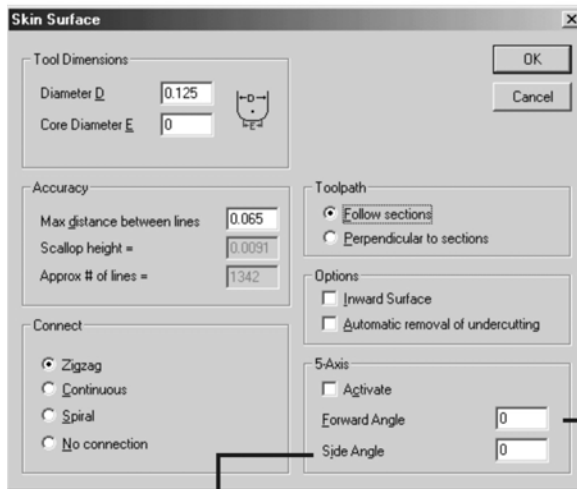


This changes the direction of the skin surface to run the length of the sides.

Dialog Image 4.



Dialog Image 5.



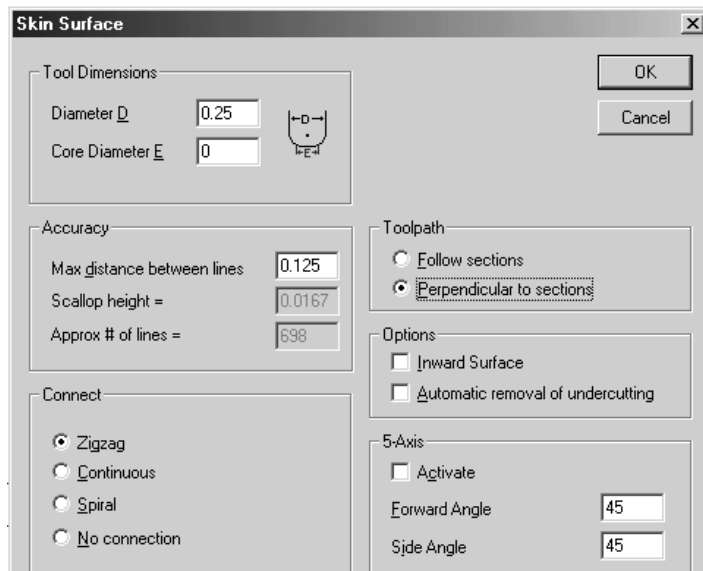
5-Axis toolpath is generated here. The Forward Angle is the B-Axis control but can be changed to another designated letter. You will enter in the degree here. If you leave it at 0, the 5th axis forward angle will be automatic.

This side angle option would typically be for the C-Axis control but the designated letter can be changed here. You will enter in the degree here. If you leave it at 0, the 5th axis forward angle will be automatic.

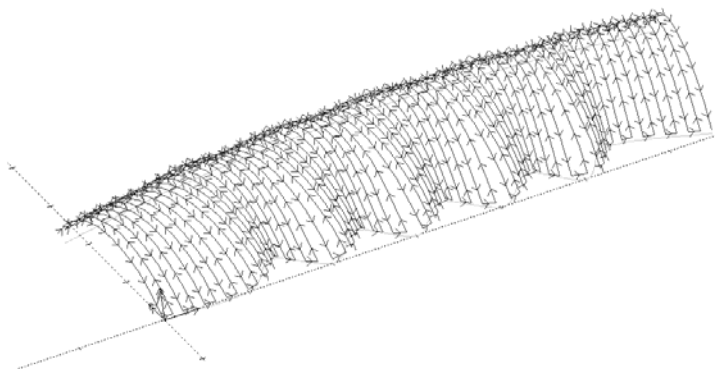
Before using the 5-axis settings you must select the ACTIVATE button.

STEP7

Now that we have the skin dialog open, we can enter the tool diameter and step over. We can also make the type of toolpath selection that we want. For this example we will use a .25-inch ball end mill. The step over is .125-inch, which is half the distance of the size tool we are using for the maximum distance between lines. See the next image.



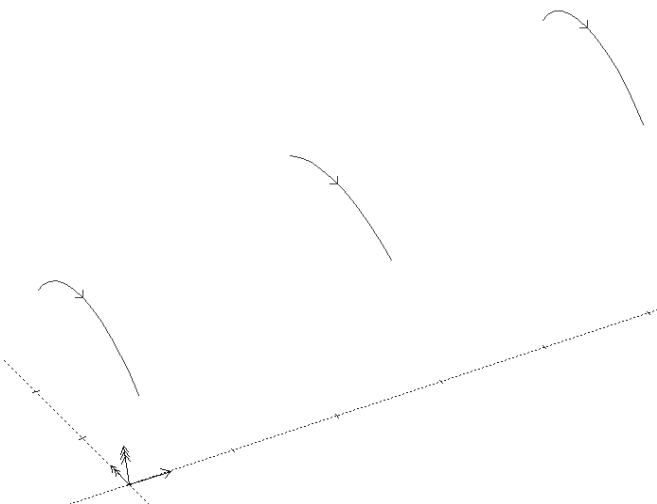
We are creating the skin surface with the Zigzag style connection and have also set the toolpath to run perpendicular to the sections. Once we select OK we get the skin surface that you can see below.



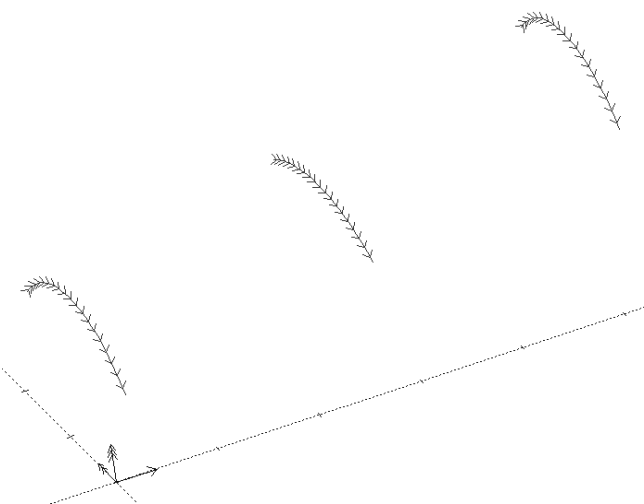
These are the basics of the skin surface function. The skin surface toolpath will always appear selected. If you make a mistake and need to use the undo function, do so and then go right back to SKIN in the 3D menu as the geometry will be selected.

CONNECT: The connect function allows you to create toolpath between selected entities that are not connected. Let's take a look below.

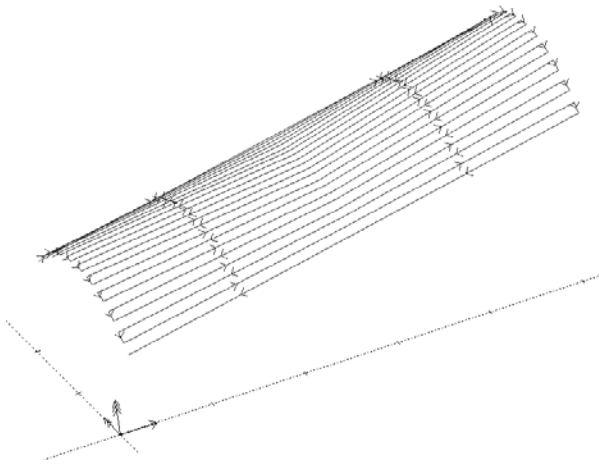
Here we have 3 selected arcs. All of them are selected with the directional arrows going in the same direction.



Then we can use the Equalize function in the 3D menu to give a length to break each of these arcs up equally.

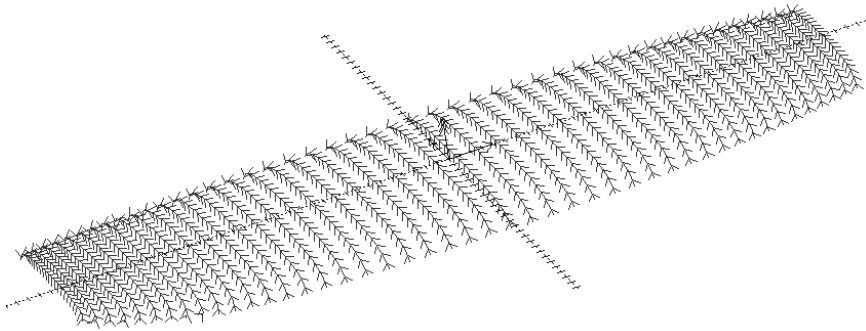


Now, by selecting the CONNECT function in the 3D menu we will get the following result.

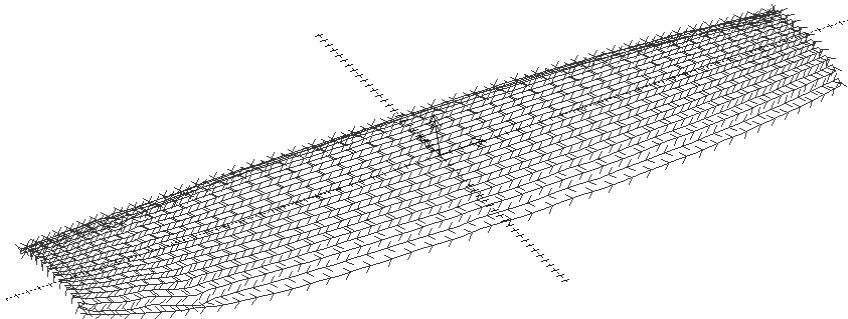


This is a fast and accurate way of generating surface toolpath as well.

Below you can see the frame of part of a surfboard. All of the directional selection arrows are going in the same direction.



You can now use the equalize function in the 3D menu to define the distance of each line, which will determine the toolpath.



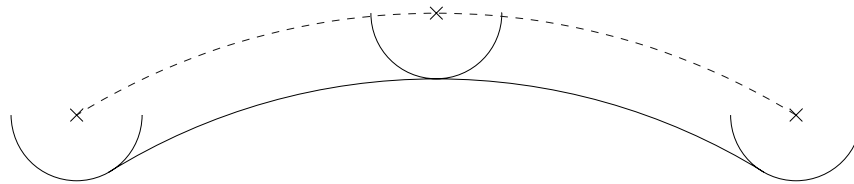
You now have a connected surface toolpath that can be used for machining. The distance that is provided in the equalize dialog will be half the distance of the size tool diameter that you will be using to cut the part. It is important to remember that rule.

TIP ON 3D SURFACING:

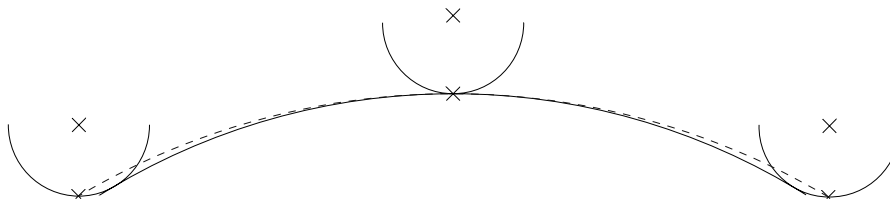
Here is **TIP** regarding the creation of 3D Toolpath directly from wireframe geometry, not using the solids or surfaces and how the tool works with the toolpath geometry.

1. With flat bottom end mills, the toolpath is programmed from the center of the tip of the tool. If you were to look at the end of the toolpath, across an end curve, for example, you will see the toolpath coming closer to the top of the end curve than at the ends – this is BobCAD adjusting the toolpath for the apparent radius of the tool at that point. This is correct, and allows the user to avoid adjusting for the length of the tool during the run of the program unnecessarily.
2. With bull nose end mills, BobCAD does a similar thing – except that the radius of the toolpath at the end curve will be offset by the radius at the side of the cutter – i.e., if you have a tool diameter of .5", with a radius on the cutter of .0625", the top dead center of the toolpath will be that same .0625" away from the finish geometry.
3. With a ball end mill, the same principle applies, as with the bull nose, only the curve at the top of the toolpath will be constant because the radius around the entire end of the tool is constant. See the following illustrations for some good examples.

Toolpath from the center of the ball end mill.



Toolpath from the tip of the ball end mill



SET SURFACE: You can define a projection surface with this choice from the 3D menu. Then you can use the Projection command to project selected elements onto this defined surface. Let's take a look at this function and the sub-menu for it.

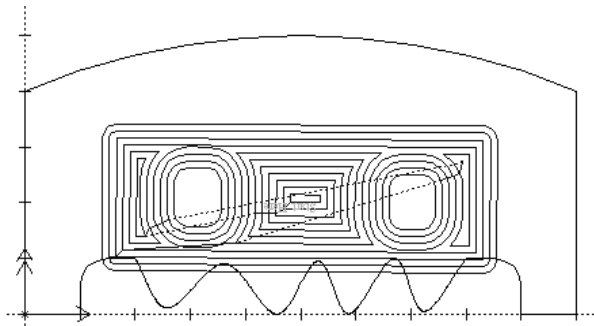


The sub-menu functions for the SET SURFACE menu are:

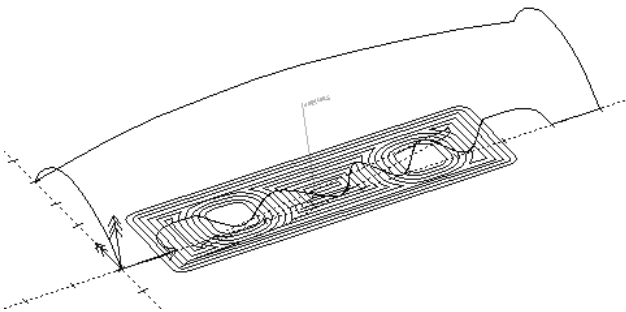
- **PLANER:** This defines the X-Y plane of the coordinate system as projection surface.
- **EXTRUDE X:** This defines the surface of a selected chain extruded in the direction of the X-axis of the coordinate system as the projection surface.
- **EXTRUDE Y:** This defines the surface of a selected chain extruded in the direction of the Y-axis of the coordinate system as the projection surface.
- **ROTATE Z:** This defines the surface rotated around the Z-axis as the projection surface.
- **INTERPOLATE:** This defines an interpolated surface as the projection surface.

Let's go ahead and take a closer look at how this all works starting below.

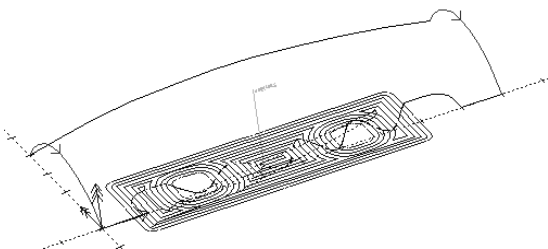
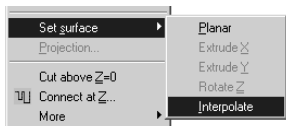
Here we have a pocket with two islands as well as 3D wireframe geometry in 2D view.



Here is the same geometry in 3D. What we are going to do is use the 3D geometry to turn the pocket toolpath into a surface that conforms to the sections/end radii.

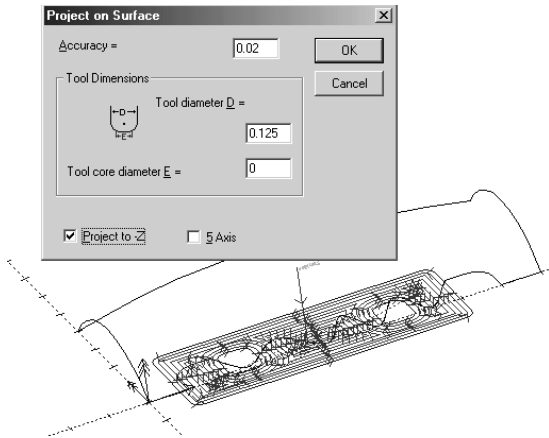


First we will select the two ends so that they are in the same direction. We do not need to use the equalize function on them as they are both 1 entity. Otherwise, we would need to use the equalize function on them as the geometry used needs to be the same number of entities. Then we select 3D, SET SURFACE and INTERPOLATE. This sets the ends and then they become unselected.

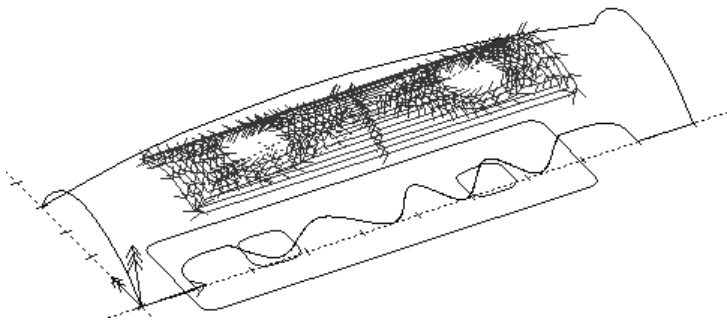


Now we need to select the pocket toolpath. It's easy to do this in 2D mode with the Select REGION function. Or you can use the selection CHAIN function if you have an easy to access starting line. Once the pocket is selected we can switch back to the 3D mode by selecting the 3D Menu and TURN 3D ON.

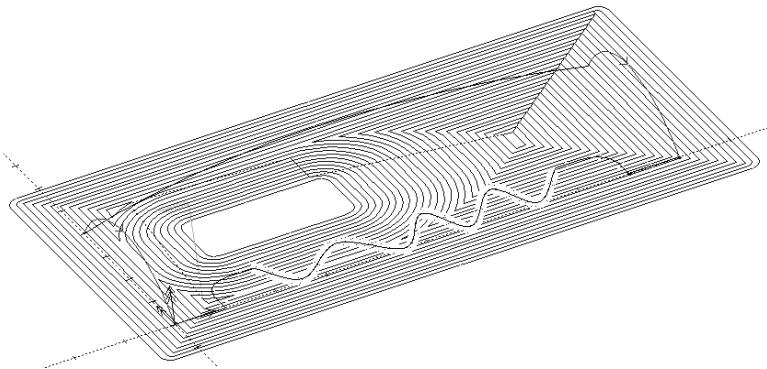
Now from the 3D menu we can select PROJECT. This brings up the Project on Surface dialog that you can see below. The accuracy setting will define the radial tolerance on how the spiral toolpath will follow along the end radii. Then we can enter in the tool diameter that we will be using and select the Project to Z option, as this will be projected into the Z-axis. This is not a 5-axis toolpath so we will leave this option unchecked and select OK.



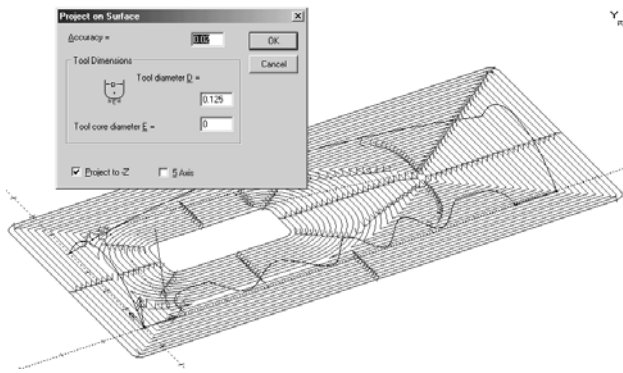
The result:



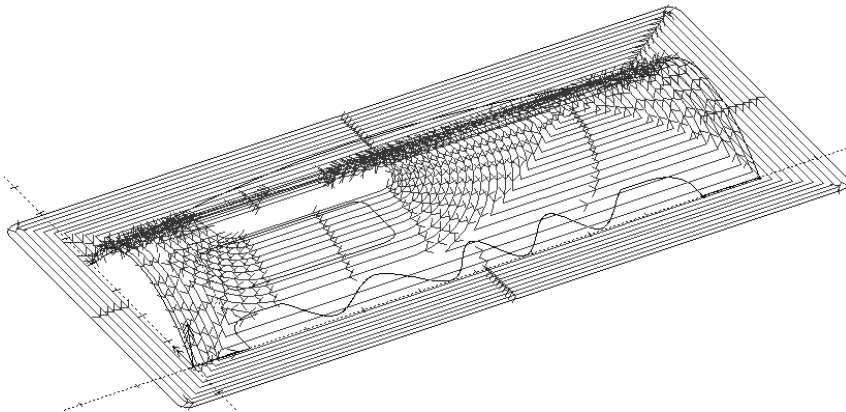
Now you can delete the 3D wireframe geometry and machine the toolpath. Here it is again with a larger pocket and the same two radii on the ends. As you can see they are already selected in the same direction in the image below.



Now we use the SET SURFACE INTERPOLATE function again. Next, we select the pocket toolpath and then go back and select PROJECT from the 3D menu.



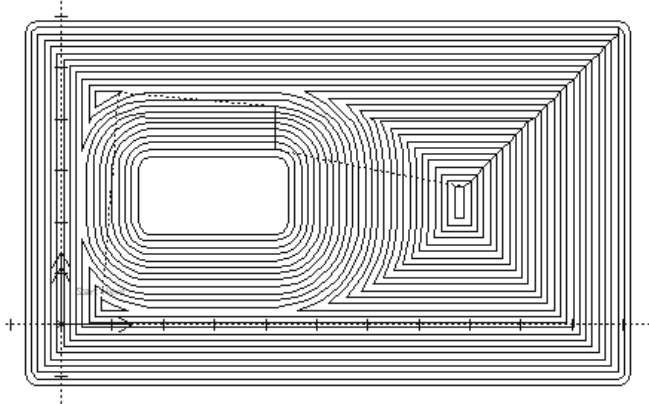
After making our projection entries, we select OK and get the result below.



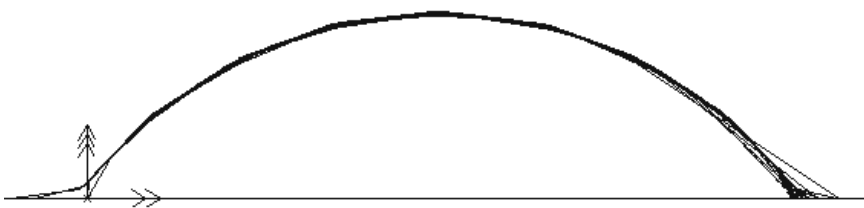
The toolpath has been modified to match those two radii and still offers the island.

Here is the part in various views.

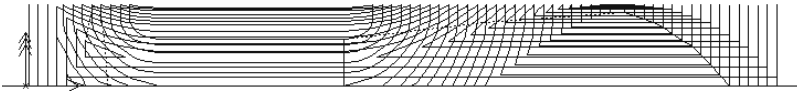
1. 2D View



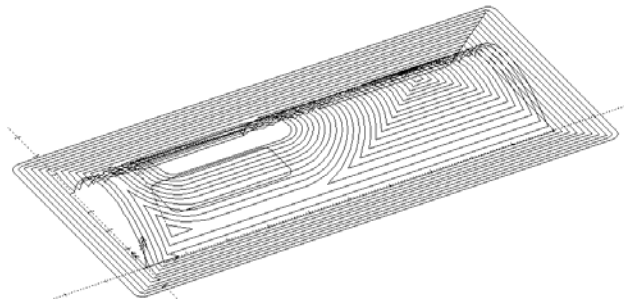
2. Side -right View.



3. Front View.



4. 3D View.

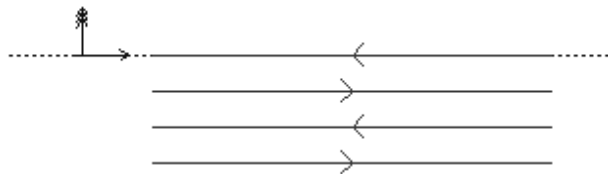


CUT ABOVE Z=0: This function allows you to remove any toolpath on the drawing screen that is above Z0 that you do not want to use to machine. In other words, if you have a wavy surface toolpath that goes above the Z 0, you can use this function to remove that toolpath for the machining process if you do not want to move the part down in Z and do not want to machine anything above the Z 0 plane.

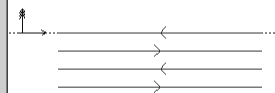
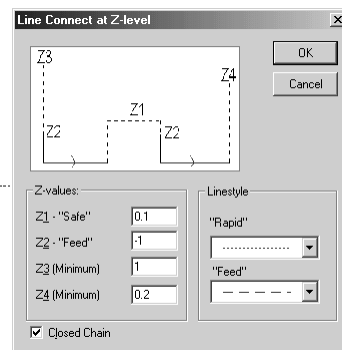
CONNECT AT Z: This function allows you to connect toolpath at varying levels in the Z-axis. Let's take a look at this function. Basically, this function is specifically intended for machining operations. You can set the three line styles separately for machining (feed and rapid style lines) at set Z levels, lowering to a smaller Z value or raising to a higher Z value. BobCAD Version 21 automatically closes the contour if the option **closed chain** is active.

See below.

1. 4 lines. Each -.25 in the Z-Axis. Take note of the direction arrows of the selected lines. Each line was selected going the opposite direction.



2. The Connect at Z function selected. The Z-Safe value is set to .1 as the clearance, the Z-FEED is set at -1 as this is the depth we will be cutting the lines, the Z-3 value is set to start at 1-inch in positive Z and will finish (Z-4) at .2. Our Rapid and Feed line styles are selected accordingly and we now simply need to select OK.



3. The new toolpath result.

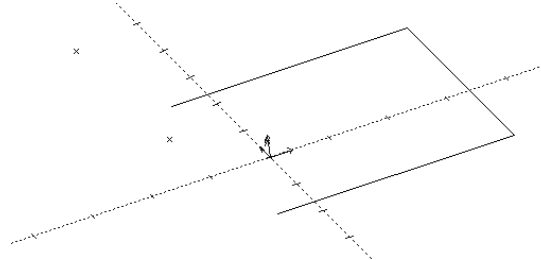


Let's take another look at this function.

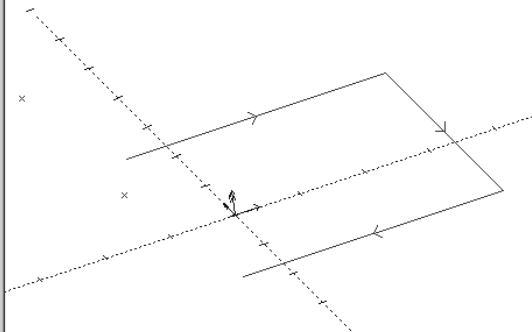
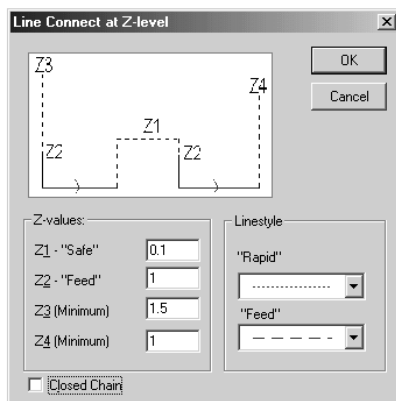
Front View



3D View



In the last images, you can see that we have an open shape at -1 in the Z-Axis and the two points at $.25$ -inches in positive Z. What we want to do is use the two points for the start and end of the path. By selecting the entities in the SAME order as the toolpath will go (do this by holding down the shift key and selecting each of the entities) we can then select the Connect at Z function in the 3D menu.



The Z value options are entered to provide information on the:

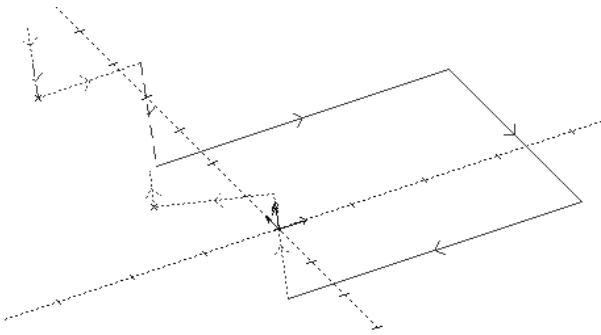
Z1 "Safe" which tells the software that if there were multiple depths, you would need to perform a rapid move to a specific Z level before feeding into the next cut.

Z2 "Feed" which tells the software that you are going to start the feed at a specific depth or at the top of your material.

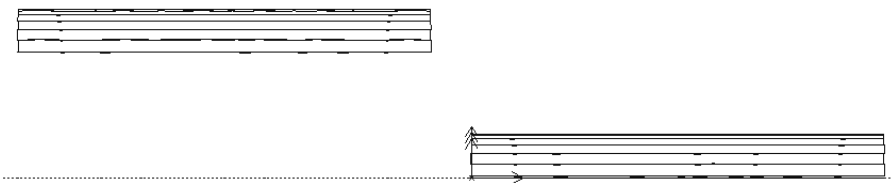
Z3 (Minimum), which tells the software the specific Z-Axis position that you will be starting the first rapid move.

Z4 (Minimum), which tells the software where the Z-Axis location will be to end the last rapid.

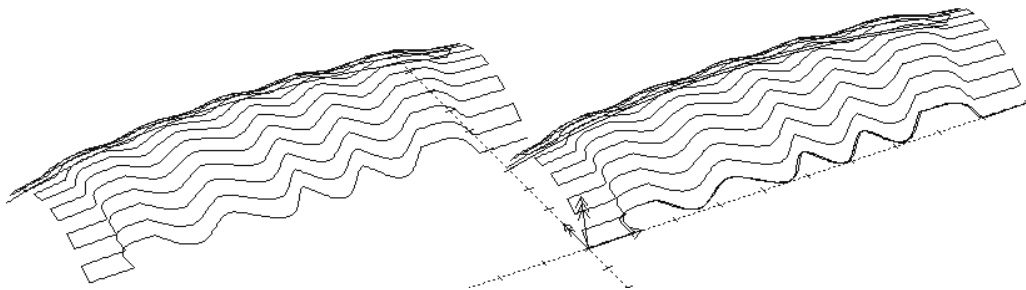
Now we can select OK.



Here we have a FRONT view of 2 surfaces, both at different levels.

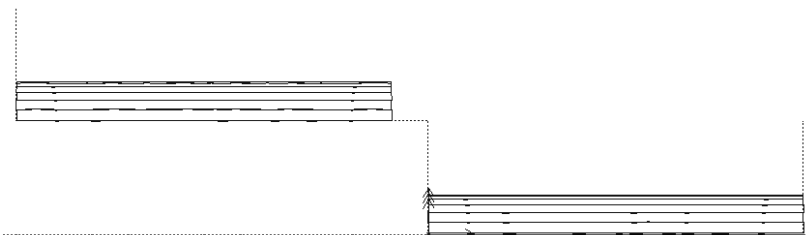


Here it is again from a 3D View.

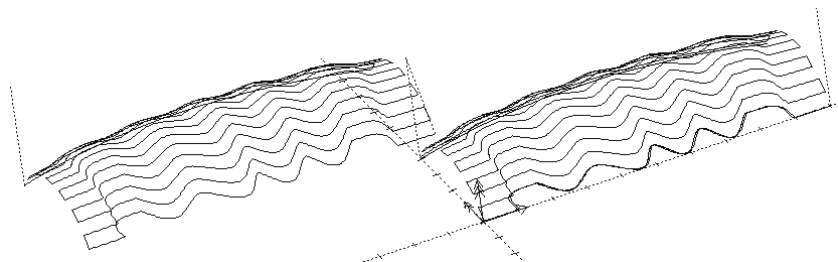


By selecting the two surfaces as chains and then the “Connect at Z” function, you would enter the values to set the connecting path properly to get the result below.

FRONT View.



3D View.



This is a fast way to connect surfaces at different Z levels.

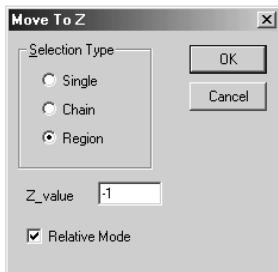
MORE: This is not a function but offers a special sub-menu with additional surfacing functions. You can see the sub-menu below.



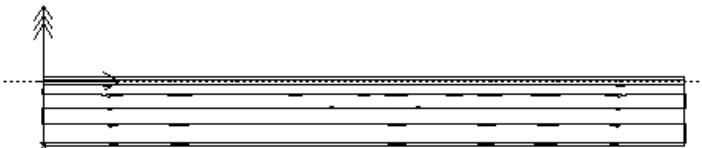
This special sub-menu offers additional 3D functions and new sub-menus as well.

This new menu consists of:

MOVE TO Z: This allows you to translate or move geometry into a positive or negative Z-axis. This function does not work with pre-selected geometry. First select the function and the Move to Z dialog will appear. Then enter the Z value that you want and choose a selection type. Single will allow you to move a single entity. Chain will allow you to select a chain for modification and region allows you to click and drag a selection box over the geometry that you want to move. Relative Mode means that you are moving the geometry in relation to its current location.

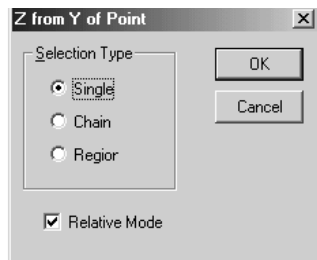


By selecting OK you would get a result like the one that you can see below.



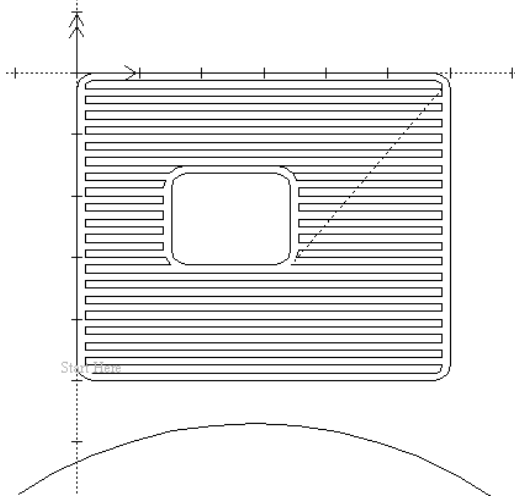
The geometry was moved -1 inch in Z.

Z FROM POINT Y: This function will allow you to move entities into the positive Z-axis by using a point along the Y-axis. The distance that the point is from the zero/zero position on the screen is the same distance that the selected entity or entities will be moved in the Z-axis transition.

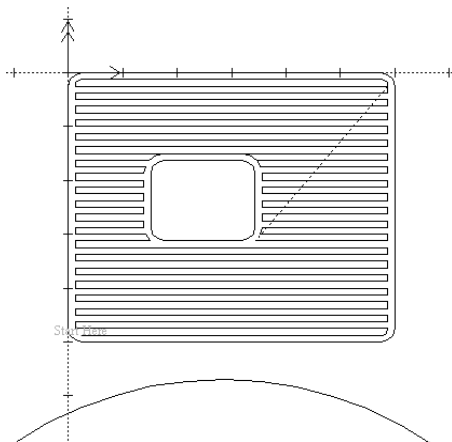


You can select a single entity, a chain of entities or an entire region. You also have the relative mode option for moving it from its current location or from the Z-Zero. Once you select OK, you will select the entity to be moved and then the point to make the change.

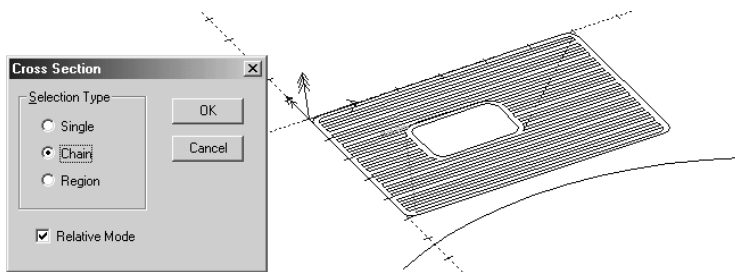
CROSS SECTION: The CROSS SECTION function gives you two options. FRONT VIEW and SIDE VIEW. These functions allow you to take flat 2D toolpath and turn the toolpath into surface toolpath through using an offset curve. Let's take a closer look starting with the FRONT VIEW cross-section function below.



The image below is a 2D image of toolpath with an island and an offset arc.

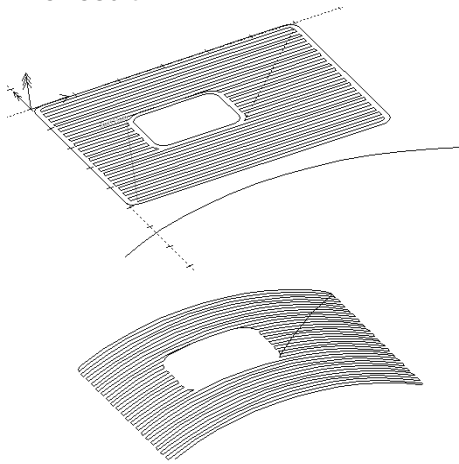


This function does not work with pre-selected geometry so you cannot have the entities to be moved selected. We would first select the CROSS-SECTION FRONT VIEW function. The selection dialog will appear asking for the selection choice. See below.

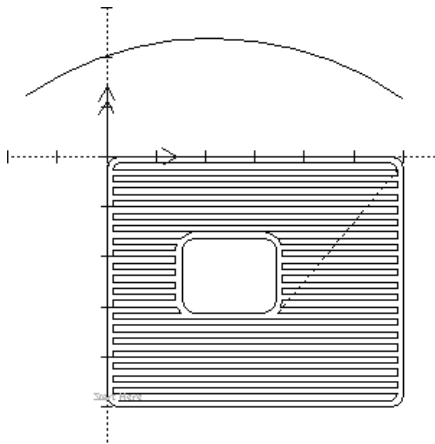


We will select CHAIN as the pocket toolpath that we are using in the example is a chain. We select the first entity of the chain, point our direction arrow in the direction of the rest of the chain, click our mouse button one time and then select the F3 key and the chain will all select. Next, we will select the offset arc, click our mouse button with the arrow in either direction and then RIGHT-click.

The result:

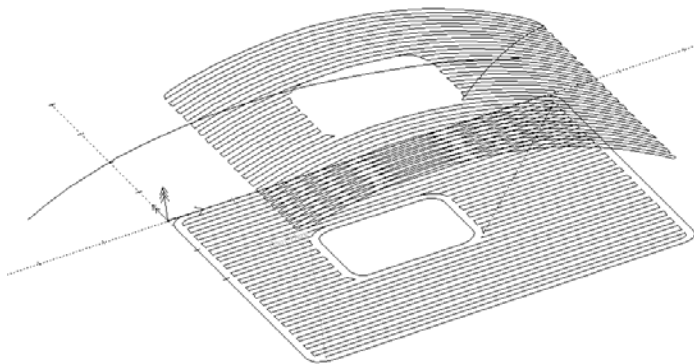


Because the offset arc was – 6-inches in the Y-axis, the geometry was moved and modified – 6-inches in Z. The offset curve in the Y-axis will determine the distance the selected entities are modified. Let's look at this again with the offset curve in a different location.

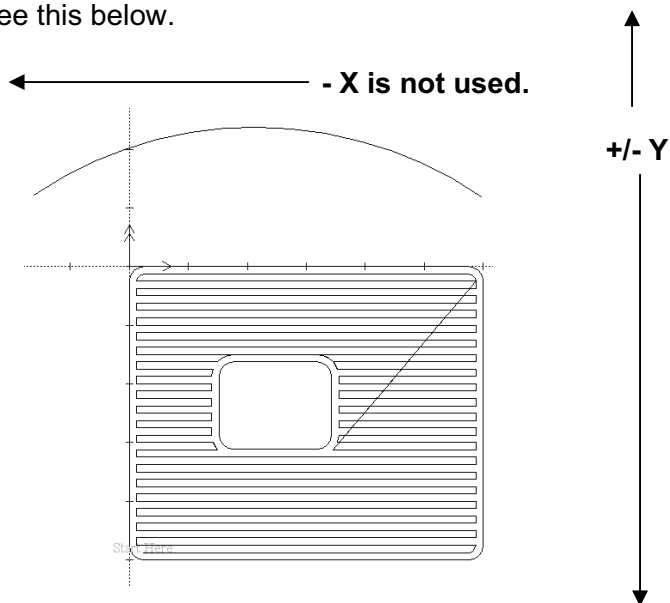


Here, the offset arc in the image above is positive in the Y-axis.

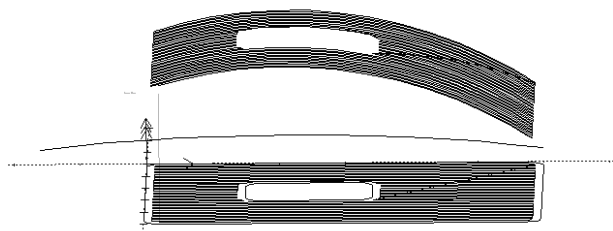
By selecting the CROSS SECTION FRONT VIEW function, selecting the chain selection option and then making the selections as we did in the earlier example, you can see the difference below.



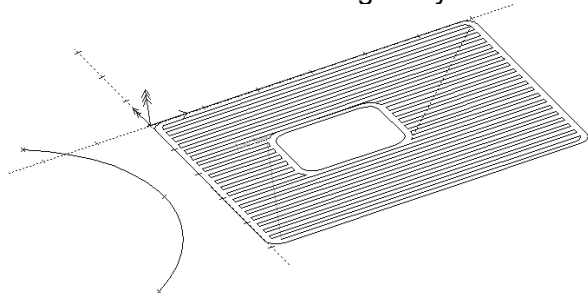
Before making the final cross-section selections, you may want to change the color that you are working with to a different color. This makes it easy to select colors for blanking or deleting. You will also notice that the result surface was also different in the X-axis, as the part of the offset arc in negative X is not used in the translation. See this below.



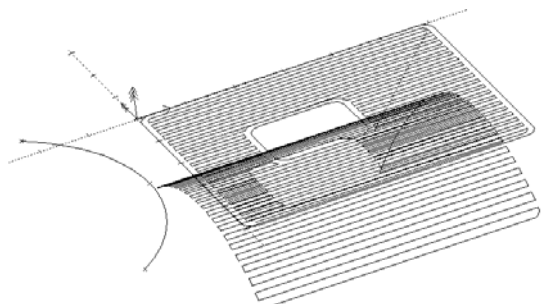
You can see the toolpath was modified relative to the arc in the Y-Axis.



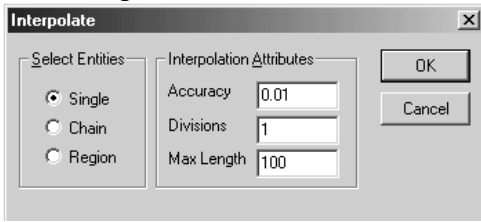
SIDE VIEW: This function gives you the same result from the side of the toolpath. See this below.



Then below, you can see the result. The X-axis is used to designate the Z-axis change in the SIDE VIEW cross-section function.



INTERPOLATE: This function is used to divide or break up entities as specified amount. Let's take a look at this dialog below.



Here you have the ability to select:

- Single entities.
- Chained entities.
- An entire region.

You divide or break up geometry by indicating:

- The accuracy, which is the length of each, divided entity.
- The number of entities that you want to divide the selected entity into.
- The maximum length of the entities once broken or divided.

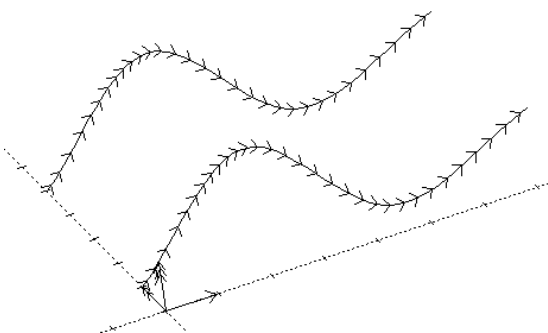
This is a simple function that can be used when dividing entities or breaking up geometry for surfacing needs.

SURFACE: This has a complete sub-menu of its own. Let's take a look below.



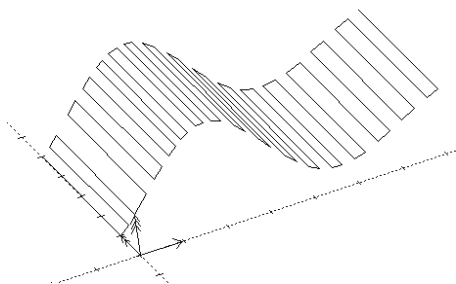
Let's go ahead and have a look at each one of these functions and what they do starting on this page.

RULED: This function allows you to create simple surface toolpath between two chains. Here we have two curved chains with an equal number of entities in both.



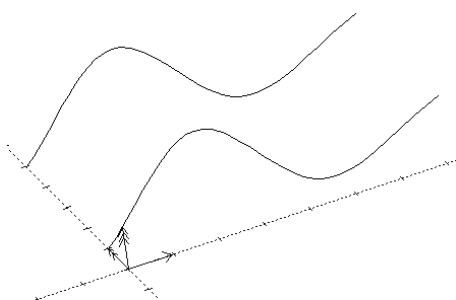
By selecting the ruled surface function you will then need to select the first chain and then the second chain with all arrows going in the same direction.

When finished with the chain selections, RIGHT-click your mouse to automatically generate the toolpath. See below.

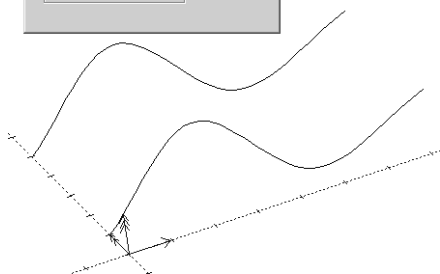
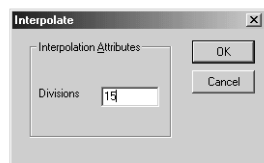


The key to using this function is to divide the curves into the same number of entities. The divisions will need to be half the distance of the size tool you will be using.

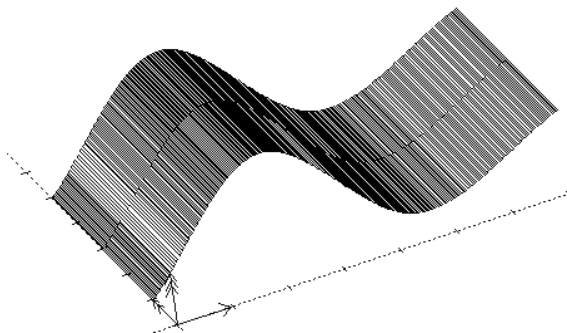
AUTO RULED: This is similar to the ruled surface function. The difference is that you can enter in the interpolation or divisions that you want the toolpath to be created by. Let's take a look.



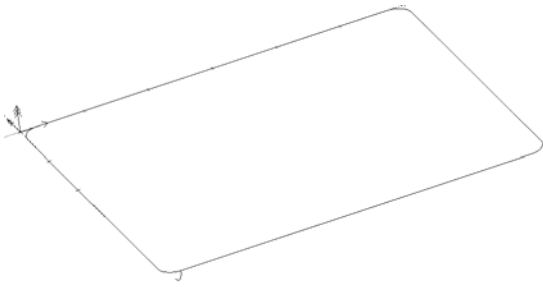
By selecting the function, you will open the interpolation dialog as seen below.



Enter the number of divisions you want both chains to be divided into. By selecting OK you will then be able to make the chain selections of both curves in the same selection directions. When finished, select your right mouse button to create the surface.



FOLLOW CURVE: This function allows you to create a surface toolpath from an offset curve. Let's take a look.



We will start this function with a profile and an offset curve that has been rotated -90 degrees in the Y-axis. Once we have a set offset curve, we will interpolate and divide the curve into the number of entities that we want there to be toolpath passes to be applied to. You can use the interpolate function in the 3D menu for this. Next we can proceed with unselected geometry to produce the surface toolpath.

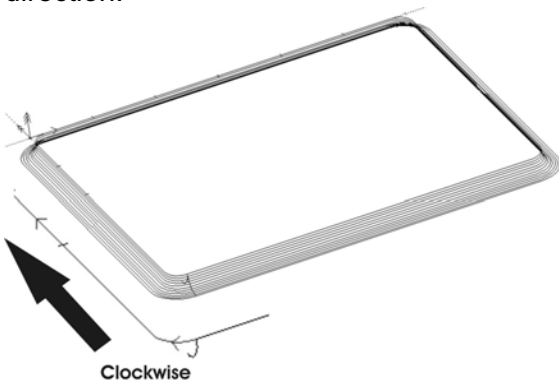
You would first select the function, and then select the profile that you want the surface to follow along. In this case it would be the rectangle geometry in the image above. You would select this geometry as a chain **AVOIDING** the offset curve. You would make your selection by clicking on the first entity of the chain, pointing your directional arrow in the direction of the chain and clicking your mouse button again to secure the arrow and last, select the last entity of the chain avoiding the offset curve. Now you would select the first entity of the interpolated offset curve, point your directional arrow toward the end of the chain, click your mouse button one time and hit the F3 key on your keyboard. This would generate the toolpath for you automatically.

When you select the first chain that the surface will follow, you have a choice of making the chain selection in a clockwise or counter clockwise direction. It makes a difference in the direction that you make this selection. Let's take a look at this.

You can see in the image below, the chain selection was made in a counter clockwise direction to produce the follow curve surface path that you can also see in the image.

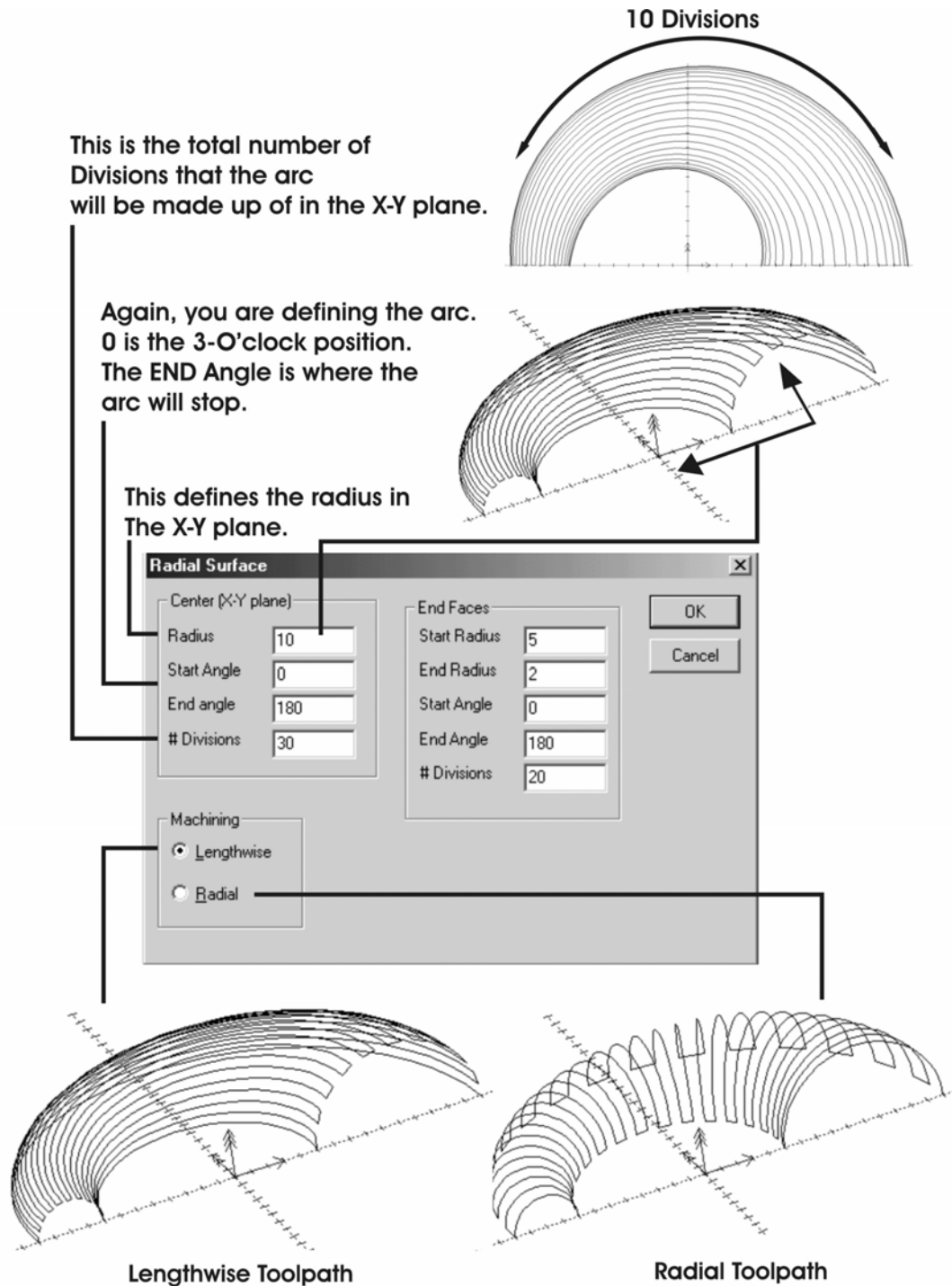


Below you can see what happens to the toolpath when the first chain selection is directed in a clockwise direction.

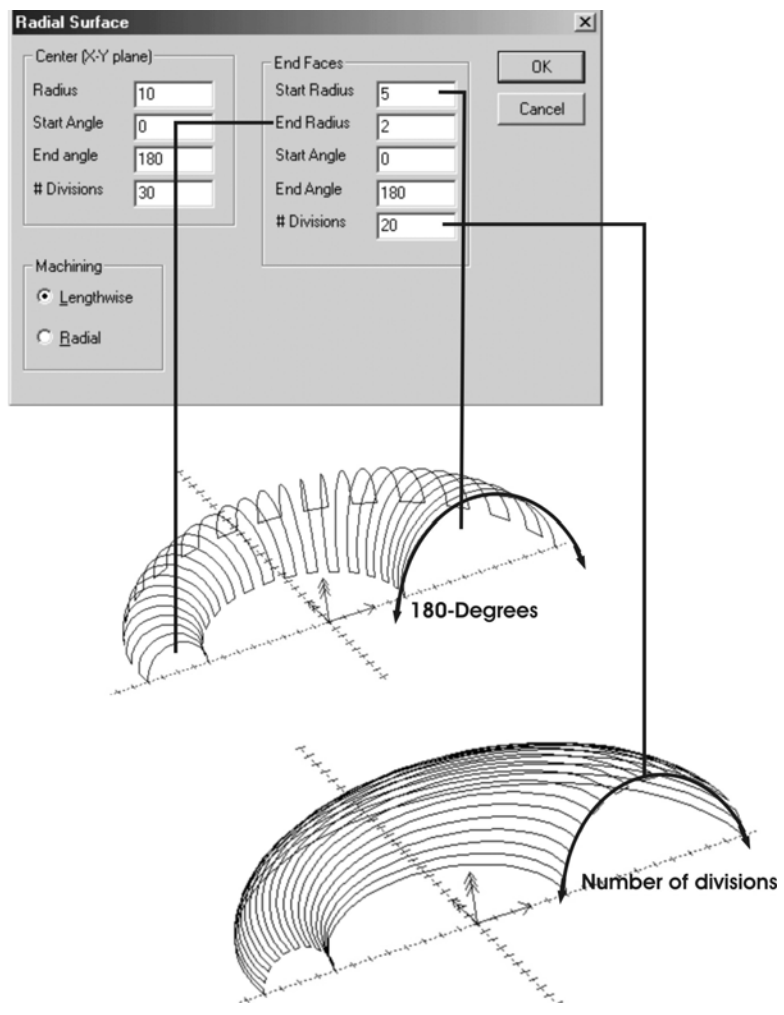


The toolpath in the image above goes in the opposite direction as the counter clockwise selected toolpath. If you make a chain selection mistake, you can always select the UNDO icon and start over.

RADIAL: This function will allow you to create a radial style surfaces. Let's take a look at this functions dialog.



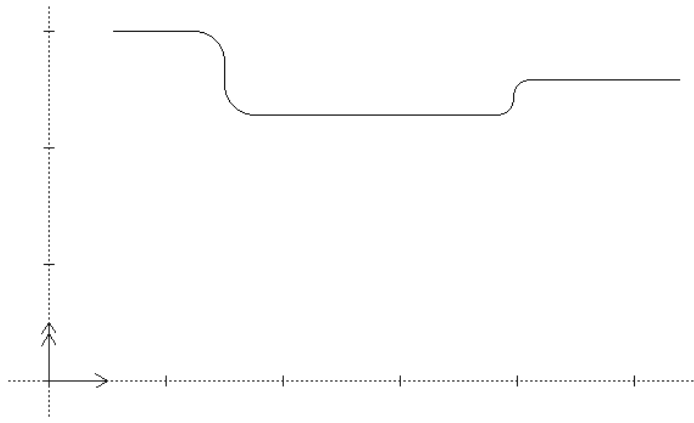
The END FACES section defines the two radii at each end of the surface. The radial surface is defined going counter clockwise. The starting radius will be the face radius on the right and the end radius is the face radius on the left. The number of divisions would define the toolpath for the radial surface itself.



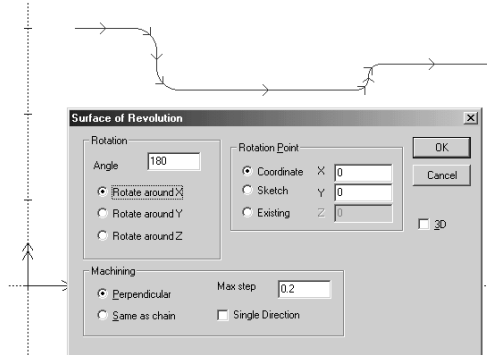
Remember that the number of divisions under the END FACES section will define the toolpath.

REVOLUTION: The revolution surface function will allow you to revolve a surface toolpath around a specified radius. Let's take a look.

You would first start with an unselected chain like the example below.

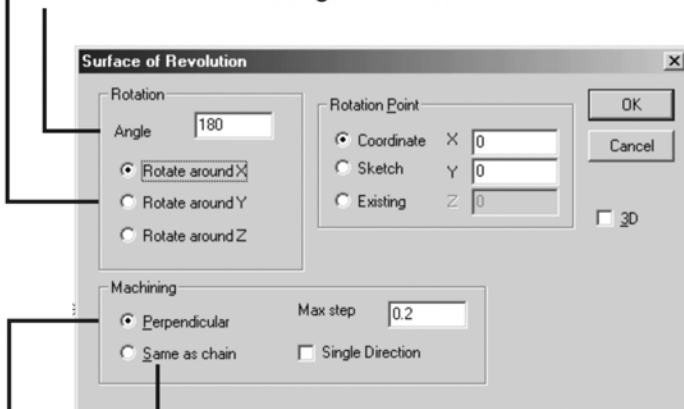


Select the function and then CHAIN SELECT the entities to open the revolved surface dialog.

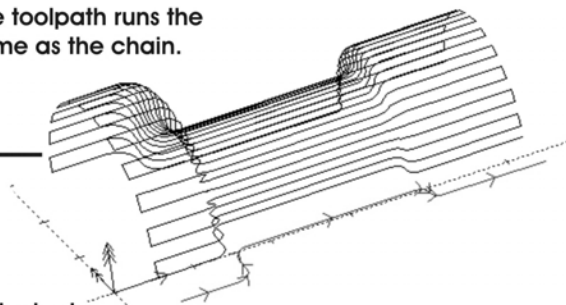


This is where you choose the designated axis for rotation.

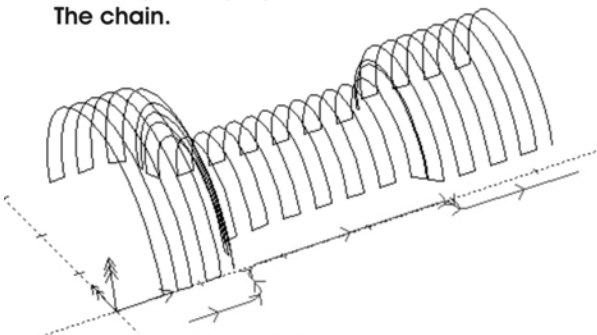
This is the degree amount that the surface will be Rotated around the designated axis.



The toolpath runs the same as the chain.



The toolpath is perpendicular to The chain.



You can also select a rotation point for use with X, Y and Z coordinates, or you can sketch a rotation point as well as use an existing entity on the screen. The maximum step determines how close or far away each toolpath line is.

CHAPTER 7

3D PROJECTION & SKIN SURFACE LESSON

Section 40. 3D Projection Lesson
Section 41. 3D Skin Surface Lesson



Section 40. 3D Projection Lesson

Let's go ahead and do a review training lesson for the 3D Menu, Projection function. You will want to start with a NEW BobCAD drawing screen.

STEP 1

With a new BobCAD screen, select 3D and TURN 3D ON.

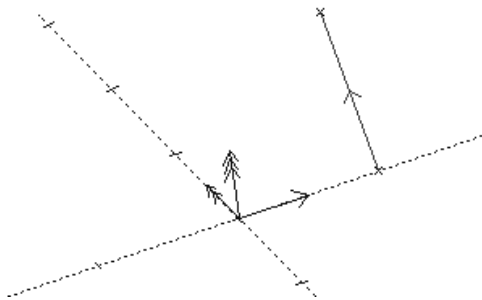
STEP 2

From the POINT menu select Coordinates and enter:

- Enter 1 for X.
- Y 0.
- Z 0.
- Select CONTINUE.
- Enter .75 for X.
- Y 0.
- Enter 1.25 for Z and select OK.

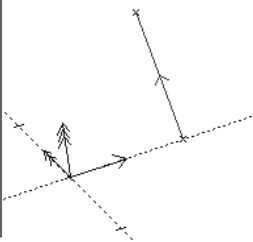
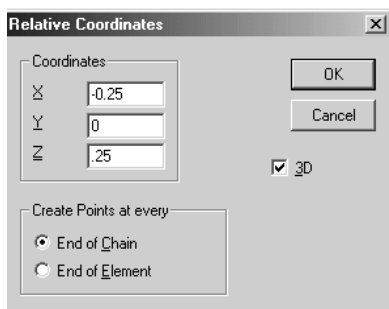
STEP 3

The two points will be selected on the screen if you have the Auto Pre-select function on from the environment menu. Now go to the LINE menu and select JOIN. A line will automatically be joined between the two points and will be selected. The two points will automatically be deselected.



STEP 4

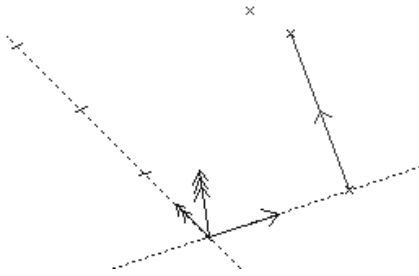
Now, with the line still selected, go to the POINT menu and select INCREMENTAL to open this dialog.



Now that the Relative Coordinates dialog is open, enter the following:

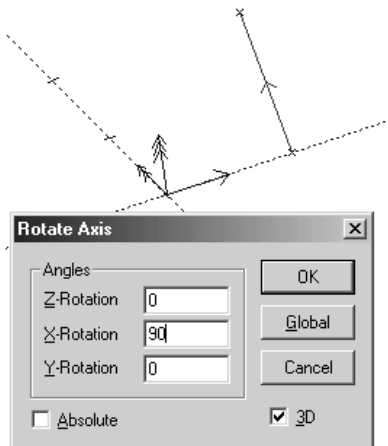
- Enter -.25 for X.
- Y 0.
- Enter .25 for Z.
- Select the END OF CHAIN option and select OK.

The result:



STEP 5

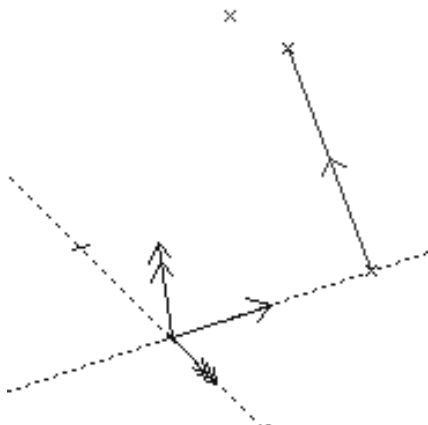
RIGHT-click your mouse at the CENTER of the UCS crosshairs. From the pop-up menu, select ROTATE.



The ROTATE AXIS dialog will appear. Enter the following:

- Z rotation 0.
- X rotation 90-degrees.
- Y rotation 0.
- Do not check Absolute and click OK to rotate the axis.

The result of the rotation:



STEP 6

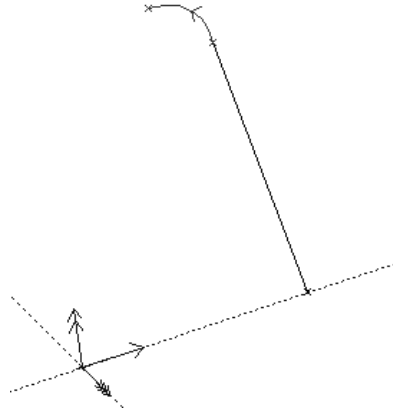
Now click your mouse off of the geometry to deselect everything. Then select the ARC menu and FILLET function. When the fillet dialog appears, enter .25 for the fillet radius and click OK. Now select the point at the end of the line and then select the point that you drew incrementally to it to produce the fillet.

Click your left mouse button to exit that function. See the result below.

The fillet dialog.



The result fillet.



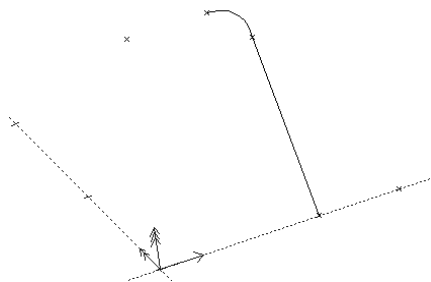
STEP 7

Now place your cursor on the UCS crosshairs and RIGHT-click your mouse. In the pop-up menu, select RESET. This will reset the UCS back to the default position.

STEP 8

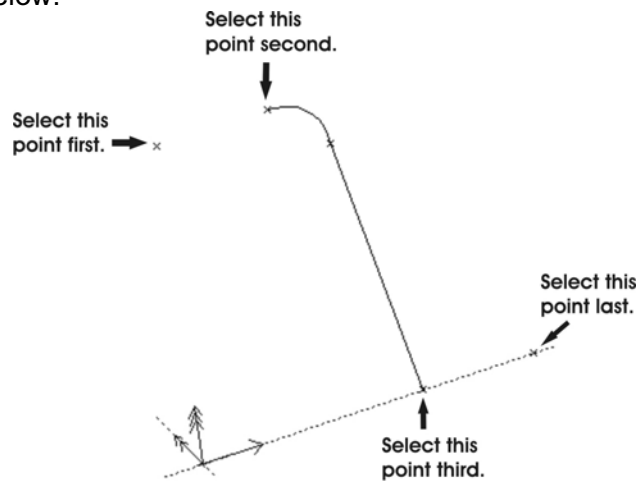
Select the POINT menu and Coordinates. Enter the following values:

- Leave X at 0.
- Leave Y at 0.
- Enter 1.5 for Z.
- Click CONTINUE.
- Enter 1.5 for X.
- Leave Y at 0.
- Leave Z at 0 and select OK. This will draw a second new point. See below.

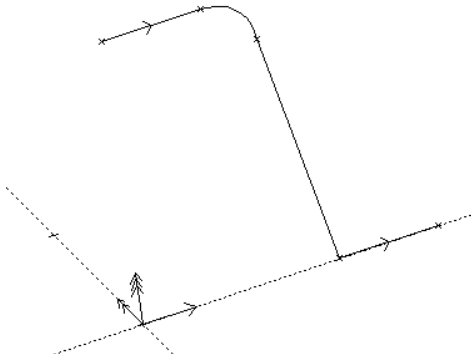


STEP 9

Click your mouse off of the geometry to deselect all of the entities. Now go to the LINE menu and select JOIN. Follow the instruction image below.



Now, RIGHT-click your mouse when finished joining the lines. Here is the result:



STEP 10

This profile that you just drew represents the finished profile of a boss in the center of our part. It has roughly, a 12-degree draft angle on it, much like the male half of a mold. We are going to cut this with a .5-inch ball end mill cutter, and in order to do this we will need to offset for the tool.

There are two things to consider at this point in time. They are:

- The bottom of this toolpath (after offsetting) needs to be on the same plane as the rest of the toolpath. The reason for this is that we can't project a curve to a toolpath that the curve doesn't meet or touch (in an abstract sort of way).
- The toolpath itself will be for a ball mill, and as in BobCAD's skinning functions, ball mill toolpaths are from the center of the ball. Therefore, offset by the radius into the side of the tool, and offset up from the tip by the radius. Therefore, the tool will have to be offset for length at the control as most machinists touch off the tip when they actually cut.

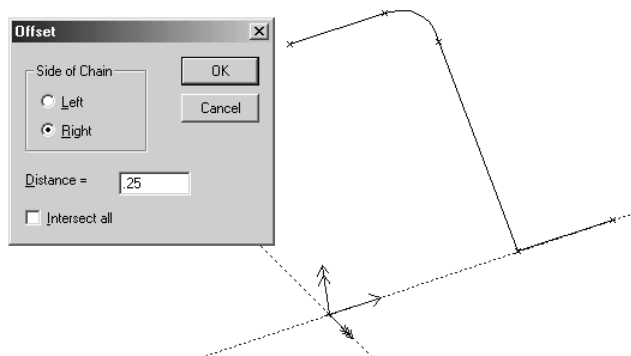
So, let's go ahead now about setting this up to do just that.

STEP 11

Place your cursor on the UCS and RIGHT-click your mouse. Select ROTATE from the pop-up menu and enter 90-degrees in the X-axis just like you did before and then select OK to rotate the UCS back to where we had it before. The offset we'll need cannot work unless the profile is flat in the X-Y plane, regardless of Z level, so we are bringing the plane up 90-degrees to the part.

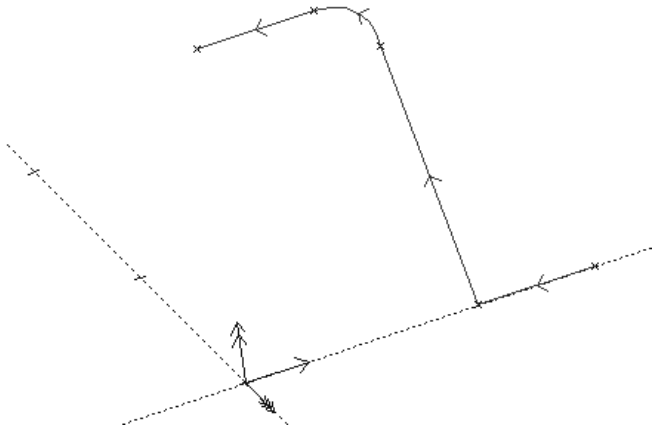
STEP 12

Select the OTHER menu and then OFFSET. Make sure that "RIGHT" is selected. Enter a DISTANCE of .25. This .25 is the radius of our tool that we are compensating for. You do not need to select the Intersect All option. Simply select OK.



STEP 13

Click on the bottom line of the profile, point your arrow in the direction of the chain which will be toward the LEFT, click your mouse button one more time to secure the direction and hit F3 on your keyboard to select the entire chain.



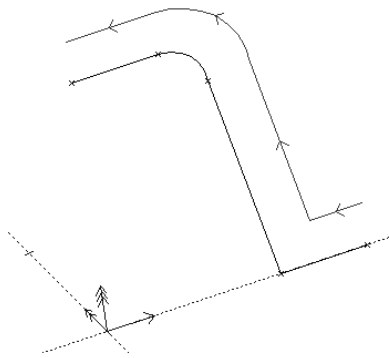
STEP 14

Place your cursor on the UCS and RIGHT-click. In the pop-up menu, select RESET to reset the UCS.

STEP 15

Here is where we do some magic. First, if the geometry is selected, go to the EDIT menu and choose DESELECT ALL or click your mouse button off of the geometry to deselect everything. If you have the auto pre-select turned on in the environment dialog, you will already have a selected offset chain. If not, go to the EDIT menu, choose Select Entities and then CHAIN or click the "Select Chain" icon.

Now click on the OFFSET chain at the bottom, point your arrow toward the left, click your mouse button one time to secure the directional arrow and then hit F3 on your keyboard to select the offset.



STEP 16

Go to the 3D menu and select SET SURFACE and ROTATE Z.

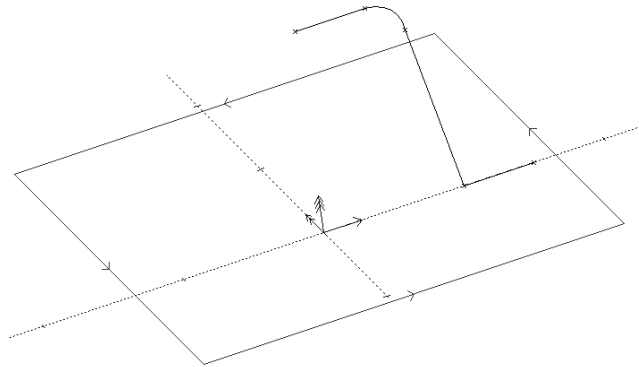
This did an important thing. It set the offset curve as a surface rotated about the Z-axis, 360-degrees centered on X0, Y0 and Z0. In effect, the one offset curve that we drew just became a thimble shape pre-offset for your tool. You will not see it yet, but BobCAD just set this as our shape for the surface projection that we are about to do.

We don't really need this offset chain any longer for now, but we may need it later. While it is still selected, go to the CHANGE menu and select BLANK to blank it out. This will hide the offset curve so that it is not in the way of our next action. Now that we have the surface defined to project to, we need to create the toolpath.

STEP 17

Go to the OTHER menu and select RECTANGLE. Set the REFERENCE POINT to BOTTOM LEFT and enter the following coordinates:

- Enter -1.5 for X.
- Enter -1.5 for Y.
- Check the 3D option box and enter .25 for Z.
- Under WIDTH & HEIGHT enter 3.
- Enter 0 for RADIUS and make sure that the DRAG box is NOT checked.
- Select OK and the rectangle will automatically be drawn.

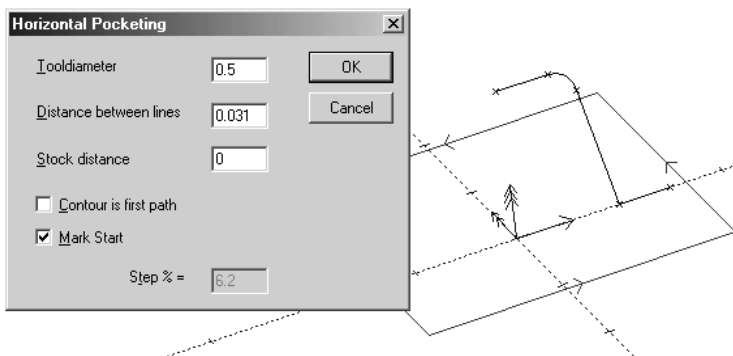


The Z value here is .25 for a reason. The offset that we used for the SET SURFACE function was offset for the tool, which meant that it was .25 up in Z. We need to match this. And so, we drew our boundary box .25 up in the Z-axis as well.

We are almost finished. Our rectangle represents the boundary box at the base of the part. To cut down to this, we will need to generate a pocket toolpath, which will also be as our toolpath to be projected to make the thimble shaped boss.

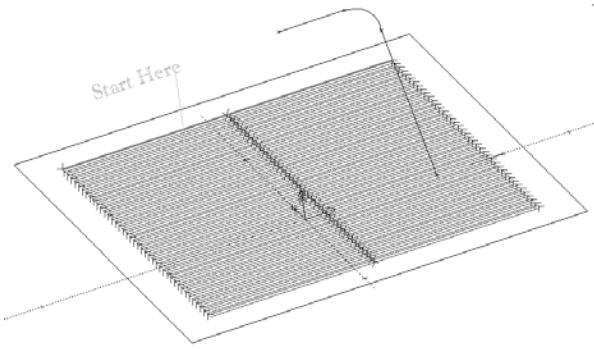
STEP 18

The rectangle should still be selected. Go to the OTHER menu and select POCKET. Select HORIZONTAL from the pocket style box to open the horizontal pocket function dialog.



Enter in the following values in the horizontal pocket dialog:

- Tool diameter: .5
- Distance Between Lines: .031
- Stock Distance 0.
- Check the "Mark Start" option.
- Select OK to generate the pocket with the pre-selected rectangle geometry.



If you had trouble and the pocket did not generate it is because the rectangle was not pre-selected and you may not have your auto pre-select option selected in the environment settings of the software. If this is the case, simply click on the rectangle and then RIGHT-click your mouse to generate the pocket toolpath as you can see in the last image.

STEP 19

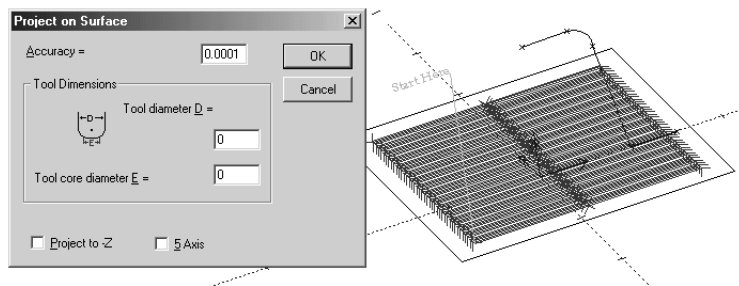
The pocket toolpath should be selected at this time. If not, use the chain selection function and start the selection at the top of the green vertical line, point your arrow down, click your mouse button one more time and hit the F3 key on your keyboard to select the pocket toolpath.

STEP 20

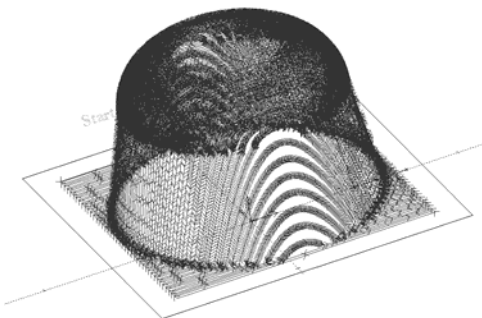
Now select the 3D menu and PROJECT. This will bring up the projection dialog. You can see this image on the next page in step 21.

STEP 21

Set the ACCURACY to .0001. Enter 0 for the tool-diameter and 0 for core-diameter as we have already offset the original shape. Make sure that the PROJECT to Z and 5-AXIS and both NOT checked and select OK to produce the new surface toolpath.



The result of your work should look like the image below.



If the toolpath doesn't come out correct, increase the tolerance setting for ACCURACY in the projection dialog.

Congratulations on completing your first 3D surface projection lesson!

Section 41. 3D Skin Surface Lesson

This lesson is dedicated to teaching you through steps, how to draw a 3D wireframe and use the Skin function from the 3D menu to create surface toolpath for machining. Start with a NEW BobCAD drawing screen.

STEP 1

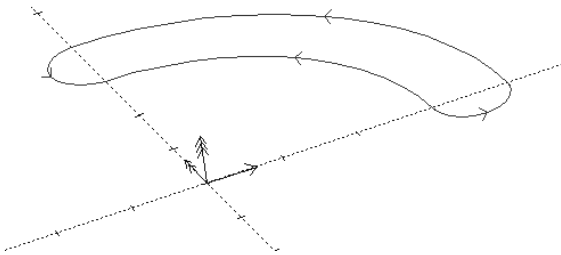
With a new screen select the 3D menu and TURN 3D ON.

STEP 2

Select the ARC menu and then COORDINATES. Enter the following values:

- Leave 0 for X.
- Leave 0 for Y.
- Leave 0 for Z.
- Enter 3 for the RADIUS.
- Start Angle is 0.
- End Angle will be 90-degrees.
- Select CONTINUE.
- CHANGE the RADIUS to 4 and select CONTINUE.
- Leave 0 for X.
- Enter 3.5 for Y.
- Leave 0 for Z.
- CHANGE the RADIUS value to .5.
- Set the Start Angel to 90 and the End Angle to 270.
- Click CONTINUE.
- Enter 3.5 for X and Y at 0 with the Radius still at .5.
- Change the Start Angle to 180 and the End Angle to 360 and click OK.

You should have the following result:

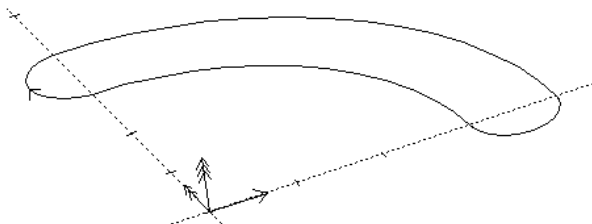


STEP 3

Click your mouse with the cursor OFF of the geometry to deselect all the entities.

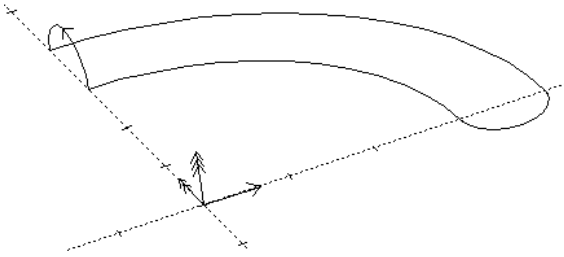
STEP 4

Click the arc at the far upper LEFT of the part to select it. It will highlight and be selected.



STEP 5

Go to the CHANGE menu and select ROTATE to open the rotate dialog. Select COORDINATES under the ROTATION POINT section. Leave X, Y and Z at 0 as we want to rotate around 0,0 because it is easier and the arc center is on Y 0. Enter 90-degrees for the Y-axis and 0 for X and Z. Make sure that "Make Copy" is NOT selected and that SCALE is set to 1. Select OK to get the result below.



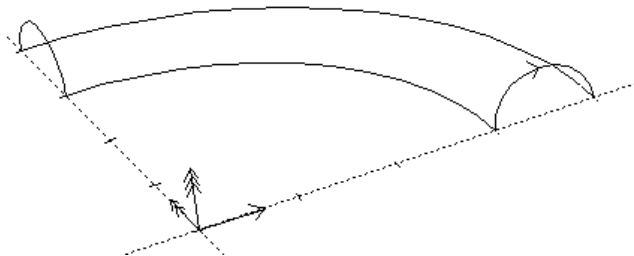
If at any time you make a mistake, select the UNDO icon and redo the last step.

STEP 6

Now click on the arc at the lower RIGHT end to select it, but make sure that you deselected the other arc first.

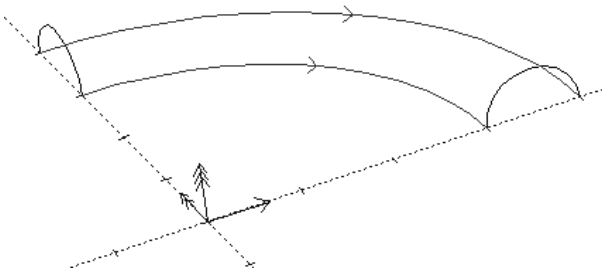
STEP 7

Select the CHANGE menu and ROTATE. Using the COORDINATES option for the rotation point, leave the X, Y and Z values at 0. This time enter -90 degrees for the Y-axis and leave the X and Z at 0. DO NOT check the "Make Copy" option and make sure that the scale is set at 1. Click OK. You should have the result as seen below.



STEP 8

Deselect all entities. Now select the first larger arc (3-inch radius) and make sure that the directional arrow is going towards the RIGHT, counter clockwise. Now hold down your CONTROL key on your keyboard and select the 3.5-inch arc so that the arrow is going toward the right as well. Then let go of the control key. Both arcs should now be selected with the directional arrows going in the same direction.



STEP 9

Select the 3D menu and then click on SET PATH. Enter the following values in the SET PATH dialog.

- Enter 0 for the Max Length of Divisions.
- Enter 1 for Minimum Length of Divisions.
- We do not need to change the arc error.

Effectively, this turns off the size for the division of the path and makes sure that the skin function will not break it into more pieces than it actually needs. Since these two arcs are not in 3D, we do not need to deal with the arc interpolation error.

Click OK. The two arcs will still be selected. Select the CHANGE menu and blank them.

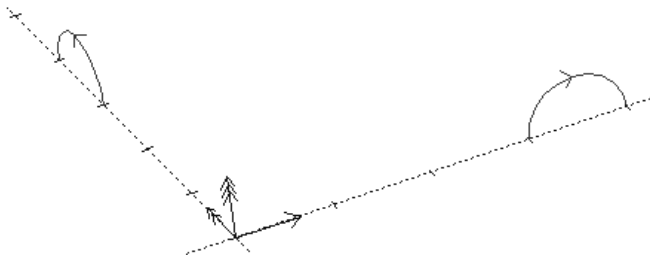
STEP 10

Select the arc on the LEFT side so that the directional arrow is point up and over the arc AWAY from you.

The reason for selecting this arc, in this direction is because we began selecting our arcs for the “Path” section of the skin from the front entity/arc. Then we selected the rear arc. Therefore, **we must select our cross sections from the beginning of the path to the end. This is important to remember when using the skin function.**

Now hold down the CONTROL key and select the arc at the right end so that the arrow goes up and over the arc away from you and toward the right.

The result:



Now let go of the control key.

STEP 11

Select the 3D menu and SKIN to open the SKIN dialog. Enter the following values:

- Enter .5 for the tool diameter.
- Leave the core diameter at 0.
- We will be using a half-inch ball end mill.
- Set the Max Distance Between Lines at .05.

This will give us a .05 step over between the passes, up and over the path.

- Under CONNECT, select Zigzag.

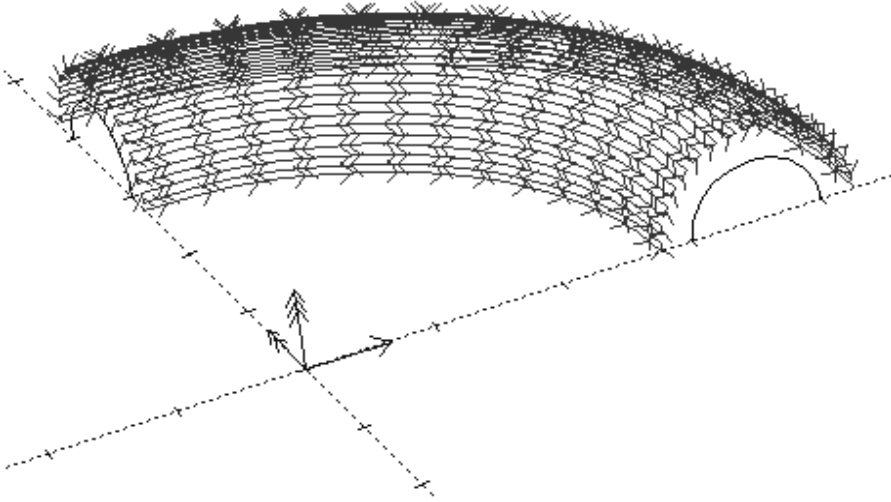
This will tell the software that we will need the tool to continuously move from the beginning of the skin to the end, obeying our cut direction up and over the top of the skin.

- Select PERPENDICULAR TO SECTIONS under “Toolpath” as this means we will be taking our passes lengthwise along the entities we assigned as the “Path.”

“Inward Surface” means that our toolpath would end up offset from the inside of our part. We want it outside, so we will leave this UNCHECKED.

- Select the option for Automatic Removal of Undercutting. This should be checked under most circumstances just as a safety precaution. This is because it will make sure that the tool does not move past the edges of our part. Since we only have one real surface patch and there are no sharp inside corners to it, this is not necessary; however, you will usually want to leave this selected anyway. Make sure that the 5-Axis option is not selected. Click OK to generate the skin surface.

See your result image below.



Congratulations on completing this lesson!

CHAPTER 8

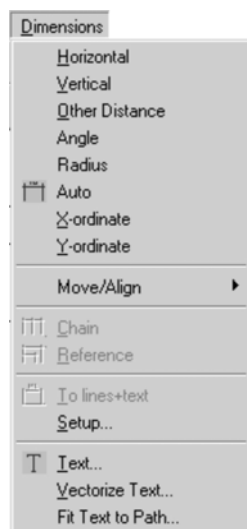
THE DIMENSIONS MENU

- Section 42. Horizontal
- Section 43. Vertical
- Section 44. Tips on Modifying a Dimension
- Section 45. Other Distance
- Section 46. Angle
- Section 47. Radius
- Section 48. Auto Dimension
- Section 49. X Ordinate
- Section 50. Y Ordinate
- Section 51. Move/Align
- Section 52. Chain
- Section 53. Reference
- Section 54. To Lines + Text
- Section 55. Setup
- Section 56. Text
- Section 57. Vectorize Text
- Section 58. Fit Text to Path
- Section 59. Tips on Text Along a Curve
- Section 60. Modifying Text
- Section 61. Creating Solid Fonts
- Section 62. Working With Text – Training Lesson



Version 21 offers a complete menu dedicated to dimensioning your parts. This chapter and each section will give you the details of how each function works so that you can fully use each feature for dimensioning.

The **Dimensions** Menu:

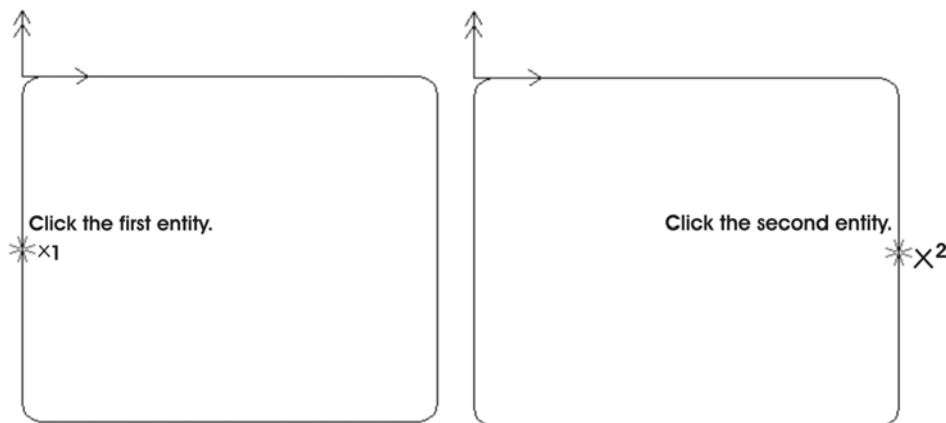


Section 42. Horizontal

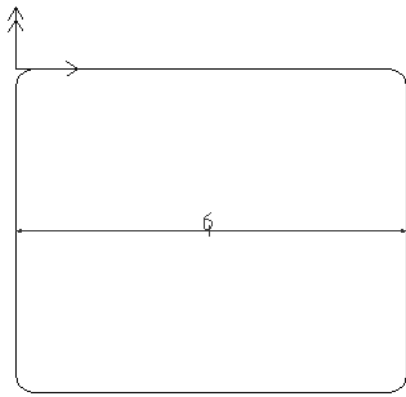
HORIZONTAL: This option allows you to create horizontal style dimensions. Let's take a closer look below.

By selecting this function, you can select the first entity and then the second entity to produce the dimension. Because this is a repetitive function, you will need to click your RIGHT mouse button to exit the function.

Example A.

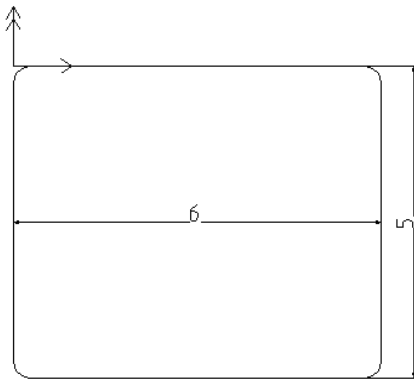


Example B. The result.



Section 43. Vertical

VERTICAL: This allows you to dimension entities in a drawing vertically. Simply select the first entity and then the second entity to produce the result. See below.

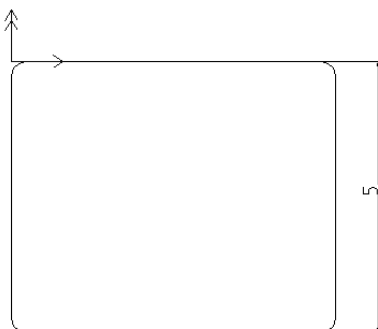


Section 44. Tips on Modifying a Dimension

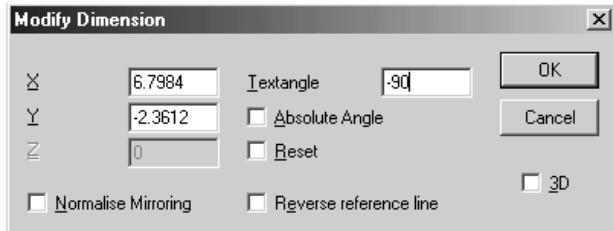
Here is a special **TIP** on how to easily **Modify** a dimension.

If you want to modify a dimension simply select the dimension and RIGHT-click your mouse. Then select the MODIFY function. See below.

In this example we have a vertical dimension that we want to change so that it is more readable on a print.

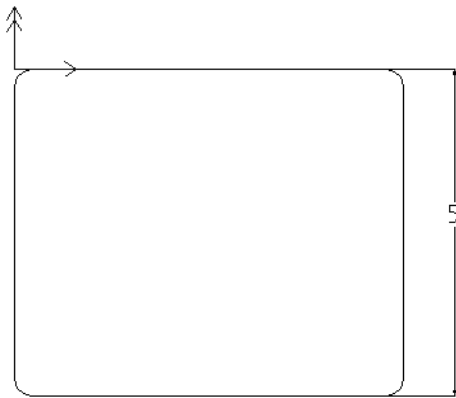


Simply click on the dimension to select it. Then RIGHT-click your mouse to bring up the modify menu. By selecting MODIFY you can see the new dialog below.



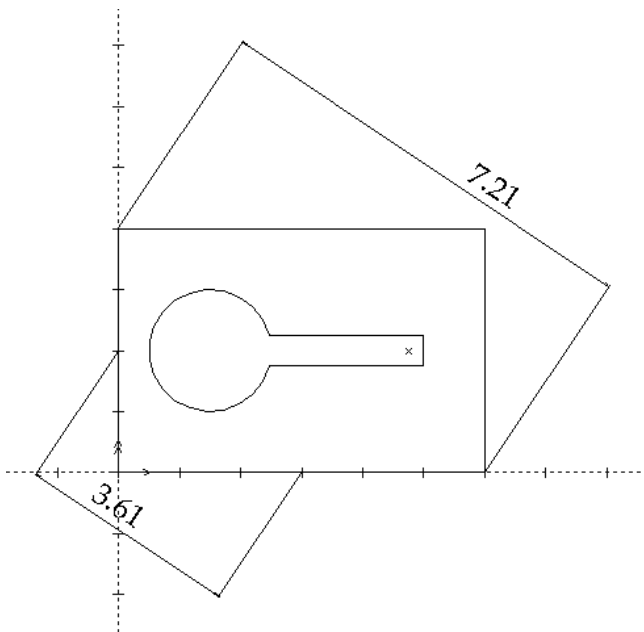
By entering -90 degrees next to TEXT ANGLE (see above image), and then clicking OK you will automatically modify the dimension to suit your needs. See this result at the top of the next page.

Result:



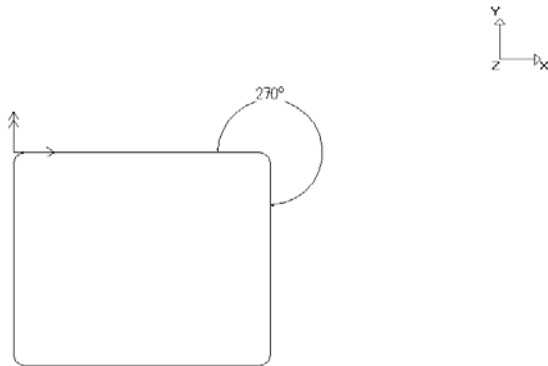
Section 45. Other Distance

OTHER DISTANCE: This dimension function allows you to add dimensions to your drawings by selecting one entity and then another. This is a repetitive function meaning that you can continue selecting entities to dimension. Click your RIGHT mouse button to exit this function when you are finished.



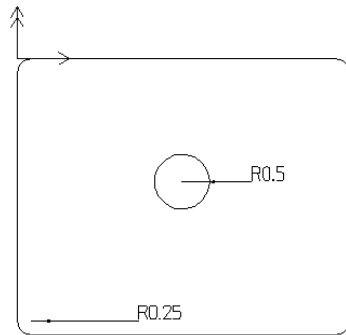
Section 46. Angle

ANGLE: The **Angle** dimension function allows you to click one entity of the angle and then the next entity of the angle to produce this specific dimension. See the example below.



Section 47. Radius

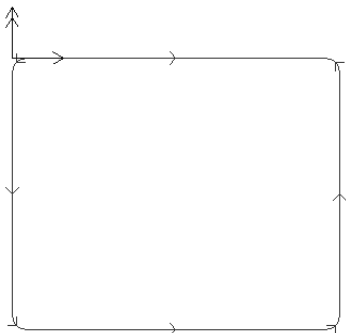
RADIUS: This dimension function will allow you to add dimensions to circles or parts of circles including fillet radii. Simply click the function and then click on a radius. See below.



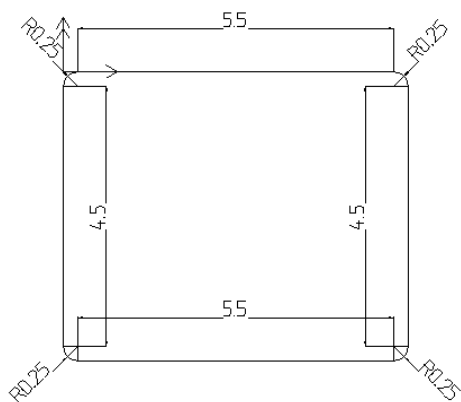
Section 48. Auto Dimension

AUTO DIMENSION: This **Dimension** function allows you to select an entire drawing first and then dimension the entire drawing at once. See below.

Example A. Selected drawing.

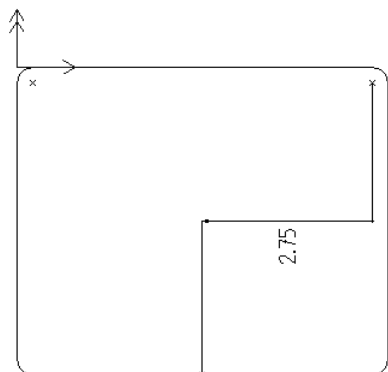


Example B. The fully dimensioned drawing.



Section 49. X Ordinate

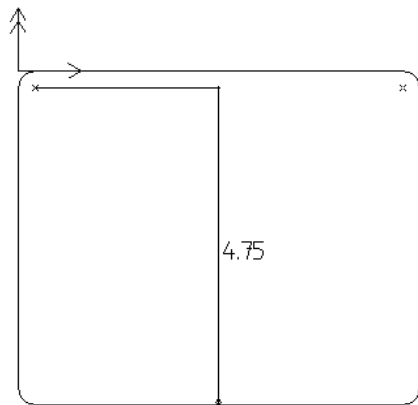
X ORDINATE: This function will allow you to add an ordinate dimension in the X-axis. This would be the distance between a selected entity to another in the X-plane. See below.



This is a good function for finding the location of points and other geometry.

Section 50. Y Ordinate

Y-ORDINATE: This allows you to add an **Ordinate** dimension in the Y-axis. This would be the distance between a selected entity to another in the X-plane. See below.



Section 51. Move/Align

MOVE/ALIGN: This dimension function has a special sub-menu. Let's take a look at this below.



MOVE: This function allows you to move a PRE-SELECTED or NON-PRESELECTED entity after you have added it to a drawing.

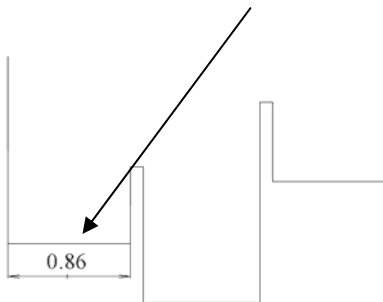
ALIGN HORIZONTAL: This function allows you to align dimensions in a drawing so that they are horizontal.

ALIGN VERTICAL: This allows you to align dimensions in a drawing so that they are all vertical.

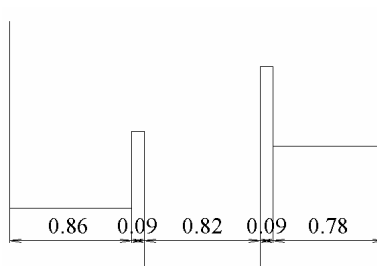
Section 52. Chain

CHAIN: This allows you to draw a chain of dimensions in a drawing. You must ensure that a dimension is already selected in the drawing (however not radius dimensions). This first dimension selected acts as a template for the new dimensions. See below.

The selected Dimension.



A **CHAIN** of dimensions.

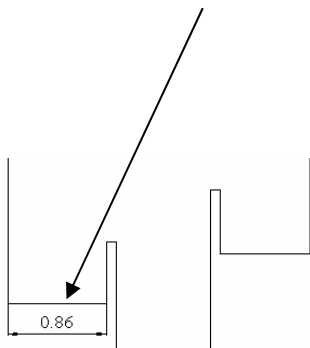


As you can see, the dimension chain is aligned.

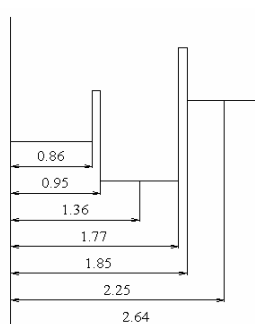
Section 53. Reference

REFERENCE: This function generates dimensions in relation to a reference line in a simple way. You must ensure that a dimension is already selected in the drawing (however not radius dimensions). If there are no dimensions present then you must first use Dimensions: Auto to make a dimension. See this example below.

The pre-selected dimension.

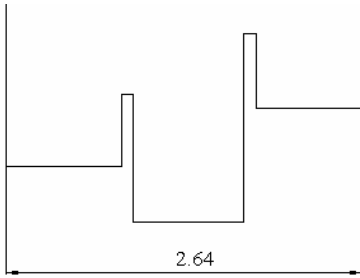


New referenced dimensions.



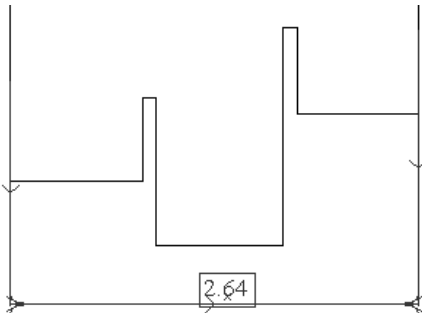
Section 54. To Lines + Text

TO LINES + TEXT: This function will allow you to break apart a dimension (text, lines & leader arrows). You may modify the text/numeral dimension value or delete it after this function easily. Let's take a look below.

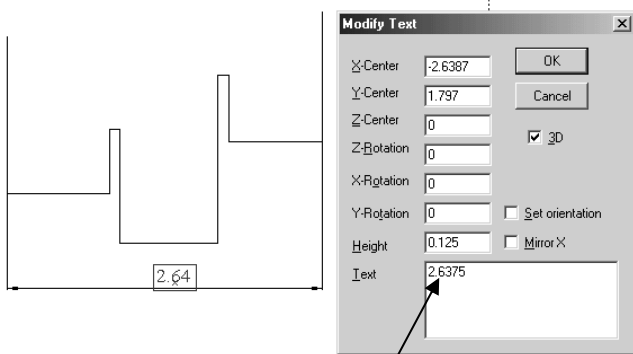


Here we have a horizontal dimension for the two outer lines. The dimension settings that we have in the software round off the number to the higher and we want to change this. Change the actual dimension itself.

We must select the dimension first and then select this function. You will see that the dimension is now inside a text box. See below.

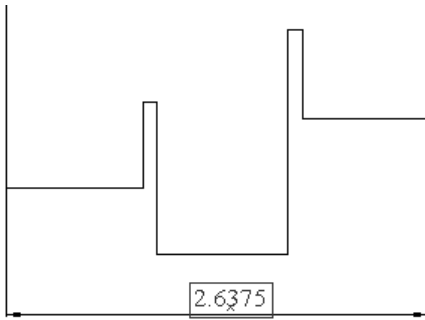


The text, dimension lines and arrows have all been broken into editable entities. Now, by clicking off the entities and then clicking only on the text, we can edit this text by RIGHT-clicking our mouse and selecting MODIFY.



We changed the text to 2.6375 in the text box and then selected OK. You can also change the size of the dimension, its location and angle.

See this result below.

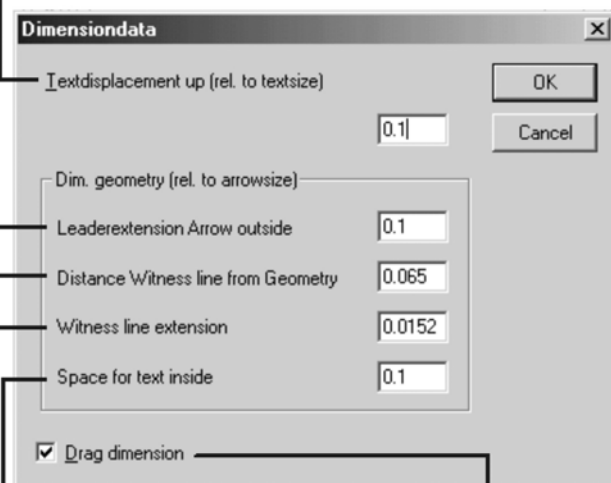


Section 55. Setup

SETUP: This function allows you to **Setup** the basic attributes of your dimensions. Let's take a closer look at this dialog.

This variable will offset the leader extension arrow.

This variable will offset the dimension text from its dimension line.

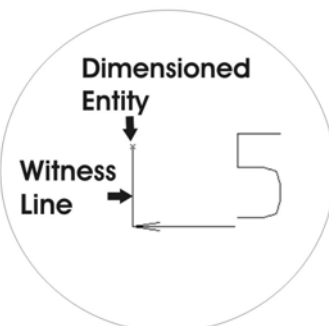


This option allows you to drag the dimension to the location you want. Then by clicking your mouse the dimension will lock into place.

This variable will provide more or less space for the dimension text so that the dimension line will not cross over the dimension text itself.

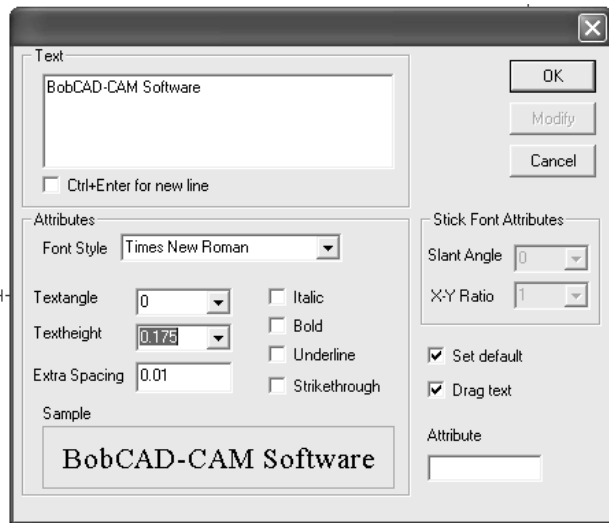
This variable will extend the witness line beyond the leader arrow.

This moves the dimension witness line closer or further away from the entity that is being dimensioned. The witness line is the line that starts at the entity itself and connects to the leader arrow.



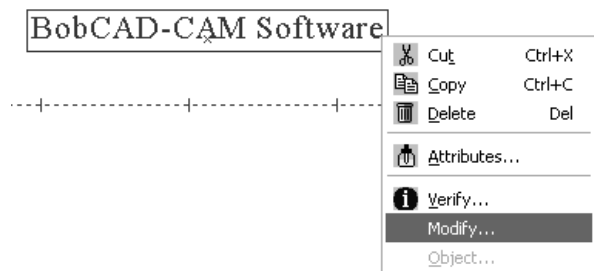
Section 56. Text

TEXT: The TEXT function and its box are located in the Dimensions menu. Here is the **Text** dialog box below.



The text dialog offers several options. They are:

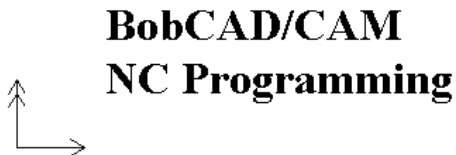
- **ITALIC**, **Underline**, **BOLD** and **Strikethrough**. These are different ways of writing text.
- Font Style allows you to pick and choose from all of the styles of font that exist on your computer.
- TEXT ANGLE refers to the angle of the text on the positive X-Axis.
- TEXT HEIGHT refers to how large the text will be.
- Extra Font Spacing.
- Stick Font Attributes. This allows you to apply a slant angle as well as an X-Y ration for changing the way the 9 available stick fonts look on the CAD screen.
- The SET DEFAULT allows you to save the settings that you have as a default. This way the next time you open the text box, your settings will be there.
- By selecting the DRAG TEXT option, you will be able to click OK and drag the text freely on the drawing screen for positioning. When satisfied with its position you would click OK.
- Control+Enter for NEW Line gives you the ability to select the CONTROL key and then the ENTER key on your keyboard to type in text on the next line down. See below.
- DRAG TEXT allows you to select OK in the text box and then drag the text to a specific location on the screen. Then, click your mouse button again to drop the text at that location.
- The MODIFY text button allows you to change pre-selected text on the screen. To use this button, click on the pre-vectorized fonts on your cad screen. You need to do this before breaking up the fonts. When selecting the fonts by clicking on them with your mouse, you will see them become selected and grouped. Simply go back to the Dimensions menu again and select text. Make your change and click on the MODIFY button to make the change. You can also click on the pre-vectorized fonts and then RIGHT-Click your mouse to access the Modify Text box from the pop-up menu.



Section 57. Vectorize Text

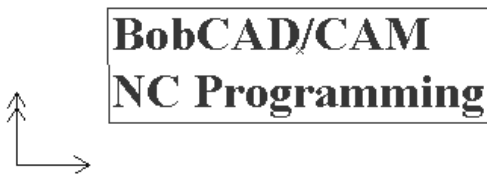
VECTORIZE TEXT: This function will give you the ability to turn True Type Fonts into entities that can be cut with V21 machine operations. First you would select the font type and attributes, then type in the text that you want into the text box and select OK. Select the text and then select the **Vectorize** function to bring up the dialog. See below.

Example A. The True Type Fonts.



BobCAD/CAM
NC Programming

Example B. Selecting the text.



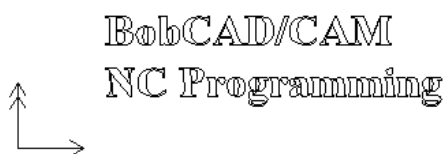
BobCAD/CAM
NC Programming

Example C. Vectorizing or smashing the text.



The accuracy is how small you want each entity to be that will make up the text once smashed or vectorized. Then you would simply select OK.

Example D. The machinable text.



BobCAD/CAM
NC Programming

The text is now made up of many entities that can be machined in the CAM.



Section 58. Fit Text to Path

FIT TEXT TO PATH: This function will allow you to take UN-VECTORIZED text and fit it to a curve. See below.

Example A. The true type fonts and the curve.



CAD/CAM

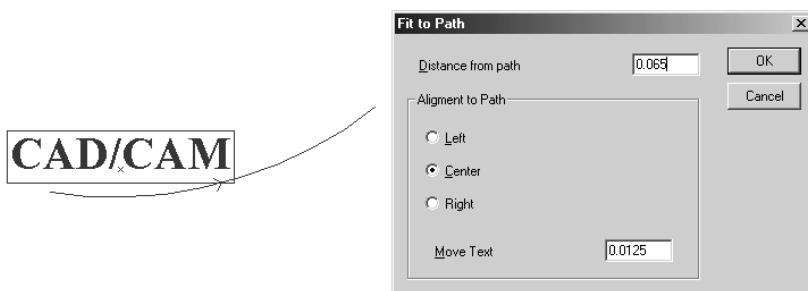
Example B. The selected text and curve.



CAD/CAM

Notice the directional arrow of the selected curve is pointing toward the RIGHT.

Example C. The FIT TEXT TO PATH DIALOG.



You can set the:

- Distance to Path: This is the distance away from the path that you want the text to be after you select OK.
- Alignment to Path: This is where you want the text to be positioned on the curve.
- Move Text: This displaces the text along the curve a specified amount.
- Once you are satisfied with the text positioning select OK.

Example D. The text along the curve.



CAD/CAM

There are some things to know about this function:

- 1) **True type fonts do not scale very well.** If you scale the text and it doesn't follow the curve anymore you can repeat this function as follows: Pick the text that is now along the curve. Next pick the curve and use function **Dimensions: Fit Text to Path** again.
- 2) **To prepare text for engraving use function Dimensions: Vectorize Text.** Once the text has been vectorized it consists of many small entities (Lines and Arcs) which now can be machined. Vectorized text scales normally.

Section 59. Tips on Text Along a Curve

TIP ON TEXT ALONG A CURVE: The direction of the selection arrow determines which side the text will be translated to. Here is an example below.

Example A. The direction arrow of the curve is going towards the RIGHT.



Example B. The result.



Example C. The direction arrow of the curve is going towards the LEFT.



Example D. The result.



This tip is important to remember when using this function.

Section 60. Single Line Fonts & Modifying Text

BobCAD-CAM Version 21 offers font selections that include new **Single-Line Font Styles**.

The “Single-Line” **Fonts**:

Font Name:

Font Preview:

- AFNT01 *The Quick Brown Fox*
- AFNT02 *The Quick Brown Fox*
- AFNT03 *The Quick Brown Fox*
- AFNT04 *The Quick Brown Fox*
- AFNT05 *The Quick Brown Fox*
- AFNT06 *The Quick Brown Fox*
- AFNT07 *The Quick Brown Fox*
- AFNT08 *THE QUICK BROWN FOX*
- AFNT09 *The Quick Brown Fox*

The **Stick Font Attributes** examples:

0 Slant
1 X-Y Ratio

Manufacturing

20 Degree Slant
2 X-Y Ratio

Manufacturing

The default attribute for the X-Y Ratio is 1.

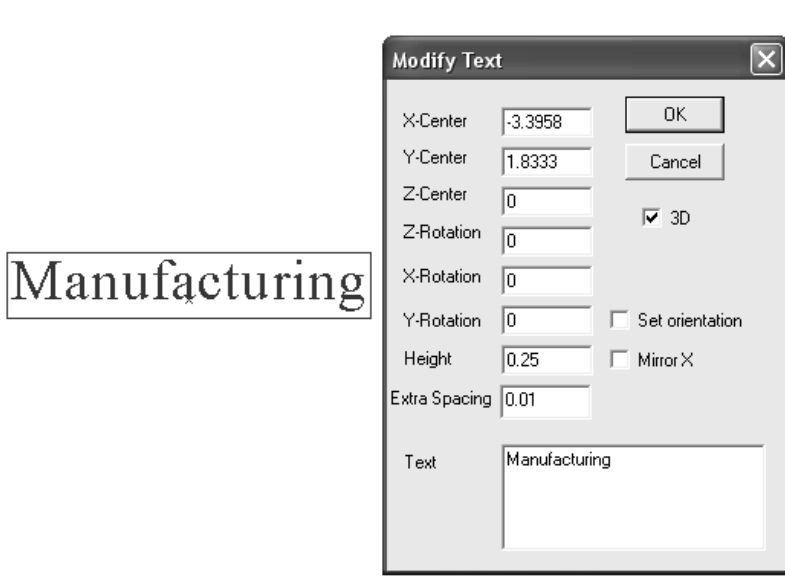
The procedure for writing Text:

1. Go to the Dimensions menu and select TEXT.
2. In the text box, write the text that you want to use and select a font style as well as assign any attributes to the text. If you are using any of the single-line fonts you can adjust the slant value or the X-Y ratio.
3. Un-check the DRAG option if you want to position the text at X, Y and Z 0. Otherwise you can check the drag option and position the text manually after you select OK.
4. When finished, click OK to position the text.
5. Once the text is on the CAD screen you will need to place your cursor on the text and click on it to select it.
6. Now go back to the Dimensions menu and select Vectorize Text to open the Smash Text box. This allows you to enter an accuracy value. The reason for this is that you cannot machine a true type font. The fonts must be broken into line and arc segments so that you can machine them. By clicking OK in the Smash Text box the fonts will be broken automatically. The accuracy value sets the length of each entity after the text is broken.
7. Now your text is ready to be machined.

MORE ON MODIFYING TEXT:

The procedure for modifying un-vectorized/smashed text: After you have placed your **Text** on the screen you can easily make modifications to the text. However, once you vectorized the text you must start over by deleting the text and going back to the text box for your changes.

1. Select the text.
2. Right-click your mouse and the command pop-up box will appear.
3. Select MODIFY.
4. Make the changes that you need to make. I.e. changing the text, repositioning the text at a different X, Y and/or Z location, changing the text size and rotation or even mirroring the text so that it is reversed. You can also add extra spacing between the letters.
5. Once you have made the desired changes, click OK.



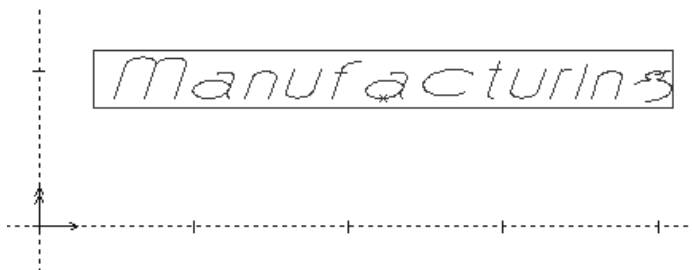
Another way to modify existing, un-vectorized text:

You can also change or modify text by selecting the un-vectorized text and going back to the Dimensions menu and select the TEXT function again. Any changes that you make will go into effect when you click OK in the box.

Example 1

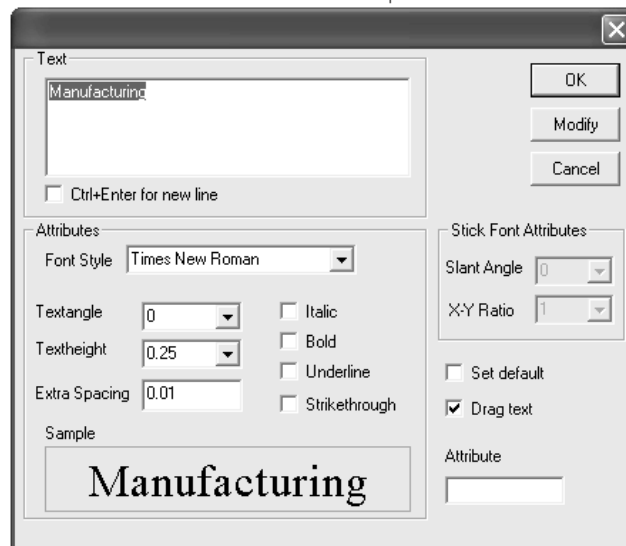
Here we have existing text that has not been vectorized.

1. First you must select it by clicking on it.

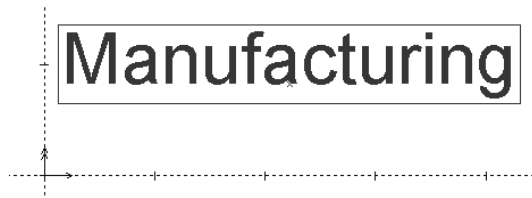


2. Go to the Dimensions menu and select TEXT.

Manufacturing



We changed the font style, text height and then selected the **Modify** button and the text was automatically changed. If you click OK instead of “Modify” you will have a second body of text that you can place on the screen. Therefore, simply click the “Modify” button.



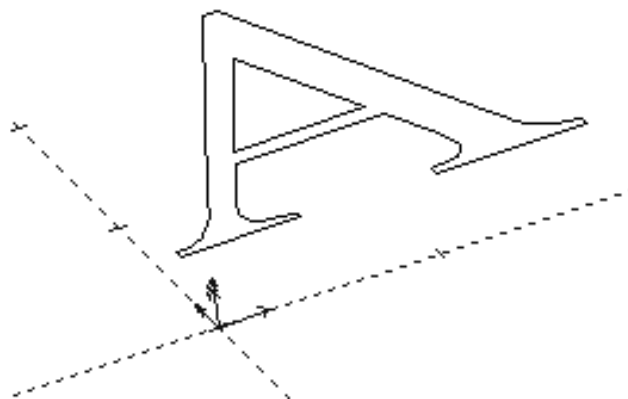
When we changed from a stick font style to a true type font style the Slant and X-Y Ratio options become unavailable for use.

Section 61. Creating Solid Fonts

CREATING SOLID FONTS:

Version 21 offers effective ways of creating solid text that can then be machined or used with Boolean operations to create various parts for solid toolpath. Let’s take a look at this.

Here we have a vectorized letter A in a 3D view.



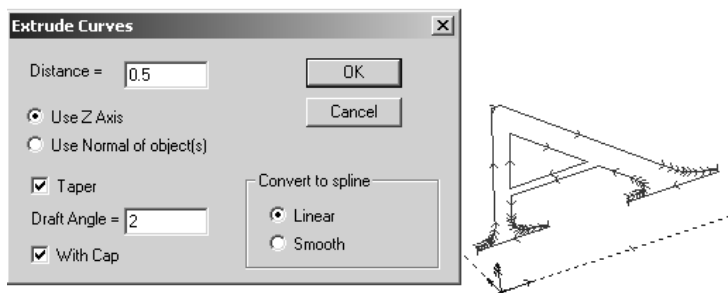
If you want to turn this into a solid font you need to:

STEP 1

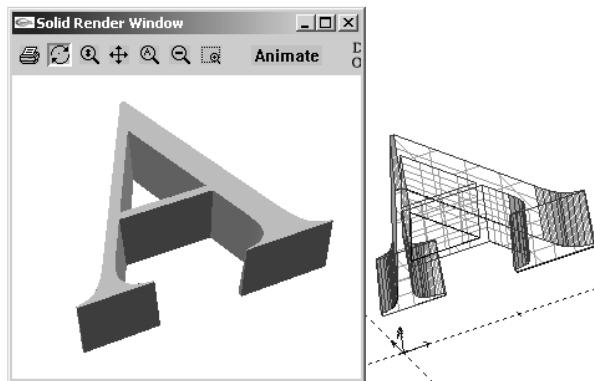
Go to the Solids menu, select Surface Creation and the Extrude Curve option. Select the NEXT button and perform a chain selection starting with the OUTSIDE contour of the letter. Then perform a chain selection operation on the INSIDE contour if there is one. The letter A has the inside chain. When finished with the last chain selection, click your RIGHT mouse button one more time to open the Extrude Curve box.

STEP 2

The Extrude Curve box will offer you settings for creating a tapered angle, Z height and more. Make your adjustments and click OK to perform the extrusion.



Typically you will use the “Use Z Axis” option unless the letter is rotated in a different plane. The result after clicking OK:

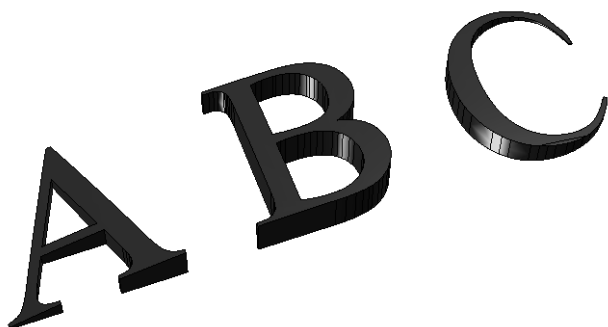


You will need to do one letter at a time.

MORE ON SOLID FONTS:

There are different effects that you can get when using **Solid Fonts** in conjunction with the Boolean operations that are available (Union, Difference & Intersect). Here is an example of solid fonts that were intersected with a sphere and then the Boolean “**Intersect**” function was used.

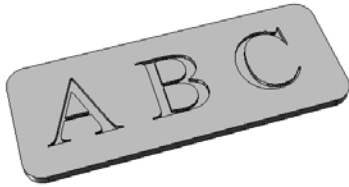
3D View:



Side View:



Here are the same fonts that were lowered to intersect a rectangular solid and then the Boolean “***Difference***” operation used by selecting the rectangular solid first and then a font and so on to get the end result.



Once you have created the solid that you want, you can then create the toolpath and complete g-code program easily.

Section 62. Working with Text – Training Lesson

Here’s a quick tutorial to walk you through an example of working with a solid font, performing a Boolean operation and then creating toolpath for it.

STEP 1

Start off with a NEW drawing screen. First close out any CAD screens that you may have open.

STEP 2

Go to the Dimensions menu and select TEXT.

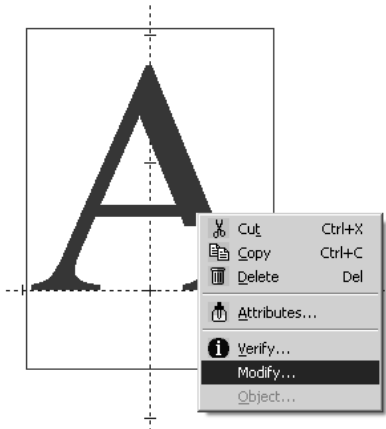
STEP 3

Select the font style “Times Roman”, type in the letter **A** and enter a Height of 2.

STEP 4

Uncheck the DRAG option if it selected and check the Set Default option.

Now select OK. The font will be dropped with its origin point at X 0, Y0 and Z0. Now click on the font to select it if it is not already selected. With the font selected and your cursor hovering over the letter, RIGHT-click your mouse and select the MODIFY function from the pop-up command menu.



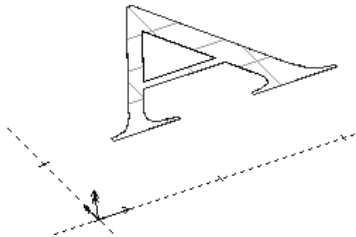
In the MODIFY box enter X 1.5 and Y 1 and click OK to reposition the font.

STEP 5

Now with the font still selected, go to the Dimensions menu and select Vectorize Text. Just click OK when the accuracy box appears. The default value will most likely be .00004. That is fine. After clicking OK the text will be broken into small line and arc segments.

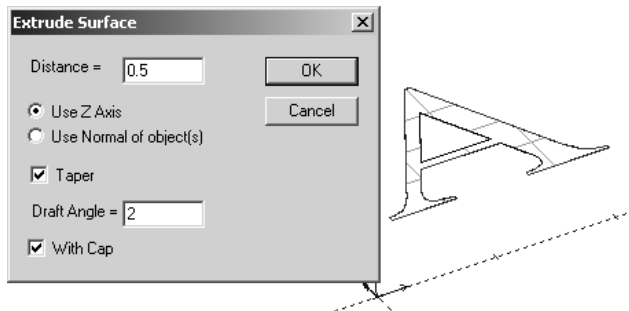
STEP 6

Go to the 3D menu and **TURN 3D ON**. Now, with the letter **SELECTED** (*make sure the text is fully selected*), go directly to the Solids menu, choose Surface Creation, the Planar option and click NEXT. This will automatically create the surface.

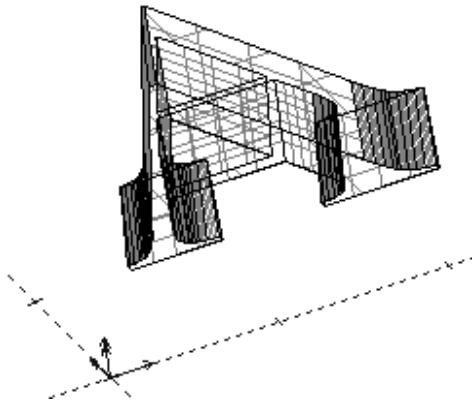


STEP 7

Now with the surface selected (make sure the surface is selected), go to the Solids menu, choose Surface Creation, Extrude Surface and click NEXT to open the Extrude Surface box.

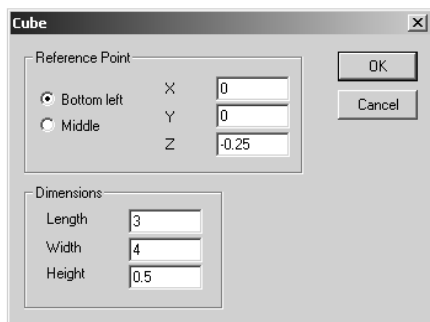


Enter a distance of .5, select the “Use Z Axis” option, select the “Taper” option and enter a value of 2 degrees for the taper. Now select the “With Caps” option and click OK to create the solid font.



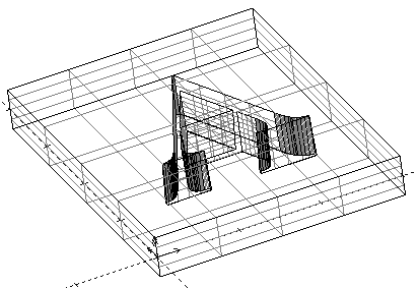
STEP 8

OK, let's create a second solid so that we can perform a Boolean operation. Go to the Solids menu and choose PRIMITIVE. Select the CUBE option and click OK to open the cube box.



Select Bottom Left as the reference point.

- X should be 0, Y should be 0 and Z should be -.25.
- Enter a length of 3.
- Enter a Width of 4.
- Enter a Height of .5.
- Select OK to draw the cube.



The cube may appear selected. If so, this is because you have the Auto Pre-select option turned on in the Environment settings. If so, that is fine. Place your cursor in an empty area of the drawing screen and click your left mouse button one time to deselect everything.

STEP 9

Now, place your cursor on an edge line of the cube and click on it to select it. Hold down your CONTROL key and click on an edge line of the letter A to select it as well.

STEP 10

With both solids selected, go to the Solids menu and choose **Booleans**. When the Boolean box appears, select UNION and click OK to unionize the two solids into one.

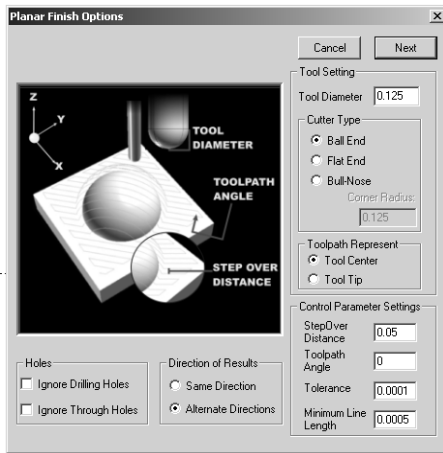
STEP 11

Go to the **Special/NC CAM** menu and select INSERT NC. In the new box, click on the post processor called, "FANUC 6M" and then click the OK button to open the CAM side. On the CAM wall you will see a toolbar. Toward the bottom there will be a button that has 3D on it. Click on this button to turn the 3D machining mode on.



STEP 12

Now, with the solid selected (if the solid is not selected, select it by clicking on it), go to the Solids menu and choose GENERATE TOOLPATH. This function should be active if the solid is selected. If not, deselect the solid and re-select it, then go back to the Solids menu and select Generate Toolpath. Select the Planar option and click NEXT to open the planar toolpath Wizard.

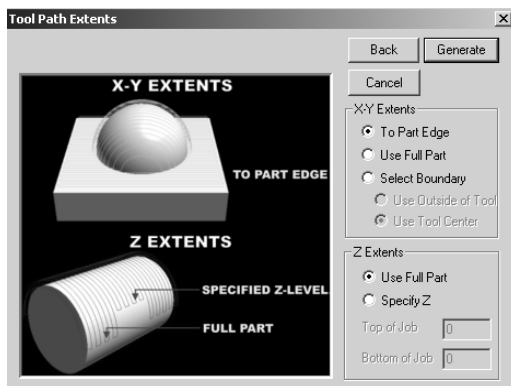


Enter the following:

- .125 for the Tool Diameter.
- Select the Ball End cutter.
- Select the Tool Center option.
- .05 as the Step Distance. (Step over)
- 0 for the Toolpath Angle.
- .001 for the Tolerance.
- .001 for the Minimum Line Length.
- Do not check either IGNORE Hole options.
- Select Alternate Directions.

Now click NEXT.

The Toolpath Extents box will appear.

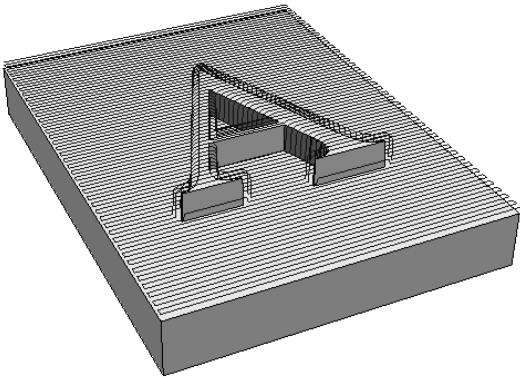


- Select the “To Part Edge” option under X-Y Extents.
- Select “Use Full Part” under Z extents.
- Select the Generate Button.

STEP 13

When the toolpath is finished the Solid Tool Depth Settings box will appear. This new box offers roughing and finishing capabilities. You created finishing toolpath in this example lesson. Therefore, you will not be roughing. Rapid Plane is basically your clearance for rapid moves and can be set to .25. The Material Top should be set to 0. Cutting Depth is really not important for this finishing toolpath as the NC Code generated is based off of the toolpath at this point. Therefore it can be left at 0 as well. If the ENABLE button is checked, un-check it as we are not roughing. Now click OK to generate the NC program in the editor.

The result showing toolpath:



The To Part Edge option keeps the toolpath from going over the edges of the solid. However, if you needed to prevent the edge of your cutting tool from crossing the edge boundary you would need to use the boundary machining option with the “Use Outside of Tool” option selected. If you didn’t have a boundary for containing the tool you could extract the edge lines of the block or simply draw your boundary.

CHAPTER 9

THE SPECIAL/NC CAM MENU

Section 63. The Special/NC CAM Menu, Verify & Close

Section 64. Generate NC & Insert NC

Section 65. Axis

Section 66. Selection

Section 67. Script & Record Script



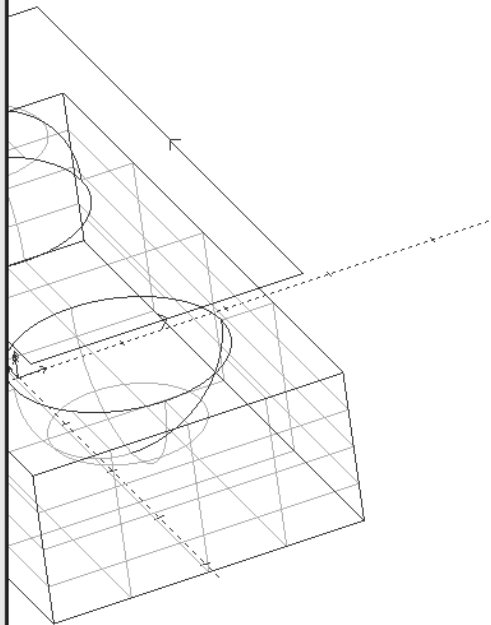
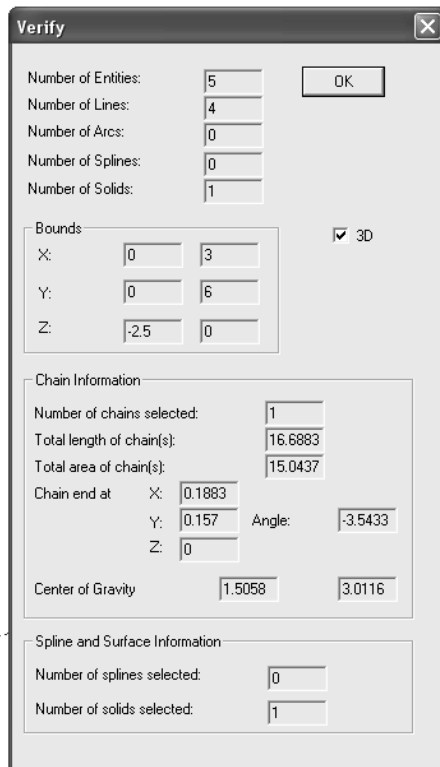
Section 63. The Special/NC-CAM Menu, Verify & Close

This chapter is dedicated to teaching you how to use the functions of the SPECIAL/NC-CAM menu. You can see the **Special/NC-CAM** Menu image below.



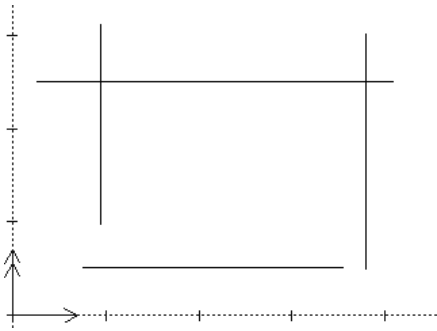
As you can see in the menu image above, the special menu has functions listed that also have their own sub-menus. Let's take a look at the VERIFY and CLOSE functions in this section.

VERIFY: This function allows you the ability to verify 2D or 3D geometry on the screen as well as solids and surfaces. Other information can be gathered regarding any wireframe geometry chains being verified. See below.



You can select the geometry on the screen first and then select this function to retain all of the coordinate information regarding the geometry. The VERIFY function is the first function located in the SPECIAL menu.

CLOSE: You can use this function to complete the closing of a chain. Let's take a closer look at this below.

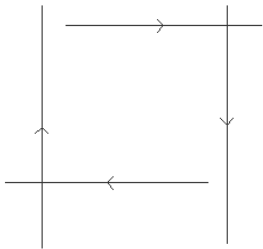


In the last image, you can see that there are four lines that are sketched on the screen. They are not connected properly to form a chain.

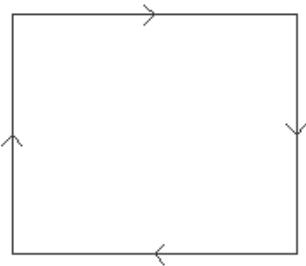
Select this function from the **SPECIAL/NC-CAM** menu and then select the **CHANGE** menu and select Trim/Extend and **TRIM STRINGS**. You will automatically trim the entities into a chain.

NOTE: This will only work if the entities have been selected properly. This means that the directional arrows in selection mode must be all going the same direction. If you have a question, you can select the **REORGANIZE ELEMENTS** icon to redirect the selection arrows so that they are all going in the correct direction. This function is also located in the **CHANGE** menu.

In the next image, you can see 4 lines that are not connected as a chain. By selecting the **CLOSE** function and then going to the **CHANGE** menu and selecting **TRIM+EXTEND** and **STRINGS**, you will automatically trim all of the lines together to form a perfect chain.



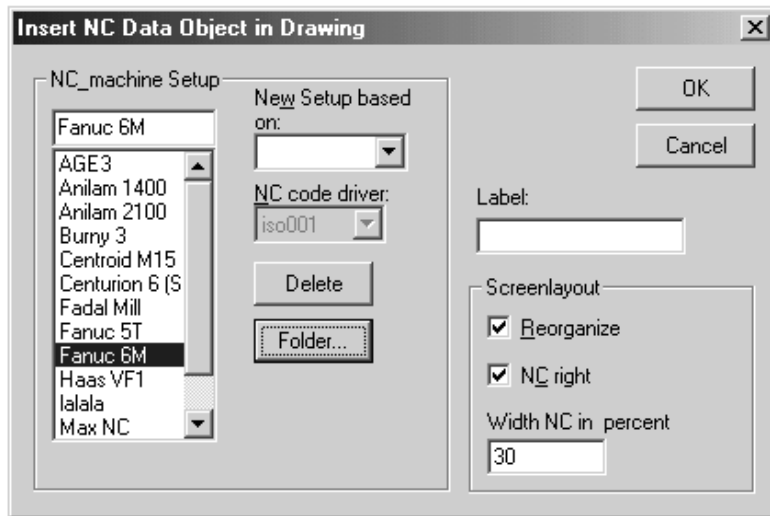
Below, you can see the result of using the close function.



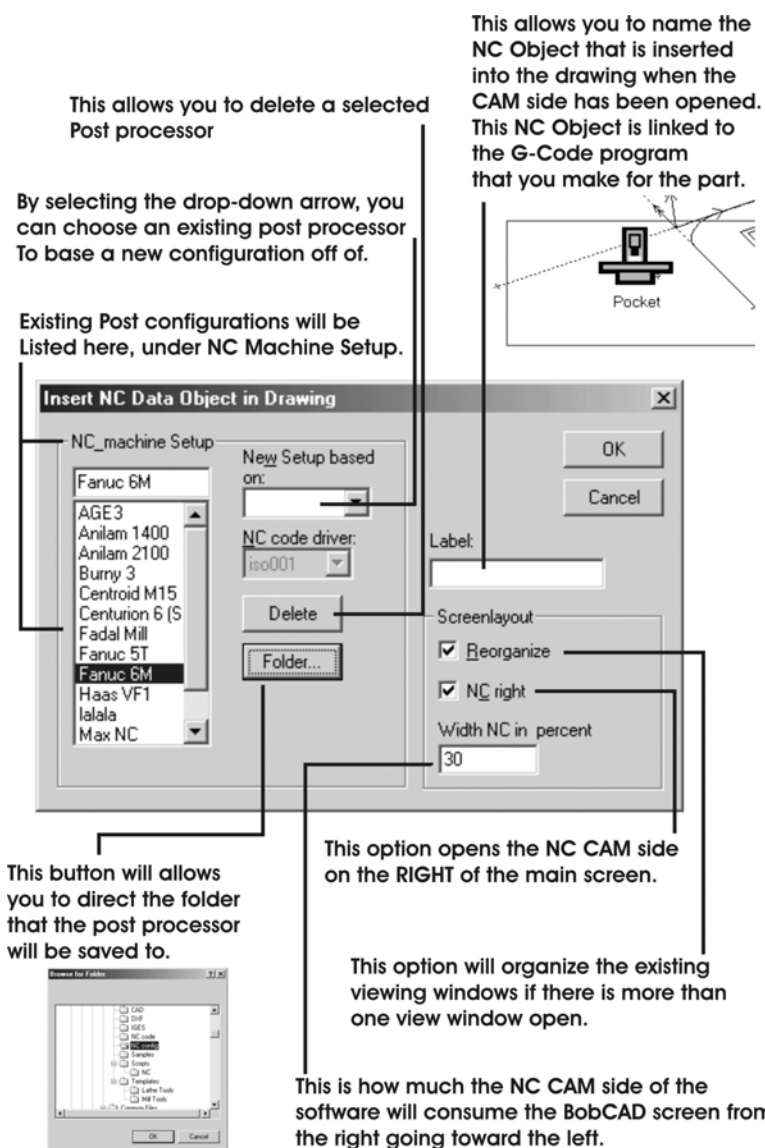
Section 64. Generate NC & INSERT NC

GENERATE NC: This function is used after the NC CAM has been opened and the cutting options and setting have been entered. After that, you can go back to the special menu and select this function as it will be available and will automatically generate the entire G-Code program for you. When we get into the CAM side of the software you will see how this function is used for fast generation of NC Code.

INSERT NC: By selecting this function you will open the INSERT NC dialog allowing you to make CAM related selections as well as direct post processor configurations to specific folders in the main BobCAD Version 21 directory. Let's take a closer look at this dialog below.



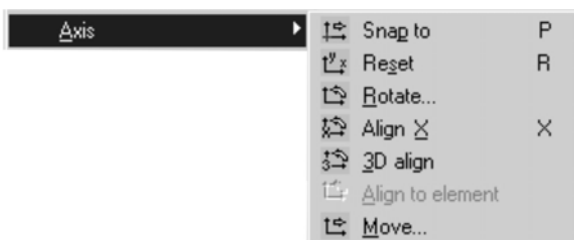
The first aspect of this dialog you will notice is that you can select various post processors here. You will be able to base new post processors off of ones that are already listed in this dialog. Let's take an even closer look at what each of these settings and options are, starting below.



By selecting OK, the NC CAM side will open on the screen.

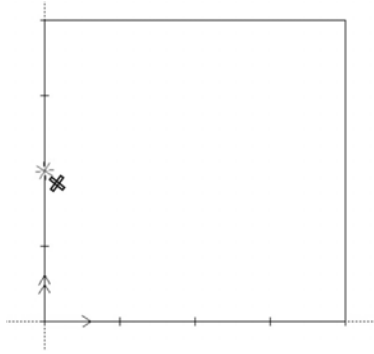
Section 65. AXIS

AXIS: This menu item offers a special sub-menu that gives you functions that deal with the UCS. Let's take a closer look at this sub-menu.

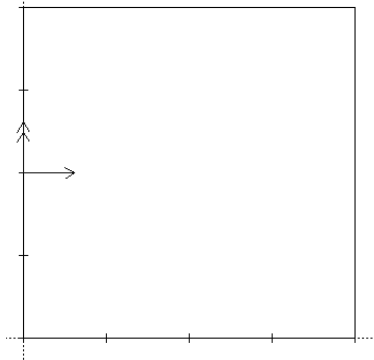


- **SNAP TO:** This allows you to snap the UCS to a specific location on your drawing. Let's take a closer look at this.

Below, we have a square and have selected the “Snap To” function. The star represents one of the snap locations on our geometry.

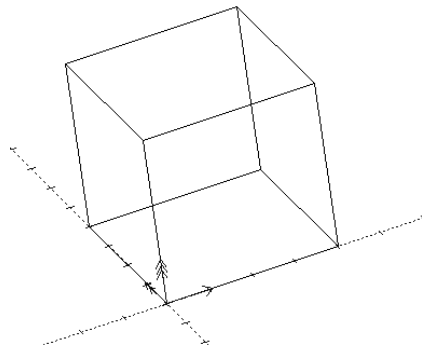


By clicking your mouse at this location, the UCS is automatically moved to its new location. Now the new location represents the ZERO position on the screen.

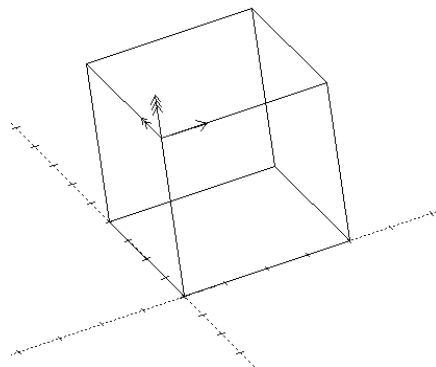


Let's take another look at this in 3D mode.

In the image below, you can see that the top of the cube is 4-inches raised in the Z-Axis.

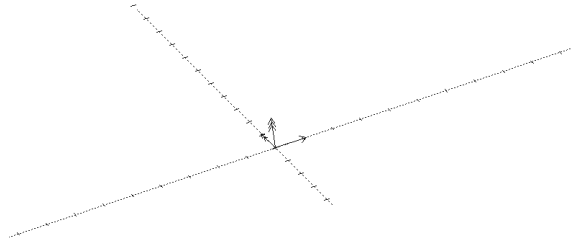


By using the SNAP TO function we were able to move the UCS to the upper corner of the cube. By doing this, we changed the coordinate system to make this corner the new Z-Zero.

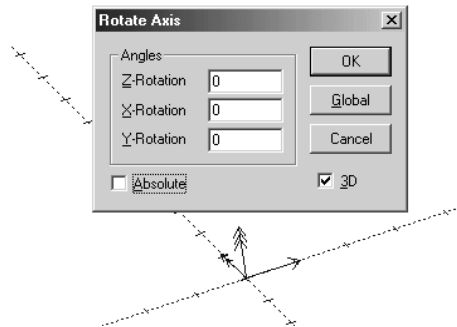


NOTE: To *RESET* the UCS, you would hover your cursor over the UCS, *RIGHT-click* your mouse and select *RESET*.

- **RESET:** If you have changed the UCS, you can select the RESET function to bring the UCS back to its native home position on the grid axis. The DEFAULT home position of the UCS is at the native Z-Zero location at the center of the grid axis as seen in the next image.

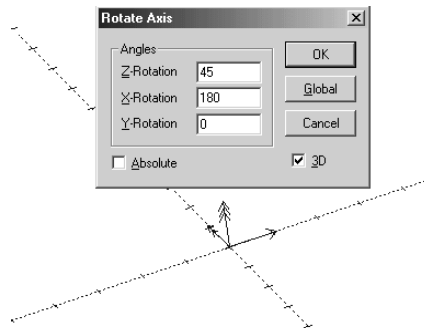


- **ROTATE THE UCS:** This allows you the ability to rotate the UCS around at its anchored location or at a designated coordinate location on the drawing screen. See this image below.

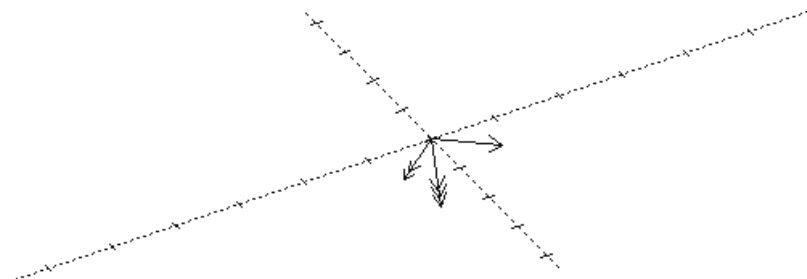


By selecting the 3D box, you can rotate in all 3-Axis at once as needed.

In the next image, you can see that we entered 45-degrees for the Z and 180-degrees for the X-Axis rotation.

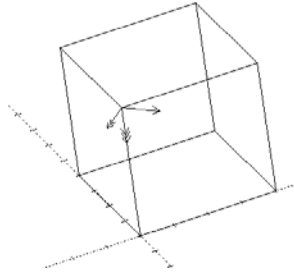


By selecting OK you can see the UCS has been changed below.

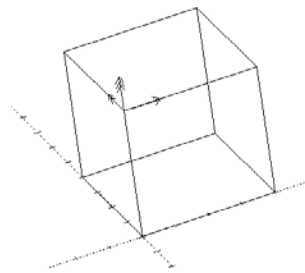


If you have snapped and applied a rotation to the UCS as seen in the image below, you can select the function and **GLOBAL** to reset the UCS at its snapped location.

The UCS snapped and rotated.

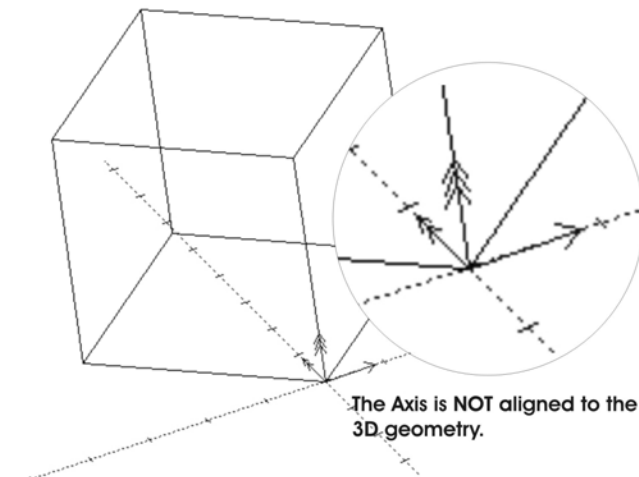


The UCS after GLOBAL was selected.

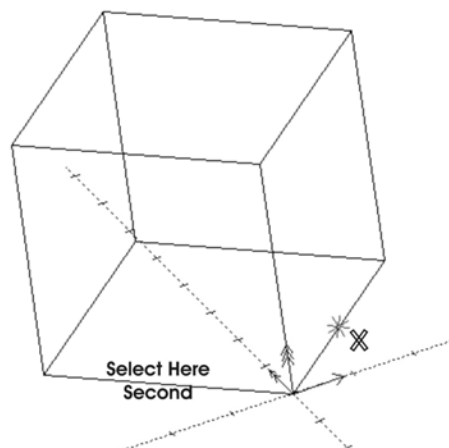


Selecting RESET would reposition the UCS at its native home-default location. **REMEMBER, the coordinate system changes when the UCS is changed.**

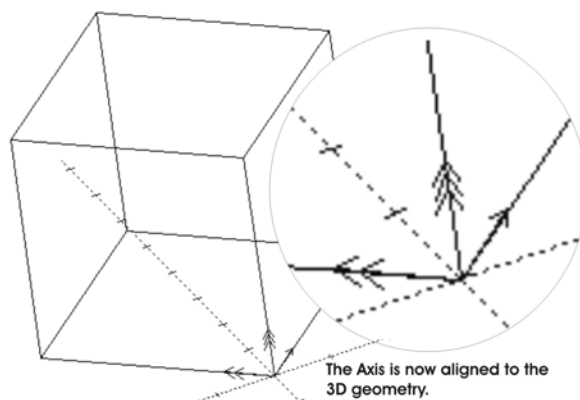
- **ALIGN THE UCS IN X-AXIS:** This function allows you to designate an X-Axis location to automatically change and align the UCS to. Once you select this function, you will have snap locations on the drawing. When your cursor comes near one of the designated snap locations, you will be able to click your mouse and automatically change the UCS to align to a new X-Axis.
- **3D ALIGN:** This allows you the ability to align the UCS in 3D. In the image below, you can see that the 3D geometry is not aligned to the UCS.



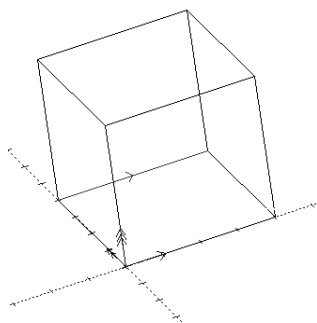
By selecting the available snap points as you can see in the image below, you can automatically align the 3D geometry.



Below is the result of using the **ALIGN** function.

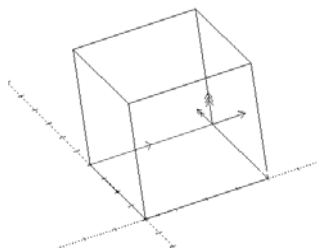


- **ALIGN TO AN ELEMENT-ENTITY:** This function allows you the ability to snap (align) the UCS to the end of a **PRE-SELECTED** entity. You must select an entity first before being able to activate and use this function. The direction arrow will determine which end of the entity the UCS is aligned and snapped to.

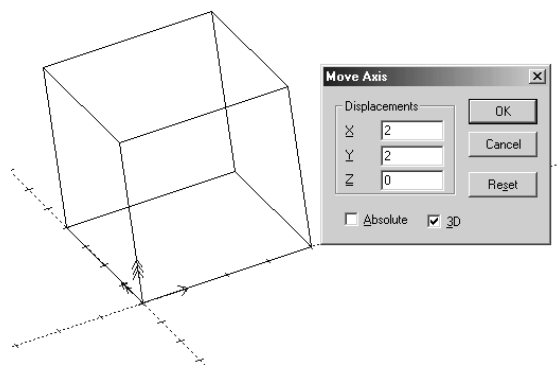


You can see in the image above that the back line of the bottom square has been selected.

By selecting this function you can see that the UCS was automatically aligned to the end of the selected line. See below.

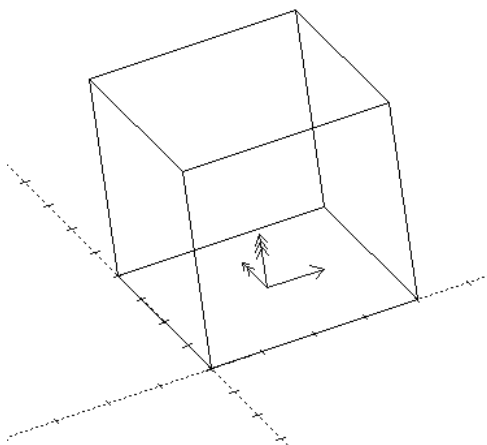


- **MOVE UCS:** When you select this function, you will open the MOVE AXIS dialog. See below.



In the last image, we have entered 2 for X and 2 for Y. By selecting OK be repositioned the UCS to its new location.

Below you can see the new UCS location. You can use X, Y and Z coordinate values to do this. The ABSOLUTE option pertains to the default ZERO position on the screen. Un-check the Absolute option and the software looks at the move as an INCREMENTAL move altogether.

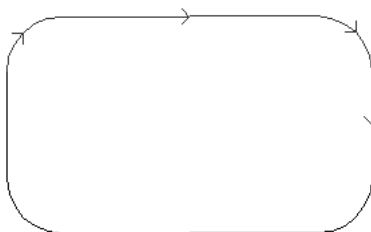


Section 66. Selection

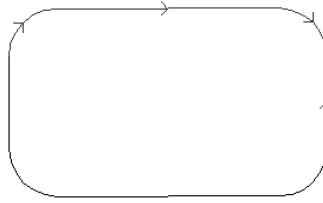
SELECTION: This Special Menu item offers a sub-menu of its own. This menu includes several specific function that deal directly with selecting geometry or selecting multiple shapes on the screen. Lets take a look at how each of the sub-menu functions work.



- **UNSELECT ALL:** This function will automatically deselect all selected entities on the screen. The other way to deselect all is to move your cursor off of the drawing and click your left mouse button to deselect everything that is selected.
- **SELECT ALL:** This function selects all entities. Chains are selected properly as well. The other way to select all is to click and drag a selection box across the screen with your cursor. You see, you are always in selection mode in BobCAD Version 18. In Chapter TWO, we discussed how to select entities in the software.
- **FLIP LAST:** This function allows you to change the direction of the last directional arrow in a selected chain automatically. See below.

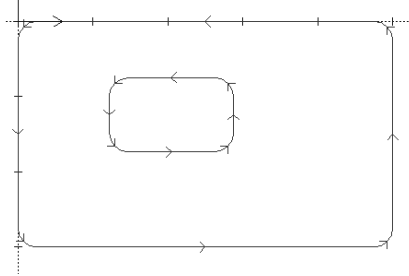


In this image you can see that there is a chain of entities selected. The first entity of the chain is on the far left and the chain is selected going toward the right. By using the FLIP LAST function you can change the direction of the last selected entity on the chain.

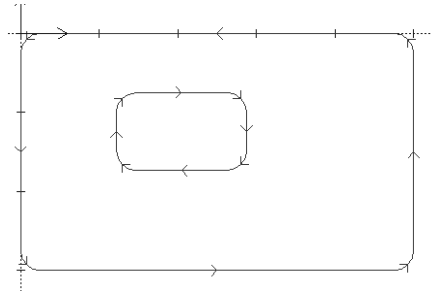


- **REVERSE:** This function will change the direction of the selection arrows automatically for you.
- **REVERSE EVERY SECOND:** This function will allow you to change the direction of every second selected entity or chain of entities. See below.

In the image below, you can see that all directional arrows are going in the same direction. You would have trouble performing the spiral pocketing function with the arrows like this.

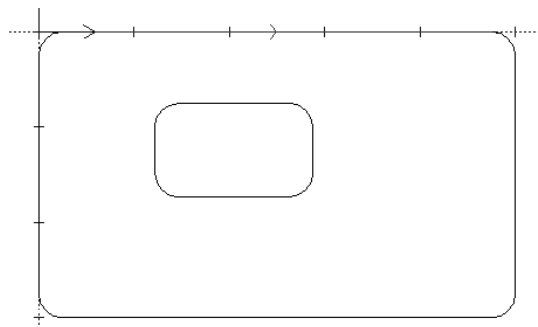


Therefore, you would use this function to get the result that you can see below.

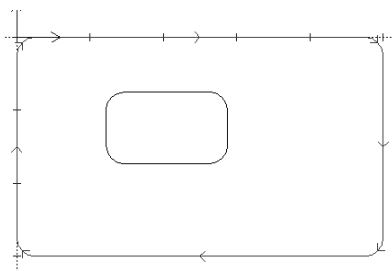


Now the spiral pocketing function would recognize the island as it is selected, going in the opposite direction. Because it is EVERY SECOND, you could only use this function for one island if using the pocketing function with this function. You would otherwise use the REORGANIZE function in the CHANGE menu to handle the arrow direction for selected entities.

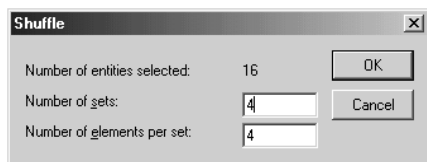
- **CHAIN SELECT:** This menu item was added for allowing you to select chained entities. This function works with ONE set of entities in a possible chain at a time. See below.



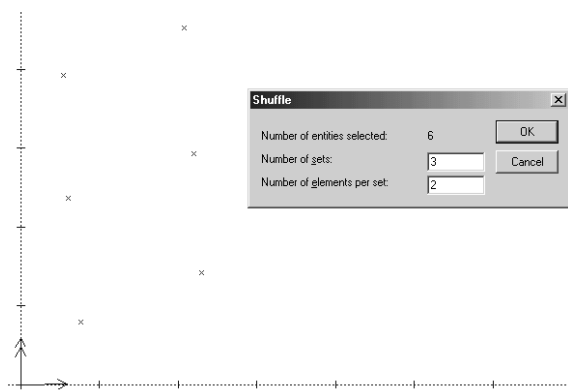
In the last image you can see that only one entity of the chain is selected. By using this function you can automatically select the entire chain without having to use the select entity CHAIN function.



- **SHUFFLE:** This function allows you to shuffle the directional arrows of selected geometry. By selecting this function you will open the SHUFFLE dialog. See this below.

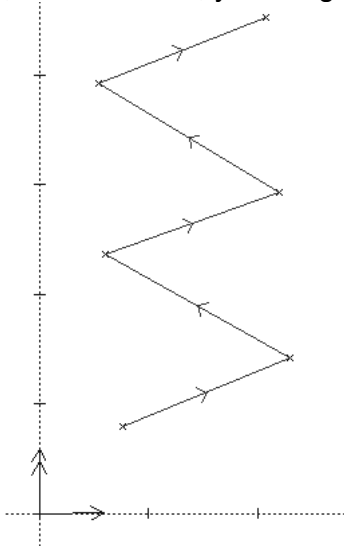


The dialog gives you the total entity count of selected entities as well as the option to enter the number of sets of entities to shuffle and the number of entities that you want per set. This function changes the selection matrix. See below.

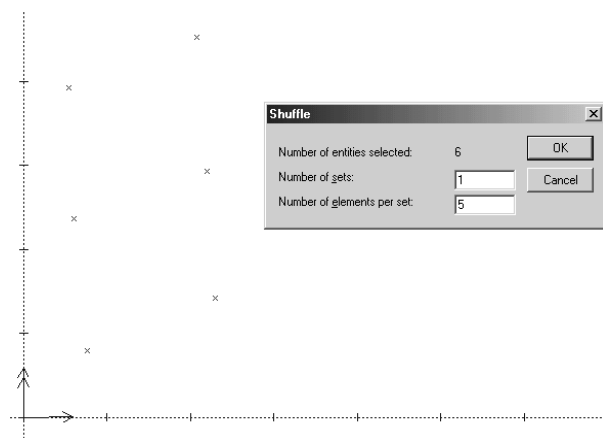


There are six selected points/entities. The number of sets is 3. The number of entities per set is 2.

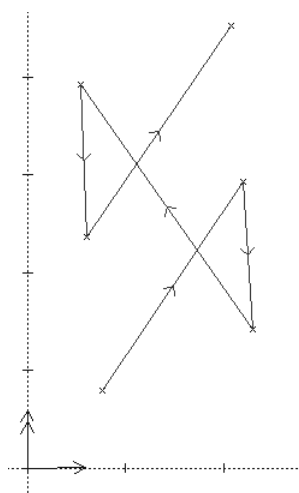
By selecting OK and then the LINE, JOIN function, you will get the following result.



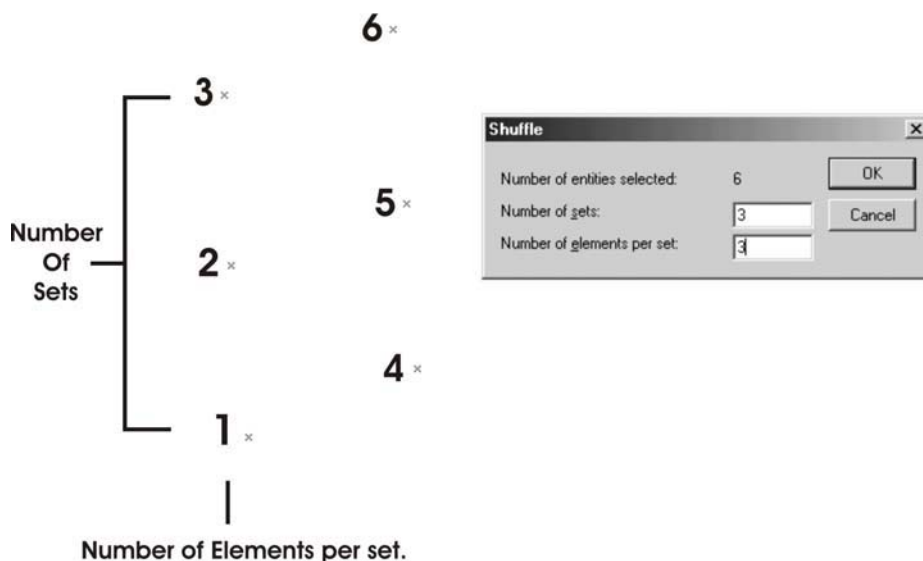
Now if we select the UNDO function to remove the lines, reselect the POINTS and go back to the SHUFFLE function we can change the matrix of the points. See below.



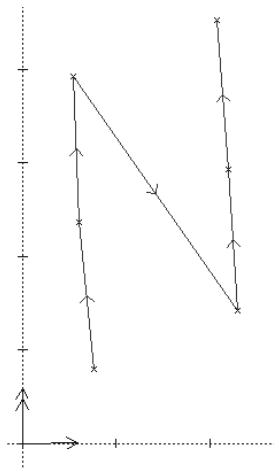
By selecting OK you can see the change in the lines.



This is a function that can be useful if understood. It is used for joining lines through construction points. Let's take another look at this below.

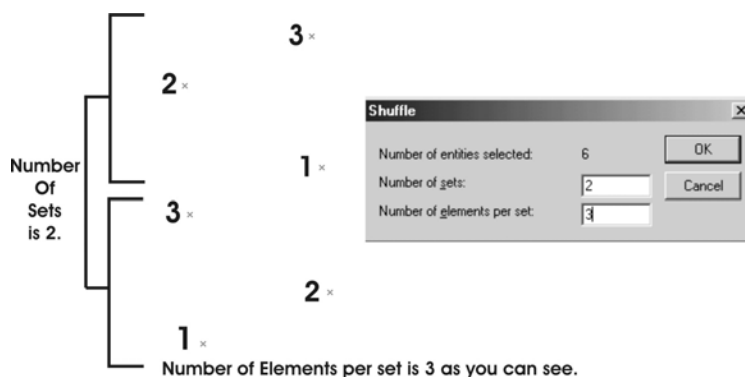


The entities in the diagram above are numbered in order and correspond to the values entered in the shuffle dialog.



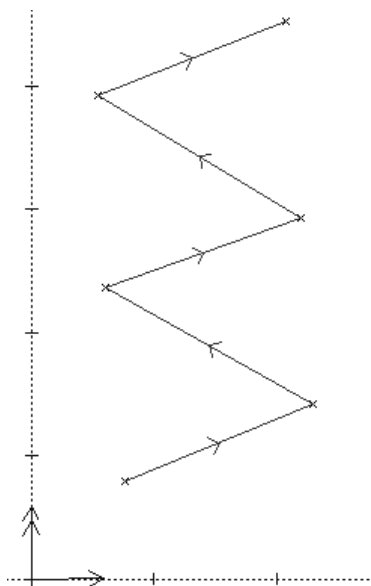
As the points are selected, we go directly to the LINE menu and select the JOIN function to get the result above.

Now let's change the SHUFFLE values. See below.



Number of Elements per set is 3 as you can see.

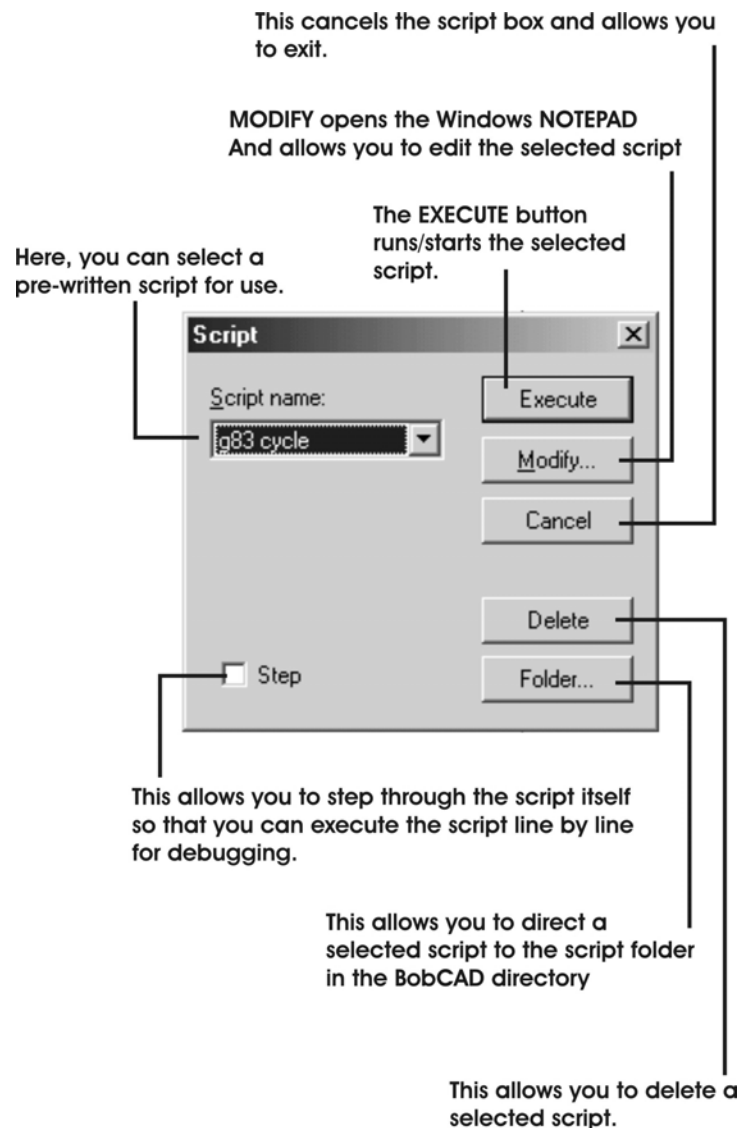
By selecting OK, choosing the LINE menu and join function, we get the result below.



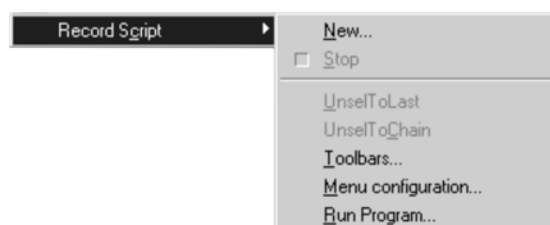
Section 67. Script & Record Script

SCRIPT: This function will allow you to create scripts for the software to execute special functions. In fact, you can customize the software for a wide variety of actions with the script function.

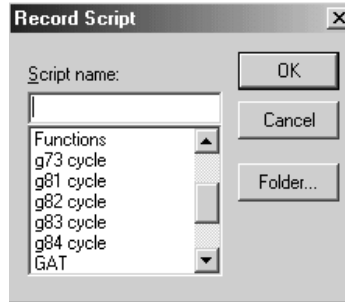
You should have an understanding of visual basic scripting before attempting to write scripts. However, you do not have to be a programmer to use this function to make the BobCAD 21 software more user-friendly and powerful for your needs. This function offers endless possibilities for the manufacturer and using BobCAD software for programming parts. It is actually a resource for creating your own custom software functions. By selecting the SCRIPT function you will open the script dialog.



RECORD SCRIPT: This menu function offers a special sub-menu that has to do with working with and creating visual basic scripts. Let take a look at this sub-menu below.

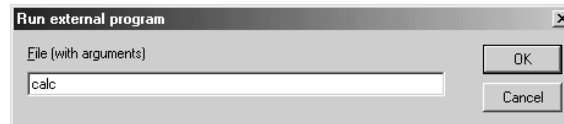


- **NEW:** This allows you to begin recording your actions so that you can save it as a script. Selecting this function opens the RECORD SCRIPT dialog as you can see below.

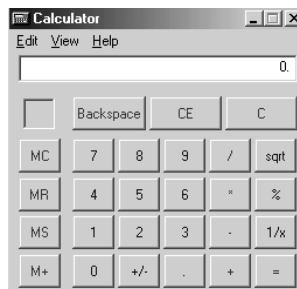


You would enter the name of the script that you wanted, select OK and begin recording drawing and / or editing actions.

- **STOP:** This allows you to stop recording when finished. It will only be available for selection if you are recording and want to stop.
- **UNSEL TO LAST:** If you are recording a script and you have drawn a series of entities, which are all selected, you can use this while in record mode to deselect all selected entities up to the last entity drawn and that will be left selected. It stands for "Unselect To Last." This function will not be active unless you are actively recording a script and need to deselect entities up to the last selected entity, which will be left selected.
- **UNSEL TO CHAIN:** This is Unselect To CHAIN. **If you have a series of entities connected to multiple chains, you can deselect all entities up to the last chain of entities. This function will not be made available unless you are actively recording a script and need to deselect entities up to the last chain, which will be left selected.**
- **TOOLBARS:** This opens the toolbar configuration dialog, which allows you to modify the toolbar icons of the BobCAD Software so that you can link an icon to a script that you may have written. Then by selecting the icon you will launch the written script.
- **MENUS:** This opens the Menu Configuration dialog and allows you to modify the software menus and add actual scripts as functions that you can make available.
- **RUN PROGRAM:** This allows you to automatically run external programs by entering their name into the dialog.



In the image above you can see that we typed in CALC, which is the code name for CALCULATOR. By selecting OK, the Windows calculator will appear.



By entering NOTEPAD, the Windows notepad will appear, etc.

CHAPTER 10

NC-CAM

- Section 68. NC CAM Basics & Icons
- Section 69. Customizing the CAM Icons & Adding Toolboxes
- Section 70. NC CAM Main Menus
- Section 71. The CAM File Menu
- Section 72. Send File
- Section 73. Receive File
- Section 74. The CAM Edit Menu
- Section 75. Simulate
- Section 76. The Machine Menu
- Section 77. The Contour Menu
- Section 78. The Cycle Menu
- Section 79. Customizing Drill Cycles Lesson
- Section 80. The CAM Tool Menu
- Section 81. The CAM Macro Menu
- Section 82. The CAM Setup Menu
- Section 83. Standard RS 232 Setup Checklist
- Section 84. The Coordinate Format Setup Checklist
- Section 85. The Conversion Function
- Section 86. Rotary Conversion
- Section 87. 4th Axis Indexing lesson
- Section 88. The CAM Help Menu



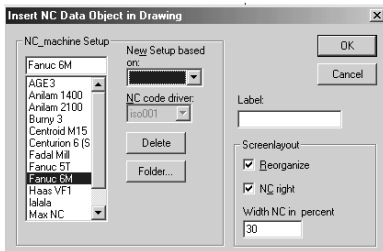
Section 68. NC-CAM Basics & Icons

To open the NC CAM side of the software you **MUST** first select **INSERT NC** from the **SPECIAL** menu on the CAD side. Then you will need to select a configuration and select **OK**. Only then will the CAM open.

Let's take a look at this procedure first.

STEP 1

Select the **SPECIAL** Menu and then **INSERT NC** to open the Insert NC Object dialog.



STEP 2

Select the post configuration that you need for your specific controller and select **OK**. This action inserts the NC Object that you see above into your drawing and opens the NC CAM. The NC Object can be named by typing in a name in the **LABEL** section of the Insert NC dialog. The NC Program will be linked to this NC Object and can be called up later if needed to run the g-code program again.

Let's take a look at each of the available **Icons** in the NC CAM and what they do.



This is the **NC SAVE** icon/button. After generating the NC Code, you can select this icon function to save the NC program to the NC Object.



This is the **SEND NC** command icon/button. This will open the SEND dialog for downloading through RS 232 to the controller.



This is the **RENUMBER THE LINES IN THE FILE** button/icon. This function allows you to customize the line numbers of the NC Code.



This is the **CUT** button/icon. This function allows you to cut lines or sections of code from the NC editor.



This is the **COPY TO CLIPBOARD** button/icon. This function allows you to copy NC Code from the editor onto your clipboard for later use.



This is the **PASTE** button/icon. This function allows you to paste information or NC Code into the NC Editor. You can copy from one program and paste into another NC Code program with this function. NC Code is actually considered text.



This is the **DELETE SELECTED TEXT** button/icon. If you want to delete a line of code you can highlight it and then select this button to delete it.



This is the **UNDOES LAST COMMAND** button/icon. If you make a change to the NC Code and want to change back or UNDO the move, you can select this icon to undo the mistake.



This is the **BRING TOOL TO CLEARANCE** button/icon. If you select this function, you will automatically bring the tool to its designated clearance. If you have just made a cutting move and selected this function, a dialog will appear offering you the speed settings for FEED and RAPID.



This is the **BRING TOOL DOWN** button/icon. By selecting this function you will be able to bring the tool down to a specified depth. This will bring up the speed settings dialog allowing you to set the speed to FEED.



This is the **TOOL DEPTH SETTINGS** button/icon. This function opens the **Auto Pocket Parameters** dialog and tool depth settings as well. You will be able to specify the clearance, total depth of your cut as well as the auto roughing parameters too. You will also be able to manually enter in speed and feed rates for the NC program. This is an important function to know when using the **CAM** to generate an NC Program.

This is the total depth of the cutting for the part. There can be multiple depths. After each tool change this will be considered. You will always have a minus symbol in front of this value. Positive values do not need a plus symbol.

This is the positive Z value for the top of the material you are working with in the program. Most of the time this will be 0.

This is your tool clearance value.

These are the feed rates along the path and for plunging into the cut. You can manually change these feed rates. Enter an F in front of the value needed.

This is the tolerance of how tight the line segments will be to the 3D arc when cutting 3D surfaces with radius values. This is how BobCAD looks at 3D arcs when machining.

This is the depth of each roughing pass going down in the Z-Axis.

This activates the automatic roughing functionality of the software and enables you to enter a maximum depth for each cut step down.



This is the **SET SPEED TO RAPID** button/icon. This lets you automatically program the set speed to RAPID wherever you are in the program at the time of the selection.



This is the **SET SPEED TO FEED** button/icon. This lets you switch to FEED mode automatically.



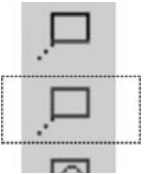
This is the **RAMP TOOL** button/icon. This will allow you to change the speed settings to FEED mode.



This is the **MOVE TO POINT** button/icon. By selecting this function and then clicking on an entity in your drawing, you will automatically create a rapid move to that location for the NC Program. The tool will rapid move from the designated home position, creating a dashed red line to the entity selected and will generate a G00 line of NC Code.



This is the **CUT SINGLE** button/icon. This allows you to cut a single entity.



This is the **AUTO CUT** button/icon. This function allows you to cut a chain of entities. You would use this function after setting your start block of NC Code and tool depth settings.



This is the **CUT ALL** button/icon. This function is a totally automatic toolpath cutting function used mainly for engraving or the cutting of entities at a single depth ONLY after certain criteria is met. Such as:

This function requires a tool change to be made so that you have the correct tool in position.

The tool depth settings are made so that the tool clears the work piece, clamps, etc.

The Z depth of the cutting is accurately set in the UP/DOWN tool depth settings dialog.

The entities that need to be cut at this depth should be the ONLY entities on the screen with all other entities blanked out.



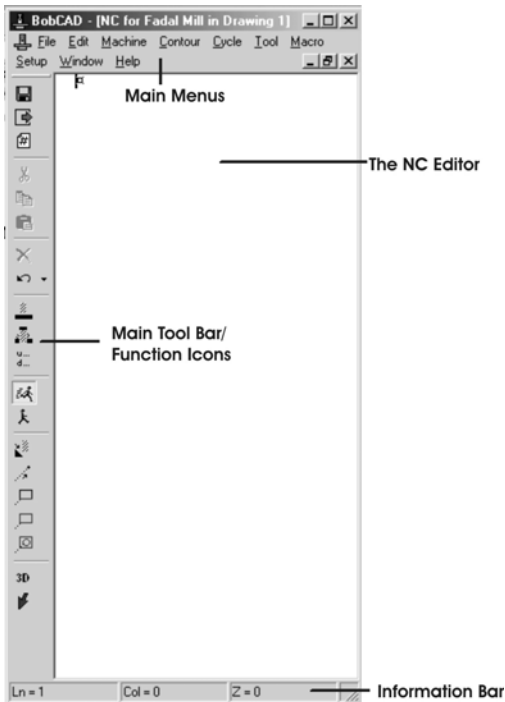
This is the **TURN 3D MODE ON/OFF** button/icon. By selecting this button, it will light up and you will be telling the software that you are cutting a 3D toolpath. You can toggle between cutting 2D shapes and 3D shapes with this icon. When you are cutting 3D toolpath, you will want to activate this function by selecting it so that you can view the roughing toolpath correctly in 3D rather than in 2D.



This is the **TURN TOOL DISPLAY ON/OFF** button/icon. This allows you to view the graphical tool rendering or a picture of the selected tool that is interactive along the generated toolpath. When the NC Program is complete, you can select a line of code and use the UP/DOWN arrow keys to scroll through your NC Code program.

As you use the arrow keys, you will see the tool follow along the toolpath, coinciding with the selected line of code that you are scrolling through. If you deactivate this function by selecting it twice, a small red arrow will replace the graphical tool representation.

Here you can see the NC **CAM** below.



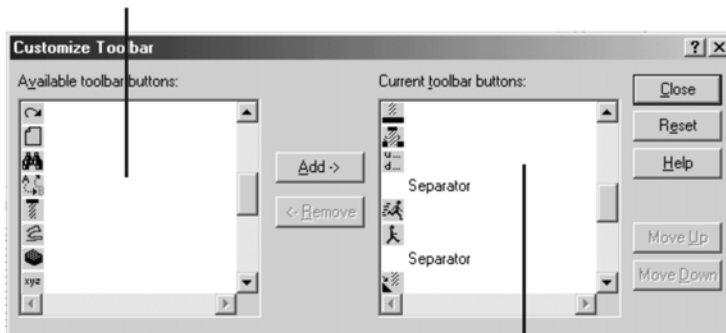
The NC CAM can be moved around the screen by clicking and holding down the left mouse button with your cursor on a corner and then moving your mouse while holding down the mouse button. You can also see the Minimize, Maximize and Exit buttons that are in the upper right corner of the NC CAM box.



Section 69. Customizing the CAM icons & adding toolboxes

You learned early in this manual about customizing the toolbar icons, adding them and removing them. You can also customize the **CAM** toolbars by adding or removing function icons as well. Let's take a quick look at this.

These are the icons that are available to select and add to the main NC CAM Toolbar. First click on the icon function and then select the ADD button to add it to the toolbar.



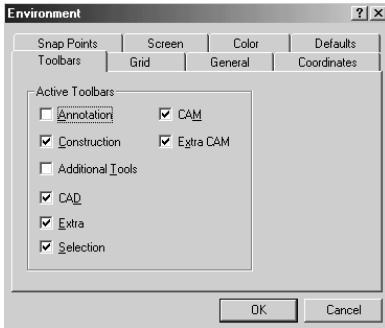
This side will list and show all of the icon functions that are currently on the main NC CAM Toolbar. If you select an icon on this side, the REMOVE button will activate allowing you to remove current icons that you may not need.

A separator is what separates the icons into sections. You can also remove a separator by selecting it and then selecting the REMOVE button. You can also move the icons up and down by selecting the icon and then the UP or the DOWN button.

This is standard Windows style functionality, making it possible for you to fully customize the way the software is arranged to suit your specific needs. Let's move on and take a look at the additional NC CAM toolbox icons and functions.

ADDITIONAL NC CAM TOOLBOXES:

In addition to the standard icon functions that are available, you can activate more function icons through the ENVIRONMENT dialog, which is located in the FILE menu on the CAD side of the software. Let's take a look.



The CAM option selected will show all of the standard NC CAM icons that we just covered. By selecting the **EXTRA CAM** option, the new toolbox will appear floating when the NC CAM is opened. Let's take a closer look at this toolbox below.

The EXTRA NC Toolbox:

This opens the SETUP TOOL LIBRARY dialog for setting up your tools.

This is the GEOMETRY FROM NC LINES or Backplot function icon.

This opens the NC Simulate dialog allowing you to enter your material attributes for solid simulation.

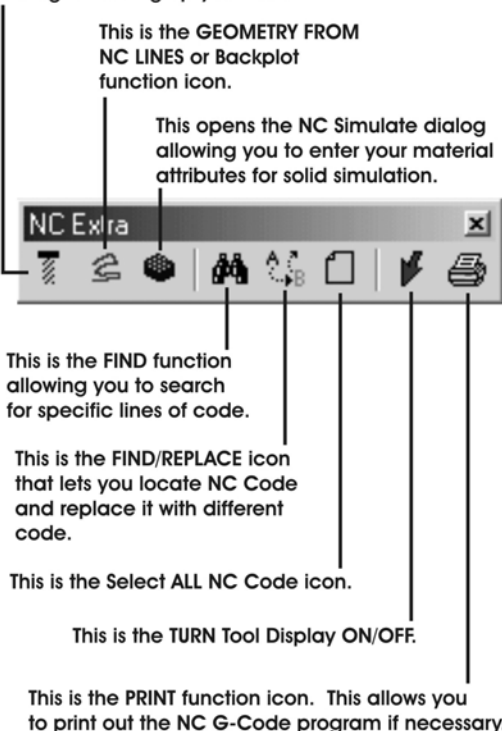
This is the FIND function allowing you to search for specific lines of code.

This is the FIND/REPLACE icon that lets you locate NC Code and replace it with different code.

This is the Select ALL NC Code icon.

This is the TURN Tool Display ON/OFF.

This is the PRINT function icon. This allows you to print out the NC G-Code program if necessary.



This toolbox can be docked and secured onto the wall of the CAM by clicking and dragging it into position while holding down the left mouse button. This toolbox can be customized by double clicking on a separator or an open space. Doing this will open the Customize Toolbar dialog that will then allow you to add or remove function icons.

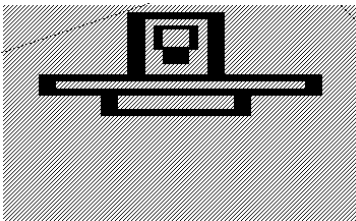
THE INSERT NC OBJECT:

After selecting your post processor and selecting OK in the Insert NC Object dialog, the CAM will open and the NC OBJECT will appear next to the UCS on the CAD side.



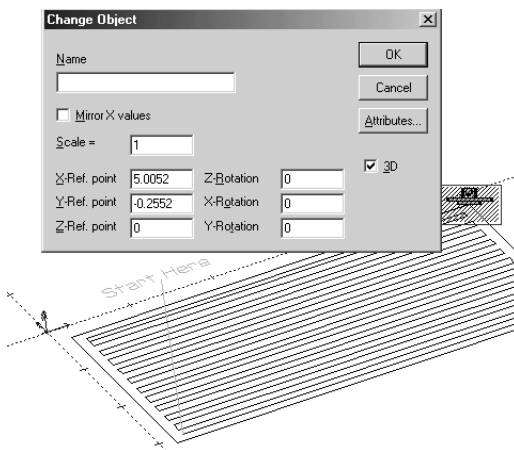
This NC Object will always tell you that there is a NC CAM file attached or linked to the drawing. After writing NC programs for drawings and then closing these programs, you will notice that the next time you open the file in the CAD, the NC Object will be apparent if there is a CAM g-code file made for the part. By double-clicking on the NC Object, the CAM program will open and be available for use. This allows you to pick CAD files for accessing NC programs easily.

When you write an NC G-code program for the first time, you will notice that there will be small hatch lines over the NC Object. This means that there is a written NC G-code program for the drawing and toolpath on the CAD side. See this below.



The NC Object can be selected and moved around the screen in MODIFY MODE so that it is not in the way of viewing your part. If you select the NC Object and then delete it, the NC CAM will go away and the NC program will be deleted unless saved. It is important to save your CAD files and your CAM NC programs.

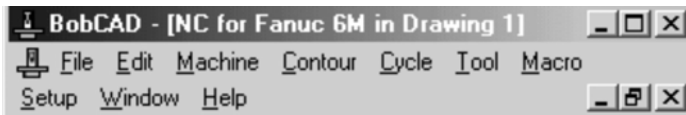
You can also change the CAM NC Object by double clicking on it to open the CHANGE OBJECT dialog, which can be seen below.



You can move the NC Object around the screen or provide a name for it as well as apply special attributes to it by selecting the ATTRIBUTES button.

Section 70. NC-CAM Main Menus

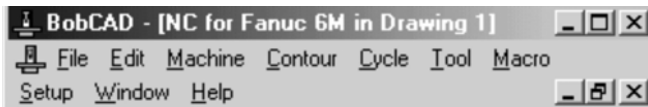
Let's take a look at the top of the NC **CAM** and the main menus.



The **NC CAM** Main Menus are:

- The **FILE** menu.
- The **EDIT** menu.
- The **MACHINE** menu.
- The **CONTOUR** menu.
- The **CYCLE** menu.
- The **TOOL** menu.
- The **MACRO** menu.
- The **SETUP** menu.
- The **WINDOW** menu.
- The **HELP** menu.

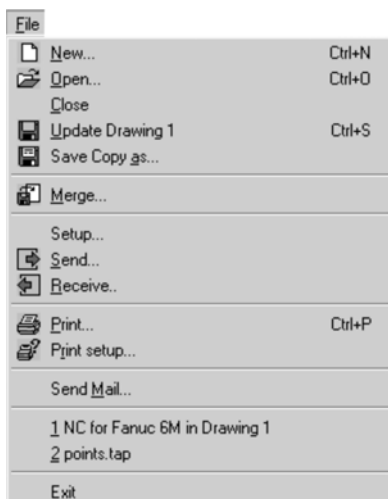
The top of the CAM will always tell you what post processor you selected as well as the name of the file once you have saved it. Because you can have more than one NC CAM window open at once, there are two sets of dialog controls.



The top set minimizes, maximizes or exits the entire NC CAM. The lower set controls the visible CAM Window. However, like the CAD side, you can tile or cascade the NC CAM widows the same way you would for the CAD side.

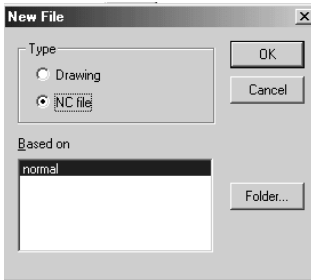
Section 71. The CAM File menu

This section is dedicated to the NC **CAM FILE** menu. Let's take a look below at this menu.



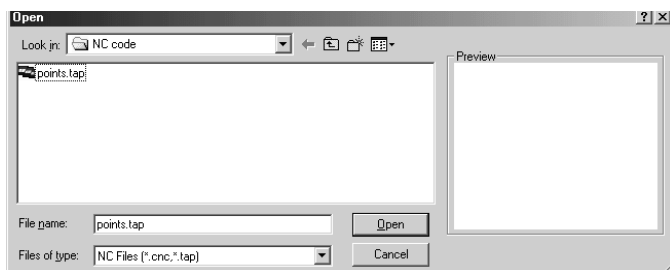
The CAM FILE menu offers the same file management functions as the CAD side, just setup for NC CAM files.

NEW: This function allows you to open a NEW NC CAM window and editor. By selecting this function, the NEW FILE dialog will appear as you can see below.



By selecting NC FILE, you will open the new window. You can also select the DRAWING option to select a CAD file. By doing this you will open a drawing on the CAD side of the software. This is slightly backwards but can be done this way. You can also select the FOLDER button to access specific folders for files previously saved.

OPEN: This function allows you to access the OPEN dialog so that you can select a specific CAM file that you previously saved.

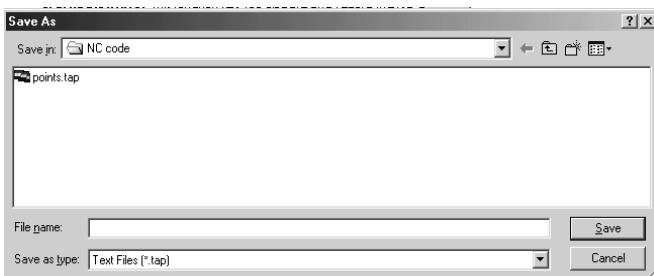


By double clicking on the file of choice, it will open for use in the NC Editor. By selecting the FILES OF TYPE drop down arrow, you will be able to access all of the BobCAD file types. This way you can better search for needed files. This dialog also contains standard Windows file controls. You can simply click on the file that you wish to open and then click the OPEN button.

CLOSE: This function allows you to close a file that is open in the NC Editor. Selecting this function will close the CAM entirely. However, you will still see the NC Object on the CAD screen. By double clicking on the NC object you can reopen the CAM.

UPDATE DRAWING: This function lets you update and record the NC Code to the NC Object on the CAD screen. This is similar to backing up the **CAM** actions to the NC Object. You still need to save your NC G-code file.

SAVE COPY AS: This lets you save a file as a copy with a different name as needed. Selecting this function will bring up the SAVE AS dialog as seen below.



Now you can select a different file extension for the same file with a different name.

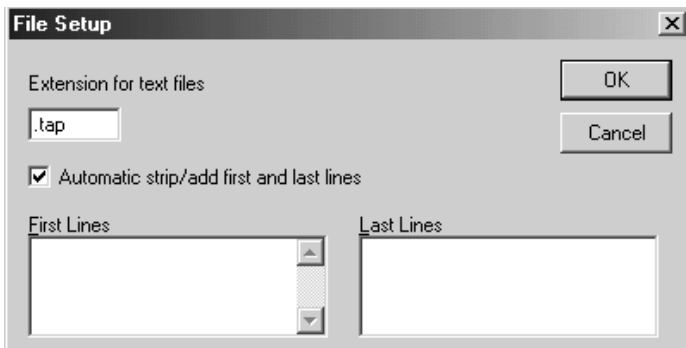
TIPS ON SAVING NC FILES:

Here is a **TIP**: When you are saving a file with a CNC extension, that file will be linked to the machine configuration that it was created with. It is a good idea to also save the file as a TEXT file that has the file extension TAP. The reason for this is because if you copy the file to another PC/Computer, or you no longer have that machine configuration down the road BobCAD will alarm out every time you try to open that file. You will NOT go into an alarm state if the file was saved as a TEXT/.TAP file. If you run into files that have been previously saved as CNC files and they give you an error or alarm, it is because that file was saved with a configuration that you no longer or do not have on your computer. You can open the problem file in Windows NOTEPAD, scroll down to the last line and see the name of the configuration that the file was originally saved with. Use that configuration to then open the CNC file.

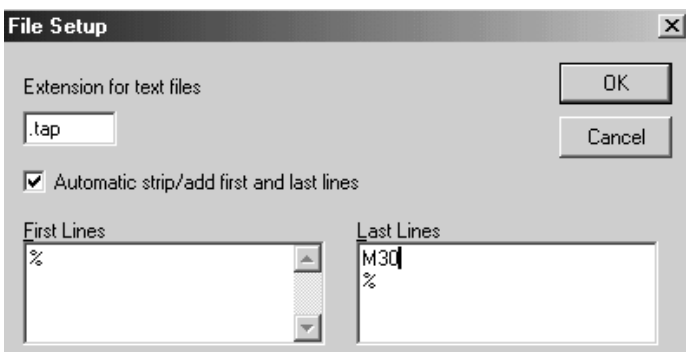
In other words, if you are opening a CNC file from another source, you will need to open your BobCAD NC CAM with the same configuration as the CNC file that you want to open.

MERGE: The merge function will allow you to merge in a CNC or TAP file that was previously saved in BobCAD into the NC Editor. You can merge NC files together through selecting the file in the OPEN dialog. Place your cursor where you want to merge in the new file and click your mouse button. Then select the merge function and bring in the new NC file at that location.

SETUP: This function will allow you to **Setup** the FIRST & LAST lines of code for your program.



The standard file extension for NC **CAM** text files is **.TAP**. By selecting the option for “Automatic Strip/Add First & Last Lines” you can enter in the start block and end block that you want for your G-Code program. For example, below you can see a percent symbol for the first lines. The rest of the NC lines may vary. However, if you have one CNC machine and need the same start block for each of your programs, you may enter it in here and it will be saved. That goes for the LAST LINES area as well. Typically, the end of a NC program will contain an M30 and a percent symbol. Entering this here will automatically add it to your NC programs when sending the file to the CNC controller. Uncheck the “Automatic Strip/Add First & Last Lines” option to deactivate the function.



Section 72. Send File

SEND FILE: This is the function that you would use to send the NC program that is complete, to the CNC controller through an **RS 232** communication cable. Let's take a closer look at this dialog.

Dialog example 1.

This relates to the speed in which you will be sending the program. Check your machine manual for the correct settings.

COM Ports are also known as Serial Ports or RS 232 Ports. Typically, COM 2 is used for sending the NC Program to the controller with RS 232.

This needs to match what your specific machine controller or controllers require. Please refer to your machine manual/s for the correct entry.

This needs to match what your specific machine controller or controllers require. Please refer to your machine manual/s for the correct entry.

The 'Send File' dialog box contains the following fields and controls:

- COM Options:**
 - Port: A dropdown menu currently showing 'COM1'.
 - Baud Rate: A dropdown menu currently showing '9600'.
 - Data Bits: A dropdown menu currently showing '7'.
 - Parity: A dropdown menu currently showing 'Even'.
 - Stop Bits: A dropdown menu currently showing '1'.
- Handshake:**
 - None (radio button)
 - Software (radio button, selected)
 - Hardware (radio button)
 - Special (radio button)
 - RTS/CTS (checkbox)
 - DTR/DSR (checkbox)
 - XON/XOFF (checkbox, checked)
- Timing (msec):**
 - Line Delay: A text input field containing '0'.
 - Add at end of line: A text input field.
- First Lines:** A text area with up and down arrows.
- Last Lines:** A text area with up and down arrows.
- Buttons:** 'OK' and 'Cancel' buttons.

Annotations with lines pointing to specific fields:

- A line from the 'This relates to the speed...' text points to the 'Baud Rate' dropdown.
- A line from the 'COM Ports...' text points to the 'Port' dropdown.
- A line from the top 'This needs to match...' text points to the 'Data Bits' dropdown.
- A line from the middle 'This needs to match...' text points to the 'Parity' dropdown.
- A line from the bottom 'This needs to match...' text points to the 'Stop Bits' dropdown.

This needs to match what your specific machine controller or controllers require. Please refer to your machine manual/s for the correct entry.

Dialog example 2.

X ON/X OFF is the FLOW. This section allows you to turn the flow on or off. By putting a check in the box next to XON/XOFF, you are turning this function ON. This must be active for software handshaking. There is also an option for starting at the machine or at the computer. We recommend starting at the machine as this confirms that the machine is communicating with the PC, and because the PC has to receive a signal from the machine before it will start to send the NC file.

This stands for DATA TERMINAL READY/DATA SET READY. This is a hardware handshake that is rarely used these days. However, it is used on some of the older machines and BTR units (Behind The Tape Reader). This must be active to use this sort of hardware handshake.

This stands for REQUEST TO SEND/CLEAR TO SEND. This must be selected and active for hardware handshaking. This can only be used if the wiring is in accordance with the requirements for hardware handshaking. Check your machine manual for this information.

The image shows a 'Send File' dialog box with the following settings:

- COM Options:**
 - Port: COM1
 - Baud Rate: 9600
 - Data Bits: 7
 - Parity: Even
 - Stop Bits: 1
- Handshake:**
 - ☐ None
 - ☒ Software
 - ☐ Hardware
 - ☐ Special
- RTS/CTS:** ☐
- DTR/DSR:** ☐
- XON/XOFF:** ☒
- Timing (msec):**
 - Line Delay: 0
 - Add at end of line: (empty text box)
- First Lines:** (empty text box)
- Last Lines:** (empty text box)

OK and Cancel buttons are located in the top right corner.

This option allows for custom settings.

The hardware/controller will make some adjustments for the communications between the software and machine tool.

The software will make some adjustments for the communications between the computer and CNC machine tool.

The machine and computer absolutely must have the same settings. This is between the computer and the CNC controller.

Dialog example 3.

The 'Send File' dialog box contains the following settings:

- COM Options:**
 - Port: COM1
 - Baud Rate: 9600
 - Data Bits: 7
 - Parity: Even
 - Stop Bits: 1
 - Handshake: ☒ Software
 - ☐ RTS/CTS
 - ☐ DTR/DSR
 - ☒ XON/XOFF
- Timing (msec):**
 - Line Delay: 0
- Add at end of line:** (empty text box)
- First Lines:** (empty text box)
- Last Lines:** (empty text box)

You can enter in the NC program start block codes for your machine controller here. An example would be a % symbol to indicate the beginning of the NC program.

You can enter in the NC program END block codes for your machine controller here. An example would be an M30 & a % symbol to indicate the END of the NC program.

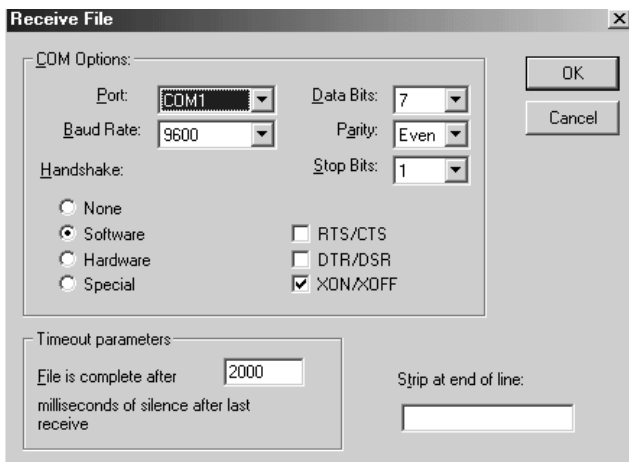
You can add text or numbers to the tail end of each line of NC Code by typing it in here.

BobCAD software send the NC program in milliseconds to the controller. The 0 value will send in normal stages. If you want to slow down the number of lines of code that is sent, you would add the millisecond time value here. This will drastically slow down the send time. This would typically be used for DNC (Direct Numeric Control) type operations.

IMPORTANT: It is important for you to review your machine handbook for verification of these settings before making selections. As this deals directly with RS 232 Communication settings, you will likely be able to find this data under that section of your machine handbook.

Section 73. Receive File

RECEIVE FILE: This function allows you to receive NC files from external locations. Let's take a closer look at this dialog below.



Just as in the SEND file function, the COM OPTIONS are the same. The HANDSHAKE options are the same in functionality as well. This dialog offers Timeout Parameters that allow you to stop receiving after a specified period of time. The STRIP AT END OF LINE allows you to add special characters at the end of each line of NC code that is received.

Special Characters:

You can use a number of special characters in the CAM part. All of these characters start with the @ symbol. They are:

@xxx represents an integer from 000 to 255: the ASCII codes. Thus @000 represents CONTROL + A and @025 represents Ctrl+Z.

The following special characters are only meaningful when defining contour, cycle and macro in the Setup menu:

@!: This character at the start of the line means that the line will not be automatically numbered.

@c: This character can only be used when defining a cycle. BobCAD enters the coordinates of the current point wherever this character occurs using the defaults from Setup: Coordinates.

@l: This character can only be used when defining a contour. BobCAD enters the line generated by the driver wherever this character occurs.

@ (followed by space): This "soft" space is only added by BobCAD if it does not occur at the start of the line.

The rest of the CAM file menu:

PRINT: This function allows you to print out NC programs from the editor.

PRINT SETUP: This function selected, opens your printer setup dialog. Here, you can customize your printer settings for printing.

EXIT: Selecting this will exit you out of the NC CAM altogether. If you have an unsaved program, the software will ask you if you would like to save it. It is always important to save your work.

Section 74.

The CAM Edit Menu

In this section we are going to take an in-depth look at the EDIT menu of the NC **CAM**. Below, you can see the menu.



The EDIT menu offers functionality that deals with the editing of NC programs as well as Tooling and Verification for NC programs. Let's go ahead and take a closer look at each of the functions offered in this menu.

UNDO GENERATE: This function allows you to UNDO the last generation of NC code in the editor. If you created G-Code and notice a mistake, you can select this function, make the changes to settings and then regenerate the code correctly.

REDO DELETE: If you accidentally make a mistake and delete NC code from the editor, you can select this function to redo the deletion.

CUT: This allows you to cut NC Code from the editor. First select the portion of NC code and then use this function to cut it from the editor.

COPY: This function allows you to COPY selected NC Code from the editor so that you can add it to other NC programs as needed.

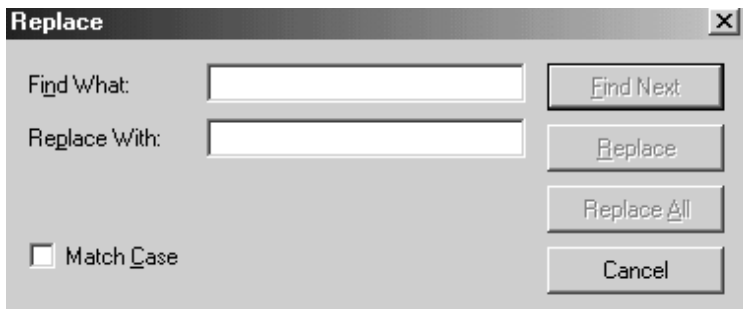
PASTE: This function will allow you to paste copied or cut NC code from the NC editor into NOTEPAD or another NC program. After copying or cutting NC code, click your mouse at the beginning of a line of code and then paste the NC code into the editor. The NC Editor is just like a spreadsheet or notepad.

DELETE: This allows you to delete selected NC Code from the editor.

SELECT ALL: This function will allow you to select and highlight ALL of the NC data in the NC Editor.

FIND: This allows you to search for specific NC lines. Simply select the function and type in the NC data you would like to search for and then select OK. The software will automatically locate the entered lines of code and you will see them highlighted in the NC Editor.

FIND/REPLACE: This function allows you to search for specific lines of code and then automatically replace those found lines with the new entry.

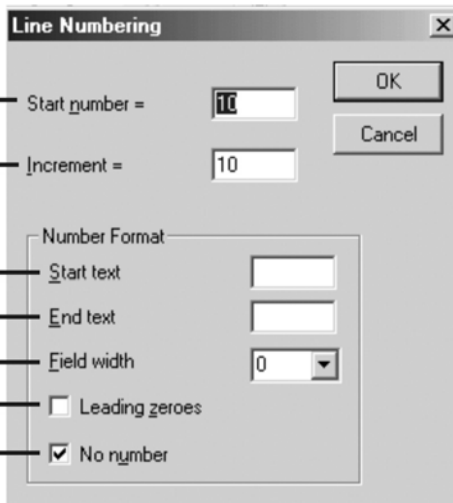


RENUMBER: This function allows you to add, remove or modify the way each line of NC Code is numbered in the **CAM NC Editor** when the code is generated.

The starting text letter will be the first letter of each line of code. Then the number will appear. For example, the letter "N" is used for some NC programs; N10, N20, N30 and so on.

This will indicate the increment of each number. If the increment is 10, the second line of code will start with 20, the third with 30 and so on.

This will be the number at the beginning of the first line of code.



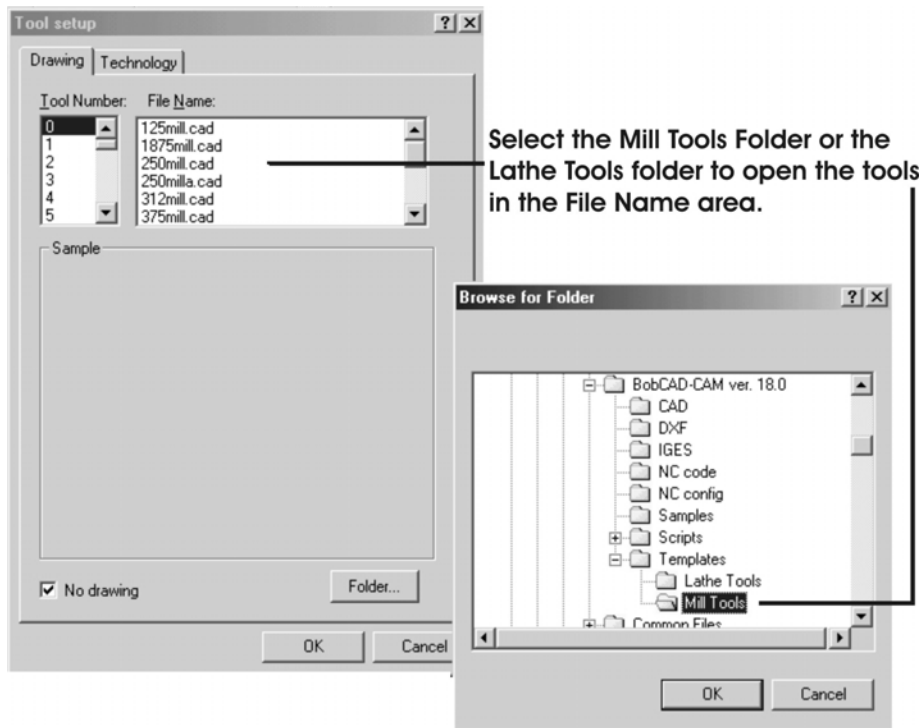
No Number allows you to completely remove line numbers from the NC Code.

This allows you to add 0's to the end of each line number.

The field width allows you to add a distance between the Start Text and the incremental line number.

If you enter a letter here, it will be placed at the end of the line number. Example: N10A, N20A, N30A and so on.

TOOL: This function allows you to access the Tool Technology dialog. This dialog has two separate tabs. They deal with graphical tool rendering setup and the actual tool attributes for each tool that you set up in the tool menu. Let's take a look below.



The DRAWING tab has two sections.

TOOL NUMBER: This simply allows you to attach tool 1, 2, 3, 4 and so on to the CAD file tool image for graphical tool rendering purposes.

FILE NAME: This is a list of all CAD tool drawings.

Select the tool number and then the CAD file that you would like to have attached to that tool number. Make sure that the "NO DRAWING" option is deselected and the dialog will change to show you an enlarged image of that tool. Now you can select OK and use that tool for graphical tool rendering.

You can proceed to the TECHNOLOGY tab to enter all of that selected tools attributes. This tab is basically used for setting up the tool for simulation purposes only. Here you can see the **Tool Technology** tab.

Enter the diameter of the tool number here.

By selecting a tool number and then entering in the tool attributes, you can save tool information in the tool menu.

You may add *compensation* to this tool as needed here. Basically, you would enter HALF the size of the tool diameter.

The *Z-Displacement* is the distance between the top of the part and the tip of the tool. This relates to the Z-Axis. If you are using cutter height compensation, this value will be represented by the appropriate G-Code, normally G43 if that code is supported by your specific machine controller. Some are different.

The *cutting height* is the length of the tool that will be used to machine the part.

Core Diameter is used in relation to the BULL NOSE END MILL cutting tool. A Flat Tool or Ball End Mill cutter would not apply. This represents the diameter of the tool tip.

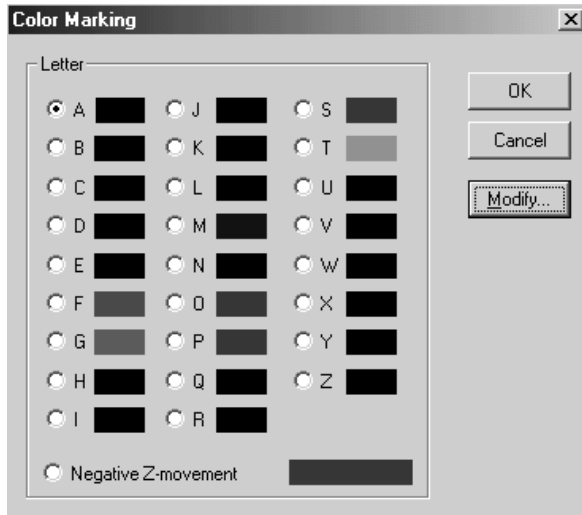
The Tool Technology dialog will also allow you to save FEED rates for each tool. The parameters section offers the ability for you to further add tool information as needed. Really, the parameters section is already covered as the tool diameter is entered and you are using the tool for the complete string of g-code. It doesn't have bearing on tool changes. Once you have added your tool information, you will still need to add tools to your tool menu. This is done through the SETUP menu and TOOL.

GEOMETRY FROM NC: This function allows you to select NC Code from the editor and backplot the code on the CAD screen. This is not associative in Version 21. Therefore, it will only backplot the NC code the way that it is in the NC editor.

SIMULATE: This allows you to simulate the G-Code program. Have a look at Section 75.

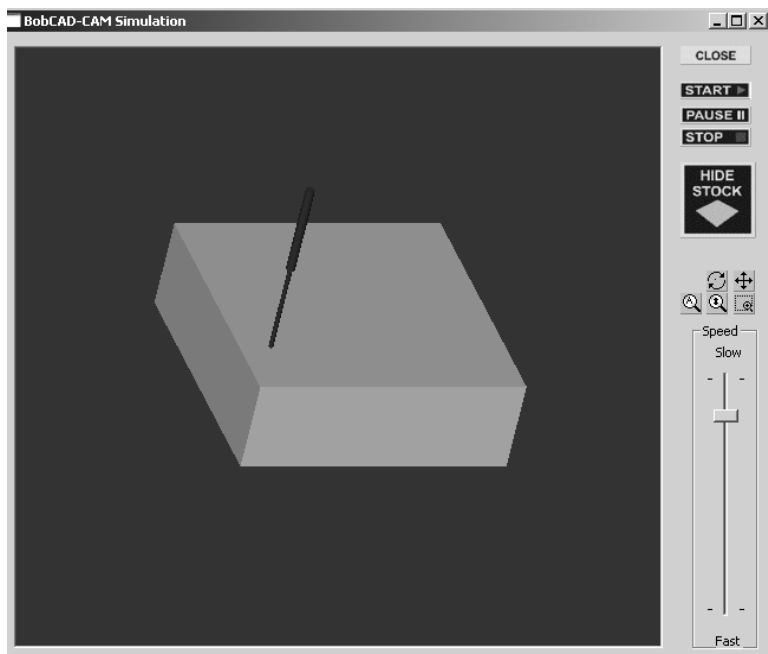
FONT: This will allow you to change font types for the NC Code. By selecting this function the FONT dialog will open.

COLOR: This function allows you to designate special colors for certain lines of code. For example, all X moves can be a certain color. All negative Z moves can be another color. Because TOOL starts with the Letter T, you can assign "T" a specific color and so on. See this dialog below.



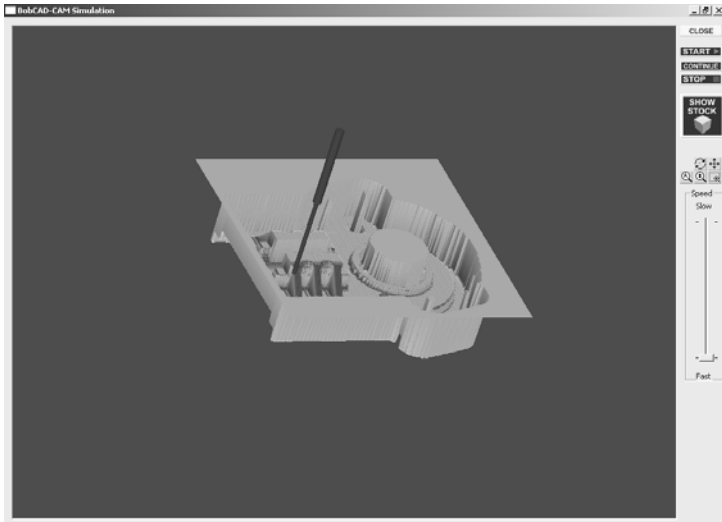
Section 75. Simulate

Version 21 provides a more sophisticated form of G-Code/part **Simulation** allowing you to view the process of your part being cut much better than in earlier software revisions. Let's take a closer look at this new function.

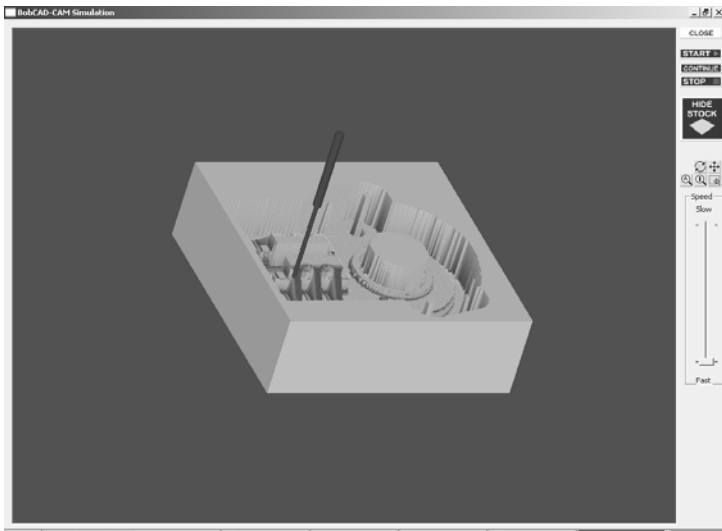


This new simulation box allows you to start, pause, stop and use viewing options for simulation. By clicking on the “Hide Stock” button you can remove the stock walls to review the under side of the simulated part cut as well.

Without Stock:



With Stock:



The speed bar allows you to speed up or slow down the simulation process. After you click on a viewing button and are finished using it, right-click your mouse to exit that viewing option. You are always in dynamic rotation mode until you select a different viewing option and then exit it.

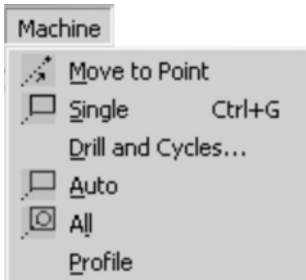
The Procedure for Simulation:

1. Enter your start block and tool at the top of the NC CAM editor.
2. Create the g-code program.
3. Select the tool and the g-code for that tool in the editor and then perform a tool setup for simulation from the CAM EDIT menu under “Tool.” Do this for each tool change in a single program. In other words, select the tool and the code for that tool for each tool setup operation.
4. Select ALL from the CAM Edit menu. (This selects all the tools and all the g-code after your tool setups for simulation.)

5. Select the Simulation function from the CAM EDIT menu. Enter the position of the stock and the length, width and height of the stock. Then click OK.
6. Use the simulation box to simulate the part program.

Section 76. The Machine Menu

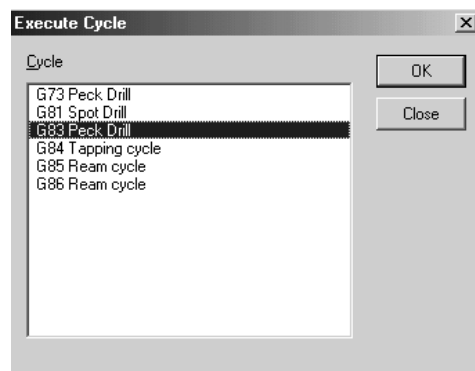
Here is the **Machine Menu** seen below.



This menu deals directly with machine functions for cutting parts with BobCAD.

The menu functions are:

- **MOVE TO POINT:** This allows you to manually select a point or entity on the screen and create a rapid move and G00 NC line in the editor.
- **SINGLE:** This allows you to select an entity, a single entity and generate NC code for cutting it. You will first need to enter your depth settings information before using this function.
- **DRILL & CYCLES:** This opens your drilling dialog to pick the cycle that you need.

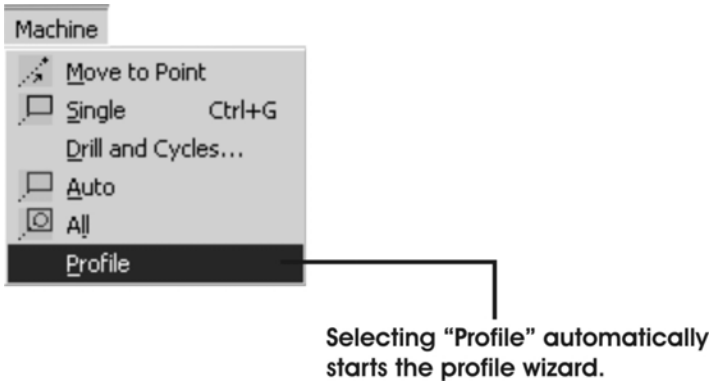


The drilling cycles are customized in the SETUP menu under CYCLE.

- **AUTO:** The auto cutting function allows you to select a chain of entities and cut them as well as cut 3D toolpath. This function will be the most common cutting option used. Before using this function you will enter your depth settings as well as tool and start block of the NC program. This is used for most cutting except drilling cycles.
- **ALL:** This option would be used for cutting everything on the screen. **Before using this function, you will need to make sure that all non-machining entities are blanked out of the drawing and that your depth settings are entered correctly for clearance and so on.** The first cut will start at the entity closest to the tools home position.

- **PROFILE:** The **Profile** function offers a complete solution to profile milling. Let's take a closer look.

Version 21 offers a new **Profile** wizard with linear and radius approach and departures for the tool as well as built-in tool compensation. The Profile wizard function is located in the Machine Menu of the CAM side.



Once you have your profile contour geometry on the CAD screen you can also draw a point that you would like to use as the **start point** for your approach and departure or you can let the software use a default position on the designated inside or outside depending on your G41 or G42 selection.

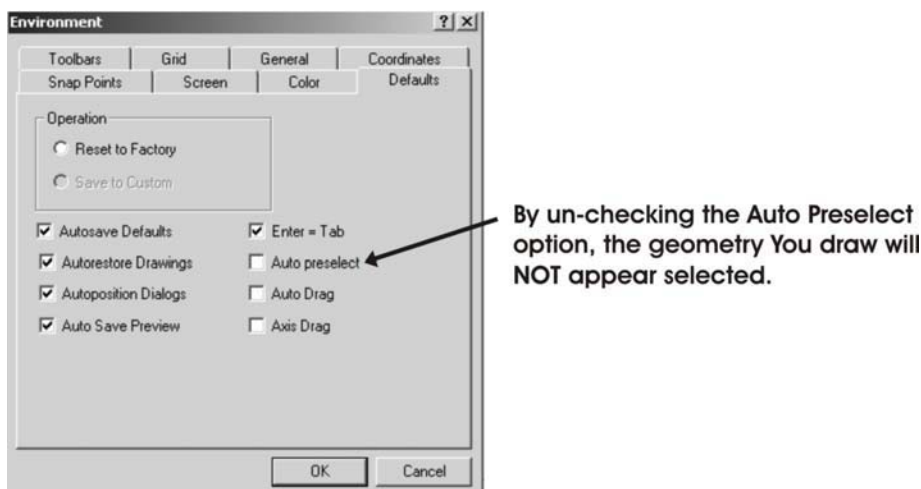
The procedure for using the Profile function:

1. First you would select the function.
2. Chain-select the open or closed shape profile.
3. Indicate if you would like to use your own start point or have the software start at the position that you initially selected for the chain selection.
4. If you clicked YES to select your own start point, you would now click on it to open the tool depth settings box.
5. Enter a clearance for rapid moves and your depth of cut. You can use roughing as well. Select Next.
6. Enter the tool diameter and other settings in the Profile Machining box and select OK.

That is the basic procedure for using the profile wizard.

PROFILE TRAINING LESSON:

NOTE: BEFORE DOING THIS LESSON, PLEASE TURN THE "AUTO-Preselect" option OFF. Do this by selecting the FILE menu and then ENVIRONMENT. Choose the DEFUALTS tab and un-check this option. Thank you.



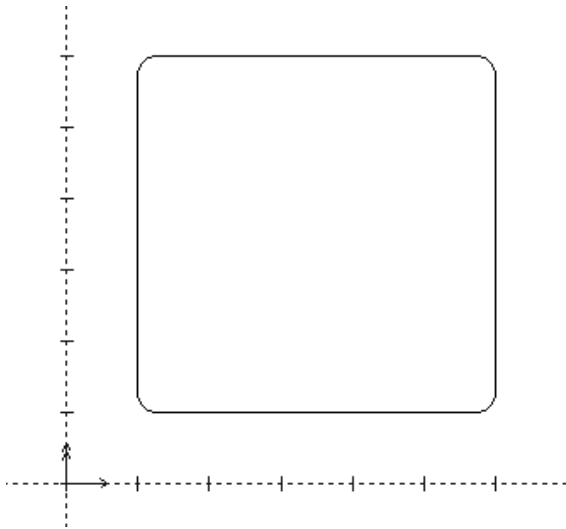
STEP 1

Start this lesson with a NEW CAD drawing screen.

Go to the OTHER menu and select RECTANGLE. Select Bottom Left under the section called, "Mode." Select Fillet as the Corner Type and enter .25 as the radius. We will be using the **bottom left** corner of the rectangle as our point of reference. Now enter 1 for X, 1 for Y and leave the Z value set at 0.

Now, go ahead and enter a width of 5 and a height of 5. This is going to be a basic square shape with radius corners.

Now click OK to draw the shape.



STEP 2

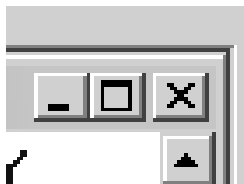
Now we are going to draw a point on the screen to indicate our start position for the tools approach into the profile cut. This is not mandatory for profiling. However it is a new option that we need you to be aware of.

Go to the POINT menu and select COORDINATES. In the box that appears, enter 2 for X and leave Y and Z at 0. Click OK to draw the point.

STEP 3

Now go to the Special/NC CAM main menu and select INSERT NC. This opens the Insert NC Data box and is where you will select the post processor. For this lesson select the FANUC 6M configuration and click OK to open the CAM.

NOTE: If your CAD window splits into 2 or more CAD windows it is because you opened a new file without fully closing out the last window. Simply use the Maximize option in the upper right corner of the CAD window we are working in. This is the middle, square looking button.

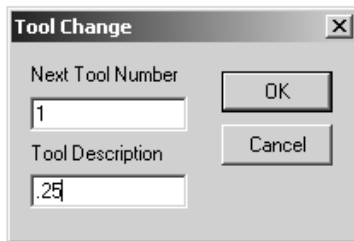


This will open the drawing window that we are working in to full view. Now select the VIEW ALL icon from the main CAD toolbar. This is the small magnifying glass with the letter, "A" inside of it.



STEP 4

Go to the CAM side of the software and select the TOOL menu and then TOOL CHANGE. Under Next Tool Number enter 1. Under Tool Description enter .25.

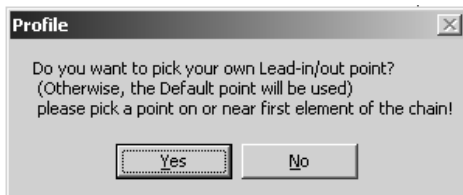


Now click OK to insert this data into the NC Editor. The NC Editor is similar to a standard word processor allowing you to type in it and make editing changes manually as needed. This is where the entire program is generated.

STEP 5

Select the MACHINE menu from the CAM side and then PROFILE. Now you need to place your cursor on the BOTTOM line of the profile closest to the point and click your left mouse button one time to select it. Now move your cursor toward the left end of the line so that the directional arrow at the midpoint of the line is pointing towards the left. Now click your left mouse button ONE time to indicate this direction and then hit the F3 key on your keyboard to select the entire profile chain.

The profile box will automatically appear asking you if you would like to pick your own lead-in/lead-out point for this profile or not. Because we drew a start point for this we will select YES.

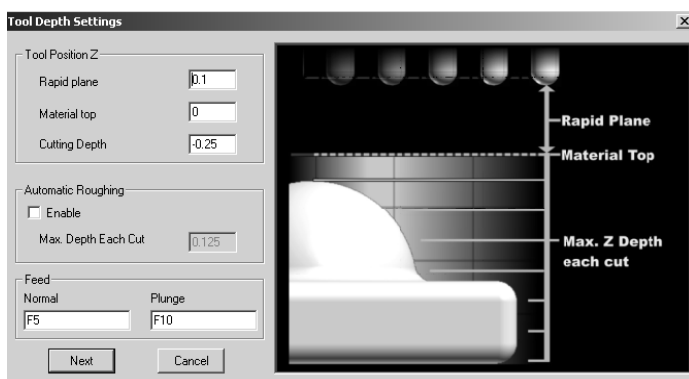


The “Default point” referred to in this box will be the starting point of the line you clicked on if you were to choose NO. In any case, select the YES button.

Go directly to the point that you created and click on it to open the Tool Depth Settings box.

STEP 6

Have a look at this new box.

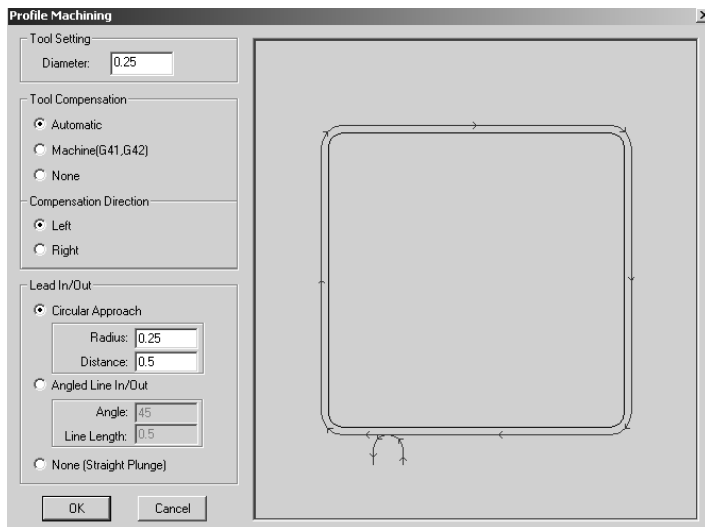


Under Tool Position Z you have your Rapid Plane. Enter .1 for this as it represents the tools clearance above the part during rapid moves. Your material top for this part is 0 and the CUTTING DEPTH is -.25. We are not going to be roughing this profile so do NOT check the Enable option under Automatic Roughing. In the FEED section enter F5 for NORMAL and F10 for the Plunge.

Now select NEXT.

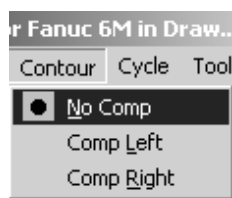
STEP 7

Now you will have the Profile Milling command box. Let's take a brief break from this lesson and have a look at this box.



The Profile Milling command box gives you control over the following:

- **The tool diameter** that you will be using. Everything in this box is associative, meaning that if you make a change, the preview will reflect the changes made. If you change the size of the tools diameter and hit your enter key the preview will reflect this modification.
- **The tool compensation.** By choosing **Automatic**, you can use the right or left options in this box and the software will automatically create an offset on the left side or right side of the original geometry. The lead-in and lead-out will be added to this offset unless you select the "Straight Plunge" option. Basically, the software does the offset for you based on half the distance of the size tool you use. By choosing **Machine G41/G42** you are telling the software that you would prefer to cut the original profile geometry using right or left compensation (G41 or G42) and that this G41 or G42 code will be added to the g-code program created in BobCAD. By choosing **NONE** you are indicating that you are not going to use compensation at all and will be using the drawn geometry for cutting with the center of the tool being used. This is the same as using the Auto Cut function in BobCAD and whatever you have selected in the Contour menu will be added to the code.



This is important to understand when using these options. Some post processors are setup differently when it comes to the Contour menu in the CAM. Some posts will add "Normal" and some will list compensation as in the image above that is setup for the Fanuc 6M post.

If you have something selected in the contour menu and use the profile wizard for compensation, the profile wizard will over ride this menu command unless you select NONE in the comp section of the profile wizard. Therefore, you should check the contour menu in the CAM to see what is selected.

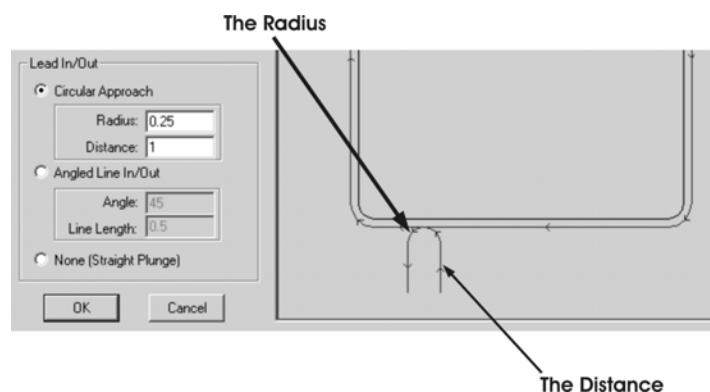
Here it is again. If you use a comp option in the contour menu and choose NONE in the profile wizard, the option selected in the contour menu will be used when the software creates the g-code program. If you use compensation in the profile wizard it will over ride the option selected in the contour menu. It's that simple.

You see, you do not always have to use the profile wizard for profiling in BobCAD. You can simply create manual offsets for your tool (half the distance of the tool size), use the Tool Depth Settings and then the Auto Cut function in the machine menu along with the comp option you need selected in the contour menu if your post processor configuration supports it.

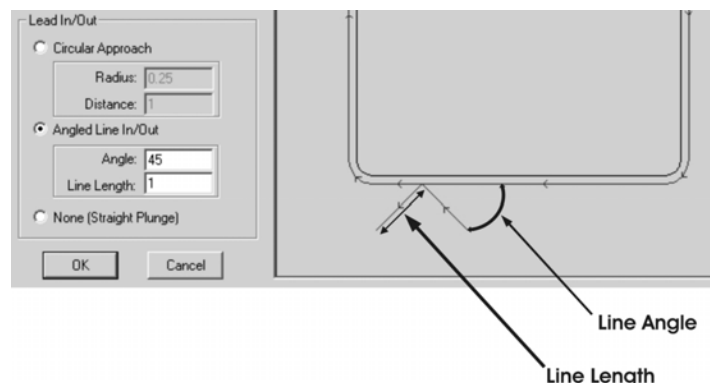
Just remember one important thing, when you are using G41 or G42 comp from the CONTOUR menu and will be using the Auto Cut option, you will want to create a lead-in and lead-out manually by using the approach/depart function from the OTHER menu of the software.

- **Compensation Direction.** You have two options, Right and Left.
- **Lead-In/Out.** This section offers you a **circular/radial** approach and depart or an angled line approach and departure for your tool. The radius value allows you to adjust the overall radial approach as well as the departure and the "Distance" option offers you the ability to shorten or extend the length of the approach and departure of the tool.

Here's an example of the **Circular approach and departure** below.



You also have the ability to create an angled line type approach and departure for the tool. The "**Angle**" value is simply the angle of the lead-in and lead-out line itself and the "Line Length" option determines how long or short you want these lines to be. Let's take a look at this in the example below.



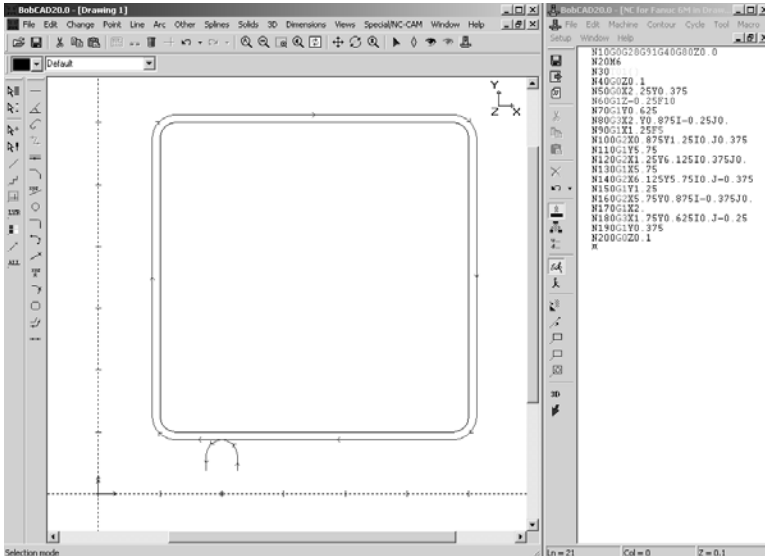
The **NONE** option refers to straight plunge ONLY.

OK, let's get back to the lesson now that you understand how to use this box. Go ahead and enter .25 for Diameter under TOOL SETTING.

STEP 8

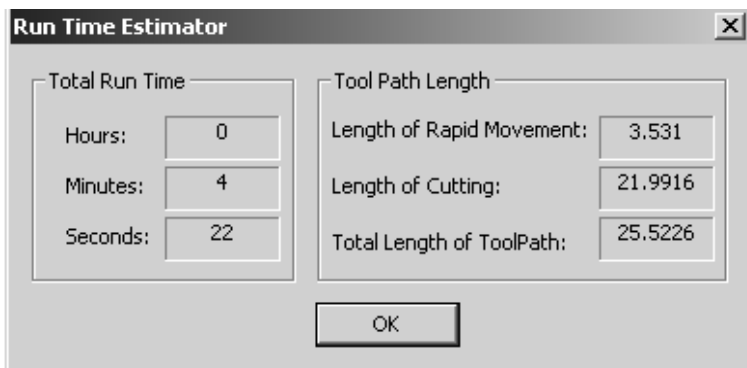
Select Automatic under the Tool Compensation. Under Compensation Direction choose the LEFT Option. Under Lead-In/Out select the CIRCULAR approach option. Now enter .25 for the Radius and .5 for the Distance.

Now select OK to produce the toolpath and your G-Code program.



STEP 9

Now go to the EDIT menu on the CAM side and choose Select ALL. This will highlight all of the code. Now go back to the EDIT menu on the CAM side and select RUN TIME ESTIMATE. A box will appear Asking if the Maximum Feed Rate is 50 inches or millimeters per minute. You can change this value if you want but if you select OK the estimator box will appear next.



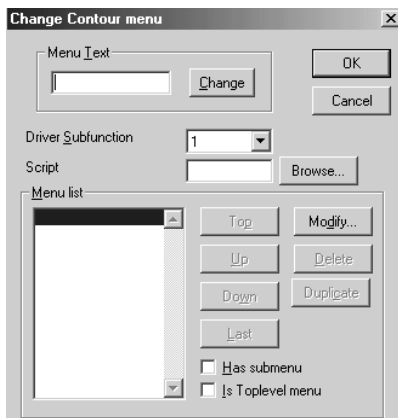
Congratulations! You have completed this lesson.

Section 77. The Contour Menu

This menu deals directly with cutter compensation for your tool. Let's take a closer look at how to setup this menu as a working function. If you want BobCAD to automatically insert cutter compensation and to cancel cutter compensation at the end of the program, you must do the following.

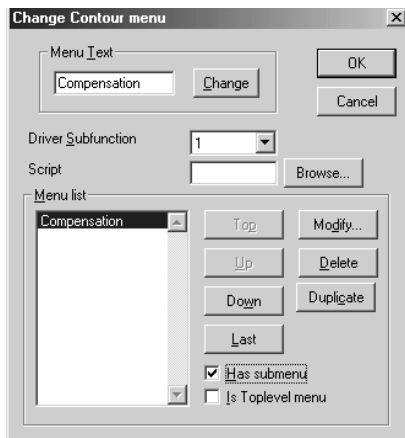
STEP 1

On the NC side of the software, select the SETUP menu and CONTOUR. This opens the "Change Contour Menu" dialog as you can see below.



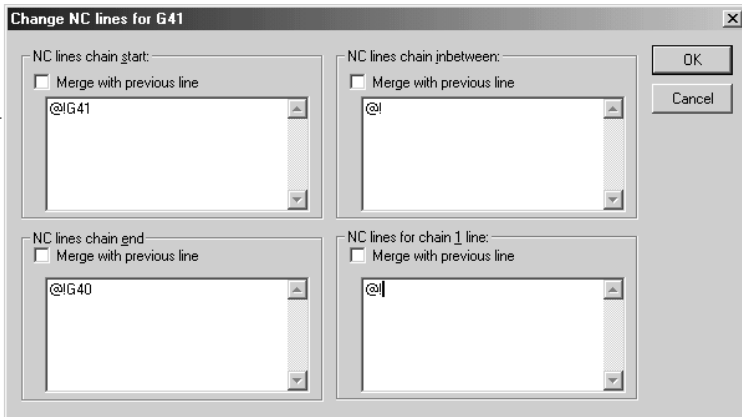
STEP 2

You will need to customize the CONTOUR function in this dialog. Enter COMPENSATION RIGHT under Menu Text and then select the CHANGE button to insert it into the Menu List section. Now select the HAS SUBMENU option.



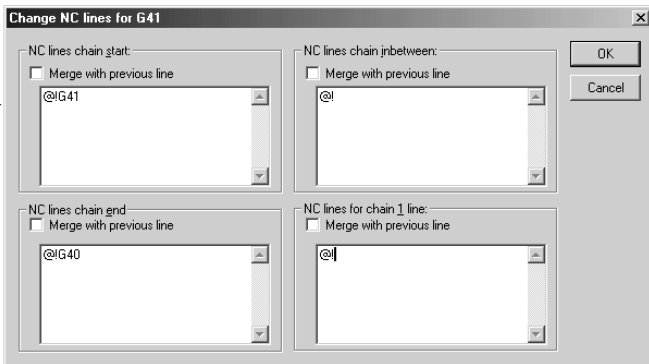
STEP 3

Now select the blank space under Compensation in the MENU LIST. The menu TEXT area will then become blank. This is correct. Type in G41 under Menu Text and select the CHANGE button. This will enter the G41 entry under COMPENSATION as a sub-menu item. Now, while G41 is still selected under MENU LIST, select the MODIFY button to open the "Change NC Lines for G41" dialog as you can see in the next image.



STEP 4

Make the following entries as you can see in the dialog below.



- **NC LINES CHANGE START:** Enter @!G41
- **NC LINES CHAIN IN-BETWEEN:** Enter @!
- **NC LINES CHAIN END:** Enter @!G40
- **NC LINES FOR CHAIN1 LINE:** Enter @!

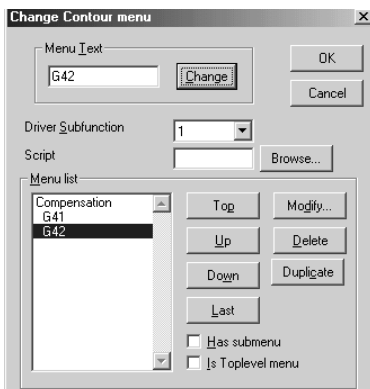
Do not check any of the Merge With Previous Lines option boxes. Select OK.

STEP 5

Now that we have customized the G41 cutter compensation menu list, we will now need to do the G42 setup.

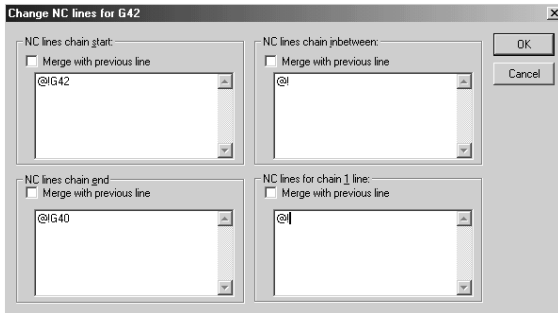
STEP 6

Now select the blank space below G41 under MENU LIST and enter G42 in the MENU TEXT space. Now select the CHANGE button to make this visible under MENU LIST and select the MODIFY button again.



STEP 7

Now you will want to make similar entries in this new dialog. Follow the entry instructions below.



- **NC LINES CHANGE START:** Enter @!G42
- **NC LINES CHAIN IN-BETWEEN:** Enter @!
- **NC LINES CHAIN END:** Enter @!G40
- **NC LINES FOR CHAIN1 LINE:** Enter @!

Now by selecting OK, you are saving these entries to be used in the CONTOUR menu.

STEP 8

Now you will want to make a menu function for turning the compensation OFF. Select the blank space under MENU LIST to blank out the space under MENU TEXT. Enter COMP OFF under menu TEXT and select the CHANGE button to insert this under the MENU LIST section and under G42. Now select the MODIFY button.

STEP 9

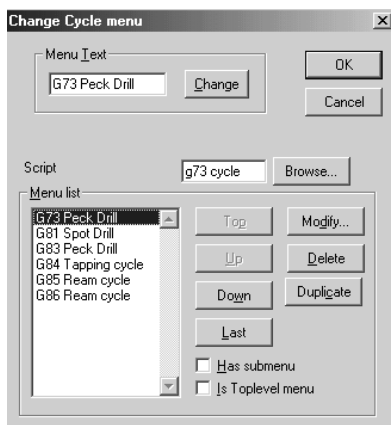
In each of the boxes you will only need to type in @!. When finished, select OK.



This is how you customize the Contour menu. This is fundamentally the procedure that you will use to customize ALL of the changeable menus in the NC CAM.

Section 78. The Cycle menu

The CYCLE menu deals directly with **Drilling** cycles for machining operations. This, like the contour menu is fully customizable. Let's take a look.



To access the CHANGE CYCLE MENU dialog, you will need to select the NC SETUP menu and then CYCLE.

It is important to understand what G-Code is and how your machine controllers recognize drilling, tapping and boring NC G-Codes. Here are standard G-Code cycles:

- **G73: Standard canned drill cycle – Chip Break.**
- **G81: Standard canned drilling cycle – Regular, Spot.**
- **G83: Standard canned drilling cycle – Deep Hole Peck.**
- **G84: Standard canned drilling cycle – Tapping.**
- **G85, G86: Standard canned drilling cycles – Boring.**
- **G87: Standard canned drilling cycle – Back Boring.**
- **G88, G89: Standard canned drilling cycles – Boring with Dwell.**

You will need to consult your machine handbook for details on the drilling cycles that are needed by your controller. Do this before customizing your drilling cycles.

Section 79. Customizing Drill Cycles Lesson

BobCAD Version 21 software has the ability for you to customize the canned drilling cycles for controllers that require this. This lesson is going to teach you how to do this in the case that you need to.

STEP 1

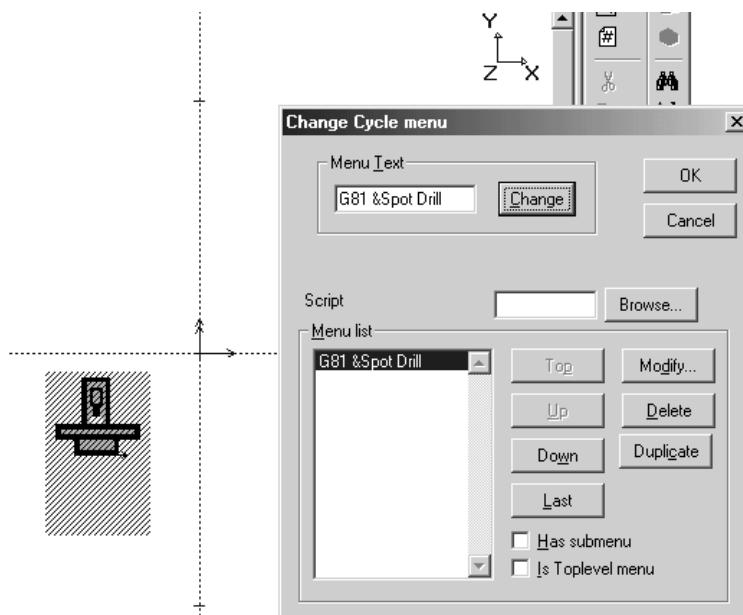
Start with a NEW BobCAD drawing screen.

STEP 2

To begin, select SPECIAL from the main menu and then choose INSERT NC. Select the configuration of your choice and then select OK to open the CAM. The large NC Object will consume the CAD screen, so you can select the REDUCE VIEW icon from the main toolbar until you have an actual drawing screen to work with.

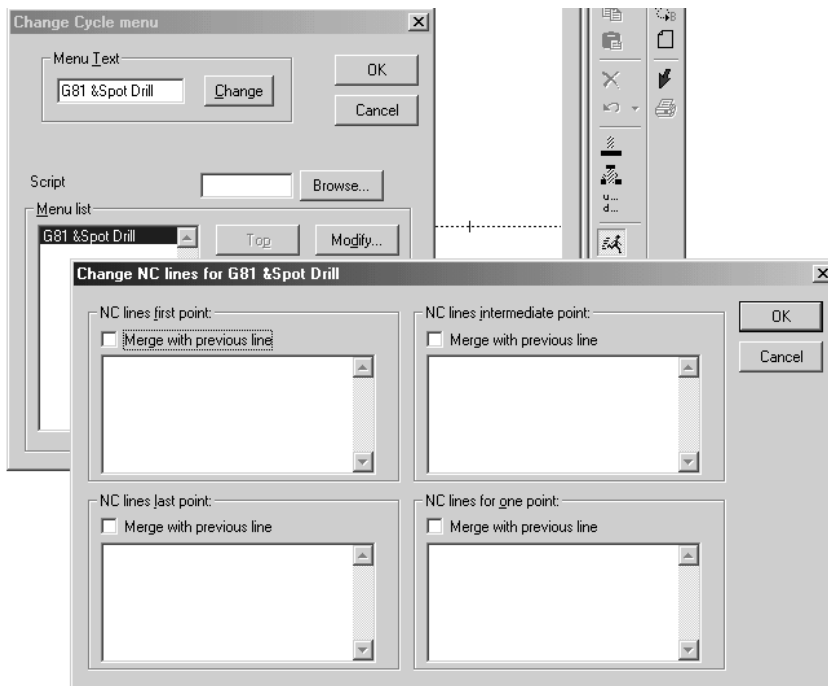
STEP 3

Now, on the CAM side of the software, select SETUP from the main CAM menus and then CYCLE. Click in the MENU LIST box. The first blank space will highlight in blue. Now, in the MENU TEXT box type in **G81& SPOT DRILL** and select the CHANGE button. The & symbol is used to designate which letter you can call the cycle by if you are using the Windows ALT+ key, keyboard short cuts.



STEP 4

Now select the MODIFY button. A new box with four sections will appear. As can be expected, each of these sections represents different points that can be selected during the course of the cycle.

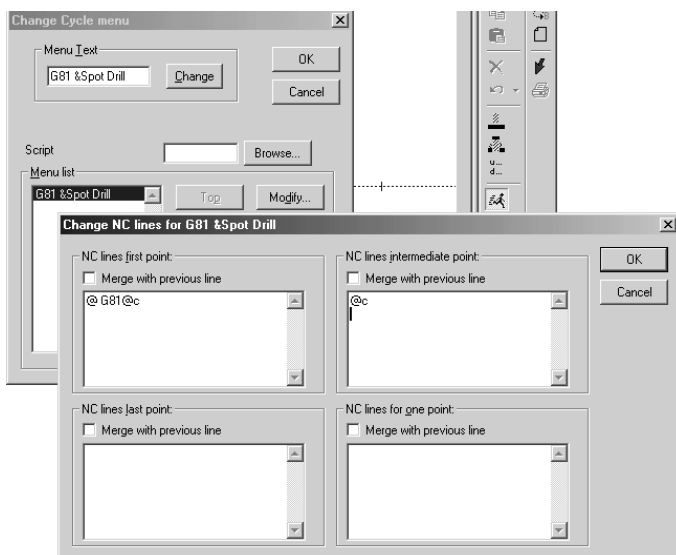


STEP 5

Under **NC LINES FIRST POINT**, type in **@ G81@c** and select ENTER. The G81 portion of this will be the name that appears at the beginning of the first line of the cycle. The "@" symbol followed by a space is a macro telling BobCAD to obey whether spacing is turned on in the code or not. The "@c" is a macro that tells BobCAD that this is where the coordinates of this first point go in the line of G-code. The ENTER key that you hit, merely brings the cursor down to the next line. **NOTE: If we did not hit the ENTER key the next line of the cycle would appear on the same line of G-code as the last cycle line of code.**

STEP 6

Under **NC LINES INTERMEDIATE POINTS**, type in the **@c** and hit your ENTER key. If our cycle doesn't need to see the G81 on each and every line, we do NOT need to include it. However, our cycle needs to see the coordinates on every line of g-code and the ENTER key does exactly the same thing here as it did for the first part of the cycle.

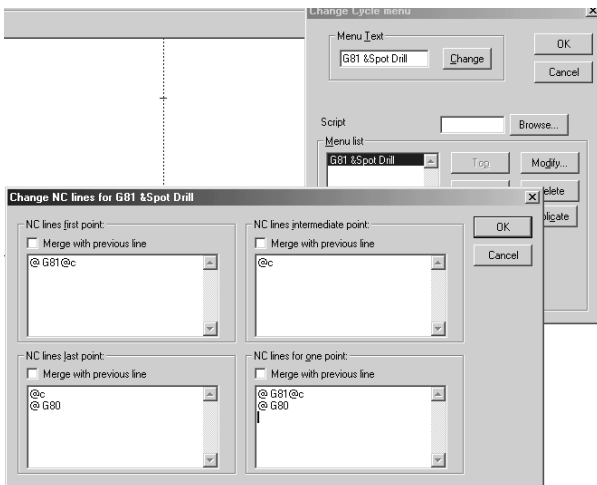


STEP 7

Under **NC LINES LAST POINT**, we will include the coordinates of the last point to be selected for our cycle and we will also end it. And so, in this box type in **@c** in the first line of the box, then hit ENTER and type in **@ G80** (remember to put a space between the @ symbol and G80) in the line below it. As before, the @c represents the coordinates of the point. @ G80, however signifies the end of the cycle here.

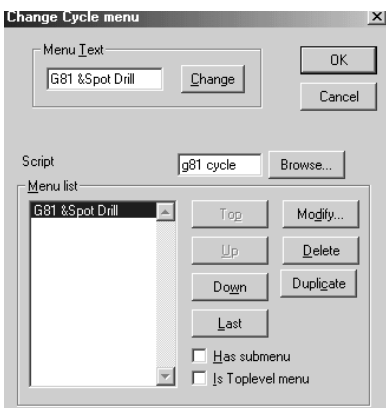
STEP 8

In the last box, **NC LINES FOR ONE POINT**, we will need to know what to type in if we only have one point to use this cycle on. In this case, we will need the cycle start, coordinates, and end all in the same box. In this box type in **@ G81@c** and hit ENTER. Now type in **@ G80** on this line followed by the ENTER key again. Now click OK.



STEP 9

Now, back in the **CHANGE CYCLE MENU** box, click your mouse in the text box marked, "SCRIPT." Type in **G81 cycle**. This will assign the pre-written script that we will use to help finish off the cycle. In this case, the script **G81 cycle.bas**, will be called up and it will present you with a box after you have selected all of your drill points on the CAD screen and will ask you for your drilling parameters. Then it will change the code inserted by BobCAD to read a correct drill cycle for the machine. It is NOT necessary to include the .bas file extension, as BobCAD already knows it is looking for the file of this type.



By selecting OK, you will be in selection mode and ready to select the drill points on the CAD side. Because we have not instructed you to do this, we will need to draw some points now.

STEP 10

Select the **POINT** menu on the CAD side and select **SKETCH**. Click your mouse 5 times dropping 5 random points on the screen. When finished, click your right mouse button to exit this function.

STEP 11

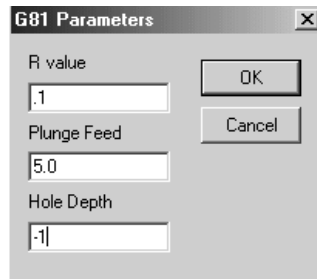
Now we need to select the points. There are various ways to select entities. Use whatever way you are comfortable with or select the EDIT menu, Select Entities and then choose SELECT ALL.

STEP 12

Now go back to the CAM side and select the MACHINE menu and then DRILL. The pop-up box will appear and our G81 drilling cycle should be listed.

STEP 13

Select the G81 cycle we created and then click OK in the pop-up box and the program will execute.



The G81 parameters box as seen above will now appear asking for the retract or clearance value, a plunge feed value and the total depth of each hole. Enter in the values as seen in the box above and click OK. Once you click OK, the G-code program will automatically be written with the G80 at the bottom. Congratulations!

NOTE: There is an alternative method to using our cycle. First we would go to the MACHINE menu and select DRILL. When we click on our cycle and then click OK and do not already have anything selected on the CAD side, the BobCAD canned cycle engine will pop a box up asking you to choose a selection method. You would select the selection method of your choice, click OK and then make the selection of your drill points and then click your RIGHT mouse button to tell BobCAD that you are finished with your selections. Then BobCAD would ask you for your drill parameters and generate the NC G-code program for you instantly.

If you decide that you would like to write your own drill cycle script, you should take a look at the BobCAD scripts already in your SCRIPT folder. By default, there are plenty of example scripts that you can actually modify and improve on for your machine. To do this, select FILE and OPEN from the CAD side or the CAM side, then choose the SCRIPTS folder to view them.

Section 80. The CAM Tool Menu

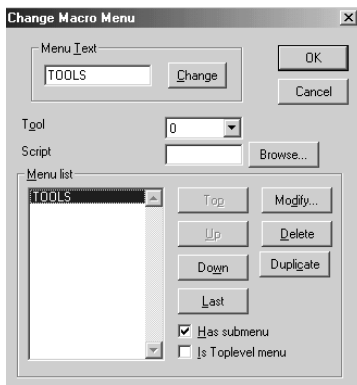
This section is dedicated to the **Tool** menu in the CAM. If by chance, you opened your BobCAD software for the first time, selected a configuration file and then discovered that there were no tools in the tool menu, it is because you will need to set them up. This is easily done and this section is going to show you how to do it.

STEP 1

Select the SPECIAL menu and choose a configuration that you will need. Then select OK to OPEN the CAM. Now select the SETUP menu and TOOL.

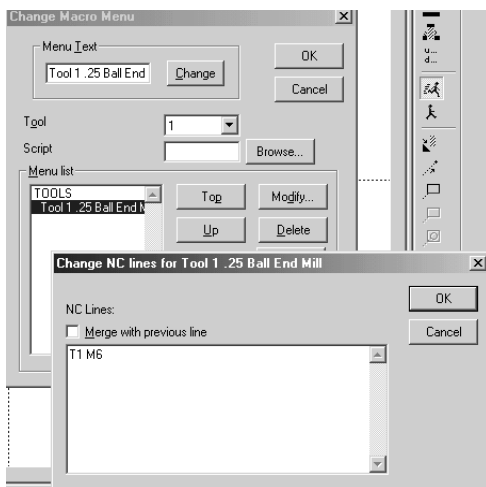
STEP 2

Now you will see the CHANGE MACRO MENU dialog for tools. Click your mouse button in the MENU LIST box and highlight the first blank slot. Now click your mouse in the MENU TEXT box and type in TOOLS. Now select the CHANGE button and check the option box called, "HAS SUBMENU."



STEP 3

Now select the next blank slot under TOOLS in the MENU LIST area. Then click the MENU TEXT slot and type in **Tool 1 .25 Ball End Mill**. Now select the CHANGE button and then select the drop down arrow next to the TOOL # and select 1. Now select the MODIFY button. See this example diagram below.



STEP 4

The “Change NC Lines for Tool 1 .25 Ball End Mill” dialog will appear. This is where you will type in the NC code used to designate tool 1 for your programs. Type in T1 M6 and select OK. Select OK again. Now go to the TOOL menu in the CAM and you will see that you have just customized the tool menu itself. By selecting tool 1, the T1 M6 code will be inserted into the NC editor.



This is the method in which you will be able to customize all of your tools. You can continue on at this time, if you would like and completely set up your tool menu.

Section 81. The CAM Macro menu

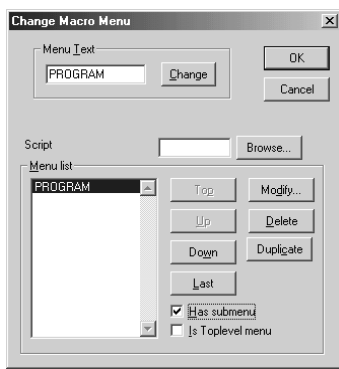
This section is going to show you how to customize the **Macro** menu of the CAM. This menu will allow you to save Start Blocks, END blocks, Spindle commands, coolant, fixture offsets and much more for use in CNC machining with BobCAD Version 21.

STEP 1

Select the SPECIAL menu and choose a configuration that you will need. Then select OK to OPEN the CAM. Now select the SETUP menu and MACRO. This will open the CHANGE MACRO MENU dialog. Just as you have learned in earlier sections on customizing menus, the same fundamental methods are used for customizing this menu.

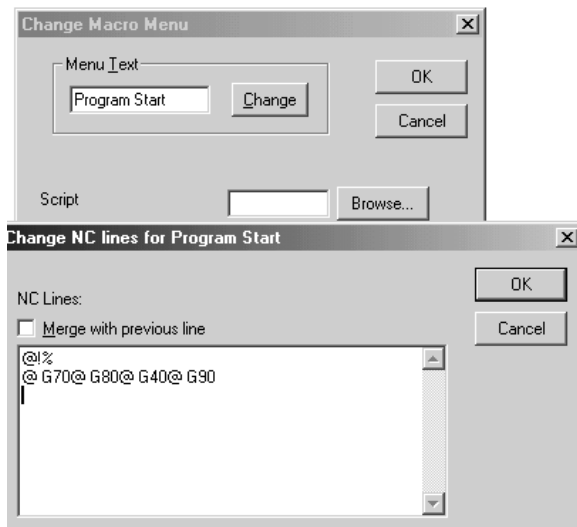
STEP 2

First, select the blank slot under MENU LIST by clicking on that slot and it will highlight. Now click your mouse in the MINU TEXT box and type in PROGRAM, select the CHANGE button and then select the HAS SUBMENU option. See the next image.



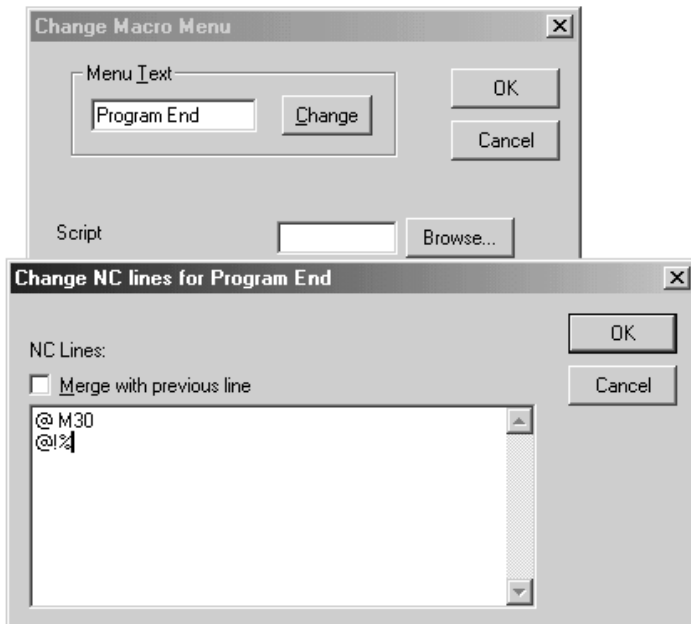
STEP 3

Now select the next blank slot under PROGRAM in the MENU LIST section so that it highlights. Now click the MENU TEXT box and type in **Program Start** and click the CHANGE button. It will appear under PROGRAM in the MENU LIST section. Now select the MODIFY button. This will open the CHANGE NC LINES for Program Start dialog. In this dialog, enter @!%, which will be the very first line of NC Code in your G-code programs. Now hit the ENTER key and on the next line down type in @ G70@ G80@ G40@ G90 and hit ENTER again. This is just an example of a start block. Your specific machine may want different information. When you go through the RS 232 Communications Setup Checklist, you will want to enter that information in this box for use with the selected configuration for your machine. You will need your machine handbook for this action.

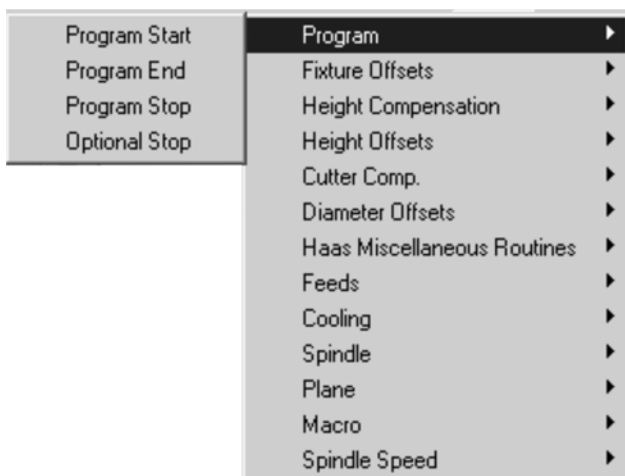


When finished entering the information, select OK and then OK again to exit the Change Macro dialog as well. You will find that this information that you just gave the software will be available for selection into your NC programs.

You can also add a Program END as well as other important machine option commands such as coolant, spindle speed, spindle clockwise and counter clockwise and so on. The next example image shows how you would enter in the PROGRAM END which would consist of @M30 and the @!% entry to indicate the stop and rewind M-code option.



Below, you can see what the MACRO menu could be customized to look like and provide. Continue using the same procedure to completely customize your Version 20 software to meet your specific machine requirements.



Section 82. The CAM Setup Menu

The **SETUP** menu offers fundamental customization functions as well as important setup functions that you will need to know about for making custom changes to your configurations and post processor settings. Let's take a look at this menu.

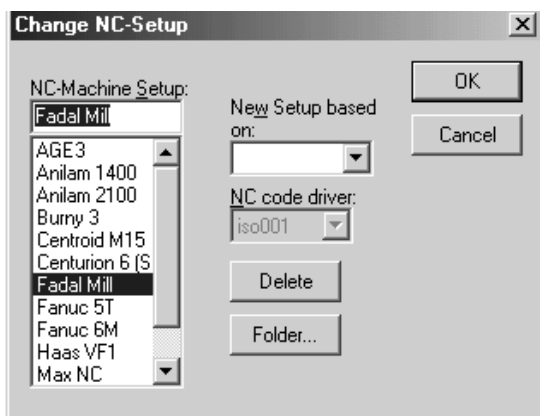


You can see from the menu image above that it is basically divided into two sections. The top four functions all lead to customizing those menus to meet your specific needs. Let's take a closer look at the bottom section of this menu. This includes:

- Configuration.
- Driver.
- Coordinates.
- Conversion.

These are all functions that deal with how the NC code will appear as you are writing programs for machining.

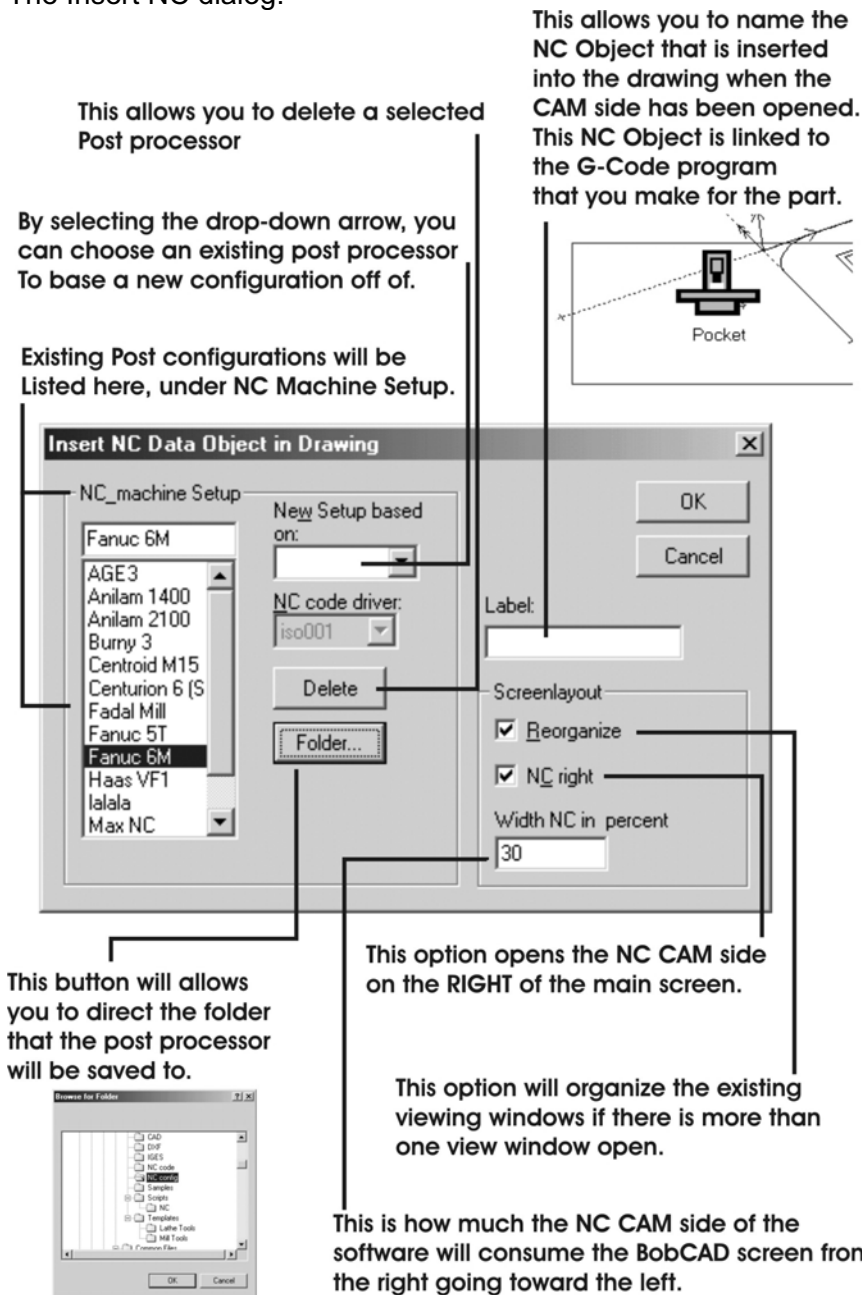
CONFIGURATION: When you open the CAM, you first select the SPECIAL menu and then INSERT NC. This opens your configuration dialog. You can also access this dialog through the **SETUP** menu of the CAM. Let's take a look.



The reason for this being available for you in the CAM as well is so that you can switch post processor configurations for different machines if you need to cut the same part on different machines. Therefore, you can have the same NC program for multiple machines in the NC editor.

Each time that you select a new configuration, a new NC Object will be inserted on the CAD screen. This is a way of showing you how many programs you may have open as well.

The Insert NC dialog.



DRIVER: The **Driver** function of the SETUP menu is important to be familiar with, as it has to do with incremental and absolute programming functionality. BobCAD Version 21 comes with 3 specific drivers as well. They are for:

1. An older Heidenhain controller for conversational code output.
2. An older Agie Wire Circa CNC 100/123 etc.
3. The most common driver that is the ISO code driver.

The ISO001 driver is the common driver used with BobCAD and offers:

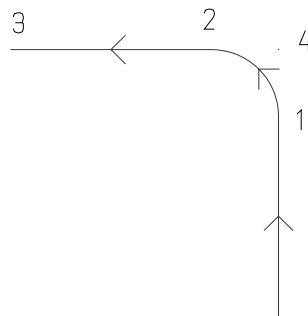
Sub-functions

- 1 - Normal mode, absolute or relative coordinates according to driver setup. (with **Options, Driver**)
- 2 - Absolute mode, all coordinates absolute.
- 3 - Relative mode, all coordinates relative.
- 4 - 4-axis Wire EDM.

Arcs

The direction of arcs (G02 or G03) is determined (as usual) by viewing the arc from the positive Z-axis. The driver generates I, J and K values for every arc. Please do not forget that arcs, which do not lie exactly in a coordinate plane, cannot be executed by the NC machine. You must first interpolate such an arc before executing it.

Options for arcs:



G02/G03 arc with R instead of I, J and K:

G02/G03 X.. Y.. Z.. R.. to (2)

Automatically divides complete circles into two semi-circles.

R positive: angle less than or equal to 180°

R negative: angle greater than 180°

G25 fillet between two lines:

G01 to intersection (4)

G25 R...

G01 to end point (3)

G06 tangential from line to point:

G01 to start of tangential arc (1)

G06 X.. Y.. Z.. to end point (2)

G26 fillet between line and line or arc:

G01 to end point fillet (2)

G26 R..

G01/G02/G03 etc.

G27 fillet between line or arc and line:

G01/G02/G03 to start point fillet (1)

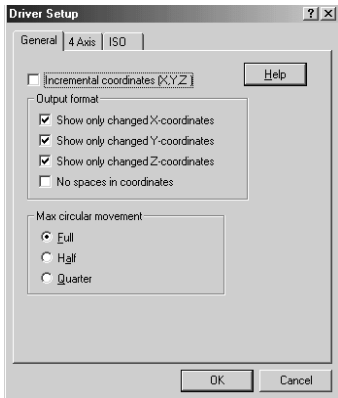
G27 R.

G01 to end point (3)

Turning

For turning with diameter input instead of radius you can set the option $Y = Y*2$. All Y values are multiplied by 2. When you copy code lines, then text as well as lines and arcs are placed on the clipboard, the driver interprets the lines according to the current setup. Only the codes G00, G01, G02 en G03 (also with radius) are recognized. The driver also recognizes the codes for absolute and relative coordinates: G90 & G91.

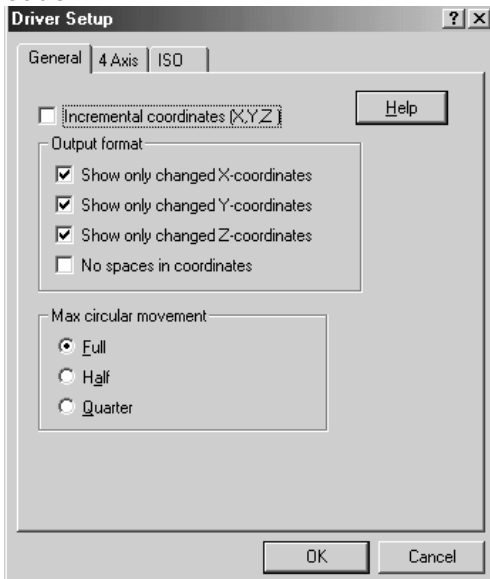
Let's take a look at the GENERAL tab of the **DRIVER SETUP** dialog.



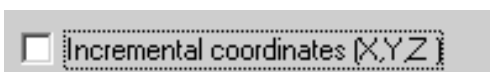
By selecting DRIVER from the SETUP menu, you will open the Driver Setup dialog as seen above. This dialog has three tabs. They are:

- General.
- 4-Axis.
- ISO.

GENERAL: This is where we set if we need to see every coordinate on every line of code or not, where we set if we need spacing, incremental coordinates and how much of a circle your machine control can cut in one code.

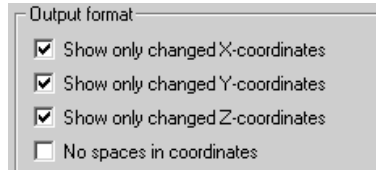


If you are programming in ABSOLUTE coordinates, you will UNCHECK the INCREMENTAL COORDINATES X, Y, Z option.



The **OUTPUT** format section provides options for:

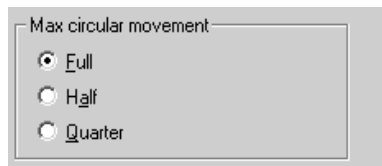
1. Showing **ONLY** changed coordinates in X. This would be checked if your controller doesn't need to see these coordinates on every line.
2. Showing **ONLY** changed coordinates in Y. This would be checked if your controller doesn't need to see these coordinates on every line.
3. Showing **ONLY** changed coordinates in Z. This would be checked if your controller doesn't need to see these coordinates on every line.
4. NO spaces between any coordinates in your NC program. Spaces make the code easier to read and most controllers accept them between X, Y and Z coordinates.



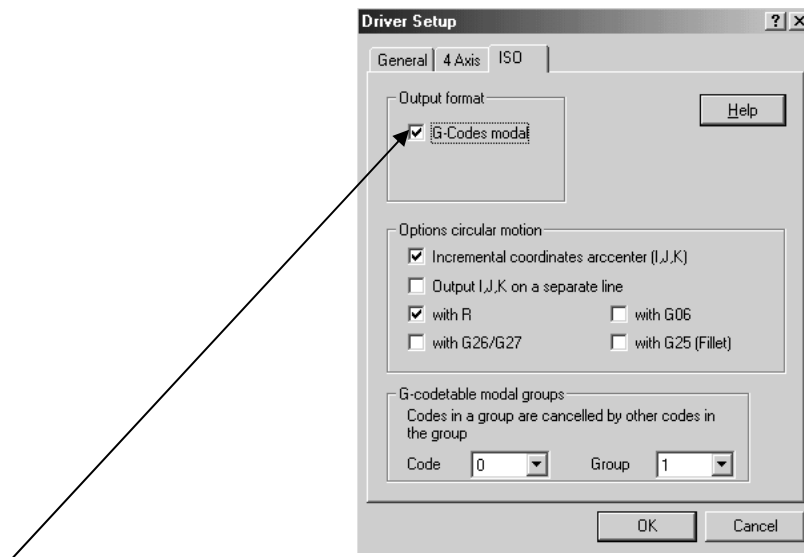
The **MAX Circular Movement** provides options for:

1. Cutting a complete 360-degree circle.
2. Cutting HALF of a circle.
3. Cutting circles in quarters.

Your machine control will require one of these settings. You will need to understand how your machine controller reacts to arcs. Do this by referring to your machine handbook as necessary.

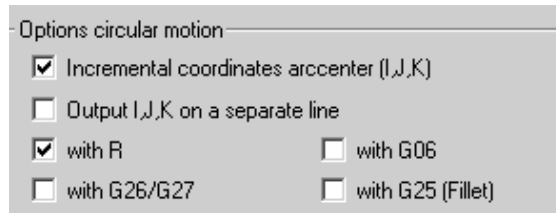


ISO: The ISO tab offers more important settings for configuring the BobCAD to communicate correctly with your machine control. You will need to verify these settings through your machine handbook. Let's take a look.



The **OUTPUT** Format section basically asks if you need your G-codes to be MODAL. This means the same thing as the SHOW ONLY CHANGED X, Y and/or Z option in the GENERAL tab. **If you need to see the g-codes repeated line after line then check this box.**

The **OPTION CIRCULAR MOTION** section requires selections regarding the ARC moves. These options are:

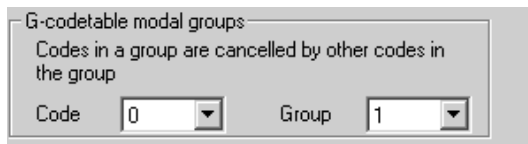


The 'Options circular motion' dialog box contains the following options:

- ☒ Incremental coordinates arccenter (I,J,K)
- ☐ Output I,J,K on a separate line
- ☒ with R
- ☐ with G06
- ☐ with G26/G27
- ☐ with G25 (Fillet)

- Incremental Coordinate ARC Center (I, J, K): If you need incremental arc coordinates then check this option. If you need ABSOLUTE arc coordinate output, then UNCHECK this option.
- Output I, J, K on a Separate Line: Some controllers require this. If so, check the option.
- Output with R: Some controllers need to see an R value after arc moves. If you need this, select this option.
- Output with G26/G27, Output with G06 and Output with G25 (fillet) all deal with corner rounding. Check your machine handbook for information on this.

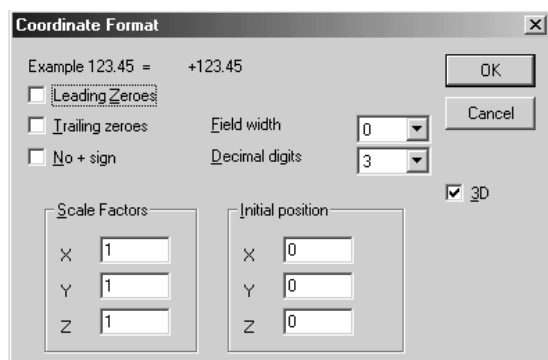
The **G-CODE Table MODAL Groups** section sets what BobCAD believes to be modal about our code. Therefore, whether a G01 cancels a G02 and so forth. If we wanted a G04 to cancel a G03 but NOT G01, we could set that by including the G03 in the same group as the G04. Note that this makes NO difference if our G-Code is set to MODAL or NON-MODAL. If we specify every code on every line, the modal grouping doesn't apply to non-modal code. BobCAD comes set with the most common modal grouping, so you will not change anything here most of the time.



The 'G-codetable modal groups' dialog box shows a text description: 'Codes in a group are cancelled by other codes in the group'. It features two dropdown menus: 'Code' set to '0' and 'Group' set to '1'.

4-AXIS: This tab deals directly with Wire EDM part programming. At this time, BobCAD Version 21 is not a Wire EDM CAD-CAM system.

COORDINATES: This section is going to allow you to modify the way the G-Codes are created and will be important when building a matching post processor you're your machine. Let's take a look at this dialog.



The 'Coordinate Format' dialog box includes the following settings:

- Example 123.45 = +123.45
- ☐ Leading zeroes
- ☐ Trailing zeroes
- ☐ No + sign
- Field width: 0
- Decimal digits: 3
- ☒ 3D
- Scale Factors: X=1, Y=1, Z=1
- Initial position: X=0, Y=0, Z=0
- Buttons: OK, Cancel

This gives you the ability to decide how many decimal places you want your code to output. You can also decide the field width of each coordinate value. You can also specify that you want your codes to have leading zeros or trailing zeros.

Initial Position means that you can change the position of where your tool's home is.

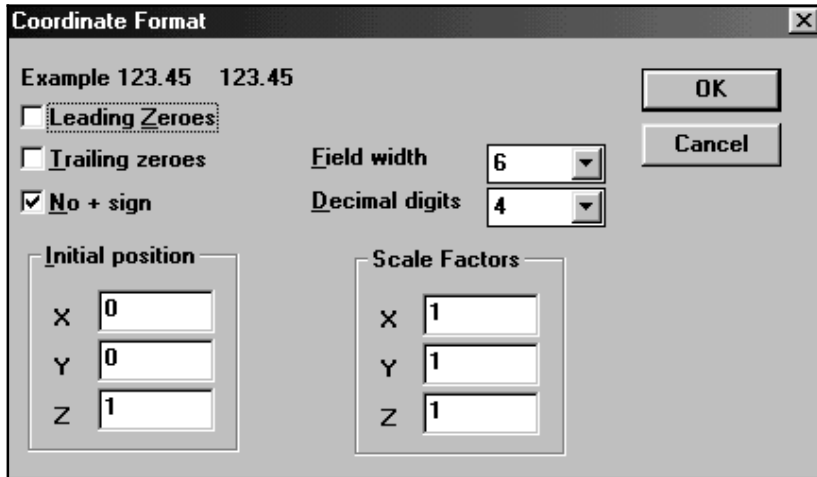
You can also modify the scale of your coordinate movements.

NOTE: FIELD WIDTH AND DECIMAL WIDTH ARE VERY IMPORTANT TO THE CALCULATION OF X, Y, Z, I, J OR R MOVEMENTS. MAKE SURE THAT THIS IS SET PROPERLY BY REFERING TO YOUR MACHINE MANUAL OR USER GUIDE.

Here is an example: If you want 4 decimal digits (places to the right of the decimal point) and two numbers to the left of the decimal place like 98.7654, you would have 6 in the **FIELD WIDTH** and 4 in **DECIMAL DIGITS**. (There are 6 total digits and 4 digits to the right of the decimal)

Also, you need a minimum of 2 for **DIGITS** added to the number of **DECIMAL DIGITS** to the right that you choose to put in **FIELD WIDTH** or your dimensions will not be calculated properly.

The **COORDINATE FORMAT** dialog box has 4 sections to it. They are:



- A. LEADING ZEROES:** This allows you to specify that there will be leading zeros in your program before the decimal points.
TRAILING ZEROES: This allows you to specify that you want to have trailing zeroes after the decimal points in your program.
NO + SIGN: This will allow you to have + and – symbols next to your X and Y moves but before the actual coordinate. Here is an example:

```
N10 G00 Z+0.  
N20 X-1.0208 Y+0.0365  
N30 G01 X+1.2396 F24.12  
N40 Y+1.8385  
N50 X-1.0208  
N60 Y+0.0365  
N70 G00 X+0. Y+0.
```

- B. FIELD WIDTH:** This allows you to choose your field width of each **COORDINATE VALUE**. This is how much room to give for the numbers in the code.



NOTE:

Field width and decimal digits are a very important to the calculation of X, Y, Z, I, J or R movements. Make sure that this is set properly by referring to your MACHINE MANUAL or user guide first.

Here is an example: If you want 4 decimal digits (places to the right of the decimal point) and two numbers to the left of the decimal place like 98.7654, you would have 6 in the FIELD WIDTH and 4 in DECIMAL DIGITS. (You see, there are 6 total digits and 4 digits to the right of the decimal.)

Also, you need a minimum of 2 for DIGITS added to the number of DECIMAL DIGITS to the right, that you choose to put in FIELD WIDTH or your dimensions will not be calculated properly.

C. INITIAL POSITION: This is where you can change the position of where your tools home is.

D. SCALE FACTOR: This sets the scale of your coordinate movements. If it is set at 1 for X, Y and Z then the software will basically mirror your X, Y and Z movements. If you have a 1-inch line and you want to have a 2-inch movement, then you would set the scale to 2 for whatever axis you were moving in. If you have a 2-inch line and set the scale factor to 2 then you are going to get a 4-inch move or 4-millimeter move depending on if you are working in inches or millimeters. You see, it multiplies the movement by the scale factor. Usually this scale will be set at 1 for X, Y and Z.

Initial position		Scale Factors	
X	0	X	1
Y	0	Y	1
Z	1	Z	1

Remember, the scale factor that you have set will multiply you movements by the value entered into this SCALE FACTOR box.

Section 83. Standard RS 232 Communications Checklist

This checklist is an aid to configuring the BobCAD-CAM Version 21 software for use with virtually any CNC controller. The checklist is used in conjunction with the machine handbook to write down settings for use in the Version 21 software once the checklist is completed.

DRIVER SECTION

This section applies to the Driver Setup that is located under the Setup menu of the NC CAM under the GENERAL TAB.

1. Does your controller require Incremental Coordinates for X, Y and Z?

YES_____ NO_____ If NO, do not check this option.

NOTE: *If you are programming in ABSOLUTE MODE you MAY need to do a special conversion to remove the K parameter from the arc moves. Here is how you do this if necessary:*

- i. You would select the SETUP menu of the CAM and then CONVERSION.
- ii. Under the ORIGINAL column, enter in: **K*[0-9]*.[0-9]***
(There are no spaces in the formula above.)
- iii. Under the CONVERT TO column, leave this space BLANK with nothing in there.
- iv. Select OK to save the settings.

2. Look at a sample G-Code program for your machine controller and determine the following.

Does your controller need to see only X changed coordinates in the code?

YES_____ NO_____ If NO, do not check this option. Generally this option will be checked for standard G-Code programming.

Does your controller need to see only Y changed coordinates in the code?

YES_____ NO_____ If NO, do not check this option. Generally this option will be checked for standard G-Code programming.

Does your controller need to see only Z changed coordinates in the code?

YES_____ NO_____ If NO, do not check this option. Generally this option will be checked for standard G-Code programming.

3. Does your controller need to see spaces between each coordinate value?

YES_____ NO_____ If NO, do not check this option. If the NC Code program needs to see spaces between every coordinate, select this option.

4. What is the MAX CIRCLE MOVEMENT of your machine?

FULL_____

HALF_____

QUARTER_____

Select the option that is required by your CNC Machine control.

This section is for the ISO TAB in the Driver Setup dialog. It is important that these options be selected correctly for an accurate Post Processor configuration to exist.

5. Does the G-Code output need to be Modal or NON-Modal?

X, Y & Z coordinates on every line of code? YES_____ NO_____

If yes, select this option. If only changed coordinates need to be in the code, do not select this option, as it will output non-modal codes.

6. Does your CNC controller require incremental coordinates of I, J & K for circular motion?

YES_____ NO_____ If NO, do not select this option.

7. If the answer to number 6 was YES, do these I, J & K coordinates need to be output on a separate line of code?

YES_____ NO_____ If NO, do not select this option.

8. Does your CNC controller require an "R" for circular moves?

YES_____ NO_____ If YES, select this option. The R stands for RADIUS.

9. Does your CNC controller require a G26 or G27 code output for circular movements? If so, select this option. Consult your machine handbook for this information as these are user defined.

10. Does your machine controller require a G06 code with circular movements?

YES_____ NO_____ If YES, select this option.

11. Does your machine controller need to see a G25 (corner rounding with R for radius) when machining fillets?

YES_____ NO_____ If YES, select this option.

Coordinate FORMAT SECTION

The Coordinate Format setup is located in the CAM under the SETUP menu and is important for your code output.

12. Does your g-code output need to show leading zeros in the code?

YES_____ NO_____ If YES, select the option

13. Does your g-code output need to show trailing zeros?

YES_____ NO_____ If YES, select this option.

14. Does your G-Code output need to show a + symbol for positive coordinate movements?

YES_____ NO_____ If YES, select this option.

15. FIELD WIDTH: How many total decimal digits does your code need to show? ***It is important to see a sample g-code program for your controller for this, so that you can review how the code looks.***

Field Width_____

16. How many digits are shown to the RIGHT of the decimal? ***It is important to see a sample g-code program for your controller for this, so that you can review how the code looks.***

Decimal Digits_____

17. For the INITIAL POSITION, where would you like your tools home position to be in the BobCAD software?

X_____ Y_____ Z_____

18. SCALE FACTORS: ***This sets the scale of your coordinate movements. The scale values that you set will multiply your movements by those values that you set. Typically, the X, Y and Z scale factors will be set at 1 for each. If X, Y and Z are set at 1, the software will basically mirror your movements. If you have a 1-inch line and want to have a 2-inch movement, then you would set the scale to 2 for whatever axis you would be moving in.***

X_____ Y_____ Z_____

SPECIAL NOTES: (If you have notes, enter them here.)

Section 84. The Coordinate Format Checklist

This set up checklist is going to help you set up the **COORDINATE FORMAT DIALOG** in the NC CAM SETUP menu. Use this checklist in conjunction with your machine handbook to accurately set up your specific post processors. There are two copies in this book for you so you may remove this checklist from the book if needed.

Does your controller need to see LEADING zeroes in the G-Code program?

YES _____ Check the function if yes.
NO _____ Do not check, if no.

Does your controller need to see TRAILING zeroes in the G-Code program?

YES _____ Check the function if yes.
NO _____ Do not check, if no.

Does there need to be + and – symbols in front of coordinates in your program?

YES _____ Check the function if yes.
NO _____ Do not check, if no.

How many digits need to be in the program for each coordinate?

FIELD WIDTH _____

How many places to the RIGHT of the decimal point does there need to be in the program? This is the DECIMAL DIGIT value.

DECIMAL DIGITS _____

Where does your INITIAL POSITION for your tools home need to be?

INITIAL POSITION:

X _____

Y _____

Z _____

What do your programming SCALE FACTORS need to be?

SCALE FACTORS:

X _____

Y _____

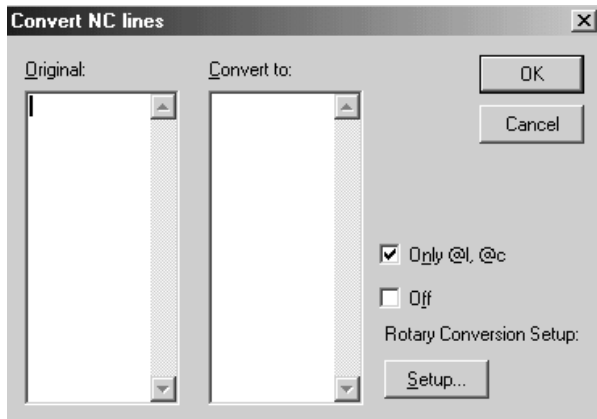
Z _____

Make sure that you select OK after you make all of the setting selections.

SPECIAL NOTES:

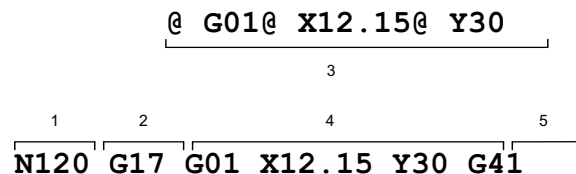
Section 85. The Conversion Function

The **Conversion** function in BobCAD 21 allows you to change one designated NC Code character to another for programming with a wide variety of machine controllers. Let's take a look at this dialog.



BobCAD offers as standard many possibilities to adjust the format of NC codes that are to be generated. Should this prove inadequate, and then you can take recourse to conversion. Conversion consists of textual replacement of certain characters in the codes by other characters. You can not only directly replace text but also swap text as BobCAD uses pattern recognition and pattern substitution. The conversion is performed just before the generated NC codes are written to the NC window. You can use conversion for instance to replace all points in a decimal number with commas. BobCAD has no other means of doing this.

The example below shows schematically how an NC code line is created for a line in a contour. We assume here that you have entered the text " G17@I G41" after using the command **Contour** in the **Options** menu:



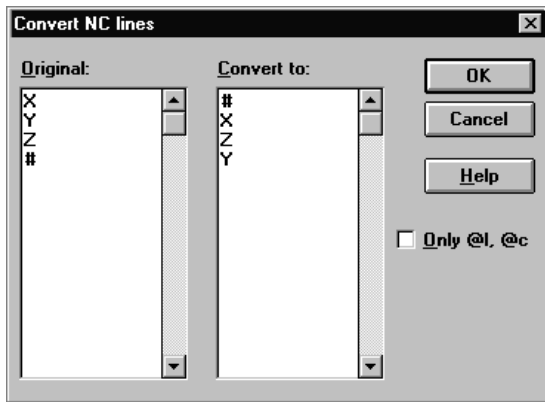
Construction of an NC line of code: an Example

1. BobCAD adds the line number (1).
2. Your text is added (2).
3. The text generated by the driver (3) is first converted and then added to the line (4).

The text (@I in the case of a contour or @c for a cycle) contains so-called soft spaces ("@ "). If the line of code is empty at the moment of conversion, then the soft space at the beginning of the converted text is not added to the final line of code. This is to cover the exceptional situation where the line numbers have been deactivated and you do not want a space at the start. Soft spaces are converted to hard spaces when added to the line of code.

4. BobCAD adds the rest of your text (5).

Important: You can find soft spaces by means of the pattern "@ " (@+ space). You can see that in the above example the @c and @I patterns always begin with a soft space.



The input to the conversion consists of two pieces of text. The first part is the pattern that is to be searched for. The second part is the text that is to be used as replacement.

The conversion process for text that has been produced by a driver proceeds as follows:

Process the target patterns line for line until a pattern has been found. Replace the pattern with the substitute text and continue with the next pattern until there are no more patterns to be replaced. Normally BobCAD only looks for one occurrence of a pattern in the text. You can instruct the software to look for all occurrences of the pattern by adding the character "&".

Special pattern characters

Pattern character: Corresponds to:

- % start of the line to be converted
- \$ end of the line to be converted
- ? an arbitrary character
- * 0 or multiple occurrences of the previous character
- [0-9] an arbitrary digit
- [A-Z] an arbitrary letter
- & When this character occurs at the end of the pattern, then the conversion must be repeated as many times as possible.

Here is an example for a lathe.

The X & Y coordinates must be converted into Z & -X, and I and J in K and -I. No other coordinates may occur because the machine only works in a flat surface. Furthermore the coordinates may contain unnecessary decimal points (not 125. but 125). The tables show each time the results of the conversions.

Remove Z and K :

"@ Z-*[0-9]*.[0-9]*" change to "nothing"

"@ K-*[0-9]*.[0-9]*" change to "nothing"

N90 G00 X40.5 Y0. Z0.	N90 G00 X40.5 Y0.
N100 G01 Y10. G41	N100 G01 Y10. G41
N110 G03 X0. Y10. I10. J0. K0.	N110 G03 X0. Y10. I10. J0.
N120 G01 X90. G40	N120 G01 X90. G40
N130 X40. Y0.	N130 X40. Y0.

X and Y becomes Z and X:

"X" to "Z"
 "Y" to "X-"
 "X--" to "X"

I and J becomes K and I:

"I" to "K"
 "J" to "I-"
 "I--" to "I"

N90 G00 X40.5 Y0.	N90 G00 Z40.5 X-0.
N100 G01 Y10. G41	N100 G01 X-10. G41
N110 G03 X0. Y10. I10. J0.	N110 G03 Z0. X-10. K10. I-0.
N120 G01 X90. G40	N120 G01 Z90. G40
N130 X40. Y0.	N130 Z40. X-0.

Switch Z and X in the line:

"{Z-*[0-9]*.[0-9]*}@ {X-*[0-9]*.[0-9]*}" to "@2 @1"

Switch K and I in the line:

"{K-*[0-9]*.[0-9]*}@ {I-*[0-9]*.[0-9]*}" to "@2 @1"

N90 G00 Z40.5 X-0.	N90 G00 X-0. Z40.5
N100 G01 X-10. G41	N100 G01 X-10. G41
N110 G03 Z0. X-10. K10. I-0.	N110 G03 X-10. Z0. I-0. K10.
N120 G01 Z90. G40	N120 G01 Z90. G40
N130 Z40. X-0.	N130 X0. X-0.

Remove the decimal point if the coordinates end with a decimal point:

".@ &" to "nothing"
 ".\$" to "nothing"
 N90 G00 X-0. Z40.5

N90 G00 X-0. Z40.5	N90 G00 X-0 Z40.5
N100 G01 X-10. G41	N100 G01 X10 G41
N110 G03 X-10. Z0. I0. K10.	N110 G03 X10 Z0 I-0 K10
N120 G01 Z90. G40	N120 G01 Z90 G40
N130 Z40. X-0.	N130 Z40 X-0

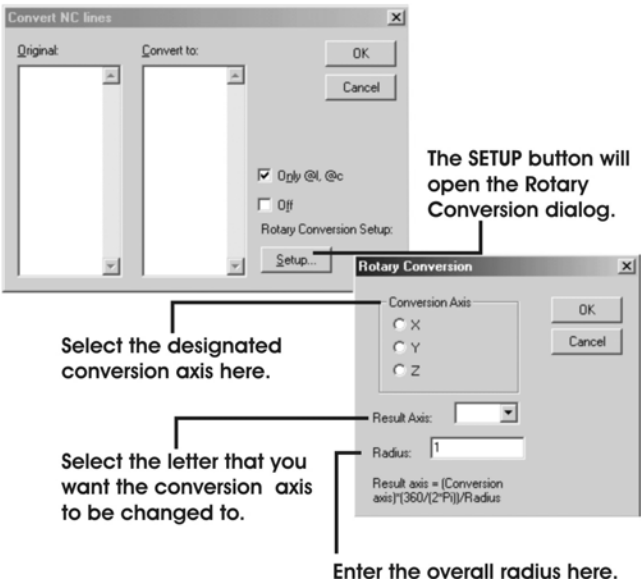
Remove the "-0"
 "-0@ &" to "0"
 "-0\$" to "0"
 N90 G00 X-0 Z40.5

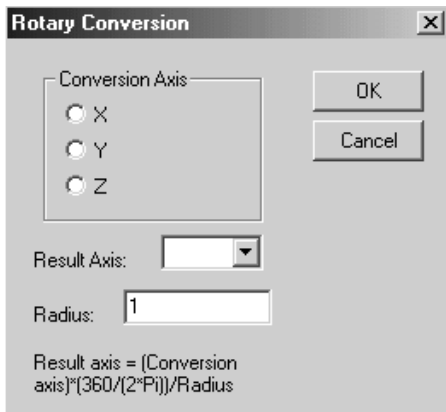
N90 G00 X-0 Z40.5	N90 G00 X0 Z40.5
N100 G01 X-10 G41	N100 G01 X10 G41
N110 G03 X-10 Z0 I0 K10	N110 G03 X10 Z0 I0 K10
N120 G01 Z90 G40	N120 G01 Z90 G40
N130 Z40 X-0	N130 Z40 X0

These are all sample conversions that will help you in different scenarios. We have also included a list of standard G & M codes for you starting below so that you can also make special conversions for them as needed.

Section 86. Rotary Conversion

The BobCAD software offers a 4th AXIS conversion that is located in the conversion dialog. Let’s take a look below.





The top portion of the dialog asks us which axis will be changed into the rotary axis. For example, choose Y and then select A for the result axis. The radius area needs to know the radius of the bar stock that you are cutting. You would type in 2 if you were cutting a 4-inch piece. You would click OK and then click OK again to exit the conversion dialog. Then you would go back to the SETUP menu and select COORDINATES. The scale factor for Y would be changed (Remember, we selected Y as the conversion axis). The number that is there will now need to be multiplied by 25.4. You would need to multiply that number and then enter the result value into the Y box. Now, when BobCAD moves in the Y-axis, it will generate A-Axis moves in degrees. It's actually quite a simple function.

CONVERSION SECTION:

The CONVERSION setup is located in the CAM under the SETUP menu and is important for your code output.

The “Convert From” section will list out the code that you want to change. The “Convert To” will list out what you want the code to be changed to. The section below will allow you to list the conversions. You will need to consult your machine handbook and will need to see a sample g-code program for your CNC controller as well, to make correct conversions as needed.

[illegible]

Section 87. 4th Axis Indexing Lesson

This function has been especially designed to simplify the Programming of the **4th Axis**. This function allows for the drawing to be made in the 2 or 3 axis environment and then easily converted suiting the 4th Axis.

These Programs can be made in the Standard ISO or Fanuc type format including I & J commands etc. or, more simply explained, just as though the shape were unfolded.

This function is used to convert 2 or 3 Axis NC programs using X, Y and Z into 4th Axis NC Programs using rotary axis A, B or C.

For this function we first need a CAD drawing with the circumference of the cylinder projected into a flat surface.

Incremental programming.

NOTE: When creating the NC Program for the 4th Axis Conversion the settings in NC-CAM need to be set to Incremental for both co-ordinates and for CIRCLE I&J positions otherwise the file will not be converted correctly.

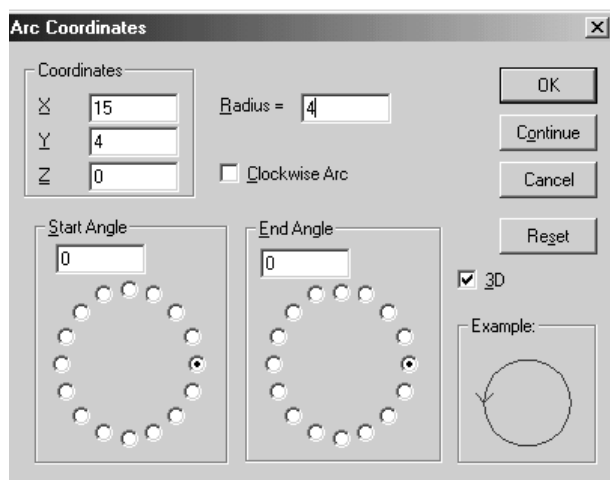
Absolute programming

For Programmers not comfortable with incremental programming the file can be created absolute in BobCAD, but all curves must be interpolated before the NC file is created for the files to be converted correctly.

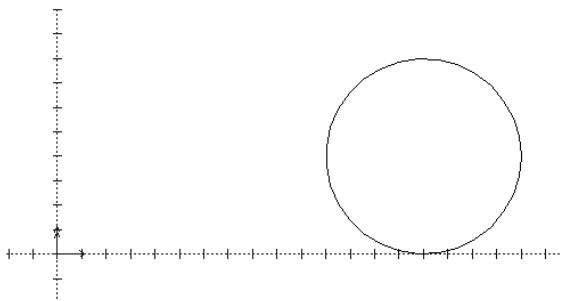
Follow the procedure below convert to 4th Axis of a cam track around a 4" shaft programmed in Fanuc type code.

STEP 1

Start a new drawing. Select ARC from the Main Menu and choose Coordinate Center. Enter 15 for X, 4 for Y, 0 for Z and 4 for the radius. Now select OK.

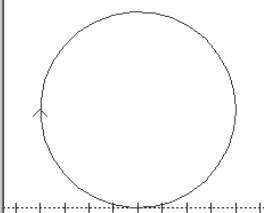
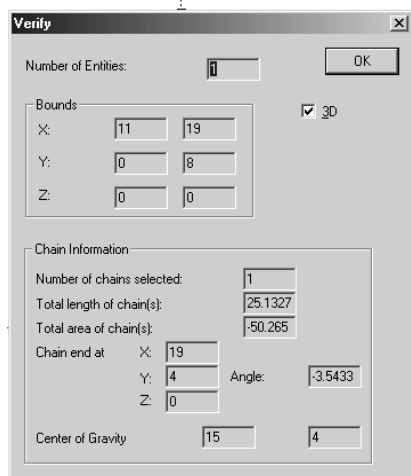


The result: An 8 diameter circle appears on the screen. If the circle is not within the viewing area click the Zoom All Icon . Then select the Zoom Out Icon  twice.



STEP 2

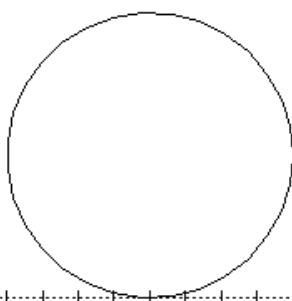
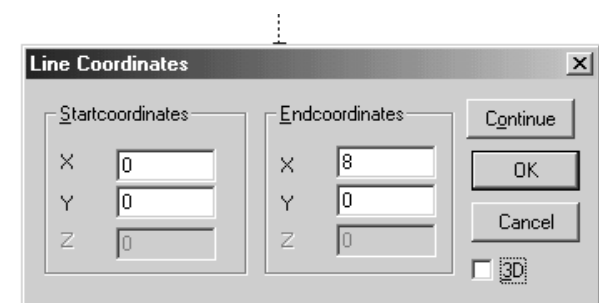
Click on the circle to select it, and then select SPECIAL from the File Menu. Now choose the VERIFY function. See the result below.



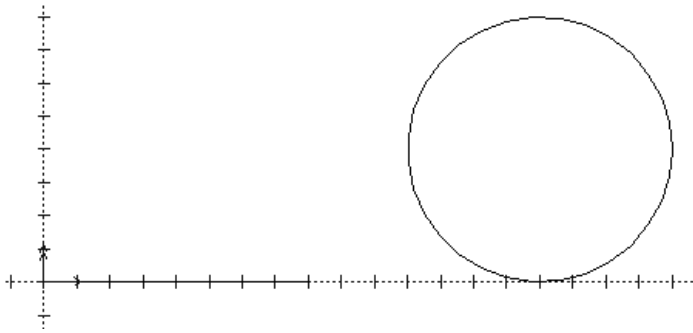
Result: The circumference is listed as 25.13274. Write this value on a notepad for future reference and Click OK.

STEP 3

Select the LINE menu and then Coordinate from the Main Menu. Enter the following values as seen in the dialogue box below and choose OK.

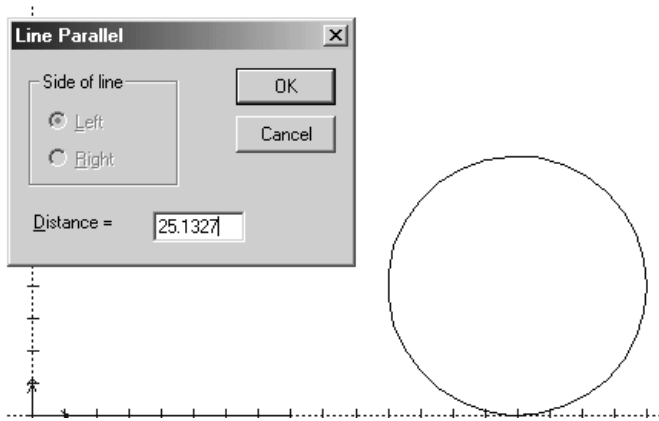


Result: A horizontal line is drawn that is 8 long in X and begins at X0, Y0.

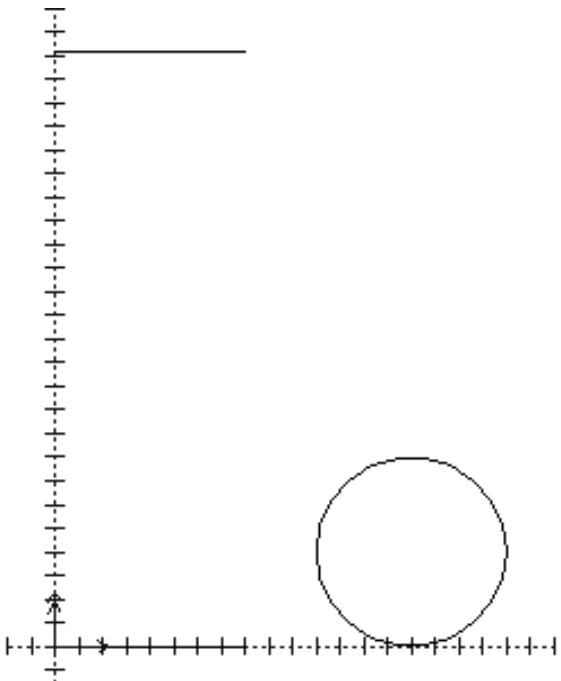


STEP 5

Select LINE from the main menu and choose PARALLEL. Enter the value obtained from the circumference earlier. (25.13274) into the Parallel dialog for DISTANCE. See below.

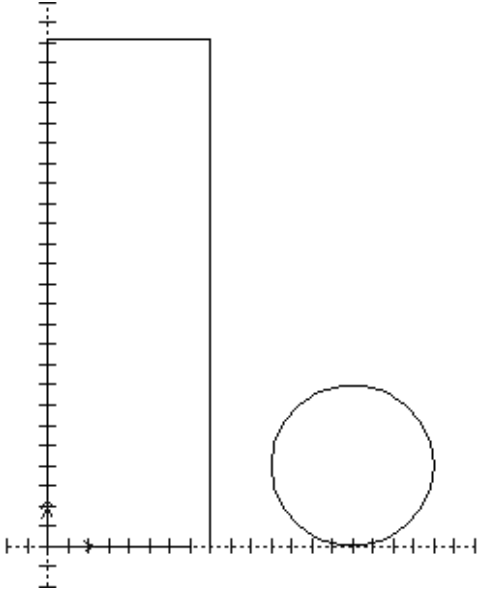


Once you have entered in the information and selected OK, you will click on the small line that you drew, move your cursor ABOVE the line and then click your left mouse button to draw the new line. When you have done this, click your right mouse button to exit the function.



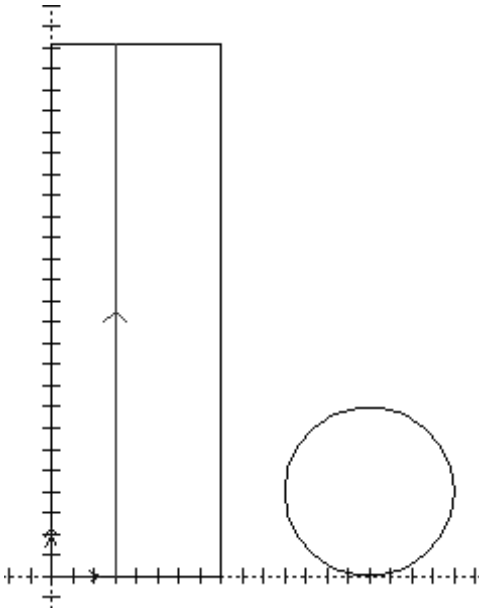
STEP 6

Now we want to join both sides of the two lines to make a rectangle box. Select the LINE menu and then JOIN. Click your mouse on one corner and drag a line to the corresponding upper or lower corner on both sides. When finished click your right mouse button to exit the function.



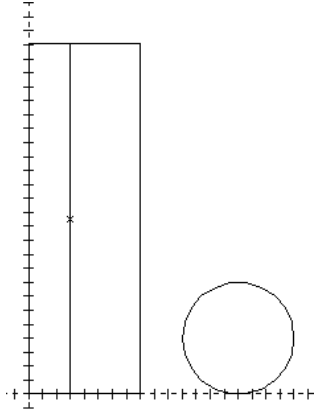
STEP 7

Select the LEFT vertical line first. It will highlight. Now choose the LINE menu and PARALLEL. Select the option RIGHT and enter a value of 3 for the distance and click OK. See the result below.



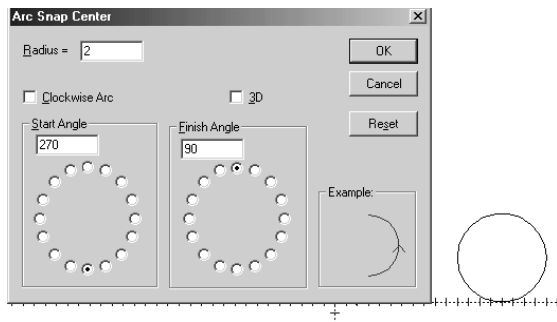
STEP 8

Select the POINT menu and ON ENTITY. Select 'Relative' under 'Mode' and enter in 50 for the ratio. Now click OK. If the line was already selected the point will automatically be drawn. If NOT, select the line and then use the ON ENTITY function to draw the point.

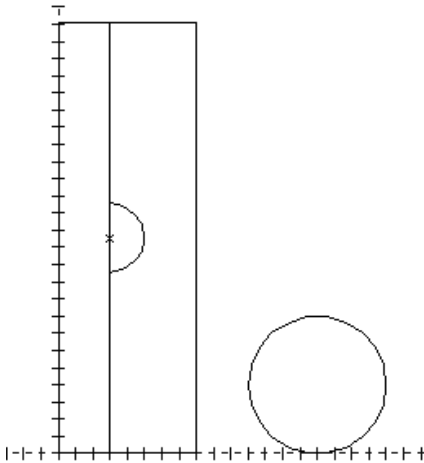


STEP 9

Now select the POINT menu and SNAP CENTER. The following dialogue box will appear. Enter the values shown and click OK. Place the cursor next to the point just drawn and left-click your mouse to snap the arc. Now right-click your mouse to exit that function.



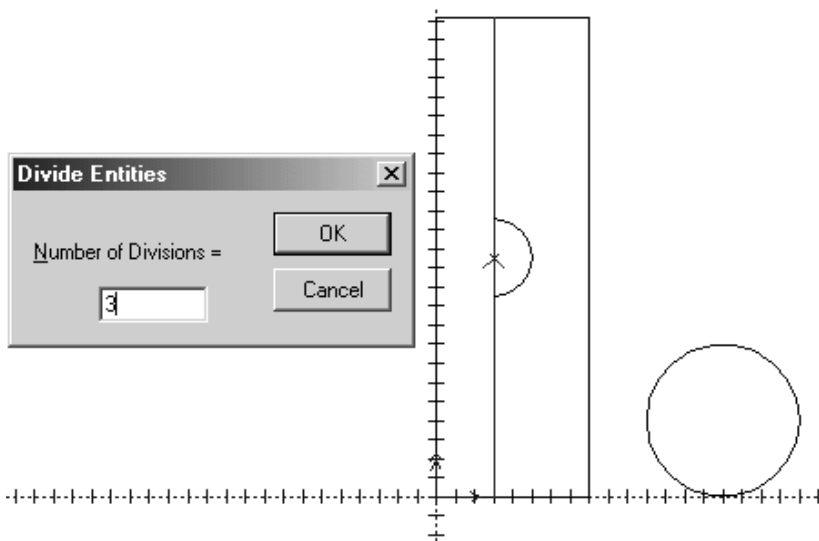
The result:



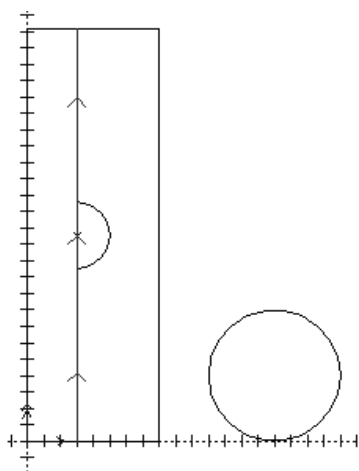
A 180-degree circle is drawn around the point. The arc begins at 270 degrees and ends at 90.

STEP 10

Place the cursor on the center vertical line and click on it to select it. From the CHANGE Menu choose BREAK and then select DIVIDE. Enter 3 for number of divisions and then select OK.



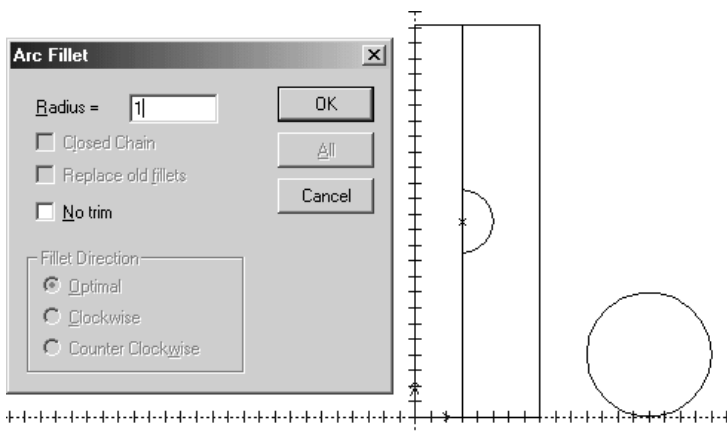
Result: The center vertical line is broken into three equal lengths. See below.



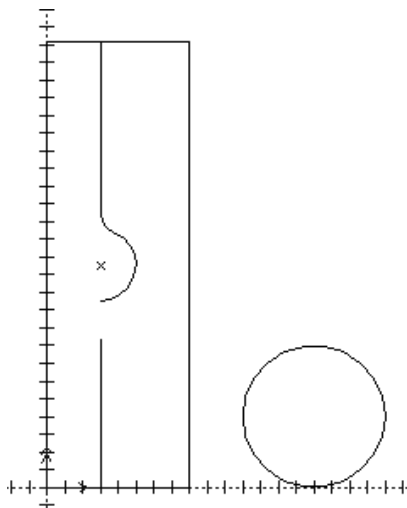
You can see that the center line is now divided into 3 segments.

STEP 11

Click your mouse off the geometry now to deselect everything if the entities are still selected. From the ARC menu choose FILLET. In the dialogue box enter 1 for the radius and select OK. With the cursor select the center vertical line ABOVE the arc and left mouse click. Then select the ARC with the left mouse button.

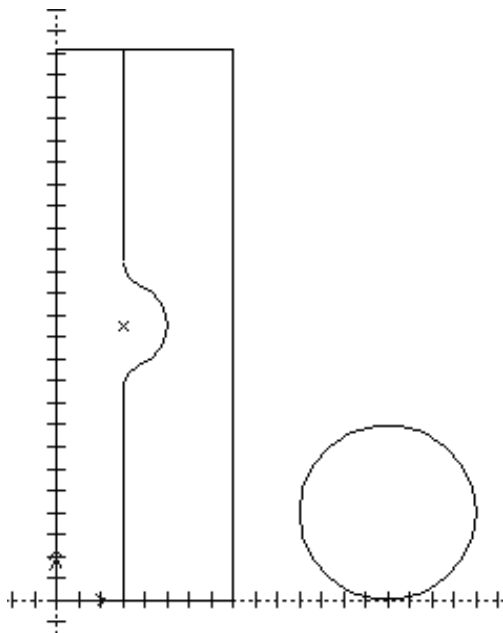


Below you can see the first fillet result.



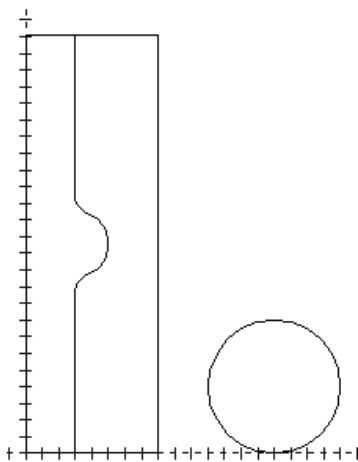
STEP 12

Now select the ARC menu again and FILLET. Leave the radius value at 1 and click OK. Select the part of the arc just before it intersects with the lower part of the line and then select the part of the line just before it intersects with the lower part of the radius to create the second fillet.



BobCAD should provide you with an actual preview of the result before you make the move to execute the function.

Click on the POINT that is left on the screen and delete it.



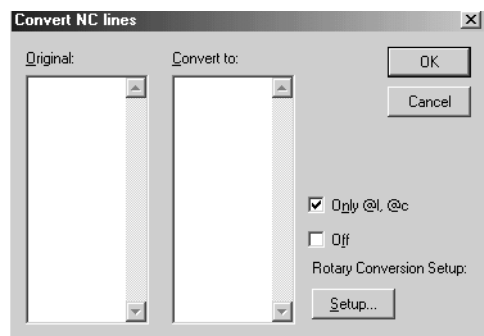
Now we are ready to generate an NC G-code program for this part.

STEP 13

Select the SPECIAL menu and INSERT NC. Select a FANUC configuration and click OK to open the CAM.

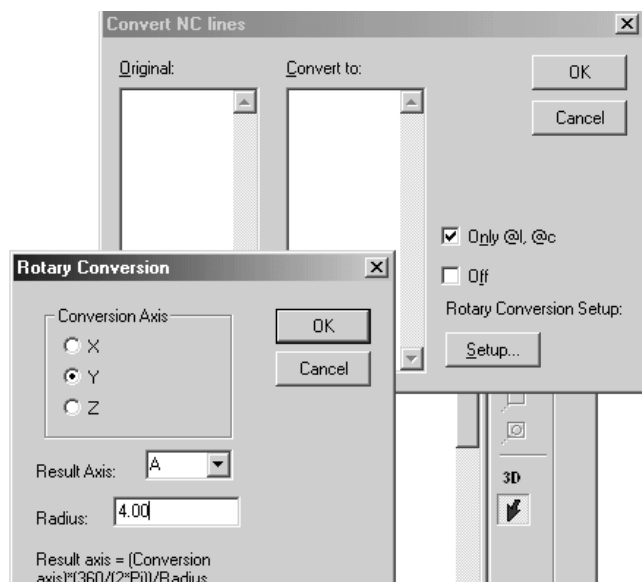
STEP 14

Now select the CAM - SETUP menu and then CONVERSION.

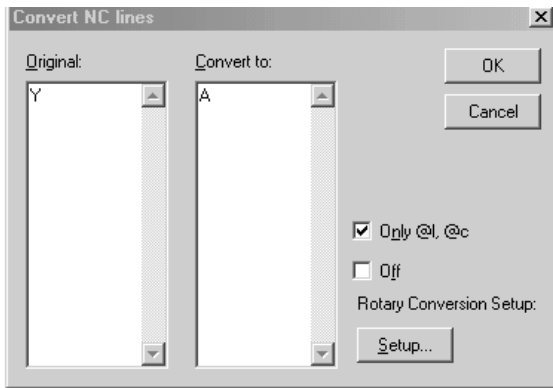


STEP 15

Select the Rotary Conversion Setup button in this dialog. Now select "A" from the RESULT AXIS drop down menu. Enter 4.00 as the radius. See below.

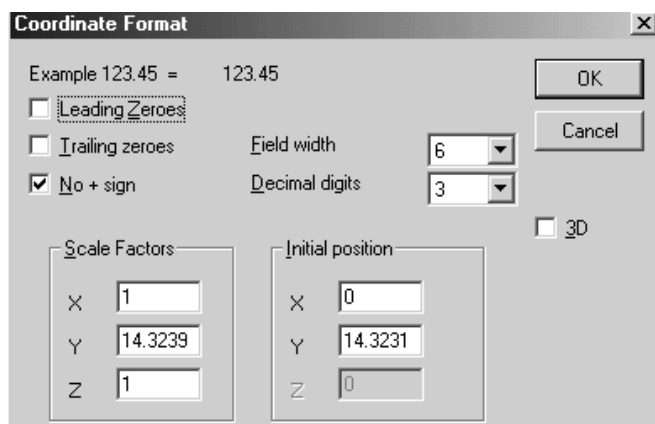


By selecting OK, you will automatically see the Conversion dialog change. See this below.



STEP 16

Now select the SETUP menu and COORDINATES.



Multiply the existing value for the Axis to be converted to 4th axis rotary by 25.4 (In this case the Y-Axis) and enter that value. $.5639 \times 25.4 = 14.3231$, which is what you can see in the image above.

Once you have done this, click OK. You have now returned to the main screen.

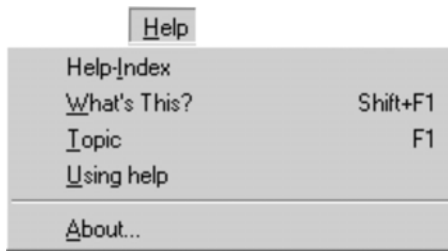
STEP 17

Select Machine Auto from the NC-CAM menu by either selecting it from the Machine Menu or by clicking the Auto-Cut Icon. . Select the Center vertical line just above the bottom horizontal line, point the arrow up by moving your cursor, click your mouse button once and then click on the last part of that line or hit F3 on your keyboard. This will generate the correct NC code for the 4th axis.

Congratulations! You have completed the 4th axis lesson.

Section 88. The Help Menu

The CAM also has a special HELP menu that can be accessed. Let's take a look.



The HELP menu offers:

The HELP – INDEX.

What is This? This allows you to select a icon or function and then use this function to gather information about it.

Topic. This opens up the main BobCAD help files.

Using Help. This instructs you on how to use the help files.

ABOUT. This gives you the software information and where it came from or who provided it.

CHAPTER 11

CNC LATHE

Section 89. Introduction & Basics



Section 89. CNC Lathe Basics

This section is dedicated to instructing you on 2-axis CNC **Lathe** programming with the BobCAD software. Let's take a look at some of the basics that you will need to know right from the start.



The BobCAD software will be set in **MILL** mode when you open the software for the first time. Above, you can see an image of the main toolbar. There is a LATHE mode function icon that if selected will change your system from MILL mode to Lathe mode automatically. This button has a BLUE DIAMOND shape on it. This icon is also seen below.

IMPORTANT: *before attempting to create a lathe program, you MUST turn the lathe mode ON FIRST. The reason for this is because dialog function boxes will change to suit lathe programming throughout the software. Therefore, remember this basic rule.*



The diamond shaped icon is the LATHE MODE icon function that you can see here. By selecting this icon, it will highlight, meaning that you are actively in Lathe Mode.

Important Note for Lathe Programming

If your machine needs the full diameter for X moves and you have “**Y=Y*2**” under the **File** menu and **Environment** checked (*on the CAD side or under Setup in the NC CAM side*), you MUST type in twice the values specified for all X coordinates in ALL lessons in this addendum. BobCAD will draw only the half profile you need, and will perform the multiplication needed for the X moves in the NC Code. If you forget to turn this on under **File** and then **Environment**, don't worry. You will NOT need to redraw each lesson if you desire to get the correct code for your machine controller. You will only need to change the scale factor for the X-axis to 2 instead of the normal 1 under **NC CAM**, **Setup** and then **Coordinates**. Please read through the RS232 Communication Setup guide on how to set up the NC CAM for your specific controller.

TOOL MERGE & TOOL SETUP:

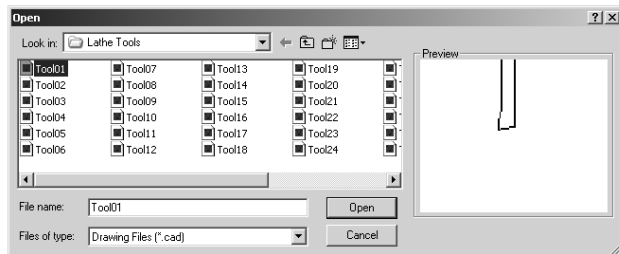
Before you draw your OD or ID profile for creating a lathe toolpath and G-Code program you can perform a **Lathe Tool Setup**. Once you have set up a lathe tool for a program, you do not have to do it again. Therefore, you can delete the lathe tool off the screen. If you **IMPORT** a profile for use in programming, you will want to do the tool setup after opening the file. All that is needed to perform a lathe program is the OD/ID profile and an outline of the stock.

After reviewing this manual you may decide that it is easier for you to just draw your part in BobCAD. The following is the instruction for where to locate your BobCAD lathe tools.

Merging & Setting Up Lathe Tools:

STEP 1

Select the FILE menu and then MERGE. Now locate the TEMPLATES folder in the BobCAD-CAM directory. Double click on it and then double click on the LATHE TOOLS folder to access your lathe tools. They will all be listed in the file window. By clicking once on a tool, you will be able to see a preview of that tool. Click on the tool you want and then click on the OK button to merge the tool into the CAD screen. You may need to click on the VIEW ALL button to see the tool at first.

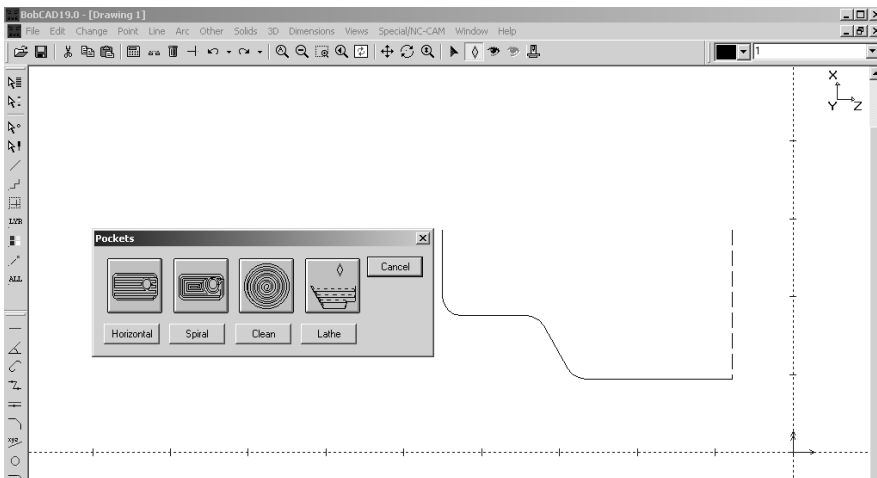


You MUST do a lathe tool setup before being able to generate roughing or finishing toolpath and G-Code.

STEP 2

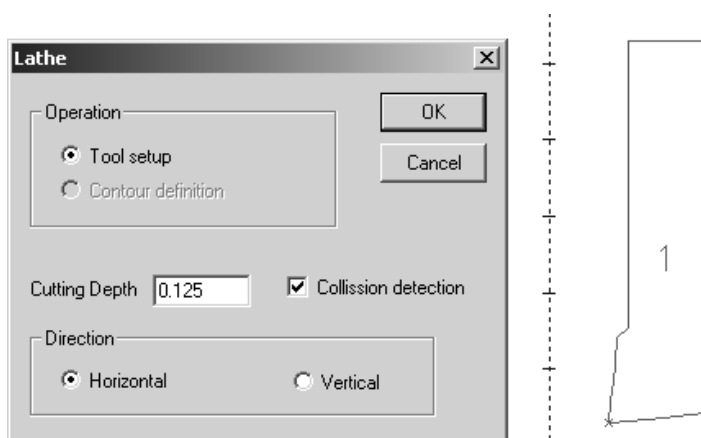
Now that you have merged in the tool, make sure that the lathe button is selected and lathe mode. Now go to the OTHER menu and select POCKET to open the pocket options box.

The POCKET option box:



STEP 3

Select the LATHE button to open the Lathe setup box. You will notice that the only option available is the Tool Setup option. This will be the case unless you have already performed a Lathe Tool Setup.



You are going to need to decide what your cutting depth will be based off of the tool size that you will be using to create roughing or finishing toolpath. Then decide the direction of the toolpath. The options are HORIZONTAL or VERTICAL.

VERTICAL OPTION: You would use the vertical option for operations like FACING or generating toolpath for GROOVING.

HORIZONTAL OPTION: This would be generally used for roughing or finishing toolpath.

COLLISION DETECTION: This should be selected to make sure that there are no collisions when creating your toolpath whether you are grooving, slotting, facing, roughing and finishing.

CUTTING DEPTH: For lathe roughing, what is the depth of each cut for the program that you are writing and the tool that you are using? Enter that value here. It is likely that you will be writing a program for roughing and then a second program for finishing. You may need to do a second tool setup for a finishing program with a different tool. For finishing, enter the total depth of cut in the CUTTING DEPTH section.

Version 21 has special scripts that allow you to bypass of this for certain controllers. There is a chance that we have scripts that we have developed for your controller model. You can contact BobCAD-CAM technical support to find out. We will cover these scripts in this manual. You can easily find out if they are built in to your post processor by reviewing that section in this manual.

STEP 4

By selecting OK you will be ready to select the tool. **As a note**, the available **lathe tools** are actually CAD drawings. They consist of lines, arcs and a point. They can be modified and are selected just like other geometry. First click on a line of the tools geometry. Point your direction arrow in either direction and click your left mouse button **ONCE**. Then hit the **F3** key to select the tools geometry as a chain. You will then notice a star on your cursor. Click on the point at the tool tip. The star will disappear and you will have completed the tool setup.

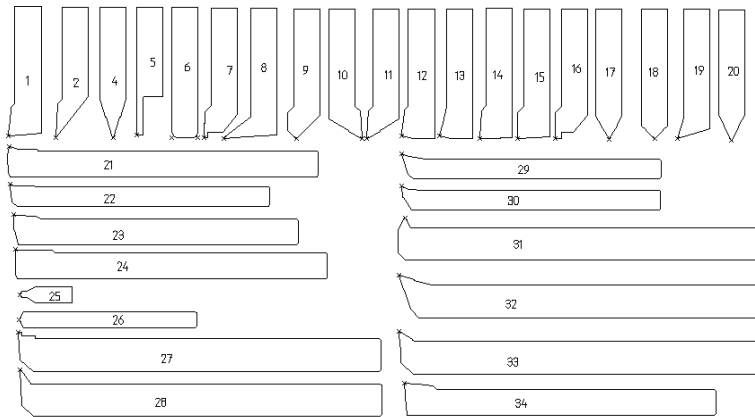
STEP 5

Now you can actually discard the lathe tool itself from the drawing screen as you no longer need it. Simply click and drag a selection box over the tool itself. When you have selected the tool, delete it.

That is how to perform a tool setup with the BobCAD software.

LATHE TOOLS LIBRARY & CUSTOMIZING A TOOL:

Here are the **Lathe Tools** that are included in the BobCAD TEMPLATES folder and in the LATHE TOOLS folder.



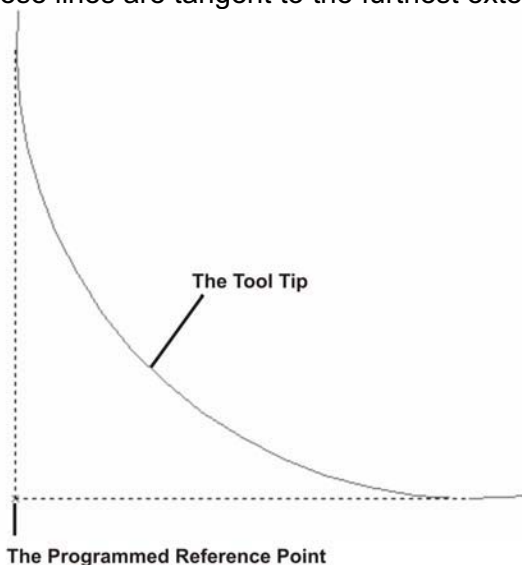
The diagram of the tools above, are all of the pre-loaded tools that are listed in the LATHE TOOLS folder. The Lathe Tools Folder is located inside of the TEMPLATES folder. This folder is found in the BobCAD-CAM directory in the same area as the CAD, DXF, IGES and other folders are found. You may want to create or customize some of these tools for your specific needs. The way that BobCAD's **lathe tool** library is assembled and used, customizing a tool can be done easily by anyone who is already familiar with the basic drawing functions of the software.

NOTE: *The tools themselves are simple CAD line drawings that are then merged in with the drawing of your lathe profile that the software uses to perform tool nose radius compensation with. Along with the full shape of the tool, BobCAD also uses the programmed point of that tool as a reference point to compensate against. This is in the case that you use the TOOL SETUP option in the lathe dialog.*

The following is an instruction on how to customize a tool once it is merged into the main CAD screen of the software.

CUSTOMIZING A LATHE TOOL:

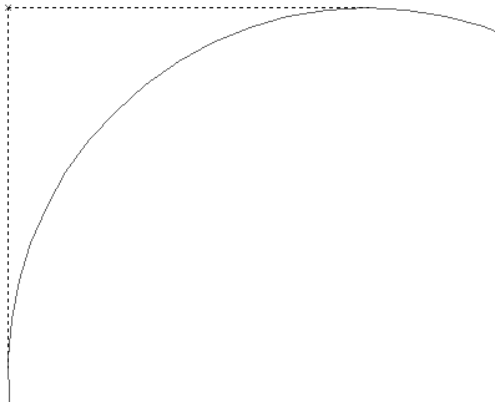
As a general rule (with a few exceptions), the programmed point is the point where two tangential lines meet. These lines are tangent to the furthest extents of the cutting radius of the tool nose.



You can see that the two tangent lines that have been changed into dashed lines, extend out to the reference point of the tool. The above image is a magnified image of tool1.cad in the lathe tool library.

Again, the solid lines and the programmed point in the illustration are part of the tool file itself; the dashed lines show the tangent lines to the extents of the cutting edges. An important thing to note here is that the tangent lines happen to be at perfect right angles, with one line at 0 degrees or 180 degrees, and the other at 90 degrees. This is how the point is placed with most angled cutters.

Another example, tool22.cad in the image below, shows an I/D tool with an angled cutter as well.

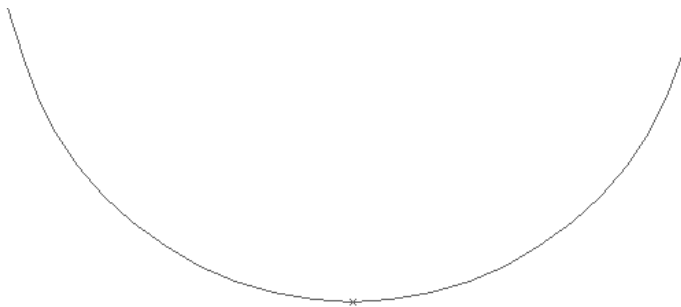


In this instance we have the same thing – the programmed point is at the intersection of two lines tangential to the extents of the cutting radius. The lines also happened to be at the same right angle to one another, at 0 degrees and (this time) -90 degrees.

These extents are very simply the points at which the cutter can meet the material. This is a general rule, but not an all-encompassing one. If you happen to be cutting on a radius, the nose radius will most definitely meet the material at different points along the nose, but the programmed point stays in the same position. This is why on some lathe parts BobCAD will draw the toolpath actually *inside* the profile. The point will cross the lines of the profile, *but the tool nose will not*, and this is one way that BobCAD's built-in nose radius compensation shows itself.

There are exceptions. The most prominent one is the straight cutter, i.e. one without a discernable angle to the nose. Now, we don't have two points at which the cutter meets the material, we have only one, and this point is not at the intersection of any tangent line whatsoever. It is directly at the tip of the tool. Let's take a look at another image of a lathe tool.

This example is a close-up of the tip of **tool04.cad**. This tool nose is in a direct line with the shank of the tool, and only meets the material *at one point*.



Let's take what we have learned here and apply it to a (slightly) more real-world example.

LATHE TOOL LESSON:

STEP 1

Start with a new drawing by selecting FILE from the main menu and then NEW. Like most of the previous lessons, suppose we have a tool that is not quite like the other tools in BobCAD's library, but we'd like BobCAD to be able to use it in a CNC program. Let's say that this tool has a 3" shank .5" wide, a nose radius of .025", a cutting angle of 30 degrees and is 1" from the tip of the nose to the center of the shank along the 30 degree angle. There is no need to know the relief angle behind the nose radius because this will not change any of the 2D lathe programming done with this tool.

STEP 2

It may be easier to draw the tool with **BobCAD's lathe functionality turned ON** for this lesson. To do this, click the BLUE DIAMOND icon that is on the main toolbar so that it is lit-up. If you were not in lathe mode earlier than you will have to do this. We'll start with a rectangle. This will represent the shank of the tool.

STEP 3

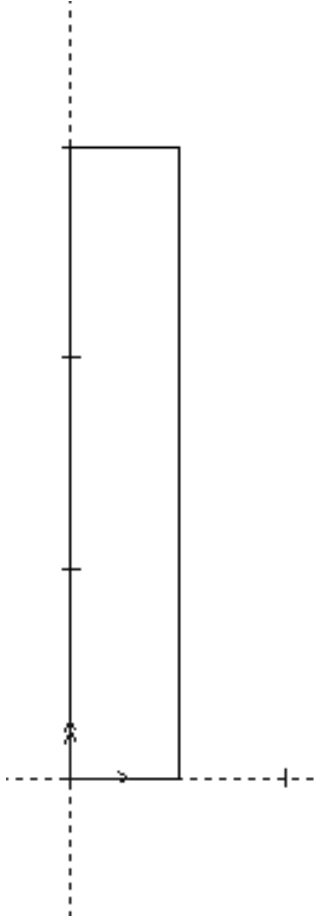
Click on **OTHER** from the main menu, and then Rectangle. Let's take a look at the rectangle box for instruction.

Select the BOTTOM LEFT option from the pull-down under Mode. X, Y and Z should all be left at 0 as our reference point.

Enter .5 for width (representing X) and 3 for the height (representing Y) and leave the radius value at 0.

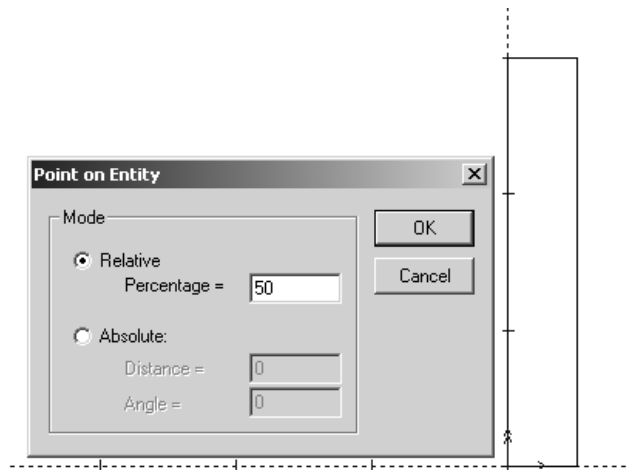
Under the Corner Type section select Straight from the pull-down.

Now select the OK button to draw the rectangle.

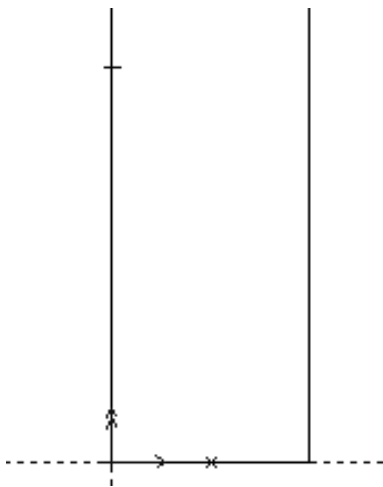


STEP 4

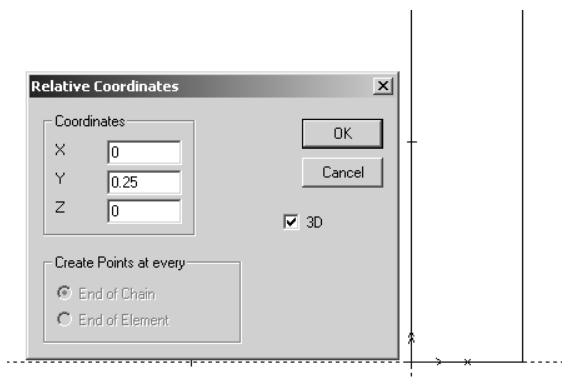
Now we want to place a point in the middle of the bottom line of the rectangle. We are doing this to get help drawing the rest of the construction lines of the tool. Select the **POINT** main menu and then the **On Entity** function.



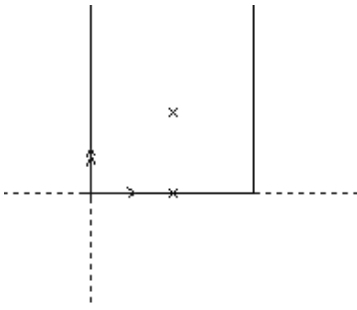
You want to be in the RELATIVE MODE and enter 50 as the percentage because we are placing a point at the middle position of the bottom line of this rectangle. Click OK and then click your mouse once with your cursor on the BOTTOM line. That will snap a point to the bottom line automatically. Now right-click your mouse to exit that function.



Now go to the POINT menu and select INCREMENTAL. X will be 0, Y will be .25 and Z will be 0. Select OK and click on the point that you just drew. This will draw the next point. Now right-click your mouse to exit this function.

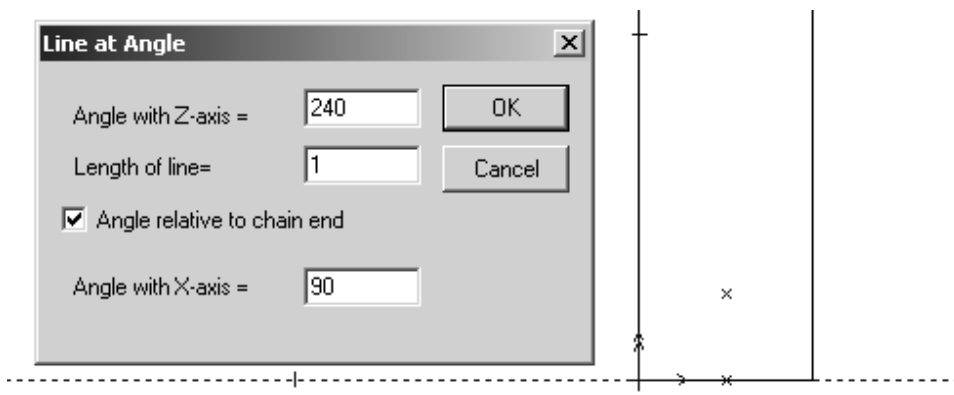


The result:



STEP 5

Now we want to draw a line out from this new point 1-inch and 30-degrees from the shank. Select **LINE** from the main menu and then **ANGLE**.



Let's start with the first option: Angle with Z axis.

NOTE: If this box doesn't look the same way as the example image, you did not turn the lathe mode on. If this is the case, select CANCEL and start over after selecting the Lathe button from the main toolbar. Once again, this is the button with the blue diamond shape on it.

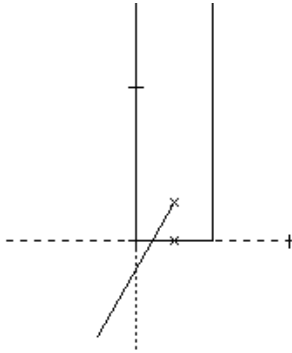
Make the following entries:

1. Enter 240 as the Angle with Z-Axis.
2. The Length of the line will be 1-inch. Therefore, enter 1.
3. Select the Angle Relative to Chain End option. Even though this is not a chain, we will do this as the software will look at this point as such.
4. Enter 90 for Angle with X-Axis.

Now select OK. You will see a representation of the line that you are about to draw starting at your cursor. Now click on the point that you drew incrementally to the first point. Then click your right mouse button to fully exit this function altogether.

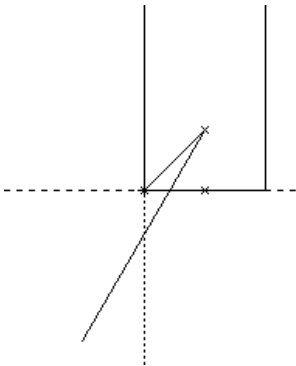
Something important to note about the LINE ANGLE box is the 'X axis option. It's talking about the view and not any part of the drawing. The reason for this is that there has to be some direction reference somewhere, and that can't be a point. A point in geometry has no 0 degree direction, so we must assign one. The X-axis of the view is something that never changes, so it is a good reference. The X axis of the view has its 0 degree direction pointing to the right, and the shank is straight up 90 degrees from this. So, in order to move straight down along the shank we'd have to specify 270 degrees (the 90 of the shank plus 180) in the box. We typed in 240 because we're 30 degrees shy of straight down.

The result:

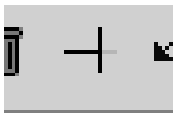


STEP 6

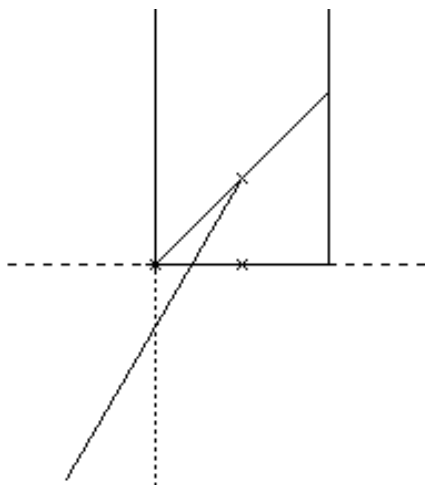
We need to draw in the profile of the rest of the tool nose now. Select **POINT** from the main menu and then **END**. Click on the **LOWER LEFT** corner of the original rectangle now to snap a point to that corner. Now select **LINE** from the main menu and then **JOIN**. Click on the NEW point you just drew and then click on the point that we drew the angled line from. See the finished image below.



OK. We now want to extend this new line to the opposite side of the shank. Select the **Trim/Extend** icon from the main toolbar.



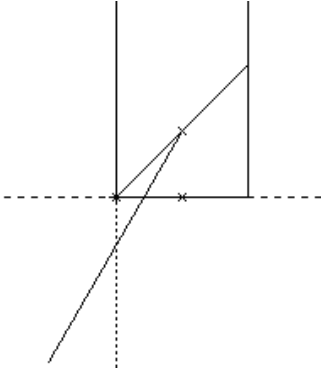
Now click on the line that you just drew/snapped between the two points and the right line of the original rectangle. See the result below.



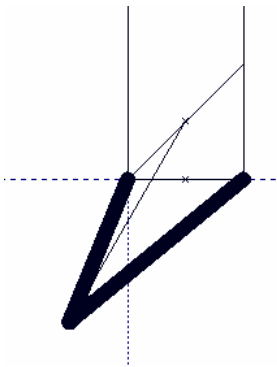
STEP 7

We are almost done with the tool profile. Now we just need to finish the tool point. Select **LINE** from the main menu and then **JOIN**. Follow the image below that has the two enlarged lines. These are the two lines that you want to create by clicking on one end and then clicking on the other end or destination. When finished with the JOIN function, click your right mouse button to exit the function

Current drawing on your screen:



After you have joined the lines (the enlarged lines are the ones that you joined)

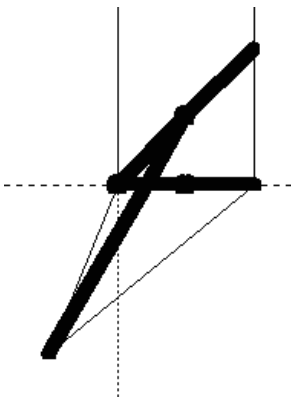


STEP 8

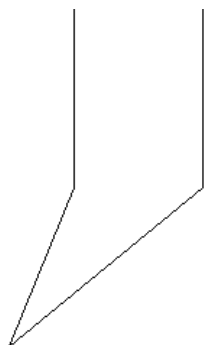
Now we need to clean up the drawing so that we can finish with making the tool tip. Select **CHANGE** from the main menu and then **BREAK, then BREAK MANY**. This brings up the option box. In that box select **ALL** and then OK. This will automatically break up all of the geometry at intersection points.

STEP 9

Select EDIT from the main menu and then Select Entities and SINGLE. Follow the diagram below by selecting ONLY the lines that you see here that are enlarged. You may need to zoom in on this area. After selecting the lines, hit the delete key on your keyboard or click on the trashcan icon on your main BobCAD toolbar.



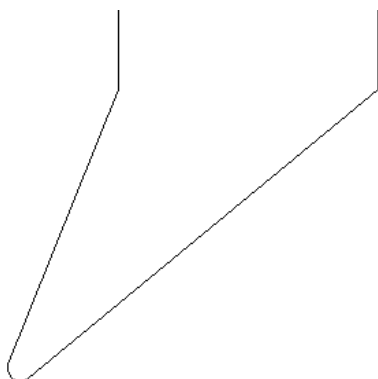
This what you should have on the screen as a result of the editing you just did.



STEP 10

We should only have the profile of the tool on the screen at this time. Select ARC from the main menu and then FILLET. Enter .025 as the radius value in the fillet box and then select OK. Now select the **LEFT** angled line first and then the **RIGHT** angled line second. This will create the fillet that we want. Now right-click your mouse to fully exit this function.

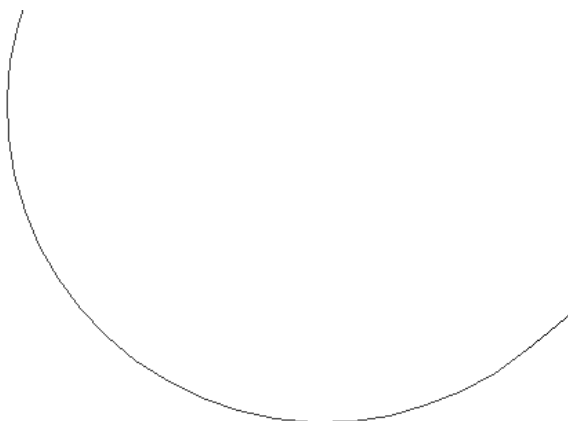
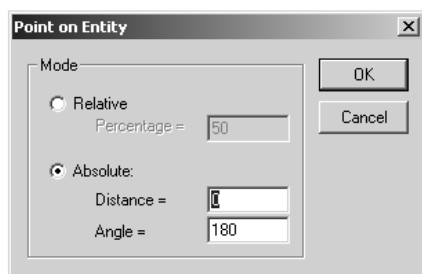
The result:



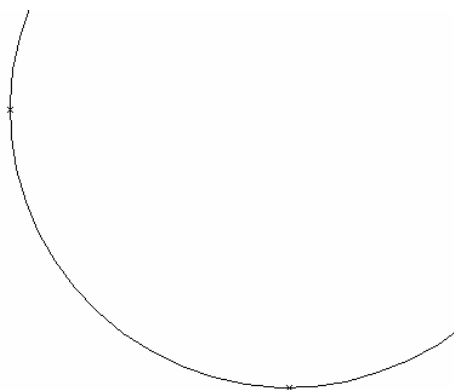
To place the programming point, we'll need to know where the tool will contact the material. In the example, that's actually fairly easy. Since the arc of the radius extends at least past both the 180 degree and 270 degree points of a circle and no other portion of the profile could be tangent to the radius at those points, we know that those are the angles on the radius that will touch the material.

STEP 11

First click on the tip radius so that it is selected. Once you have done this go to the point menu and select ON Entity. Select the ABSOLUTE option and enter 180 as the angle value. Leave DISTANCE at 0. Now click OK to draw the first point.

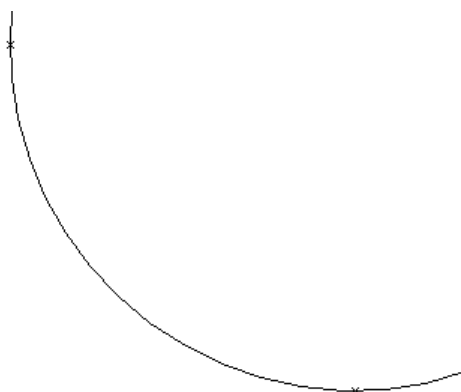
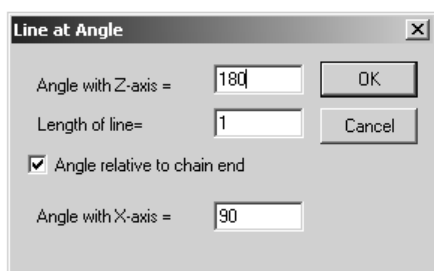


Now go back to the POINT, ON Entity function and enter 270 as the next angle value and click OK to draw the second point.

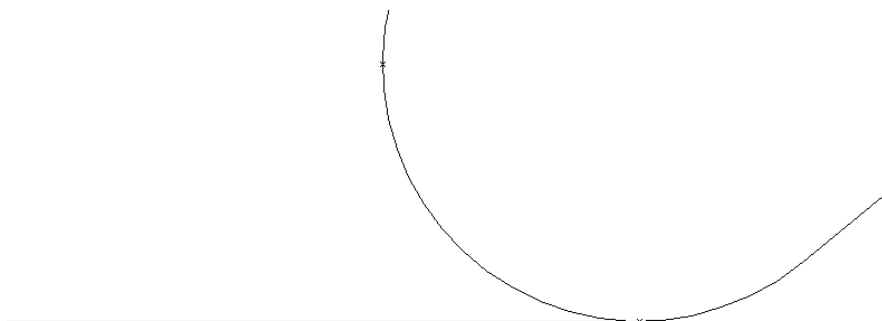


STEP 12

Now go to the LINE menu and choose ANGLE.

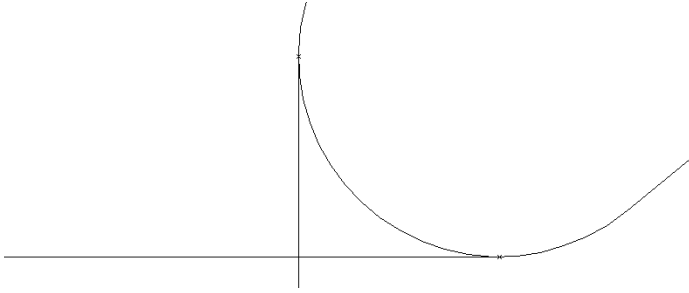


Enter 180 for the Angle with Z-Axis, Line Length needs to be 1. Select the Angle Relative to Chain End option and enter 90 as the Angle with X-Axis as we want this to be a flat 2D line. Click OK and then click the lower RIGHT point. This will draw the first line. Once the line is drawn, right-click your mouse to exit the function.



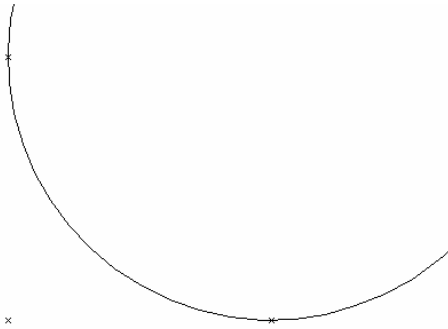
STEP 13

Now go back to the LINE menu and select the ANGLE function again. This time enter 270 as the Angle with Z-Axis. Leave everything else the same and click OK. Now click on the upper LEFT point to draw the second intersecting line.



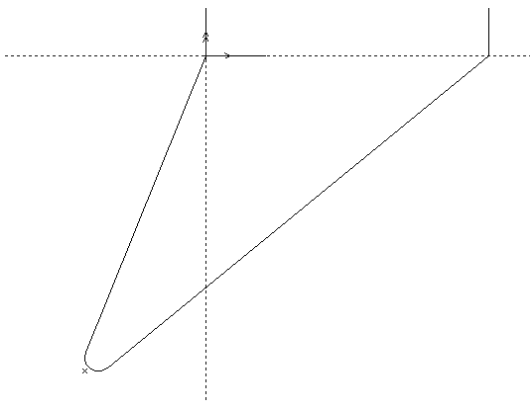
STEP 14

Now we have a new intersection that we can use to place our last point. This will represent the cutting point of the tools tip. Go to the POINT menu and select INTERSECT. Then click on one of the two lines you just drew. Then the other and the point will automatically be drawn. Now right-click your mouse to exit this function. Now hold down your CONTROL key and click on the 2 lines you just drew and the two points that are actually on the arc. Then delete them.



This may seem like a lot of steps to create a lathe tool. Practice this lesson until it becomes faster and easier for you. You can also use existing tools and use the "Save As" function to add and create new tools based off of existing ones.

The result:



That's all it takes to draw a brand new lathe tool in BobCAD. Save this tool in the lathe tool folder and try using it for other O/D lessons that are in this manual. You'll find it comes out just a little bit different, and this is how BobCAD uses each tool to find the right toolpath.

Quick Lathe Lesson 1: Simple O/D Profile

STEP 1

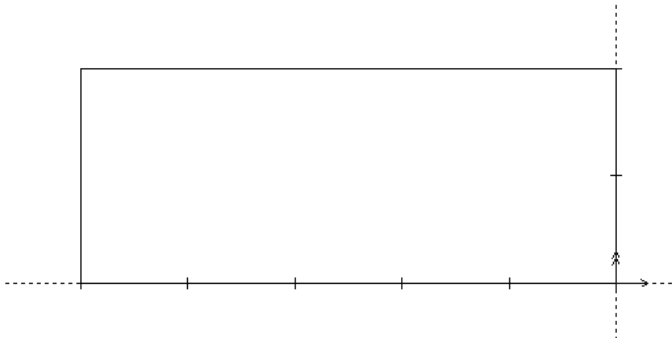
Start with a new drawing screen and make sure that you have turned the lathe mode ON by selecting the Lathe ON/OFF button. This is the button on the main toolbar that has a blue diamond on it.

STEP 2

The first thing that we need to do is draw our stock profile. Select the OTHER menu and then RECTANGLE.

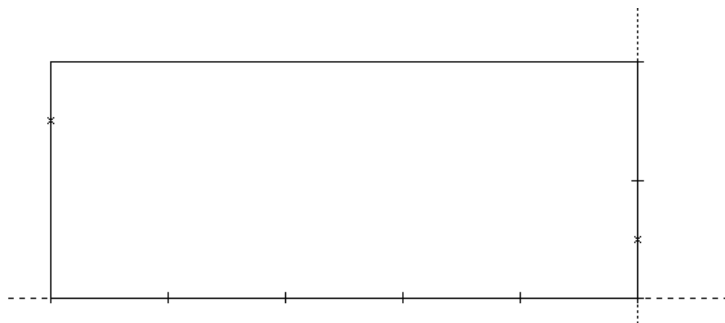
Select the Bottom Left option from the pull down under MODE. Z, X and Y will all be 0. Enter -5 for the width (z-axis) and 2 for the height (x-axis). Select Straight under Corner type and click OK. This will draw our first shape.

The result:



STEP 3

Select **POINT** from the main menu and then **Coordinates**. In the box enter 0 for Z and .5 for X and select CONTINUE. This will bring the Point Coordinates box back on the screen. We are placing the point on the face of the stock 1/2-inch from the center. We are doing this because the taper on our part requires a 1-inch diameter on the face. Now enter -5 for Z and 1.5 for X and select OK. The screen should look exactly like the example picture below.



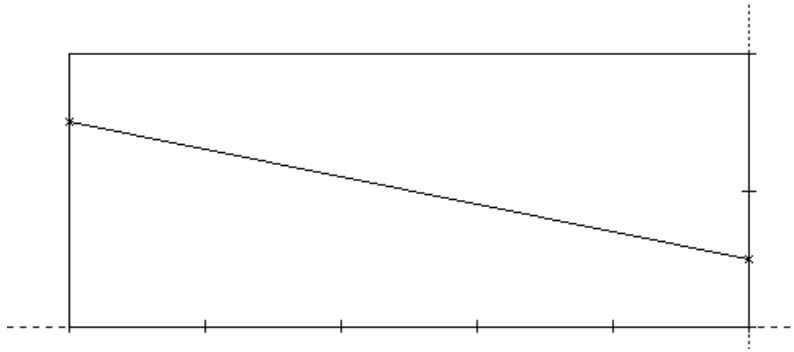
If your Auto Pre-select option is turned on in your environment settings, both points will currently be selected.

IMPORTANT NOTE: This lesson is taking into consideration that the Auto Pre-select option is NOT turned on. If it is, please go to the file menu, select environment and UN-CHECK this option that is located under the DEFAULTS tab and select OK.

STEP 4

Select **LINE** from the main menu and then **Join**. Now select the point at Z -5, X 1.5 and drag the cursor with purple line over to the other point and select that point to snap the line between the two points. Click your RIGHT mouse button to exit that function.

The result:



Now we are almost finished drawing the profile of the part. To finish it off, we will need to get rid of the extra entities that will not be a part of our profile.

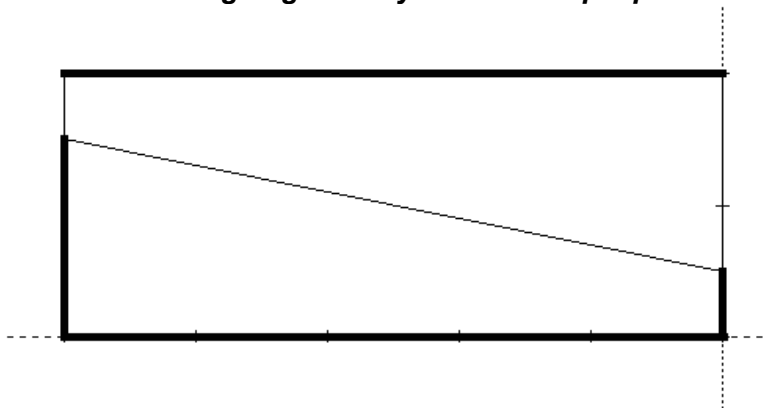
STEP 5

Select the **CHANGE** menu and then **BREAK**. A sub-menu will appear listing the different ways to break up entities. Select **BREAK MANY**. The Select Cross-Section box will appear with the break-entity selections. Select **ALL** and click OK. This action will instantly break up all of the geometry at intersection points and where the points are on the lines.

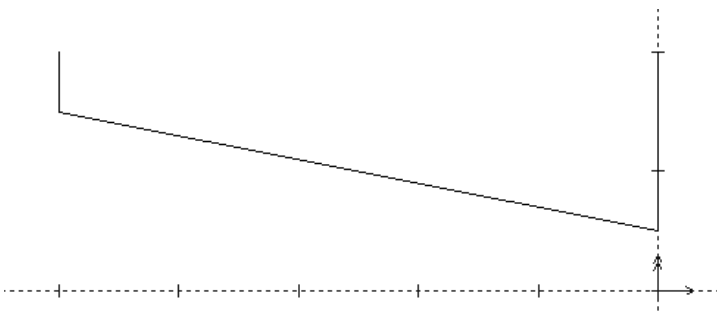
STEP 6

Select **EDIT** from the main menu, then **Select Entities** and **SINGLE**. You can use the EDIT Toolbox if you are familiar with it already from earlier lessons. The lines that we want to select are the same lines in the drawing below that are **LARGER** than the others. After selecting each line, they will turn red. Also make sure that you select the 2 points. Once you are finished with the selections hit the delete key on the keyboard or click on the Trash Can icon. If you make a mistake, you can always select the undo icon and start over.

Select the enlarged geometry in the example picture below.

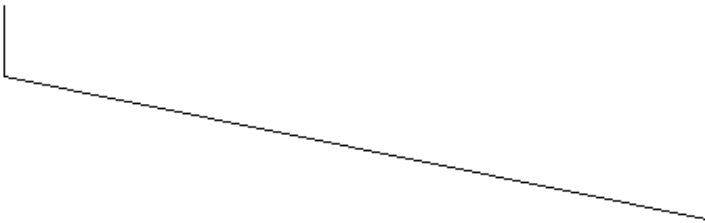


The result:



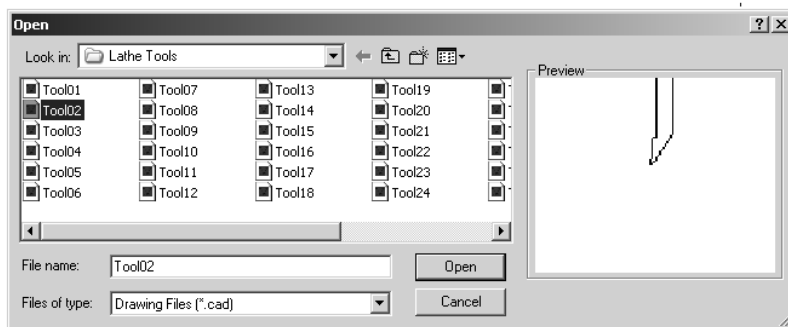
STEP 7

There is only one more thing that we need to do on the profile before we can use it to generate a toolpath. Choose **EDIT** from the main menu, **Select Entities** and **SINGLE**. This time select the line on the face of the stock above where it meets the diagonal line. The line will now be selected and will be red. Go to the **CHANGE** menu and select **ATTRIBUTES**. This brings up the ATTRIBUTES box with a number of selection options. All we are concerned with is the Line Style section under the CONTOUR TAB. Select the DASHED line style preferably the one with the longer dash marks that is located in the lower left section of this tab. Click OK in the upper right corner of the box. Now go to the **EDIT** menu and select **DESELECT ALL**. The line will now be changed to the selected dashed line style. See the example below.



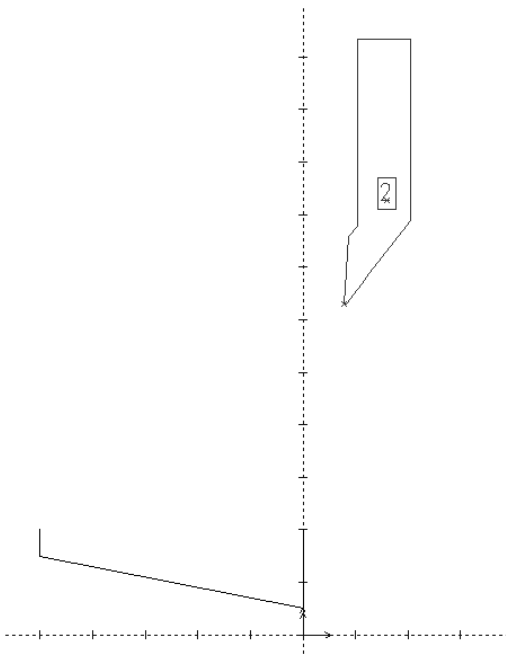
STEP 8

Now that the profile is finished, we will merge in an O/D tool so that BobCAD can do the nose radius compensation for the toolpath. Go to the **FILE** menu and select **MERGE**. Locate the BobCAD-CAM folder and double-click on it. Then double-click on the TEMPLATES folder. Then double-click on the Lathe Tools folder to access the lathe tools. They are all listed as .cad files. Click on TOOL02.cad and then click ok to merge it in.



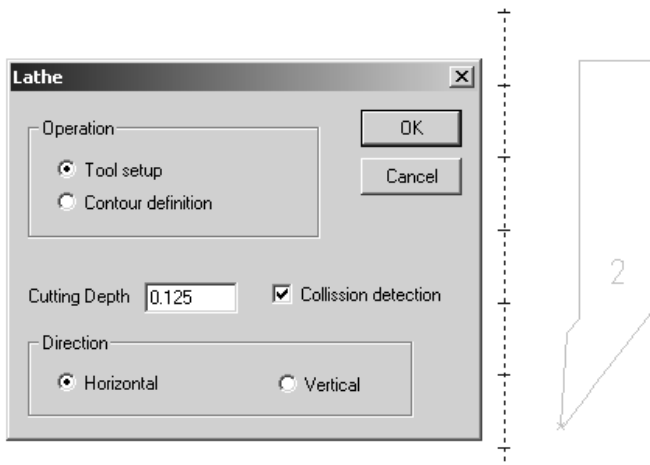
Now select the view all button from the main toolbar so that you can see the part and the tool.





STEP 9

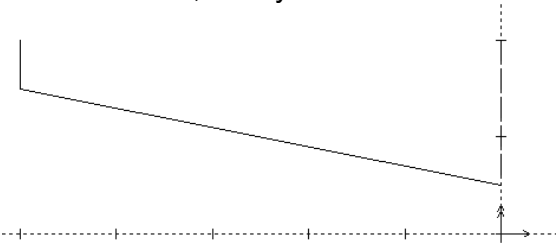
Now you will need to perform a tool setup. Go to the OTHER menu and select POCKET. This will open the Pockets box. Select the LATHE button to open the lathe box.



Select the Tool Setup option and enter a cutting depth of .125 as this is what we want to generate the roughing toolpath. Set the direction to HORIZONTAL and click OK. Now click on a geometry line of the tool, click your left mouse button once and then click on the point at the tool tip. Good. Now the tool setup is complete.

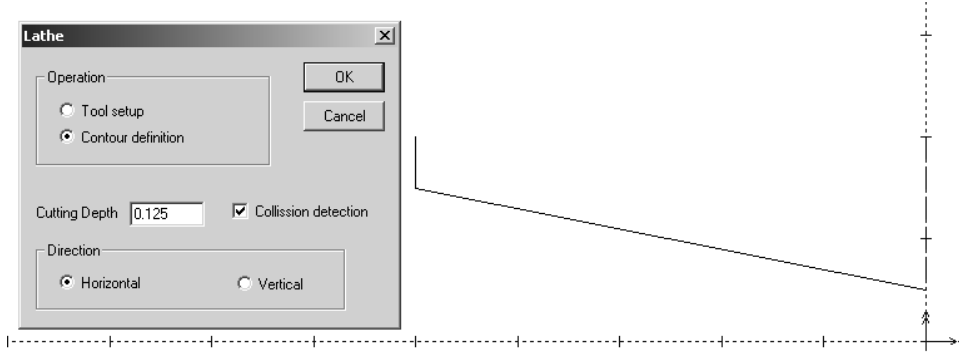
STEP 10

Now click and drag a selection window with your mouse, over the entire tool. Just click and while holding down your left mouse button, drag a window over the tool. When the selection window covers the tool let go and the entire tool will be selected in red. Now delete the tool. You don't need it on the screen anymore. Once you have done this, click your View All button again to zoom in on the O/D profile.



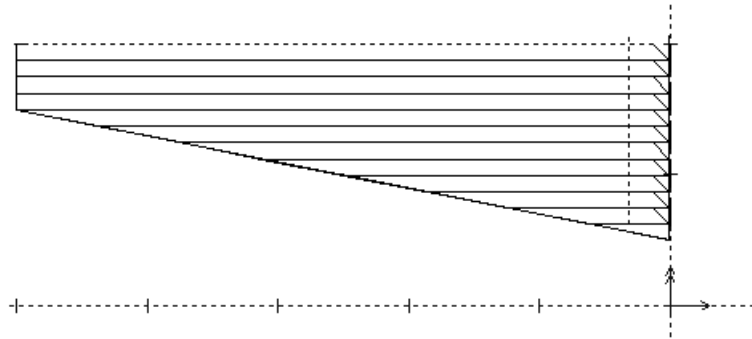
STEP 11

Go back to the OTHER menu and select the POCKET function. Now select the Lathe button to open the lathe box. Now select the CONTOUR DEFINITION option in the lathe box. The other values and settings will be the same.

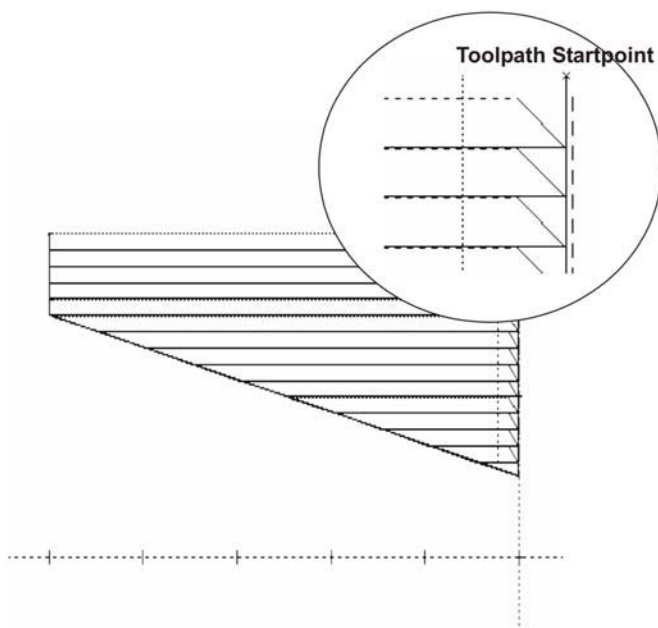


Go ahead and select OK. Once you select OK, the software will be live and ready for you to select the contour.

Select the TOP part of the dashed line by clicking on it once. Move your cursor down toward the bottom of the dashed line and click your left mouse button one more time and then hit the F3 button on your keyboard. This will generate the complete roughing toolpath for the part.



Take a closer look at where the start point of this toolpath is.



NOTE: You can snap a point to the end of the line as seen in the magnified view above. This will allow you to zoom out and still see the toolpath start point better. You may also change the color of this line or point to green or red if you choose.

Let's go ahead and create the g-code program now.

STEP 12

Select the Special/NC-CAM menu and then INSERT NC. This will open the box where you would choose a post processor and open the CAM.

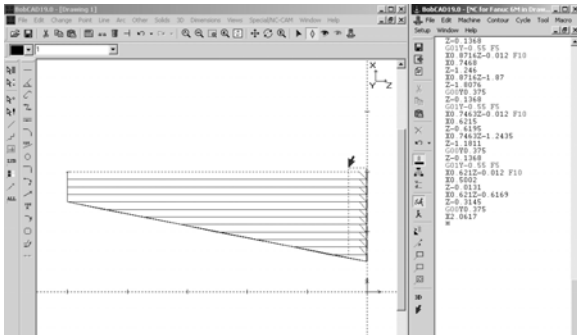
For demonstration purposes, go ahead and select a **FANUC** post configuration in this box and then select OK to fully open the CAM. Lathe mode is turned on so we will generate the correct X and Z lines.

Special note: If you CAD screen splits and you have more than one drawing up, it is because you didn't close out of previous drawings and they were in the background. If this has occurred just exit out of the other drawings so that the lathe part you just did consumes the entire CAD screen. You can have multiple drawing screen open at the same time. All we want open right now is the lesson that you are working on.

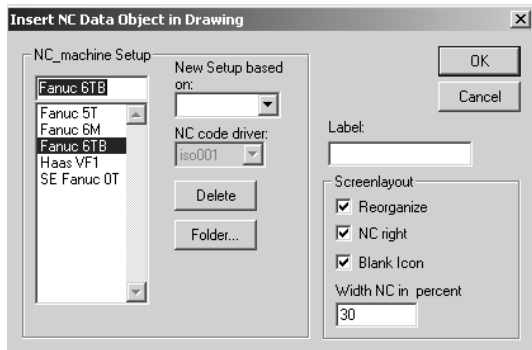
Select the View All button from your main toolbar on the CAD screen so that you can see the entire toolpath easily.

STEP 13

We are now going to focus briefly on the CAM side of the software. Go to the MACHINE menu on the CAM side and select the AUTO cut option. Just select AUTO. Now take your cursor right to the TIP of that toolpath. Click on the start of that toolpath and move your cursor downward and click your left mouse button again. Then hit the F3 key and the software will generate the entire G-Code program.



We simply select the SPECIAL NC-CAM menu and then INSERT NC to open the next box.



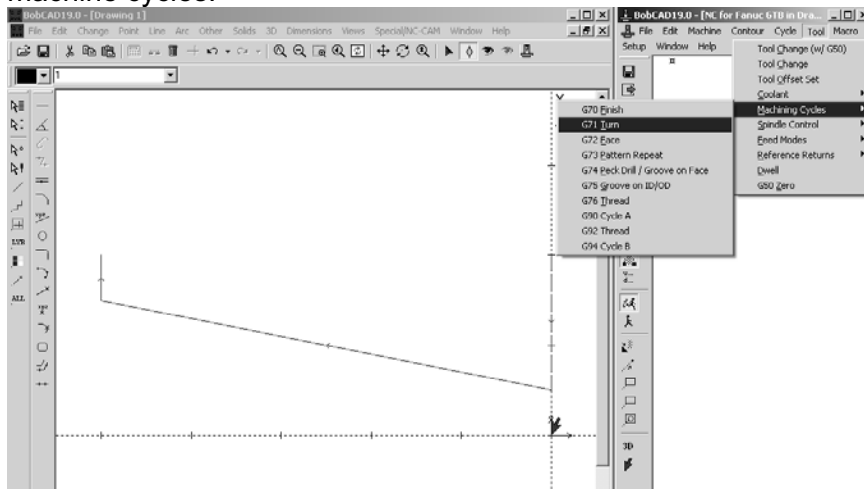
As you can see in this image we are going to select a Fanuc 6TB post processor that has been setup with special functionality for lathe programming.

By selecting the post and then the OK button the CAM will fully open up on the right side of the software.

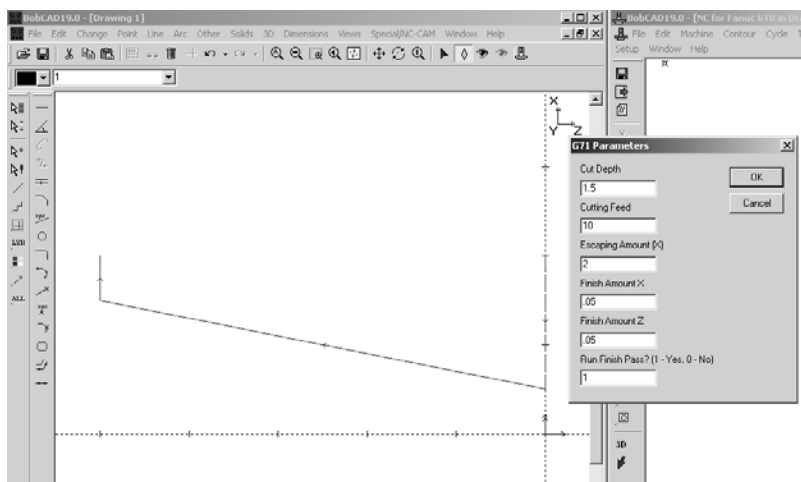
STEP 2

By holding down the SHIFT key and clicking the last entity of the O/D profile, the chain of entities will be selected with the direction going from right to left. **THIS IS IMPORTANT.** You may want to turn your **directional arrows** on in the environment settings. Do this by selecting the file menu, environment and then by choosing the SCREEN tab. This option is located in the lower left corner of this tab. Select the option and then click OK to activate the directional arrows.

Once you have selected the profile chain correctly, go to the TOOLS menu on the CAM side to locate your machine cycles.



For this example we are selecting the G 71 Turn cycle. By doing this a special box will appear asking us for the important variable entries. See this in the next image.



The G 71 Turn cycle that we selected is going to ask us for:

- A cutting depth.
- A Cutting Feed.
- An Escaping amount X.
- A finishing amount in X. Basically how much material do you want to leave behind.
- A finishing amount for Z.
- An option for running a finish pass or not.

By selecting OK, the software will automatically generate the program. It will not display the toolpath but will write the correct program. You will find that through the use of scripts, it will take less time to generate a program and the code will be just as accurate. In fact, there will also be a lot less code generated.

Continue on with this training guide to learn more about programming your CNC lathe with BobCAD-CAM software.

Quick Lathe Lesson 2: Simple I/D Profile

For this lesson, we will use the same rectangle as we did in the first lesson for an O/D profile.

STEP 1

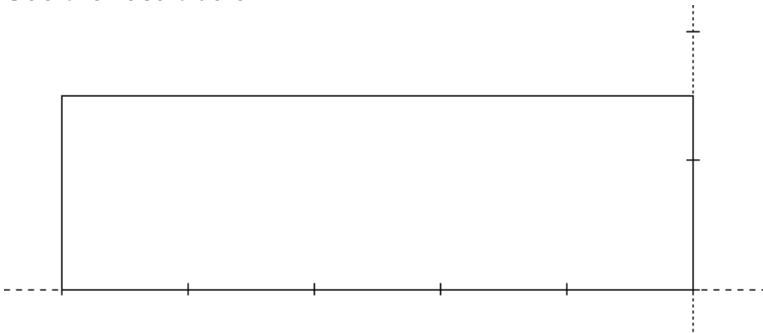
Start with a new drawing screen and make sure that you have turned the lathe mode **ON** by selecting the Lathe ON/OFF button. This is the button on the main toolbar that has a blue diamond on it.

STEP 2

The first thing that we need to do is draw our stock profile. Select the OTHER menu and then RECTANGLE.

Select the Bottom Left option under MODE. Z, X and Y will all be 0. Enter -5 for the width (z-axis) and 2 for the height (x-axis). Choose the STRAIGHT corner type and select OK. This will draw our first shape.

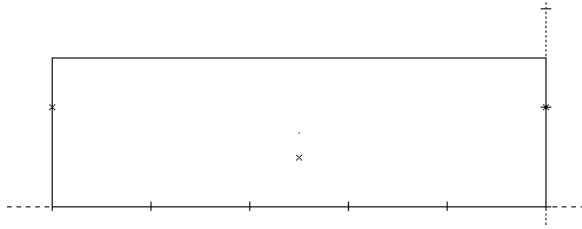
See the result below.



STEP 3

Next, we need to place some points on the screen to indicate where the corners of the profile are. Select **POINT** from the main menu and then Coordinates. This will bring up the Point/Coordinate dialog box. We are going to place 3 points on the screen. For the first point, enter 0 for Z and 1 for X and select **CONTINUE**. Now enter -2.5 for Z and .5 for X and select **CONTINUE**.

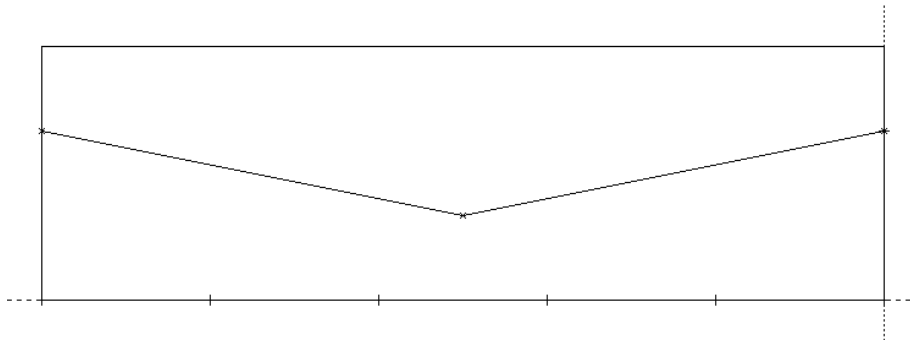
For the last point enter -5 for Z and 1 for X, then select **OK**. You can see the result in the next image.



IMPORTANT NOTE: This lesson is taking into consideration that the Auto Pre-select option is NOT turned on. If it is, please go to the file menu, select environment and UN-CHECK this option that is located under the DEFAULTS tab and select OK.

STEP 4

Now that you have placed the points at their correct locations, select **LINE** from the main menu and then **JOIN**. Select the first point that you placed on the face of the part first, then click on the second point. This will snap a line between both points. Now click on the **SECOND** point and then the last point on the left side. Now click your **RIGHT** mouse button to exit the function. You should have 2 diagonal lines from the sides coming down to the center point now as you can see below.

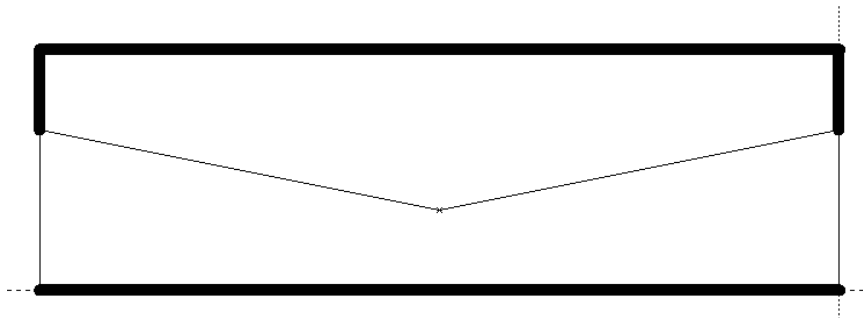


STEP 5

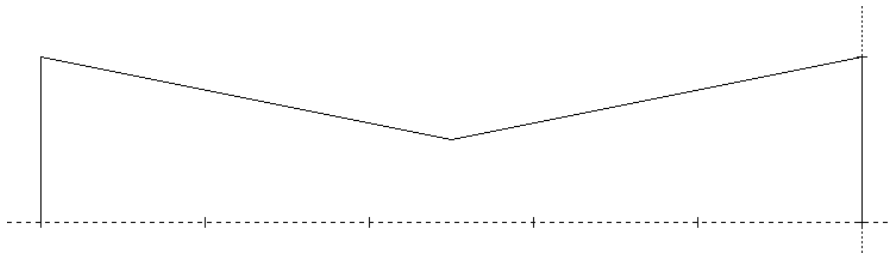
Now we need to break the profile so that we can edit the drawing for toolpath creation. Select **CHANGE** from the main menu and then **BREAK**. Select **BREAK MANY** from the sub-menu and this will bring up another box. This "Select Cross-Section" box offers several break options. Select **ALL** and click **OK**. The entire drawing will automatically be broken up at all of the intersections and locations of points that are on entities.

STEP 6

In the diagram below, you can see that the lines that we want to get rid of are the ones that are enlarged.



Select **EDIT** from the main menu, then **Select Entities** and **SINGLE**. (You can also use the edit toolbox if you are already familiar with the selection buttons.) Now hold down your CONTROL key and select the lines that are enlarged as you can see in the diagram above. After you select a line, it will highlight. When you are finished you can select the trashcan icon or hit the delete key on your keyboard. Finally, you want to select the 3 points and delete them as well.

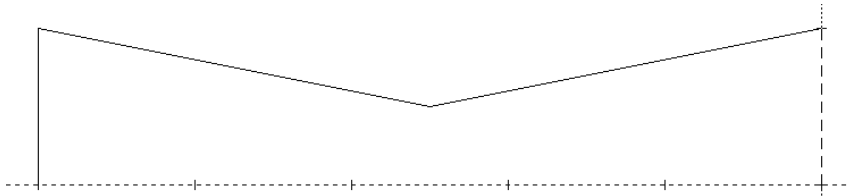


STEP 7

Now there is only one more change that we need to make before we are ready to generate the toolpath. Since the tool will be passing through the face of this part, we do not want BobCAD to avoid the face with the collision detection. Go back and select **EDIT** from the main menu, Select Entities and **SINGLE**. Click on the line of the face **BELOW** the point (RIGHT side). This will highlight the line. Now go and select **CHANGE** from the main menu and then **ATTRIBUTES**.

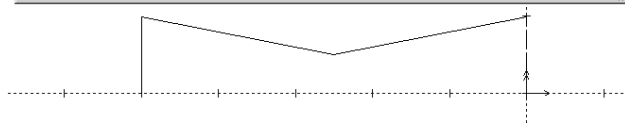
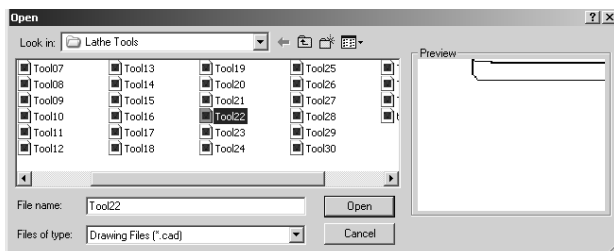
Select the CONTOUR tab. In the lower LEFT corner you will see the **LINE STYLE** section. Select the **DASHED** pattern with the longer dash marks and then click OK. Now select **EDIT** from the main menu and **DESELECT ALL** or simply click your mouse off of the geometry on the screen. This will change the style of the line automatically. You can see this in the next example image.

The result:



STEP 8

Now we are going to merge in the tool. Go to the **FILE** menu and select **MERGE**. Locate the BobCAD-CAM folder and double-click on it. Then double-click on the TEMPLATES folder. Then double-click on the Lathe Tools folder to access the lathe tools. They are all listed as .cad files. Click on **TOOL22.cad** and then click **OPEN** to merge it in.



Now select the view all button from the main toolbar so that you can see the part and the tool.



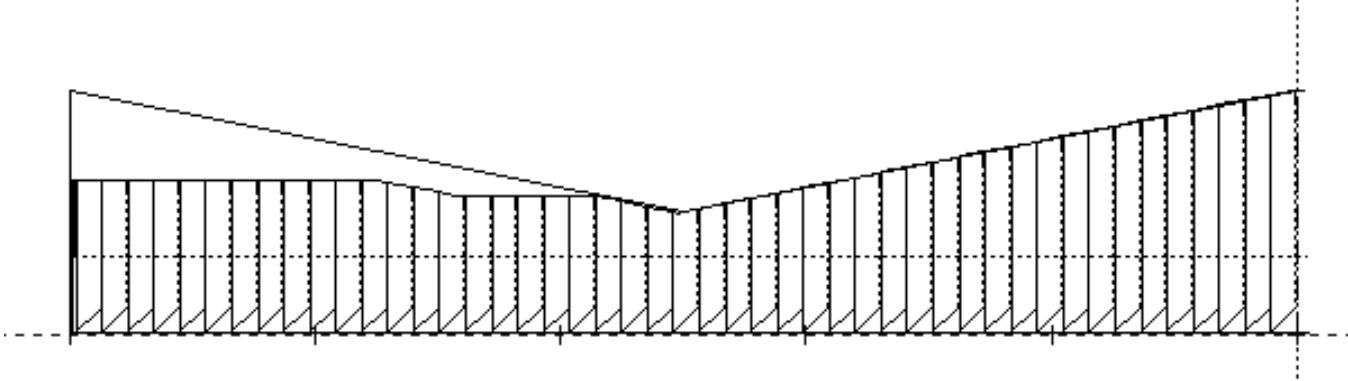
STEP 9

NOTE: We are choosing tool 22 with this profile on purpose. Because it CANNOT cut the full profile that we have drawn. We need to do this to show you the collision detection features of BobCAD's Lathe toolpath functions to you. At this point you will need to do a tool setup if you haven't done this already. To do this, select the **OTHER** menu and then **Pocket**.

Click the LATHE button. Check the Tool Setup option and enter .1 for the cutting depth. Check the **COLLISION DETECTION** option and make sure that you select the VERTICAL option as well. Now click OK. Click your mouse on any geometry line of the tool, click your left mouse button again and hit the F3 key on your keyboard to select the tool geometry as a chain. Now click on the tool tip POINT. OK. Now the tool has been setup. You can click and drag a selection window over the entire tool and completely delete it from the drawing screen altogether. Once you have done this, click the View All button from the main toolbar.

STEP 10

Now go back to the OTHER menu, select Pocket and then click on the Lathe button. Now select the Contour Definition option. Make sure that the values and selections are the same for everything else. Now click OK. At this point the software is live for selecting. Click on the lower end of the right side, dashed line. It will highlight. Move your cursor UP toward the top part of that dashed line and click your left mouse button one more time and then hit the F3 key on your keyboard. This will generate the toolpath.



Notice that the entire profile is NOT cut. This is very important because BobCAD will not put the tool where it does not belong as long as the Collision Detection feature is checked and turned on. This will prevent tool crashes and the destruction of your tools.

Quick Lathe Lesson 3: Simple Facing Operations

Facing operations are actually very easy with BobCAD. In Fact, there is no need to even draw all of the stock that you wish to face, just the face portion of the part.

Begin this lesson by first, turning on the Lathe mode. This is done by clicking on the icon from the main menu that has a blue diamond on it. This is the Lathe ON/OFF button. **IMPORTANT NOTE:** This lesson is taking into consideration that the Auto Pre-select option is NOT turned on. If it is, please go to the file menu, select environment and UN-CHECK this option that is located under the DEFAULTS tab and select OK.

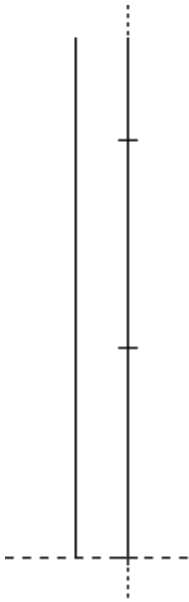
Let's suppose that we have a piece of stock with a diameter of 5-inches and we need to take .25-inch off of the face. First we will draw a line that represents the face of the stock and then we will draw the line that represents the finished face.

STEP 1

Select **LINE** from the main menu and then **Coordinates**. Leave the **Z-start value** and **X-Start** values both at **0 as well as 0 for Y**. Enter **0 for Z-End** and **2.5 for X-End** and leave **Y** at **0**. Now select CONTINUE. You just drew the line of the face of the current stock.

Now for the finish face enter **-.25 for Z-Start** and **0 for X-Start**. For enter **Z-End** enter **-.25** and **2.5 for X-End** leaving the Y end value at 0. Now select OK.

See the example below.



STEP 2

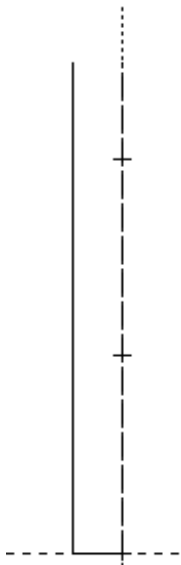
Now we need to join these two lines to represent the continuity of the path. Select **LINE** from the main menu and then **JOIN**. Select the **BOTTOM** of one of the lines you just drew and then click on the other **BOTTOM** end of the second line. This will snap a line between the two bottom ends. We are going to leave the top open. Therefore, click your right mouse button to exit this function now.

STEP 3

Now we need to change the face of the stock to show that the tool will pass through it to get to the finished dimension. You will notice that this is a recurring concept through most of the lathe lessons. Any time you need the tool to remove material from the face of a part, be it an O/D, I/D or a facing operation, the FRONT line representing the face of the stock will need to be changed to a DASHED line.

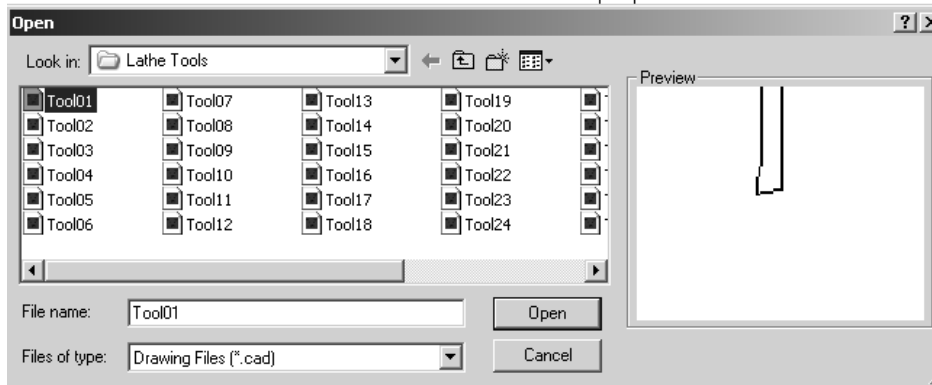
STEP 4

Select **EDIT** from the main menu and then **Select Entities** and **SINGLE**. Click on the **FRONT** (right) line to select it. Now select **CHANGE** and then **ATTRIBUTES**. Select the CONTOUR tab then select the **DASHED** line style and click OK. Now select **EDIT** from the main menu and then **DESELECT ALL**. The result:



STEP 5

Now that the profile is done, we need to merge in a tool so that BobCAD can compensate for the tool nose. Note that merging in a tool is also a recurring action for lathe part programming. Select **FILE** from the main menu and then **MERGE**. Locate **tool01.cad** and merge it into the drawing by clicking once on the tool and then selecting the OPEN button.

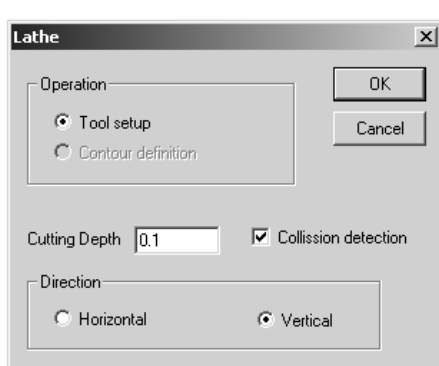


Now select the view all button from the main toolbar so that you can see the part and the tool.



STEP 6

Now you need to perform a tool setup. Go to the **OTHER** menu and select **POCKET**. Now select the **LATHE** button to open the lathe box.



Tool Setup should be selected unless you have already performed a tool setup action for this tool.

Enter .1 for the cutting depth and make sure that the **VERTICAL** option is selected.

Check the Collision Detection option and now click OK. The first thing that you will need to do is click on one of the lines of the tool itself. Just place your cursor right on one of the lines and click your left mouse button **ONE** time. Move your cursor up or down depending on what line you clicked and then click your left mouse button once more. Now hit the F3 key on your keyboard to select the entire chain. Your cursor will change and you will notice a star. Click your left mouse button on the point at the tip of the tool. OK. Now you can delete the entire tool from the CAD screen by clicking and dragging a selection window over the tool itself and then clicking on the trash can icon.

Now hit the View All button again to zoom in.

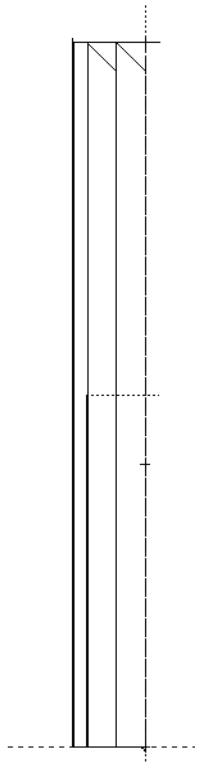
STEP 7

Now we can start on creating the toolpath. Select **OTHER** from the main menu and then **POCKET**. Remember to make sure that the Lathe Mode is activated at this time. By selecting **POCKET** and then the **Lathe** button, you will bring up the Lathe box again. Select the **CONTOUR DEFINITION** option. For **CUTTING DEPTH** we will typically enter the amount that we want taken off for each pass. This is based off of the material that you are cutting, what the tool is made of, how fast you want to go and how many passes you want to have in the program etc. For training purposes enter .1 for the cutting depth or just leave it as it should already be set to this. Make sure that you have Collision Detection checked and select **VERTICAL** under direction as moving in a straight line down the face is the normal way of doing this operation. Now select OK.

STEP 8

After selecting OK in the lathe box, the software will be live and ready to make the first geometry selection. Click your left mouse button at the **TOP** tip of the **DASHED** line and point your arrow or move your mouse **DOWN** and click your **LEFT** mouse button once to secure the direction. Now hit F3 on your keyboard. This will automatically generate the toolpath for the facing operation. If you can't see a **directional arrow** it is because the direction arrow option in the environment setting is turned off. In earlier version of the software you would always see the direction arrows when you select an entity. In Version 21 you can turn these on and off by selecting the file menu and then environment. Select the SCREEN tab and you will see this option listed.

The result toolpath.



Version 21 also has an environmental setting called, “**Auto Pre-select.**” If this is turned on, the toolpath will be generated and will be automatically selected. You may like this as it allows you to go straight to opening the CAM and generating the g-code program by selecting the Auto Cut option. You may not want to use this because you may want to have more control over what you want selected for machining purposes. This option is up to you to decide upon.

Quick Lathe Lesson 4: An O/D Profile with a Built-in Facing Operation

Again, we will begin this lesson with a rectangle. This time however, we will not be using the face end of the rectangle itself as part of the profile. We will be using the entire rectangle to represent the blank stock.

STEP 1

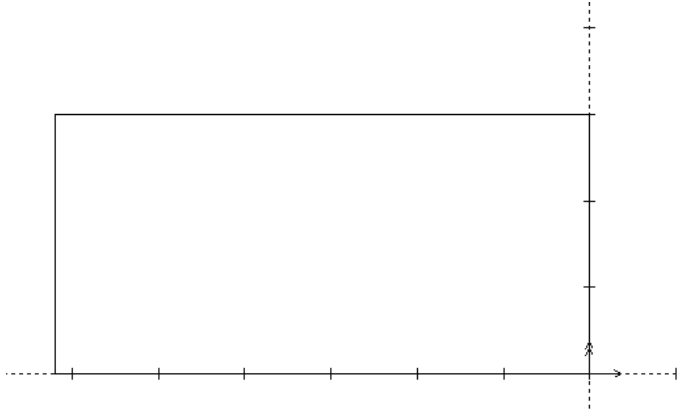
Start with a new drawing screen and make sure that you have turned the lathe mode **ON** by selecting the Lathe ON/OFF button. This is the button on the main toolbar that has a blue diamond on it.

STEP 2

The first thing that we need to do is draw our stock profile. Select the OTHER menu and then RECTANGLE.

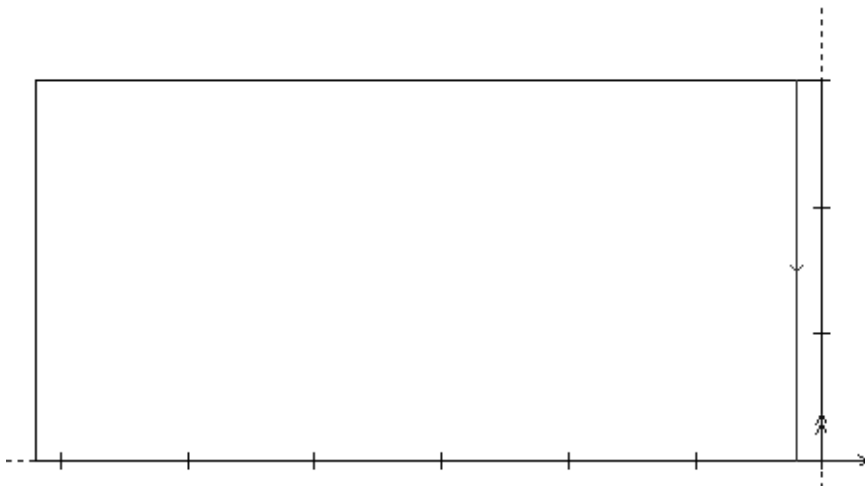
Select the Bottom Left option under MODE. Z, X and Y will all be 0. Enter -6.2 for the width (z-axis) and 3 for the height (x-axis). Choose the STRAIGHT corner type and click OK. This will draw our first shape.

The result:



STEP 3

Now that we have the stock, we can draw the face of the real part. First select the far RIGHT line by clicking on it. Select **LINE** from the main menu and then **PARALLEL**. You now need to indicate which side of this line you want to put the other line. Select the RIGHT option for the side of line as we are in Lathe Mode. Now, because we planned to remove .2 from the face, enter .2 and click OK. This will automatically generate the parallel line for you. **IMPORTANT NOTE:** *The distance that we entered is an absolute number. If we had entered -.2 as some might think, (as it is to the negative direction of Z) we put the parallel line on the OTHER side of the original.*



STEP 4

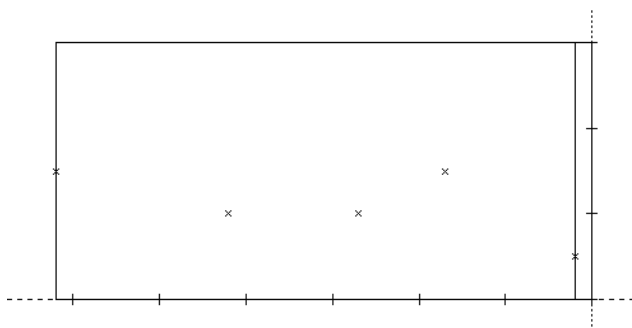
Now we can draw the profile that we want to cut. Let's place the points on the screen now. Select POINT from the main menu and then Coordinates. For the first point enter -.2 for Z and .5 for X. Select CONTINUE. Now enter -1.7 for Z and 1.5 for X and click CONTINUE. Continue as directed below.

Enter Z -2.7 and X 1. Select CONTINUE.

Enter Z -4.2 and X 1. Select CONTINUE.

Enter Z -6.2 and X 1.5. Select OK.

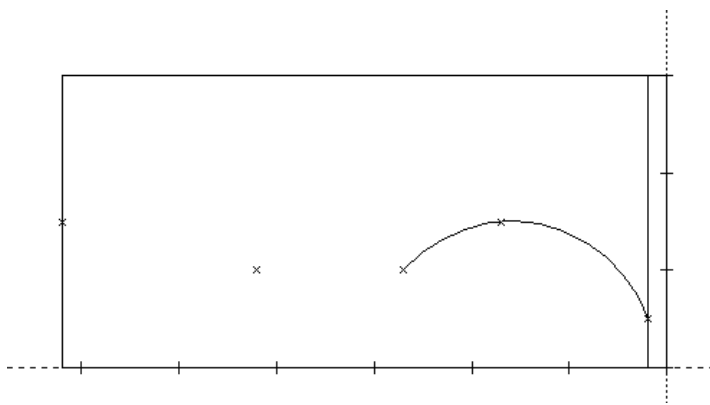
See the example below of what your screen should now look like.



Our shape is going to look something like a doorknob with a taper toward the door side. Now we want to put an arc between the first 3 points. **Make sure that these points are NOT selected before moving on to the next step. If they are selected, click your left mouse button with your cursor in an empty area of the drawing screen.**

STEP 5

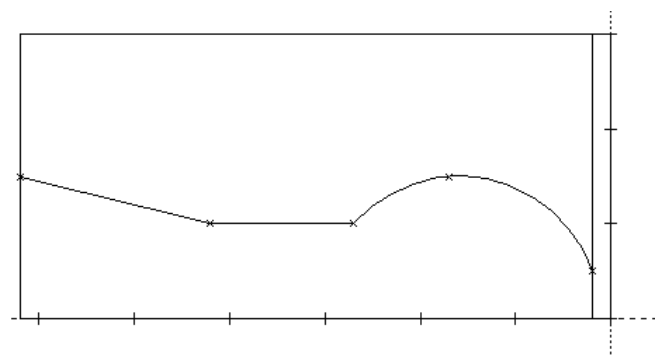
Select ARC from the main menu and then 3 ENTITIES. First select the 1st point that we drew, then the 2nd point and finally the 3rd point that we drew. This will automatically generate an arc between these 3 points. Now right-click your mouse to exit the function. Your screen should look like the example below.



Remember, we select the arcs in a counter clockwise manner in BobCAD. If your selection went the other way, the arc would be generated in the other direction, making a mess of the drawing. If this happens, select the undo button and start over.

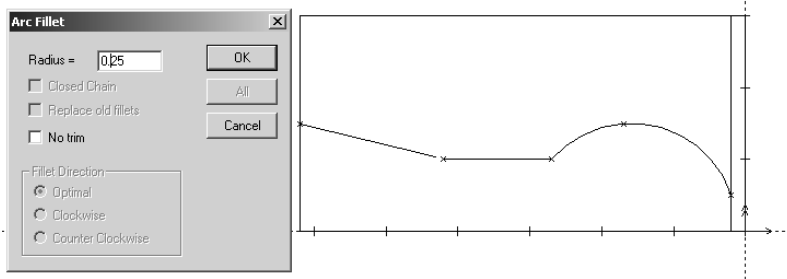
STEP 6

Next we are going to draw some lines to finish the profile. Select LINE from the main menu and then JOIN. Select the 3rd point that we drew (the last point of the arc). Drag the purple line over to the 4th point and click on it. Now click on the 4th point and then click on the last point. Click your RIGHT mouse button to exit this function. See below.



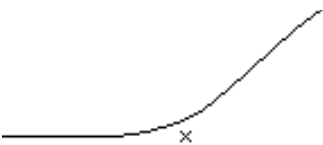
STEP 7

Now we want to round out the profile to complete it by adding two fillets. Select ARC from the main menu and then FILLET. Enter .25 in the fillet box for the radius. Leave the NO TRIM option un-checked.



Now click OK and make the following selections to produce the fillets.

Click on the line just to the LEFT of the 3rd point from the right and then click on the last part of the arc just to the RIGHT of the 3rd point. This will produce the fillet that we want. The result:

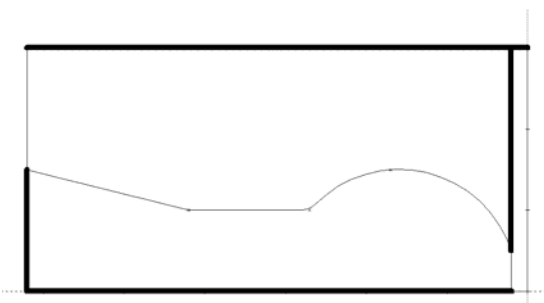


STEP 8

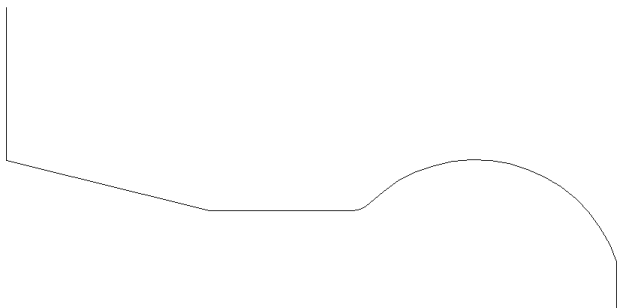
Now we need to edit out the geometry/drawing parts that are not necessary to have on the screen. As in earlier lessons, select **CHANGE** from the main menu and then **BREAK**. From the Break sub-menu select **BREAK MANY**. The Section box will appear with the break options. Select **ALL** and click OK. Now the drawing has been broken up at every intersecting piece of geometry.

STEP 9

Use the next diagram and select **ONLY** the enlarged entities for deletion. Do this by holding down your **CONTROL** key and click on them. Make sure to select each of the 5 points as well.



After selecting these parts *and* all 5 points, you can click on the trashcan icon or hit the delete key on your keyboard. After the editing, your drawing should look exactly like the image below.

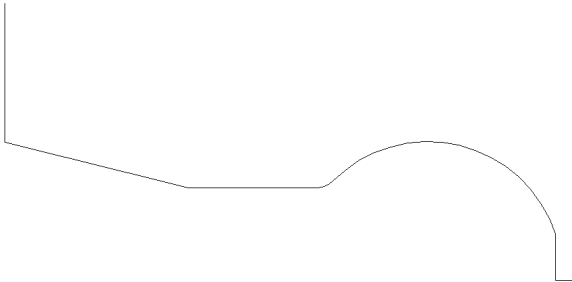


STEP 10

OK. Now that we have finished the profile, we need to change the front line on the face of the part to a dashed line as we have done in each lesson. Click on the **FRONT** (far right) line to select it. Now select **CHANGE** and then **ATTRIBUTES** and the **CONTOUR** tab. Select the **DASHED** line style and then click OK. Now click on the REFRESH button from the main toolbar.

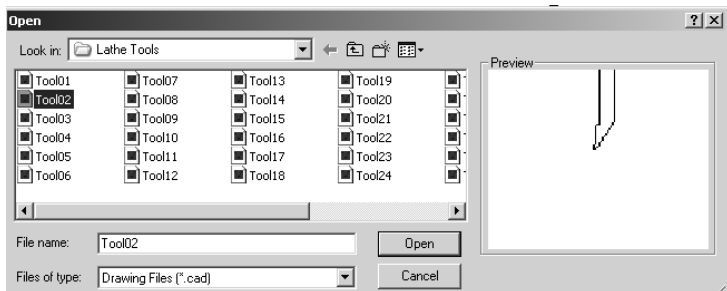


The result:



STEP 11

Now you are ready to merge in the tool and perform a tool setup to create the toolpath. Select **FILE** from the main menu and then **MERGE**. Locate **tool02.cad** and merge it into the drawing by clicking once on the tool and then selecting the OPEN button.



Now select the view all button from the main toolbar so that you can see the part and the tool.



STEP 12

Now we need to perform a tool setup. Select the **OTHER** menu and then **POCKET**. Now click on the lathe button to open the lathe box. The Tool Setup option should be selected now. If not, select it. Enter .1 for the Cutting Depth, select the Collision detection box and check the **HORIZONTAL** option. Now click OK.

The first thing that you will need to do is click on one of the lines of the tool itself. Just place your cursor right on one of the lines and click your left mouse button **ONE** time. Move your cursor up or down depending on what line you clicked and then click your left mouse button once more. Now hit the F3 key on your keyboard to select the entire chain. Your cursor will change and you will notice a star. Click your left mouse button on the point at the tip of the tool. OK. Now you can delete the entire tool from the CAD screen by clicking and dragging a selection window over the tool itself and then clicking on the trash can icon.

Now hit the View All button again to zoom in.

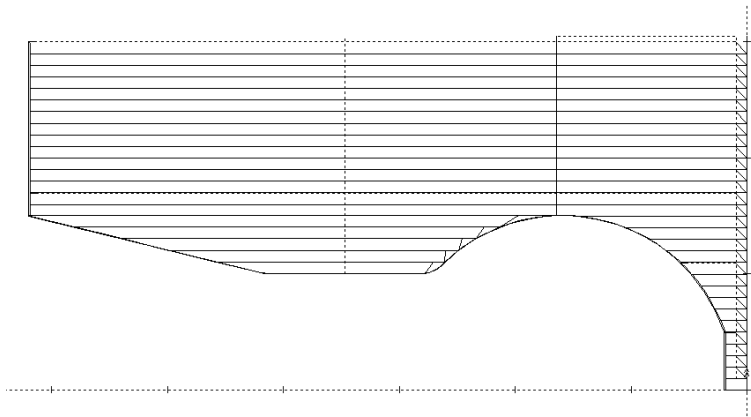
STEP 13

Now we can proceed with the toolpath generation. Select **OTHER** from the main menu and then **POCKET**. Remember to make sure that the Lathe Mode is activated at this time. By selecting **POCKET** and then the **Lathe** button, you will bring up the Lathe box again. Select the CONTOUR DEFINITION option. For CUTTING DEPTH we will typically enter the amount that we want taken off for each pass. This is based off of the material that you are cutting, what the tool is made of, how fast you want to go and how many passes you want to have in the program etc. For training purposes enter .1 for the cutting depth or just leave it as it should already be set to this. Make sure that you have Collision Detection checked and select **HORIZONTAL** under direction. Now select OK.

STEP 14

After selecting OK in the lathe box, the software will be live and ready to make the first geometry selection. Click your left mouse button at the **TOP** tip of the **DASHED** line and point your arrow or move your mouse **DOWN** and click your **LEFT** mouse button once to secure the direction. Now hit F3 on your keyboard. This will automatically generate the toolpath for the facing operation.

Once again as mentioned in the last lesson, if you can't see a **directional arrow** it is because the direction arrow option in the environment setting is turned off. In earlier version of the software you would always see the direction arrows when you select an entity. In Version 21 you can turn these on and off by selecting the file menu and then environment. Select the SCREEN tab and you will see this option listed. The result toolpath:

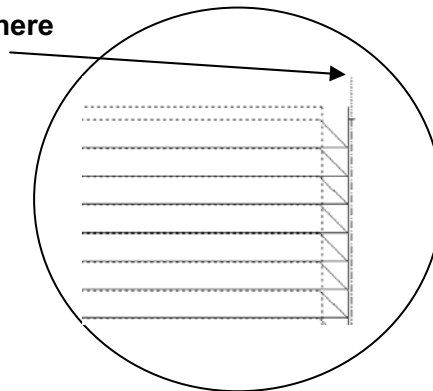


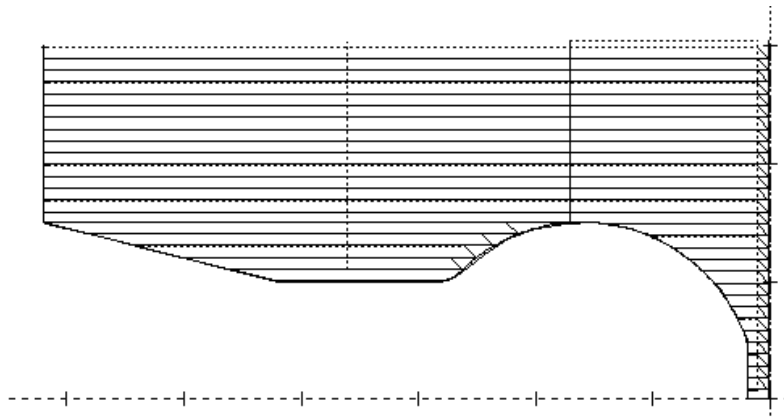
STEP 15

Now, let's go ahead and generate the program for this part. Select the SPECIAL/NC-CAM menu and then INSERT NC. Click on a FANUC post and then click OK to fully open the CAM on the right side of the screen.

Now you need to select the AUTO CUT icon or Machine from the main CAM menu and then Auto Cut. Now you need to click on the start of the Lathe toolpath that we had generated earlier that is on the CAD side at this time. See the next image.

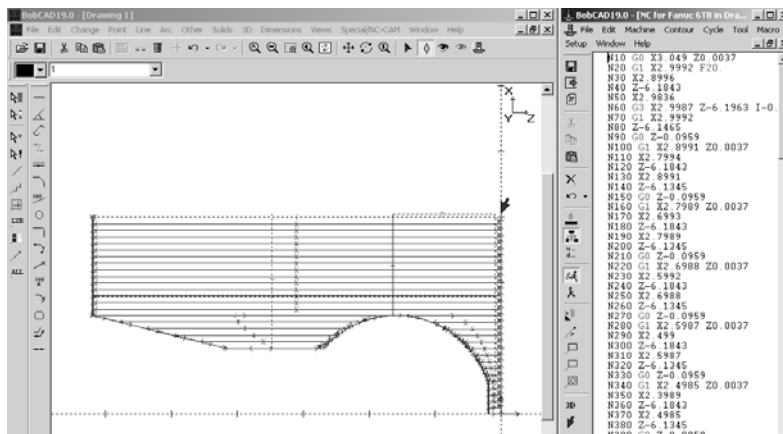
Click here





NOT the DASHED LINE, the solid line just to the left of the dashed line.

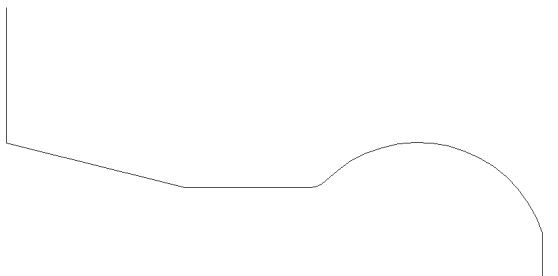
Point your arrow down and click your left mouse button one time. Now hit the F3 button on your keyboard. This will generate all of the code for the program.



As mentioned earlier in this manual, there is an alternative way to generate the program for the above part through the use of Version 21 Scripting. Follow along with the next example on how this is accomplished.

STEP 1

Here you can see the same profile that we started with in the beginning of the last lesson you just completed.



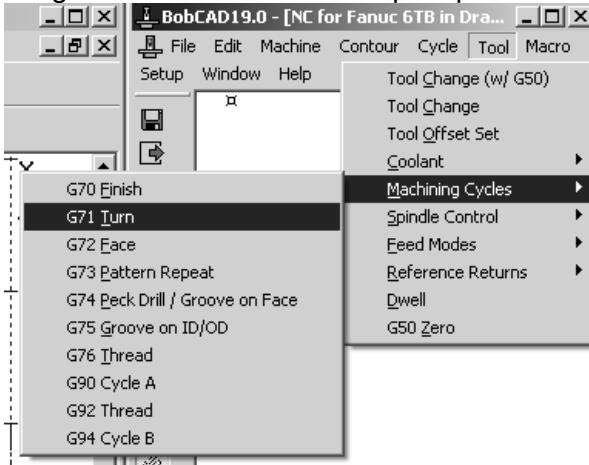
You would go ahead and open the CAM by selecting the SPECIAL/NC CAM menu and then INSERT NC. Note that special scripts have been created for specific post processors that are available through BobCAD-CAM technical support or you may also create them if you know Visual Basic.

STEP 2

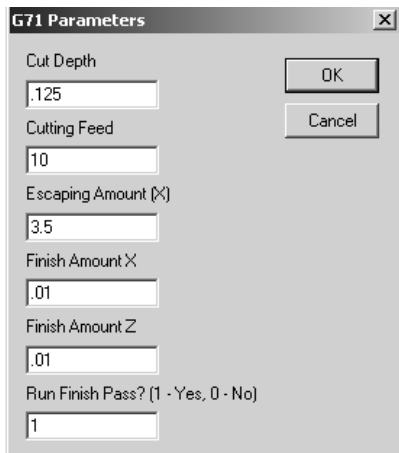
You would need to hold down your SHIFT key and click on the tip of the last entity of this profile chain. This is on the far left. The reason for this is so that the directional arrows all end up going toward the left.

STEP 3

Now you would go to the CAM Tool menu and select the Machine Cycles. This Tool menu in the example image below is based off of a post processor that was setup for a FANUC Lathe.



Simply select the script that would be needed. In this case for example, we selected the G71 Turn script. This will bring up a special box.



You would simply fill in the important information:

- Cutting Depth.
- Cutting Feed.
- The retract/escaping amount for X.
- The finish amount in X.
- Finish amount for Z.
- Indicate if you want to also run a finish pass.

Then by selecting OK, the software will generate the complete program. You will not be able to view the toolpath, only the G-Code program and profile.

That is how easy it is to program lathe parts through the use of scripts.

Quick Lathe Lesson 5: Internal Lathe Cutting

STEP 1

Start with a new drawing screen and make sure that you have turned the lathe mode **ON** by selecting the Lathe ON/OFF button. This is the button on the main toolbar that has a blue diamond on it.

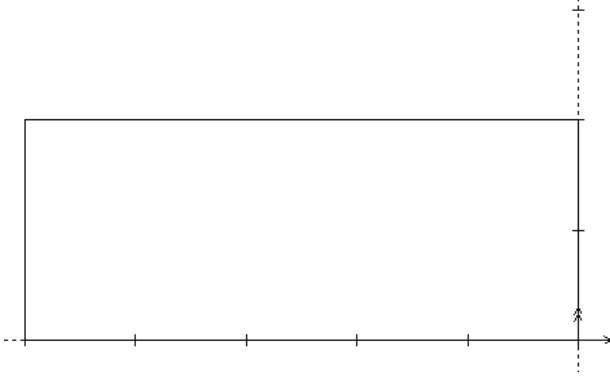
STEP 2

The first thing that we need to do is draw our stock profile. Select the OTHER menu and then RECTANGLE.

Select the Bottom Left option under MODE.

- Enter -5 for Z and leave both X and Y at 0.
- Enter a width of 5, a height of 2.
- Choose the STRAIGHT corner type and click OK.

The result:



The values that we are using for the width will represent the raw block material placed on the machine. Therefore, we are using material that is 4-inches in diameter.

STEP 3

Now go to the POINT menu and select Coordinates. Enter the following coordinates.

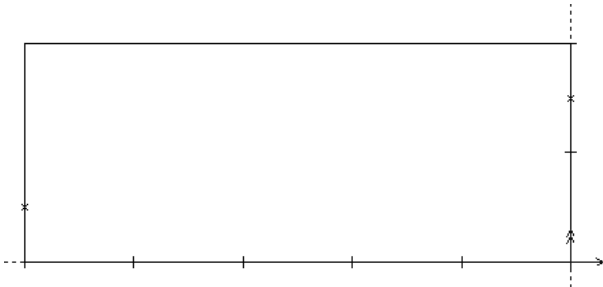
- Z 0
- X 1.5
- Y 0

Now click CONTINUE. Now enter:

- Z -5
- X .5
- Y 0

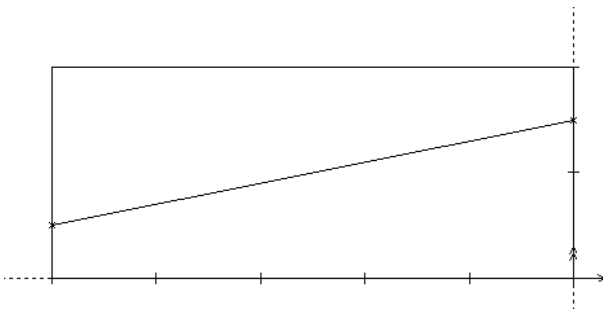
Select OK.

The result:



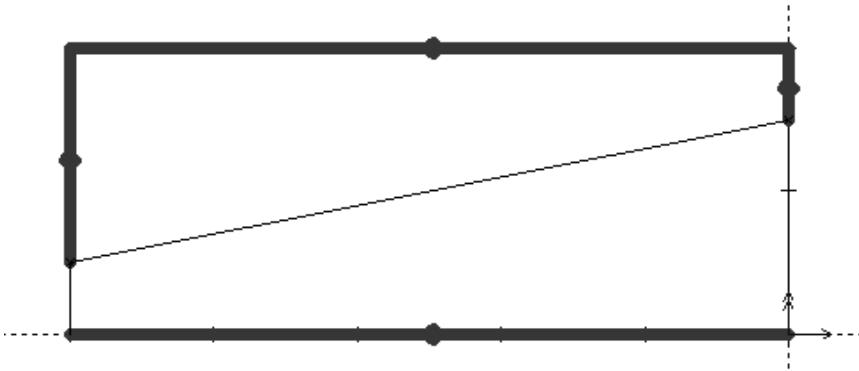
STEP 4

Now go to the line menu and select the JOIN function. Go ahead and click on one point and then click on the other to draw the new line. When you have done this, right-click your mouse to exit the function.

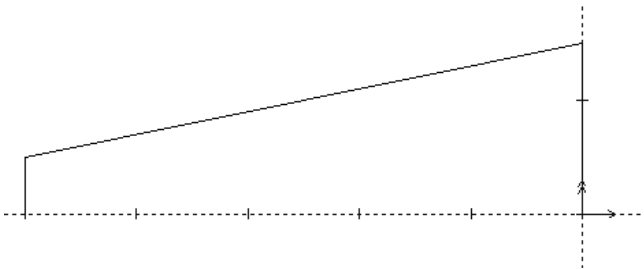


STEP 5

Now we want to break up the geometry that you have on the screen. As in earlier lessons, select **CHANGE** from the main menu and then **BREAK**. From the Break sub-menu select **BREAK MANY**. The Section box will appear with the break options. Select **ALL** and click OK. Now the drawing has been broken up at every intersecting piece of geometry. Use the next image to see the enlarged lines. You want to hold down the **CONTROL** key and click on each one of them, including the 2 points that you drew. Once you have made all of the correct selections click the trash can icon to delete them.



The result:



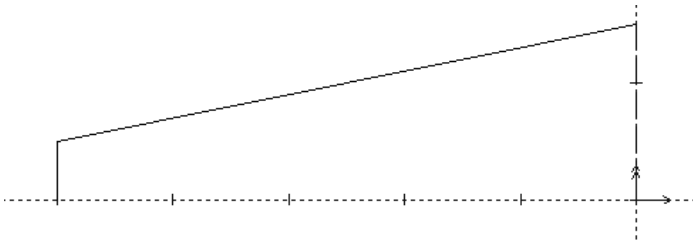
STEP 6

Now we are going to slightly modify this part by changing the far RIGHT line to a dashed line. The reason for this is because BobCAD looks at a solid line as a part of the contour. By changing this line to a dashed line you are telling the software that it is OK to cut down that wall.

To do this, click on the RIGHT vertical wall to select it. Now select **CHANGE** and then **ATTRIBUTES** and then the **CONTOUR** tab. Select the **DASHED** line style and then click OK. Now click on the REFRESH button from the main toolbar.

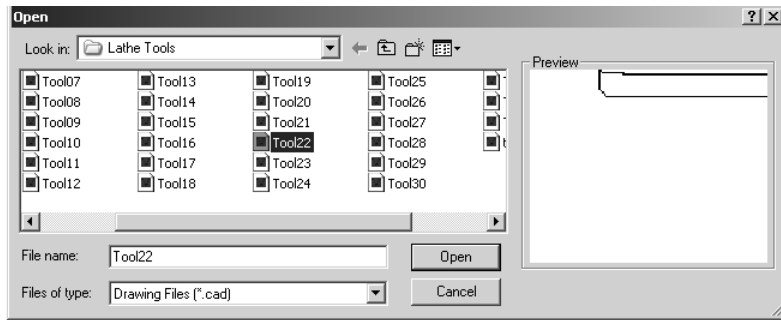


The result:



STEP 7

Now you are ready to merge in a lathe tool and perform a tool setup. Go to the **FILE** menu and select **MERGE**. Locate the BobCAD-CAM folder and double-click on it. Then double-click on the TEMPLATES folder. Then double-click on the Lathe Tools folder to access the lathe tools. They are all listed as .cad files. Click on **TOOL22.cad** and then click OPEN to merge it in.

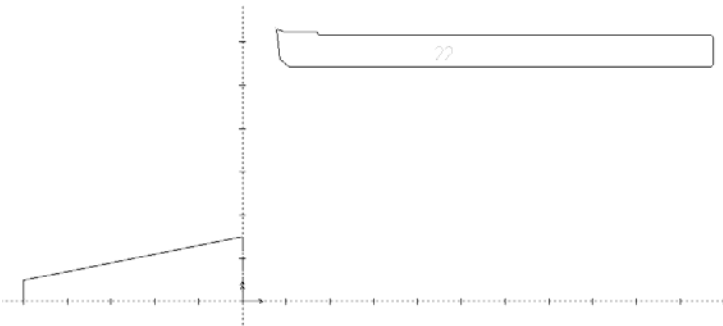


Lathe tools are located at: **C:\Program Files\BobCAD-CAM\BobCAD-CAM Version 21\Templates\Lathe Tools**

Now select the view all button from the main toolbar so that you can see the part and the tool on the screen.



The result:



STEP 8

Now we need to perform a tool setup. Select the OTHER menu and then POCKET. Now click on the lathe button to open the lathe box. The Tool Setup option should be selected now. If not, select it. Enter .125 for the Cutting Depth, select the Collision detection box and check the **HORIZONTAL** option. Now click OK.

The first thing that you will need to do is click on one of the lines of the tool itself. Just place your cursor right on one of the lines and click your left mouse button **ONE** time. Move your cursor up or down depending on what line you clicked and then click your left mouse button once more. Now hit the F3 key on your keyboard to select the entire chain. Your cursor will change and you will notice a star. Click your left mouse button on the point at the tip of the tool. OK. Now you can delete the entire tool from the CAD screen by clicking and dragging a selection window over the tool itself and then clicking on the trash can icon.

Now hit the View All button again to zoom in.

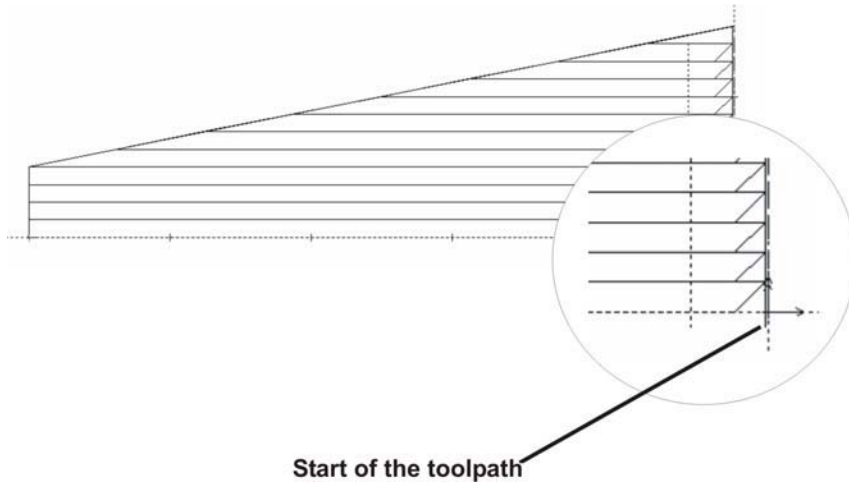


STEP 9

Now you can go back to the OTHER menu and select POCKET. Click the lathe button to open the Lathe box. Now Contour Definition should be selected. If not, select it. The Cutting Depth should be .125, collision detection should be checked and the HORIZONTAL option should also be checked. Now select OK.

At this point the software will be live and ready for you to make the next selection. Click on the **BOTTOM** tip of the **RIGHT** vertical line. Move your cursor upward toward the top of that line and then click your left mouse button one more time to secure the selection direction. Now hit the F3 key on your keyboard or click on the bottom tip of the left vertical line. This will generate the complete toolpath.

The result:



That completes this lesson.

CHAPTER 12

TROUBLE SHOOTING & FAQ

Section 90. Troubleshooting BobCAD-CAM

- ⇒ I cannot install my post processor.
- ⇒ I lost my toolbars.
- ⇒ My Booleans aren't working.
- ⇒ My tool depth is not correct.
- ⇒ I cannot Chain-Select my geometry.
- ⇒ I get an error when I merge a file.
- ⇒ My machine cannot see my g-code file.
- ⇒ My machine is cutting circles where they shouldn't be in the NC Program.
- ⇒ How do I transfer my software license to another computer?
- ⇒ How do I download and install my post processor?
- ⇒ When I generate G-Code for my 3D toolpath there is no Z-Movement in the code.
- ⇒ When I generate pocket toolpath, my toolpath over-runs my islands.
- ⇒ When I open my file, un-blank and deselect everything, there is nothing on my screen.
- ⇒ When I click on my geometry there is a red box around everything.
- ⇒ Will BobCAD-CAM software interface directly with my 3D digitizer?
- ⇒ My controller requires that my X, Y and Z Axis be reversed. How do I set this up?



Section 90. Troubleshooting BobCAD-CAM

This section will list out solutions to situations that may come about.

I CANNOT INSTALL MY POST PROCESSOR:

STEP 1

Open BobCAD-CAM.

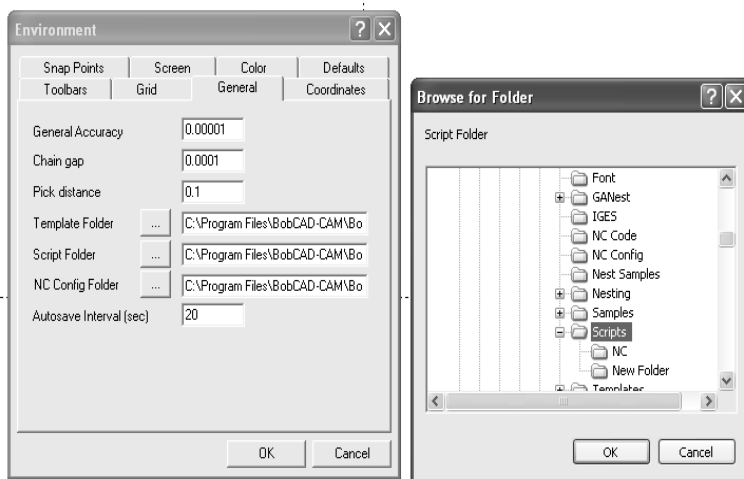
STEP 2

Go to the File menu, choose Environment and then click on the General Tab.



STEP 3

Click on the button next to “Scripts Folder” in the General Tab. When the “Browse for Folder box appears, locate and make sure that the **Scripts** folder is highlighted. Then click OK in the Browse for Folder box.



STEP 4

Now click on the button next to “NC Config Folder” make sure that NCConfig folder is highlighted in the new Browse for Folder box. You highlight the folder by clicking on it one time. When you have done this, click OK.

Now click OK at the bottom of the Environment box and your post processor should load properly now.

I LOST MY TOOLBARS:

STEP 1

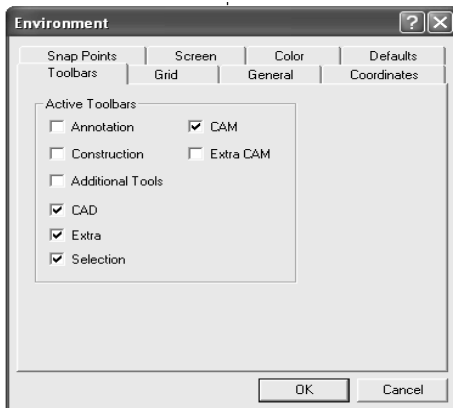
Open BobCAD-CAM.

STEP 2

Go to the File menu, choose Environment and then click on the Toolbars Tab.

STEP 3

Click on the button next to the toolbar names you need and then click OK at the bottom. You should now see them again.



MY BOOLEANS AREN'T WORKING:

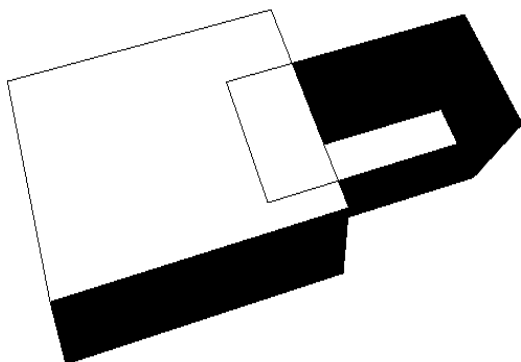
Certain commands require that geometry is selected before you *can* select the command.

- ⇒ If it is the **translate** command any geometry can be selected.
- ⇒ If it is the **Booleans** command two solids must be selected.
- ⇒ The Booleans command must be used on two **Intersecting** solids.

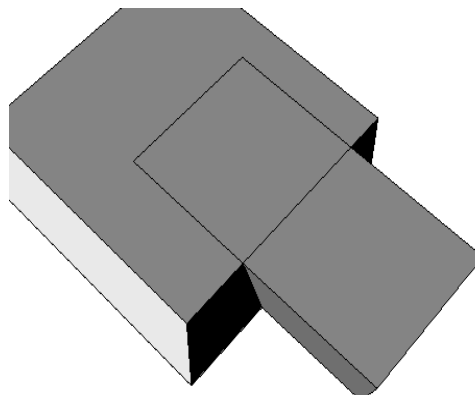
STEP 1

Check that the solids have caps and are not skins.

Incorrect



Correct



MY TOOL DEPTH IS NOT CORRECT:

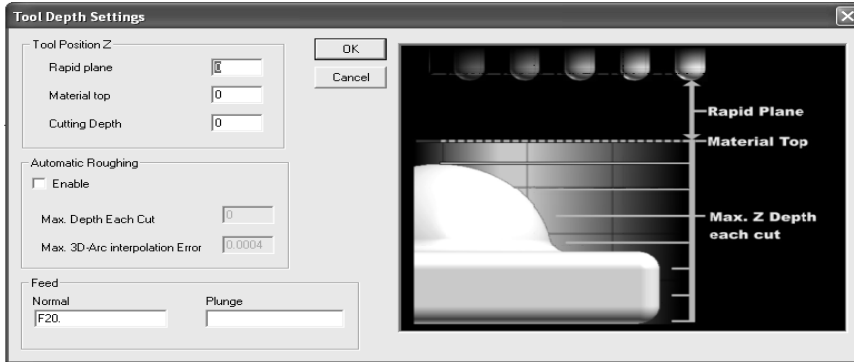
STEP 1

Click your tool depth settings button which is located on the CAM main toolbar. This is the button that has the small UD (up down) on it.



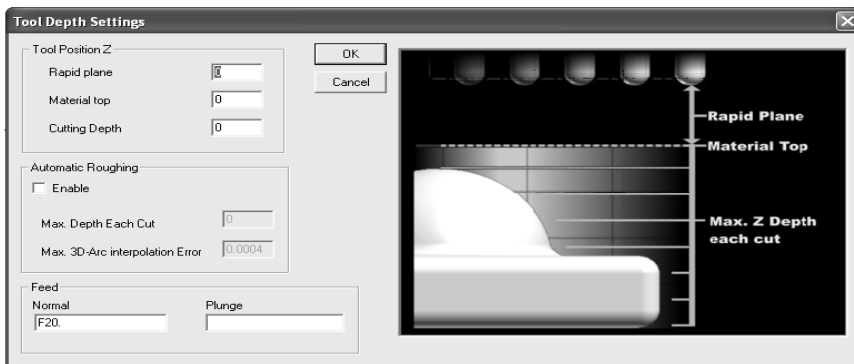
STEP 2

Set your cutting depth.



STEP 3

Set your feed rate with a capitol F as this must be in front of the value. Example: F20. If there is no F in the feed rate, the number will be inserted as part of the z value.



STEP 4

If cutting 3D toolpath, make sure that the 3D button is selected. When you click on it, it highlights



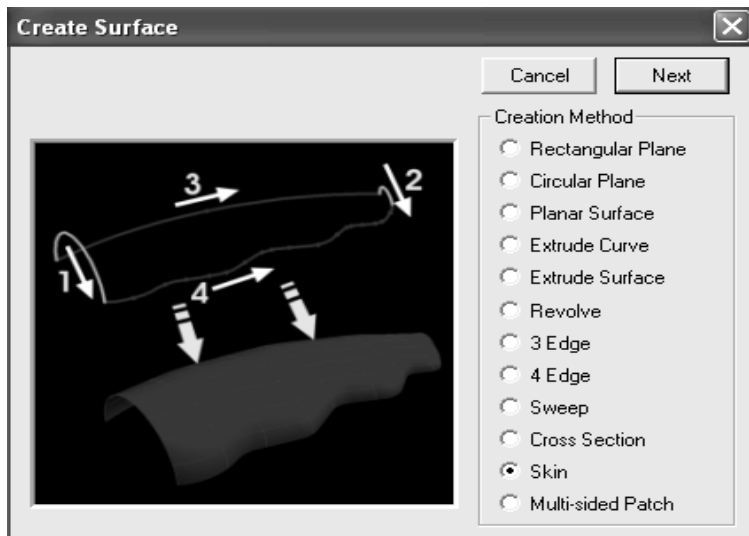
MY SKIN FUNCTION DOES NOT WORK:

STEP 1

Check that the lines and arcs being used have no overlaps and gaps.

STEP 2

Check that the geometry is being selected in the order and direction as illustrated in the Create Surface dialog box after selecting the Skin option.



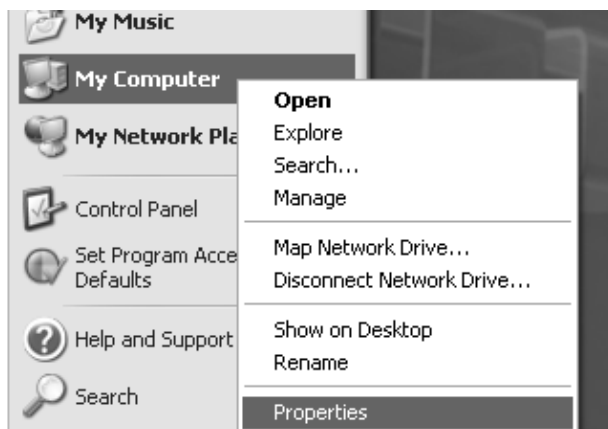
I CANNOT SEE MY RENDER WINDOW:

STEP 1

Make sure your video card meets the minimum requirements. If it does go to step 2

STEP 2

Go to your windows "My Computer" icon.

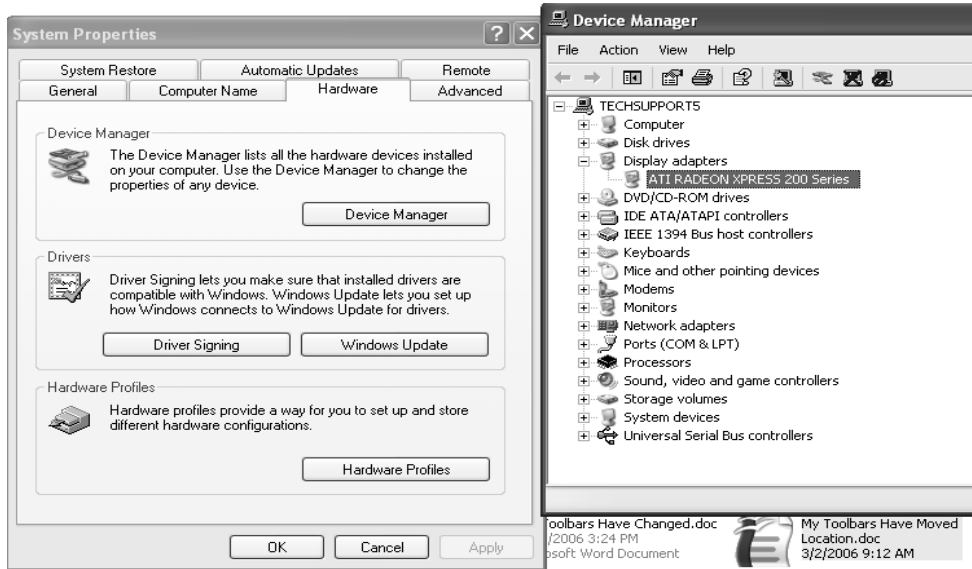


STEP 3

Right click on "my computer" and then left click on **properties**.

STEP 4

Click on the Hardware tab.



STEP 5

Click on the Device Manager button. Now click the plus sign next to where it says “display adapters” in the new “Device Manager” box that appears.

STEP 6

Note the manufactures name that is displayed.

STEP 7

Find the manufactures website, download and install the latest drivers.

I CANNOT CHAIN SELECT MY GEOMETRY:

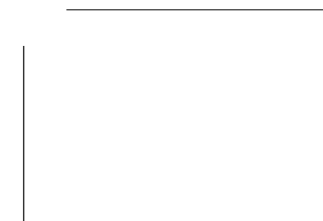
STEP 1

Zoom in on the geometry using the Zoom Window icon from the main CAD toolbar.



STEP 2

Check for any breaks or overlaps. You can use the Erase Doubles function by going to the CHANGE menu and selecting it from the Reorganize sub-menu.



STEP 3

If that doesn't work you can close any breaks or overlaps using the trim/extend feature.



STEP 4

Hold the shift key and select the geometry again.

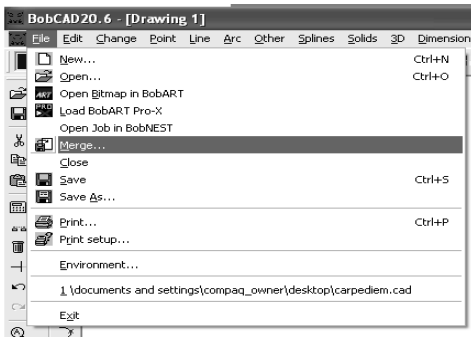
STEP 5

Continue closing any breaks or overlaps until all of profile will chain select.

I GET AN ERROR WHEN I MERGE A FILE:

STEP 1

Go to the File menu and choose Merge.



STEP 2

When the Open box appears, go to the “Files of Type” pull-down and choose the type of file that you are trying to merge into BobCAD-CAM. Then select the file and click the Open button to bring it in. By choosing “All Files” under Files of Type, you will generate an error.

MY MACHINE CANNOT SEE MY G-CODE FILE:

STEP 1

First check your machine manual to find out what file extension your machine requires.

STEP 2

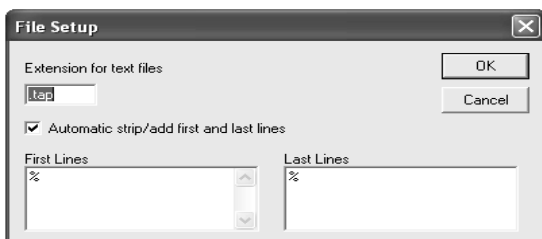
Open the NC side of your software.

STEP 3

Go to the CAM File menu and choose Setup.

STEP 4

Type the file extension in the “Extension for text files” box. Example: **.TAP**



STEP 5

Save the file as a text file and it should now have the proper extension and your machine should recognize the file now.

MY MACHINE IS CUTTING CIRCLES WHERE THEY SHOULDN'T BE IN THE NC PROGRAM:

Sometimes due to the geometry offset BobCAD can create what we call “crop circles.” This is usually due to a mathematic rounding problem being encountered by the software.

The way to correct this problem is to go to the File menu, choose Environment and select the General Tab.

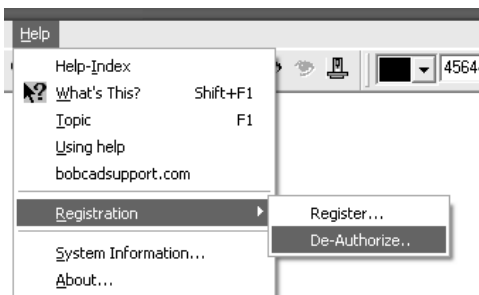
Set the general accuracy at .0001 then recreate the toolpath and g-code. This is the optimum accuracy setting for BobCAD and should correct the problem and prevent this from happening in the future.

NOTE: If the above does not solve the “crop circles” then open the NC side of the software. Go to the Special/NC CAM menu and select “Insert NC” and then open the CAM with your post processor. Then choose the Setup menu in the CAM and DRIVER. Set the Max. Circular Movement to ¼ arcs and recreate the G-Code program.

HOW DO I TRANSFER MY SOFTWARE LICENSE TO ANOTHER COMPUTER?

STEP 1

Open BobCAD-CAM and go to the Help menu. Choose Registration and then De-authorize. You will then see two user codes.

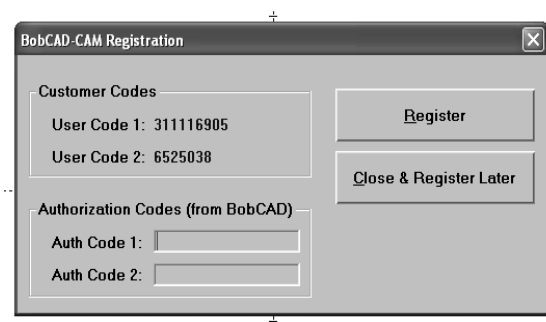


STEP 2

You will need to call BobCAD-CAM at 877-262-2231 or 727-442-3554 Monday-Friday 9am-5pm EST to speak with the registration department. Please note that you will need to be at your computer.

STEP 3

Have your user codes ready to give to the registration department as they will give you a De-Authorize code to enter. After entering this code you will be able to register another computer.



HOW DO I DOWNLOAD AND INSTALL MY POST PROCESSOR?

STEP 1

Go to the www.bobcad.com website.

STEP 2

Then click on the **Support** link at the top part of the site.

STEP 3

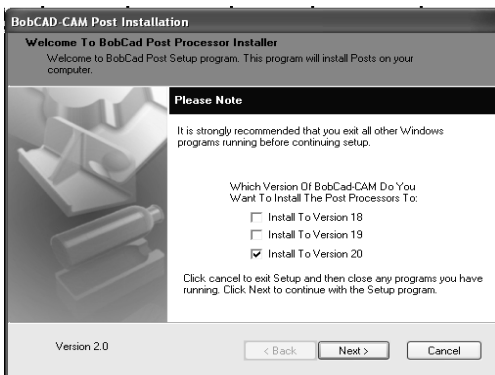
Click on the V18, V19 & V20 Series Post Processors link.

STEP 4

Double-Click on the post for your machine and then save the file to your computer desktop.

STEP 5

Double-Click on the saved post file and it will begin to install. Follow the on screen instructions.



You can also go to the following link directly and view a complete list of post processor configurations:

www.bobcad.com/posts

WHEN I GENERATE G-CODE FOR MY 3D TOOLPATH THERE IS NO Z-MOVEMENT IN THE CODE:

There are 3 ways to resolve this. They are:

1. Check your tool depth settings and make sure that the material top is the same as the top of your part on the drawing screen.
2. Move the part below Z-0. You can do this by selecting the part and using the translate feature to lower it in the Z axis.
3. Make sure that when you generate the G-Code program, you have the 3D button selected so that it is turned ON, on the CAM toolbar.

That should resolve this issue.

WHEN I GENERATE POCKET TOOLPATH, THE TOOLPATH OVER-RUNS MY ISLANDS:

Make sure that all of your profiles are on the SAME Z level on the CAD screen. Also, make sure that the islands are CLOSED chains. You should make sure that the outside profiles and inside profiles (islands) are selected going in opposite directions.

WHEN I OPEN MY FILE, UN-BLANK AND DESELECT EVERYTHING, THERE IS NOTHING ON MY SCREEN:

You can fix this by changing the background color of your drawing screen or by selecting ALL and then using the Attributes>Contour functions to change the color of all geometry on the CAD screen.

WHEN I CLICK ON MY GEOMETRY THERE IS A RED BOX AROUND EVERYTHING:

Your geometry is grouped. With your geometry selected, use the UN-GROUP function from the CHANGE menu.

WILL BOBCAD-CAM INTERFACE DIRECTLY WITH MY 3D DIGITIZER?

No. BobCAD-CAM will not directly interface with a 3D digitizer. You can however use a text file from the digitizer to create your profiles. This means that you would import the text file into the CAM editor, select all of the text and then use the "Geometry From NC" function in the CAM file menu. Before selecting all the text, make sure that the first line of code has G01 on it.

MY CONTROLLER REQUIRES THE X, Y AND Z AXIS TO BE REVERSED. HOW DO I SET THIS UP?

There are ways that you can easily accomplish this. They are:

STEP 1

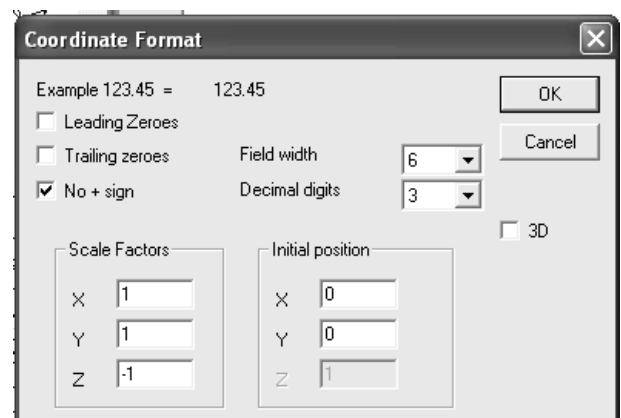
Open the NC side of the software.

STEP 2

Go to Setup and select Coordinates.

STEP 3

Set the axis that the coordinates need to be reversed as a -1 instead of 1 under Scale Factors.



CHAPTER 13

SCRIPTING

Section 91. Scripting



INTRODUCTION:

Welcome to the BobCAD-CAM **Scripting** manual. In this section of the manual we are going to cover creating, storing, running and troubleshooting scripts written for BobCAD-CAM.

Before we get started we want to clarify a few things. The BobCAD-CAM scripting engine is a proprietary scripting format for creating and executing Macros within BobCAD-CAM. These Macros can only be used within BobCAD-CAM because the Objects and References are not supported by any other programs. Also, because BobCAD-CAM does not have an Open API (Application Programming Interface), the Objects and References can not be accessed outside of the BobCAD-CAM user interface.

The BobCAD-CAM Scripting Engine can be used for many different tasks. It can be used to obtain information from the user as well as display information to the user. This information can be used to output code to the NC-CAM window or modify the code that is already present. The scripting engine can also be used to perform tasks on the drawing screen, including modifying and/or creating new geometry.

You will find several scripts already installed in your software. The files are located in "C:\Program Files\BobCAD-CAM\BobCAD-CAM V21\Scripts". You will notice that all the scripts that were included with the software installation have a .bas extension. Script files must always have a .bas extension in order for BobCAD-CAM to make use of them.

Due to the lack of a Macro editor in BobCAD-CAM scripts can be written in any text application including WordPad, Note Pad and MSWord. When saving the file you must make sure that you have selected 'All Files', so that you can define the extension saved with the file.

You will also want to make sure that all of your scripts are in the same folder so that they can be easily found. Once all of the scripts are being stored in the same location you will want to set the path for your Scripts folder in BobCAD-CAM.

TO SET THE PATH FOR YOUR SCRIPTS FOLDER:

STEP 1

Open BobCAD-CAM to a blank Drawing screen.

STEP 2

Then go to the 'File' menu and click on the 'Environment' option. In the 'Environment' window click on the 'General' tab.

STEP 3

Next, click on the button next to the 'Scripts Folder' option. Make sure the folder that your Scripts are stored in is highlighted and click OK. Then click OK in the 'Environment' window and close BobCAD-CAM.

(The reason for the last step is to store the data for your scripts folder so that you do not have to repeat this procedure every time you open the software.)

We will be covering all of the capabilities available to the scripting engine in this section. There are some limitations to the scripting engine but most of those limitations can be overcome by creativity on the programmers part.

SCRIPTING VARIABLES:

Variables

Variable(definition) - A named memory location in which a program can store intermediate results and from which it can read it them.

Variables are used for storing information. BobCAD-CAM supports three types of variables, GLOBAL variables, Local variables and Arrays. A GLOBAL variable will hold the data that is assigned until the software has been closed or until the data is replaced. A Local variable will only hold its value as long as the Macro is running. Upon exiting the Macro the data for that variable is lost. Arrays are used to store data in groups, this will be covered much more thoroughly later in this section. Variable names are user defined and can be any combination of letters, symbols and numbers. The only stipulation is that a variable must start with a letter and not a number. If a variable name starts with a number the scripting engine will produce an error.

A. Local Variables

There are three types of LOCAL variables supported by the BobCAD-CAM Scripting Engine; **double**, **integer** and **string**.

A **double** is a number that can contain decimal values(e.g. 2.1, 4.75, 3.9999).

An **integer** can be any whole number (e.g. 1,2,3,etc.).

Mathematical calculations can be performed on double and integer variable types.

A **string** can be any type of information, numbers, letters or a combination of the two. Mathematical calculations can not be performed on a string variable.

Local variables are defined prior to their use in the Macro. The syntax for defining a variable is located below.

```
dim (variable name) as (type)
```

```
dim tool_number as integer
```

Here we are defining 'tool_number' as a whole number value. The word dim is used to declare the variable. You will find that the dim command is used only on Local variables and Arrays, a GLOBAL variable will use a different declaration command. It is helpful to name the variables so that the name(s) coincide with the function or purpose for the variable. This way if there is a problem with one or more calculations locating the cause is much more simple.

B. GLOBAL Variables

Unlike the **Local** Variable a **GLOBAL** variable requires no type definition. That means that a GLOBAL variable can be used as an integer, double or string. GLOBAL variables are normally used to pass a value from one Macro to another.

GLOBAL variables, like Local variables, are defined prior to their use in the Macro. The syntax for defining a GLOBAL variable is located below.

```
GLOBAL (variable name)
```

```
GLOBAL tool_number
```

C. Arrays

An Array is used as similar to listing for the scripting engine. If you have several values that you want to store to different variables but only want to name one variable, an array would be the tool to use. Like a GLOBAL variable an Array requires no type definition. However, like the Local variable, an Array will lose its value once you have exited the Macro.

As with Local and GLOBAL variables, the Array must be defined prior to its use in the Macro. The syntax for defining an Array is located below.

```
dim (array name)()
```

```
dim tool_number()
```

Notice that, with the Array, there are parenthesis"()" at the end of the variable name. These parenthesis are used as a place holder for calling up the values stored to the array. Each position in the array is given a value from 0 to **any number** needed by the user, these values in the parenthesis are whole numbers only.

For example: To call the fifth value stored to the array: (*variable_name*) = *tool_number*(5)

These are the ways a variable can be defined. They all have different purposes in the scripting engine and can be used in various ways. We will cover the use of the different variable types later in this section of the manual. Right now it is enough to simply know of their existence.

COMMANDS:

Existing Commands

Command (definition) - A character string which tells a program to perform a specific action.

Query (definition) - A user's (or agent's) request for information

In this section we will be covering the commands available in the BobCAD-CAM scripting engine. Commands refer to the windows in BobCAD-CAM and the input options for that window.

A. Inputting Values to the Command windows

For example, the Line Coordinates window has input fields for the beginning and end coordinates of a line. To create a line that starts at X0, Y0, Z0 and goes to X0, Y1, Z0 using the Line Coordinates option, the method would be as follows:

LineCoordinates XStart = 0, YStart = 0, ZStart = 0, XEnd = 0, YEnd = 1, ZEnd = 0

You can see that in the command "LineCoordinates" is italicized. This is the main window for the Line Coordinates command. The input fields for the command are underlined. You can not use an input field without identifying the main window that field is present in. If you try to issue a field without issuing the main window command first an error will result.

Using all the input fields in the command is not necessary. The previous value input into the field will be used if that field option is omitted. This allows the programmer to assume values when it is possible to do so. The following example would still work assuming that the values for the Start Coordinates were input previously as 0.

LineCoordinates XEnd = 0, YEnd = 1, ZEnd = 0

B. Querying the Command windows for Information

The programmer can also query the software for information that was previously input into a field within a command window. The structure for this will be slightly different because the value from the input field will need to be stored in a variable. The variable is placed where a number would normally go because we as programmers may not know what the user has input into the field.

LineCoordinates Output, XStart = (variable_name), YStart = (variable_name), ZStart = (variable_name), XEnd = (variable_name), Yend = (variable_name), Zend = (variable_name)

Notice in this command line we have added the word 'output' between the main window and the input fields. In order for the values from the input fields to be assigned to the variables, we must use the Output statement prior to the issuance of the input fields. As with variable assignment and creation, the variable names are determined prior to their use in the scripting engine and must follow the guidelines for naming convention.

As with assigning a value to the command window, it is not necessary to use all of the input fields within this command. This is helpful when the programmer only needs to gather one or more of the inputs from the command window. The following sample would output the values for the Start Coordinates to variables.

LineCoordinates Output, XStart = (variable_name), YStart = (variable_name), ZStart = (variable_name)

C. The Available Commands In The BobCAD-CAM Drawing Screen

In order to provide you with a clear understanding of the Commands we have formatted them so that each element of the command can be easily identified. The main command window will be displayed in Bold Italics. The input fields will be presented in Bold Underlined text. First we will present the Commands available on the drawing screen.

The following command allows the user to open a new drawing screen. The main command is CADNew. The Name input field refers to templates that may be setup in the software. A template file must have the .cct file extension in order for it to be used in this fashion. The Type input field is set to 0 to indicate a CAD screen. The Name input field may be omitted from the command in a script. The Type may also be omitted. If one or both is omitted the script will open a window using the last input values.

CADNew Name = "normal", Type = 0

This command will open a new NC-CAM screen.

CADNew Name = "normal", Type = 1

This command Prints the current Drawing screen. If one or more drawings are open only the active drawing will be printed.

CADPrint

In scripting you can also select Geometry according to Color. There are two lines of commands that must be executed in order for this to occur. The color_value will be a string value equivalent to the color you wish to select. For example royal blue would have a value of 16711680, black would be 0 and red would be 255.

CADColor

XSelect Colors = (color_value)

You can also select Geometry according to the Layer the geometry was assigned to. Like the color selection this requires two lines of commands in order to function properly. The layer_name is a string value for the name the layer was given.

CADLayer

Xselect Layers = (layer_name)

The following command will select All Geometry that is visible on the drawing screen.

SelectAll

To Undo the last action performed use the following command.

Undo

To Repeat an action that was previously undone use this command.

Redo

To Cut the selected Geometry from the drawing screen use this command.

Cut

To Delete Selected Geometry use this command. If there is no geometry selected, the selection type dialogue box will appear.

Delete

The Paste command will Paste Geometry from the clipboard to the drawing screen. The Scale input is the scaling factor that is to be used on the part, if needed.

The Mirror X and Mirror Y inputs are either true or false. If they are set to true then the Geometry will be mirrored on the selected axis. Translate first is also true or false and will move the selected geometry to the location designated by the X, Y and Z inputs prior to any of the other options being performed on the geometry.

The NoScaleX, NoScaleY and NoScaleZ inputs are true or false data as well. If they are set to true then any change to the Scale value will not be applied to that axis of the selected geometry. This is helpful when you want to stretch or skew an entity on the drawing screen.

Paste Scale = (scale_size), **MirrorX** = (true or false), **MirrorY** = (true or false), **TranslateFirst** = (true or false), **NoScaleX** = (true or false), **NoScaleY** = (true or false), **NoScaleZ** = (true or false), **XRot** = (variable_name), **YRot** = (variable_name), **ZRot** = (variable_name), **X** = (variable_name), **Y** = (variable_name), **Z** = (variable_name), **NCopies** = (variable_name), **OnPoints** = (true or false)

This command allows the user to select geometry that is not hidden on the drawing screen by entity type. The option will be set to either true or false. If set to true the specified entities will be selected on the drawing screen.

Select Point = (true or false), **Line** = (true or false), **Arc** = (true or false), **LinDim** = (true or false), **AngDim** = (true or false), **RadDim** = (true or false), **Text** = (true or false), **Pgon** = (true or false), **Group** = (true or false), **Solids** = (true or false)

You can also Mirror selected Geometry on the drawing screen. The Type input has to be a value from 0 to 5. 0 is mirror around an existing line. This will place the user in selection mode before any mirroring occurs. It is best to use the command for existing line at the end of a script due to the lack of a wait command.

A Type input of 1 will mirror around specified coordinates. The coordinates are input into the X1, X2, Y1, Y2, Z1 and Z2 input fields.

A Type input of 2 will place the user in Selection mode. The user will have to select two points that will be used as the coordinates for mirroring the part, so it is best to use this method at the end of a Script.

A Type input of 3, 4 or 5 will mirror the part without user intervention. 3 will mirror the part around the X axis. 4 will mirror around the Y axis and 5 mirrors the part around the Z axis.

The Copy input is true or false. If set to true the part will be copied during the mirror operation.

Mirror Type = (0 to 5), **X1** = (variable_name), **Y1** = (variable_name), **Z1** = (variable_name), **X2** = (variable_name), **Y2** = (variable_name), **Z2** = (variable_name), **Copy** = (true or false)

To Rotate Selected Geometry the following command is used. Rotation will occur in one of two methods determined by the Type input. Type 0 is rotation around specified coordinates. The coordinates are designated by X2, Y2 and Z2. The amount of rotation and direction are designated by the XRot, YRot and ZRot values and are input as degrees of rotation. The rotation can also scale the geometry by changing the Scale factor. Copy is a true or false input and if set to true will output the number of copies assigned in NCopies.

Rotate Type = (0 or 1), **XRot** = (variable_name), **YRot** = (variable_name), **ZRot** = (variable_name), **X2** = (variable_name), **Y2** = (variable_name), **Z2** = (variable_name), **Copy** = (true or false), **Scale** = (variable_name), **NCopies** = (variable_name)

The next command will Translate or Move the selected geometry. The Type option controls how the geometry will be moved. If the Type value is 0 then the user will be placed in Sketch mode for the new location of the part, this is similar to the Drag option. The Type value of 1 is used for the Snap Method. The user will be prompted to select the beginning point and the end point for the movement. These options are best used at the end of a script due to the required user intervention.

If the Type is 2 then the part will be moved the distance specified in the X2, Y2 and Z2 values. Type value of 3 will result in the part being moved from X1, Y1, Z2 to X2, Y2, Z2 or the difference between the values.

The Scale, Copies and NCopies are used in the same fashion as with the rotation option.

Translate Type = (variable_name), **X1** = (variable_name), **Y1** = (variable_name), **Z1** = (variable_name), **X2** = (variable_name), **Y2** = (variable_name), **Z2** = (variable_name), **Copy** = (true or false), **Scale** = (variable_name), **NCopies** = (variable_name), **Drag** = (true or false)

This command will Optimize the selected geometry. Basically it creates a path through the selected points that reduces travel time. For lines and arcs this option will reduce the number of entities where possible and reduce travel time.

Reorganize

The next option will help Smooth out selected arc intersections or improve the tangency of intersected arc segments. MinAng is the minimum variation from tangency that is acceptable. The MaxAng is the maximum amount of correction that will be performed on the intersections.

Smooth MinAng = (variable_name), **MaxAng** = (variable_name)

The Unblank command will allow you to bring previously hidden geometry back onto the drawing screen. You can either Unblank All geometry that is hidden, a Layer that was previously hidden or place the user in Unblank Region mode. The region mode is best left as the final command in a script due to the need for user intervention. The other two options can be used at anytime within a script.

Unblank Layers = (layer_name), **All** = (true or false), **Region** = (true or false)

To Break selected geometry at intersection points use the following command. It will only work for selected geometry. If no geometry is selected the Selection Type window will be presented to the user for selection.

BreakIntersectAll

This command is used to Place all selected entities into one object. This object can not be machined, pocketed or otherwise modified while the group is maintained.

Group

This command will allow you to Break a selected group into individual entities. After the Group is broken you may generate code for the geometry or perform any of the other functions you need to.

Ungroup

This command will place the user in Point Sketch Mode.

PointSketch

This command is used to Place a point on defined coordinates in X Y and Z. If any of the coordinate addresses are omitted from the command the previous values for that axis will be used.

PointCoordinates X = (variable_name), Y = (variable_name), Z = (variable_name)

This command places a point relative to the selected point. Like the PointCoordinates function, any omitted value results in the previous entry being used.

PointRelative X = (variable_name), Y = (variable_name), Z = (variable_name), **OnAllEntities** = (true or false)

This command places a point at the intersection of two entities. This command can be used with an entire chain to produce points at every intersection. If no geometry is selected then the user is placed in selection mode.

PointIntersect

To place a point in the center of the selected arcs the following command is used. This function can be used for an assortment of selected arcs. This command will also place the user in selection mode if there are no selected arcs on the drawing screen.

PointArccenter

To place a point on a selected entity perpendicular to an existing selected point, use this command. This too will place the user in selection mode if there are no previously selected entities.

PointPerpendicular

To place a point on a previously selected entity we use this command. If the Relative input is set to 0 then the Percentage input will be used for generating the point on the entity. If Relative is set to 1 then the Distance and Angle inputs are used. The Angle input is used primarily for arc segments because they are measured in degrees, while the Distance input is used for line segments. If there is no selected geometry on the screen the user is placed in selection mode.

PointOnEntity Distance = (variable_name), **Percentage** = (variable_name), **Angle** = (variable_name), **Relative** = (0 or 1)

To change the appearance of selected geometry on the drawing screen the following command is used.

StyleP, ColorP, WidthP, SizeP, RelP and PStyleP **control the appearance of Points on the drawing screen.**

StyleC, Color, LineWidth, LineType, LineCaps and RelC **all control the appearance of lines and arcs on the drawing screen.**

StyleF, ColorF, OffsetF, DistanceF, AngleF, LineTypeF, LineWidthF, RelF and MaterialF **control the appearance of Faces placed on the drawing screen.**

StyleD, ColorD, DecPlc, UpperTol, LowerTol, DimScale, Witness, TolPlc, PreText, PostText, LineWidthD, ValType, ValZer, TolType, TolZer, RelD, ArrowSize, ArrowHeight, Arrowtype1, Arrowtype2, ArrowVis and ArrowMirror **all control the appearance of Dimensions on the drawing screen.**

StyleT, Bold, Italic, Underline, Strikeout, Font, ColorT, TextHeight, TextAngle, NorotT, RelT, LineWidthT, Slant and X-Y Ratio **all control the appearance of Text on the drawing screen.**

StyleS, ColorS, MaterialS, RelS, NumInteriorLineS and InteriorLineBrightnessS **control the appearance of Solids on the drawing screen.**

The Layer and Blank inputs allow the user to control visibility and the Layer selected geometry may be associated with.

The Style Inputs control the style of the object as depicted in the Attributes window of the main drawing screen. We will not go into all of these available inputs because they have already been covered earlier in this manual. Please refer to earlier sections regarding the style available in BobCAD-CAM V21.

Attributes Layer = "Default", **Blank** = (true or false), **StyleP** = (variable_name), **ColorP** = (variable_name), **WidthP** = (variable_name), **SizeP** = (variable_name), **RelP** = (true or false), **PStyleP** = (0 to 19), **StyleC** = (variable_name), **Color** = (variable_name), **LineWidth** = (variable_name), **LineType** = (variable_name), **LineCaps** = (variable_name), **RelC** = (variable_name), **StyleF** = (variable_name), **ColorF** = (variable_name), **OffsetF** = (variable_name), **DistanceF** = (variable_name), **AngleF** = (variable_name), **LineTypeF** = (variable_name), **LineWidthF** = (variable_name), **RelF** = (variable_name), **MaterialF** = (variable_name), **StyleD** = (variable_name), **ColorD** = (variable_name), **DecPlc** = (variable_name), **UpperTol** = (variable_name), **LowerTol** = (variable_name), **DimScale** = (variable_name), **Witness** = (variable_name), **TolPlc** = (variable_name), **PreText** = (variable_name), **PostText** = (variable_name), **LineWidthD** = (variable_name), **ValType** = (variable_name), **ValZer** = (variable_name), **TolType** = (variable_name), **TolZer** = (variable_name), **RelD** = (variable_name), **ArrowSize** = (variable_name), **ArrowHeight** = (variable_name), **Arrowtype1** = (0 to 5), **Arrowtype2** = (0 to 5), **ArrowVis** = (0 to 5), **ArrowMirror** = (0 or 1), **StyleT** = (variable_name), **Bold** = FALSE, **Italic** = FALSE, **Underline** = (true or false), **Strikeout** = (true or false), **Font** = (font_type), **ColorT** = (variable_name), **TextHeight** = (variable_name), **TextAngle** = (variable_name), **NorotT** = (variable_name), **RelT** = (variable_name), **LineWidthT** = (variable_name), **Slant** = (variable_name), **X-Y Ratio** = (variable_name), **StyleS** = (variable_name), **ColorS** = (variable_name), **MaterialS** = (variable_name), **RelS** = (variable_name), **NumInteriorLineS** = (variable_name), **InteriorLineBrightnessS** = (variable_name)

To place the user in Line Sketch Mode the following command is issued.

LineSketch

This command is used to create a line between coordinates. If any of the coordinates are omitted from the command the values input during the last use of this option will be used.

LineCoordinates **XStart** = (variable_name), **YStart** = (variable_name), **ZStart** = (variable_name), **XEnd** = (variable_name), **Yend** = (variable_name), **Zend** = (variable_name)

We use this command to create a line joining two selected entities which can be lines, arcs and points. The direction of vector arrows on lines and arcs control how effectively this command works because it will connect the end of the first entity to the beginning of the second entity. Like a lot of the other commands if nothing is selected the user will be placed in selection mode.

LineJoin

This command will Create a line at an angle from a selected entity. Length will determine the length of the line of course. The Angle input copntrols the angle of the line relative to the X axis. The AngleZ controls the angle relative to the Z axis of the drawing screen.

LineAngle **Length** = (variable_name), **Angle** = (variable_name), **AngleZ** = (variable_name), **Absolute** = (true or false)

To Create a Chamfer between two selected entities use the following command. The fixed input controls weather D1, D2 or Angle are being used. With fixed set to 0 the D1 and the Angle values are used. With Fixed set to1, D1 and D2 are used and with Fixed set to 2 D2 and the Ange values are used for generating the Chamfer.

LineChamfer **D1** = (variable_name), **D2** = (variable_name), **Angle** = (variable_name), **Fixed** = (0 to 2)

This command will Place the user in Line Continuous Mode. This one is another command that is best used at the end of a script or not at all in scripting. The reason is because this function requires large amounts of user intervention. The same function can be handled with a Loop and the Line coordinates command.

LineContinuous **PointType** = (0 to 2)

To place the user in Arc Sketch Mode use the following command. Angle1 and Angle2 refer to the start and end angles, respectively, for an arc segment. Reverse controls the direction of the arc. Reverse set to False produces a counter-clockwise arc segment while setting Rverse to True would produce a clockwise arc segment.

ArcSketch **Radius** = (variable_name), **Angle1** = (variable_name), **Angle2** = (variable_name), **Reverse** = (true or false)

To create an Arc at designated coordinates use this command. Again, any omitted values will result in the last input value being used in the operation.

ArcCoordinates **X** = (variable_name), **Y** = (variable_name), **Z** = (variable_name), **Radius** = (variable_name), **Angle1** = (variable_name), **Angle2** = (variable_name), **Reverse** = (true or false)

To Snap an Arc to selected geometry use the following command. If nothing is selected on the drawing screen then the user will be placed in selection mode at the end of the script.

ArcOnPoint **Radius** = (variable_name), **Angle1** = (variable_name), **Angle2** = (variable_name), **Reverse** = (true or false)

To Create an Arc between two selected entities use this command, more than two selected entities will not create the fillets. If nothing is selected then the user will be placed in selection mode at the end of the script. The Direction option control weather the arc will be created counter-clockwise (1) or clockwise (2) or allow the software to pick the shortest distance to create the fillet(0). Replace set to True will replace the existing fillet with the new radius. The Closed input does not influence arcs created in the scripting engine. NoTrim will remove excess geometry from the entities used to create the fillet if set to False.

ArcFillet Radius = (variable_name), Direction = (0 or 1), Replace = (true or false), FilletAll = (true or false), Closed = (true or false), NoTrim = (true or false)

To Create an Arc using three selected entities use this command. If there is nothing selected or not enough entities are selected this command will place the user in selection mode.

ArcOn3Entities

To create an Ellipse centered on designated coordinates use this command. X, Y and Z control where the center of the ellipse is located. The Width controls the X dimension of the arc the Height controls the Y dimension of the arc. Accuracy controls the number of arc segments used to create the Ellipse. Angle1 is the start angle of the ellipse and Angle2 is the end angle of the ellipse. Reverse controls weather the ellipse is created in a Clockwise or Counter-Clockwise direction, False indicates counter-clockwise.

Ellipse X = (variable_name), Y = (variable_name), Z = (variable_name), Width = (variable_name), Height = (variable_name), Accuracy = (variable_name), Angle1 = (variable_name), Angle2 = (variable_name), Revers = (true or false)

To create a Rectangle use this command. X, Y and Z indicate the reference point for the Rectangle. Width is the X dimension of the rectangle and Height is the Y dimension of the rectangle. A Radius value other than 0 will place fillets in the corner of the resulting rectangle. Reference set to 0 uses the bottom left hand corner as the reference point, while 1 will use the center of the rectangle as the reference point and 2 will place the user in Drag mode. CornerType refers to the type of corners that are produced on the rectangle. 0 will produce straight corners on the rectangle, 1 will create fillets and 2 will produce a chamfer designated by the Length option. The Angle option is an internal value.

Rectangle X = (variable_name), Y = (variable_name), Z = (variable_name), Width = (variable_name), Height = (variable_name), Radius = (variable_name), Length = (variable_name), Angle = (variable_name), Reference = (0 to 2), CornerType = (0 to 2)

To Offset selected entities this command is used. Direction indicates the side of the chain to offset to, 0 is left and 1 is right. Offset indicates the distance from the original shape to offset. IntersectAll set to True will ensure that the offsets that are created will not overlap past an intersection with another offset. Setting this option to False will result in all selected geometry being offset including some overlaps as they occur.

Offset Offset = (variable_name), Direction = (0 or 1), Intersect = (true or false)

This command will place Hatching on selected shapes. The Distance controls the distance between lines in the Hatch. Angle indicates the angle of the lines produced by the hatching function. Cross set to True will create a Cross Hatch while settings this value to False will result in Hatching at only one angle. The Layer option allows you to assign the Hatch to its' own layer upon creation.

Hatch Distance = (variable_name), Angle = (variable_name), Cross = (true or false), Layer = (variable_name)

To create a Hole Pattern use this command. X, Y and Z indicate the reference points for the pattern. The Type option controls weather the pattern is a Circular Hole Pattern (0) or a Grid Pattern (1). Diam indicates the Diameter of the Hole pattern, from the center of the pattern to the center of the holes created. DiamHole controls the size of the resulting Holes. Angle1 indicates the starting angle for the Hole Pattern, while Angle2 controls the End Angle for the pattern.

Width controls the X dimension of the Hole Grid and Height controls the Y dimension of the Hole Grid. Reference controls whether the Center of the Grid (1) or the Bottom Left corner of the Grid (0) will be used as the reference point. The NX controls the number of holes in the X direction, and NY controls the number of Holes in the Y direction. Drag set to True will allow the user to Drag the Hole Pattern on the screen, this will be the last executed command.

Hole X = (variable_name), Y = (variable_name), Z = (variable_name), N = (variable_name), Diam = (variable_name), DiamHole = (variable_name), Ang1 = (variable_name), Ang2 = (variable_name), Width = (variable_name), Height = (variable_name), Reference = (variable_name), NX = (variable_name), NY = (variable_name), Drag = (true or false), Type = (0 or 1)

This command will open the Pocket type selection window. It will not create a pocket because the user will need to select a pocket style.

Pocket

To create a Horizontal Pocket within a selected shape we use this command. The Distance input refers to the Step Over Distance for the pocket being generated. The Diameter input is the Tool diameter that will be used for the pocket operation. Stock is the material that will be left for finishing.

UseContour is set to true or false. If set to true then the original geometry will be included as part of the cutter path.

Start is also true or false. If set to true, when the pocket is generated, a line will be generated that indicates the beginning of the cutter path.

PocketHorizontal Distance = (variable_name), Diameter = (variable_name), Stock = (variable_name), UseContour = (true or false), Start = (true or false)

To create a Spiral Pocket within the selected shape this command is used. The Distance, Diameter, UseContour, Stock and Start inputs here are used the same as the Horizontal pocket function. The DiameterFinish input refers to the Diameter of the tool used for finishing. The PercFinish is the percentage of the Tool used for the Step Over Distance on the finishing passes. LastFinish is the "Contour is Last Pass" option, which makes the last pass for finishing a clean up cut around the perimeter of the profile. Operation 0 is a roughing pocket, while Operation set to 1 is the finishing option.

AutoConnect refers to the connection of the toolpaths. If set to 0 then the paths will not be connected, this results in clearance moves between each pass around the profile. If AutoConnect is set to 1 Then the toolpath will be connected. If the value is 2 then the toolpath is connected and the final move brings the tool back to the beginning of the pocket.

PocketSpiral Distance = (variable_name), Diameter = (variable_name), DiameterFinish = (variable_name), PercFinish = (variable_name), Stock = (variable_name), UseContour = (true or false), Start = (true or false), LastFinish = (true or false), AutoConnect = (0 to 2), Operation = (0 or 1), Connect = (true or false)

To create a Clean Circle Pocket within selected arcs use the following command. The Distance indicates the distance between each arc.

CleanCircle Distance = (variable_name)

The next command is used to create a Spiral Pipe Thread. The X Y and Z values indicate the location of the thread that will be generated. Radius1 is the radius at the beginning of the thread. Radius2 is the radius at the end of the thread. The Second Z value is the depth of the thread(positive height results in a Z negative thread, while a negative value results in a Z positive thread.) The Accuracy value controls the tangency and curvature of the Arcs. Turns refers to the number of turns will be generated over the length of the thread.

Arcs is a True or False input. If set to true then arcs will be generated. If set to false the spiral pipe thread will be generated as line segments.

Spiral X = (variable_name), Y = (variable_name), Z = (variable_name), Radius1 = (variable_name), Radius2 = (variable_name), Z = (variable_name), Accuracy = (variable_name), Turns = (variable_name), Arcs = (true or false)

This command is used to Create a Gear. The Nteeth is the number of teeth on the gear. Pdiam is the Pitch Diameter of the gear teeth. Pangle is the Pressure Angle. The Odiam is the Outer Diameter of the Gear. RAngle is the Root Diameter. The RFilletRad is the radius of the fillet that is applied at the transition between the Gear Tooth and the Root Diameter.

Gear NTeeth = (variable_name), PDiam = (variable_name), PAngle = (variable_name), Odiam = (variable_name), RAngle = (variable_name), RFilletRad = (variable_name)

To Create a Spline, or curves, from selected geometry use the following command. The accuracy controls how closely the splined geometry will follow the entities that are already selected. Closed set to true indicates that the resulting curves will be a closed chain, false will not close the chain. The Arcs option indicates that the spline will be generated with arcs if set to true.

Spline Accuracy = (variable_name), Closed = (true or false), Arcs = (true or false)

This function is used to create an Approach/Depart. One point and a Chain of lines and/or arcs must be selected, otherwise the user will be placed in selection mode for this function. The Radius indicates the radius of the ramp into the contour. Distance1 is the overlapping distance for the approach, while Distance2 is the overlapping distance for the departure from the profile. The AutoDepart set to true will use the same point for the approach and the departure, setting this to false will require a second point for the departure.

Approach Radius = (variable_name), Distance1 = (variable_name), Distance2 = (variable_name), AutoDepart = (true or false)

This command will place the user in Spline creation mode or generate a spline from the selected entities. This is quite similar to the Spline command covered on the previous page. The difference is that the CreationMethod is included in this command. The creation method refers to the four types of splines that can be performed in software (please see Spline Creation" in your manual).

SplineCreate Creation Method = (0 to 3), Accuracy = (variable_name), Closed = (true or false), Arcs = (true or false)

The following command is used to create a Primitive Solid. The type can be 0 to 4, 0 being a Sphere and 4 being a Torus. If an option is input but no values are presented within the script the script will stop at the input window for the specified Primitive type.

SolidPrimitives Type = (0 to 4)

For creating a Sphere(0). X, Y and Z are the center of the sphere. Radius is the radius of the sphere.

SolidPrimitives X = (variable_name), Y = (variable_name), Z = (variable_name), Radius = (variable_name)

For creating a Cube(1). X, Y and Z are the reference location. Length is the X dimension, Width is the Y dimension and Height is the Z dimension. Reference 0 uses the bottom left hand corner of the cube as the reference point while 1 will use the center of the Cube as the reference point.

SolidPrimitives X = (variable_name), Y = (variable_name), Z = (variable_name), Length = (variable_name), Width = (variable_name), Height = (variable_name), Reference = (0 or 1)

For creating a Cone(2). X, Y and Z are the reference points for the center of the BaseRadius. BaseRadius is the radius at the bottom of the Cone. TopRadius is the radius at the top of the cone. Height controls the Distance between the BaseRadius and the TopRadius.1

SolidPrimitives X = (variable_name), Y = (variable_name), Z = (variable_name), RadiusBase = (variable_name), RadiusTop = (variable_name), Height = (variable_name)

For creating a cylinder(3). X, Y and Z are again the reference points for the center of the bottom of the cylinder. Radius is of course the Radius of the cylinder. Height is the z extents of the cylinder.

SolidPrimitives X = (variable_name), Y = (variable_name), Z = (variable_name), Radius = (variable_name), Height = (variable_name)

For creating a Torus(4). X, Y and Z are the center of the solid to be created. RadiusMajor is the distance from the center of the Torus to the center of the Minor Radius. RadiusMinor is the size of the radius that follows the Major Radius.

SolidPrimitives X = (variable_name), Y = (variable_name), Z = (variable_name), RadiusMajor = (variable_name), RadiusMinor = (variable_name)

To create a Surface there are several different methods. Most of these methods, while accessible through the scripting engine, are better performed outside of scripting because of the need for user intervention. The CreationMethod can be a number from 0 to 11, 0 being a rectangular plane and 11 being a Multi-Sided patch.

SurfaceCreation Creation Method = (0 to 11)

CreationMethod 0 is a rectangular plane. The X, Y and Z values indicate the reference point for the rectangle. Width is the dimension for the X axis and Height is the Y dimension of the Rectangle. Reference set to 0 uses the bottom left hand corner as the reference point, while 1 will use the center of the rectangle as the reference point and 2 will place the user in Drag mode. CornerType refers to the type of corners that are produced on the rectangle. 0 will produce straight corners on the rectangle, 1 will create fillets and 2 will produce a chamfer designated by the Length option. The Angle option is an internal value.

RectangularPlane X = (variable_name), Y = (variable_name), Z = (variable_name), Width = (variable_name), Height = (variable_name), Radius = (variable_name), Length = (variable_name), Angle = (variable_name), Reference = (0 to 2), CornerType = (0 to 2)

CreationMethod 1 is a circular plane. The X, Y and Z values are the center of the circle. Radius is the radius of the resulting circle. Sketch set to true will allow the user to choose a location for the center of the circle by clicking somewhere on the drawing screen (this does not use snap points).

CircularPlane Radius = (variable_name), X = (variable_name), Y = (variable_name), Z = (variable_name), Sketch = (true or false)

CreationMethod 2 is a planar surface. This option will only work with selected entities. The entities must form a closed chain that is 2 dimensional. Unlike the previous two options, this way of creating a surface only requires the type definition.

SurfaceCreation Creation Method = 2

Creation Method 3 is the Extrude Curve function. This however does not work within the scripting engine because user intervention is mandated by this function. If you use the ExtrudeCurve option that is recorded, when run through again it will cause BobCAD-CAM to close. You can place the user in selection mode for the Extrude Curve command.

SurfaceCreation Creation Method = 3(can use)

SolidExtrudeCurve **Extrude Direction** = (0 or 1), **Extrude Distance** = (variable_name), **Taper** = (true or false), **Draft Angle** = (variable_name), **Extrude Caps** = (true or false), **Smoothness** = (0 or 1) (can not use)

Creation Method 4 is the Extrude Surface function. This function will work within the scripting engine but it must include both lines of code displayed below and the surface must be selected prior to execution. The Extrude Direction refers to either using the Z axis for the extrusion or using the normal of the selected surface. Extrude Distance is how far the resulting extrusion will reach. Taper is the true or false, if set to true then the extruded part will have a drafted angle on the sides. Draft Angle is the taper that is applied when using the Tapered extrusion option. Extrude Caps set to true will place a surface on the top and bottom of the resulting surfaces. Set to False the part will be extruded with out a top and bottom surface.

SurfaceCreation Creation Method = 3(can use)

SolidExtrudeSurface **Extrude Direction** = (0 or 1), **Extrude Distance** = (variable_name), **Taper** = (true or false), **Draft Angle** = (variable_name), **Extrude Caps** = (true or false)

Creation Method 5 is the Revolved surface function. This function will also work within the scripting engine provided the profile has been selected before execution of the command. Angle is how many degrees of revolution will be applied to the resulting surface. The point input refers to using either the Pick option, which requires user intervention, or using coordinates for the rotation point. The first X, Y and Z values are the Coordinates that will be used to rotate the part around. The second X, Y and Z value are the rotation axis direction. If one or more of these values is set to something other than 0 the part will be rotated around those axis'. Caps controls weather caps will be placed on the open ends of the resulting surface. The Smoothness input controls how the resulting contours are revolved. If set to 2 then it will create arcs similar to the way the spline option creates arcs.

SolidRevolve **Angle** = (variable_name), **Point** = (0 or 1), **X** = (variable_name), **Y** = (variable_name), **Z** = (variable_name), **X** = (variable_name), **Y** = (variable_name), **Z** = (variable_name), **Caps** = (true or false), **Smoothness** = (0 or 1)

You can use the Boolean Function on two selected surfaces with the scripting engine. As long as the solids are selected prior to execution of this command it will work properly. Type 0 is a Union of the two solids, basically this adds them together to make one solid. Type 1 will create a Difference, it takes the first selected solid and subtracts the second selected solid. Type 2 is an Intersection, this will leave only what is common between both of the selected Solids.

SolidBooleans **Type** = (0 to 2)

To create Fillets or Chamfers on selected Solids use this command. Fillet Type 1 is constant radius, 2 is variable radius and 0 is the chamfer option. The Variable radius option can not be used within the scripting engine due to the need for user intervention. The Fillet Radius is both the radius if using type 1 and the fillet length if using type 0.

CircularFillet **Fillet Type** = (0 to 2), **Fillet Radius** = (variable_name)

Create or Open the Solid Toolpath Generation window:

SolidMachine **Machine Method** = (0 to 2)

SolidPlanarFinish **StepDistance** = (variable_name), **Radius** = (variable_name), **Angle** = (variable_name), **Tolerance** = (variable_name), **Alternate** = (0 or 1), **MinLineLength** = (variable_name), **ToolTips** = (0 or 1), **ToolType** = (0 to 2), **CornerRadius** = (variable_name), **DHoles** = (true or false), **THoles** = (true or false)
SolidPlanarFinish **UseXYExtentsOfPart** = (0 to 2), **UseZExtentsOfPart** = (0 to 1), **TopZ** = (variable_name), **BottomZ** = (variable_name), **ToolCenter** = (0 or 1), **Back** = (true or false)

Offset Selected Surfaces:

SolidOffset **Distance** = (variable_name), **Offset Method** = (0 or 1)

Extract Edges of Selected Solid:

SolidExtractEdges **Extract Method** = (0 to 2)

To Cover Holes on Selected Surfaces use this command. Using Options 0 will remove the holes without saving the file. Using 1 will prompt the user to save the file before the Holes are removed.

SolidRemoveHoles **Options** = (0 or 1)

This command is used to set a Selected shape as a imaginary revolved surface that can later be used to project geometry onto. There must be some geometry on the screen selected prior to execution of this command.

SurfaceRotateZ

To create an imaginary surface between two selected entities use the following command. The entities must be selected prior to executing this command

SurfaceInterpolate

To create an imaginary surface that parallels the Y axis use the following command. At least one entity must be selected prior to execution.

SurfaceExtrudeY

To create an imaginary surface that parallels the X axis use the following command. Like the ExtrudeY command this requires at least one entity be selected.

SurfaceExtrudeX

To create an imaginary Planar surface for projection this command is used. This will create a flat imaginary surface on Z0. Nothing needs to be selected prior to execution of this command.

SurfacePlanar

To Project selected geometry to previously set surface use this command. The ToolDia will be commonly set to 0 as well as the ToolCoreDia should also be set to 0. The reason we do this is to avoid causing the toolpath to be created with erroneous tool movements, but you can always try adding a value in other than 0. The Accuracy is the accuracy of the projected toolpath. The larger the value here the longer the movements will be and the less the tool path will conform to the specified surface. The Zneg option allows for the projection to occur below the surface provided you have entered a tool diameter. The Axis5 option will create 5 Axis vectors to the projected geometry.

Project ToolDia = (variable_name), **ToolCoreDia** = (variable_name), **Accuracy** = (variable_name), **ZNeg** = (true or false), **Axis5** = (true or false)

The following commands will either dimension the selected entities or place the user in dimension mode. These should be used sparingly if needed as they can cause a lot of data to be added to the drawing screen.

DimHorizontal
DimVertical
DimAligned
DimAngle
DimRadius
Dimension
DimXOrdinate
DimYOrdinate
DimMove
DimensionChain
DimensionReference

This command will place the user in Face mode.

Face

This command allows us to change dimension appearance values. The DeltaY is the displacement of the text in reference to the leader lines. The ArExt is the distance the leader arrow will extend past the witness lines. The Gap option is the distance from the witness lines to the geometry it is dimensioning. The TextGap is the space inside left for the dimension text. The DimExt is the length the witness lines will exceed past the leader line. AutoDrag controls whether the dimension will simply be displayed (no user intervention) or the user has to drag it to the proper location (user intervention) on the screen.

DimSetup DeltaY = (variable_name), **ArExt** = (variable_name), **Gap** = (variable_name), **TextGap** = (variable_name), **DimExt** = (variable_name), **Autodrag** = (true or false)

To Create Text the following command is used.

Text Attribute = (variable_name), **Modify** = (true or false), **TextHeight** = (variable_name), **TextAngle** = (variable_name), **Default** = (true or false), **Bold** = (true or false), **Italic** = (true or false), **Underline** = (true or false), **Strikeout** = (true or false), **Drag** = (true or false), **Enter** = (true or false), **Font** = (font available on your computer), **Slant** = (variable_name), **X-Y Ratio** = (variable_name)

Fit text to selected shape contour:

TextToPath Accuracy = (variable_name) **TextAlongPath Y** = (variable_name), **X** = (variable_name), **Align** = (variable_name)

Change current drawing view:

ViewSelectView Front = (true or false), **Top** = (true or false), **SideRight** = (true or false), **Iso** = (true or false), **Default** = (true or false), **Sideleft** = (true or false), **Bottom** = (true or false), **Back** = (true or false)

This command will bring all geometry on the screen into view on the drawing screen.

ViewAll

This command will place the user in Zoom Region mode.

ViewZoom

This command will place the user in Pan mode.

ViewDynPan

This command will place the user in Dynamic Rotation mode.

ViewDynRot

To place the user in view scale mode, use this command. This function allows the user to control the Scale of their drawing by holding the left mouse button down and dragging the cursor across the screen.

ViewDynScale

This command is used to Refresh the current drawing screen.

ViewRepaint

This command will Return to previous view scale.

ViewPrevious

This command will Reduce the current view scale.

ViewReduce

This command allows you to Change specific View window values within the scripting engine. The Scale is of course the Scale of the drawing screen. The X, Y and Z targets are the center of the view screen. X, Y and Z Rot values are the amount the drawing screen will be rotated around that specific axis. The rest of the values pertain to Faces in the drawing screen. We will not be defining them here because we seldom need to use faces when working with solids.

ViewOptions Scale = (variable_name), XTarget = (variable_name), YTarget = (variable_name), ZTarget = (variable_name), XRot = (variable_name), YRot = (variable_name), ZRot = (variable_name), ZSort = (variable_name), Shade = (variable_name), Freeze = (true or false), Wire = (true or false), BackFaceCulling = (true or false), Shade = (true or false)

This command will rotate the drawing window around the axis the amount specified in the Xrot, Yrot and the Zrot.

ViewRotate XRot = (variable_name), YRot = (variable_name), ZRot = (variable_name)

This command will change to a top view of drawing.

ViewOrient

This option allows us to change the view on the drawing screen. Only one input needs to be used and it should be set to true.

ViewSelectView Front = (true or false), Top = (true or false), SideRight = (true or false), Iso = (true or false), Default = (true or false), Sideleft = (true or false), Bottom = (true or false), Back = (true or false)

To Gather general data from drawing window we use this command. Nent is the total number of entities on the drawing screen (not the hidden entities). Nsel is the number of selected entities. Nlines is the total number of line segments on the drawing screen (selected and unselected). Narcs is the number of arcs. Nsplines is the number of splines on the drawing screen. NselSplines is the number of selected splines. Nsolids is the number of solids on the drawing screen, while NselSolids is the number of selected solids on the drawing screen. Xmin, Ymin and Zmin are the lowest (smallest) x, y and Z values on the drawing screen. Xmax, Ymax and Zmax are the largest values on the drawing screen. Nchains is the total number of connected chains that are selected. LinDim, AngDim and RadDim are the number of dimensions on the drawing screen, this includes linear, angular and radius dimensions. Text is the number of Text objects on the Drawing screen. Aside from the Area option the rest of these refer to the types of entities visible on the drawing screen. The Area is used to determine the overall area of your geometry.

Verify NEnt = (variable_name), NSel = (variable_name), NLines = (variable_name), NARcs = (variable_name), NSplines = (variable_name), NselSplines = (variable_name), NSolids = (variable_name), NselSolids = (variable_name), XMin = (variable_name), YMin = (variable_name), ZMin = (variable_name), XMax = (variable_name), YMax = (variable_name), ZMax = (variable_name), NChains = (variable_name), LastType = (variable_name), Point = (variable_name), Line = (variable_name), Arc = (variable_name), LinDim = (variable_name), AngDim = (variable_name), RadDim = (variable_name), Text = (variable_name), Polygon = (variable_name), OLE = (variable_name), Group = (variable_name), Caption = (variable_name), Area = (variable_name)

To open the NC-CAM window use this command. Config is the .cfg file that you want to use, if it does not exist the file will be created. Template can be used when creating a new post processor because this is what the post processor will be based on. Driver is the type of driver file that will be used, you can check the BobCAD-CAM V21 folder to see what driver files are available for selection. AutoArrange set to true will rearrange the Drawing and NC-CAM windows so that they both fit within your computer screen, set to False your NC-CAM window would overlap your drawing screen. NCRight places the NC-CAM window on the right hand side of the drawing screen when set to true and places it on the left when set to False. BlankIcon controls the visibility of the NC-CAM icon that is placed on a drawing, true will hide the icon while false will display it on the drawing. Label is a way to identify more than one NC icon on a drawing screen. Width is the percentage of your screen that the NC-CAM window will encompass.

NCInsert Config = (cfg file name), Template = (template file name), Driver = (driver file name), AutoArrange = (true or false), NCRight = (true or false), BlankIcon = (true or false), Label = (variable_name), Width = (variable_name)

This command will place the user in UCS Snap Mode.

AxisToPoint

This command will reset the UCS to the original position.

AxisReset

The following command will Rotate the UCS a specified amount. Zrot is the amount that the UCS will be rotated around the Z axis. Xrot is the amount on the X axis and Yrot is the amount around the Y axis. If Absolute is true then the values within the Zrot, Xrot and Yrot will be applied as absolute positions. If Absolute is set to false then the rotation amounts will be applied to the current orientation of the UCS. The Global option does not have any real affect on the rotation of the UCS.

AxisRotate ZRot = (variable_name), XRot = (variable_name), YRot = (variable_name), Absolute = (true or false), Global = (true or false)

This command will place the user in Align UCS to the X axis mode. They will then need to select a point or intersection that they want to be the X positive direction.

AxisAlignX

This command will place the user in 3D Align mode. They will then need to select an X positive position and a Y positive position.

AxisAlign

To place the user in Align to element mode use this command. The user will then be required to pick an entity to align the axis to.

AxisAlignG

To move the UCS a specified amount, use this command. The X, Y and Z values are the distance or positions to move to. If Absolute is true then the X, Y and Z values are viewed as the new positions. If Absolute is false then the X, Y and Z values are viewed as incremental distances.

AxisMove X = (variable_name), Y = (variable_name), Z = (variable_name), **Absolute** = (true or false)

This command is used to run a specified Script already in the scripts folder. The script name comes after the equal sign and should be contained within quotes. If Edit is set to true then the file is brought up in a NotePad window for editing the script. If Edit is set to False then the script is executed.

Run Macro = (script_name), **Edit** = (true or false)

This final set of commands allows you to control the general software environment.

The OptionsGeneral has several important inputs. The DisTol controls the accuracy with which your code is generated from your geometry. The ChainGap controls how far apart two entities have to be in order for them not to be viewed as a Chain. The PickAperture is the furthest distance away from an entity the cursor has to be in order to select that entity. The TemplateDir is the Directory for saving and opening files. The ScriptDir is the location from which the Scripts are executed. The ConfigDir is the location where the cfg files are located. AutoSave is the interval in which BobCAD-CAM saves the file(s) you are working on, timed in milliseconds.

OptionsGeneral **DistTol** = (variable_name), **Chaingap** = (variable_name), **PickAperture** = (variable_name), **TemplateDir** = "(variable_name)", **ScriptDir** = "(variable_name)", **ConfigDir** = "(variable_name)", **AutoSave** = (variable_name)

The OptionsUnits window controls the coordinate values on the Drawing screen as well as the output values to the NC-CAM window. Unit is the current unit of measurement, 1 is mm 25.4 is inches. Digits is the number of decimal places that will be displayed on the CAD screen. Turn is a true or false option, if set to true then Lathe mode has been turned on. TurnDia will output code in Diameter mode rather than radius mode if set to true.

OptionsUnits **Unit** = (variable_name), **Digits** = (variable_name), **Turn** = (true or false), **TurnDia** = TRUE

This next option controls the active grid on the drawing window. If "Active" is set to true then a dot will be displayed separated by the given Distance. "Every" controls how often the dot is displayed, such as every 2 dots are displayed rather than every one dot.

OptionsGrid **Active** = (true or false), **Distance** = (variable_name), **Every** = (variable_name)

In this option we are controlling the size of the object on the drawing screen and their visibility. **AxisDim** controls the size of the UCS on the drawing screen (not the unit size). The **ObjectAxisDim** controls how big the UCS in the top right corner of the drawing screen will appear. The **TriadDim** controls the size of the Global Coordinate system (the cross hairs that have the hatch marks indicating distance). The **PreviewPixels** indicates how many pixels will be used in the preview window of the file open Dialogue box. The **ArowSize** controls the size of the vector arrows that appear on lines and arcs. **ScreenAxis** controls the visibility of the Global coordinate system. **ToolTips** controls whether or not a ToolTip or pop-up window will appear when hovering the cursor over an item on the screen. **DisplayDirectionArows** controls whether or not the vector arrows are visible.

OptionsScreen **AxisDim** = (variable_name), **ObjectAxisDim** = (variable_name), **TriadDim** = (variable_name), **PreviewPixels** = (variable_name), **ArrowSize** = (variable_name), **ScreenAxis** = (true or false), **Tooltips** = (true or false), **DisplayDirectionArrows** = (true or false)

This option controls what toolbars are available in BobCAD-CAM. Bar1 is the Annotation toolbar. Bar2 is the Construction toolbar. Bar3 is the Additional Tools toolbar. Bar4 is the CAD toolbar. Bar5 is the Extra toolbar, Bar6 is the Selection toolbar, Bar7 is the CAM and Bar8 is the ExtraCAM toolbar. Bar9 is not yet used within BobCAD-CAM.

OptionsToolbars **Bar1** = (true or false), **Bar2** = (true or false), **Bar3** = (true or false), **Bar4** = (true or false), **Bar5** = (true or false), **Bar6** = (true or false), **Bar7** = (true or false), **Bar8** = (true or false), **Bar9** = (true or false)

The next option controls snap points. They are either set to true or false. If set to true then that option is used as a snap point in the drawing window. They should be pretty self explanatory so we will not go into defining them at this time.

OptionsSnap **Existing** = (true or false), **Endpoint** = (true or false), **MidPoint** = (true or false), **CenterPoint** = (true or false), **IntersectionPoint** = (true or false), **Axis** = (true or false), **ReferencePoint** = (true or false), **ScreenPosition** = (true or false), **Arcs** = (true or false)

This is the final option available on the Drawing window through the NC-CAM window. This option controls the Defaults for the general Environment and are set to either true or false, with the exception of the Type input. The type input should never be modified as this could result in instability of the BobCAD-CAM software. Like the Snap points these defaults are self explanatory, if you require further definition please refer to the earlier chapters on the Environment window.

OptionsDefaults **Type** = (0 or 1), **AutoSave** = (true or false), **AutoRestore** = (true or false), **AutoPos** = (true or false), **EnterTab** = (true or false), **AutoPre** = (true or false), **AutoDrag** = (true or false), **AxisDrag** = (true or false), **AutoSavePre** = (true or false)

SCRIPTING COMMAND AVAILABLE ON THE NC-CAM WINDOW:

Now we will cover the scripting commands available on the NC-CAM window.

This command will generate NC-Code for selected geometry on the Drawing screen. If nothing is selected the user will be placed in selection mode for the Machine Auto command.

NCAuto

This command will generate NC-Code for all the geometry on the drawing screen regardless of whether it is selected or not.

NCCutAll

This command will generate NC-Code based on the selected Geometry. This command is used to create 3D arc commands as well as point to point positions. The type of code generated is based on the selected geometry. As a special note this command does not utilize any of the settings in the Tool Depth Settings window except the Rapid Plane, so no roughing will occur if it is set and the Z values will match the Z positions on the drawing screen.

Generate:

NCRenumber

This command will generate a G00/rapid positioning move to a selected point or intersection.

NCPoint

The next command is the bread and butter of the NC-CAM window. This command is used for everything from assigning focus to a line of code to outputting new lines of code. The main command is NCEdit. NumLines is the total number of lines in the program. Line is the current line position or the line that has been assigned focus. Text is the text that is present on the current line(or the code you see on the NC-CAM editor). X, Y and Z are the positions stored on the NC-CAM window so that the red arrow follows the code properly. Nx, Ny and Nz are the Normals of the current entity, these are used for 5 Axis positioning. NoDelete is used to clean out empty lines of code, if set to True and the line has no text the line is deleted.

NCEdit NumLines = (variable_name), Line = (variable_name), Text = (variable_name), X = (variable_name), Y = (variable_name), Z = (variable_name), Nx = (variable_name), Ny = (variable_name), Nz = (variable_name), NoDelete = (true or false)

This command will delete the current active line of code, or the line that has been assigned focus.

NCDelete

To generate a Move to clearance (the Rapid Plane) the following command is used. If the Feed input is set to 0 the move will be a G01/Feed move. If the Feed input is set to 1 the resulting move will be a G00/rapid move.

NCToolup Feed = (0 or 1)

To generate a Move to depth (Cutting Depth) the following command is used. If the Feed input is set to 0 the move will be a G01/Feed move. If the Feed input is set to 1 the resulting move will be a G00/rapid move.

NCTooldown Feed = (0or1)

This command will turn the 3D option on the NC-CAM window on or off. If the command is issued the opposite value is assigned to the 3D button(i.e. If 3D is off when the command is executed then 3D will then be turned on, if 3D was on then 3D will be turned off)

NC3D

This command is used to generate code for a single selected entity. If more than one entity is selected, only the last entity will be coded. If nothing is selected then it will place the user in select mode for the Single Cut function.

NCSingle

This command is used to modify or gather information from the Tool Depth Settings window. Zup is the Material Top, Zdown is the Cutting Depth and ZFree is the Rapid Plane. The ArcError is the arc interpolation error for 3D arcs. AutoRough controls whether the Automatic Roughing option is turned on, if set to true then the Automatic roughing option is turned on. MaxZ is the maximum depth of cut per pass. The Feed and Plunge inputs are the Feed-rates. These values are viewed as strings.

NCUpDown ZUp = (variable_name), ZDown = (variable_name), MaxZ = (variable_name), ArcError = (variable_name), ZFree = (variable_name), AutoRough = (true or false), Feed = (variable_name), Plunge = (variable_name)

This command will undo the last action on the NC-CAM window.

NCUndo

This command will Cut the currently selected line of code and place it in the Clipboard for pasting.

NCCut

This command will paste the information last added to the Clipboard from the ClipBoard to the NC-CAM window.

NCPaste

This command will Copy the currently active line of code to the Clipboard for pasting.

NCCopy

This command will select all the code on the NC-CAM window.

NCSelectAll

The following command controls the output of the line numbering. NCRenumber is the main command. First is the starting number for your numbering scheme and Delta is the increment that will be used to advance the line numbers. Width is the total number of spaces that need to be occupied by the line number, this is used if you need every line number to consist of a specific number of characters. NoNumber is either true or false, if set to true then no numbers will be output on the lines. LoadZero controls the leading zeros on the line numbers, if set to true then the line numbers will be generated with enough zeros at the beginning of the line number to get the desired Width. PreText is any text that you want to be on the line number, such as the letter "N". PostText is the text that will be displayed at the end of a line number.

NCRenumber First = (variable_name), Delta = (variable_name), Width = (variable_name), NoNumber = (true or false), LoadZero = (true or false), PreText = (variable_name), PostText = (variable_name)

This command is used to assign a tool to the NC-CAM window. This does not output any code it simply assigns the Tool from the ToolSetup window to the code window. This is more commonly used for simulation purposes.

NCTool Toolnumber = (true or false)

This command will open the simulation window. X,Y and Z indicate the bottom left and corner of the workpiece. DX, DY and DZ indicate the dimensions of the material block. The Shade controls the display accuracy of the simulation, 0 being fine and 2 being coarse. CALSTEP is not yet used in the simulation window and may be omitted from the command line.

NCSimulate X = (variable_name), Y = (variable_name), Z = (variable_name), DX = (variable_name), DY = (variable_name), DZ = (variable_name), Shade = (variable_name), CALSTEP = (variable_name)

This command will open the Color Dialogue for the NC-CAM window.

OptionsColor

This command will open the Drill And Cycles window.

NCCycle

This command will allow you to change the current post processor on the NC-CAM window.

OptionsSetup Config = (post_name), Template = (variable_name), Driver = (driver_file)

The following command controls the output format of the NC-CAM window. Rel indicates weather the code will be incremental (TRUE) or absolute (FALSE). Xchg, Ychg and Zchg indicate weather the X, Y and Z values will be modal (TRUE) or non-modal (FALSE). NoSpace controls the spacing of the coordinates, if set to true there will be no spaces between coordinate values.

ArcFull controls the maximum circular motion output by the software. 0 is a full circle, 1 is a half circle and 2 is quarter circles.

The Uvrel, Uvcoupled, ZXYGuide, ZUVGuide and Arcerr options all refer to fourth axis operation. This is not the Rotary fourth axis it is used for four axis wire positioning. At this time we are not going to define or use these options. It is enough to know that they exist.

Gmodal controls weather or not a G code will be displayed on every line of code. If set to true G codes, will only be displayed when they change from G00 to G01 for example. Slash actually has no purpose in the scripting engine. IJRel controls weather the I and J values are incremental (TRUE) or absolute (FALSE). The I and J values can also be output on a separate line by setting IJSep to true. Rad indicates weather the radius value of an arc is output, if set to true the radius option will override the IJ options. With G26, G25 or G06 set to true the respective command will be issued in place of G02 or G03.

OptionsConfiguration Rel = (true or false), Xchg = (true or false), Ychg = (true or false), Zchg = (true or false), ArcsFull = (0 to 2), NoSpace = (true or false), UVrel = (true or false), Uvcoupled = (true or false), ZXYGuide = (variable_name), ZUVGuide = (variable_name), Arcerr = (variable_name), GModal = (true or false), Slash = (true or false), IJRel = (true or false), IJSep = (true or false), Rad = (true or false), G26 = (true or false), G25 = (true or false), G06 = (true or false)

This option is used to control the coordinate values that are output on the NC-CAM window.

ScaleX, ScaleY and ScaleZ control the size values that are output. A value of 1 will output the coordinate values as they are on the drawing screen. The X, Y and Z values control the initial position of the tool (inside the software). This allows us to control where the software assumes the tool is starting.

The Width input controls how many characters must be output for each coordinate value. The Precision controls how many decimal places will be output in the code. LeadZero is either true or false and will add zeros to the beginning of the coordinate values so that the Width criteria is met. TrailZero is also a true or false option, if set to true then the coordinate values will have zeros added to the end so that the Precision value is met. NoPlus controls weather a plus sign will be output on positive coordinate, if set to true no plus sign will be issued.

OptionsCoordinates ScaleX = (variable_name), ScaleY = (variable_name), ScaleZ = (variable_name), X = (variable_name), Y = (variable_name), Z = (variable_name), Width = (variable_name), Precision = (variable_name), LeadZero = (true or false), Trailzero = (true or false), NoPlus = (true or false)

SAMPLE 1:

Concatenate (definition) – To connect or link in a series.

For our first lesson we are going to create a Script for a program start. The purpose of this script is to make sure that all the coordinate values on the Drawing and NC-CAM window are set properly for the code to work on the machine. To make this easier to read we are going to make all the standard scripting commands Bold and the defined variables will be Bold and Italicized.

To make this simple we will be asking the user for a program number as well as a description of the program. We are also going to make some assumptions as to the type of code the machine accepts as well as the setup codes that are input at the beginning of the program.

```
dim program_number as integer  
dim program_description as string  
dim current_line as integer
```

NCEdit Output, Line = 1 'this command tells us what line we are starting on.

***current_line* = 1** 'then we assign that value to an integer for later use.

OptionsUnits Unit = 25.4 'setting the units to 25.4 places us in inch output mode.

OptionsConfiguration Rel = FALSE 'setting the Rel option to False places us in Absolute positioning mode.

Ask Header = “Program Start”, Output, Cancel = **canc, “Program Number;0” = *program_number*, “Program Description” = *program_description*** 'notice the “;0” after Program Number inside the quotes. This allows us to input a value into this window without converting it to the Unit equivalent. If this is omitted from the command the input value will be multiplied by the Unit value. This is fine if you are always working in millimeters but if you are working in inches the resulting output for the number 1 would be 25.4. We do not need to use this method for String variables however because multiplication can not be performed on a String.

If **canc Then**

Exit

End If

(This checks to see if the Cancel button was pressed. If the cancel button was pressed then the script will exit.)

NCEdit Line = *current_line*

(This assigns focus to the first line so that it can be edited.)

txt* = “P” + *program_number

(Here we concatenate the string to produce the code we want to output and assign that value to a variable called txt.)

NCEdit Line = *current_line*, Text = *txt*, Output, NumLines = *nl*

(On this line we output the code to the NC-CAM window by making Text equal to our txt variable and placing it before the word Output. Then we output the total number of lines in the NC-CAM window to the nl variable.)

***current_line* = *current_line* + 1**

(Here we advance our counter by 1 so that we can move the focus to the next line in the NC-CAM window.)

Now we are going to output the description to the NC-CAM window in the same fashion. After the description is output to the screen we are going to output our commands for the machine.

```
NCEdit Line = current_line txt = "(" + program_description + ")"  
NCEdit Line = current_line, Text = txt, Output, NumLines = nl  
current_line = current_line + 1
```

```
NCEdit Line = current_line  
txt = "G00 G90 G80 G70"  
NCEdit Line = current_line, Text = txt, Output, NumLines = nl  
current_line = current_line + 1
```

(Next we move the focus of the editor to the first empty line on the NC-CAM window so that we can renumber the whole program.)

```
NCEdit Line = nl + 2
```

(Now we renumber the program because the lines we have added have not yet been numbered. Because we have omitted all but the main command for renumbering the values that were previously input into the Numbering window will be used.)

NCRenumber

(Last we assign general focus to the NC-CAM window.)

window2

That is all for this first lesson. You should now have a fully functioning Program Start Macro.

SAMPLE 2:

For our next example we are going to create a simple drawing script. The purpose of this script will be to generate a polygon. The size of the polygon as well as the number of sides will be determined by the user.

We will also add a bitmap to the dialogue box.

OptionsUnits Output, Unit = UN

(Here we output the current unit value to a variable.)

```
dim polygon_size as double  
dim polygon_sides as double  
dim rotation_amount as double  
dim current_rotation as double  
dim new_x_position as double, new_y_position as double  
dim previous_x as double, previous_y as double
```

(You can see above that we are able to define more than one variable per dim statement. The variables must be separated by a comma and each must have a declared type of its' own.)

```
Ask Header = "Polygon Creation", BitMap = "polygonpic", Output, Cancel = canc, "Size of Polygon;0" =  
polygon_size, "Number of  
Sides;0" = polygon_sides
```

(The change that you can see above is the use of the BitMap input. When assigning a bitmap image it done this way. The BitMap command must occur prior to the Output statement. Also, the file name is placed within quotes and the extension is removed.

**If cancel Then
Exit**

End If

(These command lines, again, check to make sure the user has not pressed the Cancel button. If they did then the function will end here.)

Attributes Layer = "polygon points"

(In this command we are creating a new layer. We assume that nothing is on the screen already, so this now becomes the active layer on the drawing screen.)

current_rotation = 90

(We are setting the current rotation amount to 90 this indicates a positive X motion from the zero position.)

rotation_amount = 360/polygon_sides

(Here we determine the number of degrees between each point on the polygon by dividing a full rotation by the number of sides the user input into the window.)

Now it is time to start creating the line segments for the polygon. We are going to use a loop structure here because there will be more than one line created. We can use the same commands over and over within a loop command and change the values as we go to get different output.

For CreatingPoints = 1 To (polygon_sides-1) Step 1

(CreatingPoints is our loop counter. A loop counter is used to keep track of how many times the loop command has been executed.)

If CreatingPoints = 1 Then

(If we are on the first time through the loop we are going to use some preset values for the beginning of the line. We need to add the rotation amount to our current angle so that we can get the proper pointy for the end of our line. We are going to use the Sin and Cos functions to do get the correct point locations. Notice that we used pi/180 in the Sin and Cos equations. The reason we do this is because BobCAD-CAM returns the Radian value from the Sin and Cos function and not the degrees. We multiply the value times pi and divide by 180 to get the degree value.

**current_rotation = current_rotation + rotation_amount
new_x_position = ((Sin(current_rotation*(pi/180)))*polygon_size)
new_y_position = ((Cos(current_rotation*(pi/180)))*polygon_size)**

(Here we are creating the line. When assigning a value we also multiply the variable and the Unit value we had gathered at the beginning of the script. This way the locations match up with the values the user had input. BobCAD-CAMs native unit is the millimeter so we multiply by UN, which was the unit value gathered at the beginning of the script, to get the proper output value for the command.)

**LineCoordinates Xstart=(polygon_size*UN),YStart=0,ZStart=0,\
Xend=(new_x_position*UN),\
YEnd=(new_y_position*UN),ZEnd=0**

Now we assign the new values to a different variable. We do this so that we can reuse the old variable name the next time the loop is executed.

```
previous_x = new_x_position  
previous_y = new_y_position  
Else
```

(If this is not the first time through the loop then we will execute the commands below.)

We again add the rotation amount to the current angle and use the Sin and Cos functions to get the new X and Y locations.

```
current_rotation = current_rotation + rotation_amount  
new_x_position = ((Sin(current_rotation*(pi/180)))*polygon_size)  
new_y_position = ((Cos(current_rotation*(pi/180)))*polygon_size)
```

(Here we are creating a new line segment. You can see that we have used the previous_x and the previous_y as the Start points for our new line. This way we have not lost track of where the beginning of the line was.)

```
LineCoordinates Xstart=(previous_x*UN),YStart=(previous_y*UN),\  
Zstart=0,XEnd=(new_x_position*UN),\  
YEnd=(new_y_position*UN),ZEnd=0
```

(Here we are again assigning the new X and Y values to our previous X and Y variables. To keep the code short we need to reuse the variables as often as possible.)

```
previous_x = new_x_position  
previous_y = new_y_position
```

```
End If
```

(This is the end of our If statement.)

(Here we are simply moving the CreatingPoints counter ahead the designated Step Value.)

Next CreatingPoints

(This last LineCoordinates command creates the last line for the polygon. This way the polygon becomes a closed shape.)

```
LineCoordinates Xstart=(previous_x*UN),YStart=(previous_y*UN),\  
ZStart=0,XEnd=(polygon_size*UN),YEnd=0,ZEnd=0  
End
```

(This ends the Macro.)

INDEX:

.cct file

Drawing Templates., 17

3 Edge

Creating a 3 Edge surface., 209, 210

Surface Creation., 209

3 Entities

Drawing arcs by using 3 entities., 128

3 ENTITIES

Creating an arc using 3 entities., 128

Drawing arcs from 3 entities., 128

3D

The 3D Menu., 318

Using the SET PATH function for creating skin toolpath., 318

3D ALIGN

The UCS., 44

3D Menu

Brief description., 26

3D Skin Box

How to use the SKIN box., 322

3D Toolpath

How the tool works with 3D toolpath geometry., 329

3DM

Information regarding opening RhinoCAD 3DM files., 23

4 EDGE

Surface Creation., 210

4th Axis, 452

Training Lesson., 452

4th AXIS

Performing a 4th axis conversion., 450

ACIS

Information regarding importing these types of files., 23

ACTION/OBJECT

In selection mode., 32

Selecting a function and then selecting the entities involved., 32

What surface creation functions are Action-Object., 219

ACTIVE COLOR

Where to automatically change it., 66

ACTIVE LAYER

Where to change it while drawing., 66

ALIGN

Aligning Dimension text., 105

The UCS in the X-Axis., 44

UCS - To an element, Entity., 45

Angle

Create a dimension for an angle., 363, 364

ANGLE

Angle with Z-Axis., 116

Drawing angled lines., 113

ANIMATE

For use in the Render window., 312

ANNOTATION

Toolbox, 64

What is the Annotation Toolbox?, 64

APPROACH & DEPARTURE

Important RULE when using the approach & depart function., 181

APPROACH / DEPART

How to create an approach or departure., 180

Manually creating an approach and departure for a tool., 180

Approximate

In reference to the "Approximate" spline option., 133

ARC

Drawing arcs using the coordinate system., 122

Sketching arcs., 121

The complete ARC menu., 120

ARC Interpolation error, 319

ARC Menu

Brief description., 26

ARCS & CIRCULAR MOVEMENTS

Changing the settings for cutting arcs and circles for your machine control., 438

Arrows

Directional arrows

Important for programming., 483

Attributes

About solid attributes, 245

For Solids & Surfaces., 245

Geometry, text, dimensions and solids attributes., 95

The Attributes dialog and functions., 95

ATTRIBUTES

The Complete Attributes dialog explained., 95

AUTO CUT, 400, 417

AUTO DIMENSION

Automatically dimension an entire part at once., 363

Auto Pocket Parameters, 399

Auto Pre-select

Selecting geometry., 490

AUTO PRESELECT, 75

AUTO RESTORE

Saving & backing up drawings., 74

AUTO RULED, 341

Creating toolpath., 341

Auto Store Drawings

Environment Setting., 27

AutoCAD DWG

Importing 2D or 3D AutoCAD DWG files into V21., 22

AXIS

Pertaining to changing the UCS., 385

The Axis Object Size., 71

Turning the Screen Axis on and off., 72

UCS Options., 385

AXIS DRAG

Environment Dialog., 49

BACKGROUND

How to change the main software colors., 72

Basic Rules for using the Boolean Difference function, 241, 243

Basic Rules for using the Boolean Union function, 240, 241, 243

BLANK

Blanking and Un-Blanking geometry, surfaces and solids., 91

BOLT CENTER

Hole Patterns., 143

Boolean, 237

The Boolean Operations., 237

Booleans, 187

BOOLEANS

The Difference Function., 302

Boundary Machining

Using extracted geometry., 298

Break

About the Break function., 94

Breaking geometry., 94

BRIGHTNESS OF INTERIOR LINES, 247

CAD

File Open, 18

CAM

Adding or customizing CANNED cycles and Macros., 432

All of the CAM icons., 398

Characters that can be used for making CAM/G-code conversions., 448

Complete CAM basics and icons., 398

Coordinate Format SETUP CHECKLIST., 443

Coordinate Formats., 440

Creating a NEW NC File., 405

Customizing & adding toolboxes and icons., 401

Customizing the CAM main toolbar., 401

Drilling cycles., 417

Driver Setup., 435

Line Renumbering., 412

Main Toolbar., 58

Making conversions for customizing G-Codes. Use the checklist., 451

Merging in CAM/G-Code files., 406

NC CAM main menus., 404

Opening a new NC File., 405

Opening an NC File., 405

Opening, Updating & saving NC files., 405

Perform G-Code conversions., 447

Receiving G-Code files from your machine to the software., 410

RENUMBERING lines of code., 412

SAVING CAM/G-Code files., 405

Scale Factor., 441

Selecting POST PROCESSORS., 434

SENDING the g-code file to the machine., 407

Setting roughing and finishing parameters., 399

Setting up the CAM/G-Code file for sending to the machine., 406

Setting up the first & last lines of your NC program., 406

Setting up the right coordinate format for your G-Codes., 439

Setup for circular motion and G-Code., 438

The CAM file menu., 404

The CAM toolboxes., 402

The CAM Window., 401

The CONTOUR menu., 424

The Edit menu., 411

The NC-CAM's main menus., 404

The TOOL menu., 430

Turning 3D Mode on or off for different types of toolpath., 400

Cardinal

In reference to the "Cardinal" spline option., 132

The Cardinal spline option., 132

Chain

Selecting a chain of entities., 34

The description for selecting a chain of entities., 77

Chain Gap

Setting the software to select chains of entities properly or to adjust this., 35

Setting your chain gap., 35

Tips., 69

What is "Chain Gap" and how to set this up when needed., 69

Chamfer

Creating a 2D Wireframe chamfer., 117

Creating independent chamfers on a model without un-stitching or stitching surfaces., 271

For Surfaces & Solids., 263

How to draw a chamfered edge with wireframe geometry., 117

Change

The Change Menu., 81

Change the solid color, 245

CIRCULAR

Circular Plane Surface., 198

Circular Movement

G-Code output., 438

Circular Plane

Creating this type of surface., 198

CIRCULAR PLANE

How to create a circular planar surface., 198

Surface Creation., 198

CLEAN CIRCLE, 161

Using the clean circle function for creating this type of toolpath., 161

CLEANUP

Cleaning up and optimizing geometry., 90

CLEARANCE

Bring Tool to Clearance., 399

CLOCKWISE, 122

In relationship to arcs., 122

CLOSE, 383

Automatically close geometry chains., 383

Close un-trimmed geometry into a chain., 383

COLLISION DETECTION

Turn ON or OFF., 466

Using Collision Detection for Lathe Programs., 487

Color

Changing general colors in the environment dialog., 72

How to modify the screen and render colors., 72

Selecting entities by their color easily., 35

Color Setup., 104

Designating specific colors for specific NC Codes., 415

Main system colors & how to modify or change them., 72

COLOR PALLET, 245

Colors

How to modify the softwares basic colors., 104

Commands

Existing Scripting commands., 516

COMPENSATION

How to add cutter compensation to a G-Code program you create through menu item selection., 424

Setting up tool compensation in the CAM Contour Menu., 424

Cone, 192

How to draw a solid cone., 192

What is a cone?, 186

CONFIGURATION

Changing post configurations in the CAM., 434

The CAM setup menu., 434

CONNECT, 327

Connecting existing geometry with 3D toolpath., 327

CONNECT AT Z, 333

CONSTANT RADIUS FILLET:

Creating a constant radius filleted part by preselecting it and then selecting the fillet function., 263

CONTINUOUS

Drawing continuous lines., 119

CONTOUR

Attributes., 97

Controls

The UCS Controls., 41

Conversion

NC Code conversion., 447

CONVERSION

Converting codes and characters so that you have a matching post processor., 447
conversion process, 448

Coordinate System

The drawing screen., 28

Basics of the coordinate system X, Y & Z Axis., 28

Coordinates

Using the UCS., 40

Coordinates

Mirroring geometry using coordinates., 83

COORDINATES

Changing from inches to millimeters etc., 69

Drawing arcs., 122

Setting up the coordinate format of the NC Program., 439

COUNTER CLOCKWISE

Selection direction for drawing fillets., 127

CROSS SECTION, 337

Surface Creation., 211

Cross Section Surface

How to create cross section surfaces., 211

CROSS-SECTION FRONT VIEW, 337

Cube, 186

How to draw a solid cube., 190

What is a cube?, 186

Cube Command box, 191

Cube Dimensions, 190

CUSTOMIZE TOOLBAR, 30

CUSTOMIZING

Creating a Lathe Tool., 467

CUT ABOVE Z=0, 333

CUT ALL, 400, 417

CUT SINGL, 400

CUTTING DEPTH

Lathe Tool Setup., 466

CYCLE

The Cycle menu in the CAM., 426

Cylinder, 187, 193

How to draw a solid cylinder., 193

What is a cylinder?, 187

Defaults

Setting up the basic software defaults., 73

The general system defaults., 73

DEMO MODE

After Installation., 13

DEPTH SETTINGS, 399

DGN Files

Importing *.dgn files from earlier revisions of

BobCAD-CAM software., 20

DIFFERENCE

Boolean Operation., 240

The Boolean Difference function., 240

Digits

Dialog Digits in the Environment settings., 69

Dimension

Auto Dimension method., 363

Automatically dimension an entire drawing., 363

How to modify a dimension., 361

DIMENSION

Attributes., 100

Dimension Attributes., 100

DIMENSION ARROWS

Attributes., 102

Dimensions

Modifying the dimension arrows through the

Attributes box located in the Change menu., 102

The complete Dimensions Menu., 360

The Dimensions menu., 360

The Dimensions Menu., 360

DIMENSIONS menu

Brief description., 26

DIRECTION ARROWS

Turn them on and off., 72

Directional arrow

Turning ON and OFF the selection directional arrows., 490, 495

DISPLACEMENT

Pasting geometry., 78

DIVIDE, 456

Drawing grid axis, 27

DRAWING OPTIONS

Render, 312

Drawing screen, 26

Drawing Screen

Basics of the drawing screen., 26

Drill, 309

DRILL & CYCLES, 417

DRILL CYCLES

How to customize them for your machine., 427

Drilling

The Cycle menu in the CAM., 426

Various ways to create toolpath and then use drilling operations., 306

Driver

CAM setup drivers., 435

DRIVER, 435

DRIVER SETUP, 437

DWG

Information regarding importing DWG files., 22

DWG Files

Importing DWG files., 22

DXF

File Open, 18

Information and tip regarding DXF files., 20

DYNAMIC PANNING & ZOOM

For use in the Render window., 312

DYNAMIC ROTATION

For use in the Render window., 311

Edge Lines, 245

Edges

Render, 314

EDIT

The Edit Menu., 76

ELEMENTS

Select Elements/Entities in a dialog., 78

Ellipse

Creating an ellipse., 130

ELLIPSE

How to create an ellipse., 130

Environment

The Environment Dialog in the File Menu., 64

The main software environment dialog box and various settings that can be made to the software., 64

ENVIRONMENT

Modifying the UCS., 54

System defaults & main Settings., 64

Equalize, 319

Explode, 371

Extending

Extending & Trimming wireframe geometry., 38

Extending & Trimming

Using Modify Mode., 38

External CAD Files

Importing them., 20

EXTRA NC, 402

EXTRACT ALL ARC EDGES

For use with drilling cycles., 299

Extract an ARC, 188

Extract Edges

Creating wireframe geometry from extracting surface and solid edges., 297

Extrude Curve

Creating extruded surfaces and solids from wireframe geometry., 200

EXTRUDE SURFACE

Surface Creation., 202

FACE

- Creating a FACE., 183
- How to create different color faces., 183

Facing

- Lathe Facing Operation Lesson., 487

Field Width, 440

File

- The File menu., 62

File Extensions

- Different file types that V21 supports., 19

FILE menu

- Brief Description., 26

FILE OPEN, 62

FILES of TYPE

- File Open., 18

Fillet

- Creating fillet arcs on wireframe geometry., 124
- Creating independent fillets on surfaces and solids., 269
- Creating independent fillets without un-stitching a model., 265
- For surfaces & Solids., 263

FILLET

- Creating arc fillets., 124

FIND/REPLACE

- CAM functionality., 411

Fit

- In regards to the "Fit" spline option., 133

Flip Last, 390

FOLLOW CURVE, 342

- 3D Toolpath., 342

Fonts

- Creating solid Fonts., 374
- Single line fonts for engraving., 372

GEAR

- Creating a gear., 176
- The GEAR Function., 176

General

- Setting up the V21 general environment settings., 68

General Accuracy

- Setting the general accuracy correctly to avoid toolpath errors., 68

GENERATE NC, 383

Generate Toolpath, 188

- Understanding the Generate Toolpath function for creating machine toolpath on solids and surfaces., 272

GENERATE TOOLPATH, 289

- Procedure steps., 287

GEOMETRY FROM NC, 415

- Also known as "Backplot.", 415

Geometry Selection

- For creating planar surfaces., 200

GLOBAL

- The UCS., 44

Grid

- Setting up a snap grid., 67

Grid Axis

- Main Screen Grid Axis., 27

GROUP

- Grouping & Un-Grouping geometry., 95

Hatch

- Creating a hatch., 141

HOLE

- Creating hole patterns for drilling., 143

HOLE PATTERN

- Bolt holes and grid patterns for making holes., 143
- Creating hole patterns for use with drilling cycles., 143

Holes

- Removing holes represented by cylinders., 188

Horizontal

- Dimension method., 360

HORIZONTAL

- Dimension., 360
- Horizontal Pocket Toolpath, 150
- Toolpath Option., 466

I/D Profile

- Quick Lathe Lesson 2., 484

Icons

- CAM Icons., 398
- Main Toolbar icons., 57
- The V21 icons., 57

IGES

- File Open., 18
- Information regarding importing IGES files., 21

IGNORE HOLES

- The option for generating toolpath that avoids holes represented by cylinders., 274

INCREMENTAL

- Drawing points incrementally., 106

Initial Position

- Coordinate Format Output., 439

INITIAL POSITION

- Changing your tools initial position through the software., 441

INSERT NC, 384

- Choosing a post processor., 434
- Selecting a Post Processor prior to generating the NC program., 384

Installation Procedure, 12

- Installing Version 21 Software., 12

Interface

- The basic software interface., 26

Interior Lines, 245

INTERIOR LINES

- The brightness of surfaces and solids., 246

Interior Lines ON/OFF, 245

INTERIOR LINES ON/OFF, 246

Internal Lathe Cutting

Quick Lathe Lesson 5., 497

INTERPOLATE, 340

Interpolated

In reference to the "Interpolated" spline option., 133

INTERPOLATED

The interpolated spline option., 133

INTERSECT

Snapping points., 107

INTERSECTION, 243

Boolean Operation., 243

The Boolean Intersection function., 243

Lathe, 464

Basics, 464

LATHE

Lathe Tool Setup., 174

Lathe Tool

Lathe Tool Library & Customizing Tools., 467

Lathe Tool Setup, 464

Lathe Tools

selecting lathe tools for a tool setup., 466

Layer

Selecting entities by what layer they are located on., 35

Saving layers is done in the ATTRIBUTES, General tab., 95

Line

Drawing angled lines., 113

Drawing parallel lines., 112

The complete Line menu., 109

The Line Menu., 109

Line Continuous

How to draw continuous lines., 119

Using the Line Continuous function., 119

LINE Menu

Brief description., 26

LINEAR

Extend Surface., 236

LIVE UPDATE

Recommended for software updates and enhancements upon installation., 13

Machine

The CAM Machine Menu., 417

MACHINE HOLES, 274

Macro

Creating customized macros in the CAM., 432

MACRO

Customizing the software for your machines specific needs., 432

The CAM Macro menu., 432

Main Toolbar, 30

How to customize the main software toolbar., 30

Max Length of Division

Using the Set Path option., 319

MERGE

Merging CAD Files., 63

Merging in a Lathe Tool., 465

Mirror

How to use the Mirror function., 81

Mirroring geometry., 81

Modify

Modify a Dimension., 361

Modifying text., 374

MODIFY

Change or modify dimension attributes., 104

Modify Mode

Using Modify Mode., 37

Using the Modify Mode function., 37

MOVE TO POINT, 417

MOVE TO Z, 336

Multi-Sided Patch

How to create multi-sided patch surfaces., 218

MULTI-SIDED PATCH

Surface Creation., 218

NC CAM

Main menus., 404

NC File

File Open., 18

Opening an NC File., 18

NC OBJECT, 403

NC SAVE, 398

NC-CAM

4th Axis Conversion, 452

Decimal Digits, 440

Field Width, 440

leading zeros, 439

Trailing zeros, 439

NC-CAM XE "CAM:Setting up the right coordinate format for your G-Codes."

Decimal places, 439

NEW

Creating a new NC File., 405

New File

Opening a new file through the initial New File box., 17

NUMBER OF INTERIOR LINES, 247

O/D Profile

Lathe Lesson 1, 477, 484, 487

Quick Lathe Lesson 1., 477

Quick Lathe Lesson 4., 490

OBJECT/ACTION, 32

Selecting entities and then a function., 32

Which surface creation functions are Object-Action., 219

Offset

- A special note on offsetting geometry for manual toolpath., 138
- Creating offset geometry for use as toolpath., 138
- Offsetting surfaces and solids., 291
- Surfaces & Solids, 292

OFFSET

- How to create offset geometry., 137
- How to offset existing geometry., 137

Offset Surfaces & Solids, 188**ON ENTITY**

- How to use the function., 108
- Placing points on existing geometry., 108

OPEN CAD FILE

- New Files, 18

Options, Contour, 447**Ordinate**

- Y Ordinate dimension method., 364

OTHER Menu

- Brief description., 26

OUTPUT

- Changing your CAM output format., 438
- G-Code output., 438

Outside Diameter.

- Aspect of making a gear., 176

Parasolids

- Opening these types of files in Version 21., 24

Pick & Match

- Method of selecting one radius size and having all of the same radius sizes select at the same time., 35
- Using this feature to click on an entity and select everything on that entities layer., 35

PICK & MATCH COLOR

- Selecting entities with the same color at once., 36

PICK & MATCH LAYER

- Selecting all entities, surfaces or solids on a specific layer at one time., 36

PICK & MATCH LINE WIDTH

- Selecting lines with the same thickness or line width., 36

PICK & MATCH RADIUS

- Selecting arcs with the same radius., 35

PIPE THREAD

- Creating spiral pipe thread style toolpath., 175

Pitch Diameter.

- The gear function., 176

PIXELS

- Changing the preview pixel size., 71

Planar

- Creating this type of surface., 199

- Using existing geometry to create surfaces in the X-Y plane., 199

PLANAR

- Creating Planar Surfaces., 198
- Planar Surface Creation., 198
- Toolpath for solids & surfaces., 272

PLANAR SURFACE

- Surface Creation., 198

Planer Cut Options, 273**Planer Finish Options, 273****Pocket**

- How to create pocket toolpath., 149
- Spiral Pocket Toolpath & lesson., 152
- Tapered wall pocketing toolpath., 162
- Tapered Wall Pocketing., 162
- Tapered wall spiral pocketing., 162
- The different pocketing functions., 149
- Creating pocketing toolpath., 149
- Spiral Pocket Lesson., 154**

Point

- Modifying the attributes of points., 97
- Point Attributes., 97
- The complete Point menu., 106

POST PROCESSOR

- How to select post processors., 434

Pressure Angle.

- Making a gear in BobCAD., 176

Primitives, 186, 187, 188

- The available primitive solids in BobCAD software, 186

Primitives Defined, 186**PRINT**

- Printing Files & Print Setup., 63

Print Template, 17**Profile**

- Automatic Profiling., 418
- The Profile Wizard., 418
- Using the Version 20 Profiling Wizard., 418

RADIAL, 343

- Creating radial toolpath for surfaces and solid models., 278
- Toolpath from Wireframe geometry., 343

Radial Cut Options

- Control options for radial toolpath., 278

RADIAL SURFACE, 343**RADIUS**

- Dimension a Radius., 363

RAM, 12**RAMP TOOL, 400****RAPID**

- Setting the speed to RAPID., 400

RECEIVE FILE

- NC CAM, 410
- Receiving an NC File in the CAM., 410

RECORD SCRIPT, 395

RECTANGLE

Drawing rectangles., 136

RECTANGULAR

How to create a rectangular planar surface., 197

Region

Selecting entities by region., 35

REGISTER

After Installation., 13

RELATIVE

Using the "ON-Entity" Function., 109

Remove File, 16

Remove Holes

Removing Holes from a part for toolpathing., 306

REMOVE HOLES

Removing holes for creating toolpath that avoids drill holes represented by cylinders., 306

Render box, 190

RENUMBER

Lines of G-Code., 412

Reorganize, 90

The reorganize options for cleaning up geometry., 90

REORGANIZE, 90

Reset

Resetting the UCS., 42

Back to factory defaults., 64

Resolution

Render, 314

Restore File, 16

Reverse Surface, 235

REVOLUTION, 344

Toolpath from wireframe geometry., 344

REVOLUTION SURFACE, 344

REVOLVED

Surface Creation., 203

Revolved Surface

How to create a revolved surface., 203

This is an Object-Action / Action -Object function., 208

Rhino CAD

Direct File support from Rhino CAD software
3DM files into V20., 23

Rhino CAD 3DM Files

Importing Rhino CAD 3DM files directly into
V20., 23, 24

Root Diameter.

Aspect of making a gear., 176

Root Fillet Radius

Aspect of making a gear., 176

Rotate

How to use the Rotate function., 86

ROTATE THE UCS, 43

RS 232

Review your machine handbook for correct RS
232 settings., 409

Sending the NC File., 407

The "Send File" box., 407

RS 232 Communications SETUP

Complete checklist to help you setup the CAM.,
442

RULED

Creating toolpath., 340

SAT/ACIS Files

Importing *.sat files in V21., 23

Save

Saving your files., 62

SAVE AS, 63

Saving CAD Files., 63

Screen

Modifying the main screen to suite your
requirements., 70

SCREEN

Environment Settings., 70

SCRIPT, 395

From the NC-CAM Menu., 395

Record Script., 395

Scripting

Alternative ways to program lathe parts., 496

Introduction to scripting in BobCAD-CAM., 514

SCRIPTS

Programming Lathe parts using scripts., 482

Select Elements

How to select multiple entity types through one
dialog., 79

Select Entities, 76

Using the pull-down menu options for selecting
entities., 76

Selection Toolbox, 33

SEND FILE

NC CAM., 407

SEND NC, 398

SET PATH, 318

Used in creating a Skin Toolpath on 3D
geometry., 318

SET SURFACE, 329

SET THE PATH

The first thing to do when creating a skin surface
toolpath., 321

Setup, 367

Setting up your dimensions., 367

Setting up your NC programs first and last lines.,
406

SETUP

CAM Setup functionality., 434

Setting up the NC File., 406

The CAM SETUP menu., 434

SIDE VIEW

Creating cross section surface toolpath., 339

Simulate

Basic 2D toolpath simulation in the CAD window., 178

Basic 2D wireframe simulation., 178

SIMULATE, 178

Simulation

Simulating the G-Code program., 415

SINGLE

Selecting single entities., 76

Single-Line Font

The added single line font styles., 372

SKIN

Creating skin surface toolpath., 320

Skin Surface

How to create a skin surface., 216

SLDPRT

Importing SolidWorks files., 24

Snap

Setting up snap locations on geometry., 69

Setting up your snap locations for geometry., 69

Snapping arcs with the "Snap Center" function in the ARC menu., 123

Using the Snap function to snap arcs to existing points that are already selected., 123

Using the snap grid for drawing., 28

Snapping arcs to points., 123

Using the Mirror function., 85

Basics of the snap grid and how to customize it., 28

SNAP GRID, 28

SNAP POINT

Environment Settings., 69

Snap To

Using this function for snapping geometry., 41

SNAP TO

UCS function., 41

Software Interface

The basic software interface., 26

Solid Fonts, 375

Examples of solid fonts when using Boolean Operations., 375

Setting up your cutting variables., 305

Solids

Attributes, 103

How to change the attributes of solids and surfaces through the V21 Attributes box in the Change menu., 103

Selecting solids and surfaces., 37

The complete solids menu., 187

The Primitives defined., 186

The Solids Menu., 187

SOLIDS

Attributes., 103

Solids & Boolean Operation icons, 315

Solids Menu, 187

SOLIDS Menu

Brief description., 26

SPECIAL

The complete Special/NC CAM menu., 382

SPECIAL/NC CAM

Brief description., 27

Selecting a Post Processor., 304

Special/NC-CAM

The Menu, 382

SPECIAL/NC-CAM, 382

SPEED

Set the tool speed to RAPID., 400

Sphere, 186

How to draw a sphere., 188

What is a sphere?, 186

Spiral

Creating a spiral pocket toolpath., 152

SPIRAL

Creating Spiral Toolpath., 175

How to make spiral pocketing toolpath., 152

Toolpath for solids and surface type parts., 284

Spiral Toolpath Options

Control options for creating Spiral toolpath., 284

Creating Spiral Toolpath., 284

Spline

Creating a simple spline-arc., 179

Creating splines by sketching continuous lines., 134

The available spline options., 130

The complete Spline menu., 130

Using spline geometry for offset and toolpath., 135

Using the spline function in the OTHER menu., 179

SPLINE, 179

Spline Creation

V21 spline options., 131

STEP

File Open., 18

Information regarding importing STEP files., 22

STEP Files

Importing Step files into V21., 22

STITCH SURFACES TO SOLID, 295

How to use., 295

Must use for surfaces before creating toolpath., 293

STITCHING

Surfaces & Solids, 295

Working with Offset Surfaces, 293

Support ON-Demand

Extended technical support services., 10

Surface

Creating Surfaces., 196

The surface creation options are located in the SOLIDS menu under "Create Surface.", 196

SURFACE, 340

Creating a surface from wireframe., 198

Surface Creation

Different methods of creating surfaces., 196

The surface creation box., 197

SURFACE/GENERATE TOOLPATH RULE

When to use the Stitching function for multiple surfaces., 293

SURFACES

Render, 313

SWEEP

Surface Creation., 211

Sweep Surface

How to create a sweep surface., 211

System Requirements, 12

For Installation., 12

Minimum and Recommended., 12

Tangent

Drawing tangent lines., 111

TANGENT

Drawing Tangent Lines., 111

TANGENTIAL

Making arcs tangential., 91

Taper Wall Pocketing

How to use the V20 system for tapered wall pocketing., 162

Technical support, 10

First 45-days., 10

Technical Support

Information on technical support and training services., 10

Information regarding technical support., 10

Technical Support ON-Demand, 10

Templates

Different design templates., 17

Opening special pre-loaded print templates in the V21 software., 24

Text

How to change the attributes of text which also applies to dimension text., 103

Modifying text., 373

Text Attributes., 103

The procedure for working with text., 372

The text box located in the dimensions menu., 368

TEXT

Attributes., 103

Breaking up text so that it can be machined., 369

Fitting text to a path., 370

The Solids Menu, 186

THREAD, 175

Through Holes

Avoiding holes when creating toolpath for surfaces and solids., 273

Toggle Grid

Tool

The CAM Tool Menu., 430

The Tool Technology tab for setting up simulation., 414

TOOL

Open up the CAM "Tool Technology" box., 413

The "Bring Tool Down" icon., 399

The CAM Tool Menu., 430

The Tool Depth Settings icon., 399

TOOL SETUP., 413

Turn the tool display on or off., 400

Tool Setup Option

For LATHE., 174

Tool Tips

Turning on and off the tool tips feature., 29

Toolbar

Customizing you main toolbar., 30

Toolbars

Environment Toolbar settings., 64

Setting up your toolbars through the environment box., 64

Toolbox

The Construction Toolbox., 57

The Selection Toolbox., 57

TOOLPATH, 291

The basic procedure for creating toolpath for solid models and surface files., 287

Toolpath Extents box, 274

Toolpath Extents box., 275, 282

Tools

Lathe Tools., 467

TOOLS

How to add tools to your CAM tool menu., 430

Tooltips, 54

Torus, 187, 195

How to draw a solid torus., 195

What is a torus?, 187

Transform

Toolbox, 58

Translate

How to use the Translate function in the Change menu., 88

How to use the Translate function., 88

Trim & Extend

The trim & extend function in the Change menu., 92

Trimming

Using Modify Mode, 38

TYPE

New File Open., 17

UCS, 40

- Changing its size in the environment settings., 71
- Modifying the UCS in the Environment Settings., 54
- More on the UCS controls., 54
- Must reset the UCS before creating toolpath., 294
- Reset the UCS before creating toolpath and G-Code for models., 294
- Reset the UCS before generating toolpath & G-Code., 56
- Resetting the UCS., 387
- The User Coordinate System, 40
- Using the UCS Controls., 41

UCS Reset

- How to quickly reset the UCS after having changed it., 42

UNDO GENERATE, 411**UNION**

- Boolean Operation., 237
- The Boolean Union function., 237

Units

- Changing Parameters for Millimeters, Centimeters, Decameters in the Environmental Settings., 28
- Changing the main coordinate system from inches to centimeters, etc., 28

Un-Stitch Solid to Surfaces, 188**UN-STITCH SOLID TO SURFACES, 296**

- How to use., 296

UN-STITCH SURFACE FROM SOLID, 296**Use Tool Center**

- For boundary machining., 276, 282

USER COORDINATE SYSTEM

- Introduction to the UCS., 40

Variable Radius Fillet

- The procedure for performing a variable radius fillet., 264

Variables

- Scripting Variables., 515

Vectorize, 371

- Breaking up true type fonts so that they can be machined., 369
- Exploding text for machine operations., 369

Vectorize Text

- Turning true type fonts into text that can be cut., 369

VERIFY, 382

- Surfaces & Solids, 315
- Verify Data., 104

Verify Data

- Different options available for verifying geometry as well as modifying existing dimensions., 104

VIEW ALL, 287**View All icon, 189****VIEWS menu**

- Brief description., 27

WINDOW menu

- Brief description., 27

Y=Y*2

- Description and why you would use this setting., 69

Z FROM POINT, 336